



US012416016B2

(12) **United States Patent**
Wilson et al.

(10) **Patent No.:** **US 12,416,016 B2**

(45) **Date of Patent:** **Sep. 16, 2025**

(54) **ADENO-ASSOCIATED VIRUS (AAV)
VECTORS, AAV VECTORS HAVING
REDUCED CAPSID DEAMIDATION AND
USES THEREFOR**

(58) **Field of Classification Search**
None
See application file for complete search history.

(71) Applicant: **The Trustees of the University of
Pennsylvania**, Philadelphia, PA (US)

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(US); **Joshua Joyner Sims**,
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Philadelphia, PA (US)

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 1151 days.

(21) Appl. No.: **16/975,545**

(22) PCT Filed: **Feb. 27, 2019**

(86) PCT No.: **PCT/US2019/019861**

§ 371 (c)(1),

(2) Date: **Aug. 25, 2020**

(87) PCT Pub. No.: **WO2019/169004**

PCT Pub. Date: **Sep. 6, 2019**

(65) **Prior Publication Data**

US 2021/0123073 A1 Apr. 29, 2021

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(57) **ABSTRACT**

A recombinant adeno-associated virus (rAAV) vector com-
prising an AAV capsid having a heterogeneous population of
vp1 proteins, a heterogeneous population of vp2 protein and
a heterogeneous population of vp3 proteins. The capsid
contains modified amino acids as compared to the encoded
VP1 amino acid sequence, the capsid containing highly
deamidated asparagine residues at asparagine-glycine pair,
and further comprising multiple other, less deamidated
asparagine and optionally glutamine residues.

Related U.S. Application Data

(60) Provisional application No. 62/722,382, filed on Aug.
24, 2018, provisional application No. 62/722,388,
filed on Aug. 24, 2018, provisional application No.
62/703,673, filed on Jul. 26, 2018, provisional
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provisional application No. 62/635,968, filed on Feb.
27, 2018, provisional application No. 62/635,964,
filed on Feb. 27, 2018.

(51) **Int. Cl.**
C12N 15/86 (2006.01)

C07K 14/005 (2006.01)

(52) **U.S. Cl.**
CPC **C12N 15/86** (2013.01); **C07K 14/005**
(2013.01); **C12N 2750/14121** (2013.01); **C12N**
2750/14122 (2013.01); **C12N 2750/14143**
(2013.01)

25 Claims, 63 Drawing Sheets

Specification includes a Sequence Listing.

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FIG. 1A

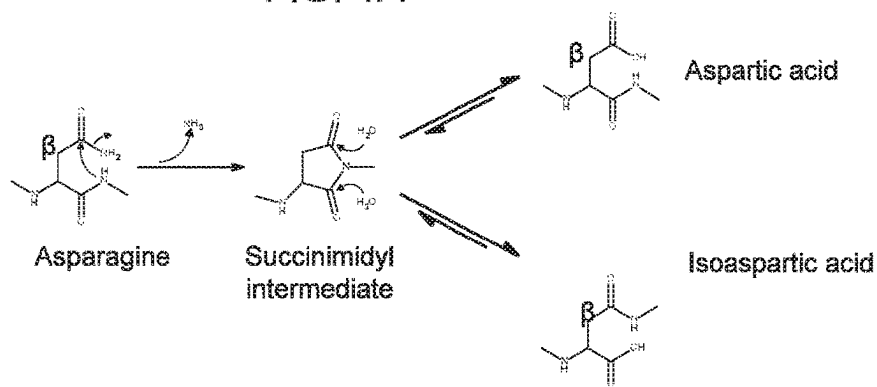


FIG. 1B

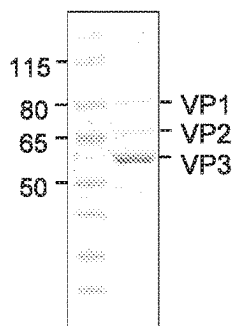


FIG. 1C

Spot	pI	Spot	pI
1	7.30	11	5.70
2	7.11	12	5.63
3	6.97	13	5.54
4	6.75	14	5.46
5	6.61	15	5.37
6	6.45	16	5.27
7	6.27	17	5.20
8	6.11	18	5.14
9	6.01	19	5.08
10	5.86	20	4.99

FIG. 1D

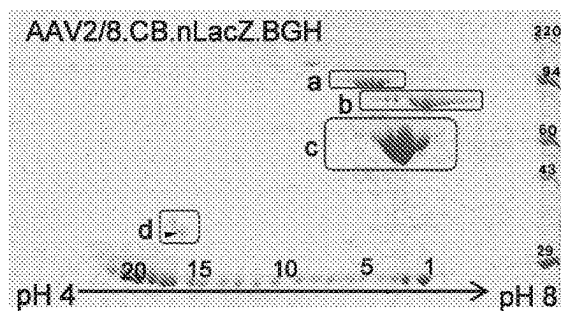
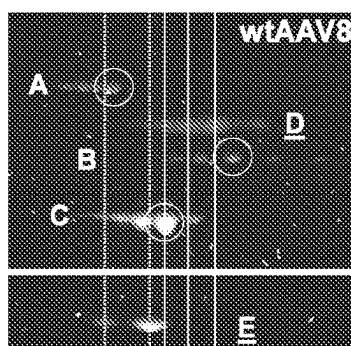
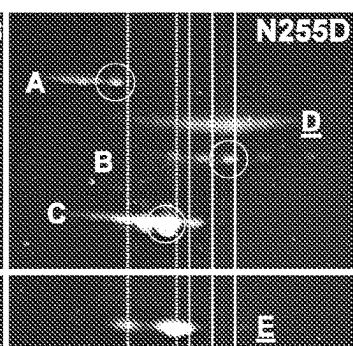


FIG. 1E



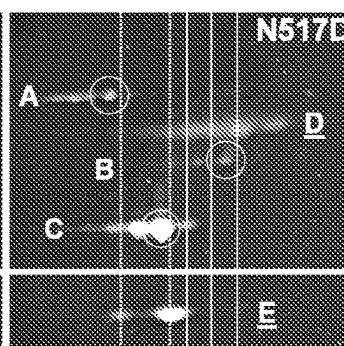
A-AAV8 VP1: 87kDa
D-Conalbumin: 76kDa

FIG. 1F



B-AAV8 VP2: 73kDa
E-Turbonuclease: 27kDa

FIG. 1G



C-AAV8 VP3: 61kDa

FIG. 2A

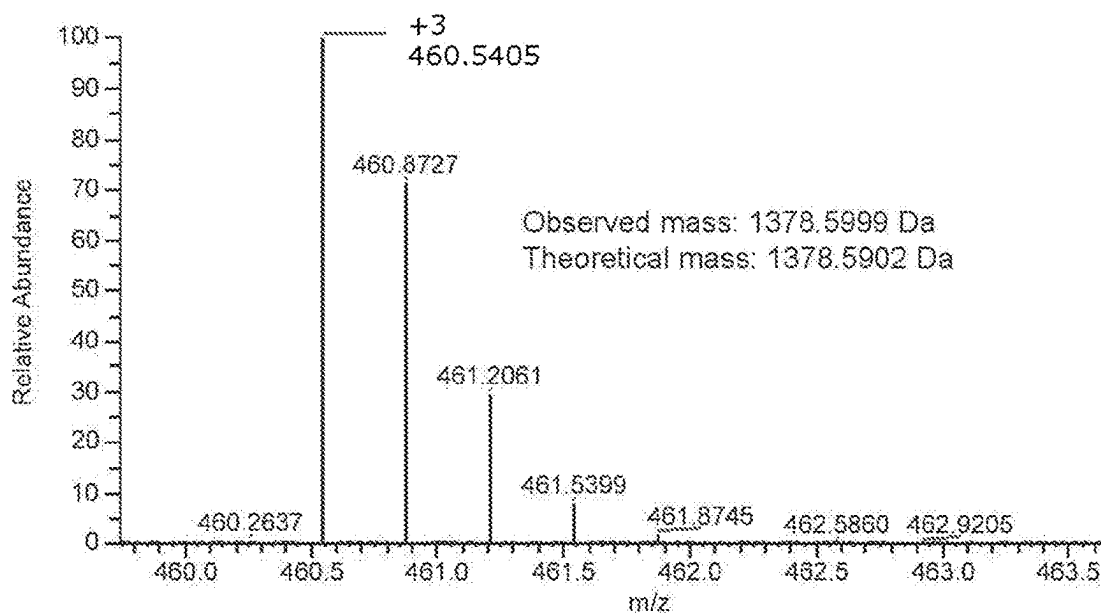


FIG. 2B

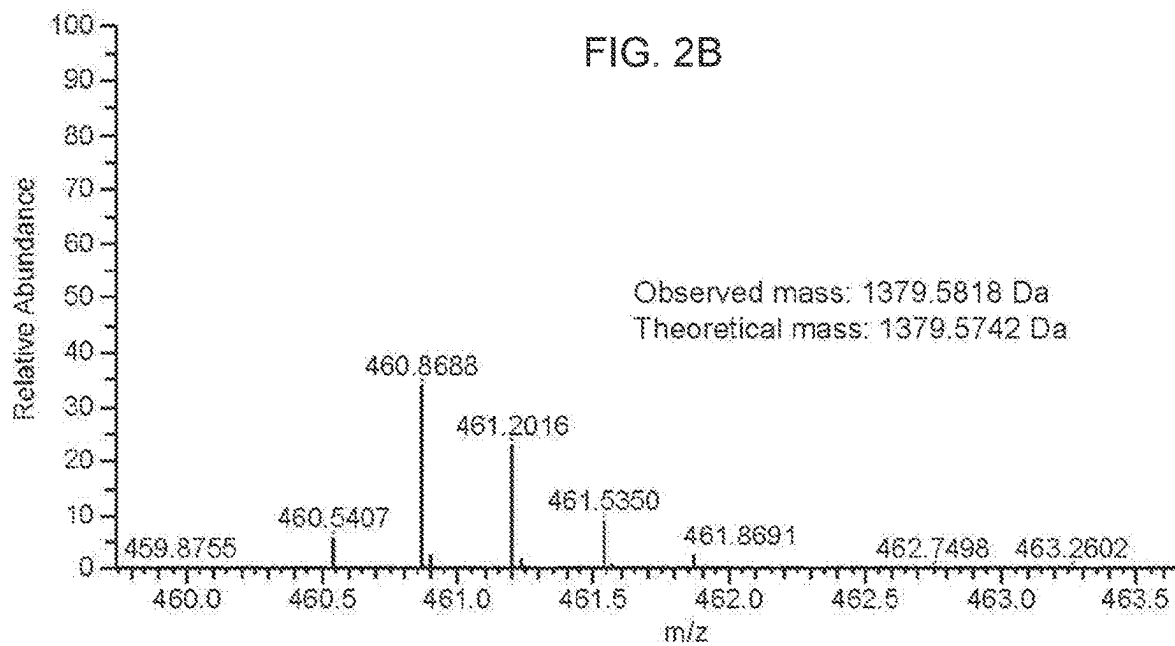
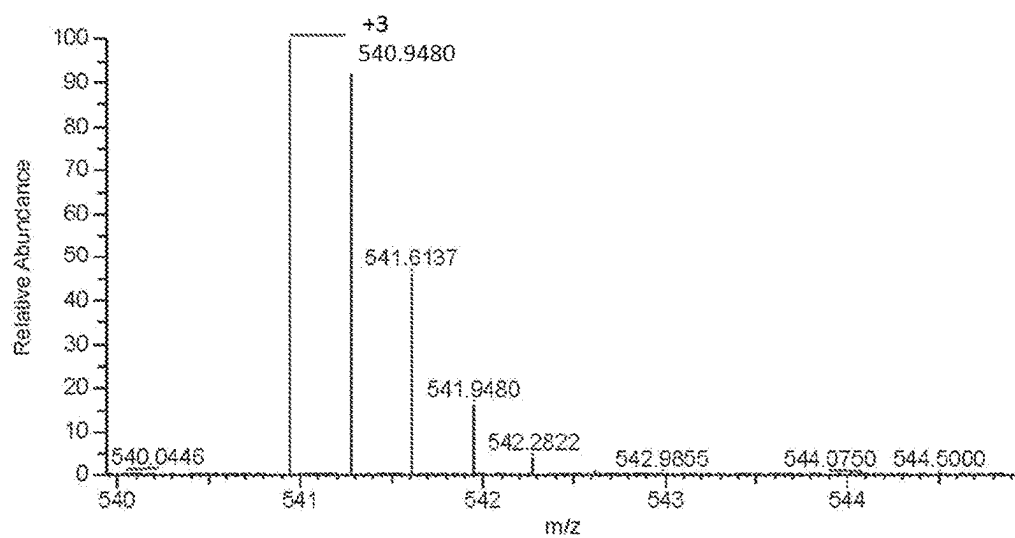


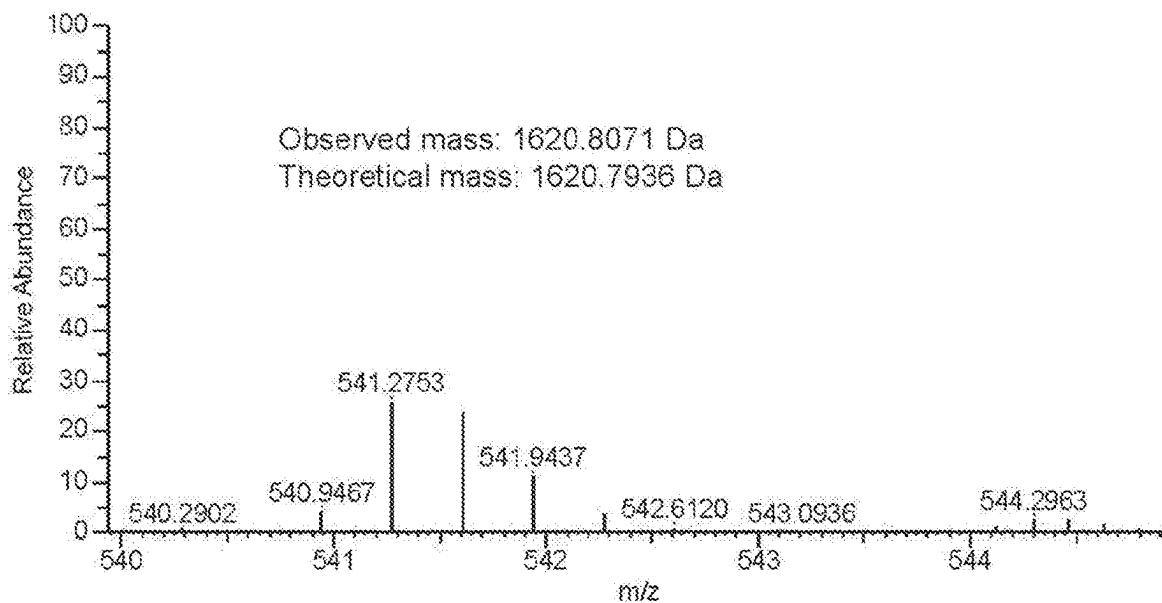
FIG. 2C



Observed mass: 1619.8213 Da

Theoretical mass: 1619.8096 Da

FIG. 2D



Observed mass: 1620.8071 Da

Theoretical mass: 1620.7936 Da

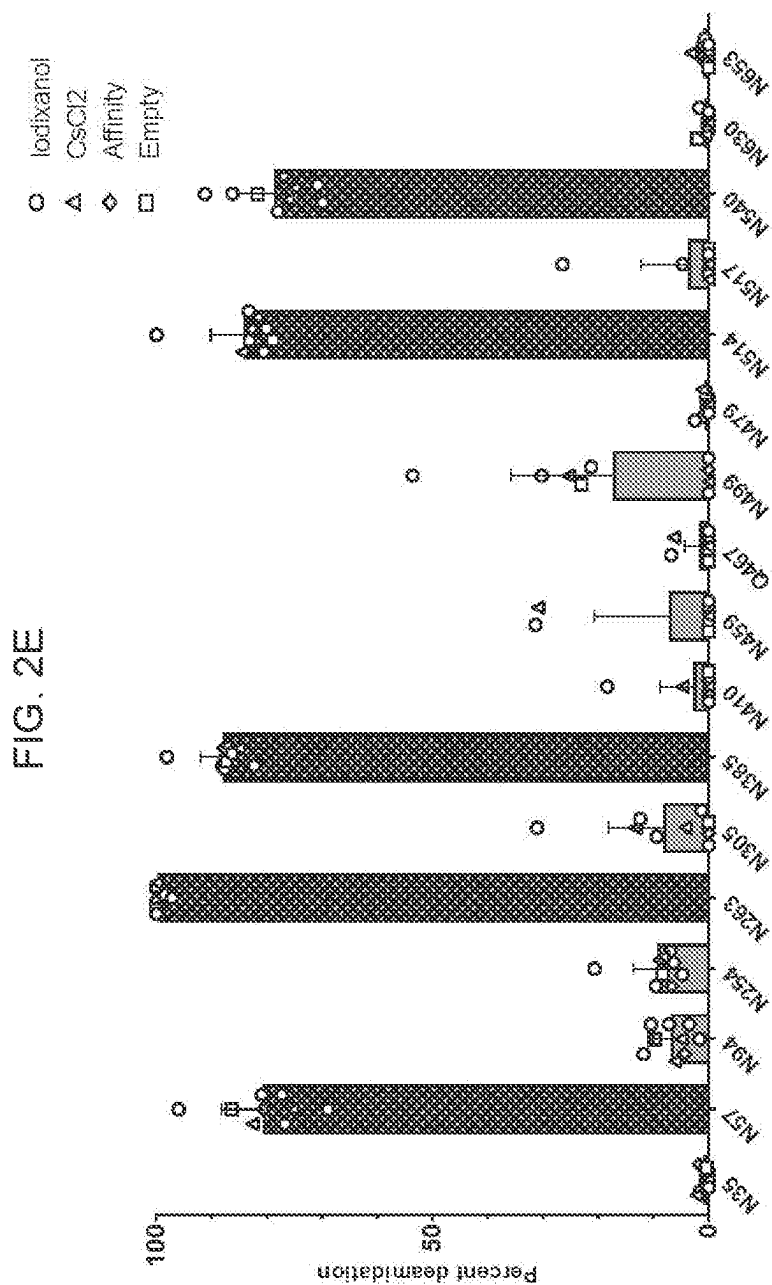


FIG. 3A

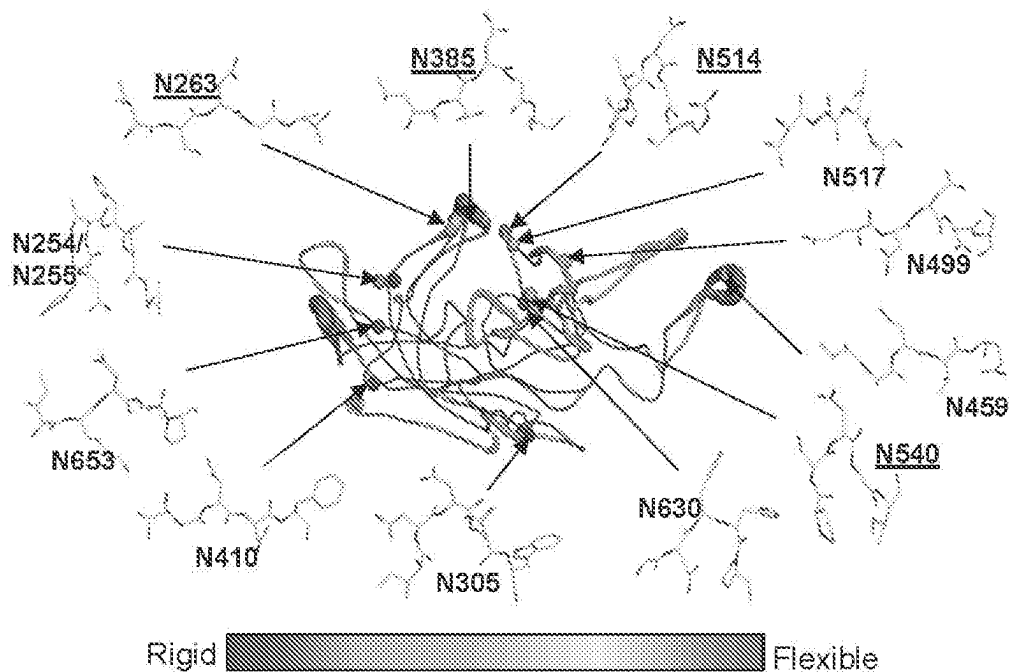


FIG. 3B

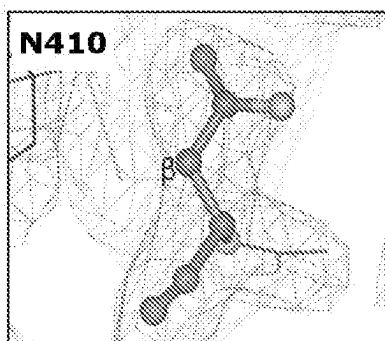


FIG. 3C

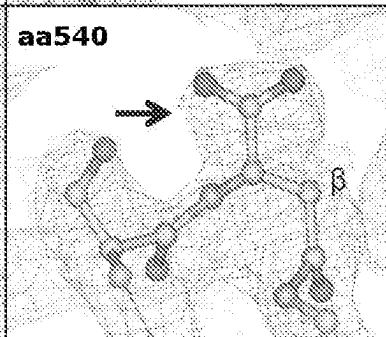
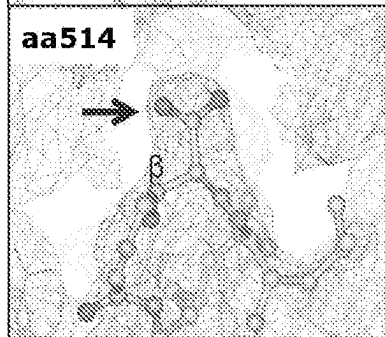
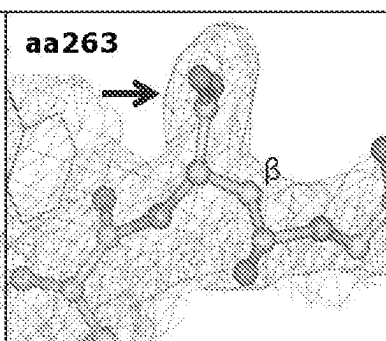


FIG. 3D

FIG. 3E

FIG. 4A

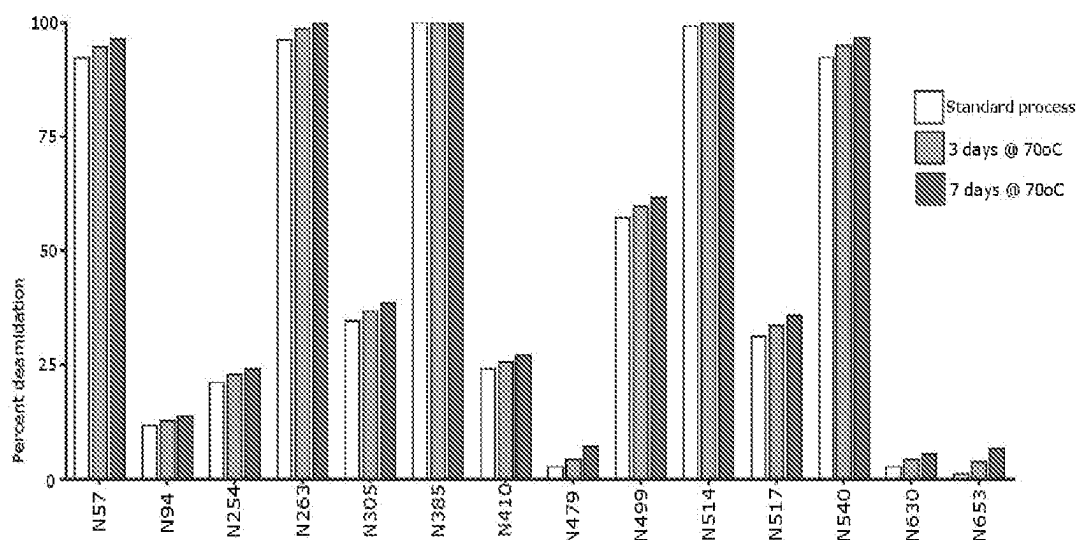


FIG. 4B

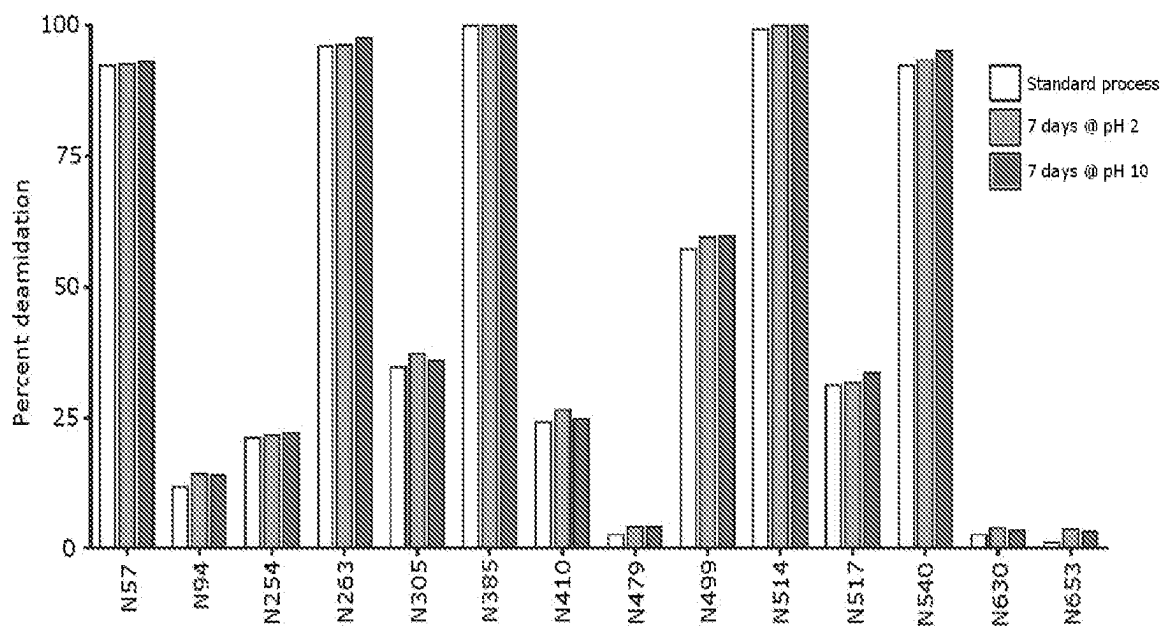


FIG. 4C

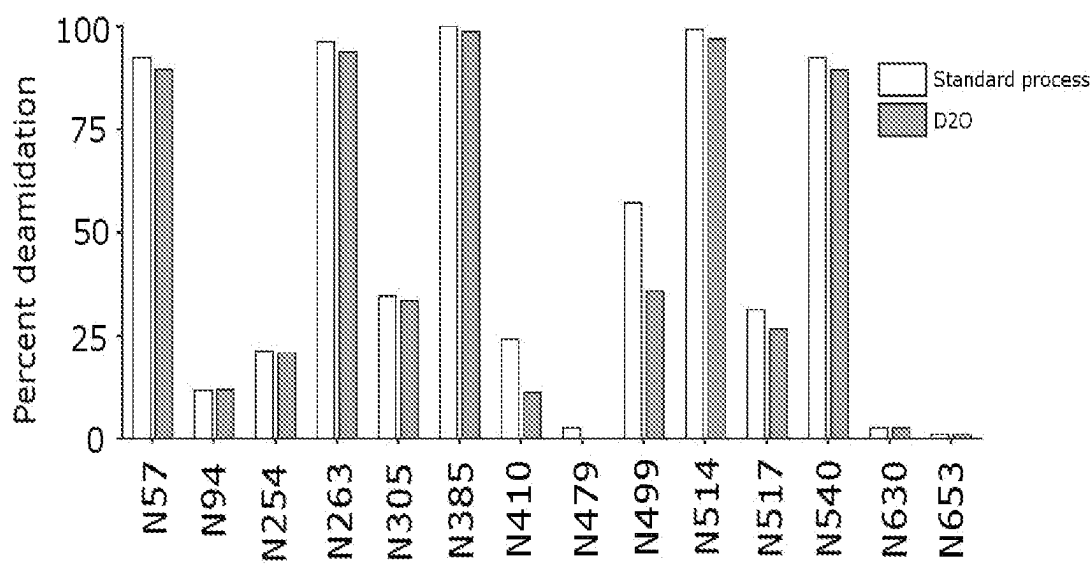


FIG. 4D

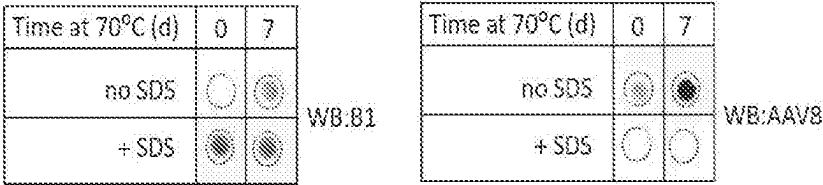


FIG. 5A

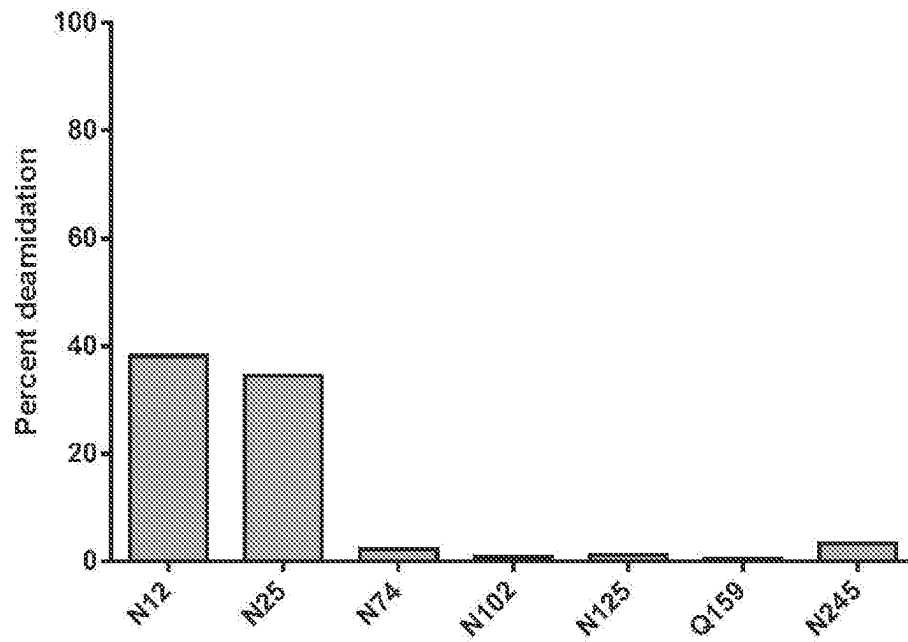
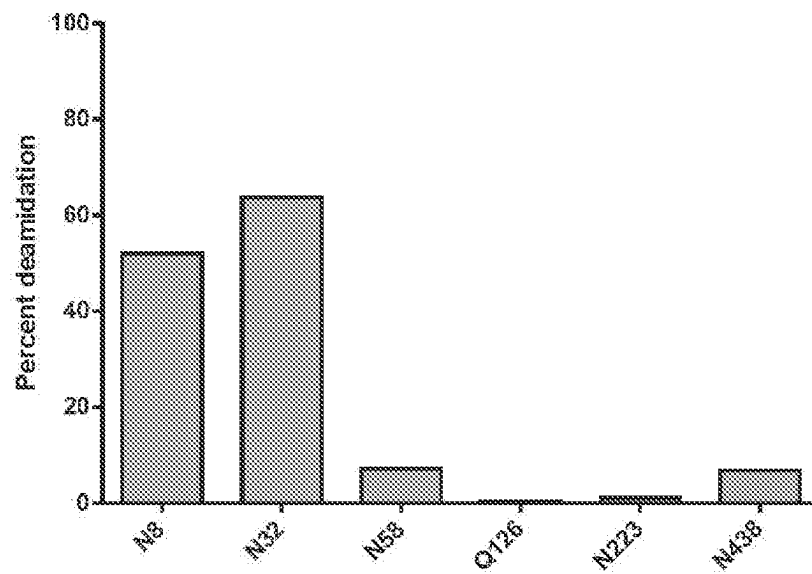


FIG. 5B



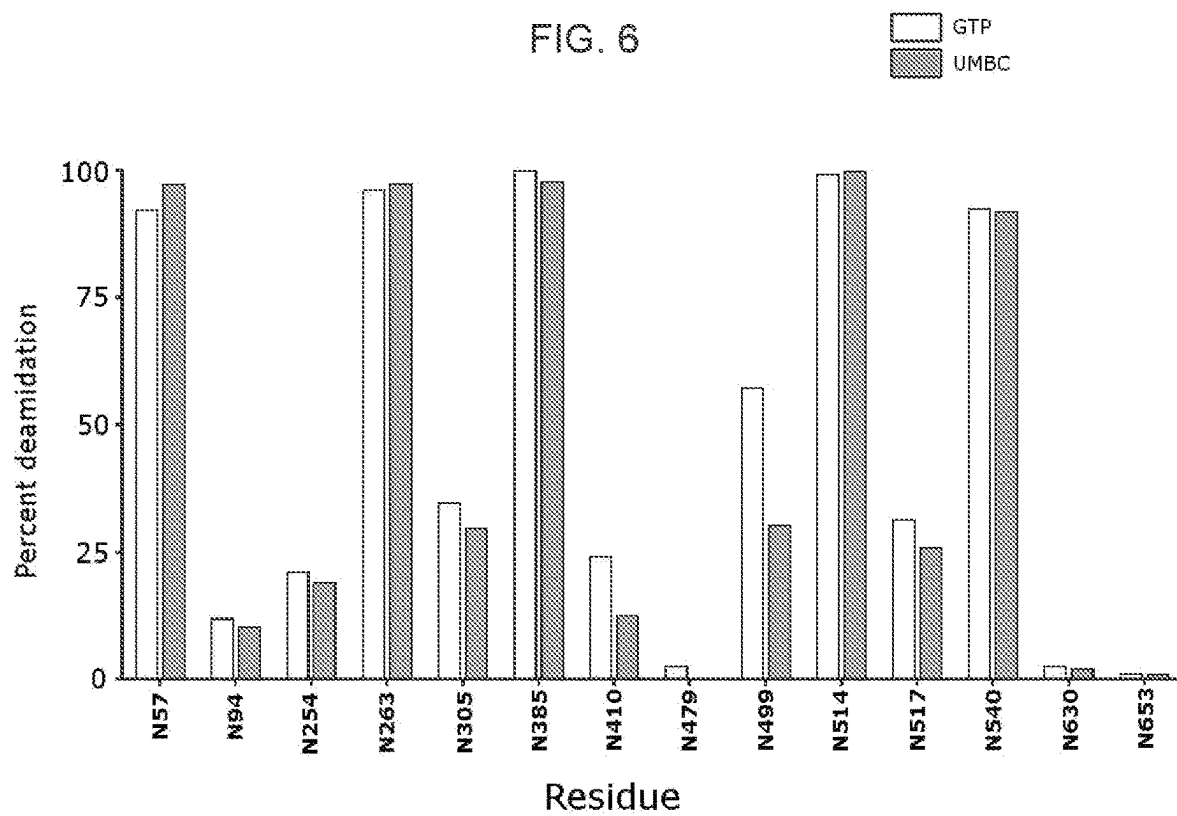


FIG. 7B

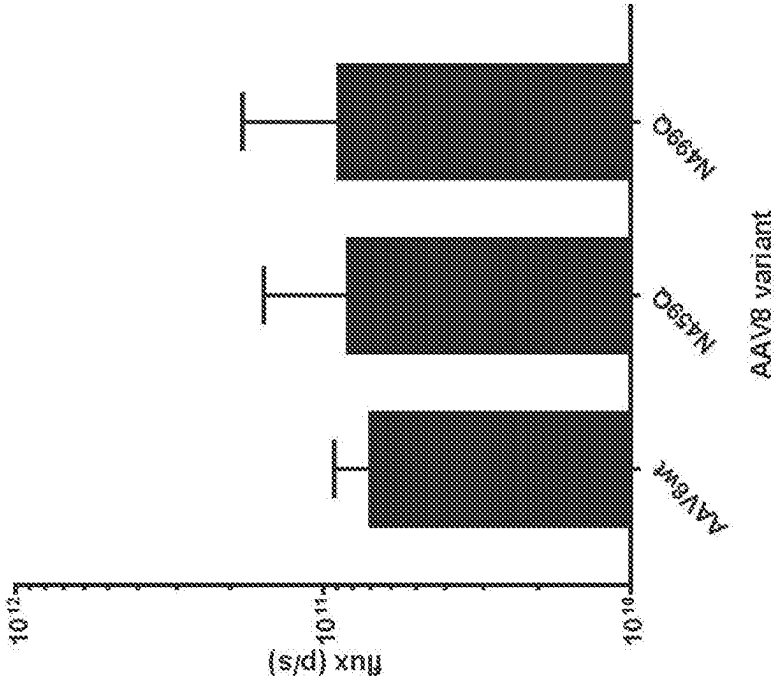
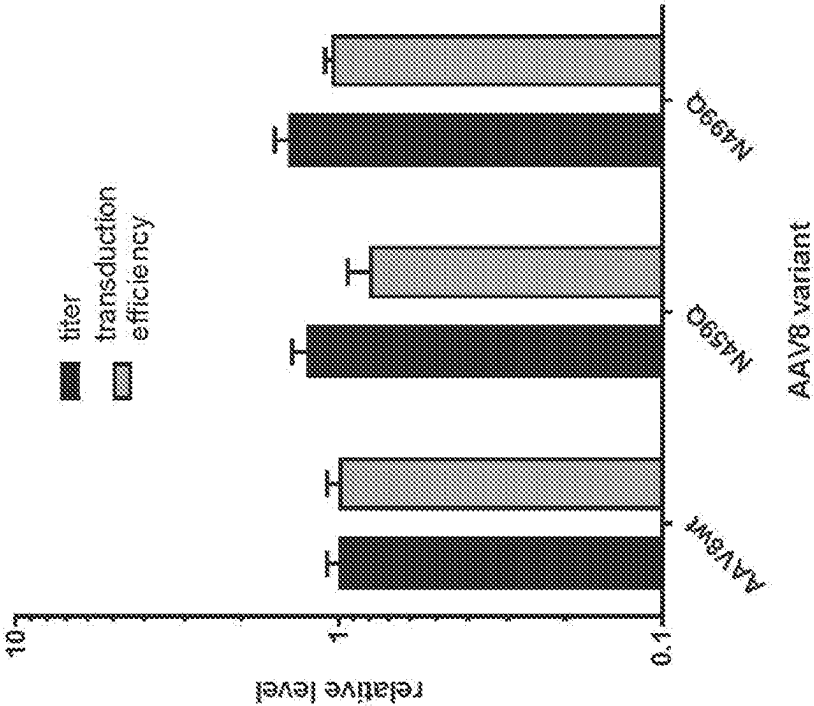


FIG. 7A



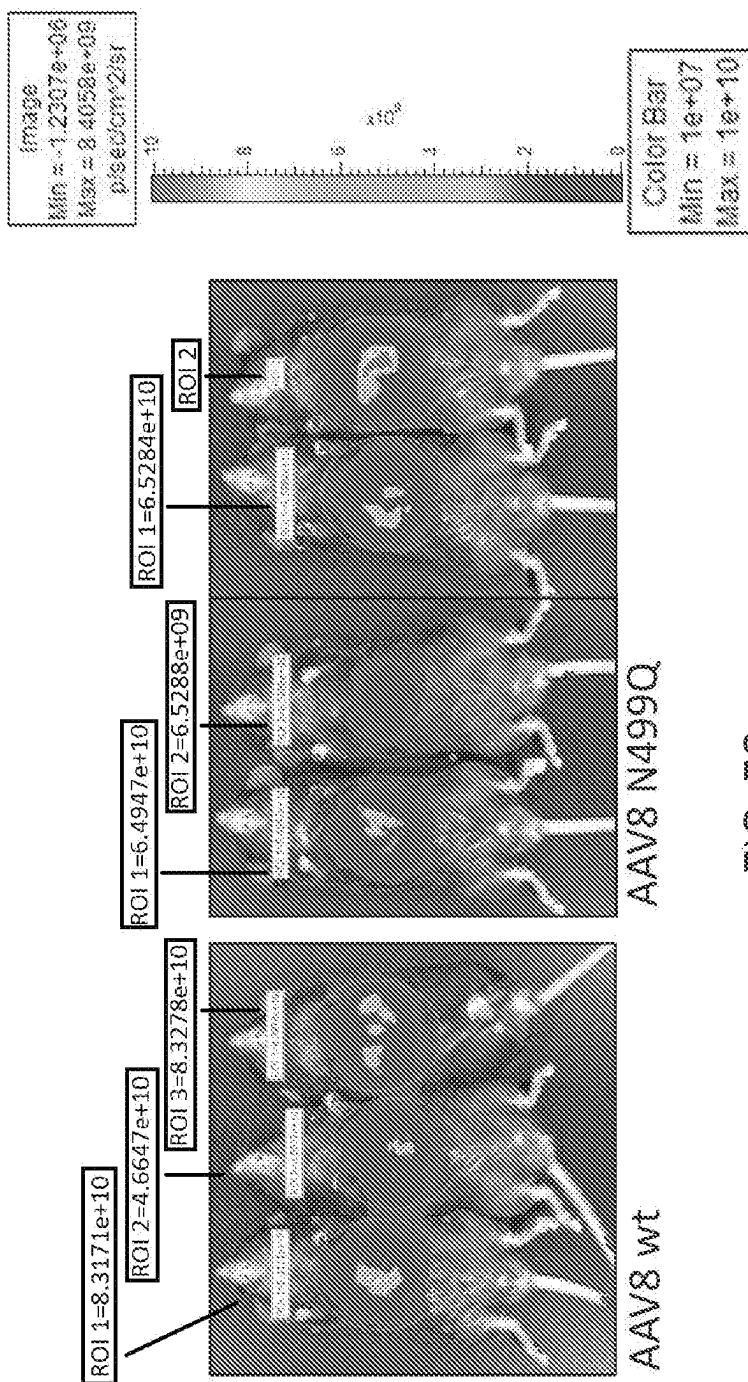


FIG. 7C

FIG. 8A

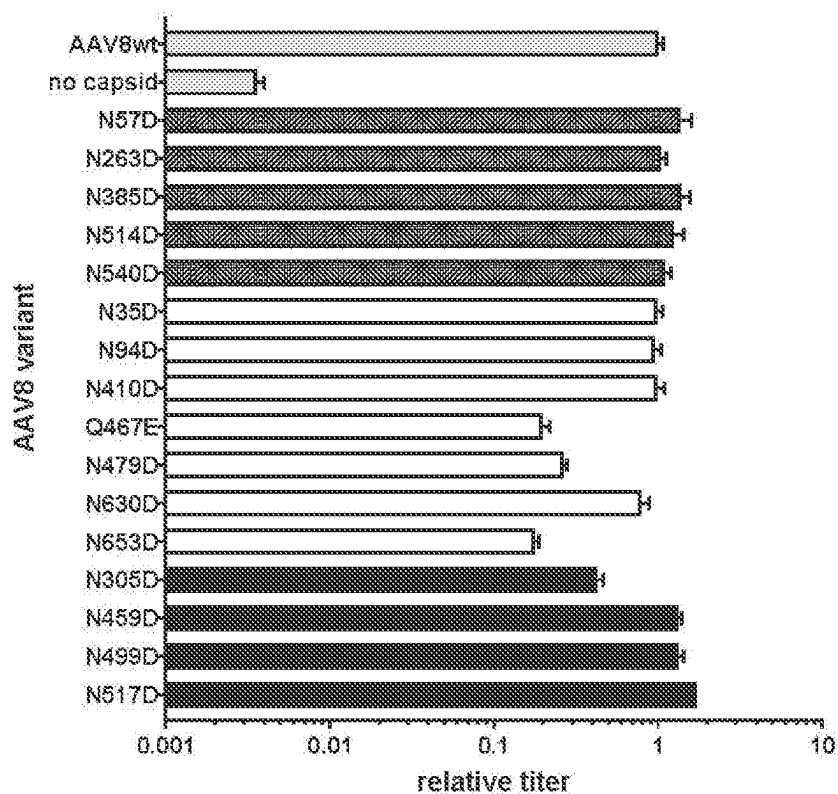


FIG. 8B

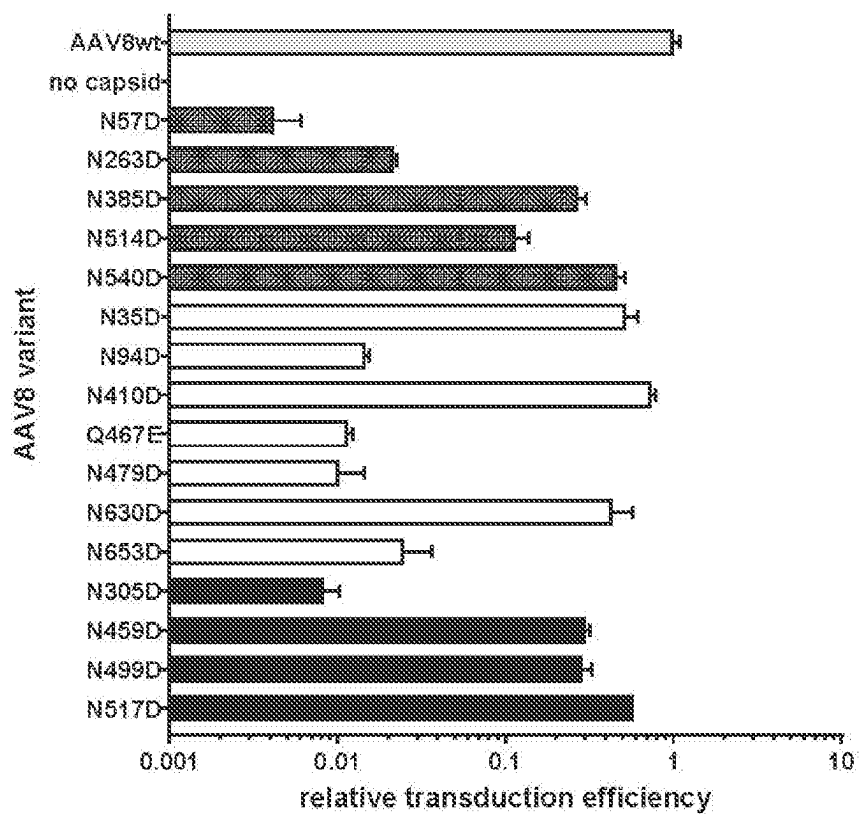


FIG. 9A

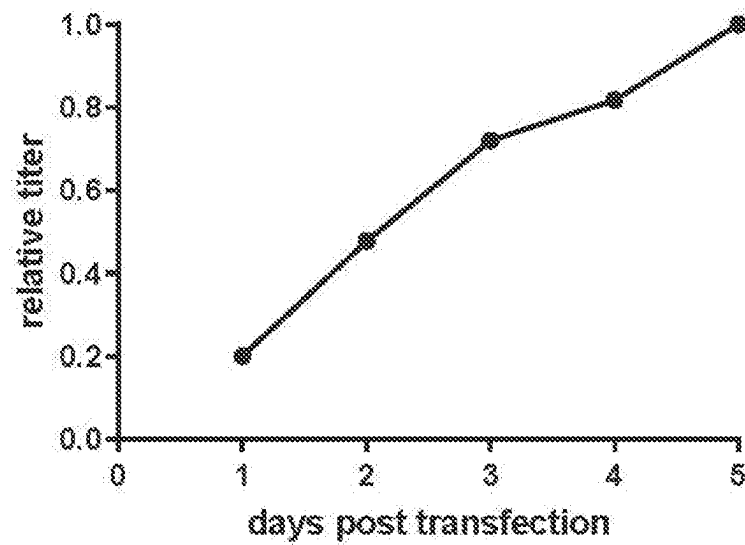


FIG. 9B

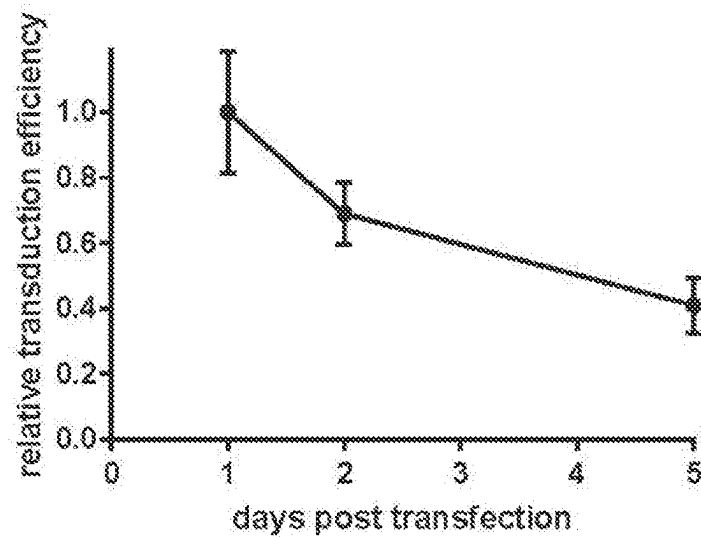


FIG. 9C

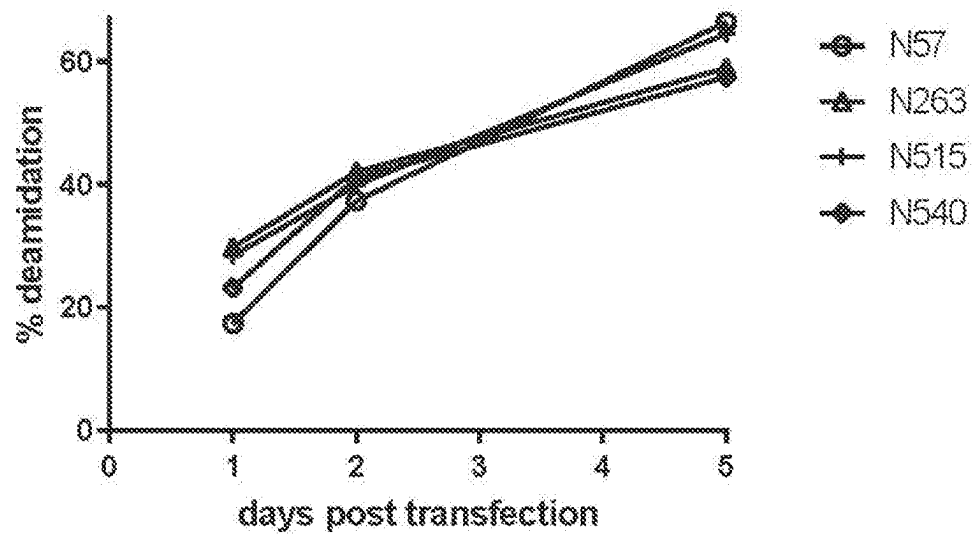
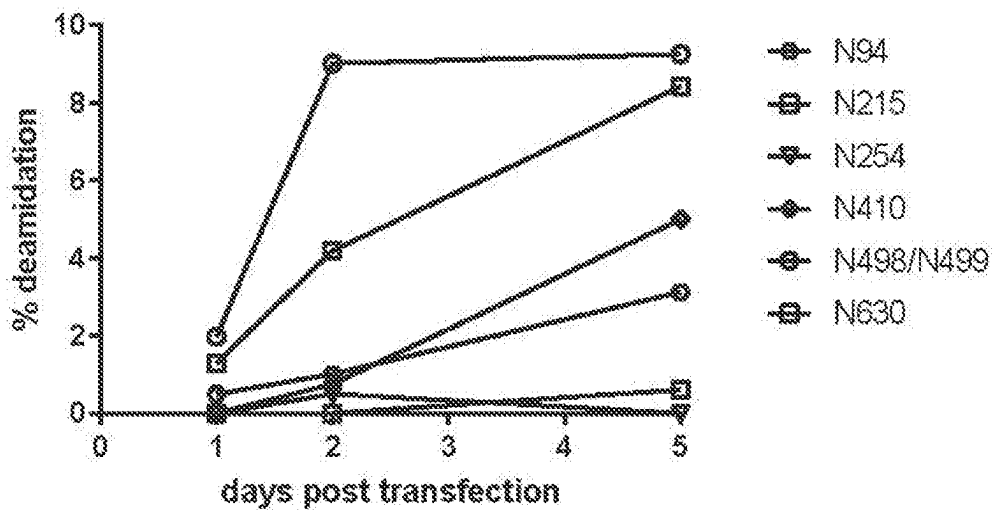
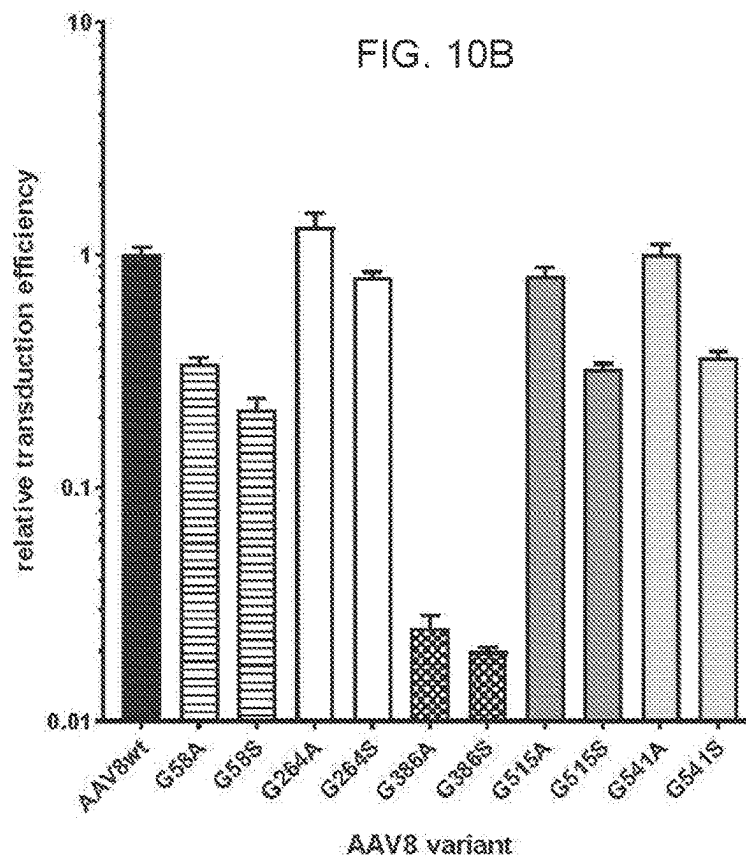
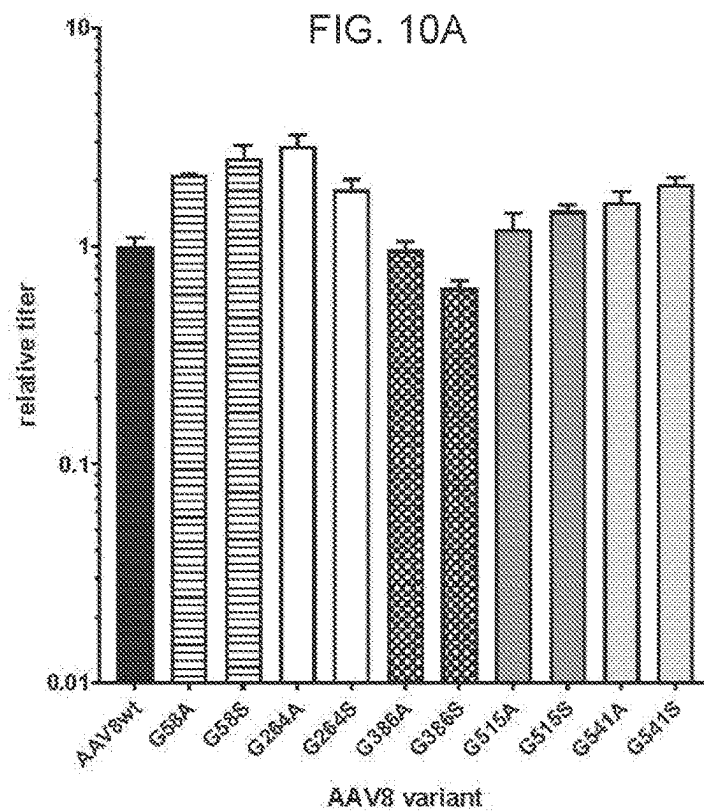


FIG. 9D





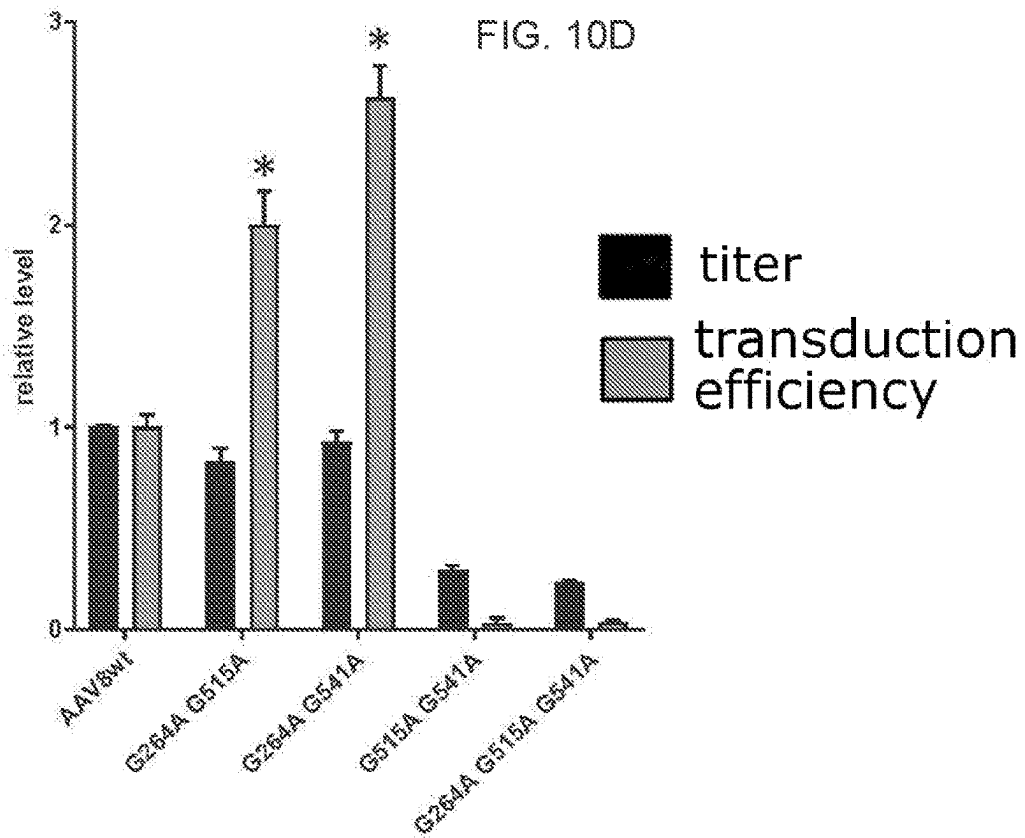
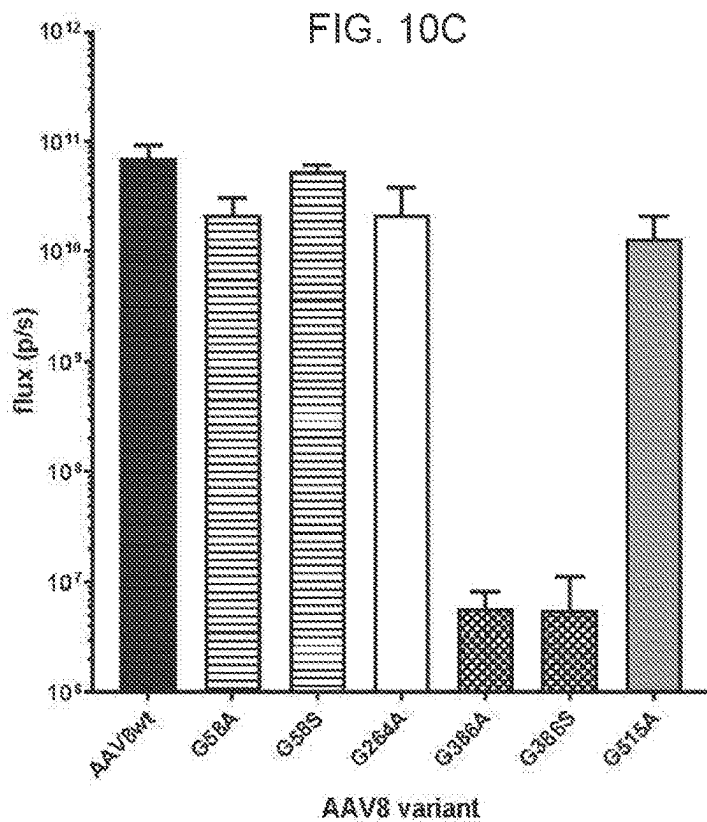


FIG. 11 A

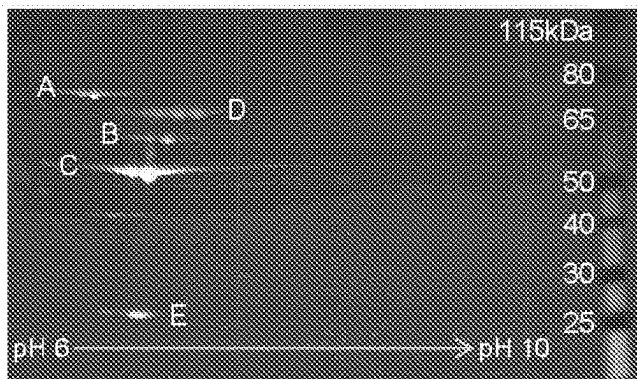


FIG. 11C

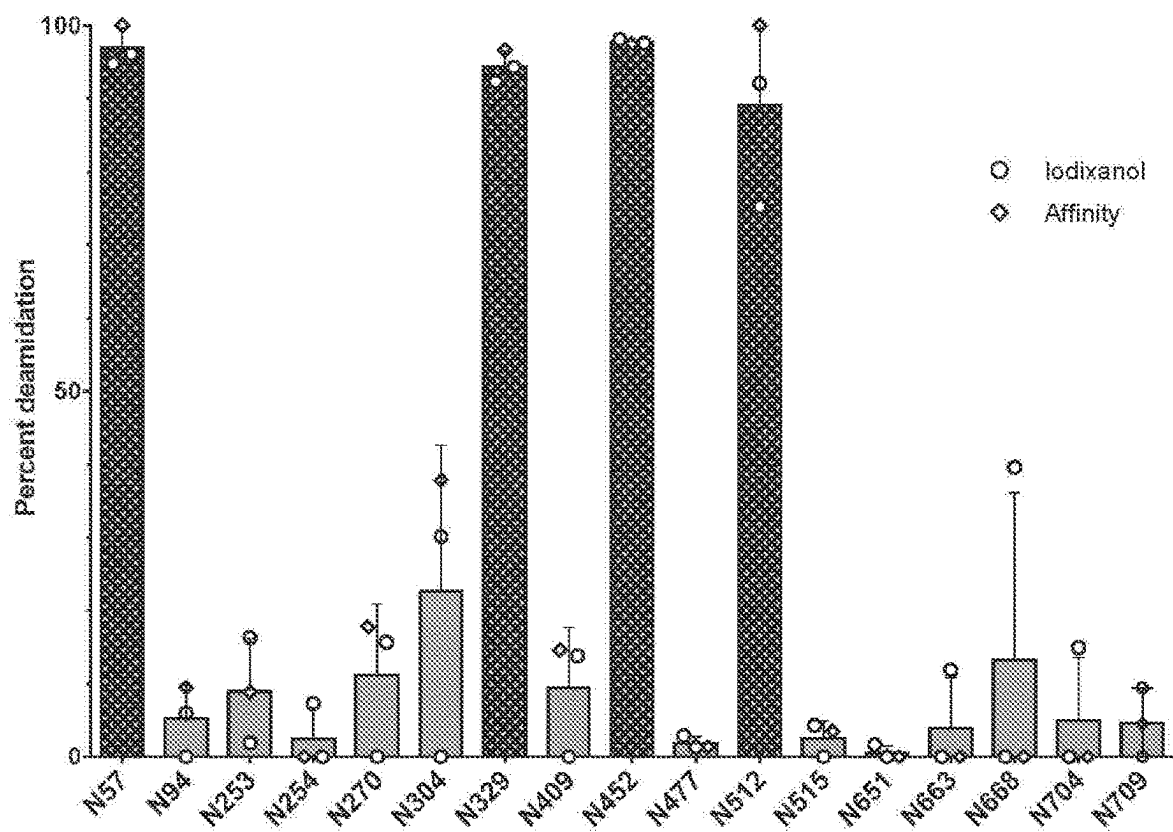


FIG. 11B

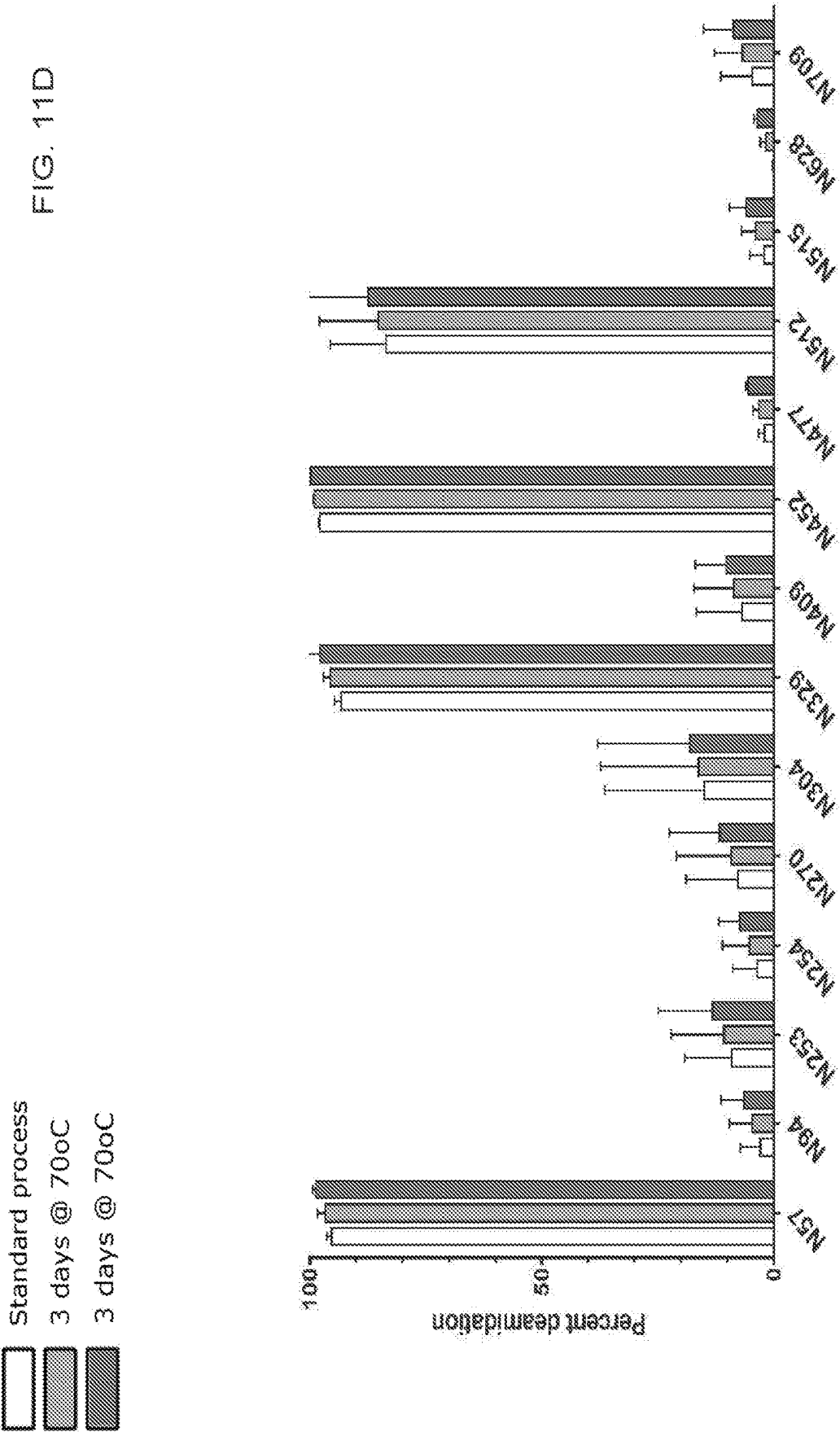


FIG. 11E

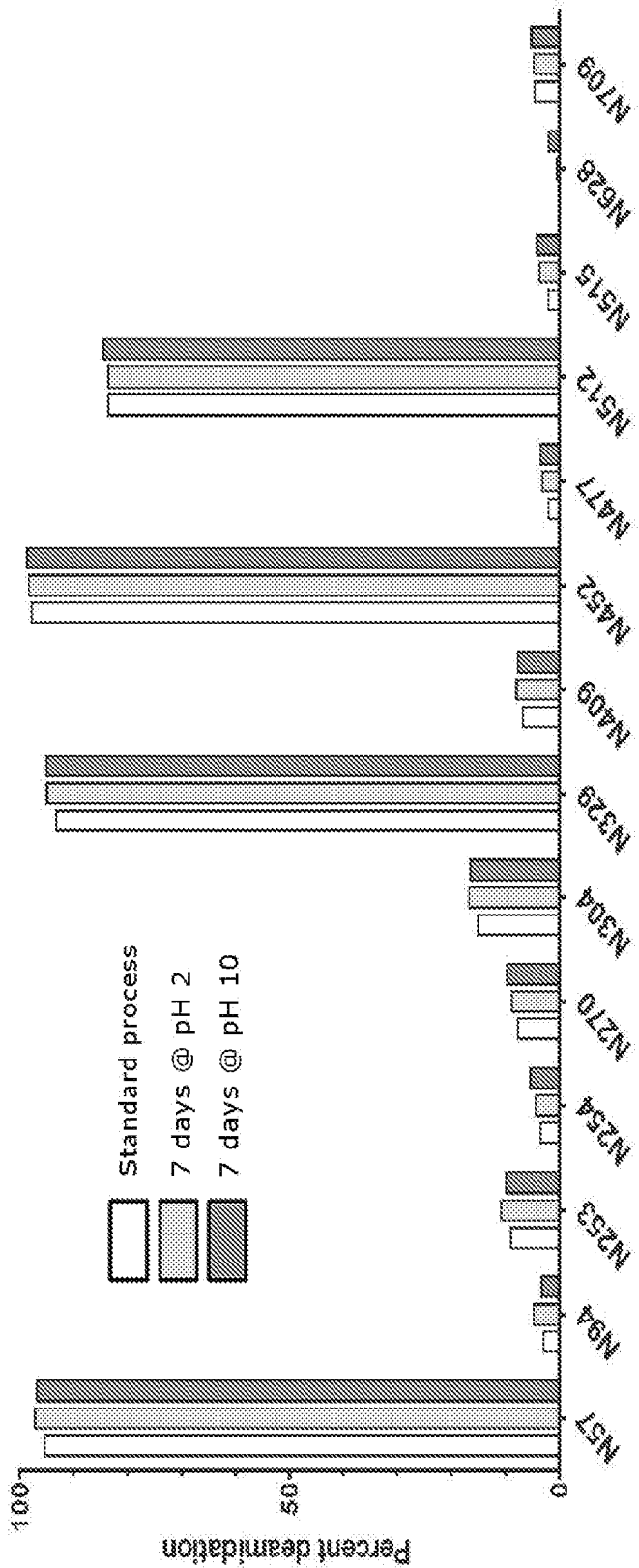


FIG. 11F

Serotype		AAV9	
Time at 70°C (d)		0	7
no SDS			
+ SDS			

FIG. 11G

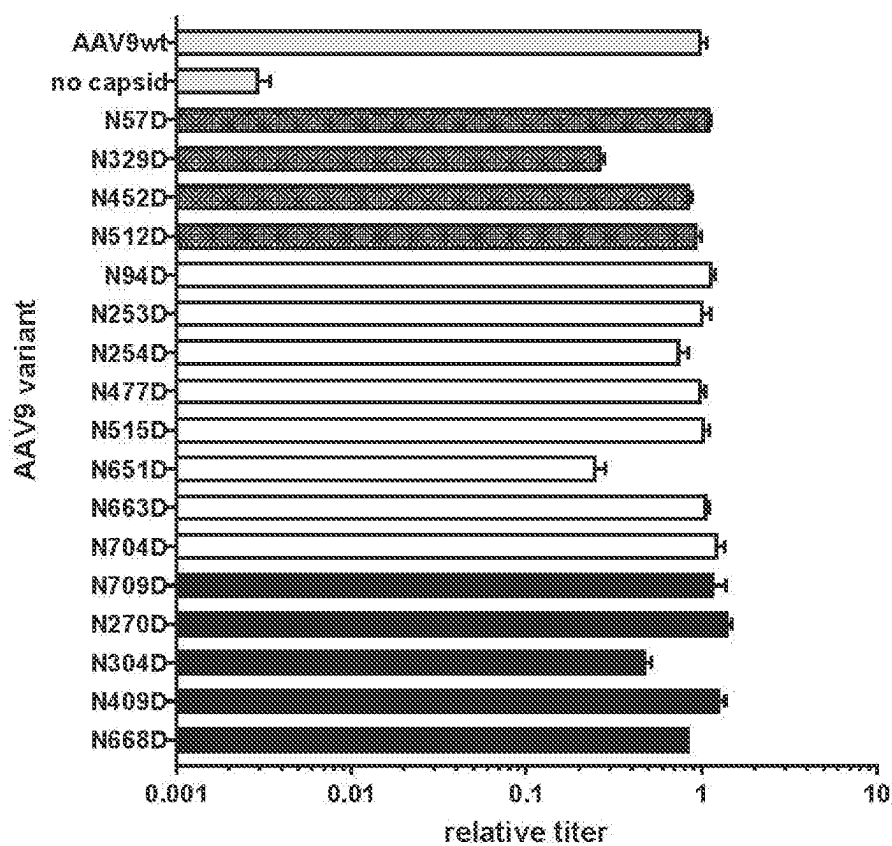


FIG. 11H

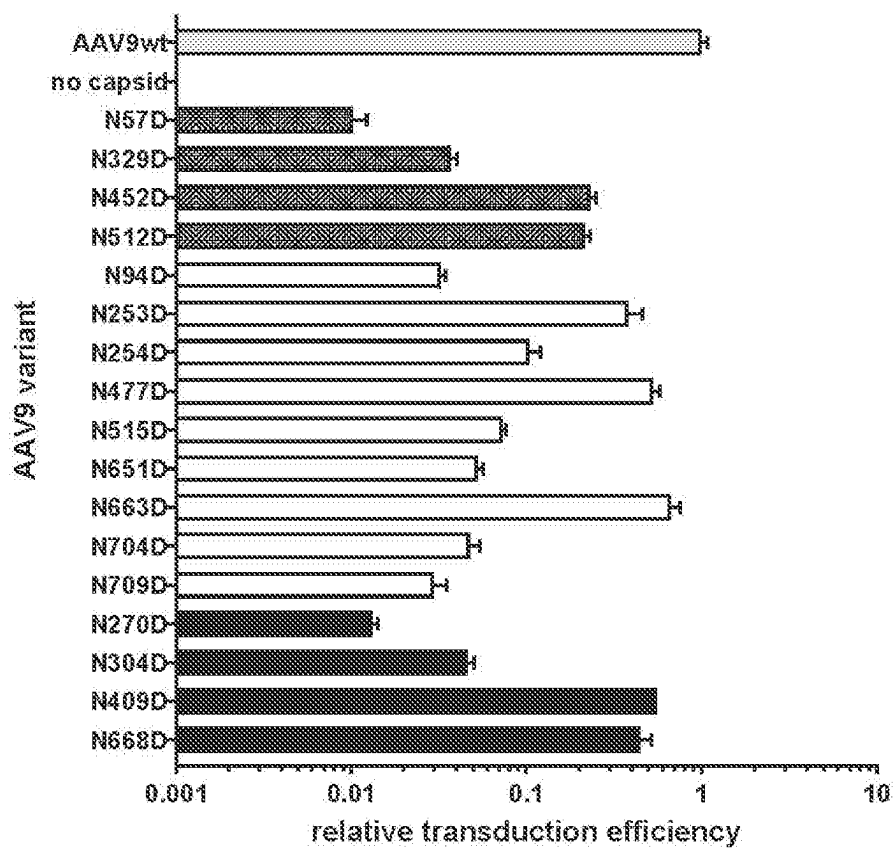


FIG. 11I

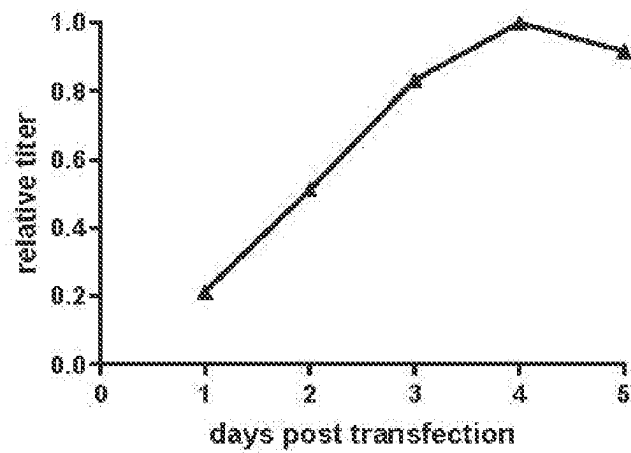


FIG. 11J

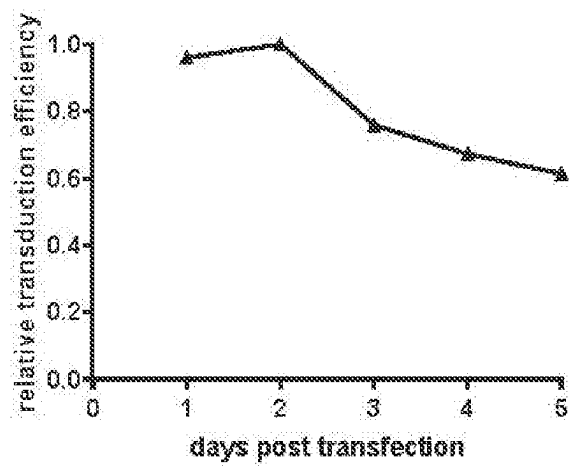


FIG. 11K

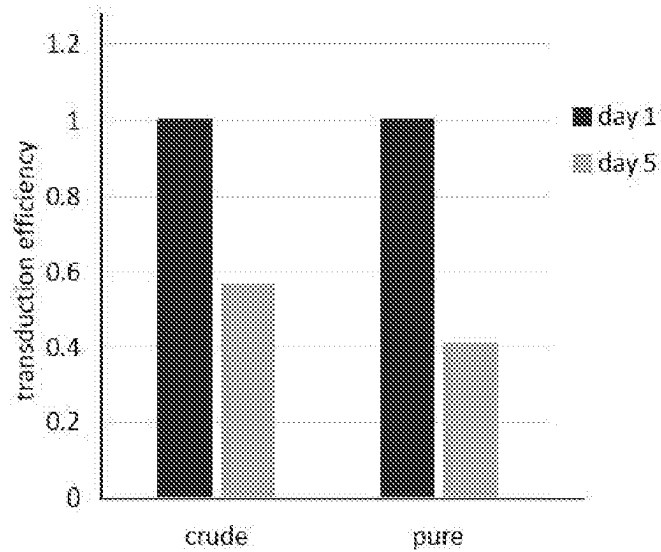


FIG. 12A (Clade F aa)

		vp1 start	
AAV5 1	MSFVDPDLEE - VEGELREFLGLIAGPPKPKPNQOQDQARGLVLPNGNGLDRGEPVNRADDEVAREHDISYNEQLEAGDNPYLKYNHADA		EF
AAV9 1	MAADGYPDWLENDLSEGIREWWALKPGAPQPKANQQHQNARGVLPGYKYLPGNGLDKGEPVNAADAAALEHDKAYDQQLKAGDNPYLKYNHADA		EF
PHP . B 1	MAADGYPDWLENDLSEGIREWWALKPGAPQPKANQQHQNARGVLPGYKYLPGNGLDKGEPVNEADAAALEHDKAYDQQLKAGDNPYLKYNHADA		EF
		vp2 start	
AAV5 101	QEKLADDTSGGNLGAVFQAKKRVLPEPFLVEEGAKTAPTGKRIDDHFFPKRKARTED - - - SKPS - - - - - TSSDAEAGPSGSQQLQIPAQPASSLG		EF
AAV9 101	QERLKEDTSFGGNLGRAVFQAKKRLLLEPLGLVEEAAKTAPGKKRPVEQSPQEPDSSAGIGKGAQPAKKRLNFGQTGDTES - VPDPPQPIGEPPAAPSGVG		EF
PHP . B 101	QERLKEDTSFGGNLGRAVFQAKKRLLLEPLGLVEEAAKTAPGKKRPVEQSPQEPDSSVIGKGAQPAKKRLNFGQTGDTES - VPDPPQPIGEPPAAPSGVG		EF
		vp3 start	
AAV5 201	ADTMASGGGAPVADNNEGADVGSSSSGNWHCDSQWLGDRTVITSTRTWALPTYNHLYKQISNSTSGSSNDNAYFGYSTPWGYFDNRFHCHFSPRDWQ		EF
AAV9 201	SLTMASGGGAPVADNNEGADVGSSSSGNWHCDSQWLGDRTVITSTRTWALPTYNHLYKQISNSTSGSSNDNAYFGYSTPWGYFDNRFHCHFSPRDWQ		EF
PHP . B 201	SLTMASGGGAPVADNNEGADVGSSSSGNWHCDSQWLGDRTVITSTRTWALPTYNHLYKQISNSTSGSSNDNAYFGYSTPWGYFDNRFHCHFSPRDWQ		EF
AAV5 301	RLINNNWGRPKRLNFKLFNIQVKEVTDNNGVKTIANNLTSTVQVFTDSDYQLPYVLGSAHEGCLPPFPADVFMIPQYGYLTINDGSQAVGRSSFYCLEY		EF
AAV9 301	RLINNNWGRPKRLNFKLFNIQVKEVTDNNGVKTIANNLTSTVQVFTDSDYQLPYVLGSAHEGCLPPFPADVFMIPQYGYLTINDGSQAVGRSSFYCLEY		EF
PHP . B 301	RLINNNWGRPKRLNFKLFNIQVKEVTDNNGVKTIANNLTSTVQVFTDSDYQLPYVLGSAHEGCLPPFPADVFMIPQYGYLTINDGSQAVGRSSFYCLEY		EF
AAV5 401	FPSQMLRTGNNFQFSYEFENVPFHSSYAHSSQSLDRLMNPLIDQYLYLSKTINGSGNQQTTLKFSVAGPSNMVQGRNYPGFSYRQQRVSTTTVTQNNNS		EF
AAV9 401	FPSQMLRTGNNFQFSYEFENVPFHSSYAHSSQSLDRLMNPLIDQYLYLSKTINGSGNQQTTLKFSVAGPSNMVQGRNYPGFSYRQQRVSTTTVTQNNNS		EF
PHP . B 401	FPSQMLRTGNNFQFSYEFENVPFHSSYAHSSQSLDRLMNPLIDQYLYLSKTINGSGNQQTTLKFSVAGPSNMVQGRNYPGFSYRQQRVSTTTVTQNNNS		EF
AAV5 501	EFAPWGASSWALNGRNSLMNPGPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNVDADKVMITNEEEIKTTNPVATESYGQVATNHQSAQ - - - - - AQAQ		EF
AAV9 501	EFAPWGASSWALNGRNSLMNPGPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNVDADKVMITNEEEIKTTNPVATESYGQVATNHQSAQ - - - - - AQAQ		EF
PHP . B 501	EFAPWGASSWALNGRNSLMNPGPAMASHKEGEDRFFPLSGSLIFGKQGTGRDNVDADKVMITNEEEIKTTNPVATESYGQVATNHQSAQ - - - - - AQAQ		EF
AAV5 601	TGWVQNQGILPGMVWQDRDVLQGPWAKIPHTDGNFHPSPLMGGFGMKHPPPPQILIKNTVPADPPTAFNKKLNSFITQYSTGQVSVEIEWELQKENS		EF
AAV9 601	TGWVQNQGILPGMVWQDRDVLQGPWAKIPHTDGNFHPSPLMGGFGMKHPPPPQILIKNTVPADPPTAFNKKLNSFITQYSTGQVSVEIEWELQKENS		EF
PHP . B 601	TGWVQNQGILPGMVWQDRDVLQGPWAKIPHTDGNFHPSPLMGGFGMKHPPPPQILIKNTVPADPPTAFNKKLNSFITQYSTGQVSVEIEWELQKENS		EF
AAV5 701	KRWNPETQYTSNYKSNVVEFAVNTGEGVSEPRPIGTRYLTRNL		EF
AAV9 701	KRWNPETQYTSNYKSNVVEFAVNTGEGVSEPRPIGTRYLTRNL		EF
PHP . B 701	KRWNPETQYTSNYKSNVVEFAVNTGEGVSEPRPIGTRYLTRNLMAADGYLPDWLEDNLSEGIREWWALKPGAPQPKANQQHQNARGVLPGYKYLGP		EF

FIG. 12B (Clade F nt)

AAVG5 1 atgtcttttgtgatcaccctccagattgg---ttggaagaagtgtgtgaaggcttcgcgagtttttgggccttgaaagcggggccaccgaaacccaaaaac
PHP.B 1 atggctgccgatggttatcttcagattggctcgaggacaaccttagtgaaggaaattcgcgagtggtggctttgaaacctggagccctcaaccccaagg
AAV9 1 atggctgccgatggttatcttcagattggctcgaggacaaccttagtgaaggaaattcgcgagtggtggctttgaaacctggagccctcaaccccaagg
hu68 1 atggctgccgatggttatcttcagattggctcgaggacaacctcagtgaaaggcattcgcgagtggtggctttgaaacctggagccctcaaccccaagg

AAVG5 101 ccaatcagcagcatcaagatcaagcccggtggtcttgtgctgctggttataactatctcggaccgggaaacggtctcgatcggagagacgtgtcaacag
PHP.B 101 caaatcaacaacatcaagacaacgctcgaggtcttgtgcttcgggttacaaataccttggaccgggcaacggactcgacaagggggagccggtcaacgc
AAV9 101 caaatcaacaacatcaagacaacgctcgaggtcttgtgcttcgggttacaaataccttggaccgggcaacggactcgacaagggggagccggtcaacgc
hu68 101 caaatcaacaacatcaagacaacgctcggggtcttgtgcttcgggttacaaataccttggaccgggcaacggactcgacaagggggagccggtcaacga

AAVG5 201 ggcagacgaggtcgcgcgagagcacgacatctcgtacaacgagcagcttgaggcgggagacaacccctacctcaagtaacaaccaacgacgcgagttt
PHP.B 201 agcagacgcggcgccctcgagcacgacaaggcctacgaccagcagctcaagccgggagacaacccctacctcaagtaacaaccaacgacgcgagttc
AAV9 201 agcagacgcggcgccctcgagcacgacaaggcctacgaccagcagctcaagccgggagacaacccctacctcaagtaacaaccaacgacgcgagttc
hu68 201 agcagacgcggcgccctcgagcacgacaaggcctacgaccagcagctcaagccgggagacaacccctacctcaagtaacaaccaacgacgcgagttc

AAVG5 301 caggagaagctcgcgcgacacacatccttcgggggaaacctcggaaaggcagctcttccaggccaagaaagggttctcgaaccttttggcctggttgaaag
PHP.B 301 caggagcggctcaaaagaagatcgtcttttgggggcaacctcggcgagcagctcttcaggccaaaagggtcttgaacctcttggctggttgagg
AAV9 301 caggagcggctcaaaagaagatcgtcttttgggggcaacctcggcgagcagctcttcaggccaaaagggtcttgaacctcttggctggttgagg
hu68 301 caggagcggctcaaaagaagatcgtcttttgggggcaacctcggcgagcagctcttcaggccaaaagggtcttgaacctcttggctggttgagg

AAVG5 401 agggtgctaagacggccctaccggaaaagcggatagacgaccactttccaaaaagaaaggcccggaagagagactcca-----a-----
PHP.B 401 aagcggctaagacggctcctggaaaagaagagccctgtagagcagctctcctcaggaaacgggactcctcgggggtattggcaaatcgggtgcacagccgc
AAV9 401 aagcggctaagacggctcctggaaaagaagagccctgtagagcagctctcctcaggaaacgggactcctcgggggtattggcaaatcgggtgcacagccgc
hu68 401 aagcggctaagacggctcctggaaaagaagagccctgtagagcagctctcctcaggaaacgggactcctcgggggtattggcaaatcgggtgcacagccgc

AAVG5 501 -----gccttccacctcgtcagacgcccgaagctggaccacgaggatccagcagctgcaaatccagcccaaccagcctcaagtttgggagct
PHP.B 501 taaaaagagactcaatttcggtcagactggcgacacagagtcagtcacagacccctcaaccaatcgagaaacctcccgagccctcaggtgtgggactt
AAV9 501 taaaaagagactcaatttcggtcagactggcgacacagagtcagtcacagacccctcaaccaatcgagaaacctcccgagccctcaggtgtgggactt
hu68 501 taaaaagagactcaatttcggtcagactggcgacacagagtcagtcacagacccctcaaccaatcgagaaacctcccgagccctcaggtgtgggactt

FIG. 12C (Clade F nt - cont'd)

AAVG5 601	gatacaatggcttcagggtggcgccaccagtggcagacaataacgaaggtgcgatggagtggttagttcctcgggaaattggcattggcgattcccaat
PHP.B 601	cttacaaatggcttcagggtggcgccaccagtggcagacaataacgaaggtgcgatggagtggttagttcctcgggaaattggcattggcgattcccaat
AAV9 601	cttacaaatggcttcagggtggcgccaccagtggcagacaataacgaaggtgcgatggagtggttagttcctcgggaaattggcattggcgattcccaat
hu68 601	cttacaaatggcttcagggtggcgccaccagtggcagacaataacgaaggtgcgatggagtggttagttcctcgggaaattggcattggcgattcccaat
AAVG5 701	ggctgggggacagagtcataccaccagcacccgaacctggggccctgccacctacaacaatacacctctacaaggcaaatctccaacagcacatcttgagg
PHP.B 701	ggctgggggacagagtcataccaccagcacccgaacctggggccctgccacctacaacaatacacctctacaaggcaaatctccaacagcacatcttgagg
AAV9 701	ggctgggggacagagtcataccaccagcacccgaacctggggccctgccacctacaacaatacacctctacaaggcaaatctccaacagcacatcttgagg
hu68 701	ggctgggggacagagtcataccaccagcacccgaacctggggccctgccacctacaacaatacacctctacaaggcaaatctccaacagcacatcttgagg
AAVG5 801	atcttcaaatgacaaacgcctactctgggtacagcacccccctgggggtatttttgacttcaacagattccactgccacttccaacacgtgactggcagcga
PHP.B 801	atcttcaaatgacaaacgcctactctgggtacagcacccccctgggggtatttttgacttcaacagattccactgccacttccaacacgtgactggcagcga
AAV9 801	atcttcaaatgacaaacgcctactctgggtacagcacccccctgggggtatttttgacttcaacagattccactgccacttccaacacgtgactggcagcga
hu68 801	atcttcaaatgacaaacgcctactctgggtacagcacccccctgggggtatttttgacttcaacagattccactgccacttccaacacgtgactggcagcga
AAVG5 901	ctcatcaacaacaactggggattccggcctaagcgactcaacttcaagctcttcaacattcaggtcaaaagggttacggacaacaataatggagtcgaagacca
PHP.B 901	ctcatcaacaacaactggggattccggcctaagcgactcaacttcaagctcttcaacattcaggtcaaaagggttacggacaacaataatggagtcgaagacca
AAV9 901	ctcatcaacaacaactggggattccggcctaagcgactcaacttcaagctcttcaacattcaggtcaaaagggttacggacaacaataatggagtcgaagacca
hu68 901	ctcatcaacaacaactggggattccggcctaagcgactcaacttcaagctcttcaacattcaggtcaaaagggttacggacaacaataatggagtcgaagacca
AAVG5 1001	tcgccaataaaccttaccagcacggtccagggtcttcacggactcagactatcagctcccgtacgtgctcgggtcgggtcacgaggggtgcctcccgcggtt
PHP.B 1001	tcgccaataaaccttaccagcacggtccagggtcttcacggactcagactatcagctcccgtacgtgctcgggtcgggtcacgaggggtgcctcccgcggtt
AAV9 1001	tcgccaataaaccttaccagcacggtccagggtcttcacggactcagactatcagctcccgtacgtgctcgggtcgggtcacgaggggtgcctcccgcggtt
hu68 1001	tcgctaataaaccttaccagcacggtccagggtcttcacggactcagactatcagctcccgtacgtgctcgggtcgggtcacgaggggtgcctcccgcggtt
AAVG5 1101	cccagcggacgtttttcatgattcctcagtcacgggtatctgacgcgttaatgatggaagccagccgctgggtcgttcgtcttttactgcctggaatatattc
PHP.B 1101	cccagcggacgtttttcatgattcctcagtcacgggtatctgacgcgttaatgatggaagccagccgctgggtcgttcgtcttttactgcctggaatatattc
AAV9 1101	cccagcggacgtttttcatgattcctcagtcacgggtatctgacgcgttaatgatggaagccagccgctgggtcgttcgtcttttactgcctggaatatattc
hu68 1101	cccagcggacgtttttcatgattcctcagtcacgggtatctaacgcgttaatgatggaagccagccgctgggtcgttcgtcttttactgcctggaatatattc
AAVG5 1201	ccgtcgcaaatgctaagaacgggtaacaactccagttcagtcacgagtttgagaacgtacctttccatagcagtcacgtcacagccaaagcctggacc
PHP.B 1201	ccgtcgcaaatgctaagaacgggtaacaactccagttcagtcacgagtttgagaacgtacctttccatagcagtcacgtcacagccaaagcctggacc
AAV9 1201	ccgtcgcaaatgctaagaacgggtaacaactccagttcagtcacgagtttgagaacgtacctttccatagcagtcacgtcacagccaaagcctggacc
hu68 1201	ccgtcgcaaatgctaagaacgggtaacaactccagttcagtcacgagtttgagaacgtacctttccatagcagtcacgtcacagccaaagcctggacc

FIG. 12D (Clade F - cont'd)

AAVG5 1301 gactaatgaatccactcatcgaccaataacttgactatctctcaaaagactattaacggttcttgacagaatcaacaaacgctaaaaatttcagtgtggccgg
PHP.B 1301 gactaatgaatccactcatcgaccaataacttgactatctctctagaactattaacggttcttgacagaatcaacaaacgctaaaaatttcagtgtggccgg
AAV9 1301 gactaatgaatccactcatcgaccaataacttgactatctctcaaaagactattaacggttcttgacagaatcaacaaacgctaaaaatttcagtgtggccgg
hu68 1301 gactcatgaatccactcatcgaccaataacttgactatctctcaaaagactattaacggttcttgacagaatcaacaaacgctaaaaatttcagtgtggccgg

AAVG5 1401 acccagcaacatggctgtccaggggaagaactacatacctggaccagctaccgacacaacacgtgtctcaaccactgtgactcaaaacaaacacagcgaa
PHP.B 1401 acccagcaacatggctgtccaggggaagaactacatacctggaccagctaccgacacaacacgtgtctcaaccactgtgactcaaaacaaacacagcgaa
AAV9 1401 acccagcaacatggctgtccaggggaagaactacatacctggaccagctaccgacacaacacgtgtctcaaccactgtgactcaaaacaaacacagcgaa
hu68 1401 acccagcaacatggctgtccaggggaagaactacatacctggaccagctaccgacacaacacgtgtctcaaccactgtgactcaaaacaaacacagcgaa

AAVG5 1501 ttgtcttggcctggagcttcttcttgggctctcaatggacgtaataagcttgatgaatcctggacctgctatggccagccacaaaagaaggagaccgtt
PHP.B 1501 ttgtcttggcctggagcttcttcttgggctctcaatggacgtaataagcttgatgaatcctggacctgctatggccagccacaaaagaaggagaccgtt
AAV9 1501 ttgtcttggcctggagcttcttcttgggctctcaatggacgtaataagcttgatgaatcctggacctgctatggccagccacaaaagaaggagaccgtt
hu68 1501 ttgtcttggcctggagcttcttcttgggctctcaatggacgtaataagcttgatgaatcctggacctgctatggccagccacaaaagaaggagaccgtt

AAVG5 1601 tctttccttctgtctggatctttaaatttttggcaaacaaaggaaactgggaagagacacacgtggatgcggacaaaagtcatgataaccaacgaagaagaaattaa
PHP.B 1601 tctttccttctgtctggatctttaaatttttggcaaacaaaggtaaccggcagagacacacgtggatgcggacaaaagtcatgataaccaacgaagaagaaattaa
AAV9 1601 tctttccttctgtctggatctttaaatttttggcaaacaaaggaaactgggaagagacacacgtggatgcggacaaaagtcatgataaccaacgaagaagaaattaa
hu68 1601 tctttccttctgtctggatctttaaatttttggcaaacaaaggaaactgggaagagacacacgtggatgcggacaaaagtcatgataaccaacgaagaagaaattaa

AAVG5 1701 aactactaaccggtagcaacggagtcctatggacaagtggcacaacaccacagagtgcccaag-----cacaggcgcagacc
PHP.B 1701 aactactaaccggtagcaacggagtcctatggacaagtggcacaacaccacagagtgcccaactttggcgggtgccttttaaggcacagcgcgagacc
AAV9 1701 aactactaaccggtagcaacggagtcctatggacaagtggcacaacaccacagagtgcccaag-----cacaggcgcagacc
hu68 1701 aactaccacaccagtagcaacggagtcctatggacaagtggcacaacaccacagagtgcccaag-----cacaggcgcgagacc

AAVG5 1801 ggctgggttcaaaaccaaaggaaataacttccgggtatggtttggcaggacagagatgtgtacctgcaaggaccatttgggccaaaattcctcacacggacg
PHP.B 1801 ggctgggttcaaaaccaaaggaaataacttccgggtatggtttggcaggacagagatgtgtacctgcaaggaccatttgggccaaaattcctcacacggacg
AAV9 1801 ggctgggttcaaaaccaaaggaaataacttccgggtatggtttggcaggacagagatgtgtacctgcaaggaccatttgggccaaaattcctcacacggacg
hu68 1801 ggctgggttcaaaaccaaaggaaataacttccgggtatggtttggcaggacagagatgtgtacctgcaaggaccatttgggccaaaattcctcacacggacg

FIG. 12E (Clade F - cont'd)

AAVG5	1901	gcaactttcaccccttctccgtgatgggaggggtttggaatgaagcaccgcctcctcagatcctcatcaaaaaacacacctgtacctgcggatcctccaac
PHP.B	1901	gcaactttcaccccttctccgtgatgggaggggtttggaatgaagcaccgcctcctcagatcctcatcaaaaaacacacctgtacctgcggatcctccaac
AAV9	1901	gcaactttcaccccttctccgtgatgggaggggtttggaatgaagcaccgcctcctcagatcctcatcaaaaaacacacctgtacctgcggatcctccaac
hu68	1901	gcaactttcaccccttctccgtgatgggaggggtttggaatgaagcaccgcctcctcagatcctcatcaaaaaacacacctgtacctgcggatcctccaac
AAVG5	2001	ggccttcaacaaggacaagctgaactctttcatcacccagttattctactggccaagtcagcgtggagatcgagtgggagctgcagaaggaaaaacagcaag
PHP.B	2001	ggccttcaacaaggacaagctgaactctttcatcacccagttattctactggccaagtcagcgtggagatcgagtgggagctgcagaaggaaaaacagcaag
AAV9	2001	ggccttcaacaaggacaagctgaactctttcatcacccagttattctactggccaagtcagcgtggagatcgagtgggagctgcagaaggaaaaacagcaag
hu68	2001	ggccttcaacaaggacaagctgaactctttcatcacccagttattctactggccaagtcagcgtggagatcgagtgggagctgcagaaggaaaaacagcaag
AAVG5	2101	cgctggaaccccgagatccagttacacttccaaactattacaagtcctaataatgttgaatttgctgttaactgaaggtgtatatagtgaacccccccca
PHP.B	2101	cgctggaaccccgagatccagttacacttccaaactattacaagtcctaataatgttgaatttgctgttaactgaaggtgtatatagtgaacccccccca
AAV9	2101	cgctggaaccccgagatccagttacacttccaaactattacaagtcctaataatgttgaatttgctgttaactgaaggtgtatatagtgaacccccccca
hu68	2101	cgctggaaccccgagatccagttacacttccaaactattacaagtcctaataatgttgaatttgctgttaactgaaggtgttatctgaacccccccca
AAVG5	2201	ttggcaccagatacctgactcgtaattctg
PHP.B	2201	ttggcaccagatacctgactcgtaattctg
AAV9	2201	ttggcaccagatacctgactcgtaattctg
hu68	2201	ttggcaccagatacctgactcgtaattctg

FIG. 13A (AAV8)

AAV8T 1 AAVG3 1 AAV8 1	vp1 start		MAADGYLPDWLEDNLSEGIREWALKPGAPKPKANQKQKDDGRGLVLPGYKYLGPFNGLDKGEPVNAADAAALEHDKAYDQQLQAGDNPYLRYNHADADEF MAADGYLPDWLEDNLSEGIREWALKPGAPKPKANQKQKDDGRGLVLPGYKYLGPFNGLDKGEPVNAADAAALEHDKAYDQQLQAGDNPYLRYNHADADEF MAADGYLPDWLEDNLSEGIREWALKPGAPKPKANQKQKDDGRGLVLPGYKYLGPFNGLDKGEPVNAADAAALEHDKAYDQQLQAGDNPYLRYNHADADEF
	vp2 start		QERLQEDTSFGGNLGRAVFOAKKRVLPLGLVEEGAKTAPGKKRPVEPSQORSDDSTGIGKKGQQPARKRLNFGQTGDSSEVPDPQPLGEPPAAPSGVG QERLQEDTSFGGNLGRAVFOAKKRVLPLGLVEEGAKTAPGKKRPVEPSQORSDDSTGIGKKGQQPARKRLNFGQTGDSSEVPDPQPLGEPPAAPSGVG QERLQEDTSFGGNLGRAVFOAKKRVLPLGLVEEGAKTAPGKKRPVEPSQORSDDSTGIGKKGQQPARKRLNFGQTGDSSEVPDPQPLGEPPAAPSGVG
	vp3 start		PNTMAAGGGAPMADNNEGADGVSSSGNWHCDSTWLGDRVITTTSTRTWALPTYNHLYKQISSGTHGATNDNTYFGYSTPWGYFDNRFHCHFSPRDWQ PNTMAAGGGAPMADNNEGADGVSSSGNWHCDSTWLGDRVITTTSTRTWALPTYNHLYKQISNGTSGGATNDNTYFGYSTPWGYFDNRFHCHFSPRDWQ PNTMAAGGGAPMADNNEGADGVSSSGNWHCDSTWLGDRVITTTSTRTWALPTYNHLYKQISNGTSGGATNDNTYFGYSTPWGYFDNRFHCHFSPRDWQ
AAV8T 301 AAVG3 301 AAV8 301			RLINNNWGFRRPKRLSFKLFNIQVKEVTQNEGTKTIANNLTSTIQVFTDSEYQLPYVLGSAHQGCLPPFPADVFMIPQYGYLTINNGSQAVGRSSFYCLEY RLINNNWGFRRPKRLSFKLFNIQVKEVTQNEGTKTIANNLTSTIQVFTDSEYQLPYVLGSAHQGCLPPFPADVFMIPQYGYLTINNGSQAVGRSSFYCLEY RLINNNWGFRRPKRLSFKLFNIQVKEVTQNEGTKTIANNLTSTIQVFTDSEYQLPYVLGSAHQGCLPPFPADVFMIPQYGYLTINNGSQAVGRSSFYCLEY
			FPSQMLRTGNNFQFTYTFEDVPFHSSYAHSSQSLDRLMNPLIDQYLYLSRTQTGTGSRPTQTLGFSQGGPNMTMANQAKNWLPGPCYRQQRVSTTTTGQNNN FPSQMLRTGNNFQFTYTFEDVPFHSSYAHSSQSLDRLMNPLIDQYLYLSRTQTGTGTANTQTLGFSQGGPNMTMANQAKNWLPGPCYRQQRVSTTTTGQNNN FPSQMLRTGNNFQFTYTFEDVPFHSSYAHSSQSLDRLMNPLIDQYLYLSRTQTGTGTANTQTLGFSQGGPNMTMANQAKNWLPGPCYRQQRVSTTTTGQNNN
			SNFAWTAGTKYHLNGRNSLANPGIAMATHKDDEERFFPNSGILIFGKQNAARDNADYSDVMLTSEEEIKTTNPVATEEYGI VGDNLQLYNTAPGVSFVNS SNFAWTAGTKYHLNGRNSLANPGIAMATHKDDEERFFPNSGILIFGKQNAARDNADYSDVMLTSEEEIKTTNPVATEEYGI VCDNLQSRNTAPREEIVNS SNFAWTAGTKYHLNGRNSLANPGIAMATHKDDEERFFPNSGILIFGKQNAARDNADYSDVMLTSEEEIKTTNPVATEEYGI VADNLQQQNTAPQIGTVNS
AAV8T 601 AAVG3 601 AAV8 601			QGALPGMVWQNRDVYLGPIWAKIPHTDGNFHPSPLMGGFGLKHPPPPQILIKNTVPADPPTTFNQSKLNSFITQYSTGQVSVEIEWELQKENS KRWNP QGALPGMVWQNRDVYLGPIWAKIPHTDGNFHPSPLMGGFGLKHPPPPQILIKNTVPADPPTTFNQSKLNSFITQYSTGQVSVEIEWELQKENS KRWNP QGALPGMVWQNRDVYLGPIWAKIPHTDGNFHPSPLMGGFGLKHPPPPQILIKNTVPADPPTTFNQSKLNSFITQYSTGQVSVEIEWELQKENS KRWNP
			IQYTSNYYKSTSVDFAVNTEGVYSEPRPIGTRYLTRNL IQYTSNYYKSTSVDFAVNTEGVYSEPRPIGTRYLTRNL IQYTSNYYKSTSVDFAVNTEGVYSEPRPIGTRYLTRNL

vp1 start

1 atggtgcgatggttatcttccagattggtctgaggacaacctctctgagggcattcgcgagtggtggcgctgaaacctggagccccgaagcccaaaag
AAV8T 1
1 atggtgcgatggttatcttccagattggtctgaggacaacctctctgagggcattcgcgagtggtggcgctgaaacctggagccccgaagcccaaaag
AAVG3 1
1 atggtgcgatggttatcttccagattggtctgaggacaacctctctgagggcattcgcgagtggtggcgctgaaacctggagccccgaagcccaaaag
AAV8 1

AAV8^T 101 ccaaccagcaaaagcaggacgagccgggggtctggtgcttctggtacaaagtacctcggaaccttcaacggactcgacaagggggagcccgccaacgc
AAVG3 101 ccaaccagcaaaagcaggacgagccgggggtctggtgcttctggtacaaagtacctcggaaccttcaacggactcgacaagggggagcccgccaacgc
AAV8 101 ccaaccagcaaaagcaggacgagccgggggtctggtgcttctggtacaaagtacctcggaaccttcaacggactcgacaagggggagcccgccaacgc

AAV8T 201 ggcgagcagcgccctcgagcacgacaaggcctacgaccagcagctgcaggcgggtgacaatcgtacctgcggtataaacacgcgcgcgagttt

AAVG3 201 ggcgagcagcgccctcgagcacgacaaggcctacgaccagcagctgcaggcgggtgacaatcgtacctgcggtataaacacgcgcgcgagttt

AAV8 201 ggcgagcagcgccctcgagcacgacaaggcctacgaccagcagctgcaggcgggtgacaatcgtacctgcggtataaacacgcgcgcgagttt

AAV8T 301 caggagcgtctgcaagaagatacgtcttttgggggcaacctctccaggccaaagaagcgggttctcgaacctctcgggtctggttgagg
AAVG3 301 caggagcgtctgcaagaagatacgtcttttgggggcaacctctccaggccaaagaagcgggttctcgaacctctcgggtctggttgagg
AAV8 301 caggagcgtctgcaagaagatacgtcttttgggggcaacctctccaggccaaagaagcgggttctcgaacctctcgggtctggttgagg

| vp2 start

AAV8T 401 aaggcgtaaagcggtcctggaagaagagacgggtagagggcatcacccacgggttctccagactcctctacggggcatcgggcaagaagggccaacagcc

AAVG3 401 aaggcgtaaagcggtcctggaagaagagacgggtagagggcatcacccacgggttctccagactcctctacggggcatcgggcaagaagggccaacagcc

AAV8 401 aaggcgtaaagcggtcctggaagaagagacgggtagagggcatcacccacgggttctccagactcctctacggggcatcgggcaagaagggccaacagcc

AAV8T 501 cgccagaaaagactcaattttggtcagactggcgactcagagtcaggttcagacccctcaactctcggagaaacctccagcagcgccctctggtgtggga
AAVG3 501 cgccagaaaagactcaattttggtcagactggcgactcagagtcaggttcagacccctcaactctcggagaaacctccagcagcgccctctggtgtggga
AAV8 501 cgccagaaaagactcaattttggtcagactggcgactcagagtcaggttcagacccctcaactctcggagaaacctccagcagcgccctctggtgtggga

|vp3 start

AAV8T 601 cctaatacaatggctgcaggcgggtggcgccaccaatggcagacacaataacgaaggcgcgcaggagtggttagttcctcgggaaattggcattgcgattccca
AAVG3 601 cctaatacaatggctgcaggcgggtggcgccaccaatggcagacacaataacgaaggcgcgcaggagtggttagttcctcgggaaattggcattgcgattccca
AAV8 601 cctaatacaatggctgcaggcgggtggcgccaccaatggcagacacaataacgaaggcgcgcaggagtggttagttcctcgggaaattggcattgcgattccca

AAV8T 701 catggctggggcgacagagtcataccaccagcaccggaacctgggcccctgccacctacaacaaccacctctacaagcaaatctcctctggtactca--

AAVG3 701 catggctggggcgacagagtcataccaccagcaccggaacctgggcccctgccacctacaacaaccacctctacaagcaaatctccaacgggacatcggg

AAV8 701 catggctggggcgacagagtcataccaccagcaccggaacctgggcccctgccacctacaacaaccacctctacaagcaaatctccaacgggacatcggg

AAV8T 801 tggagccaccaacgacaacacactacttcgggtacagcacccccctgggggtattttgacttttaacagattccactgccacttttcaccacgtgactggcag

AAVG3 801 aggagccaccaacgacaacacactacttcgggtacagcacccccctgggggtattttgacttttaacagattccactgccacttttcaccacgtgactggcag

AAV8 801 aggagccaccaacgacaacacactacttcgggtacagcacccccctgggggtattttgacttttaacagattccactgccacttttcaccacgtgactggcag

FIG. 13C (AAV8 - cont'd)

AAV8T 901	cgactcatcaacaacaactgggggattccggcccaagagactcagcttcaagctcttcaacatccagggtcaaggaggtcacgcagaatgaaggcacccaaga
AAVG3 901	cgactcatcaacaacaactgggggattccggcccaagagactcagcttcaagctcttcaacatccagggtcaaggaggtcacgcagaatgaaggcacccaaga
AAV8 901	cgactcatcaacaacaactgggggattccggcccaagagactcagcttcaagctcttcaacatccagggtcaaggaggtcacgcagaatgaaggcacccaaga
AAV8T 1001	ccatcgccaataaacctaccagcaccatccagggtgtttacgggactcggagtaccagctgccgtacgtttctcggctctgccaccagggtgcctgcctcc
AAVG3 1001	ccatcgccaataaacctaccagcaccatccagggtgtttacgggactcggagtaccagctgccgtacgtttctcggctctgccaccagggtgcctgcctcc
AAV8 1001	ccatcgccaataaacctaccagcaccatccagggtgtttacgggactcggagtaccagctgccgtacgtttctcggctctgccaccagggtgcctgcctcc
AAV8T 1101	gttccccggcgacgtgttcattgattccccagtcacggctacctaactcaacaacggtagtcaggccgtgggacgctcctcttactgcctggaatac
AAVG3 1101	gttccccggcgacgtgttcattgattccccagtcacggctacctaactcaacaacggtagtcaggccgtgggacgctcctcttactgcctggaatac
AAV8 1101	gttccccggcgacgtgttcattgattccccagtcacggctacctaactcaacaacggtagtcaggccgtgggacgctcctcttactgcctggaatac
AAV8T 1201	tttccttcgcagatgctgagaacccggcaacaacttccagtttacttacccttcgaggacgtgcctttccacagcagctacgccacagccagagcttgg
AAVG3 1201	tttccttcgcagatgctgagaacccggcaacaacttccagtttacttacccttcgaggacgtgcctttccacagcagctacgccacagccagagcttgg
AAV8 1201	tttccttcgcagatgctgagaacccggcaacaacttccagtttacttacccttcgaggacgtgcctttccacagcagctacgccacagccagagcttgg
AAV8T 1301	accggctgatgaatcctctgattgaccagtcacctgtactacttgtctcgggactcaaacacaggtgggagtaggctacgcagactctgggcttcagcca
AAVG3 1301	accggctgatgaatcctctgattgaccagtcacctgtactacttgtctcgggactcaaacacaggtgggagtaggctacgcagactctgggcttcagcca
AAV8 1301	accggctgatgaatcctctgattgaccagtcacctgtactacttgtctcgggactcaaacacaggtgggagtaggctacgcagactctgggcttcagcca
AAV8T 1401	aggtgggcctaatacaaatggccaatcaggcaagaactggctgccaggacacctgtaccgccacaacacgcgtctcaacgacaacccgggcaaaaacaacaat
AAVG3 1401	aggtgggcctaatacaaatggccaatcaggcaagaactggctgccaggacacctgtaccgccacaacacgcgtctcaacgacaacccgggcaaaaacaacaat
AAV8 1401	aggtgggcctaatacaaatggccaatcaggcaagaactggctgccaggacacctgtaccgccacaacacgcgtctcaacgacaacccgggcaaaaacaacaat
AAV8T 1501	agcaactttgcctggactgctgggaccacaaataccatctgaatggaagaaattcattggctaattcctggcatcgctatggcaacacacaaagacgacgagg
AAVG3 1501	agcaactttgcctggactgctgggaccacaaataccatctgaatggaagaaattcattggctaattcctggcatcgctatggcaacacacaaagacgacgagg
AAV8 1501	agcaactttgcctggactgctgggaccacaaataccatctgaatggaagaaattcattggctaattcctggcatcgctatggcaacacacaaagacgacgagg
AAV8T 1601	agcgtttttttccccagtaacgggacacctgattttttggcaaacaaaatgctgccagagacaaatcgggattacagcgatgtcatgtcaccagcgagggaaga
AAVG3 1601	agcgtttttttccccagtaacgggacacctgattttttggcaaacaaaatgctgccagagacaaatcgggattacagcgatgtcatgtcaccagcgagggaaga
AAV8 1601	agcgtttttttccccagtaacgggacacctgattttttggcaaacaaaatgctgccagagacaaatcgggattacagcgatgtcatgtcaccagcgagggaaga

FIG. 13D (AAV8 - cont'd)

AAV8T 1701	aatcaaaaccactaacccctgtggctacagaggaatacgggtatcgtgggtgataaacttgcagttgtataaacacgggctcctgggttcggtgttgcacacagc
AAVG3 1701	aatcaaaaccactaacccctgtggctacagaggaatacgggtatcgtgtgtataaacttgcagagtgcgaaacacgggctcctcgtgaggagattgtcaacagc
AAV8 1701	aatcaaaaccactaacccctgtggctacagaggaatacgggtatcgtggcagataaacttgcagcagcaaaacacgggctcctcctcaaatggaaactgtcaacacagc
AAV8T 1801	cagggggccttacccgggtatggctggcagaaccgggacgtgtacctgcaggggtcccatctgggccaagattcctcaacggacggcaacttccacccgt
AAVG3 1801	cagggggccttacccgggtatggctggcagaaccgggacgtgtacctgcaggggtcccatctgggccaagattcctcaacggacggcaacttccacccgt
AAV8 1801	cagggggccttacccgggtatggctggcagaaccgggacgtgtacctgcaggggtcccatctgggccaagattcctcaacggacggcaacttccacccgt
AAV8T 1901	ctccgctgatggcggttbbggcctgaaacatcctccgctcagatcctgatcaagaacacgcctgtacctgcggatcctccgaccaccttcaaccagtc
AAVG3 1901	ctccgctgatggcggttbbggcctgaaacatcctccgctcagatcctgatcaagaacacgcctgtacctgcggatcctccgaccaccttcaaccagtc
AAV8 1901	ctccgctgatggcggttbbggcctgaaacatcctccgctcagatcctgatcaagaacacgcctgtacctgcggatcctccgaccaccttcaaccagtc
AAV8T 2001	aaagctgaactcttttcacacgcaatacacagcaccggacaggtcagcgtgggaaattgaatgggagctgcagaaggaaaaacagcaagcgtggaacccccgag
AAVG3 2001	aaagctgaactcttttcacacgcaatacacagcaccggacaggtcagcgtgggaaattgaatgggagctgcagaaggaaaaacagcaagcgtggaacccccgag
AAV8 2001	aaagctgaactcttttcacacgcaatacacagcaccggacaggtcagcgtgggaaattgaatgggagctgcagaaggaaaaacagcaagcgtggaacccccgag
AAV8T 2101	atccagtacacctccaactactacaaaatctacaagtgtggacttttgctgttaatacacagaaggcgtgtactctgaacccccgccccattggcacccttacc
AAVG3 2101	atccagtacacctccaactactacaaaatctacaagtgtggacttttgctgttaatacacagaaggcgtgtactctgaacccccgccccattggcacccttacc
AAV8 2101	atccagtacacctccaactactacaaaatctacaagtgtggacttttgctgttaatacacagaaggcgtgtactctgaacccccgccccattggcacccttacc
AAV8T 2201	tcacccgtaaatctg
AAVG3 2201	tcacccgtaaatctg
AAV8 2201	tcacccgtaaatctg

FIG. 14A (AAVrh10)

vp1 start			
AAVG2 1	1	MAADGYLPDWLENDLSEGIREWDLKPGAPKPKANQQKQDDGRGLVLPGYKYLGPFNGLDKGEPVNEADAAALEHDKAYDQQLKAGDNPYLRYNHADADEF	
rh10 1	1	MAADGYLPDWLENDLSEGIREWDLKPGAPKPKANQQKQDDGRGLVLPGYKYLGPFNGLDKGEPVNEADAAALEHDKAYDQQLKAGDNPYLRYNHADADEF	
hu37 1	1	MAADGYLPDWLENDLSEGIREWDLKPGAPKPKANQQKQDDGRGLVLPGYKYLGPFNGLDKGEPVNEADAAALEHDKAYDQQLKAGDNPYLRYNHADADEF	
vp2 start			
AAVG2 101	101	QERLQEDTSFGGNLGRAVFQAKKRVLLEPLGLVEEAAKTA PGKKRPVEPSQRS PDSSTGIGKKGQQPARKRLNFGQTGDSVDPDPQPIGEPPAGPSGLG	
rh10 101	101	QERLQEDTSFGGNLGRAVFQAKKRVLLEPLGLVEEAAKTA PGKKRPVEPSQRS PDSSTGIGKKGQQPARKRLNFGQTGDSVDPDPQPIGEPPAGPSGLG	
hu37 101	101	QERLQEDTSFGGNLGRAVFQAKKRVLLEPLGLVEEAAKTA PGKKRPVEPSQRS PDSSTGIGKKGQQPARKRLNFGQTGDSVDPDPQPIGEPPAGPSGLG	
vp3 start			
AAVG2 201	201	SGTMAAGGGAPMADNNEGADGVGSSSGNWHCHDS TWLGD RVITTSRTWALPTYNHLYKQISNGTSGGSTNDNTYFGYSTPWGYFDNRFHCHFSPRDWQ	
rh10 201	201	SGTMAAGGGAPMADNNEGADGVGSSSGNWHCHDS TWLGD RVITTSRTWALPTYNHLYKQISNGTSGGSTNDNTYFGYSTPWGYFDNRFHCHFSPRDWQ	
hu37 201	201	SGTMAAGGGAPMADNNEGADGVGSSSGNWHCHDS TWLGD RVITTSRTWALPTYNHLYKQISNGTSGGSTNDNTYFGYSTPWGYFDNRFHCHFSPRDWQ	
AAVG2 301	301	RLINNNWGFRRPKRLSFKLENIQVKEVTQNEGTKTIANNLSTIQVFTDSEYQLPYVLGSAHQGCLPPFPADVFMIPQYGYLTLNNGSQAVGRSSFYCLEY	
rh10 301	301	RLINNNWGFRRPKRLSFKLENIQVKEVTQNEGTKTIANNLSTIQVFTDSEYQLPYVLGSAHQGCLPPFPADVFMIPQYGYLTLNNGSQAVGRSSFYCLEY	
hu37 301	301	RLINNNWGFRRPKRLSFKLENIQVKEVTQNEGTKTIANNLSTIQVFTDSEYQLPYVLGSAHQGCLPPFPADVFMIPQYGYLTLNNGSQAVGRSSFYCLEY	
AAVG2 401	401	FPSQMLRTGNNEFFSYTFEDVFPFHSSYAHSSQSLDRLMNPLIDQYLXYLSRTQSTGGTQGTQQLLFSQAGPANMSAQAKNWLPGPCYRQQRVSTTLSQNNN	
rh10 401	401	FPSQMLRTGNNEFFSYTFEDVFPFHSSYAHSSQSLDRLMNPLIDQYLXYLSRTQSTGGTQGTQQLLFSQAGPANMSAQAKNWLPGPCYRQQRVSTTLSQNNN	
hu37 401	401	FPSQMLRTGNNEFFSYTFEDVFPFHSSYAHSSQSLDRLMNPLIDQYLXYLSRTQSTGGTQGTQQLLFSQAGPANMSAQAKNWLPGPCYRQQRVSTTLSQNNN	
AAVG2 501	501	SNFAWTGATKYHLNGRDSLVPVGVAMATHKDDDEERFFPSSSGVLMFGKQAGARDNVDYSSVMLTSEEEIKTTNPVATEQYGVVADNLQQTNTGPIGVNVS	
rh10 501	501	SNFAWTGATKYHLNGRDSLVPVGVAMATHKDDDEERFFPSSSGVLMFGKQAGARDNVDYSSVMLTSEEEIKTTNPVATEQYGVVADNLQQTNTGPIGVNVS	
hu37 501	501	SNFAWTGATKYHLNGRDSLVPVGVAMATHKDDDEERFFPSSSGVLMFGKQAGARDNVDYSSVMLTSEEEIKTTNPVATEQYGVVADNLQQTNTGPIGVNVS	
AAVG2 601	601	QGALPGMVWQNRDVYLQGFIAWKIPHTDGNFHPSPMLGGFGLKHPPQILLIKNTVPADPPTTFSQAKLASFITQYSTGQVSVEIEWELQKENS KRWNP	
rh10 601	601	QGALPGMVWQNRDVYLQGFIAWKIPHTDGNFHPSPMLGGFGLKHPPQILLIKNTVPADPPTTFSQAKLASFITQYSTGQVSVEIEWELQKENS KRWNP	
hu37 601	601	QGALPGMVWQNRDVYLQGFIAWKIPHTDGNFHPSPMLGGFGLKHPPQILLIKNTVPADPPTTFSQAKLASFITQYSTGQVSVEIEWELQKENS KRWNP	
AAVG2 701	701	IQYTSNYYKSTNVDFAVNTEGTYSEPRPIGTRYLTRNL	
rh10 701	701	IQYTSNYYKSTNVDFAVNTEGTYSEPRPIGTRYLTRNL	
hu37 701	701	IQYTSNYYKSTNVDFAVNTEGTYSEPRPIGTRYLTRNL	

FIG. 14B (AAVrh10)

AAVG2 AAVhu37 AAVrh10	1 1 1	vp1 start		atggctgctgacgggttatcttccagattggctcgaggacaacctctctgagggcattcgcgagtggtgggacctgaaacctggagcccccaagcccccaagg
				atggctgctgacgggttatcttccagattggctcgaggacaacctctctgagggcattcgcgagtggtgggacctgaaacctggagcccccaagcccccaagg
				atggctgccgatgggttatcttccagattggctcgaggacaacctctctgagggcattcgcgagtggtgggacctgaaacctggagcccccaagcccccaagg
AAVG2 AAVhu37 AAVrh10	101 101 101			ccaaccagcagaagcagcagcagcgccggggtctggtgcttctctggctacaagtacctcgacccttcaacgggactcgacaagggggagccccgtcaacga
				ccaaccagcagaagcagcagcagcgccggggtctggtgcttctctggctacaagtacctcgacccttcaacgggactcgacaagggggagccccgtcaacgc
				ccaaccagcagaagcagcagcagcgccggggtctggtgcttctctggctacaagtacctcgacccttcaacgggactcgacaagggggagccccgtcaacgc
AAVG2 AAVhu37 AAVrh10	201 201 201			ggcggacgcgcggccctcgagcaccagcagctcaaaagggtgacaatccgtacctgcggtataaacccagccgacgcgcgagttt
				ggcggacgcgcggccctcgagcaccagcagctcaaaagggtgacaatccgtacctgcggtataaacccagccgacgcgcgagttt
				ggcggacgcgcggccctcgagcaccagcagctcaaaagggtgacaatccgtacctgcggtataaacccagccgacgcgcgagttt
AAVG2 AAVhu37 AAVrh10	301 301 301			caggagcgtctgcaagaagatacgtcttttgggggcaacctcgggcgagcagtccttccaggccaaagaacgggttctcgaaacctctcgttctggttgagg
				caggagcgtctgcaagaagatacgtcttttgggggcaacctcgggcgagcagtccttccaggccaaagaacgggttctcgaaacctctcgttctggttgagg
				caggagcgtctgcaagaagatacgtcttttgggggcaacctcgggcgagcagtccttccaggccaaagaacgggttctcgaaacctctcgttctggttgagg
AAVG2 AAVhu37 AAVrh10	401 401 401			aaagcgttaagacgggtcctcctggaaagaagagaccgggtagaaacgggtcacctcagcgcgactcccccgactcctccacggggcatcgggcaaaaaaggccagcgc
				aaagcgttaagacgggtcctcctggaaagaagagaccgggtagaaacgggtcacctcagcgcgactcccccgactcctccacggggcatcgggcaaaaaaggccagcgc
				aaagcgttaagacgggtcctcctggaaagaagagaccgggtagaaacgggtcacctcagcgcgactcccccgactcctccacggggcatcgggcaaaaaaggccagcgc
AAVG2 AAVhu37 AAVrh10	501 501 501			cgcgagaaaagagactgaactttgggcagactggcgactcagagtcagtcctcccgacacctcaaccaatcgggagaaccaccagcaggccccctctggtctgggga
				cgcgagaaaagagactgaactttgggcagactggcgactcagagtcagtcctcccgacacctcaaccaatcgggagaaccaccagcaggccccctctggtctgggga
				cgcgagaaaagagactgaactttgggcagactggcgactcagagtcagtcctcccgacacctcaaccaatcgggagaaccaccagcaggccccctctggtctgggga
AAVG2 AAVhu37 AAVrh10	601 601 601			tctggtacaaatggctgcaggcgggtggcgctccaatggcagacaataaacgaaggcgccgacggagtggtgggtagttcctcagggaattggcattgcgattcca
				tctggtacaaatggctgcaggcgggtggcgctccaatggcagacaataaacgaaggcgccgacggagtggtgggtagttcctcagggaattggcattgcgattcca
				tctggtacaaatggctgcaggcgggtggcgctccaatggcagacaataaacgaaggcgccgacggagtggtgggtagttcctcagggaattggcattgcgattcca
AAVG2 AAVhu37 AAVrh10	701 701 701			catggctgggcgacagagtcataccaccagcaccccgaaacctgggccccctgccaccctacaacaaccacctctacaagcaaatctccaatgggacatcgggg
				catggctgggcgacagagtcataccaccagcaccccgaaacctgggccccctgccaccctacaacaaccacctctacaagcaaatctccaatgggacatcgggg
				catggctgggcgacagagtcataccaccagcaccccgaaacctgggccccctgccaccctacaacaaccacctctacaagcaaatctccaatgggacttcggg

| vp2 start

| vp3 start

FIG. 14C (AAVrh10)

AAVG2	801	aggaagcaccacgacaaacacctaacttgggtacagcaccctcctgggggtattttgacttcaacagattccactgtcacttctcaccacgtgactggcag
AAVhu37	801	aggaagcaccacgacaaacacctaacttgggtacagcaccctcctgggggtattttgacttcaacagattccactgtcacttctcaccacgtgactggcag
AAVrh10	801	aggaagcaccacgacaaacacctaacttgggtacagcaccctcctgggggtattttgacttcaacagattccactgtcacttctcaccacgtgactggcag
AAVG2	901	agactcatcaacaacaactggggaattccgggcccgaagagactcagcttcaagctcttcaacatccagggttaaggaggtcacgcagaatgaaggccaccaaga
AAVhu37	901	cgactcatcaacaacaactggggaattccgggcccgaagagactcagcttcaagctcttcaacatccagggttaaggaggtcacgcagaatgaaggccaccaaga
AAVrh10	901	cgactcatcaacaacaactggggaattccgggcccgaagagactcagcttcaagctcttcaacatccagggttaaggaggtcacgcagaatgaaggccaccaaga
AAVG2	1001	ccatcgccaataaaccttaccagcacgattcagggtattttacggactcgggaataccagctgccgtacgtcctcggtccgcgcaccagggtgcctgcctcc
AAVhu37	1001	ccatcgccaataaaccttaccagcacgattcagggtattttacggactcgggaataccagctgccgtacgtcctcggtccgcgcaccagggtgcctgcctcc
AAVrh10	1001	ccatcgccaataaaccttaccagcacgattcagggtattttacggactcgggaataccagctccccgtacgtcctcggtccgcgcaccagggtgcctgcctcc
AAVG2	1101	gttccccggcgatgtcttcattgatgtccocagtagcggtagctgacactgaacaaacgggaagtcaagccgtaggccgttcctcattctactgcctgggaatat
AAVhu37	1101	gttccccggcgatgtcttcattgatgtccocagtagcggtagctgacactgaacaaacgggaagtcaagccgtaggccgttcctcattctactgcctgggaatat
AAVrh10	1101	gttccccggcgatgtcttcattgatgtccocagtagcggtagctgacactgaacaaacgggaagtcaagccgtaggccgttcctcattctactgcctgggaatat
AAVG2	1201	tttccatctcaaatgctgcggactggaaacaaactttgaatttagctacacctttgaggacgtgcccttccacagcagctacgcacacagccagagcctgg
AAVhu37	1201	tttccatctcaaatgctgcggactggaaacaaactttgtgaatttagctacacctttcgaggacgtgcccttccacagcagctacgcacacagccagagcctgg
AAVrh10	1201	tttcccttctcaaatgctgagaacgggcaacaactttgagttcagctaccagtttgaggacgtgcccttttcaacagcagctacgcgcacagccaaagcctgg
AAVG2	1301	accggctgatgaacccctctcaccagtagtacctgtattacctatccagaactcagtcacacaggaggaactcaagggtacacacagcaattgtttttctca
AAVhu37	1301	accgactgatgaatcctctcaccagtagtacctgtattacctatccagaactcagtcacacaggaggaactcaagggtacacacagcaattgtttttctca
AAVrh10	1301	accggctgatgaacccctctcaccagtagtacctgtattacctatccagaactcagtcacacaggaggaactcaagggtacacacagcaattgtttttctca
AAVG2	1401	agccggcctgcataatgtcgggtcaggccaaagaactgggtacctggaccttaccgggcagcagcagagctctccacgacactgtcgcaaaaacaacaac
AAVhu37	1401	agctgggcctgcataatgtcgggtcaggccaaagaactgggtacctggaccttaccgggcagcagcagagctctctacgacactgtcgcaaaaacaacaac
AAVrh10	1401	ggccggcctgaataataacatgtcgggtcaggccaaagaactgggtacctggaccttaccgggcagcagcagagctctccacgacactgtcgcaaaaataacaac
AAVG2	1501	agcaactttgcttgactggtgccacgaaatatcatctgaacgggaagagactctttgggtgaatccccgggtgtgtctatggcaacgcataaaggacgacgagg
AAVhu37	1501	agcaactttgcttgactggtgccacgaaatatcatctgaacgggaagagactctttggtaaatccccgggtgtgtctatggcaacgcataaaggacgacgagg
AAVrh10	1501	agcaactttgcttgactggtgccacgaaatatcatctgaatggcagagactctcttggttaaatccccgggtgtgtctatggcaacgcataaaggacgacgagg

FIG. 14D (AAVrh10)

AAVG2	1601	aacgtttcttccatcgagcggagtcctgatgtttggaaaaacacaggggtgctggaagagagacaatgtggactatagcagcgttatgctaaccacgagcaggaaga
AAVhu37	1601	aacgcttctccgtcagtgagtcctgatgttcggaaaaacaggggtgctggaagagacaatgtggactacagcagcgttatgctaaccacgagcaggaaga
AAVrh10	1601	agcgatttttccgtccagcggagtccttaattgtttggaaaacacagggagctggaaaagacaacgtggactatagcagcgttatgctaaccacgtgaggaaga
AAVG2	1701	aatcaagaccactaacccctgtagccactgaacaatacggcgtgggtggtgataaacttgacagcaaaacccaatacagggacctaactcgtgggaaatgtcaacagc
AAVhu37	1701	aattaaaaaccactaaccccgtagccacagaacaatacgggtgtgggtgacctgacaaacttgacagcaaaacccaatacagggcctaattgtgggaaatgtcaacagc
AAVrh10	1701	aattaaaaaccacaaacccagtgccacagaacagtagcggcgtgggtggcgataaacctgcaacagcaaaaacgcgcgtcctaattgtaggggccgtcaacagt
AAVG2	1801	caaggagccttacctggcgtggtctggcagaacccgagacgtgtacctgcagggtcccaatttggggccaagattcctcacacgggacggccaactttcacccgt
AAVhu37	1801	caaggagccttacctggcgtggtctggcagaacccgagacgtgtacctgcagggtcccaatttggggccaagattcctcacacgggacggccaactttcacccgt
AAVrh10	1801	caaggagccttacctggcgtggtctggcagaacccggacgtgtacctgcagggtcctatctggggccaagattcctcacacgggacgggaaactttcatccct
AAVG2	1901	ctcctctgatggggcgttttggactgaaacaccccgccctcctcaaatcctgatcaagaacacactcccggttctcgtcggtatcctccaaacgagcttcagccaggc
AAVhu37	1901	caccgctaattgggaggttttggactgaagcaccacccctcctcagatcctgatcaagaacacgcgcgtacctcgtcggtatcctccaaacacgttcagccaggc
AAVrh10	1901	cgcgctgatgggaggttttggactgaaacaccccgccctcctcagatcctgattaaagaatacacctgttccgcggatcctccaaactaccttcagtcgaagc
AAVG2	2001	gaaattggccttccttcatacgcagtagtataccggccagggtcagcgtggagatcgagctgggagctgcagaaggagaaacagcaagcgtggaacccagaa
AAVhu37	2001	gaaattggccttccttcatacgcagtagtataccggccagggtcagcgtggaaatcgagctgggagctgcagaaggagaaacagcaaaacgtggaacccagag
AAVrh10	2001	taagctggcgtcgttcatacgcagtagtataccggccagggtcagcgtggaaattgaaatggagctgcagaaagaaaaacagcaaaacgtggaacccagag
AAVG2	2101	attcagtagatacttccaaactactacaaaatctacaaaatgtggactttgctgtcaataacgaggggtacatatattcagagcctcgccccattggaaactcgttacc
AAVhu37	2101	attcagtagatacttccaaactactacaaaatctacaaaatgtggactttgctgtcaatacagagggaaactattctgagcctcgccccattgggtactcgttacc
AAVrh10	2101	attcaatacacttccaaactactacaaaatctacaaaatgtggactttgctgttaaacacagatggcacttattctgagcctcgccccattcggcaccccggttacc
AAVG2	2201	tcaccccgtaaatctg
AAVhu37	2201	tcaccccgtaaatctg
AAVrh10	2201	tcaccccgtaaatctg

FIG. 15A

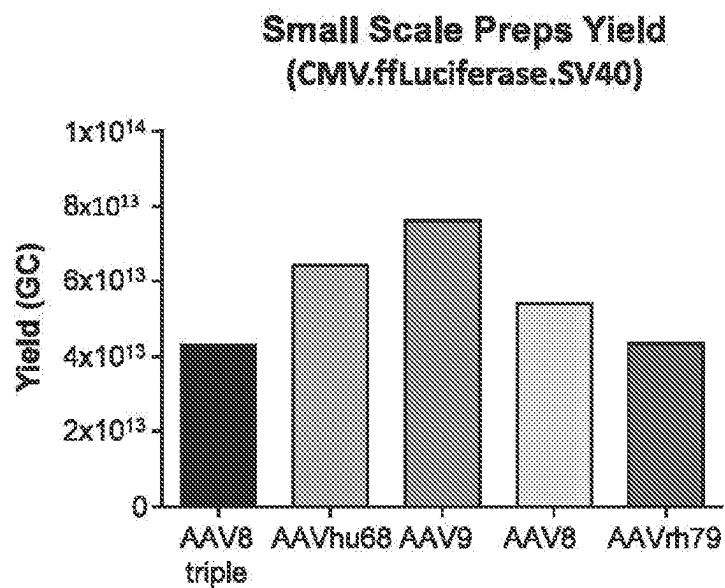


FIG. 15B

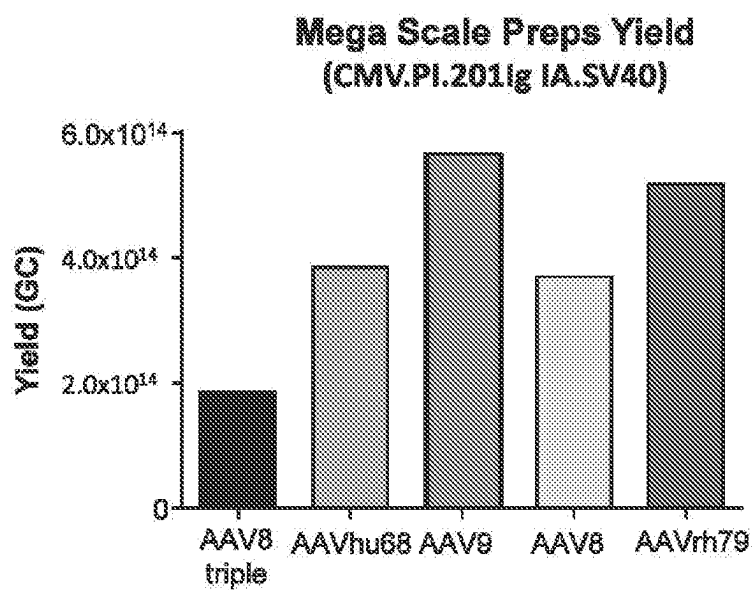


FIG. 16

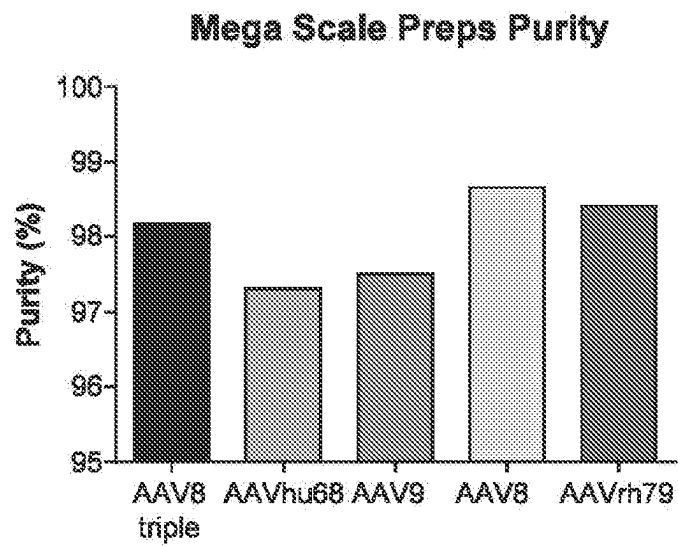


FIG. 17A

Liver

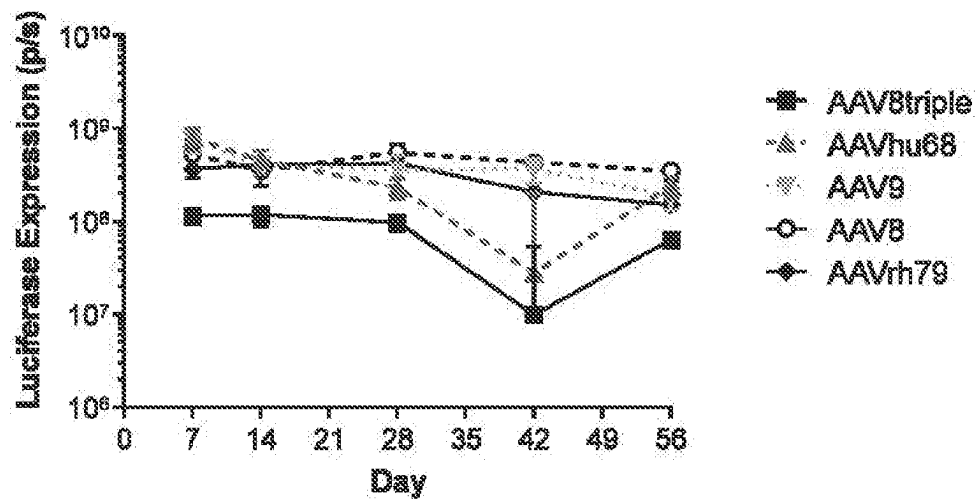


FIG. 17B

Muscle

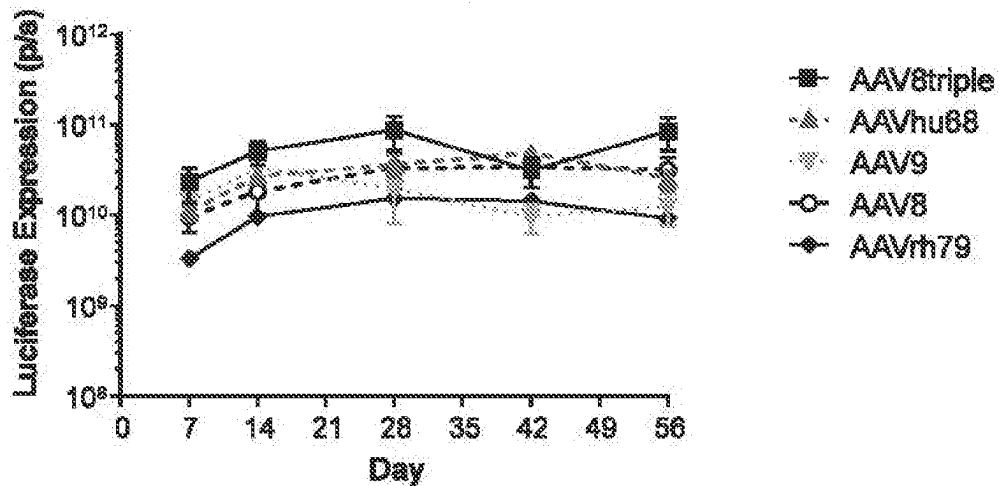


FIG. 17C

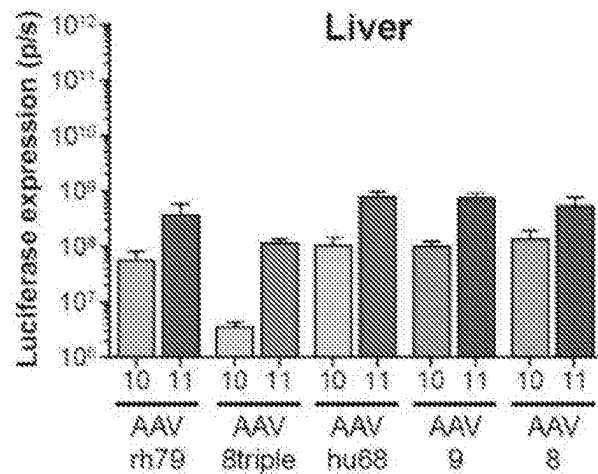


FIG. 17D

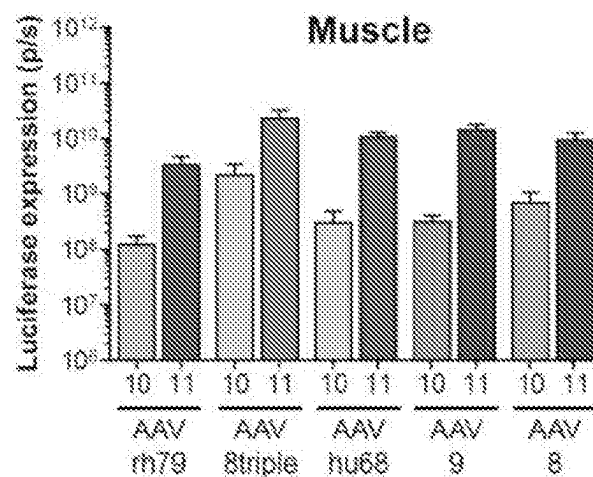


FIG. 17E

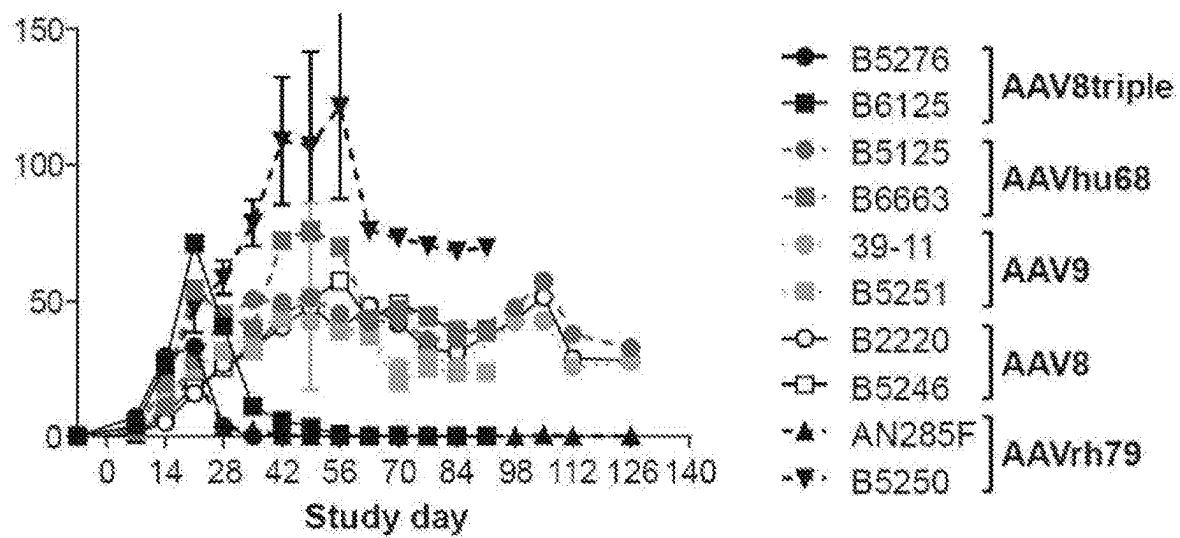
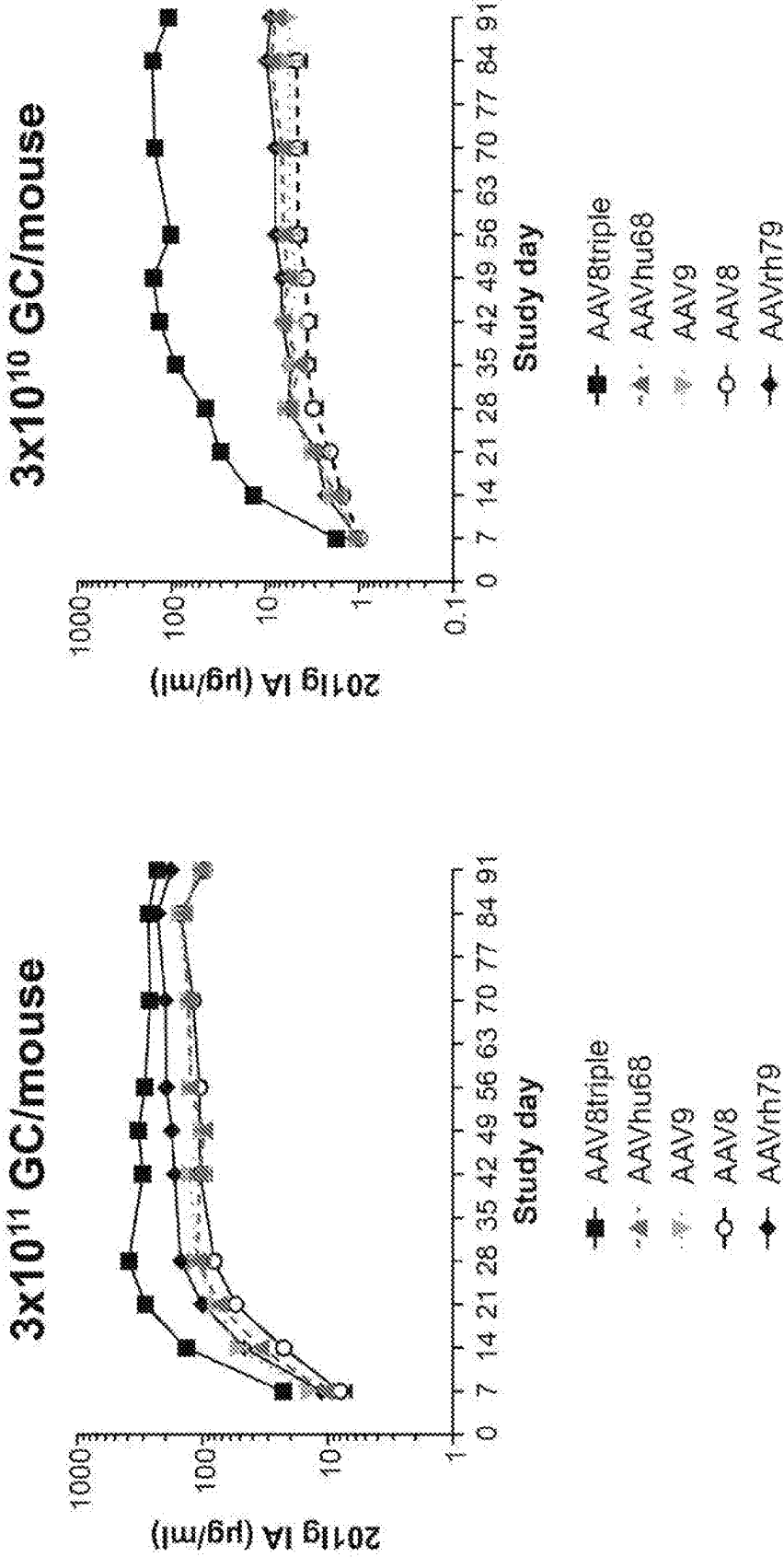


FIG. 17F



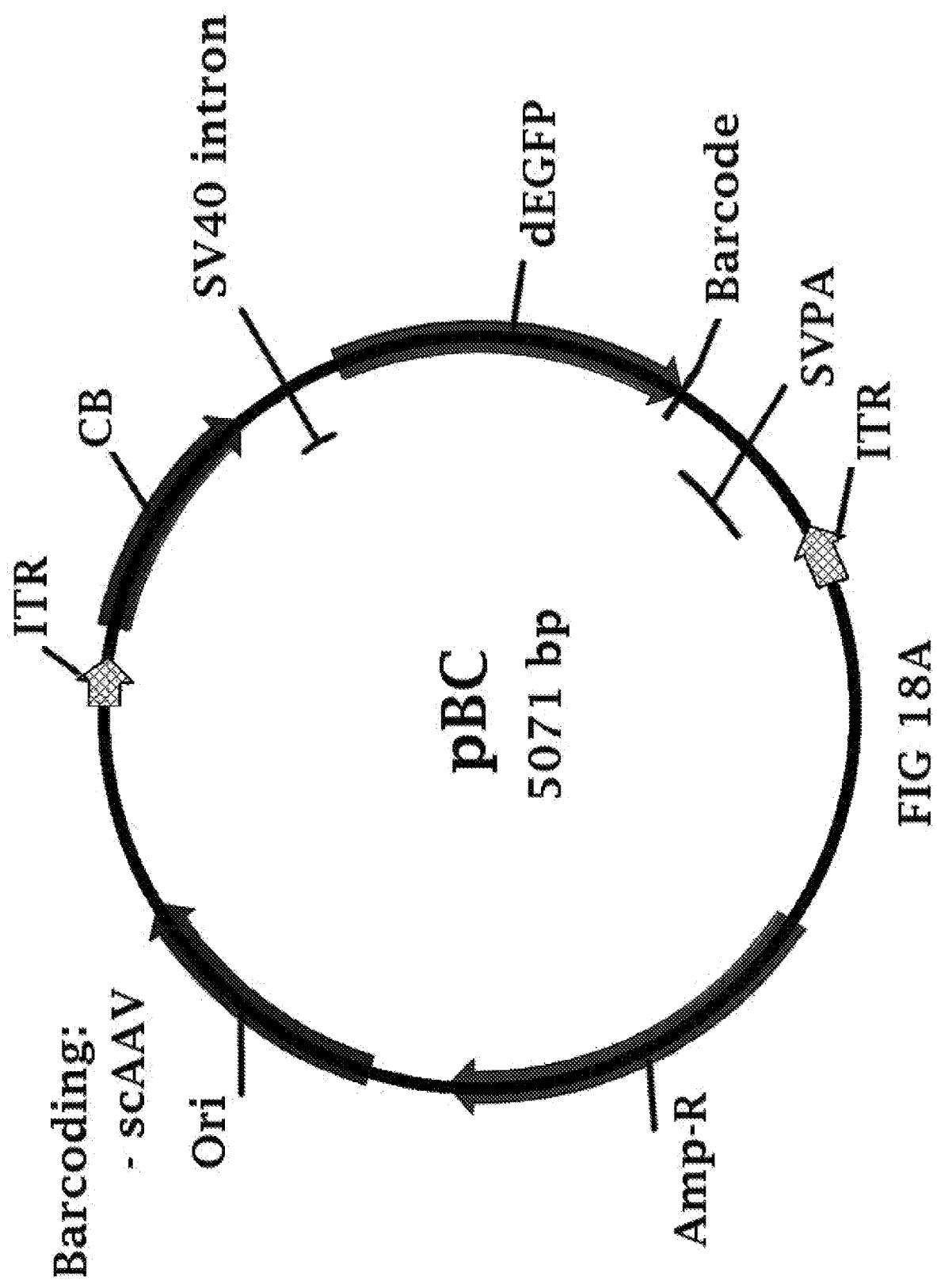


FIG 18A

- In vivo testing, B6 mice
 - AAV8.AR2.08, 2e12 gc/mouse, 2 weeks, i.v.

Code	Amount per mouse (GC)
BC01	6.87E+09
BC02	1.04E+10
BC03	1.59E+10
BC04	2.41E+10
BC05	3.67E+10
BC06	5.57E+10
BC07	8.47E+10
BC08	1.29E+11
BC09	1.96E+11
BC10	2.97E+11
BC11	4.52E+11
BC12	6.87E+11
Total gc of the mix	2.00E+12

FIG 18B

FIG 18C
• Vector genomic DNA distribution
Vector biodistribution

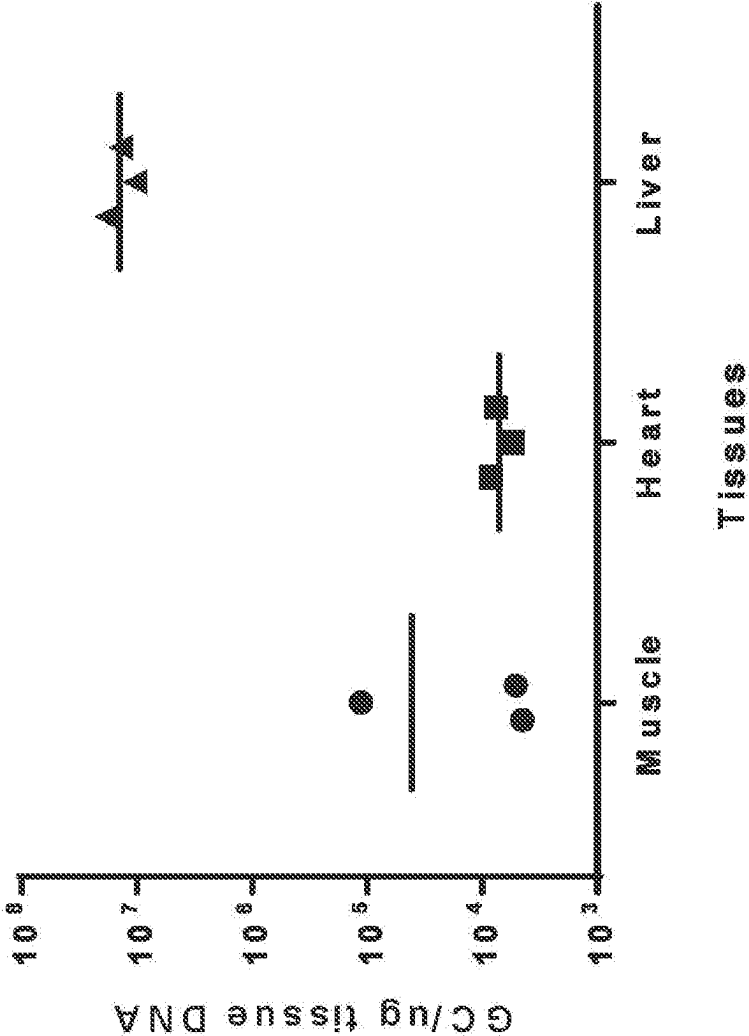


FIG 18D

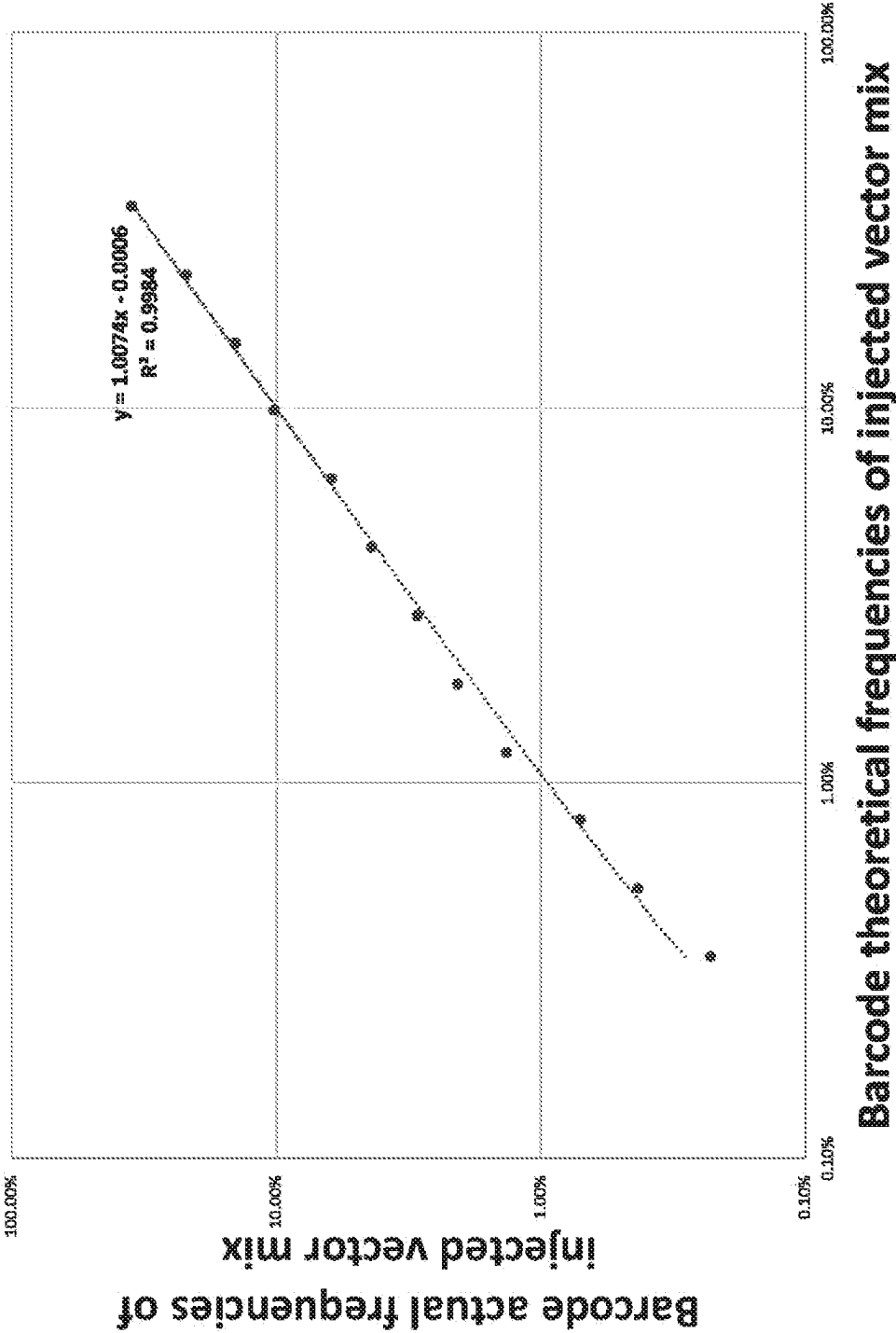


FIG. 19A

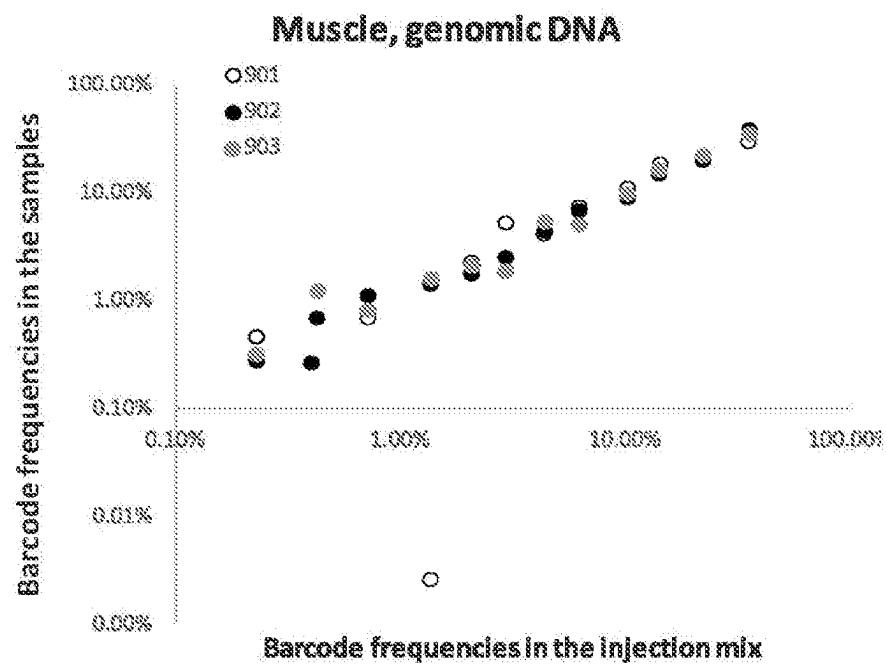
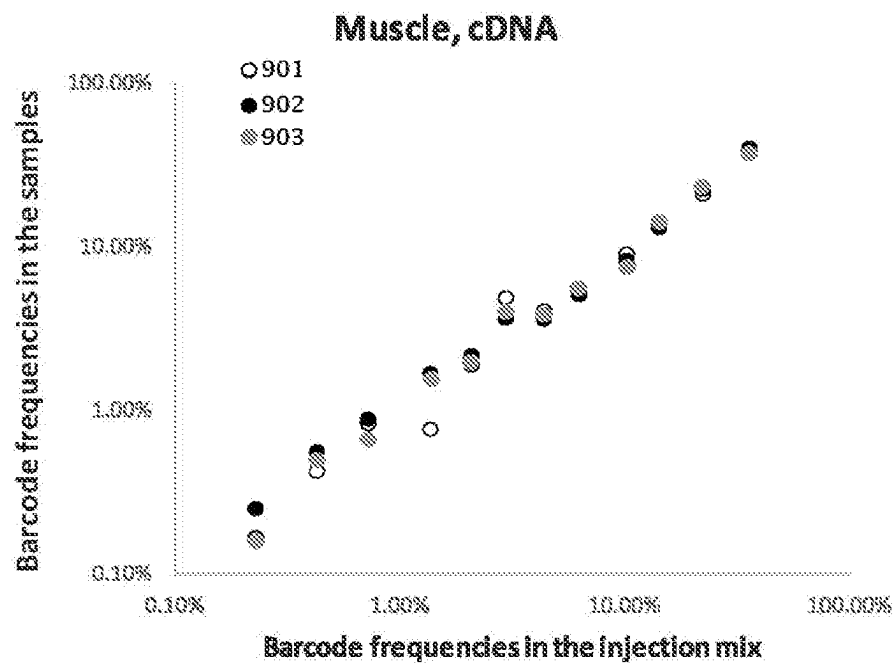


FIG. 19B



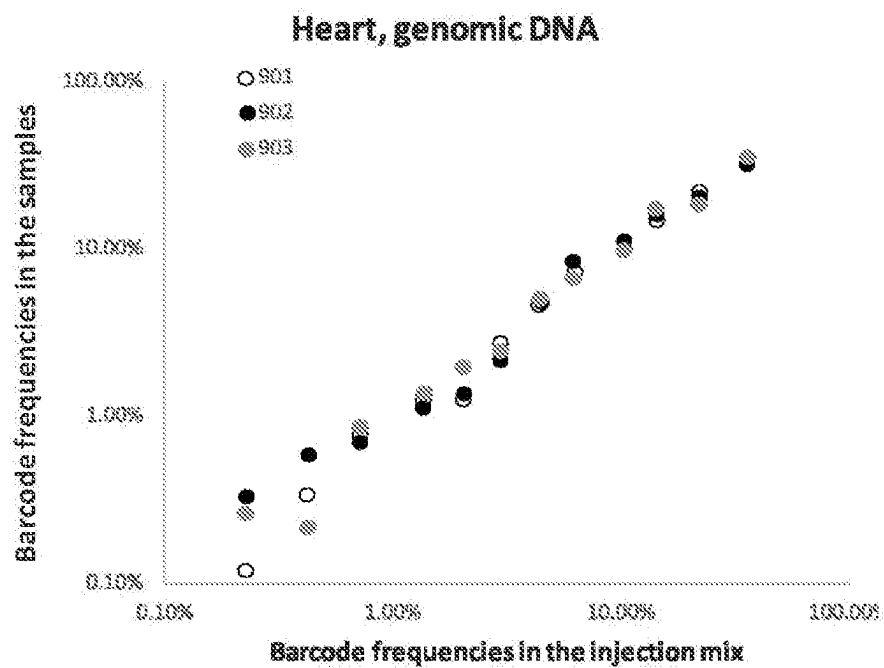


FIG. 19C

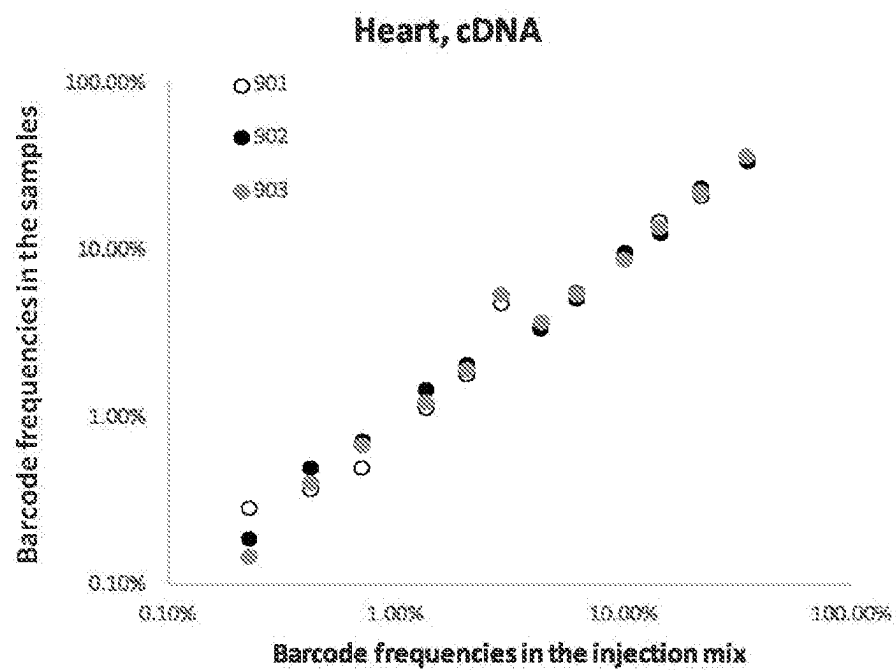


FIG. 19D

FIG. 19E

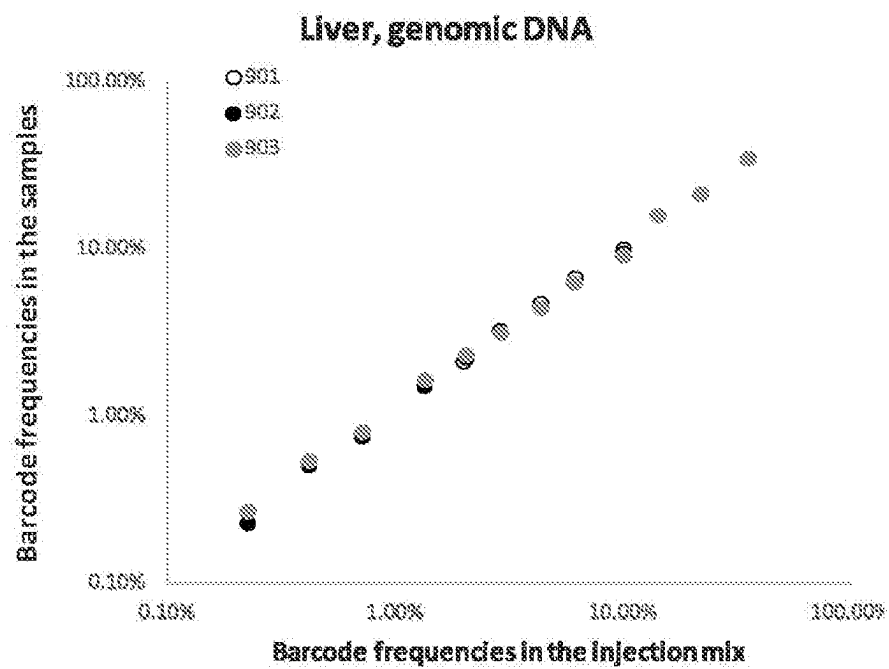
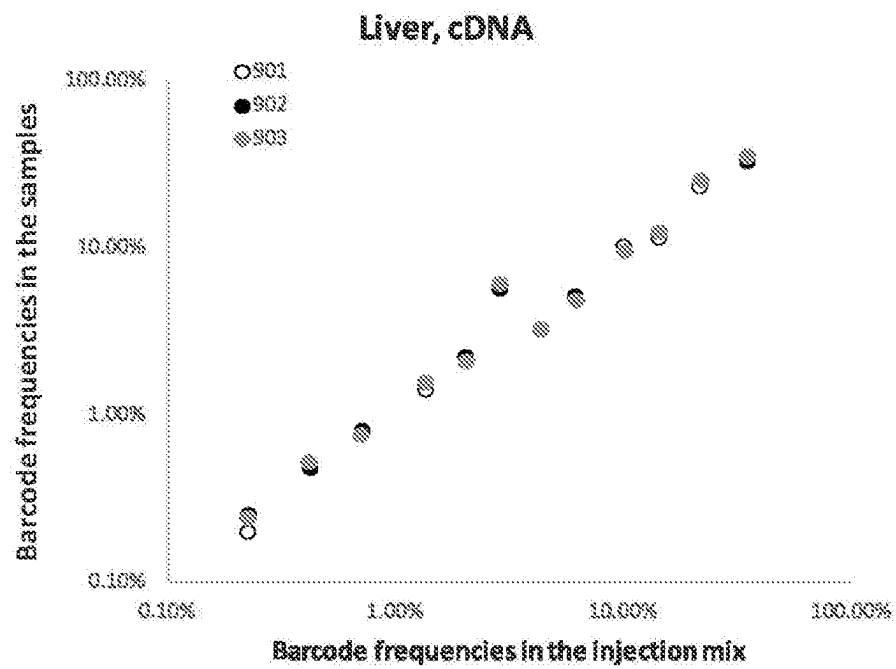
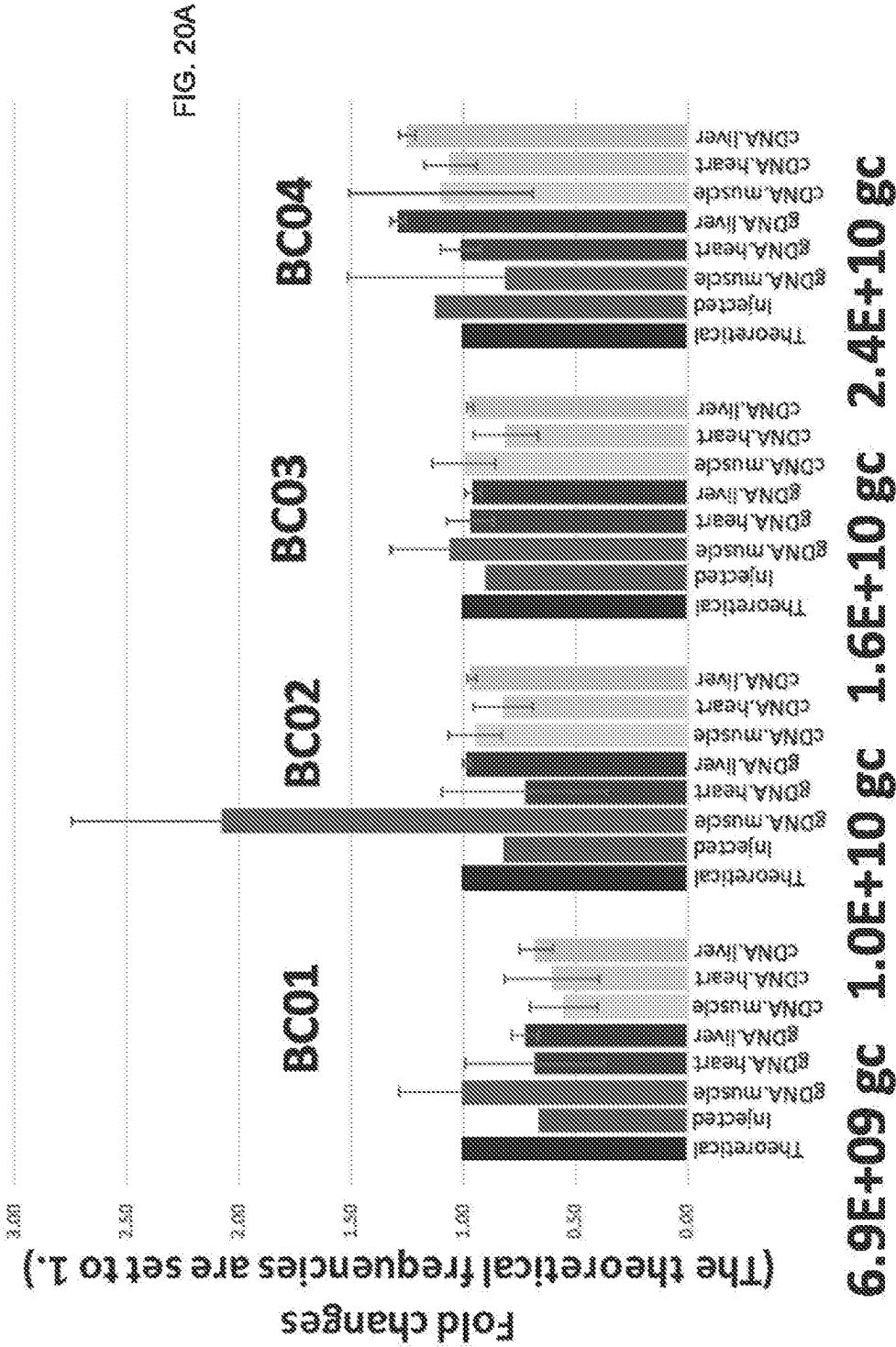


FIG. 19F

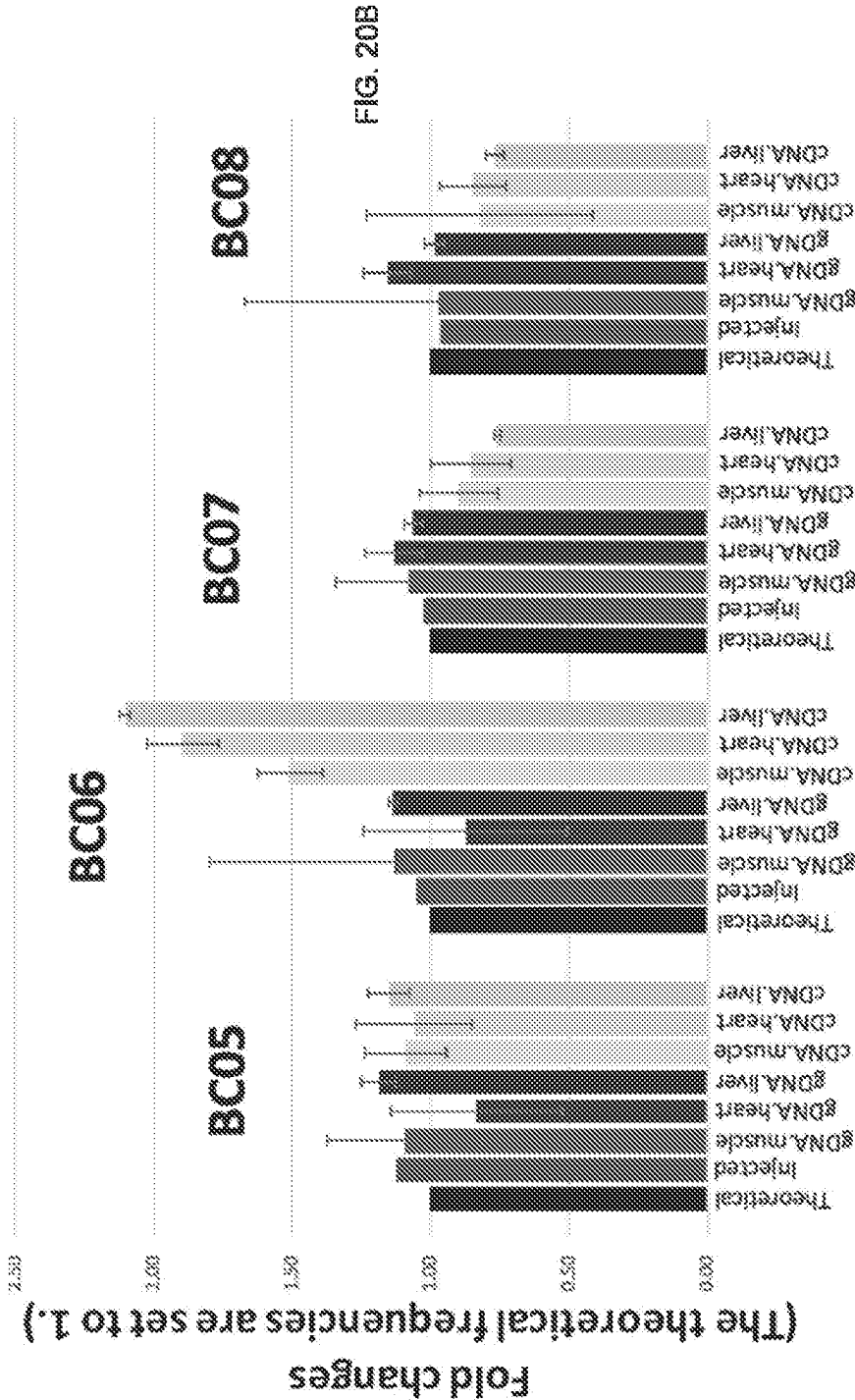


Fold changes (compared to the theoretical frequencies)



The barcodes and their genome frequencies injection.

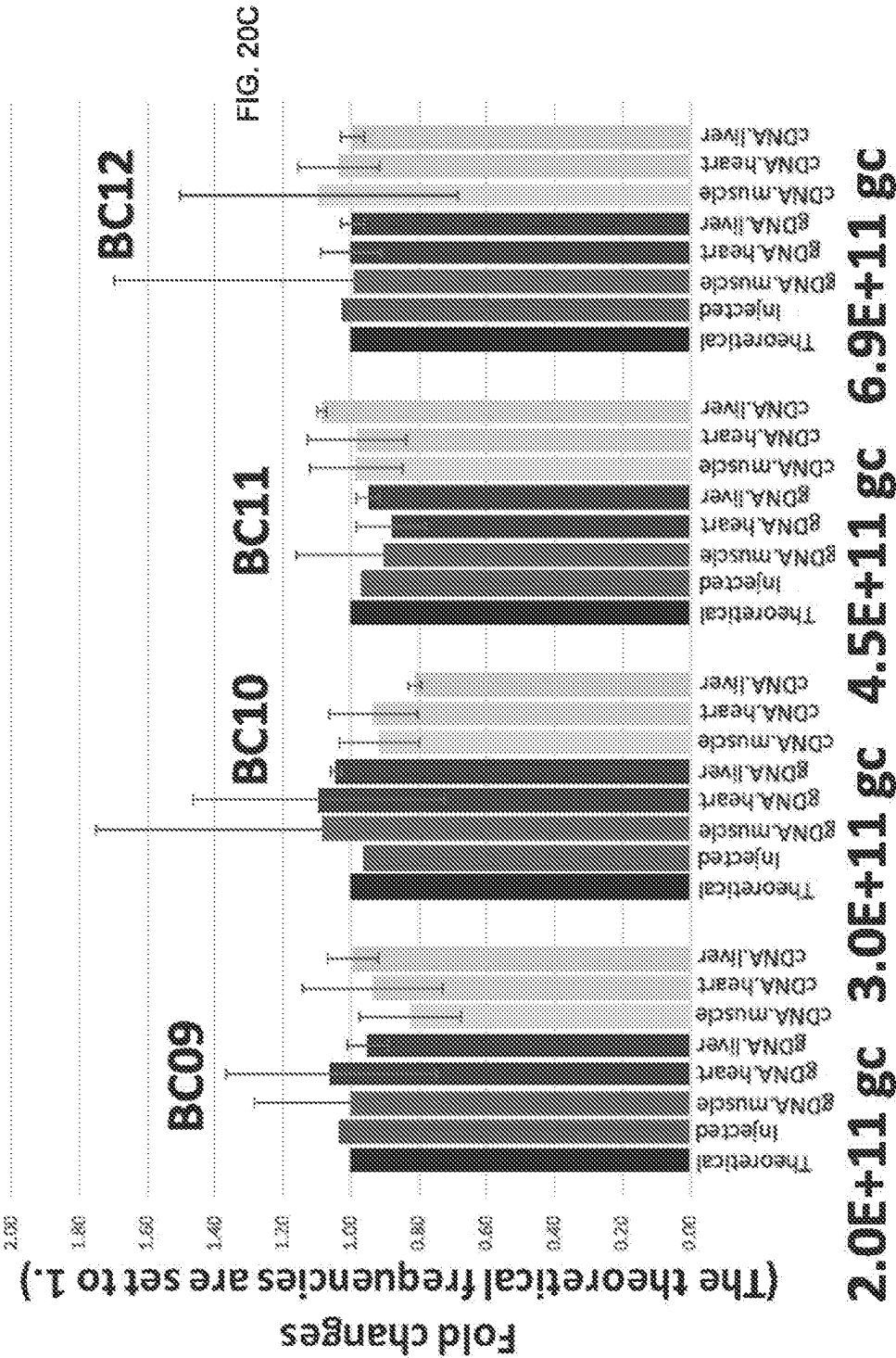
Fold changes (compared to the theoretical frequencies)



3.7E+10 gc 5.6E+10 gc 8.5E+10 gc 1.3E+11 gc

The barcodes and their genome frequencies injection.

Fold changes (compared to the theoretical frequencies)



The barcodes and their genome frequencies injection.

Biodistribution in Mice

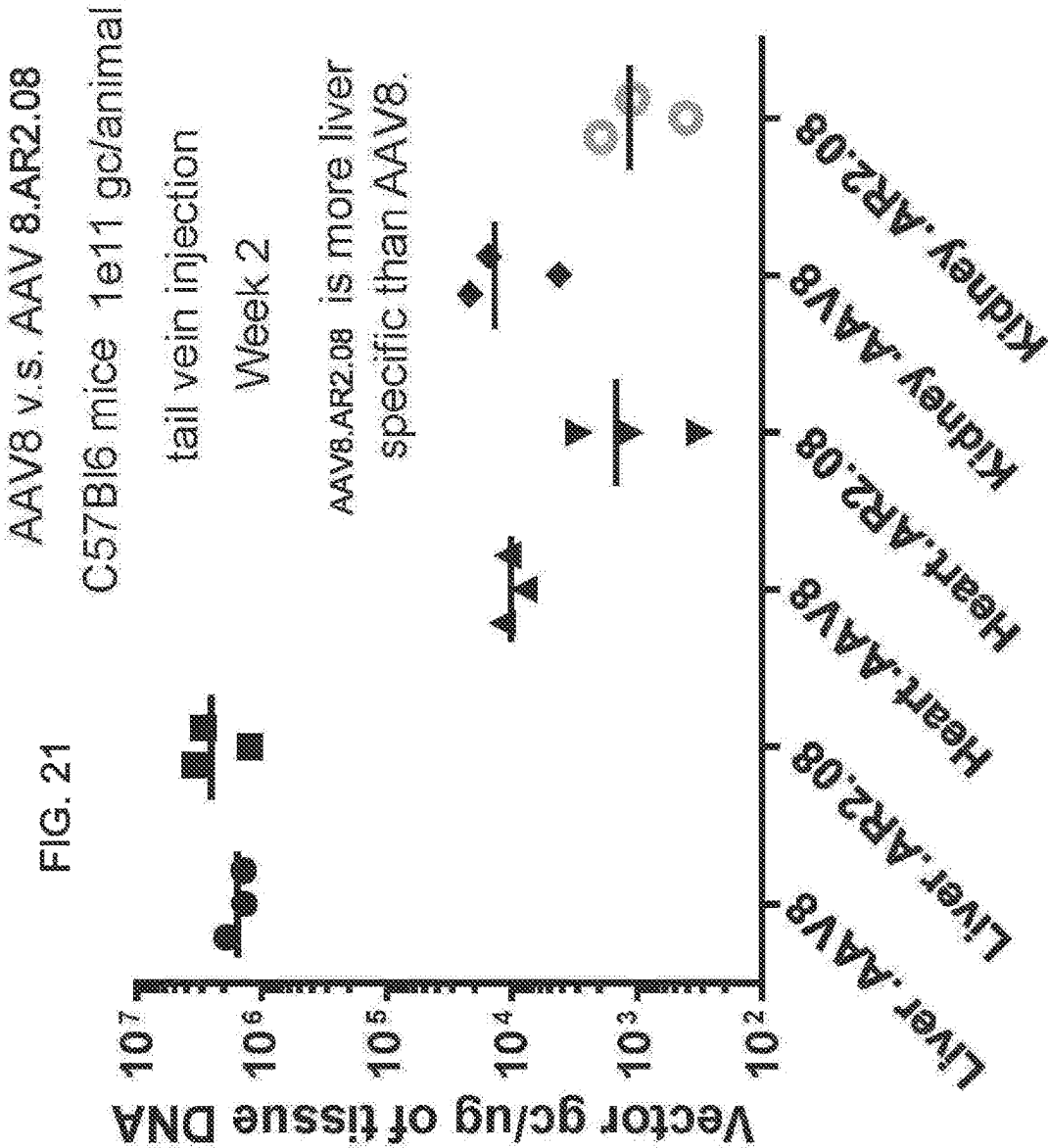


FIG. 22

- AAV8 vs AAV8.AR2.08
 - CB7.C1.eGFP.W PRE.rBG
 - ddPCR(PK), 1e9 and 1e10 gc/animal
 - B6, male, i.v., week 2 liver

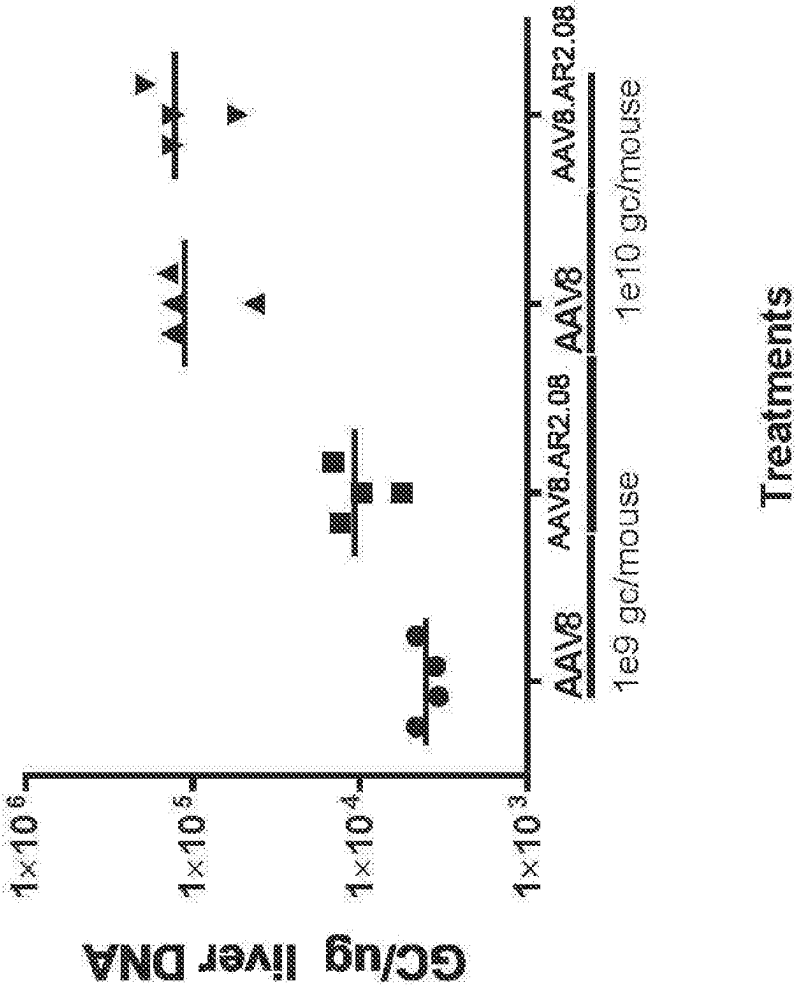


FIG. 23

Lot #	Vector Name	Genome Titer ddPCR- GC/ml	ddPCR-Total Yield	Purity
WL184C S	AAVG3.TBG.PI.eGFP.WPRE.bGH(p0080:p0146)	1.141e14 GC/ml	6.104e14 GC	100.000
WL1549C S	AAV8.TBG.PI.eGFP.WPRE.bGH	1.11e14	5.578e14	98.9
WL1956C S	AAVrh79.TBG.PI.eGFP.WPRE.bGH(p3889:p0146)	1.609e14	8.568e14	

FIG. 24

Vector Biodistribution of AAV8.AR2.08 (17-07)

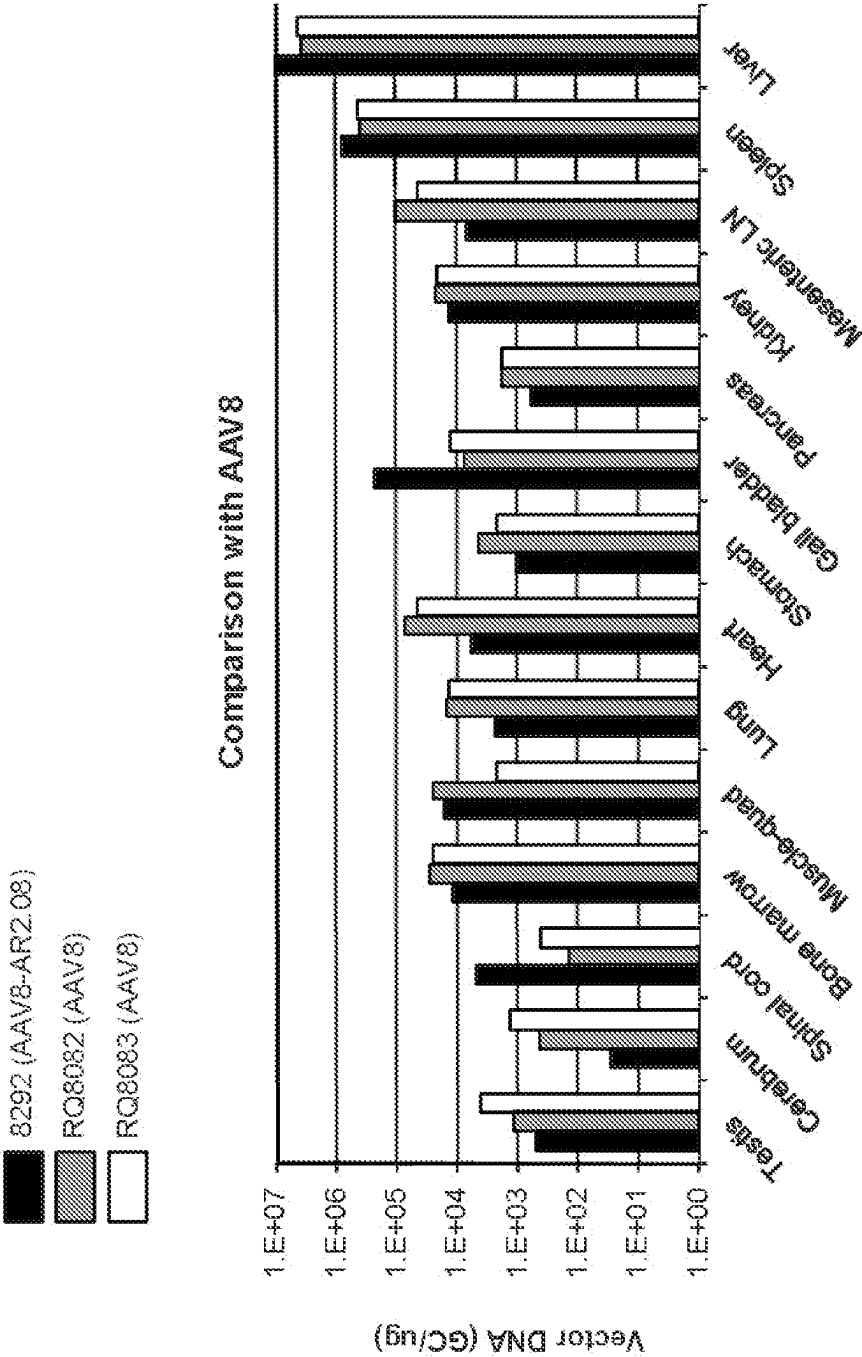


FIG. 25

17-07: Liver Transduction by New Serotypes

AAV.TBG.EGFP.WPRE (1x10¹³ ddGC/kg), day 7 necropsy

Serotype	ID	Species	Sex	Weight (kg)
AAV8	RA1475	rhesus	F	4.60
AAVrh79	RA2362	cyno	F	4.10
AAV8.AR2.08	8292	cyno	M	8.55
AAV8.AR2.08	RQ9471	rhesus	F	9.26

17-07: Liver Transduction by New Serotypes

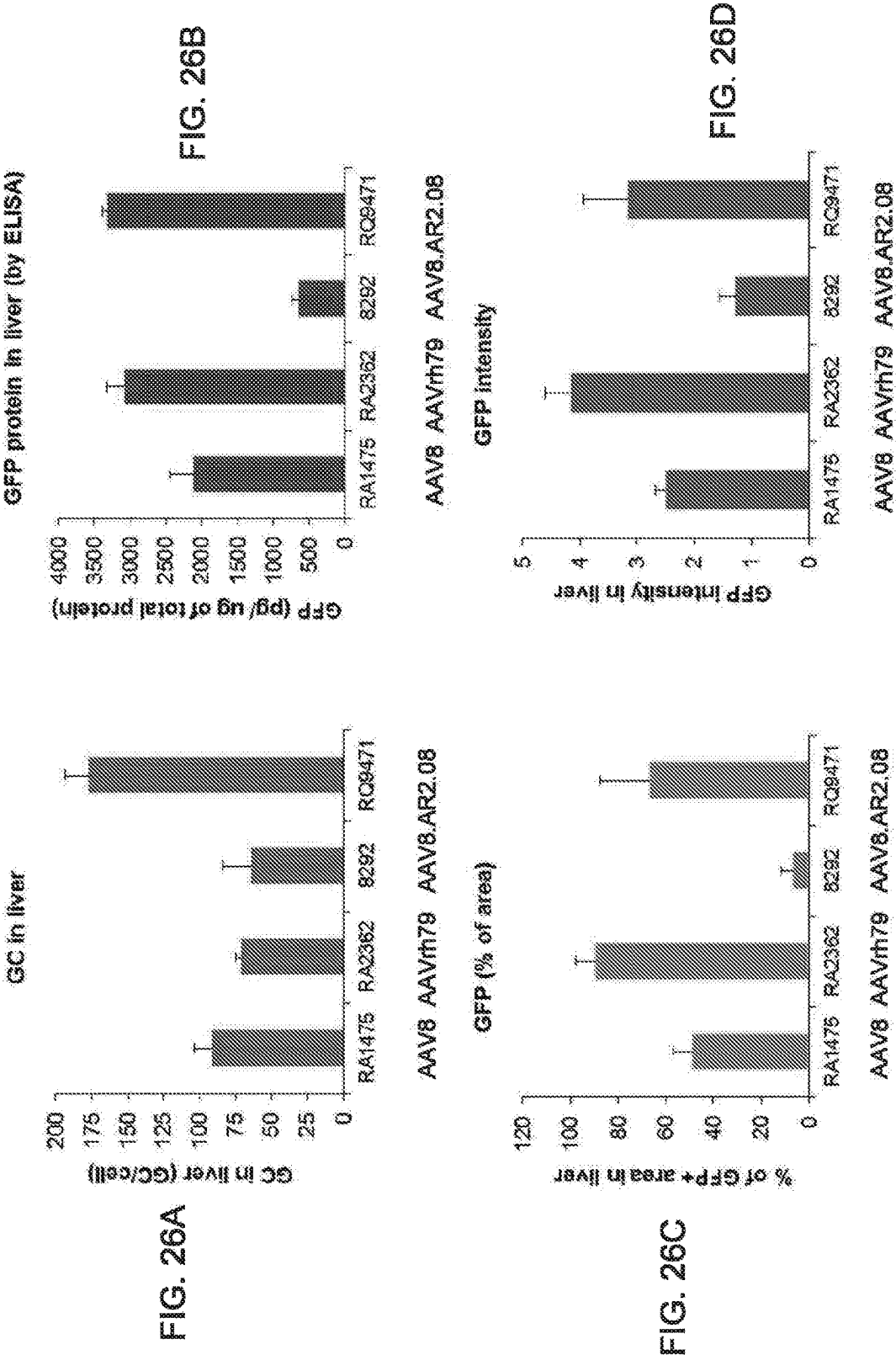


FIG. 27A

GFP Expression in NHP Liver

GFP intensity

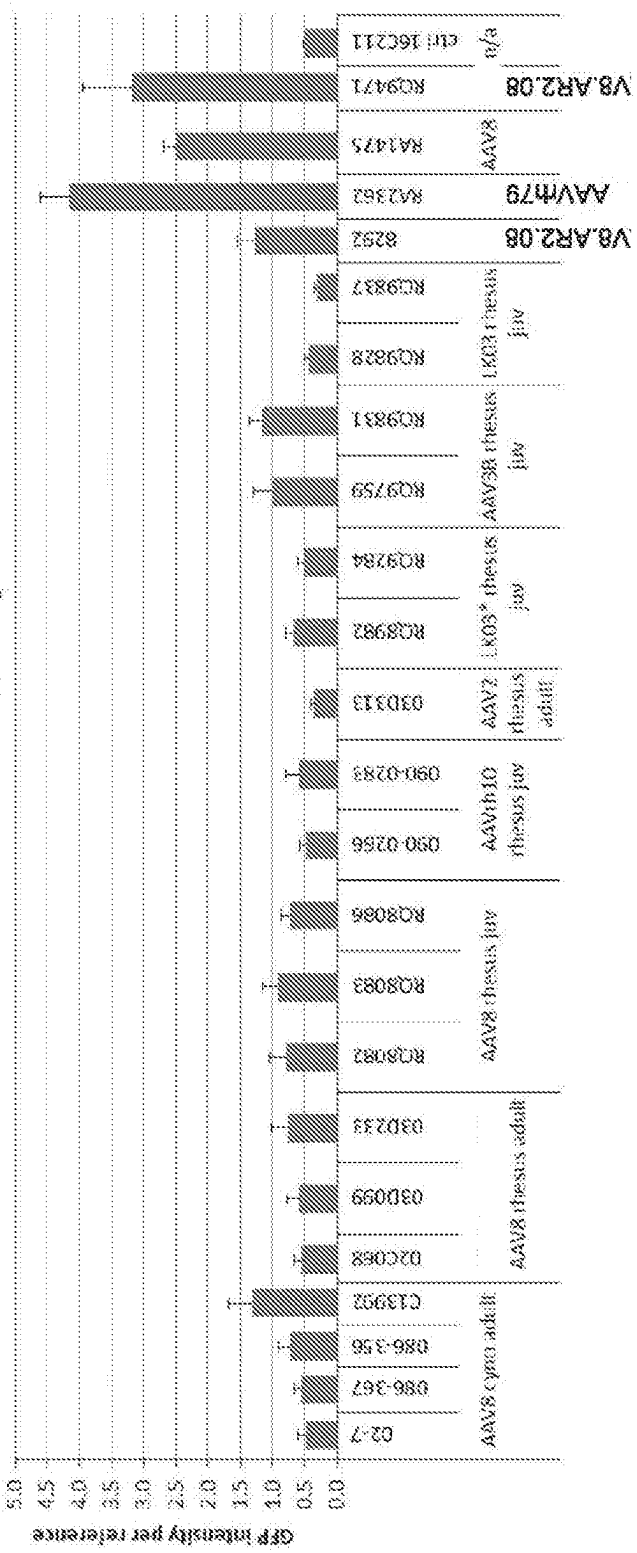


FIG. 28

17-07: Vector Biodistribution

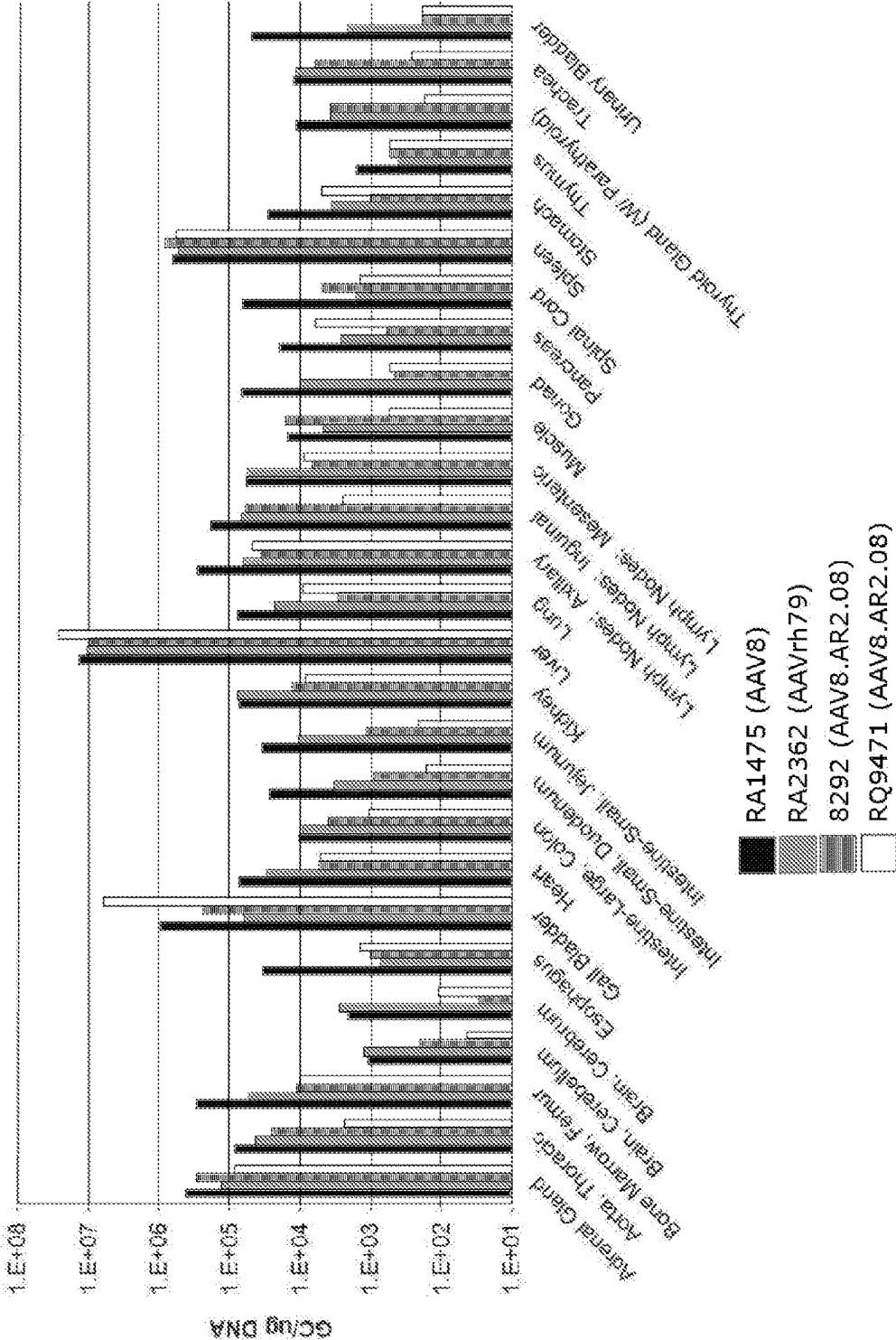
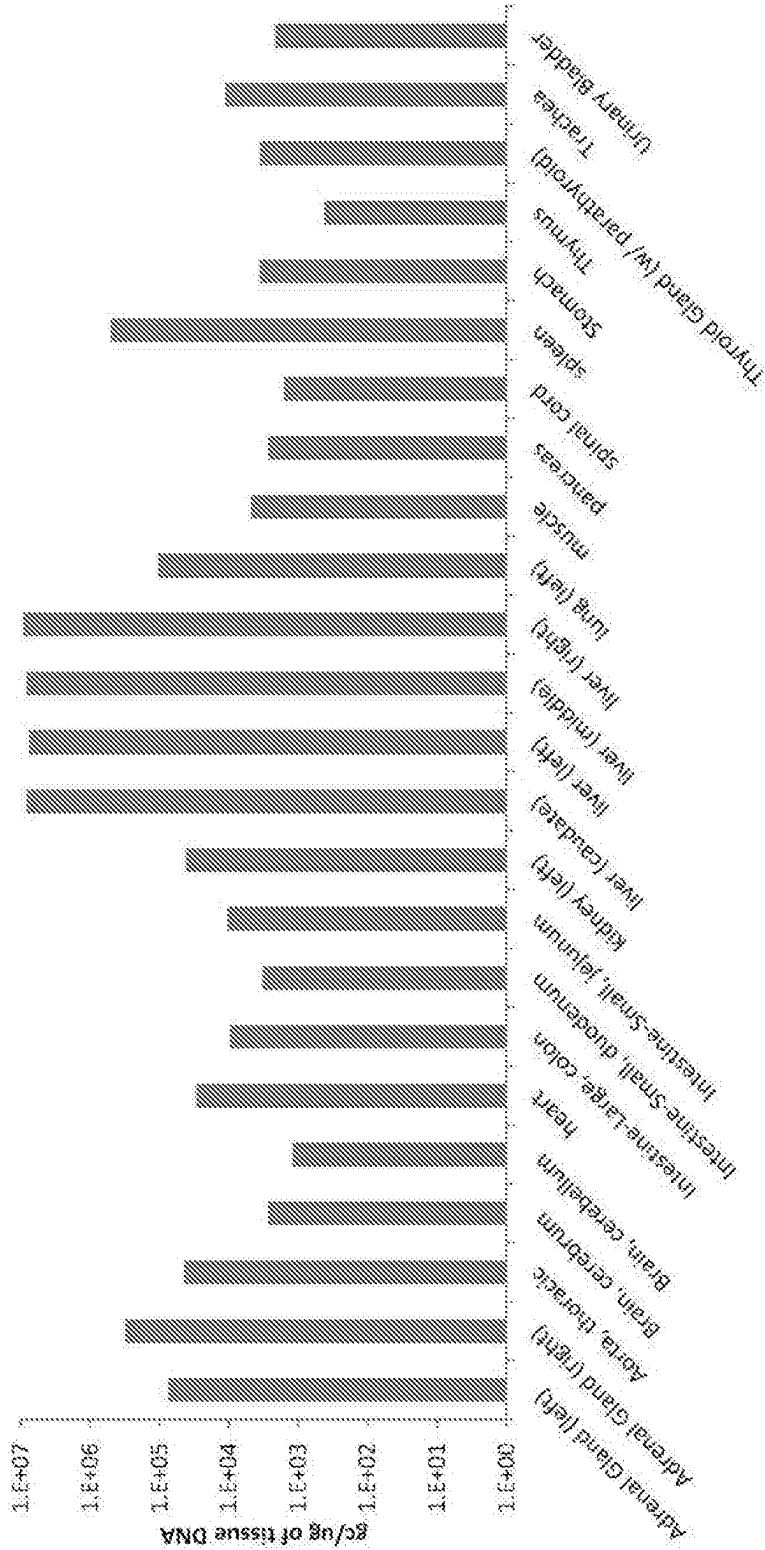


FIG. 29

17-07c RA2362 Biodistribution



1

**ADENO-ASSOCIATED VIRUS (AAV)
VECTORS, AAV VECTORS HAVING
REDUCED CAPSID DEAMIDATION AND
USES THEREFOR**

**CROSS-REFERENCE TO RELATED
APPLICATIONS**

This is a National Stage Entry under 35 U.S.C. 371 of International Patent Application No. PCT/US2019/019861, filed Feb. 27, 2019, which claims priority to U.S. Provisional Patent Application No. 62/635,964, filed Feb. 27, 2018, U.S. Provisional Patent Application No. 62/635,968, filed Feb. 27, 2018, U.S. Provisional Patent Application No. 62/663,788, filed Apr. 27, 2018, U.S. Provisional Patent Application No. 62/663,797, filed Apr. 27, 2018, U.S. Provisional Patent Application No. 62/667,587, filed May 6, 2018, U.S. Provisional Patent Application No. 62/667,888, filed May 7, 2018, U.S. Provisional Patent Application No. 62/667,881, filed May 7, 2018, U.S. Provisional Patent Application No. 62/677,471, filed May 29, 2018, U.S. Provisional Patent Application No. 62/677,474, filed May 29, 2018, U.S. Provisional Patent Application No. 62/667,585, filed May 29, 2018, U.S. Provisional Patent Application No. 62/703,670, filed Jul. 26, 2018, U.S. Provisional Patent Application No. 62/703,673, filed Jul. 26, 2018, U.S. Provisional Patent Application No. 62/722,382, filed Aug. 24, 2018, and U.S. Provisional Patent Application No. 62/722,388, filed Aug. 24, 2018. These applications are incorporated by reference herein.

**STATEMENT OF FEDERALLY SPONSORED
RESEARCH**

This invention was made with government support under P01HL059407 awarded by the National Institutes of Health. The government has certain rights in the invention.

**REFERENCE TO A SEQUENCE LISTING
SUBMITTED VIA EFS-WEB**

The content of the text filed of the sequence listing named "18-8592PCT_Sequence_Listing_ST25" which was created on Feb. 27, 2019 and electronically submitted via EFS-Web herewith the application is incorporated herein by reference in its entirety.

BACKGROUND OF THE INVENTION

The adeno-associated virus (AAV) capsid is icosahedral in structure and is comprised of 60 of viral protein (VP) monomers (VP1, VP2, and VP3) in a 1:1:10 ratio (Xie Q, et al. *Proc Natl Acad Sci USA*. 2002; 99 (16): 10405-10). The entirety of the VP3 protein sequence (~535aa) is contained within the C-terminus of both VP1 and VP2, and the shared VP3 sequences are primarily responsible for the overall capsid structure. Due to the structural flexibility of the VP1/VP2 unique regions and the low representation of VP1 and VP2 monomers relative to VP3 monomers in the assembled capsid, VP3 is the only capsid protein to be resolved via x-ray crystallography (Nam H J, et al. *J Virol*. 2007; 81 (22): 12260-71). VP3 contains nine hypervariable regions (HVRs) that are the primary source of sequence variation between AAV serotypes (Govindasamy L, et al. *J Virol*. 2013; 87 (20): 11187-99). Given their flexibility and location on the capsid surface, HVRs are largely responsible for interactions with target cells as well as with the immune

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system (Huang L Y, et al. *J Virol*. 2016; 90 (11): 5219-30; Raupp C, et al. *J Virol*. 2012; 86 (17): 9396-408). While the structures of a number of serotypes are published (Protein Data Bank (PDB) IDs 1LP3, 4RSO, 4V86, 3UX1, 3KIC, 2QA0, 2G8G from the Research Collaboratory for Structural Bioinformatics (RCSB) database) for the structure entries for AAV2, AAVrh.8, AAV6, AAV9, AAV3B, AAVS, and AAV4, respectively), there is very little information in the literature regarding modifications on the surface of these capsids. Research suggests that intracellular phosphorylation of the capsid occurs at specific tyrosine residues (Zhong L, et al. *Virology*. 2008; 381 (2): 194-202). Despite putative glycosylation sites in the primary VP3 sequence, no glycosylation events have been identified in AAV2 (Murray S, et al. *J Virol*. 2006; 80 (12): 6171-6; Jin X, et al. *Hum Gene Ther Methods*. 2017; 28 (5): 255-267); other AAV serotypes have not yet been evaluated for capsid glycosylation.

AAV gene therapy vectors have undergone less of the molecular-level scrutiny that typically accompanies the development and manufacturing of recombinant protein therapeutics. AAV capsid post-translational modifications (PTM) have largely been unexplored, so accordingly, little is known about their potential to impact function, or about strategies to control PTM levels in manufactured AAV therapies.

Variations in post-translational modifications of non-gene therapy protein therapeutics have complicated their development as drugs. Jenkins, N, Murphy, L, and Tyther, R (2008). Post-translational modifications of recombinant proteins: significance for biopharmaceuticals. *Mol Biotechnol* 39:113-118; Houde, D, Peng, Y, Berkowitz, S A, and Engen, JR (2010). Post-translational modifications differentially affect IgG1 conformation and receptor binding. *Mol Cell Proteomics* 9:1716-1728. For example, deamidation of selected amino acids modulates the stability of and the immune response to the recombinant protective antigen-based anthrax vaccine. (Powell B S, et al. *Proteins*. 2007; 68 (2): 458-79; Verma A, et al. *Clin Vaccine Immunol*. 2016; 23 (5): 396-402). In some instances, this process is catalyzed by viral or bacterial deamidases to modulate host cell signaling pathways or innate immune responses (Zhao J, et al. *J Virol*. 2016; 90 (9): 4262-8; Zhao J, et al. *Cell Host Microbe*. 2016; 20 (6): 770-84). More commonly, endogenous deamidation is an enzyme-independent spontaneous process. Although the purpose of spontaneous deamidation has not been fully elucidated, previous studies have suggested that this event serves as a molecular clock to indicate the relative age of a protein and regulate its turnover (Robinson N E and Robinson A B. *Proc Natl Acad Sci USA*. 2001; 98 (3): 944-9).

Deamidation occurs when the amide group of asparagine or less frequently glutamine undergoes nucleophilic attack from an adjacent nitrogen atom and the amide group is lost. This process leads to a succinimide intermediate (Yang H and Zubarev R A. *Electrophoresis*. 2010; 31 (11): 1764-72) that, via hydrolysis, resolves into a mixture of aspartic acid and isoaspartic acid (or glutamic acid and isoglutamic acid) (Catak S, et al. *J Phys Chem A*. 2009; 113 (6): 1111-20). Studies of short, synthetic peptides estimate that this hydrolysis results in a 3:1 mixture of isoaspartic acid to aspartic acid (Geiger T. and Clarke S. *J Biol Chem*. 1987; 262 (2): 785-94).

There continues to be a need for compositions comprising AAV-based constructs for delivery of heterologous molecules which have stable receptor binding and/or stable capsids, avoid neutralizing antibodies and/or retain purity on storage.

SUMMARY OF THE INVENTION

In one embodiment, a composition is provided which includes a mixed population of recombinant adeno-associated virus (rAAV), each of said rAAV comprising: (a) an AAV capsid comprising about 60 capsid vp1 proteins, vp2 proteins and vp3 proteins, wherein the vp1, vp2 and vp3 proteins are: a heterogeneous population of vp1 proteins which are produced from a nucleic acid sequence encoding a selected AAV vp1 amino acid sequence, a heterogeneous population of vp2 proteins which are produced from a nucleic acid sequence encoding a selected AAV vp2 amino acid sequence, a heterogeneous population of vp3 proteins which produced from a nucleic acid sequence encoding a selected AAV vp3 amino acid sequence, wherein: the vp1, vp2 and vp3 proteins contain subpopulations with amino acid modifications comprising at least two highly deamidated asparagines (N) in asparagine-glycine pairs in the AAV capsid and optionally further comprising subpopulations comprising other deamidated amino acids, wherein the deamidation results in an amino acid change; and (b) a vector genome in the AAV capsid, the vector genome comprising a nucleic acid molecule comprising AAV inverted terminal repeat sequences and a non-AAV nucleic acid sequence encoding a product operably linked to sequences which direct expression of the product in a host cell.

In certain embodiments, the deamidated asparagines are deamidated to aspartic acid, isoaspartic acid, an interconverting aspartic acid/isoaspartic acid pair, or combinations thereof. In certain embodiments, the capsid further comprises deamidated glutamine(s) which are deamidated to (α)-glutamic acid, γ-glutamic acid, an interconverting (α)-glutamic acid/γ-glutamic acid pair, or combinations thereof.

In a further aspect, a recombinant adeno-associated virus (rAAV) is provided which comprises: (A) an AAVrh79 capsid comprising one or more of: (1) AAVrh79 capsid proteins comprising: a heterogeneous population of AAVrh79 vp1 proteins selected from: vp1 proteins produced by expression from a nucleic acid sequence which encodes the predicted amino acid sequence of 1 to 738 of SEQ ID NO:2, vp1 proteins produced from SEQ ID NO: 1, or vp1 proteins produced from a nucleic acid sequence at least 70% identical to SEQ ID NO: 1 which encodes the predicted amino acid sequence of 1 to 738 of SEQ ID NO:2; a heterogeneous population of AAVrh79 vp2 proteins selected from: vp2 proteins produced by expression from a nucleic acid sequence which encodes the predicted amino acid sequence of at least about amino acids 138 to 738 of SEQ ID NO:2, vp2 proteins produced from a sequence comprising at least nucleotides 412 to 2214 of SEQ ID NO:1, or vp2 proteins produced from a nucleic acid sequence at least 70% identical to at least nucleotides 412 to 2214 of SEQ ID NO:1 which encodes the predicted amino acid sequence of at least about amino acids 138 to 738 of SEQ ID NO:2; a heterogeneous population of AAVrh79 vp3 proteins selected from: vp3 proteins produced by expression from a nucleic acid sequence which encodes the predicted amino acid sequence of at least about amino acids 204 to 738 of SEQ ID NO:2, vp3 proteins produced from a sequence comprising at least nucleotides 610 to 2214 of SEQ ID NO:1, or vp3 proteins produced from a nucleic acid sequence at least 70% identical to at least nucleotides 610 to 2214 of SEQ ID NO:1 which encodes the predicted amino acid sequence of at least about amino acids 204 to 738 of SEQ ID NO:2; and/or (2) a heterogeneous population of vp1 proteins which are the product of a nucleic acid sequence encoding the amino acid

sequence of SEQ ID NO: 2, a heterogeneous population of vp2 proteins which are the product of a nucleic acid sequence encoding the amino acid sequence of at least about amino acids 138 to 738 of SEQ ID NO: 2, and a heterogeneous population of vp3 proteins which are the product of a nucleic acid sequence encoding at least amino acids 204 to 738 of SEQ ID NO:2, wherein: the vp1, vp2 and vp3 proteins contain subpopulations with amino acid modifications comprising at least two highly deamidated asparagines (N) in asparagine-glycine pairs in SEQ ID NO: 2 and optionally further comprising subpopulations comprising other deamidated amino acids, wherein the deamidation results in an amino acid change; and (B) a vector genome in the AAVrh79 capsid, the vector genome comprising a nucleic acid molecule comprising AAV inverted terminal repeat sequences and a non-AAV nucleic acid sequence encoding a product operably linked to sequences which direct expression of the product in a host cell.

In another aspect, a method of transducing a target tissue is provided. In one embodiment, the method includes administering an AAV having an AAVrh79 capsid as described herein. In one embodiment, a method of transducing liver tissue is provided, comprising administering an AAV having the AAVrh79 capsid. In another embodiment, a method of transducing muscle tissue is provided, comprising administering an AAV having the AAVrh79 capsid.

In yet another aspect, a method of reducing deamidation of an AAVrh79 capsid is provided. In one embodiment, the method includes producing an AAVrh79 capsid from a nucleic acid sequence containing modified AAVrh79 VP codons, the nucleic acid sequence comprising independently modified glycine codons at one to four of the asparagine-glycine pairs located at position N57, N263, N385 and/or N514 in SEQ ID NO: 2, such that the modified codon encodes an amino acid other than glycine. In another embodiment, the method includes producing an AAVrh79 capsid from a nucleic acid sequence containing modified AAVrh79 vp codons, the nucleic acid sequence comprising independently modified glycine codons at one to four of the asparagine-glycine pairs located at position N94, N254, N305, N410, and/or N479 of SEQ ID NO: 2.

In a further embodiment, a rAAV8.AR2.08 is provided which comprises: (A) an AAV8.AR2.08 capsid comprising one or more of: (1) AAV8.2.08 capsid proteins comprising: a heterogeneous population of AAV8.AR2.08 vp1 proteins selected from: vp1 proteins produced by expression from a nucleic acid sequence which encodes the predicted amino acid sequence of 1 to 738 of SEQ ID NO: 18, vp1 proteins produced from SEQ ID NO: 17, or vp1 proteins produced from a nucleic acid sequence at least 70% identical to SEQ ID NO: 17 which encodes the predicted amino acid sequence of 1 to 738 of SEQ ID NO:18, a heterogeneous population of AAV8.AR2.08 vp2 proteins selected from: AAV8.AR2.08 vp2 proteins produced by expression from a nucleic acid sequence which encodes the predicted amino acid sequence of at least about amino acids 138 to 738 of SEQ ID NO:18, vp2 proteins produced from a sequence comprising at least nucleotides 412 to 2214 of SEQ ID NO: 17, or vp2 proteins produced from a nucleic acid sequence at least 70% identical to at least nucleotides 412 to 2214 of SEQ ID NO:17 which encodes the predicted amino acid sequence of at least about amino acids 138 to 738 of SEQ ID NO:18; a heterogeneous population of AAV8.AR2.08 vp3 proteins selected from: vp3 proteins produced by expression from a nucleic acid sequence which encodes the predicted amino acid sequence of at least about amino acids 204 to 738 of SEQ ID NO:18, vp3 proteins produced from a sequence

comprising at least nucleotides 607 to 2214 of SEQ ID NO:17, or vp3 proteins produced from a nucleic acid sequence at least 70% identical to at least nucleotides 607 to 2214 of SEQ ID NO:17 which encodes the predicted amino acid sequence of at least about amino acids 204 to 738 of SEQ ID NO:18; and/or (2) a heterogeneous population of vp1 proteins which are the product of a nucleic acid sequence encoding the amino acid sequence of SEQ ID NO: 18, a heterogeneous population of vp2 proteins which are the product of a nucleic acid sequence encoding the amino acid sequence of at least about amino acids 138 to 738 of SEQ ID NO: 18, and a heterogeneous population of vp3 proteins which are the product of a nucleic acid sequence encoding at least amino acids 204 to 738 of SEQ ID NO: 18 wherein: the vp1, vp2 and vp3 proteins contain subpopulations with amino acid modifications comprising at least two highly deamidated asparagines (N) in asparagine-glycine pairs in SEQ ID NO: 18 and optionally further comprising subpopulations comprising other deamidated amino acids, wherein the deamidation results in an amino acid change; and (B) a vector genome in the AAV8.AR2.08 capsid, the vector genome comprising a nucleic acid molecule comprising AAV inverted terminal repeat sequences and a non-AAV nucleic acid sequence encoding a product operably linked to sequences which direct expression of the product in a host cell.

In another aspect, a method of transducing a target tissue is provided. In one embodiment, the method includes administering an AAV having an AAV8.AR2.08 capsid as described herein. In one embodiment, a method of transducing liver tissue is provided, comprising administering an AAV having the AAV8.AR2.08 capsid. In another embodiment, a method of transducing muscle tissue is provided, comprising administering an AAV having the AAV8.AR2.08 capsid.

In yet another aspect, a method of reducing deamidation of an AAV8.AR2.08 capsid is provided. In one embodiment, the method includes producing an AAV8.AR2.08 capsid from a nucleic acid sequence containing modified AAV8.AR2.08 vp codons, the nucleic acid sequence comprising independently modified glycine codons at one to four of the asparagine—glycine pairs located at position N57, N263, N385, N514, and/or N540 in SEQ ID NO: 18, such that the modified codon encodes an amino acid other than glycine. In another embodiment, the method includes producing an AAV8.AR2.08 capsid from a nucleic acid sequence containing modified AAV8.AR2.08 vp codons, the nucleic acid sequence comprising independently modified glycine codons at one to four of the asparagine-glycine pairs located at position N94, N254, N305, N521, N590, Q601, N653, and/or N665 of SEQ ID NO: 18.

In certain embodiments, a rAAV5.5.9 is provided which comprises: (A) an AAV5.5.9 capsid comprising one or more of: (1) AAVG5 capsid proteins comprising: a heterogeneous population of AAV5.5.9 vp1 proteins selected from: vp1 proteins produced by expression from a nucleic acid sequence which encodes the predicted amino acid sequence of 1 to 726 of SEQ ID NO: 10, vp1 proteins produced from SEQ ID NO:9, or vp1 proteins produced from a nucleic acid sequence at least 70% identical to SEQ ID NO:9 which encodes the predicted amino acid sequence of 1 to 726 of SEQ ID NO:1, a heterogeneous population of AAV5.5.9 vp2 proteins selected from: AAVG5 vp2 proteins produced by expression from a nucleic acid sequence which encodes the predicted amino acid sequence of at least about amino acids 137 to 726 of SEQ ID NO: 10, vp2 proteins produced from a sequence comprising at least nucleotides 409 to 2178 of

SEQ ID NO:9, or vp2 proteins produced from a nucleic acid sequence at least 70% identical to at least nucleotides 409 to 2178 of SEQ ID NO:9 which encodes the predicted amino acid sequence of at least about amino acids 137 to 726 of SEQ ID NO:10, a heterogeneous population of AAV5.5.9 vp3 proteins selected from: AAV5.5.9 vp3 proteins produced by expression from a nucleic acid sequence which encodes the predicted amino acid sequence of at least about amino acids 203 to 726 of SEQ ID NO:10, vp3 proteins produced from a sequence comprising at least nucleotides 607 to 2178 of SEQ ID NO:9, or vp3 proteins produced from a nucleic acid sequence at least 70% identical to at least nucleotides 607 to 2178 of SEQ ID NO: 9 which encodes the predicted amino acid sequence of at least about amino acids 203 to 726 of SEQ ID NO:10; and/or (2) a heterogeneous population of vp1 proteins which are the product of a nucleic acid sequence encoding the amino acid sequence of SEQ ID NO: 10, a heterogeneous population of vp2 proteins which are the product of a nucleic acid sequence encoding the amino acid sequence of at least about amino acids 137 to 726 of SEQ ID NO: 10, and a heterogeneous population of vp3 proteins which are the product of a nucleic acid sequence encoding at least amino acids 203 to 726 of SEQ ID NO:10 wherein: the vp1, vp2 and vp3 proteins contain subpopulations with amino acid modifications comprising at least two highly deamidated asparagines (N) in asparagine-glycine pairs in SEQ ID NO: 10 and optionally further comprising subpopulations comprising other deamidated amino acids, wherein the deamidation results in an amino acid change; and (B) a vector genome in the AAV5.5.9 capsid, the vector genome comprising a nucleic acid molecule comprising AAV inverted terminal repeat sequences and a non-AAV nucleic acid sequence encoding a product operably linked to sequences which direct expression of the product in a host cell.

In another aspect, a method of transducing a target tissue is provided. In one embodiment, the method includes administering an AAV having an AAV5.5.9 capsid as described herein. In one embodiment, a method of transducing liver tissue is provided, comprising administering an AAV having the AAV5.5.9 capsid. In another embodiment, a method of transducing muscle tissue is provided, comprising administering an AAV having the AAV5.5.9 capsid.

In yet another aspect, a method of reducing deamidation of an AAV5.5.9 capsid is provided. In one embodiment, the method includes producing an AAV5.5.9 capsid from a nucleic acid sequence containing modified AAV5.5.9 vp codons, the nucleic acid sequence comprising independently modified glycine codons at one to four of the asparagine-glycine pairs located at position N57, N319, N442, and/or N502 in SEQ ID NO: 10, such that the modified codon encodes an amino acid other than glycine. In another embodiment, the method includes producing an AAV5.5.9 capsid from a nucleic acid sequence containing modified AAV5.5.9 vp codons, the nucleic acid sequence comprising independently modified glycine codons at one to four of the asparagine-glycine pairs located at position N35, N113, N204, N217, N243, N249, N293/294, N304, N399/400, N505, Q589, N618, N641, N653, N658, and/or N699 of SEQ ID NO: 10.

In another aspect, a composition comprising a mixed population of recombinant AAVrh79, AAV8.AR2.08, or AAV5.5.9, as described herein, is provided.

In yet another aspect, a recombinant AAV (rAAV) as described herein is provided, for delivering a desired gene product to a subject in need thereof.

In another aspect, a rAAV production system useful for producing a rAAV as described herein is provided. In one embodiment, the system includes (a) an AAVrh79, AAV8.AR2.08, or AAV5.5.9 capsid nucleic acid sequence encoding the predicted amino acid sequence of SEQ ID NO: 2, 10, or 18; (b) a nucleic acid molecule suitable for packaging into the AAV capsid, said nucleic acid molecule comprising at least one AAV inverted terminal repeat (ITR) and a non-AAV nucleic acid sequence encoding a gene product operably linked to sequences which direct expression of the product in a host cell; and (c) sufficient AAV rep functions and helper functions to permit packaging of the nucleic acid molecule into the recombinant AAV capsid.

These and other aspects of the invention will be apparent from the following detailed description of the invention.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1A-FIG. 1G. Electrophoretic analysis of AAV8 VP isoforms. (FIG. 1A) Diagram illustrating the mechanism by which asparagine residues undergo nucleophilic attack by adjacent nitrogen atoms, forming a succinimidyl intermediate. This intermediate then undergoes hydrolysis, resolving into a mixture of aspartic acid and isoaspartic acid. The beta carbon is labeled as such. The diagram was generated in BIOVIA Draw 2018. (FIG. 1B) 1 μ g of AAV8 vector was run on a denaturing one-dimensional SDS-PAGE. (FIG. 1C) Isoelectric points of carbonic anhydrase pI marker spots are shown. (FIG. 1D) 5 μ g of AAV8 vector was analyzed by two-dimensional gel electrophoresis and stained with Coomassie Blue. Spots 1-20 are carbamylated carbonic anhydrase pI markers. Boxed regions are as follows: a=VP1, b=VP2, c=VP3, d=internal tropomyosin marker (arrow: tropomyosin spot of MW=33 kDa, pI=5.2). Isoelectric focusing was performed with a pI range of 4-8. (FIG. 1E-FIG. 1G) Results of isoelectric focusing performed with a pI range of 4-8. 1e11 GC of wtAAV8 (FIG. 1E) or mutant (FIG. 1F and FIG. 1G) vector, which were analyzed by 2D gel electrophoresis and stained with Sypro Ruby. Protein labeling: A=VP1; B=VP2; C=VP3, D=chicken egg white conalbumin marker, E=turbonuclease marker. Isoelectric focusing was performed with a pI range of 6-10. Primary VP1/2/3 isoform spots are circled, and migration distance of major spots of markers are indicated by vertical lines (turbonuclease=dashed, conalbumin=solid).

FIG. 2A-FIG. 2E. Analysis of asparagine and glutamine deamidation in AAV8 capsid proteins. (FIG. 2A-FIG. 2B) Electrospray ionization (ESI) mass spectrometry and theoretical and observed masses of the 3+ peptide (93-103) containing Asn-94 (FIG. 2A) and Asp-94 (FIG. 2B) are shown. (FIG. 2C-FIG. 2D) ESI mass spectrometry and theoretical and observed masses of the 3+ peptide (247-259) containing Asn-254 (FIG. 2C) and Asp-254 (FIG. 2D) are shown. The observed mass shifts for Asn-94 and Asn-254 were 0.982 Da and 0.986 Da, respectively, versus a theoretical mass shift of 0.984 Da. (FIG. 2E) Percent deamidation at specific asparagine and glutamine residues of interest are shown for AAV8 tryptic peptides purified by different methods. Bars indicating deamidation at asparagine residues with N+1 glycines are crosshatched. Residues determined to be at least 2% deamidated in at least one prep analyzed were included. Data are represented as mean $\pm$ standard deviation.

FIG. 3A-FIG. 3E. Structural modeling of the AAV8 VP3 monomer and analysis of deamidated sites. (FIG. 3A) The AAV8 VP3 monomer (PDB identifier: 3RA8) is shown in a coil representation. The ribbon indicates the relative degree of flexibility. Spheres indicate residues of interest. Expanded

diagrams are ball and stick representations of residues of interest and their surrounding residues to demonstrate local protein structure. Underlined residues are those in NG motifs. FIG. 3B-FIG. 3E: Isoaspartic models of deamidated asparagines with N+1 glycines are shown. The 2FoFc electron density map (1 sigma level) generated from refinement of the AAV8 crystal structure (PDB ID: 3RA8) with (FIG. 3B) an asparagine model of N410 in comparison with isoaspartic acid models of (FIG. 3C) N263, (FIG. 3D) N514, and (FIG. 3E) N540. Electron density map is shown. The beta carbon is labeled as such. Arrow indicates electron density corresponding to the R group of the residue of interest.

FIG. 4A-FIG. 4D. Determination of factors influencing AAV8 capsid deamidation. An AAV8 prep was (FIG. 4A) incubated at 70° C. for three or seven days, (FIG. 4B) exposed to pH 2 or pH 10 for seven days, or (FIG. 4C) prepared for mass spectrometry using D₂O in place of H₂O to determine possible sources of deamidation not intrinsic to AAV capsid formation. (FIG. 4D) A dot blot of vector treated as in FIG. 4A using the B1 antibody (reacts to denatured capsid) and an AAV8 conformation specific antibody (reacts to intact capsids) to assess capsid structural integrity.

FIG. 5A-FIG. 5B. Deamidation frequencies in non-AAV proteins. Deamidation percentages are shown for two non-AAV recombinant proteins containing NG motifs likely to be deamidated, human carbonic anhydrase (FIG. 5A) and rat phenylalanine-hydroxylase (FIG. 5B), for comparison with AAV deamidation percentages.

FIG. 6. Comparison of AAV8 percent deamidation calculated using data analysis pipelines from two institutions. Percent deamidation at specific asparagine and glutamine residues of interest are shown for AAV8 tryptic peptides evaluated at two different institutions.

FIG. 7A-FIG. 7C illustrate functional asparagine substitutions at non-NG sites with high variability between lots. (FIG. 7A) Titers of wtAAV8 and mutant vectors were produced by small-scale triple transfection in 293 cells, as measured by quantitative PCR (qPCR). Titers are reported relative to the wtAAV8 control. Transduction efficiencies were measured as described in FIG. 3B. Titers and transduction efficiencies are normalized to the value for the wtAAV8 control. (FIG. 7B) Representative luciferase images at day 14 post-injection are shown for mice receiving wtAAV8.CB7.fluc and N499Q capsid mutant vector. (FIG. 7C) Luciferase expression on day 14 of the study periods from C57BL/6 mice injected intravenously with wtAAV8 or mutant vectors (n=3 or 4) was measured by luciferase imaging and reported in total flux units. All data are represented as mean+standard deviation.

FIG. 8A and FIG. 8B show the results of in vitro analysis of the impact of genetic deamidation on vector performance. (FIG. 8A) Titers of wtAAV8 and genetic deamidation mutant vectors produced by small-scale triple transfection in 293 cells, as measured by quantitative PCR (qPCR). Titers are reported relative to the wtAAV8 control. NG sites with high deamidation (patterned bars), sites with low deamidation (white bars) and highly variable sites (black bars) are presented with wtAAV8 and a negative control. (FIG. 8B) Transduction efficiency of mutant AAV8 vectors producing firefly luciferase reported relative to the wtAAV8 control. Transduction efficiency is measured in luminescence units generated per GC added to HUH17 cells, and is determined by performing transductions with crude vector at multiple

dilutions. Transduction efficiency data are normalized to the wt reference. All data are represented as mean±standard deviation.

FIG. 9A-FIG. 9D) illustrate that vector activity loss through time is correlated to progressive deamidation. (FIG. 9A) Vector production (DNaseI resistant Genome Copies, GC) for a timecourse of triple-transfected HEK 293 cells producing AAV8 vector packaging a luciferase reporter gene. GC levels are normalized to the maximum observed value. (FIG. 9B) Purified timecourse vector was used to transduce Huh7 cells. Transduction efficiency (luminescence units per GC added to target cells) was measured as in FIG. 8B using multiple dilutions of purified timecourse vector samples. Error bars represent the standard deviation of at least 10 technical replicates for each sample time. Deamidation of AAV8 NG sites (FIG. 9C) and non-NG sites (FIG. 9D) for vector collected 1, 2 and 5 days post transfection.

FIG. 10A-FIG. 10D illustrates the impact of stabilizing asparagines on vector performance. FIG. 10A shows titers of wtAAV8 and +1 position mutant vectors produced by small-scale triple transfection in 293 cells, as measured by quantitative PCR (qPCR). Titers are reported relative to the wtAAV8 control. FIG. 10B shows the transduction efficiency of mutant AAV8 vectors producing firefly luciferase reported relative to the wtAAV8 control. Transduction efficiency was measured as in FIG. 8B using crude vector material. A two-sample t-test (*p<0.005) was run to determine significance between wtAAV8 and mutant transduction efficiency for G264A/G515A and G264A/G541A. FIG. 10C shows luciferase expression on day 14 of the study period in the liver region from C57BL/6 mice injected intravenously with wtAAV8 or mutant vectors (n=3 to 5) measured by luciferase imaging and reported in total flux units. FIG. 10D shows the titers and transduction efficiency of multi-site AAV8 mutant vectors producing firefly luciferase reported relative to the wtAAV8 control. All data are represented as mean±standard deviation.

FIG. 11A-FIG. 11C. Analysis of asparagine and glutamine deamidation in AAV9 capsid proteins. (FIG. 11A) 1e11 GCs of wtAAV9 were analyzed by 2D gel electrophoresis and stained with Sypro Ruby. Protein labeling: A=VP1; B=VP2; C=VP3, D=chicken egg white conalbumin marker, E=turbonuclease marker. Isoelectric focusing was performed with a pI range of 6-10. (FIG. 11B) Percent deamidation at specific asparagine and glutamine residues of interest are shown for AAV9 tryptic peptides purified by different methods. Bars indicating deamidation at asparagine residues with N+1 glycines are crosshatched. Residues determined to be at least 2% deamidated in at least one prep analyzed were included. Data are represented as mean±standard deviation. (FIG. 11C) Isoaspartic model of N512 is shown in the 2FoFc electron density map generated by non-biased refinement of the AAV9 crystal structure (PDB ID: 3UX1). Arrow indicates electron density corresponding to the R group of residue N512.

FIG. 11D-FIG. 11F. Determination of factors influencing AAV9 capsid deamidation. (FIG. 11D) Two AAV9 preps were incubated at 70° C. for three or seven days or (FIG. 11E) exposed to pH 2 or pH 10 for seven days to determine possible sources of deamidation not intrinsic to AAV capsid formation. Data are represented as mean±standard deviation. (FIG. 11F) A dot blot of vector treated as in FIG. 11D using the B1 antibody (reacts to denatured capsid) to assess capsid structural integrity.

FIG. 11G and FIG. 11H illustrate in vitro analysis of the impact of genetic deamidation on vector performance for

AAV9. (FIG. 11G) Titers of wtAAV9 and genetic deamidation mutant vectors were produced by small-scale triple transfection in 293 cells, as measured by quantitative PCR (qPCR). Titers are reported relative to the wtAAV9 control. NG sites with high deamidation (patterned bars), sites with low deamidation (white bars) and highly variable sites (black bars) are presented with wtAAV8 and a negative control. (FIG. 11H) The transduction efficiency of mutant AAV9 vectors producing firefly luciferase are reported relative to the wtAAV9 control. All data are represented as mean±standard deviation.

FIG. 11I-FIG. 11K show AAV9 vector in vitro potency through time. (FIG. 11I) Vector production (DNaseI resistant Genome Copies, GC) for a timecourse of triple-transfected HEK 293 cells producing AAV9 vector packaging a luciferase reporter gene. GC levels are normalized to the maximum observed value. (FIG. 11J) Crude timecourse vector was used to transduce Huh7 cells. (FIG. 11K) Transduction efficiencies of vector collected 1 day post transfection vs 5 days post transfection are shown for crude and purified vector samples. Transduction efficiency is expressed as luciferase activity/GC, normalized to the value at day 1.

FIG. 12A provides an alignment of the amino acid sequences of AAV5.5.9 [SEQ ID NO: 10] (also sometimes called AAVG5), AAV9 [SEQ ID NO: 4], and AAVPHP.B [SEQ ID NO: 12], prepared using Clustal Omega 1.2.2 and its default parameters for alignment. FIGS. 12B-12E provide an alignment of the nucleotide sequences of AAV5.5.9 [SEQ ID NO: 9], PHP.B [SEQ ID NO: 11], AAV9 [SEQ ID NO: 3], and AAVhu68 [SEQ ID NO: 14].

FIG. 13A provides an alignment of the amino acid sequences of AAV8Triple mutant (AAV8T) [SEQ ID NO: 16], AAV8.AR2.08 [SEQ ID NO: 18] (also sometimes called AAVG3 or AR2 or AAV.AR2), and AAV8 [SEQ ID NO: 20], prepared using Clustal Omega 1.2.2 and its default parameters for alignment. FIGS. 13B-13D provide an alignment of the nucleotide sequences of AAV8 Triple Mutant [SEQ ID NO: 15], AAV8.AR2.08 [SEQ ID NO: 17], AAV8 [SEQ ID NO: 19].

FIG. 14A provides an alignment of the amino acid sequences of AAVrh79 [SEQ ID NO: 2] (also sometimes called AAVG2), AAVrh10 [SEQ ID NO: 24] and AAVhu37 [SEQ ID NO: 22], using Clustal Omega 1.2.2 and its default parameters for alignment. FIGS. 14B-14D provide an alignment of the nucleotide sequences of AAVrh79 [SEQ ID NO: 1], AAVrh10 [SEQ ID NO: 22] and AAVhu37 [SEQ ID NO: 21].

FIGS. 15A and 15B illustrate the production yield for AAV8triple, AAVhu68, AAV9, AAV9 and AAVrh79 in small scale or mega scale preps of the referenced vector.

FIG. 16 provides the production purity of the mega scale preps of FIG. 15B.

FIGS. 17A to 17D show expression of luciferase in liver and muscle tissue following intramuscular (IM) administration of 3×10¹¹ GC/mouse into the gastrocnemius muscle of male C57BL/6 mice (n=5/group) using vectors expressing firefly luciferase. Expression of AAV8triple, AAVhu68, AAV9, AAV8, and AAVrh79 vectors was compared following intramuscular administration of 10¹³ GC/kg AAVrh79 into male and female cynomolgus macaques (FIG. 17E). Vectors expressing a secreted transgene (201Ig IA) were administered intramuscularly into the gastrocnemius muscle of male RAG KO mice (n=5/group) (3×10¹⁰ or 3×10¹¹ GC/mouse) (FIG. 17F). The results indicated that AAV8triple expresses better following IM injection and at the lower dose tested the difference in expression from

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AAV8triple was substantial. At higher the higher dose, AAVrh79 expressed at levels at comparable to the other vectors tested.

FIG. 18A shows the plasmid used for the barcode experiments in Example 5. FIG. 18B shows the amounts of each AAV barcode variant injected into black 6 mice. The animals were sacrificed, tissue samples harvested, and DNA isolated from each of them. Total vector distribution for the three animals is shown in FIG. 18C. Actual vs. theoretical frequency of injected vector mix is shown in FIG. 18D.

FIGS. 19A-FIG. 20C show the results of the barcode biodistribution experiments of Example 5. Individual tissue samples were analyzed for individual barcode frequency in the sample vs. injection mix for genomic and cDNA. Results are shown for muscle (FIG. 19A, 19B); heart (FIG. 19C and FIG. 19D) and liver (FIG. 19E and FIG. 19F). Fold changes as compared to theoretical frequencies are shown in FIG. 20A-FIG. 20C.

FIGS. 21 and 22 show AAV8.AR2.08 biodistribution in mice as compared to AAV8. The results show that AAV8.AR2.08 is more liver specific than AAV8.

FIG. 23 compares AAV8 vs. AAV8.AR2.08 vs. AAVrh79 for titer and yield relating to manufacturability.

FIG. 24 shows AAV8.AR2.08 biodistribution in tissues (left most bar) compared with AAV8 (middle and right bars).

FIGS. 25-28 show results following administration of AAV vectors to non-human primates. FIG. 25 provides details for vectors and animals used for studies. FIG. 26 quantifies levels of GC and GFP detected in liver from animals that received AAV8, AAVrh79, or AAV8.AR2.08 vectors. FIG. 27 summarizes levels of GFP expression in HNP livers. FIG. 28 shows level of vector detected in tissue from HNP that were administered AAV8, AAVrh79, or AAV8.AR2.08 vectors.

FIG. 29 shows biodistribution of AAVrh79 vector detected in various tissues.

DETAILED DESCRIPTION OF THE INVENTION

Provided herein are recombinant adeno-associated virus (rAAV) having sequence and charge heterogeneity in each of the three populations of capsid proteins VP1, VP2, and VP3 found within the capsid of a recombinant AAV and compositions containing same. Provided herein are novel rAAV, as well as methods for reducing the deamidation, and optionally other capsid monomer modifications. Further provided herein are modified rAAV having decreased modifications, which are useful for providing rAAV having capsids which retain greater stability, potency, and/or purity.

A “recombinant AAV” or “rAAV” is a DNase-resistant viral particle containing two elements, an AAV capsid and a vector genome containing at least non-AAV coding sequences packaged within the AAV capsid. Unless otherwise specified, this term may be used interchangeably with the phrase “rAAV vector”. The rAAV is a “replication-defective virus” or “viral vector”, as it lacks any functional AAV rep gene or functional AAV cap gene and cannot generate progeny. In certain embodiments, the only AAV sequences are the AAV inverted terminal repeat sequences (ITRs), typically located at the extreme 5' and 3' ends of the vector genome in order to allow the gene and regulatory sequences located between the ITRs to be packaged within the AAV capsid.

As used herein, a “vector genome” refers to the nucleic acid sequence packaged inside the rAAV capsid which forms a viral particle. Such a nucleic acid sequence contains AAV

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inverted terminal repeat sequences (ITRs). In the examples herein, a vector genome contains, at a minimum, from 5' to 3', an AAV 5' ITR, coding sequence(s), and an AAV 3' ITR. ITRs from AAV2, a different source AAV than the capsid, or other than full-length ITRs may be selected. In certain embodiments, the ITRs are from the same AAV source as the AAV which provides the rep function during production or a transcomplementing AAV. Further, other ITRs may be used. Further, the vector genome contains regulatory sequences which direct expression of the gene products. Suitable components of a vector genome are discussed in more detail herein.

A rAAV is composed of an AAV capsid and a vector genome. An AAV capsid is an assembly of a heterogeneous population of vp1, a heterogeneous population of vp2, and a heterogeneous population of vp3 proteins. As used herein when used to refer to vp capsid proteins, the term “heterogeneous” or any grammatical variation thereof, refers to a population consisting of elements that are not the same, for example, having vp1, vp2 or vp3 monomers (proteins) with different modified amino acid sequences.

As used herein, the term “heterogeneous population” as used in connection with vp1, vp2 and vp3 proteins (alternatively termed isoforms), refers to differences in the amino acid sequence of the vp1, vp2 and vp3 proteins within a capsid. The AAV capsid contains subpopulations within the vp1 proteins, within the vp2 proteins and within the vp3 proteins which have modifications from the predicted amino acid residues. These subpopulations include, at a minimum, certain deamidated asparagine (N or Asn) residues. For example, certain subpopulations comprise at least one, two, three or four highly deamidated asparagines (N) positions in asparagine-glycine pairs and optionally further comprising other deamidated amino acids, wherein the deamidation results in an amino acid change and other optional modifications.

As used herein, a “subpopulation” of vp proteins refers to a group of vp proteins which has at least one defined characteristic in common and which consists of at least one group member to less than all members of the reference group, unless otherwise specified. For example, a “subpopulation” of vp1 proteins may be at least one (1) vp1 protein and less than all vp1 proteins in an assembled AAV capsid, unless otherwise specified. A “subpopulation” of vp3 proteins may be one (1) vp3 protein to less than all vp3 proteins in an assembled AAV capsid, unless otherwise specified. For example, vp1 proteins may be a subpopulation of vp proteins; vp2 proteins may be a separate subpopulation of vp proteins, and vp3 are yet a further subpopulation of vp proteins in an assembled AAV capsid. In another example, vp1, vp2 and vp3 proteins may contain subpopulations having different modifications, e.g., at least one, two, three or four highly deamidated asparagines, e.g., at asparagine-glycine pairs.

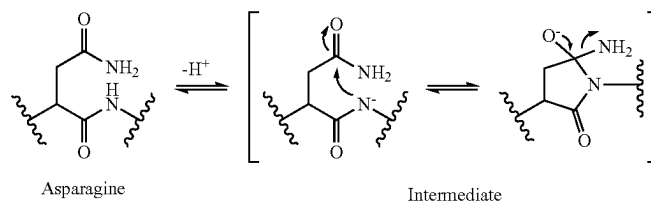
Unless otherwise specified, highly deamidated refers to at least 45% deamidated, at least 50% deamidated, at least 60% deamidated, at least 65% deamidated, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, at least 99%, or up to about 100% deamidated at a referenced amino acid position, as compared to the predicted amino acid sequence at the reference amino acid position (e.g., at least 80% of the asparagines at amino acid 57 of SEQ ID NO:2 may be deamidated based on the total vp1 proteins may be deamidated based on the total vp1, vp2 and vp3 proteins). Such percentages may be determined using 2D-gel, mass spectrometry techniques, or other suitable techniques.

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Without wishing to be bound by theory, the deamidation of at least highly deamidated residues in the vp proteins in the AAV capsid is believed to be primarily non-enzymatic in nature, being caused by functional groups within the capsid protein which deamidate selected asparagines, and to a lesser extent, glutamine residues. Efficient capsid assembly of the majority of deamidation vp1 proteins indicates that either these events occur following capsid assembly or that

deamidation in individual monomers (vp1, vp2 or vp3) is well-tolerated structurally and largely does not affect assembly dynamics. Extensive deamidation in the VP1-unique (VP1-u) region (~aa 1-137), generally considered to be located internally prior to cellular entry, suggests that VP deamidation may occur prior to capsid assembly.

Without wishing to be bound by theory, the deamidation of N may occur through its C-terminus residue's backbone nitrogen atom conducts a nucleophilic attack to the Asn's



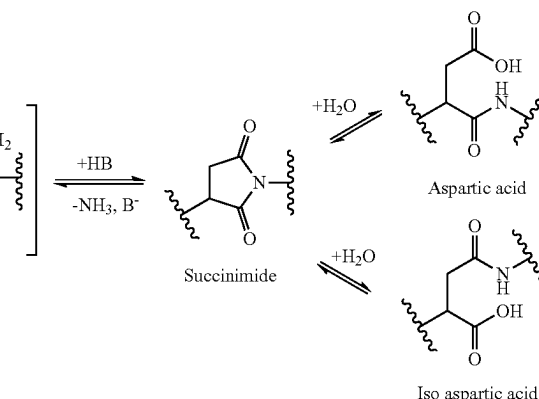
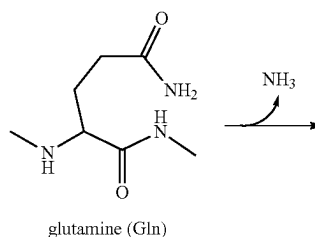
side chain amide group carbon atom. An intermediate ring-closed succinimide residue is believed to form. The succinimide residue then conducts fast hydrolysis to lead to the final product aspartic acid (Asp) or iso aspartic acid (IsoAsp). Therefore, in certain embodiments, the deamidation of asparagine (N or Asn) leads to an Asp or IsoAsp, which may interconvert through the succinimide intermediate e.g., as illustrated below.

As provided herein, each deamidated N in the VP1, VP2 or VP3 may independently be aspartic acid (Asp), isoaspartic acid (isoAsp), aspartate, and/or an interconverting blend of Asp and isoAsp, or combinations thereof. Any suitable ratio of α - and isoaspartic acid may be present. For example, in certain embodiments, the ratio may be from 10:1 to 1:10 aspartic to isoaspartic, about 50:50 aspartic: isoaspartic, or about 1:3 aspartic: isoaspartic, or another selected ratio.

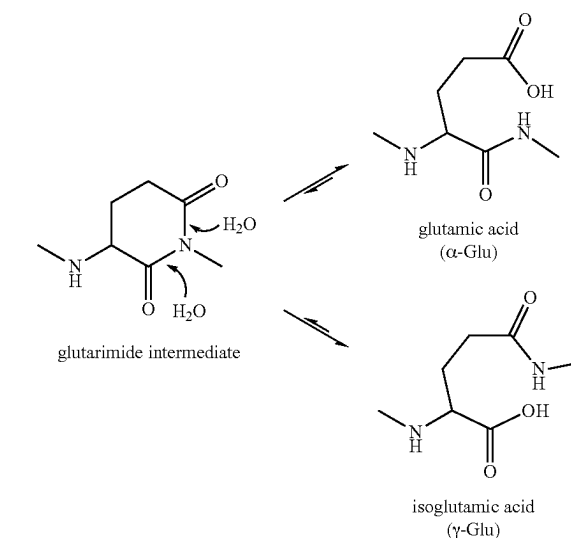
In certain embodiments, one or more glutamine (Q) may deamidates to glutamic acid (Glu), i.e., α -glutamic acid, γ -glutamic acid (Glu), or a blend of α - and γ -glutamic acid, which may interconvert through a common glutarimide intermediate. Any suitable ratio of α - and γ -glutamic acid may be present. For example, in certain embodiments, the

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ratio may be from 10:1 to 1:10 α to γ , about 50:50 α : γ , or about 1:3 α : γ , or another selected ratio.



-continued



Thus, an rAAV includes subpopulations within the rAAV capsid of vp1, vp2 and/or vp3 proteins with deamidated amino acids, including at a minimum, at least one subpopulation comprising at least one highly deamidated asparagine.

In addition, other modifications may include isomerization, particularly at selected aspartic acid (D or Asp) residue positions. In still other embodiments, modifications may include an amidation at an Asp position.

In certain embodiments, an AAV capsid contains subpopulations of vp1, vp2 and vp3 having at least 1, at least 2, at least 3, at least 4, at least 5 to at least about 25 deamidated amino acid residue positions, of which at least 1 to 10%, at least 10 to 25%, at least 25 to 50%, at least 50 to 70%, at least 70 to 100%, at least 75 to 100%, at least 80-100% or at least 90-100% are deamidated as compared to the encoded amino acid sequence of the vp proteins. The majority of these may be N residues. However, Q residues may also be deamidated.

As used herein, "encoded amino acid sequence" refers to the amino acid which is predicted based on the translation of a known DNA codon of a referenced nucleic acid sequence being translated to an amino acid. The following table illustrates DNA codons and twenty common amino acids, showing both the single letter code (SLC) and three letter code (3LC).

Amino Acid	SLC	3LC	DNA codons
Isoleucine	I	Ile	ATT, ATC, ATA
Leucine	L	Leu	CTT, CTC, CTA, CTG, TTA, TTG
Valine	V	Val	GTT, GTC, GTA, GTG
Phenylalanine	F	Phe	TTT, TTC
Methionine	M	Met	ATG
Cysteine	C	Cys	TGT, TGC
Alanine	A	Ala	GCT, GCC, GCA, GCG
Glycine	G	Gly	GGT, GGC, GGA, GGG
Proline	P	Pro	CCT, CCC, CCA, CCG
Threonine	T	Thr	ACT, ACC, ACA, ACG
Serine	S	Ser	TCT, TCC, TCA, TCG, AGT, AGC
Tyrosine	Y	Tyr	TAT, TAC
Tryptophan	W	Trp	TGG
Glutamine	Q	Gln	CAA, CAG
Asparagine	N	Asn	AAT, AAC
Histidine	H	His	CAT, CAC
Glutamic acid	E	Glu	GAA, GAG
Aspartic acid	D	Asp	GAT, GAC
Lysine	K	Lys	AAA, AAG
Arginine	R	Arg	CGT, CGC, CGA, CGG, AGA, AGG
Stop codons	Stop		TAA, TAG, TGA

In certain embodiments, a rAAV has an AAV capsid having vp1, vp2 and vp3 proteins having subpopulations comprising combinations of two, three, four, five or more deamidated residues at the positions set forth in the tables provided herein and incorporated herein by reference.

Deamidation in the rAAV may be determined using 2D gel electrophoresis, and/or mass spectrometry, and/or protein modelling techniques. Online chromatography may be performed with an Acclaim PepMap column and a Thermo UltiMate 3000 RSLC system (Thermo Fisher Scientific) coupled to a Q Exactive HF with a NanoFlex source (Thermo Fisher Scientific). MS data is acquired using a data-dependent top-20 method for the Q Exactive HF, dynamically choosing the most abundant not-yet-sequenced precursor ions from the survey scans (200-2000 m/z). Sequencing is performed via higher energy collisional dissociation fragmentation with a target value of 1e5 ions determined with predictive automatic gain control and an isolation of precursors was performed with a window of 4 m/z. Survey scans were acquired at a resolution of 120,000 at m/z 200. Resolution for HCD spectra may be set to 30,000

at m/z200 with a maximum ion injection time of 50 ms and a normalized collision energy of 30. The S-lens RF level may be set at 50, to give optimal transmission of the m/z region occupied by the peptides from the digest. Precursor ions may be excluded with single, unassigned, or six and higher charge states from fragmentation selection. BioPharma Finder 1.0 software (Thermo Fischer Scientific) may be used for analysis of the data acquired. For peptide mapping, searches are performed using a single-entry protein FASTA database with carbamidomethylation set as a fixed modification; and oxidation, deamidation, and phosphorylation set as variable modifications, a 10-ppm mass accuracy, a high protease specificity, and a confidence level of 0.8 for MS/MS spectra. Examples of suitable proteases may include, e.g., trypsin or chymotrypsin. Mass spectrometric identification of deamidated peptides is relatively straightforward, as deamidation adds to the mass of intact molecule +0.984 Da (the mass difference between OH and —NH₂ groups). The percent deamidation of a particular peptide is determined by mass area of the deamidated peptide divided by the sum of the area of the deamidated and native peptides. Considering the number of possible deamidation sites, isobaric species which are deamidated at different sites may co-migrate in a single peak. Consequently, fragment ions originating from peptides with multiple potential deamidation sites can be used to locate or differentiate multiple sites of deamidation. In these cases, the relative intensities within the observed isotope patterns can be used to specifically determine the relative abundance of the different deamidated peptide isomers. This method assumes that the fragmentation efficiency for all isomeric species is the same and independent on the site of deamidation. It will be understood by one of skill in the art that a number of variations on these illustrative methods can be used. For example, suitable mass spectrometers may include, e.g. a quadrupole time of flight mass spectrometer (QTOF), such as a Waters Xevo or Agilent 6530 or an orbitrap instrument, such as the Orbitrap Fusion or Orbitrap Velos (Thermo Fisher). Suitably liquid chromatography systems include, e.g., Acquity UPLC system from Waters or Agilent systems (1100 or 1200 series). Suitable data analysis software may include, e.g., MassLynx (Waters), Pinpoint and Peppfinder (Thermo Fischer Scientific), Mascot (Matrix Science), Peaks DB (Bioinformatics Solutions). Still other techniques may be described, e.g., in X. Jin et al, Hu Gene Therapy Methods, Vol. 28, No. 5, pp. 255-267, published online Jun. 16, 2017.

In addition to deamidations, other modifications may occur do not result in conversion of one amino acid to a different amino acid residue. Such modifications may include acetylated residues, isomerizations, phosphorylations, or oxidations.

Modulation of Deamidation: In certain embodiments, the AAV is modified to change the glycine in an asparagine-glycine pair, to reduce deamidation. In other embodiments, the asparagine is altered to a different amino acid, e.g., a glutamine which deamidates at a slower rate; or to an amino acid which lacks amide groups (e.g., glutamine and asparagine contain amide groups); and/or to an amino acid which lacks amine groups (e.g., lysine, arginine and histidine contain a: nine groups). As used herein, amino acids lacking amide or amine side groups refer to, e.g., glycine, alanine, valine, leucine, isoleucine, serine, threonine, cystine, phenylalanine, tyrosine, or tryptophan, and/or proline. Modifications such as described may be in one, two, or three of the asparagine-glycine pairs found in the encoded AAV amino acid sequence. In certain embodiments, such modifications

are not made in all four of the asparagine-glycine pairs. Thus, a method for reducing deamidation of AAV and/or engineered AAV variants having lower deamidation rates. Additionally, or alternatively one or more other amide amino acids may be changed to a non-amide amino acid to reduce deamidation of the AAV. In certain embodiments, a mutant AAV capsid as described herein contains a mutation in an asparagine-glycine pair, such that the glycine is changed to an alanine or a serine. A mutant AAV capsid may contain one, two or three mutants where the reference AAV natively contains four NG pairs. In certain embodiments, an AAV capsid may contain one, two, three or four such mutants where the reference AAV natively contains five NG pairs. In certain embodiments, a mutant AAV capsid contains only a single mutation in an NG pair. In certain embodiments, a mutant AAV capsid contains mutations in two different NG pairs. In certain embodiments, a mutant AAV capsid contains mutation in two different NG pairs which are located in structurally separate location in the AAV capsid. In certain embodiments, the mutation is not in the VP1-unique region. In certain embodiments, one of the mutations is in the VP1-unique region. Optionally, a mutant AAV capsid contains no modifications in the NG pairs, but contains mutations to minimize or eliminate deamidation in one or more asparagines, or a glutamine, located outside of an NG pair.

In certain embodiments, a method of increasing the potency of a rAAV vector is provided which comprises engineering an AAV capsid which eliminating one or more of the NGs in the wild-type AAV capsid. In certain embodiments, the coding sequence for the "G" of the "NG" is engineered to encode another amino acid. In certain examples below, an "S" or an "A" is substituted. However, other suitable amino acid coding sequences may be selected. See, e.g., the tables below in which based on the numbering of AAV8, the coding sequence for at least one of the following positions: N57+1, N263+1, N385+1, N514+1, N540+1, is modified, or as shown in the tables below. In certain embodiments, AAV8 mutants avoid changing the NG pairs at positions N57, N94, N263, N305, Q467, N479, and/or N653. In certain embodiments, other AAVs avoid mutation at corresponding N positions as determined based on an alignment with AAV8, using AAV8 numbering as a reference.

These amino acid modifications may be made by conventional genetic engineering techniques. For example, a nucleic acid sequence containing modified AAV vp codons may be generated in which one to three of the codons encoding glycine in arginine-glycine pairs are modified to encode an amino acid other than glycine. In certain embodiments, a nucleic acid sequence containing modified arginine codons may be engineered at one to three of the arginine-glycine pairs, such that the modified codon encodes an amino acid other than arginine. Each modified codon may encode a different amino acid. Alternatively, one or more of the altered codons may encode the same amino acid. In certain embodiments, these modified AAVrh79, AAV8.AR2.08 or AAV5.5.9 nucleic acid sequences may be used to generate a mutant rAAV having a capsid with lower deamidation than the native AAVrh79, AAV8.AR2.08 or AAV5.5.9 capsid. Such mutant rAAV may have reduced immunogenicity and/or increase stability on storage, particularly storage in suspension form.

Also provided herein are nucleic acid sequences encoding the AAV capsids having reduced deamidation. It is within the skill in the art to design nucleic acid sequences encoding this AAV capsid, including DNA (genomic or cDNA), or RNA (e.g., mRNA). Such nucleic acid sequences may be

codon-optimized for expression in a selected system (i.e., cell type) can be designed by various methods. This optimization may be performed using methods which are available on-line (e.g., GencArt), published methods, or a company which provides codon optimizing services, e.g., DNA2.0 (Menlo Park, CA). One codon optimizing method is described, e.g., in International Patent Publication No. WO 2015/012924, which is incorporated by reference herein in its entirety. See also, e.g., US Patent Publication No. 2014/0032186 and US Patent Publication No. 2006/0136184. Suitably, the entire length of the open reading frame (ORF) for the product is modified. However, in some embodiments, only a fragment of the ORF may be altered. By using one of these methods, one can apply the frequencies to any given polypeptide sequence and produce a nucleic acid fragment of a codon-optimized coding region which encodes the polypeptide. A number of options are available for performing the actual changes to the codons or for synthesizing the codon-optimized coding regions designed as described herein. Such modifications or synthesis can be performed using standard and routine molecular biological manipulations well known to those of ordinary skill in the art. In one approach, a series of complementary oligonucleotide pairs of 80-90 nucleotides each in length and spanning the length of the desired sequence are synthesized by standard methods. These oligonucleotide pairs are synthesized such that upon annealing, they form double stranded fragments of 80-90 base pairs, containing cohesive ends, e.g., each oligonucleotide in the pair is synthesized to extend 3, 4, 5, 6, 7, 8, 9, 10, or more bases beyond the region that is complementary to the other oligonucleotide in the pair. The single-stranded ends of each pair of oligonucleotides are designed to anneal with the single-stranded end of another pair of oligonucleotides. The oligonucleotide pairs are allowed to anneal, and approximately five to six of these double-stranded fragments are then allowed to anneal together via the cohesive single stranded ends, and then they ligated together and cloned into a standard bacterial cloning vector, for example, a TOPO® vector available from Invitrogen Corporation, Carlsbad, Calif. The construct is then sequenced by standard methods. Several of these constructs consisting of 5 to 6 fragments of 80 to 90 base pair fragments ligated together, i.e., fragments of about 500 base pairs, are prepared, such that the entire desired sequence is represented in a series of plasmid constructs. The inserts of these plasmids are then cut with appropriate restriction enzymes and ligated together to form the final construct. The final construct is then cloned into a standard bacterial cloning vector, and sequenced. Additional methods would be immediately apparent to the skilled artisan. In addition, gene synthesis is readily available commercially.

In certain embodiments, AAV capsids are provided which have a heterogeneous population of AAV capsid isoforms (i.e., VP1, VP2, VP3) which contain multiple highly deamidated "NG" positions. In certain embodiments, the highly deamidated positions are in the locations identified below, with reference to the predicted full-length VP1 amino acid sequence. In other embodiments, the capsid gene is modified such that the referenced "NG" is ablated and a mutant "NG" is engineered into another position.

In certain embodiments, the mixed population of rAAV results from a production system using a single AAV capsid nucleic acid sequence encoding a predicted AAV VP1 amino acid sequence of one AAV type. However, the production and manufacture process provides the heterogenous population of capsid proteins described above.

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In certain embodiments, a novel isolated AAVrh79 capsid is provided. The nucleic acid sequence encoding the AAV is provided in SEQ ID NO:1 and the encoded amino acid sequence is provided in SEQ ID NO: 2.

In certain embodiments, a rAAV comprises a AAVrh79 capsid. An AAVrh79 capsid comprises a heterogeneous population of AAVrh79 vp1 proteins, AAVrh79 vp2 proteins, and AAVrh79 vp3 proteins. In one embodiment, the AAVrh79 capsid is produced by expression from a nucleic acid sequence which encodes the predicted amino acid sequence of 1 to 738 of SEQ ID NO:2. Optionally, sequences co-expressing the vp3 protein from a nucleic acid sequence excluding the vp1-unique region (about aa 1 to 137) or the vp2-unique region (about aa 1 to 203), vp1 proteins produced from SEQ ID NO:1, or vp1 proteins produced from a nucleic acid sequence at least 70% identical to SEQ ID NO:1 which encodes the predicted amino acid sequence of 1 to 738 of SEQ ID NO:2. In other embodiments, the AAVrh79 vp2 proteins produced by expression from a nucleic acid sequence which encodes the predicted amino acid sequence of at least about amino acids 138 to 738 of SEQ ID NO:2, vp2 proteins produced from a sequence comprising at least nucleotides 412 to 2214 of SEQ ID NO:1, or vp2 proteins produced from a nucleic acid sequence at least 70% identical to at least nucleotides 412 to 2214 of SEQ ID NO: 1 which encodes the predicted amino acid sequence of at least about amino acids 138 to 738 of SEQ ID NO: 2, AAVrh79 vp3 proteins produced by expression from a nucleic acid sequence which encodes the predicted amino acid sequence of at least about amino acids 204 to 738 of SEQ ID NO:2, vp3 proteins produced from a sequence comprising at least nucleotides 610 to 2214 of SEQ ID NO:1, or vp3 proteins produced from a nucleic acid sequence at least 70% identical to at least nucleotides 610 to 2214 of SEQ ID NO:1 which encodes the predicted amino acid sequence of at least about amino acids 204 to 738 of SEQ ID NO:2.

In certain embodiments, an AAVrh79 capsid comprises: a heterogeneous population of vp1 proteins which are the product of a nucleic acid sequence encoding the amino acid sequence of SEQ ID NO: 2, a heterogeneous population of vp2 proteins which are the product of a nucleic acid sequence encoding the amino acid sequence of at least about amino acids 138 to 738 of SEQ ID NO: 2, and a heterogeneous population of vp3 proteins which are the product of a nucleic acid sequence encoding at least amino acids 204 to 738 of SEQ ID NO: 2.

The AAVrh79 vp1, vp2 and vp3 proteins contain subpopulations with amino acid modifications comprising at least two highly deamidated asparagines (N) in asparagine-glycine pairs in SEQ ID NO: 2 and optionally further comprising subpopulations comprising other deamidated amino acids, wherein the deamidation results in an amino acid change. High levels of deamidation at N-G pairs N57, N263, N385 and/or N514 are observed, relative to the number of SEQ ID NO:2. Deamidation has been observed in other residues, as shown in the table below and in the examples. In certain embodiments, AAVrh79 may have other residues deamidated, e.g., typically at less than 10% and/or may have other modifications, including methylations (e.g., ~R487) (typically less than 5%, more typically less than 1% at a given residue), isomerization (e.g., at D97) (typically less than 5%, more typically less than 1% at a given residue, phosphorylation (e.g., where present, in the range of about 10 to about 60%, or about 10 to about 30%, or about 20 to about 60%) (e.g., at one or more of S149, ~S153, ~S474, ~T570, ~S665), or oxidation (e.g., at one or

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more of W248, W307, W307, M405, M437, M473, W480, W480, W505, M526, M544, M561, W621, M637, and/or W697). Optionally the W may oxidize to kynurenine.

TABLE A

AAVrh79 Deamidation	
AAVrh79 Deamidation based on VP1 numbering	% Deamidation
N57 + Deamidation	65-90, 70-95, 80-95, 75-100, 80-100, or 90-100
N94 + Deamidation	5-15, about 10
~N254 + Deamidation	10-20
~N263 + Deamidation	75-100
~N305 + Deamidation	1-5
~N385 + Deamidation	65-90, 70-95, 80-95, 75-100, 80-100, or 90-100
~N410 + Deamidation	1-25,
N479 + Deamidation	1-5, 1-3
~N514 + Deamidation	65-90, 70-95, 80-95, 75-100, 80-100, or 90-100
~Q601 + Deamidation	0-1
N653 + Deamidation	0-2

In certain embodiments, an AAVrh79 capsid is modified in one or more of the positions identified in the preceding table, in the ranges provided below, as determined using mass spectrometry with a trypsin enzyme. In certain embodiments, one or more of the following positions, or the glycine following the N is modified as described herein. Residue numbers are based on the AAVrh79 sequence provided herein. See, SEQ ID NO: 2.

In certain embodiments, the nucleic acid sequence encoding the AAVrh79 vp1 capsid protein is provided in SEQ ID NO: 1. In other embodiments, a nucleic acid sequence of 70% to 99.9% identity to SHQ ID NO: 1 may be selected to express the AAVrh79 capsid proteins. In certain other embodiments, the nucleic acid sequence is at least about 75% identical, at least 80% identical, at least 85%, at least 90%, at least 95%, at least 97% identical, or at least 99% to 99.9% identical to SEQ ID NO: 1. However, other nucleic acid sequences which encode the amino acid sequence of SEQ ID NO: 2 may be selected for use in producing rAAV capsids. In certain embodiments, the nucleic acid sequence has the nucleic acid sequence of SEQ ID NO: 1 or a sequence at least 70% to 99% identical, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, at least 99%, identical to SEQ ID NO: 1 which encodes SEQ ID NO: 2. In certain embodiments, the nucleic acid sequence has the nucleic acid sequence of SEQ ID NO: 1 or a sequence at least 70% to 99. %, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, at least 99%, identical to about nt 412 to about nt 2214 of SEQ ID NO: 1 which encodes the vp2 capsid protein (about aa 138 to 738) of SEQ ID NO: 2. In certain embodiments, the nucleic acid sequence has the nucleic acid sequence of about nt 610 to about nt 2214 of SEQ ID NO:1 or a sequence at least 70% to 99. %, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, at least 99%, identical to nt SEQ ID NO: 1 which encodes the vp3 capsid protein (about aa 204 to 738) of SEQ ID NO: 2.

The invention also encompasses nucleic acid sequences encoding mutant AAVrh79, in which one or more residues has been altered in order to decrease deamidation, or other modifications which are identified herein. Such nucleic acid sequences can be used in production of mutant rAAVrh79 capsids.

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In certain embodiments, a novel AAV8.AR2.08 capsid is provided. The nucleic acid sequence encoding the AAV is provided in SEQ ID NO: 17 and the encoded amino acid sequence is provided in SEQ ID NO:18. In one embodiment, a recombinant adeno-associated virus (rAAV) has an AAV8.AR2.08 capsid. An alignment of the amino acid sequences of AAV8T, AAV8.AR2.08 and AAV8 are provided in FIG. 13A. An alignment of the nucleic acid sequences of AAV8T, AAV8.AR2.08 and AAV8 are provided in FIG. 13B-13D.

In certain embodiments, an AAV8.AR2.08 capsid comprises AAV8.AR2.08 capsid proteins comprising: AAV8.AR2.08 vp1 proteins produced by expression from a nucleic acid sequence which encodes the predicted amino acid sequence of 1 to 738 of SEQ ID NO: 18, vp1 proteins produced from SEQ ID NO: 17, or vp1 proteins produced from a nucleic acid sequence at least 70% identical to SEQ ID NO: 17 which encodes the predicted amino acid sequence of 1 to 738 of SEQ ID NO:18, AAV8.AR2.08 vp2 proteins produced by expression from a nucleic acid sequence which encodes the predicted amino acid sequence of at least about amino acids 138 to 738 of SEQ ID NO:18, vp2 proteins produced from a sequence comprising at least nucleotides 412 to 2214 of SEQ ID NO:17, or vp2 proteins produced from a nucleic acid sequence at least 70% identical to at least nucleotides 412 to 2214 of SEQ ID NO:17 which encodes the predicted amino acid sequence of at least about amino acids 138 to 738 of SEQ ID NO:18, AAV8.AR2.08 vp3 proteins produced by expression from a nucleic acid sequence which encodes the predicted amino acid sequence of at least about amino acids 204 to 738 of SEQ ID NO:18, vp3 proteins produced from a sequence comprising at least nucleotides 610 to 2214 of SEQ ID NO:17, or vp3 proteins produced from a nucleic acid sequence at least 70% identical to at least nucleotides 610 to 2214 of SEQ ID NO:17 which encodes the predicted amino acid sequence of at least about amino acids 204 to 738 of SEQ ID NO:18.

Additionally, or alternatively an AAV8.AR2.08 capsid comprises: a heterogeneous population of vp1 proteins which are the product of a nucleic acid sequence encoding the amino acid sequence of SEQ ID NO: 18, a heterogeneous population of vp2 proteins which are the product of a nucleic acid sequence encoding the amino acid sequence of at least about amino acids 138 to 738 of SEQ ID NO: 18, and a heterogeneous population of vp3 proteins which are the product of a nucleic acid sequence encoding at least amino acids 204 to 738 of SEQ ID NO:18 wherein: the vp1, vp2 and vp3 proteins contain subpopulations with amino acid modifications comprising at least two highly deamidated asparagines (N) in asparagine-glycine pairs in SEQ ID NO: 18 and optionally further comprising subpopulations comprising other deamidated amino acids, wherein the deamidation results in an amino acid change. AAV8.AR2.08 is characterized by having highly deamidated residues, e.g., at positions N57, N263, N385, N514 and N540 based on the numbering of the AAV8.AR2.08 VP1 [SEQ ID NO: 18]. Additionally, residues at the positions following table and the detailed table in the application show the deamidations which have been observed in the AAV8.AR2.08 capsid.

In certain embodiments, an AAV8.AR2.08 capsid is modified in one or more of the following positions, in the ranges provided below, as determined using mass spectrometry with a trypsin enzyme. In certain embodiments, one or more of the following positions, or the glycine following the N is modified as described herein. For example, in certain

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deamidation is observed when NG57/58 is altered to NS 57/58 or NA57/58. However, in certain embodiments, an increase in deamidation is observed when NG is altered to NS or NA. In certain embodiments, an N of an NG pair is modified to a Q while retaining the G. In certain embodiments, both amino acids of an NG pair are modified. In certain embodiments, N385Q results in significant reduction of deamidation in that location. In certain embodiments, N499Q results in significant increase of deamidation in that location.

In addition to deamidation, other modifications may include isomerization (e.g. at one or more of D442 and/or D584) (1-15%), phosphorylations (e.g. at one or more of ~S149, ~T417, ~T454, ~T493, S600, and/or ~T663), and/or oxidations (e.g., at one or more of positions ~W22, ~M204, ~M212, W248, W307, M405, M437, M473, W480, W505, M526, M561, M607, ~W609, W621, M637, W697). Still other positions may have such these or other modifications (e.g., acetylation or further deamidations).

TABLE B

AAV8.AR2.08 Deamidation	
AAV8.AR2.08 Deamidation based on VP1 numbering SEQ ID NO: 18	%
N57 + Deamidation	65-90, 70-95, 80-95, 75-100, 80-100, or 90-100
N94 + Deamidation	1-15
~N254 + Deamidation	1-15
~N263 + Deamidation	65-90, 70-95, 80-95, 75-100, 80-100, or 90-100
~N305 + Deamidation	1-15
~N385 + Deamidation	65-90, 70-95, 80-95, 75-100, 80-100, or 90-100
~N514 + Deamidation	65-100
~N521 + Deamidation	1-10
~N540 + Deamidation	65-90, 70-95, 80-95, 75-100, 80-100, or 90-100
N590 + Deamidation	0-5
Q601 + Deamidation	0-5
N653 + Deamidation	0-5
N665 + Deamidation	0-5

In certain embodiments, the nucleic acid sequence encoding the AAV8.AR2.08 vp1 capsid protein is provided in SEQ ID NO: 17. In other embodiments, a nucleic acid sequence of 70% to 99.9% identity to SEQ ID NO: 17 may be selected to express the AAV8.AR2.08 capsid proteins. In certain other embodiments, the nucleic acid sequence is at least about 75% identical, at least 80% identical, at least 85%, at least 90%, at least 95%, at least 97% identical, or at least 99% to 99.9% identical to SEQ ID NO: 17. However, other nucleic acid sequences which encode the amino acid sequence of SEQ ID NO: 18 may be selected for use in producing rAAV8.AR2.08 capsids. In certain embodiments, the nucleic acid sequence has the nucleic acid sequence of SEQ ID NO: 17 or a sequence at least 70% to 99% identical, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, at least 99%, identical to SEQ ID NO: 17 which encodes SEQ ID NO: 18. In certain embodiments, the nucleic acid sequence has the nucleic acid sequence of SEQ ID NO: 17 or a sequence at least 70% to 99.%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, at least 99%, identical to about nt 412 to about nt 2214 of SEQ ID NO: 17 which encodes the vp2 capsid protein (about aa 138 to 738) of SEQ ID NO: 18. In certain embodiments, the nucleic acid sequence has the nucleic acid sequence of about nt 607 to about nt 2214 of SEQ ID NO:

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17 or a sequence at least 70% to 99. %, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, at least 99%, identical to nt SEQ ID NO: 17 which encodes the vp3 capsid protein (about aa 204 to 738) of SEQ ID NO:18.

The invention also encompasses nucleic acid sequences encoding mutant AAV8.2.08, in which one or more residues has been altered in order to decrease deamidation, or other modifications which are identified herein. Such nucleic acid sequences can be used in production of mutant rAAV8.2.08.

In certain embodiments, a novel AAV5.5.9 capsid is provided. The nucleic acid sequence encoding the AAV is provided in SEQ ID NO:9 and the encoded amino acid sequence is provided in SEQ ID NO: 10. An alignment of the amino acid sequences of AAV5.5.9, AAV9, and AAVPHP.B is shown in FIG. 12A. An alignment of the nucleic acid sequences of AAV5.5.9, AAV9, and AAVPHP.B is shown in FIG. 12B-12E. In one embodiment, a recombinant adeno-associated virus (rAAV) has an AAV5.5.9 capsid comprising: AAV5.5.9 capsid proteins comprising: a heterogeneous population of AAV5.5.9 vp1 proteins produced by expression from a nucleic acid sequence which encodes the predicted amino acid sequence of 1 to 726 of SEQ ID NO: 10, vp1 proteins produced from SEQ ID NO:9, or vp1 proteins produced from a nucleic acid sequence at least 70% identical to SEQ ID NO:9 which encodes the predicted amino acid sequence of 1 to 726 of SEQ ID NO: 10; a heterologous population of AAV5.5.9 vp2 proteins produced by expression from a nucleic acid sequence which encodes the predicted amino acid sequence of at least about amino acids 137 to 726 of SEQ ID NO:10, vp2 proteins produced from a sequence comprising at least nucleotides 409 to 2178 of SEQ ID NO:9, or vp2 proteins produced from a nucleic acid sequence at least 70% identical to at least nucleotides 577 to 2178 of SEQ ID NO: 9 which encodes the predicted amino acid sequence of at least about amino acids 137 to 726 of SEQ ID NO:10, and a heterologous population of AAV5.5.9 vp3 proteins produced by expression from a nucleic acid sequence which encodes the predicted amino acid sequence of at least about amino acids 193 to 726 of SEQ ID NO:10, vp3 proteins produced from a sequence comprising at least nucleotides 577 to 2178 of SEQ ID NO:9, or vp3 proteins produced from a nucleic acid sequence at least 70% identical to at least nucleotides 577 to 2178 of SEQ ID NO:9 which encodes the predicted amino acid sequence of at least about amino acids 193 to 726 of SEQ ID NO:10.

Additionally or alternatively, an AAV5.5.9 capsid comprises: a heterogeneous population of vp1 proteins which are the product of a nucleic acid sequence encoding the amino acid sequence of SEQ ID NO: 10, a heterogeneous population of vp2 proteins which are the product of a nucleic acid sequence encoding the amino acid sequence of at least about amino acids 137 to 726 of SEQ ID NO: 10, and a heterogeneous population of vp3 proteins which are the product of a nucleic acid sequence encoding at least amino acids 193 to 726 of SEQ ID NO:10 wherein: the vp1, vp2 and vp3 proteins contain subpopulations with amino acid modifications comprising at least two highly deamidated asparagines (N) in asparagine-glycine pairs in SEQ ID NO: 10 and optionally further comprising subpopulations comprising other deamidated amino acids, wherein the deamidation results in an amino acid change.

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TABLE C

AAV5.5.9 Deamidation		
	Deamidation based on VP1 numbering SEQ ID NO: 10	%
5	AAV5.5.9	
	N35 + Deamidation	0-15, 1-10
	~N57 + Deamidation	65-90, 70-95, 80-95, 75-100, 80-100, or 90-100
10	N113 + Deamidation	0-15
	~N204 + Deamidation	0-20, 1-20
	N217 + Deamidation	0-5, 1-5
	~N243 + Deamidation	0-25, 1-25
	Q249 + Deamidation	1-20
	N293/294 + Deamidation	10-45, 15-40
15	N304 + Deamidation	1-10
	N309 + Deamidation	1-2
	Q311 + Deamidation	1-2
	~N319 + Deamidation	65-90, 70-95, 80-95, 75-100, 80-100, or 90-100
	N399/400 + Deamidation	5-40, 10-40, 15-35
20	~N442 + Deamidation	65-90, 70-95, 80-95, 75-100, 80-100, or 90-100
	N467 + Deamidation	1-5
	N502 + Deamidation	65-90, 70-95, 80-95, 75-100, 80-100, or 90-100
	N505 + Deamidation	5-25, 10-25
	~Q589 + Deamidation	5-30, 10-30
25	N618 + Deamidation	1-15, 5-10
	~N641 + Deamidation	1-15, 5-10
	N653 + Deamidation	1-15, 5-10
	~N658 + Deamidation	5-40, 10-30
	N694 + Deamidation	0-5
30	~N699 + Deamidation	1-10

In certain embodiments, the nucleic acid sequence encoding the AAV5.5.9 vp1 capsid protein is provided in SEQ ID NO: 9. In other embodiments, a nucleic acid sequence of 70% to 99.9% identity to SEQ ID NO: 9 may be selected to express the AAV5.5.9 capsid proteins. In certain other embodiments, the nucleic acid sequence is at least about 75% identical, at least 80% identical, at least 85%, at least 90%, at least 95%, at least 97% identical, or at least 99% to 99.9% identical to SEQ ID NO: 9. However, other nucleic acid sequences which encode the amino acid sequence of SEQ ID NO: 10 may be selected for use in producing rAAV5.5.9 capsids. In certain embodiments, the nucleic acid sequence has the nucleic acid sequence of SEQ ID NO: 10 or a sequence at least 70% to 99% identical, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, at least 99%, identical to SEQ ID NO: 9 which encodes SEQ ID NO: 10. In certain embodiments, the nucleic acid sequence has the nucleic acid sequence of SEQ ID NO: 9 or a sequence at least 70% to 99. %, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, at least 99%, identical to about nt 409 to about nt 2178 of SEQ ID NO: 9 which encodes the vp2 capsid protein (about aa 137 to 726) of SEQ ID NO: 10. In certain embodiments, the nucleic acid sequence has the nucleic acid sequence of about nt 577 to about nt 2178 of SEQ ID NO:9 or a sequence at least 70% to 99%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, at least 99%, identical to nt 577 to 2178 SEQ ID NO: 9 which encodes the vp3 capsid protein (about aa 193 to 726) of SEQ ID NO: 10.

The invention also encompasses nucleic acid sequences encoding mutant AAV5.5.9, in which one or more residues has been altered in order to decrease deamidation, or other modifications which are identified herein. Such nucleic acid sequences can be used in production of mutant rAAV5.5.9.

I. rAAV Vectors

As indicated above, the novel AAV sequences and proteins are useful in production of rAAV and are also useful in recombinant AAV vectors which may be antisense delivery vectors, gene therapy vectors, or vaccine vectors. Additionally, the engineered AAV capsids described herein may be used to engineer rAAV vectors for delivery of a number of suitable nucleic acid molecules to target cells and tissues.

Genomic sequences which are packaged into an AAV capsid and delivered to a host cell are typically composed of, at a minimum, a transgene and its regulatory sequences, and AAV inverted terminal repeats (ITRs). Both single-stranded AAV and self-complementary (sc) AAV are encompassed with the rAAV. The transgene is a nucleic acid coding sequence, heterogeneous to the vector sequences, which encodes a polypeptide, protein, functional RNA molecule (e.g., miRNA, miRNA inhibitor) or other gene product, of interest. The nucleic acid coding sequence is operatively linked to regulatory components in a manner which permits transgene transcription, translation, and/or expression in a cell of a target tissue.

The AAV sequences of the vector typically comprise the cis-acting 5' and 3' inverted terminal repeat sequences (See, e.g., B. J. Carter, in "Handbook of Parvoviruses", ed., P. Tijsser, CRC Press, pp. 155-168 (1990)). The ITR sequences are about 145 bp in length. Preferably, substantially the entire sequences encoding the ITRs are used in the molecule, although some degree of minor modification of these sequences is permissible. The ability to modify these ITR sequences is within the skill of the art. (See, e.g., texts such as Sambrook et al., "Molecular Cloning. A Laboratory Manual", 2d ed., Cold Spring Harbor Laboratory, New York (1989); and K. Fisher et al., *J. Virol.*, 70:520-532 (1996)). An example of such a molecule employed in the present invention is a "cis-acting" plasmid containing the transgene, in which the selected transgene sequence and associated regulatory elements are flanked by the 5' and 3' AAV ITR sequences. In one embodiment, the ITRs are from an AAV different than that supplying a capsid, resulting in a pseudotyped vector. In one embodiment, the ITR sequences from AAV2. A shortened version of the 5' ITR, termed AITR, has been described in which the D-sequence and terminal resolution site (trs) are deleted. In other embodiments, the full-length AAV 5' and 3' ITRs are used. However, ITRs from other AAV sources may be selected. Where the source of the ITRs is from AAV2 and the AAV capsid is from another AAV source, the resulting vector may be termed pseudotyped. However, other configurations of these elements may be suitable.

In addition to the major elements identified above for the recombinant AAV vector, the vector also includes conventional control elements necessary which are operably linked to the transgene in a manner which permits its transcription, translation and/or expression in a cell transfected with the plasmid vector or infected with the virus produced by the invention. As used herein, "operably linked" sequences include both expression control sequences that are contiguous with the gene of interest and expression control sequences that act in trans or at a distance to control the gene of interest.

The regulatory control elements typically contain a promoter sequence as part of the expression control sequences, e.g., located between the selected 5' ITR sequence and the coding sequence. Constitutive promoters, regulatable promoters [see, e.g., WO 2011/126808 and WO 2013/04943], tissue specific promoters, or a promoter responsive to physiologic cues may be used may be utilized in the vectors

described herein. The promoter(s) can be selected from different sources. e.g., human cytomegalovirus (CMV) immediate-early enhancer/promoter, the SV40 early enhancer/promoter, the JC polyomavirus promoter, myelin basic protein (MBP) or glial fibrillary acidic protein (GFAP) promoters, herpes simplex virus (HSV-1) latency associated promoter (LAP), rouse sarcoma virus (RSV) long terminal repeat (LTR) promoter, neuron-specific promoter (NSE), platelet derived growth factor (PDGF) promoter, hSYN, melanin-concentrating hormone (MCH) promoter, CBA, matrix metalloprotein promoter (MPP), and the chicken beta-actin promoter. In one embodiment, the promoter is a liver specific promoter, such as that termed LSP exemplified herein.

In addition to a promoter a vector may contain one or more other appropriate transcription initiation, termination, enhancer sequences, efficient RNA processing signals such as splicing and polyadenylation (polyA) signals; sequences that stabilize cytoplasmic mRNA for example WPRE; sequences that enhance translation efficiency (i.e., Kozak consensus sequence); sequences that enhance protein stability; and when desired, sequences that enhance secretion of the encoded product. An example of a suitable enhancer is the CMV enhancer. Other suitable enhancers include those that are appropriate for desired target tissue indications. In one embodiment, the expression cassette comprises one or more expression enhancers. In one embodiment, the expression cassette contains two or more expression enhancers. These enhancers may be the same or may differ from one another. For example, an enhancer may include a CMV immediate early enhancer. This enhancer may be present in two copies which are located adjacent to one another. Alternatively, the dual copies of the enhancer may be separated by one or more sequences. In still another embodiment, the expression cassette further contains an intron, e.g., the chicken beta-actin intron. Other suitable introns include those known in the art, e.g., such as are described in WO 2011/126808. Examples of suitable polyA sequences include, e.g., SV40, SV50, bovine growth hormone (bGH), human growth hormone, and synthetic polyAs. Optionally, one or more sequences may be selected to stabilize mRNA. An example of such a sequence is a modified WPRE sequence, which may be engineered upstream of the polyA sequence and downstream of the coding sequence [see, e.g., MA Zanta-Boussif, et al, *Gene Therapy* (2009) 16:605-619].

These rAAVs are particularly well suited to gene delivery for therapeutic purposes and for immunization, including inducing protective immunity. Further, the compositions of the invention may also be used for production of a desired gene product in vitro. For in vitro production, a desired product (e.g., a protein) may be obtained from a desired culture following transfection of host cells with a rAAV containing the molecule encoding the desired product and culturing the cell culture under conditions which permit expression. The expressed product may then be purified and isolated, as desired. Suitable techniques for transfection, cell culturing, purification, and isolation are known to those of skill in the art.

Therapeutic Transgenes

Useful products encoded by the transgene include a variety of gene products which replace a defective or deficient gene, inactivate or "knock-out", or "knock-down" or reduce the expression of a gene which is expressing at an undesirably high level, or delivering a gene product which has a desired therapeutic effect. In most embodiments, the therapy will be "somatic gene therapy", i.e., transfer of genes to a cell of the body which does not produce sperm or eggs. In certain embodiments, the transgenes express pro-

teins have the sequence of native human sequences. However, in other embodiments, synthetic proteins are expressed. Such proteins may be intended for treatment of humans, or in other embodiments, designed for treatment of animals, including companion animals such as canine or feline populations, or for treatment of livestock or other animals which come into contact with human populations.

Examples of suitable gene products may include those associated with familial hypercholesterolemia, muscular dystrophy, cystic fibrosis, and rare or orphan diseases. Examples of such rare disease may include spinal muscular atrophy (SMA), Huntingdon's Disease, Rett Syndrome (e.g., methyl-CpG-binding protein 2 (MeCP2); UniProtKB-P51608), Amyotrophic Lateral Sclerosis (ALS), Duchenne Type Muscular dystrophy, Friedrichs Ataxia (e.g., frataxin), progranulin (PRGN) (associated with non-Alzheimer's cerebral degenerations, including, frontotemporal dementia (FTD), progressive non-fluent aphasia (PNFA) and semantic «lmental), among others. See, e.g., www.orpha.net/consor/cgi-bin/Disease_Search_List_php; rarediseases.info.nih.gov/diseases.

Examples of suitable genes may include, e.g., hormones and growth and differentiation factors including, without limitation, insulin, glucagon, glucagon-like peptide-1 (GLP1), growth hormone (GH), parathyroid hormone (PTH), growth hormone releasing factor (GRF), follicle stimulating hormone (FSH), luteinizing hormone (LH), human chorionic gonadotropin (hCG), vascular endothelial growth factor (VEGF), angiopoietins, angiostatin, granulocyte colony stimulating factor (GCSF), erythropoietin (EPO) (including, e.g., human, canine or feline epo), connective tissue growth factor (CTGF), neutrophilic factors including, e.g., basic fibroblast growth factor (bFGF), acidic fibroblast growth factor (aFGF), epidermal growth factor (EGF), platelet-derived growth factor (PDGF), insulin growth factors I and II (IGF-I and IGF-II), any one of the transforming growth factor α superfamily, including TGF α , activins, inhibins, or any of the bone morphogenic proteins (BMP) BMPs 1-15, any one of the heregluin/neuregulin/ARIA/neu differentiation factor (NDF) family of growth factors, nerve growth factor (NGF), brain-derived neurotrophic factor (BDNF), neurotrophins NT-3 and NT-4/5, ciliary neurotrophic factor (CNTF), glial cell line derived neurotrophic factor (GDNF), neurturin, agrin, any one of the family of semaphorins/collapsins, netrin-1 and netrin-2, hepatocyte growth factor (HGF), ephrins, noggin, sonic hedgehog and tyrosine hydroxylase.

Other useful transgene products include proteins that regulate the immune system including, without limitation, cytokines and lymphokines such as thrombopoietin (TPO), interleukins (IL) IL-1 through IL-36 (including, e.g., human interleukins IL-1, IL-1 α , IL-1 β , IL-2, IL-3, IL-4, IL-6, IL-8, IL-12, IL-11, IL-12, IL-13, IL-18, IL-31, IL-35), monocyte chemoattractant protein, leukemia inhibitory factor, granulocyte-macrophage colony stimulating factor, Fas ligand, tumor necrosis factors α and β , interferons α , β , and γ , stem cell factor, flk-2/flt3 ligand. Gene products produced by the immune system are also useful in the invention. These include, without limitations, immunoglobulins IgG, IgM, IgA, IgD and IgE, chimeric immunoglobulins, humanized antibodies, single chain antibodies, T cell receptors, chimeric T cell receptors, single chain T cell receptors, class I and class II MHC molecules, as well as engineered immunoglobulins and MHC molecules. For example, in certain embodiments, the rAAV antibodies may be designed to delivery canine or feline antibodies, e.g., such as anti-IgE, anti-IL31, anti-CD20, anti-NGF, anti-GnRH. Useful gene

products also include complement regulatory proteins such as complement regulatory proteins, membrane cofactor protein (MCP), decay accelerating factor (DAF), CR1, CF2, CD59, and C1 esterase inhibitor (C1-INH).

Still other useful gene products include any one of the receptors for the hormones, growth factors, cytokines, lymphokines, regulatory proteins and immune system proteins. The invention encompasses receptors for cholesterol regulation and/or lipid modulation, including the low density lipoprotein (LDL) receptor, high density lipoprotein (HDL) receptor, the very low density lipoprotein (VLDL) receptor, and scavenger receptors. The invention also encompasses gene products such as members of the steroid hormone receptor superfamily including glucocorticoid receptors and estrogen receptors, Vitamin D receptors and other nuclear receptors. In addition, useful gene products include transcription factors such as jun, fos, max, mad, serum response factor (SRF), AP-1, AP2, myb, MyoD and myogenin, ETS-box containing proteins, TFE3, E2F, ATF1, ATF2, ATF3, ATF4, ZF5, NFAT, CREB, HNF-4, C/EBP, SP1, CCAAT-box binding proteins, interferon regulation factor (IRF-1), Wilms tumor protein, ETS-binding protein, STAT, GATA-box binding proteins, e.g., GATA-3, and the forkhead family of winged helix proteins.

Other useful gene products include, carbamoyl synthetase I, ornithine transcarbamylase (OTC), arginosuccinate synthetase, arginosuccinate lyase (ASL) for treatment of arginosuccinate lyase deficiency, arginase, fumarylacetate hydrolase, phenylalanine hydroxylase, alpha-1 antitrypsin, rhesus alpha-fetoprotein (AFP), rhesus chorionic gonadotropin (CG), glucose-6-phosphatase, porphobilinogen deaminase, cystathione beta-synthase, branched chain ketoacid decarboxylase, albumin, isovaleryl-coA dehydrogenase, propionyl CoA carboxylase, methyl malonyl CoA mutase, glutaryl CoA dehydrogenase, insulin, beta-glucosidase, pyruvate carboxylate, hepatic phosphorylase, phosphorylase kinase, glycine decarboxylase, H-protein, T-protein, a cystic fibrosis transmembrane regulator (CFTR) sequence, and a dystrophin gene product [e.g., a mini- or micro-dystrophin]. Still other useful gene products include enzymes such as may be useful in enzyme replacement therapy, which is useful in a variety of conditions resulting from deficient activity of enzyme. For example, enzymes that contain mannose-6-phosphate may be utilized in therapies for lysosomal storage diseases (e.g., a suitable gene includes that encoding β -glucuronidase (GUSB)).

In certain embodiments, the rAAV may be used in gene editing systems, which system may involve one rAAV or co-administration of multiple rAAV stocks. For example, the rAAV may be engineered to deliver SpCas9, SaCas9, ARCUS, Cpf1, and other suitable gene editing constructs.

Still other useful gene products include those used for treatment of hemophilia, including hemophilia B (including Factor IX) and hemophilia A (including Factor VIII and its variants, such as the light chain and heavy chain of the heterodimer and the B-deleted domain; U.S. Pat. Nos. 6,200,560 and 6,221,349). In some embodiments, the minigene comprises first 57 base pairs of the Factor VIII heavy chain which encodes the 10 amino acid signal sequence, as well as the human growth hormone (hGH) polyadenylation sequence. In alternative embodiments, the minigene further comprises the A1 and A2 domains, as well as 5 amino acids from the N-terminus of the B domain, and/or 85 amino acids of the C-terminus of the B domain, as well as the A3, C1 and C2 domains. In yet other embodiments, the nucleic acids encoding Factor VIII heavy chain and light chain are pro-

vided in a single minigene separated by 42 nucleic acids coding for 14 amino acids of the B domain [U.S. Pat. No. 6,200,560].

Other useful gene products include non-naturally occurring polypeptides, such as chimeric or hybrid polypeptides having a non-naturally occurring amino acid sequence containing insertions, deletions or amino acid substitutions. For example, single-chain engineered immunoglobulins could be useful in certain immunocompromised patients. Other types of non-naturally occurring gene sequences include antisense molecules and catalytic nucleic acids, such as ribozymes, which could be used to reduce overexpression of a target.

Reduction and/or modulation of expression of a gene is particularly desirable for treatment of hyperproliferative conditions characterized by hyperproliferating cells, as are cancers and psoriasis. Target polypeptides include those polypeptides which are produced exclusively or at higher levels in hyperproliferative cells as compared to normal cells. Target antigens include polypeptides encoded by oncogenes such as myb, myc, fyn, and the translocation gene bcr/abl, ras, src, P53, neu, trk and EGFR. In addition to oncogene products as target antigens, target polypeptides for anti-cancer treatments and protective regimens include variable regions of antibodies made by B cell lymphomas and variable regions of T cell receptors of T cell lymphomas which, in some embodiments, are also used as target antigens for autoimmune disease. Other tumor-associated polypeptides can be used as target polypeptides such as polypeptides which are found at higher levels in tumor cells including the polypeptide recognized by monoclonal antibody 17-1A and folate binding polypeptides.

Other suitable therapeutic polypeptides and proteins include those which may be useful for treating individuals suffering from autoimmune diseases and disorders by conferring a broad based protective immune response against targets that are associated with autoimmunity including cell receptors and cells which produce "self"-directed antibodies. T cell mediated autoimmune diseases include Rheumatoid arthritis (RA), multiple sclerosis (MS), Sjögren's syndrome, sarcoidosis, insulin dependent diabetes mellitus (IDDM), autoimmune thyroiditis, reactive arthritis, ankylosing spondylitis, scleroderma, polymyositis, dermatomyositis, psoriasis, vasculitis, Wegener's granulomatosis, Crohn's disease and ulcerative colitis. Each of these diseases is characterized by T cell receptors (TCRs) that bind to endogenous antigens and initiate the inflammatory cascade associated with autoimmune diseases.

Further illustrative genes which may be delivered via the rAAV include, without limitation, glucose-6-phosphatase, associated with glycogen storage disease or deficiency type 1A (GSD1), phosphoenolpyruvate-carboxykinase (PEPCK), associated with PEPCK deficiency; cyclin-dependent kinase-like 5 (CDKL5), also known as serine/threonine kinase 9 (STK9) associated with seizures and severe neurodevelopmental impairment; galactose-1 phosphate uridyl transferase, associated with galactosemia; phenylalanine hydroxylase, associated with phenylketonuria (PKU); branched chain alpha-ketoacid dehydrogenase, associated with Maple syrup urine disease; fumarylacetoacetate hydroxylase, associated with tyrosinemia type 1; methylmalonyl-CoA mutase, associated with methylmalonic acidemia; medium chain acyl CoA dehydrogenase, associated with medium chain acetyl CoA deficiency; ornithine transcarbamylase (OTC), associated with ornithine transcarbamylase deficiency; argininosuccinic acid synthetase (ASS1), associated with citrullinemia; lecithin-cholesterol acyltrans-

ferase (LCAT) deficiency; amethylmalonic acidemia (MMA); Niemann-Pick disease, type C1); propionic acidemia (PA); low density lipoprotein receptor (LDLR) protein, associated with familial hypercholesterolemia (FH); UDP-glucouronosyltransferase, associated with Crigler-Najjar disease; adenosine deaminase, associated with severe combined immunodeficiency disease; hypoxanthine guanine phosphoribosyl transferase, associated with Gout and Lesch-Nyan syndrome; biotinidase, associated with biotinidase deficiency; alpha-galactosidase A (a-Gal A) associated with Fabry disease); ATP7B associated with Wilson's Disease; beta-glucocerebrosidase, associated with Gaucher disease type 2 and 3; peroxisome membrane protein 70 kDa, associated with Zellweger syndrome; arylsulfatase A (ARSA) associated with metachromatic leukodystrophy, galactocerebrosidase (GALC) enzyme associated with Krabbe disease, alpha-glucosidase (GAA) associated with Pompe disease; sphingomyelinase (SMPD1) gene associated with Nieman Pick disease type A; argininosuccinate synthase associated with adult onset type II citrullinemia (CTLN2); carbamoyl-phosphate synthase 1 (CPS1) associated with urea cycle disorders; survival motor neuron (SMN) protein, associated with spinal muscular atrophy; ceramidase associated with Farber lipogranulomatosis; b-hexosaminidase associated with GM2 gangliosidosis and Tay-Sachs and Sandhoff diseases; aspartylglucosaminidase associated with aspartylglucosaminuria; α -fucosidase associated with fucosidosis; α -mannosidase associated with alpha-mannosidosis; porphobilinogen deaminase, associated with acute intermittent porphyria (AIP); alpha-1 antitrypsin for treatment of alpha-1 antitrypsin deficiency (emphysema); erythropoietin for treatment of anemia due to thalassemia or to renal failure; vascular endothelial growth factor, angiopoietin-1, and fibroblast growth factor for the treatment of ischemic diseases; thrombomodulin and tissue factor pathway inhibitor for the treatment of occluded blood vessels as seen in, for example, atherosclerosis, thrombosis, or embolisms; aromatic amino acid decarboxylase (AADC), and tyrosine hydroxylase (TH) for the treatment of Parkinson's disease; the beta adrenergic receptor, anti-sense to, or a mutant form of, phospholamban, the sarco (endo) plasmic reticulum adenosine triphosphatase-2 (SERCA2), and the cardiac adenyl cyclase for the treatment of congestive heart failure; a tumor suppressor gene such as p53 for the treatment of various cancers; a cytokine such as one of the various interleukins for the treatment of inflammatory and immune disorders and cancers; dystrophin or minidystrophin and utrophin or miniutrophin for the treatment of muscular dystrophies; and, insulin or GLP-1 for the treatment of diabetes.

Additional genes and diseases of interest include, e.g., dystonia gene related diseases such as Hereditary Sensory and Autonomic Neuropathy Type VI (the DST gene encodes dystonin; dual AAV vectors may be required due to the size of the protein (~7570 aa); SCN9A related diseases, in which loss of function mutants cause inability to feel pain and gain of function mutants cause pain conditions, such as erythromelalgia. Another condition is Charcot-Marie-Tooth type 1F and 2E due to mutations in the NEFL gene (neurofilament light chain). characterized by a progressive peripheral motor and sensory neuropathy with variable clinical and electrophysiologic expression.

In certain embodiments, the rAAV described herein may be used in treatment of mucopolysaccharidoses (MPS) disorders. Such rAAV may contain carry a nucleic acid sequence encoding α -L-iduronidase (IDUA) for treating MPS I (Hurler, Hurler-Scheie and Scheie syndromes); a

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nucleic acid sequence encoding iduronate-2-sulfatase (IDS) for treating MPS II (Hunter syndrome); a nucleic acid sequence encoding sulfamidase (SGSH) for treating MPSIII A, B, C, and D (Sanfilippo syndrome); a nucleic acid sequence encoding N-acetylgalactosamine-6-sulfate sulfatase (GALNS) for treating MPS IV A and B (Morquio syndrome); a nucleic acid sequence encoding arylsulfatase B (ARSB) for treating MPS VI (Maroteaux-Lamy syndrome); a nucleic acid sequence encoding hyaluronidase for treating MPSI IX (hyaluronidase deficiency) and a nucleic acid sequence encoding beta-glucuronidase for treating MPS VII (Sly syndrome).

Immunogenic Transgenes

In some embodiments, an rAAV vector comprising a nucleic acid encoding a gene product associated with cancer (e.g., tumor suppressors) may be used to treat the cancer, by administering a rAAV harboring the rAAV vector to a subject having the cancer. In some embodiments, an rAAV vector comprising a nucleic acid encoding a small interfering nucleic acid (e.g., shRNAs, miRNAs) that inhibits the expression of a gene product associated with cancer (e.g., oncogenes) may be used to treat the cancer, by administering a rAAV harboring the rAAV vector to a subject having the cancer. In some embodiments, an rAAV vector comprising a nucleic acid encoding a gene product associated with cancer (or a functional RNA that inhibits the expression of a gene associated with cancer) may be used for research purposes, e.g., to study the cancer or to identify therapeutics that treat the cancer. The following is a non-limiting list of exemplary genes known to be associated with the development of cancer (e.g., oncogenes and tumor suppressors): AARS, ABCB1, ABCC4, ABI2, ABL1, ABL2, ACK1, ACP2, ACY1, ADL1, AK1, AKRIC2, AKT1, ALB, ANPEP, ANXA5, ANXA7, AP2M1, APC, ARHGAP5, ARHGEF5, ARID4A, ASNS, ATF4, ATM, ATP5B, ATP50, AXL, BARD1, BAX, BCL2, BHLHB2, BLMH, BRAF, BRCA1, BRCA2, BTK, CANX, CAPI, CAPN1, CAPNS1, CAV1, CBFB, CBLB, CCL2, CCND1, CCND2, CCND3, CCNE1, CCT5, CCYR61, CD24, CD44, CD59, CDC20, CDC25, CDC25A, CDC25B, CDC25L, CDK10, CDK4, CDK5, CDK9, CDKL1, CDKN1A, CDKN1B, CDKN1C, CDKN2A, CDKN2B, CDKN2D, CEBPG, CENPC1, CGRRF1, CHAF1A, CIB1, CKMT1, CLK1, CLK2, CLK3, CLNSIA, CLTC, COL1A1, COL6A3, COX6C, COX7A2, CRAT, CRHR1, CSF1R, CSK, CSNK1G2, CTNNA1, CTNNB1, CTPS, CTSC, CTSB, CUL1, CYR61, DCC, DCN, DDX10, DEK, DHCR7, DHRS2, DHX8, DLG3, DVL1, DVL3, E2F1, E2F3, E2F5, EGFR, EGR1, EIF5, EPHA2, ERBB2, ERBB3, ERBB4, ERCC3, ETV1, ETV3, ETV6, F2R, FASTK, FBN1, FBN2, FES, FGFR1, FGR, FKBP8, FN1, FOS, FOSL1, FOSL2, FOXG1A, FOXO1A, FRAP1, FRZB, FTL, FZD2, FZD5, FZD9, G22P1, GAS6, GCN5L2, GDF15, GNA13, GNAS, GNB2, GNB2L1, GPR39, GRB2, GSK3A, GSPT1, GTF21, HDAC1, HDGF, HMMR, HPR1, HRB, HSPA4, HSPA5, HSPA8, HSPB1, HSPH1, HYL1, HYOU1, ICAM1, ID1, ID2, IDUA, IER3, IFITM1, IGF1R, IGF2R, IGFBP3, IGFBP4, IGFBP5, IL1B, ILK, ING1, IRF3, ITGA3, ITGA6, ITGB4, JAK1, JARIDIA, JUN, JUNB, JUND, K-ALPHA-1, KIT, KITLG, KLK10, KPNA2, KRAS2, KRT18, KRT2A, KRT9, LAMB1, LAMP2, LCK, LCN2, LEP, LITAF, LRPAP1, LTF, LYN, LZTR1, MADH1, MAP2K2, MAP3K8, MAPK12, MAPK13, MAPKAPK3, MAPRE1, MARS, MAS1, MCC, MCM2, MCM4, MDM2, MDM4, MET, MGST1, MICB, MLLT3, MME, MMP1, MMP14, MMP17, MMP2, MNDA, MSH2, MSH6, MT3, MYB, MYBL1, MYBL2, MYC, MYCL1, MYCN, MYD88, MYL9, MYLK,

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NEO1, NF1, NF2, NFKB1, NFKB2, NFSF7, NID, NINE, NMBR, NME1, NME2, NME3, NOTCH1, NOTCH2, NOTCH4, NPM1, NQO1, NR1D1, NR2F1, NR2F6, NRAS, NRG1, NSE1, OSM, PA2G4, PABPC1, PCNA, PCTK1, PCTK2, PCTK3, PDGFA, PDGFB, PDGFRA, PDPK1, PEA15, PFDN4, PFDN5, PGAM1, PHB, PIK3CA, PIK3CB, PIK3CG, PIM1, PKM2, PKMYT1, PLK2, PPARD, PPARG, PPIH, PPPICA, PPP2R5A, PRDX2, PRDX4, PRKARIA, PRKCBP1, PRNP, PRSS15, PSMA1, PTCH, PTEN, PTGS1, PTMA, PTN, PTPRN, RABSA, RAC1, RAD50, RAF1, RALBP1, RAPIA, RARA, RARB, RASGRF1, RB1, RBBP4, RBI 2, REA, REL, RELA, RELB, RET, RFC2, RGS19, RHOA, RHOB, RHOC, RHOD, RIPK1, RPN2, RPS6 KB1, RRM1, SARS, SELENBP1, SEMA3C, SEMA4D, SEPP1, SERPINH1, SFN, SFPQ, SFRS7, SHB, SHH, SIAH2, SIVA, SIVA TP53, SKI, SKIL, SLC16A1, SLC1A4, SLC20A1, SMO, sphingomyelin phosphodiesterase 1 (SMPD1), SNAI2, SND1, SNRPB2, SOCS1, SOCS3, SOD1, SORT1, SPINT2, SPRY2, SRC, SRPX, STAT1, STAT2, STAT3, STAT5B, STC1, TAF1, TBL3, TBRG4, TCF1, TCF7L2, TFAP2C, TFDPI1, TFDPI2, TGFA, TGFB1, TGFB1, TGFB2, TGFB3, THBS1, TIE, TIMP1, TIMP3, TJP1, TK1, TLE1, TNF, TNFRSF10A, TNFRSF10B, TNFRSF1A, TNFRSF1B, TNFRSF6, TNFRSF7, TNK1, TOB1, TP53, TP53BP2, TP53I3, TP73, TPBG, TPT1, TRADD, TRAM1, TRRAP, TSG101, TUFM, TXNRD1, TYRO3, UBC, UBE2L6, UCHL1, USP7, VDAC1, VEGF, VHL, VIL2, WEE1, WNT1, WNT2, WNT2B, WNT3, WNT5A, WT1, XRCC1, YES1, YWHAB, YWHAZ, ZAP70, and ZNF9.

A rAAV vector may comprise as a transgene, a nucleic acid encoding a protein or functional RNA that modulates apoptosis. The following is a non-limiting list of genes associated with apoptosis and nucleic acids encoding the products of these genes and their homologues and encoding small interfering nucleic acids (e.g., shRNAs, miRNAs) that inhibit the expression of these genes and their homologues are useful as transgenes in certain embodiments of the invention: RPS27A, ABL1, AKT1, APAF1, BAD, BAG1, BAG3, BAG4, BAK1, BAX, BCL10, BCL2, BCL2A1, BCL2L1, BCL2L10, BCL2L11, BCL2L12, BCL2L13, BCL2L2, BCLAF1, BFAR, BID, BIK, NAIP, BIRC2, BIRC3, XIAP, BIRC5, BIRC6, BIRC7, BIRC8, BNIPI, BNIP2, BNIP3, BNIP3L, BOK, BRAF, CARD10, CARD11, NLRC4, CARD14, NOD2, NOD1, CARD6, CARDS, CARDS, CASPI, CASP10, CASP14, CASP2, CASP3, CASP4, CASP5, CASP6, CASP7, CASP8, CASP9, CFLAR, CIDEA, CIDEB, CRADD, DAPK1, DAPK2, DFFA, DFFB, FADD, GADD45A, GDNF, HRK, IGFR, LTA, LTBR, MCL1, NOL3, PYCARD, RIPK1, RIPK2, TNF, TNFRSF10A, TNFRSF10B, TNFRSF10C, TNFRSF10D, TNFRSF11B, TNFRSF12A, TNFRSF14, TNFRSF19, TNFRSF1A, TNFRSF1B, TNFRSF21, TNFRSF25, CD40, FAS, TNFRSF6B, CD27, TNFRSF9, TNFRSF10, TNFRSF14, TNFRSF18, CD40LG, FASLG, CD70, TNFRSF8, TNFRSF9, TP53, TP53BP2, TP73, TP63, TRADD, TRAF1, TRAF2, TRAF3, TRAF4, and TRAF5.

Useful transgene products also include miRNAs. miRNAs and other small interfering nucleic acids regulate gene expression via target RNA transcript cleavage/degradation or translational repression of the target messenger RNA (mRNA). miRNAs are natively expressed, typically as final 19-25 non-translated RNA products. miRNAs exhibit their activity through sequence-specific interactions with the 3' untranslated regions (UTR) of target mRNAs. These endogenously expressed miRNAs form hairpin precursors which are subsequently processed into a miRNA duplex, and

further into a “mature” single stranded miRNA molecule. This mature miRNA guides a multiprotein complex, miRISC, which identifies target site, e.g., in the 3' UTR regions, of target mRNAs based upon their complementarity to the mature miRNA.

The following non-limiting list of miRNA genes, and their homologues, are useful as transgenes or as targets for small interfering nucleic acids encoded by transgenes (e.g., miRNA sponges, antisense oligonucleotides, TUD RNAs) in certain embodiments of the methods: hsa-let-7a, hsa-let-7a*, hsa-let-7b, hsa-let-7b*, hsa-let-7c, hsa-let-7c*, hsa-let-7d, hsa-let-7d*, hsa-let-7e, hsa-let-7e*, hsa-let-7f, hsa-let-7f-1*, hsa-let-7f-2*, hsa-let-7g, hsa-let-7g*, hsa-let-71, hsa-let-71*, hsa-miR-1, hsa-miR-100, hsa-miR-100*, hsa-miR-101, hsa-miR-101*, hsa-miR-103, hsa-miR-105, hsa-miR-105*, hsa-miR-106a, hsa-miR-106a*, hsa-miR-106b, hsa-miR-106b*, hsa-miR-107, hsa-miR-10a, hsa-miR-10a*, hsa-miR-10b, hsa-miR-10b*, hsa-miR-1178, hsa-miR-1179, hsa-miR-1180, hsa-miR-1181, hsa-miR-1182, hsa-miR-1183, hsa-miR-1184, hsa-miR-1185, hsa-miR-1197, hsa-miR-1200, hsa-miR-1201, hsa-miR-1202, hsa-miR-1203, hsa-miR-1204, hsa-miR-1205, hsa-miR-1206, hsa-miR-1207-3p, hsa-miR-1207-5p, hsa-miR-1208, hsa-miR-122, hsa-miR-122*, hsa-miR-1224-3p, hsa-miR-1224-5p, hsa-miR-1225-3p, hsa-miR-1225-5p, hsa-miR-1226, hsa-miR-1226*, hsa-miR-1227, hsa-miR-1228, hsa-miR-1228*, hsa-miR-1229, hsa-miR-1231, hsa-miR-1233, hsa-miR-1234, hsa-miR-1236, hsa-miR-1237, hsa-miR-1238, hsa-miR-124, hsa-miR-124*, hsa-miR-1243, hsa-miR-1244, hsa-miR-1245, hsa-miR-1246, hsa-miR-1247, hsa-miR-1248, hsa-miR-1249, hsa-miR-1250, hsa-miR-1251, hsa-miR-1252, hsa-miR-1253, hsa-miR-1254, hsa-miR-1255a, hsa-miR-1255b, hsa-miR-1256, hsa-miR-1257, hsa-miR-1258, hsa-miR-1259, hsa-miR-125a-3p, hsa-miR-125a-5p, hsa-miR-125b, hsa-miR-125b-1*, hsa-miR-125b-2*, hsa-miR-126, hsa-miR-126*, hsa-miR-1260, hsa-miR-1261, hsa-miR-1262, hsa-miR-1263, hsa-miR-1264, hsa-miR-1265, hsa-miR-1266, hsa-miR-1267, hsa-miR-1268, hsa-miR-1269, hsa-miR-1270, hsa-miR-1271, hsa-miR-1272, hsa-miR-1273, hsa-miR-127-3p, hsa-miR-1274a, hsa-miR-1274b, hsa-miR-1275, hsa-miR-127-5p, hsa-miR-1276, hsa-miR-1277, hsa-miR-1278, hsa-miR-1279, hsa-miR-128, hsa-miR-1280, hsa-miR-1281, hsa-miR-1282, hsa-miR-1283, hsa-miR-1284, hsa-miR-1285, hsa-miR-1286, hsa-miR-1287, hsa-miR-1288, hsa-miR-1289, hsa-miR-129*, hsa-miR-1290, hsa-miR-1291, hsa-miR-1292, hsa-miR-1293, hsa-miR-129-3p, hsa-miR-1294, hsa-miR-1295, hsa-miR-129-5p, hsa-miR-1296, hsa-miR-1297, hsa-miR-1298, hsa-miR-1299, hsa-miR-1300, hsa-miR-1301, hsa-miR-1302, hsa-miR-1303, hsa-miR-1304, hsa-miR-1305, hsa-miR-1306, hsa-miR-1307, hsa-miR-1308, hsa-miR-130a, hsa-miR-130a*, hsa-miR-130b, hsa-miR-130b*, hsa-miR-132, hsa-miR-132*, hsa-miR-1321, hsa-miR-1322, hsa-miR-1323, hsa-miR-1324, hsa-miR-133a, hsa-miR-133b, hsa-miR-134, hsa-miR-135a, hsa-miR-135a*, hsa-miR-135b, hsa-miR-135b*, hsa-miR-136, hsa-miR-136*, hsa-miR-137, hsa-miR-138, hsa-miR-138-1*, hsa-miR-138-2*, hsa-miR-139-3p, hsa-miR-139-5p, hsa-miR-140-3p, hsa-miR-140-5p, hsa-miR-141, hsa-miR-141*, hsa-miR-142-3p, hsa-miR-142-5p, hsa-miR-143, hsa-miR-143*, hsa-miR-144, hsa-miR-144*, hsa-miR-145, hsa-miR-145*, hsa-miR-146a, hsa-miR-146a*, hsa-miR-146b-3p, hsa-miR-146b-5p, hsa-miR-147, hsa-miR-147b, hsa-miR-148a, hsa-miR-148a*, hsa-miR-148b, hsa-miR-148b*, hsa-miR-149, hsa-miR-149*, hsa-miR-150, hsa-miR-150*, hsa-miR-151-3p, hsa-miR-151-5p, hsa-miR-152, hsa-miR-153, hsa-miR-154, hsa-miR-154*, hsa-miR-155, hsa-miR-155*, hsa-miR-15a,

hsa-miR-15a*, hsa-miR-15b, hsa-miR-15b*, hsa-miR-16, hsa-miR-16-1*, hsa-miR-16-2*, hsa-miR-17, hsa-miR-17*, hsa-miR-181a, hsa-miR-181a+, hsa-miR-181a-2*, hsa-miR-181b, hsa-miR-181c, hsa-miR-181c*, hsa-miR-181d, hsa-miR-182, hsa-miR-182*, hsa-miR-1825, hsa-miR-1826, hsa-miR-1827, hsa-miR-183, hsa-miR-183*, hsa-miR-184, hsa-miR-185, hsa-miR-185*, hsa-miR-186, hsa-miR-186*, hsa-miR-187, hsa-miR-187*, hsa-miR-188-3p, hsa-miR-188-5p, hsa-miR-18a, hsa-miR-18a*, hsa-miR-18b, hsa-miR-18b*, hsa-miR-190, hsa-miR-190b, hsa-miR-191, hsa-miR-191*, hsa-miR-192, hsa-miR-192*, hsa-miR-193a-3p, hsa-miR-193a-5p, hsa-miR-193b, hsa-miR-193b*, hsa-miR-194, hsa-miR-194*, hsa-miR-195, hsa-miR-195*, hsa-miR-196a, hsa-miR-196a*, hsa-miR-196b, hsa-miR-197, hsa-miR-198, hsa-miR-199a-3p, hsa-miR-199a-5p, hsa-miR-199b-5p, hsa-miR-19a, hsa-miR-19a*, hsa-miR-19b, hsa-miR-19b-1*, hsa-miR-19b-2*, hsa-miR-200a, hsa-miR-200a*, hsa-miR-200b, hsa-miR-200b*, hsa-miR-200c, hsa-miR-200c*, hsa-miR-202, hsa-miR-202*, hsa-miR-203, hsa-miR-204, hsa-miR-205, hsa-miR-206, hsa-miR-208a, hsa-miR-208b, hsa-miR-20a, hsa-miR-20a*, hsa-miR-20b, hsa-miR-20b*, hsa-miR-21, hsa-miR-21*, hsa-miR-210, hsa-miR-211, hsa-miR-212, hsa-miR-214, hsa-miR-214*, hsa-miR-215, hsa-miR-216a, hsa-miR-216b, hsa-miR-217, hsa-miR-218, hsa-miR-218-1*, hsa-miR-218-2*, hsa-miR-219-1-3p, hsa-miR-219-2-3p, hsa-miR-219-5p, hsa-miR-22, hsa-miR-22*, hsa-miR-220a, hsa-miR-220b, hsa-miR-220c, hsa-miR-221, hsa-miR-221*, hsa-miR-222, hsa-miR-222*, hsa-miR-223, hsa-miR-223*, hsa-miR-224, hsa-miR-23a, hsa-miR-23a*, hsa-miR-23b, hsa-miR-23b*, hsa-miR-24, hsa-miR-24-1*, hsa-miR-24-2*, hsa-miR-25, hsa-miR-25*, hsa-miR-26a, hsa-miR-26a-1*, hsa-miR-26a-2*, hsa-miR-26b, hsa-miR-26b*, hsa-miR-27a, hsa-miR-27a*, hsa-miR-27b, hsa-miR-27b*, hsa-miR-28-3p, hsa-miR-28-5p, hsa-miR-296-3p, hsa-miR-296-5p, hsa-miR-297, hsa-miR-298, hsa-miR-299-3p, hsa-miR-299-5p, hsa-miR-29a, hsa-miR-29a*, hsa-miR-29b, hsa-miR-296-1*, hsa-miR-296-2*, hsa-miR-29c, hsa-miR-29c*, hsa-miR-300, hsa-miR-301a, hsa-miR-301b, hsa-miR-302a, hsa-miR-302a*, hsa-miR-302b, hsa-miR-302b*, hsa-miR-302c, hsa-miR-302c*, hsa-miR-302d, hsa-miR-302d*, hsa-miR-302e, hsa-miR-302f, hsa-miR-30a, hsa-miR-30a*, hsa-miR-30b, hsa-miR-30b*, hsa-miR-30c, hsa-miR-30c-1*, hsa-miR-30c-2*, hsa-miR-30d, hsa-miR-30d*, hsa-miR-30e, hsa-miR-30) e*, hsa-miR-31, hsa-miR-31*, hsa-miR-32, hsa-miR-32*, hsa-miR-320a, hsa-miR-320b, hsa-miR-320c, hsa-miR-320d, hsa-miR-323-3p, hsa-miR-323-5p, hsa-miR-324-3p, hsa-miR-324-5p, hsa-miR-325, hsa-miR-326, hsa-miR-328, hsa-miR-329, hsa-miR-330-3p, hsa-miR-330-5p, hsa-miR-331-3p, hsa-miR-331-5p, hsa-miR-335, hsa-miR-335*, hsa-miR-337-3p, hsa-miR-337-5p, hsa-miR-338-3p, hsa-miR-338-5p, hsa-miR-339-3p, hsa-miR-339-5p, hsa-miR-33a, hsa-miR-33a*, hsa-miR-33b, hsa-miR-33b*, hsa-miR-340, hsa-miR-340*, hsa-miR-342-3p, hsa-miR-342-5p, hsa-miR-345, hsa-miR-346, hsa-miR-34a, hsa-miR-34a*, hsa-miR-34b, hsa-miR-34b*, hsa-miR-34c-3p, hsa-miR-34c-5p, hsa-miR-361-3p, hsa-miR-361-5p, hsa-miR-362-3p, hsa-miR-362-5p, hsa-miR-363, hsa-miR-363*, hsa-miR-365, hsa-miR-367, hsa-miR-367*, hsa-miR-369-3p, hsa-miR-369-5p, hsa-miR-370, hsa-miR-371-3p, hsa-miR-371-5p, hsa-miR-372, hsa-miR-373, hsa-miR-373*, hsa-miR-374a, hsa-miR-374a*, hsa-miR-374b, hsa-miR-374b*, hsa-miR-375, hsa-miR-376a, hsa-miR-376a*, hsa-miR-376b, hsa-miR-376c, hsa-miR-377, hsa-miR-377*, hsa-miR-378, hsa-miR-378*, hsa-miR-379, hsa-miR-379*, hsa-miR-380, hsa-miR-380*, hsa-miR-381, hsa-miR-382, hsa-miR-383, hsa-miR-384, hsa-miR-409-3p, hsa-miR-409-5p, hsa-miR-410, hsa-miR-411, hsa-

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miR-411*, hsa-miR-412, hsa-miR-421, hsa-miR-422a, hsa-miR-423-3p, hsa-miR-423-5p, hsa-miR-424, hsa-miR-424*, hsa-miR-425, hsa-miR-425*, hsa-miR-429, hsa-miR-431, hsa-miR-431*, hsa-miR-432, hsa-miR-432*, hsa-miR-433, hsa-miR-448, hsa-miR-449a, hsa-miR-449b, hsa-miR-450a, hsa-miR-450b-3p, hsa-miR-450b-5p, hsa-miR-451, hsa-miR-452, hsa-miR-452*, hsa-miR-453, hsa-miR-454, hsa-miR-454*, hsa-miR-455-3p, hsa-miR-455-5p, hsa-miR-483-3p, hsa-miR-483-5p, hsa-miR-484, hsa-miR-485-3p, hsa-miR-485-5p, hsa-miR-486-3p, hsa-miR-486-5p, hsa-miR-487a, hsa-miR-487b, hsa-miR-488, hsa-miR-488*, hsa-miR-489, hsa-miR-490-3p, hsa-miR-490-5p, hsa-miR-491-3p, hsa-miR-491-5p, hsa-miR-492, hsa-miR-493, hsa-miR-493*, hsa-miR-494, hsa-miR-495, hsa-miR-496, hsa-miR-497, hsa-miR-497*, hsa-miR-498, hsa-miR-499-3p, hsa-miR-499-5p, hsa-miR-500, hsa-miR-500*, hsa-miR-501-3p, hsa-miR-501-5p, hsa-miR-502-3p, hsa-miR-502-5p, hsa-miR-503, hsa-miR-504, hsa-miR-505, hsa-miR-505*, hsa-miR-506, hsa-miR-507, hsa-miR-508-3p, hsa-miR-508-5p, hsa-miR-509-3-5p, hsa-miR-509-3p, hsa-miR-509-5p, hsa-miR-510, hsa-miR-511, hsa-miR-512-3p, hsa-miR-512-5p, hsa-miR-513a-3p, hsa-miR-513a-5p, hsa-miR-513b, hsa-miR-513c, hsa-miR-514, hsa-miR-515-3p, hsa-miR-515-5p, hsa-miR-516a-3p, hsa-miR-516a-5p, hsa-miR-516b, hsa-miR-517*, hsa-miR-517a, hsa-miR-517b, hsa-miR-517c, hsa-miR-518a-3p, hsa-miR-518a-5p, hsa-miR-518b, hsa-miR-518c, hsa-miR-518c*, hsa-miR-518d-3p, hsa-miR-518d-5p, hsa-miR-518e, hsa-miR-518e*, hsa-miR-518f, hsa-miR-518f*, hsa-miR-519a, hsa-miR-519b-3p, hsa-miR-519c-3p, hsa-miR-519d, hsa-miR-519e, hsa-miR-519e*, hsa-miR-520a-3p, hsa-miR-520a-5p, hsa-miR-520b, hsa-miR-520c-3p, hsa-miR-520d-3p, hsa-miR-520d-5p, hsa-miR-520e, hsa-miR-520f, hsa-miR-520g, hsa-miR-520h, hsa-miR-521, hsa-miR-522, hsa-miR-523, hsa-miR-524-3p, hsa-miR-524-5p, hsa-miR-525-3p, hsa-miR-525-5p, hsa-miR-526b, hsa-miR-526b*, hsa-miR-532-3p, hsa-miR-532-5p, hsa-miR-539, hsa-miR-541, hsa-miR-541*, hsa-miR-542-3p, hsa-miR-542-5p, hsa-miR-543, hsa-miR-544, hsa-miR-545, hsa-miR-545*, hsa-miR-548a-3p, hsa-miR-548a-5p, hsa-miR-548b-3p, hsa-miR-548b-5p, hsa-miR-548c-3p, hsa-miR-548c-5p, hsa-miR-548d-3p, hsa-miR-548d-5p, hsa-miR-548e, hsa-miR-548f, hsa-miR-548g, hsa-miR-548h, hsa-miR-548i, hsa-miR-548j, hsa-miR-548k, hsa-miR-548l, hsa-miR-548m, hsa-miR-548n, hsa-miR-548o, hsa-miR-548p, hsa-miR-549, hsa-miR-550, hsa-miR-550*, hsa-miR-551a, hsa-miR-551b, hsa-miR-551b*, hsa-miR-552, hsa-miR-553, hsa-miR-554, hsa-miR-555, hsa-miR-556-3p, hsa-miR-556-5p, hsa-miR-557, hsa-miR-558, hsa-miR-559, hsa-miR-561, hsa-miR-562, hsa-miR-563, hsa-miR-564, hsa-miR-566, hsa-miR-567, hsa-miR-568, hsa-miR-569, hsa-miR-570, hsa-miR-571, hsa-miR-572, hsa-miR-573, hsa-miR-574-3p, hsa-miR-574-5p, hsa-miR-575, hsa-miR-576-3p, hsa-miR-576-5p, hsa-miR-577, hsa-miR-578, hsa-miR-579, hsa-miR-580, hsa-miR-581, hsa-miR-582-3p, hsa-miR-582-5p, hsa-miR-583, hsa-miR-584, hsa-miR-585, hsa-miR-586, hsa-miR-587, hsa-miR-588, hsa-miR-589, hsa-miR-589*, hsa-miR-590-3p, hsa-miR-590-5p, hsa-miR-591, hsa-miR-592, hsa-miR-593, hsa-miR-593*, hsa-miR-595, hsa-miR-596, hsa-miR-597, hsa-miR-598, hsa-miR-599, hsa-miR-600, hsa-miR-601, hsa-miR-602, hsa-miR-603, hsa-miR-604, hsa-miR-605, hsa-miR-606, hsa-miR-607, hsa-miR-608, hsa-miR-609, hsa-miR-610, hsa-miR-611, hsa-miR-612, hsa-miR-613, hsa-miR-614, hsa-miR-615-3p, hsa-miR-615-5p, hsa-miR-616, hsa-miR-616*, hsa-miR-617, hsa-miR-618, hsa-miR-619, hsa-miR-620, hsa-miR-621, hsa-miR-622, hsa-miR-623, hsa-miR-624, hsa-miR-624*, hsa-miR-625, hsa-miR-

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625*, hsa-miR-626, hsa-miR-627, hsa-miR-628-3p, hsa-miR-628-5p, hsa-miR-629, hsa-miR-629*, hsa-miR-630, hsa-miR-631, hsa-miR-632, hsa-miR-633, hsa-miR-634, hsa-miR-635, hsa-miR-636, hsa-miR-637, hsa-miR-638, hsa-miR-639, hsa-miR-640, hsa-miR-641, hsa-miR-642, hsa-miR-643, hsa-miR-644, hsa-miR-645, hsa-miR-646, hsa-miR-647, hsa-miR-648, hsa-miR-649, hsa-miR-650, hsa-miR-651, hsa-miR-652, hsa-miR-653, hsa-miR-654-3p, hsa-miR-654-5p, hsa-miR-655, hsa-miR-656, hsa-miR-657, hsa-miR-658, hsa-miR-659, hsa-miR-660, hsa-miR-661, hsa-miR-662, hsa-miR-663, hsa-miR-663b, hsa-miR-664, hsa-miR-664*, hsa-miR-665, hsa-miR-668, hsa-miR-671-3p, hsa-miR-671-5p, hsa-miR-675, hsa-miR-7, hsa-miR-708, hsa-miR-708*, hsa-miR-7-1*, hsa-miR-7-2*, hsa-miR-720, hsa-miR-744, hsa-miR-744*, hsa-miR-758, hsa-miR-760, hsa-miR-765, hsa-miR-766, hsa-miR-767-3p, hsa-miR-767-5p, hsa-miR-768-3p, hsa-miR-768-5p, hsa-miR-769-3p, hsa-miR-769-5p, hsa-miR-770-5p, hsa-miR-802, hsa-miR-873, hsa-miR-874, hsa-miR-875-3p, hsa-miR-875-5p, hsa-miR-876-3p, hsa-miR-876-5p, hsa-miR-877, hsa-miR-877*, hsa-miR-885-3p, hsa-miR-885-5p, hsa-miR-886-3p, hsa-miR-886-5p, hsa-miR-887, hsa-miR-888, hsa-miR-888*, hsa-miR-889, hsa-miR-890, hsa-miR-891a, hsa-miR-891b, hsa-miR-892a, hsa-miR-892b, hsa-miR-9, hsa-miR-9*, hsa-miR-920, hsa-miR-921, hsa-miR-922, hsa-miR-923, hsa-miR-924, hsa-miR-92a, hsa-miR-92a-1*, hsa-miR-92a-2*, hsa-miR-92b, hsa-miR-92b*, hsa-miR-93, hsa-miR-93*, hsa-miR-933, hsa-miR-934, hsa-miR-935, hsa-miR-936, hsa-miR-937, hsa-miR-938, hsa-miR-939, hsa-miR-940, hsa-miR-941, hsa-miR-942, hsa-miR-943, hsa-miR-944, hsa-miR-95, hsa-miR-96, hsa-miR-96*, hsa-miR-98, hsa-miR-99a, hsa-miR-99a*, hsa-miR-99b, and hsa-miR-99b*. For example, miRNA targeting chromosome 8 open reading frame 72 (C9orf72) which expresses superoxide dismutase (SOD1), associated with amyotrophic lateral sclerosis (ALS) may be of interest.

A miRNA inhibits the function of the mRNAs it targets and, as a result, inhibits expression of the polypeptides encoded by the mRNAs. Thus, blocking (partially or totally) the activity of the miRNA (e.g., silencing the miRNA) can effectively induce, or restore, expression of a polypeptide whose expression is inhibited (derepress the polypeptide). In one embodiment, derepression of polypeptides encoded by mRNA targets of a miRNA is accomplished by inhibiting the miRNA activity in cells through any one of a variety of methods. For example, blocking the activity of a miRNA can be accomplished by hybridization with a small interfering nucleic acid (e.g., antisense oligonucleotide, miRNA sponge, TuD RNA) that is complementary, or substantially complementary to, the miRNA, thereby blocking interaction of the miRNA with its target mRNA. As used herein, a small interfering nucleic acid that is substantially complementary to a miRNA is one that is capable of hybridizing with a miRNA, and blocking the miRNA's activity. In some embodiments, a small interfering nucleic acid that is substantially complementary to a miRNA is a small interfering nucleic acid that is complementary with the miRNA at all but 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, or 18 bases. A "miRNA Inhibitor" is an agent that blocks miRNA function, expression and/or processing. For instance, these molecules include but are not limited to microRNA specific antisense, microRNA sponges, tough decoy RNAs (TuD RNAs) and microRNA oligonucleotides (double-stranded, hairpin, short oligonucleotides) that inhibit miRNA interaction with a Drosha complex.

Still other useful transgenes may include those encoding immunoglobulins which confer passive immunity to a

pathogen. An “immunoglobulin molecule” is a protein containing the immunologically-active portions of an immunoglobulin heavy chain and immunoglobulin light chain covalently coupled together and capable of specifically combining with antigen. Immunoglobulin molecules are of any type (e.g., IgG, IgE, IgM, IgD, IgA and IgY), class (e.g., IgG1, IgG2, IgG3, IgG4, IgA1 and IgA2) or subclass. The terms “antibody” and “immunoglobulin” may be used interchangeably herein.

An “immunoglobulin heavy chain” is a polypeptide that contains at least a portion of the antigen binding domain of an immunoglobulin and at least a portion of a variable region of an immunoglobulin heavy chain or at least a portion of a constant region of an immunoglobulin heavy chain. Thus, the immunoglobulin derived heavy chain has significant regions of amino acid sequence homology with a member of the immunoglobulin gene superfamily. For example, the heavy chain in a Fab fragment is an immunoglobulin-derived heavy chain.

An “immunoglobulin light chain” is a polypeptide that contains at least a portion of the antigen binding domain of an immunoglobulin and at least a portion of the variable region or at least a portion of a constant region of an immunoglobulin light chain. Thus, the immunoglobulin-derived light chain has significant regions of amino acid homology with a member of the immunoglobulin gene superfamily.

An “immunoadhesin” is a chimeric, antibody-like molecule that combines the functional domain of a binding protein, usually a receptor, ligand, or cell-adhesion molecule, with immunoglobulin constant domains, usually including the hinge and Fc regions.

A “fragment antigen-binding” (Fab) fragment” is a region on an antibody that binds to antigens. It is composed of one constant and one variable domain of each of the heavy and the light chain.

The anti-pathogen construct is selected based on the causative agent (pathogen) for the disease against which protection is sought. These pathogens may be of viral, bacterial, or fungal origin, and may be used to prevent infection in humans against human disease, or in non-human mammals or other animals to prevent veterinary disease.

The rAAV may include genes encoding antibodies, and particularly neutralizing antibodies against a viral pathogen. Such anti-viral antibodies may include anti-influenza antibodies directed against one or more of Influenza A, Influenza B, and Influenza C. The type A viruses are the most virulent human pathogens. The serotypes of influenza A which have been associated with pandemics include, H1N1, which caused Spanish Flu in 1918, and Swine Flu in 2009; H2N2, which caused Asian Flu in 1957; H3N2, which caused Hong Kong Flu in 1968; H5N1, which caused Bird Flu in 2004; H7N7; H1N2; H9N2; H7N2; H7N3; and H10N7. Other target pathogenic viruses include, arenaviruses (including funin, machupo, and Lassa), filoviruses (including Marburg and Ebola), hantaviruses, picornoviridae (including rhinoviruses, echovirus), coronaviruses, paramyxovirus, morbillivirus, respiratory syncytial virus, togavirus, coxsackievirus, JC virus, parvovirus B19, parainfluenza, adenoviruses, reoviruses, variola (Variola major (Smallpox)) and Vaccinia (Cowpox) from the poxvirus family, and varicella-zoster (pseudorabies). Viral hemorrhagic fevers are caused by members of the arenavirus family (Lassa fever) (which family is also associated with Lymphocytic choriomeningitis (LCM)), filovirus (ebola virus), and hantavirus (pseudomala). The members of picornavirus (a subfamily of rhinoviruses), are associated with the common cold in humans.

The coronavirus family, which includes a number of non-human viruses such as infectious bronchitis virus (poultry), porcine transmissible gastroenteric virus (pig), porcine hemagglutinin encephalomyelitis virus (pig), feline infectious peritonitis virus (cat), feline enteric coronavirus (cat), canine coronavirus (dog). The human respiratory coronaviruses have been putatively associated with the common cold, non-A, B or C hepatitis, and sudden acute respiratory syndrome (SARS). The paramyxovirus family includes parainfluenza Virus Type 1, parainfluenza Virus Type 3, bovine parainfluenza Virus Type 3, rubulavirus (mumps virus, parainfluenza Virus Type 2, parainfluenza virus Type 4, Newcastle disease virus (chickens), rinderpest, morbillivirus, which includes measles and canine distemper, and pneumovirus, which includes respiratory syncytial virus (RSV). The parvovirus family includes feline parvovirus (feline enteritis), feline panleukopenia virus, canine parvovirus, and porcine parvovirus. The adenovirus family includes viruses (EX, AD7, ARD, O.B.) which cause respiratory disease. Thus, in certain embodiments, a rAAV vector as described herein may be engineered to express an anti-ebola antibody, e.g., 2G4, 4G7, 13C6, an anti-influenza antibody, e.g., FI6, CR8033, and anti-RSV antibody, e.g., palivizumab, motavizumab. A neutralizing antibody construct against a bacterial pathogen may also be selected for use in the present invention. In one embodiment, the neutralizing antibody construct is directed against the bacteria itself. In another embodiment, the neutralizing antibody construct is directed against a toxin produced by the bacteria. Examples of airborne bacterial pathogens include, e.g., *Neisseria meningitidis* (meningitis), *Klebsiella pneumonia* (pneumonia), *Pseudomonas aeruginosa* (pneumonia), *Pseudomonas pseudomallei* (pneumonia), *Pseudomonas mallei* (pneumonia), *Acinetobacter* (pneumonia), *Moraxella catarrhalis*, *Moraxella lacunata*, *Alkaligenes*, *Cardiobacterium*, *Haemophilus influenzae* (flu), *Haemophilus parainfluenzae*, *Bordetella pertussis* (whooping cough), *Francisella tularensis* (pneumonia/fever), *Legionella pneumonia* (Legionnaires disease), *Chlamydia psittaci* (pneumonia), *Chlamydia pneumoniae* (pneumonia), *Mycobacterium tuberculosis* (tuberculosis (TB)), *Mycobacterium kansasii* (TB), *Mycobacterium avium* (pneumonia), *Nocardia asteroides* (pneumonia), *Bacillus anthracis* (anthrax), *Staphylococcus aureus* (pneumonia), *Streptococcus pyogenes* (scarlet fever), *Streptococcus pneumoniae* (pneumonia), *Corynebacteria diphtheria* (diphtheria), *Mycoplasma pneumoniae* (pneumonia).

The rAAV may include genes encoding antibodies, and particularly neutralizing antibodies against a bacterial pathogen such as the causative agent of anthrax, a toxin produced by *Bacillus anthracis*. Neutralizing antibodies against protective agent (PA), one of the three peptides which form the toxoid, have been described. The other two polypeptides consist of lethal factor (LF) and edema factor (EF). Anti-PA neutralizing antibodies have been described as being effective in passively immunization against anthrax. See, e.g., U.S. Pat. No. 7,442,373; R. Sawada-Hirai et al, J Immune Based Ther Vaccines. 2004; 2:5. (on-line 2004 May 12). Still other anti-anthrax toxin neutralizing antibodies have been described and/or may be generated. Similarly, neutralizing antibodies against other bacteria and/or bacterial toxins may be used to generate an AAV-delivered anti-pathogen construct as described herein.

Antibodies against infectious diseases may be caused by parasites or by fungi, including, e.g., *Aspergillus* species, *Absidia corymbifera*, *Rhizopus stolonifer*, *Mucor plumbeus*, *Cryptococcus neoformans*, *Histoplasma capsulatum*, *Blasto-*

myces dermatitidis, *Coccidioides immitis*, *Penicillium* species, *Micropolyspora faeni*, *Thermoactinomyces vulgaris*, *Alternaria alternate*, *Cladosporium species*, *Helminthosporium*, and *Stachybotrys* species.

The rAAV may include genes encoding antibodies, and particularly neutralizing antibodies, against pathogenic factors of diseases such as Alzheimer's disease (AD), Parkinson's disease (PD), GBA-associated-Parkinson's disease (GBA-PD), Rheumatoid arthritis (RA), Irritable bowel syndrome (IBS), chronic obstructive pulmonary disease (COPD), cancers, tumors, systemic sclerosis, asthma and other diseases. Such antibodies may be, without limitation, e.g., alpha-synuclein, anti-vascular endothelial growth factor (VEGF) (anti-VEGF), anti-VEGFA, anti-PD-1, anti-PDL1, anti-CTLA-4, anti-TNF-alpha, anti-IL-17, anti-IL-23, anti-IL-21, anti-IL-6, anti-IL-6 receptor, anti-IL-5, anti-IL-7, anti-Factor XII, anti-IL-2, anti-HIV, anti-IgE, anti-tumour necrosis factor receptor-1 (TNFR1), anti-notch 2/3, anti-notch 1, anti-OX40, anti-erb-b2 receptor tyrosine kinase 3 (ErbB3), anti-ErbB2, anti-beta cell maturation antigen, anti-B lymphocyte stimulator, anti-CD20, anti-HER2, anti-granulocyte macrophage colony-stimulating factor, anti-oncostatin M (OSM), anti-lymphocyte activation gene 3 (LAG3) protein, anti-CCL20, anti-serum amyloid P component (SAP), anti-prolyl hydroxylase inhibitor, anti-CD38, anti-glycoprotein IIb/IIIa, anti-CD52, anti-CD30, anti-IL-1beta, anti-epidermal growth factor receptor, anti-CD25, anti-RANK ligand, anti-complement system protein C5, anti-CD11a, anti-CD3 receptor, anti-alpha-4 (α_4) integrin, anti-RSV F protein, and anti-integrin $\alpha_4 \beta_7$. Still other pathogens and diseases will be apparent to one of skill in the art. Other suitable antibodies may include those useful for treating Alzheimer's Disease, such as, e.g., anti-beta-amyloid (e.g., crenezumab, solanezumab, aducanumab), anti-beta-amyloid fibril, anti-beta-amyloid plaques, anti-tau, a bapineuzumab, among others. Other suitable antibodies for treating a variety of indications include those described, e.g., in PCT/US2016/058968, filed 27 Oct. 2016, published as WO 2017/075119A1.

II. rAAV Vector Production

For use in producing an AAV viral vector (e.g., a recombinant (r) AAV), the expression cassettes can be carried on any suitable vector, e.g., a plasmid, which is delivered to a packaging host cell. The plasmids useful in this invention may be engineered such that they are suitable for replication and packaging in vitro in prokaryotic cells, insect cells, mammalian cells, among others. Suitable transfection techniques and packaging host cells are known and/or can be readily designed by one of skill in the art.

Methods for generating and isolating AAVs suitable for use as vectors are known in the art. See generally, e.g., Grieger & Samulski, 2005, "Adeno-associated virus as a gene therapy vector: Vector development, production and clinical applications," *Adv. Biochem. Engin/Biotechnol.* 99:119-145; Buning et al., 2008, "Recent developments in adeno-associated virus vector technology," *J. Gene Med.* 10:717-733; and the references cited below, each of which is incorporated herein by reference in its entirety. For packaging a transgene into virions, the ITRs are the only AAV components required in cis in the same construct as the nucleic acid molecule containing the expression cassettes. The cap and rep genes can be supplied in trans.

In one embodiment, the expression cassettes described herein are engineered into a genetic element (e.g., a shuttle plasmid) which transfers the transgene construct sequences carried thereon into a packaging host cell for production of a viral vector. In one embodiment, the selected genetic ele-

ment may be delivered to an AAV packaging cell by any suitable method, including transfection, electroporation, liposome delivery, membrane fusion techniques, high velocity DNA-coated pellets, viral infection and protoplast fusion.

Stable AAV packaging cells can also be made. Alternatively, the expression cassettes may be used to generate a viral vector other than AAV. The methods used to make such constructs are known to those with skill in nucleic acid manipulation and include genetic engineering, recombinant engineering, and synthetic techniques. See, e.g., *Molecular Cloning: A Laboratory Manual*, cd. Green and Sambrook, Cold Spring Harbor Press, Cold Spring Harbor, NY (2012).

The term "AAV intermediate" or "AAV vector intermediate" refers to an assembled rAAV capsid which lacks the desired genomic sequences packaged therein. These may also be termed an "empty" capsid. Such a capsid may contain no detectable genomic sequences of an expression cassette, or only partially packaged genomic sequences which are insufficient to achieve expression of the gene product. These empty capsids are non-functional to transfer the gene of interest to a host cell.

The recombinant adeno-associated virus (AAV) described herein may be generated using techniques which are known. See, e.g., WO 2003/042397; WO 2005/033321, WO 2006/110689; U.S. Pat. No. 7,588,772 B2. Such a method involves culturing a host cell which contains a nucleic acid sequence encoding an AAV capsid protein; a functional rep gene; an expression cassette composed of, at a minimum, AAV inverted terminal repeats (ITRs) and a transgene; and sufficient helper functions to permit packaging of the expression cassette into the AAV capsid protein. Methods of generating the capsid, coding sequences therefor, and methods for production of rAAV viral vectors have been described. See, e.g., Gao, et al, *Proc. Natl. Acad. Sci. U.S.A.* 100 (10), 6081-6086 (2003) and US 2013/0045186A1.

In one embodiment, a production cell culture useful for producing a recombinant AAV is provided. Such a cell culture contains a nucleic acid which expresses the AAV capsid protein in the host cell; a nucleic acid molecule suitable for packaging into the AAV capsid, e.g., a vector genome which contains AAV ITRs and a non-AAV nucleic acid sequence encoding a gene product operably linked to sequences which direct expression of the product in a host cell; and sufficient AAV rep functions and adenovirus helper functions to permit packaging of the nucleic acid molecule into the recombinant AAV capsid. In one embodiment, the cell culture is composed of mammalian cells (e.g., human embryonic kidney 293 cells, among others) or insect cells (e.g., baculovirus).

Optionally the rep functions are provided by an AAV other than the AAV providing the capsid. For example the rep may be, but is not limited to, AAV1 rep protein, AAV2 rep protein, AAV3 rep protein, AAV4 rep protein, AAV5 rep protein, AAV6 rep protein, AAV7 rep protein, AAV8 rep protein; or rep 78, rep 68, rep 52, rep 40, rep68/78 and rep40/52; or a fragment thereof; or another source. Optionally, the rep and cap sequences are on the same genetic element in the cell culture. There may be a spacer between the rep sequence and cap gene. Any of these AAV or mutant AAV capsid sequences may be under the control of exogenous regulatory control sequences which direct expression thereof in a host cell.

In one embodiment, cells are manufactured in a suitable cell culture (e.g., HEK 293) cells. Methods for manufacturing the gene therapy vectors described herein include methods well known in the art such as generation of plasmid DNA used for production of the gene therapy vectors,

generation of the vectors, and purification of the vectors. In some embodiments, the gene therapy vector is an AAV vector and the plasmids generated are an AAV cis-plasmid encoding the vector genome including the gene of interest, an AAV trans-plasmid containing AAV rep and cap genes, and an adenovirus helper plasmid. The vector generation process can include method steps such as initiation of cell culture, passage of cells, seeding of cells, transfection of cells with the plasmid DNA, post-transfection medium exchange to serum free medium, and the harvest of vector-containing cells and culture media. The harvested vector-containing cells and culture media are referred to herein as crude cell harvest. In yet another system, the gene therapy vectors are introduced into insect cells by infection with baculovirus-based vectors. For reviews on these production systems, see generally, e.g., Zhang et al., 2009, "Adenovirus-adenovirus-associated virus hybrid for large-scale recombinant adeno-associated virus production," Human Gene Therapy 20:922-929, the contents of each of which is incorporated herein by reference in its entirety. Methods of making and using these and other AAV production systems are also described in the following U.S. patents, the contents of each of which is incorporated herein by reference in its entirety: U.S. Pat. Nos. 5,139,941; 5,741,683; 6,057,152; 6,204,059; 6,268,213; 6,491,907; 6,660,514; 6,951,753; 7,094,604; 7,172,893; 7,201,898; 7,229,823; and 7,439,065.

The crude cell harvest may thereafter be subject method steps such as concentration of the vector harvest, diafiltration of the vector harvest, microfluidization of the vector harvest, nuclease digestion of the vector harvest, filtration of microfluidized intermediate, crude purification by chromatography, crude purification by ultracentrifugation, buffer exchange by tangential flow filtration, and/or formulation and filtration to prepare bulk vector.

A two-step affinity chromatography purification at high salt concentration followed anion exchange resin chromatography are used to purify the vector drug product and to remove empty capsids. These methods are described in more detail in International Patent Publication No. WO 2017/160360, which is incorporated by reference herein. Purification methods for AAV8 Publication No. WO2017/100676, and rh10, International Patent Publication No. WO 2017/100704 and for AAV1, International Patent Publication No. WO 2017/100674, are all incorporated by reference herein.

To calculate empty and full particle content, VP3 band volumes for a selected sample (e.g., in examples herein an iodixanol gradient-purified preparation where # of GC=# of particles) are plotted against GC particles loaded. The resulting linear equation ($y=mx+c$) is used to calculate the number of particles in the band volumes of the test article peaks. The number of particles (pt) per 20 μ L loaded is then multiplied by 50 to give particles (pt)/mL. Pt/mL divided by GC/mL gives the ratio of particles to genome copies (pt/GC). Pt/mL-GC/mL gives empty pt/mL. Empty pt/mL divided by pt/mL and $\times 100$ gives the percentage of empty particles.

Generally, methods for assaying for empty capsids and AAV vector particles with packaged genomes have been known in the art. See, e.g., Grimm et al., *Gene Therapy* (1999) 6:1322-1330; Sommer et al., *Molec. Ther.* (2003) 7:122-128. To test for denatured capsid, the methods include subjecting the treated AAV stock to SDS-polyacrylamide gel electrophoresis, consisting of any gel capable of separating the three capsid proteins, for example, a gradient gel containing 3-8% Tris-acetate in the buffer, then running the gel until sample material is separated, and blotting the gel onto nylon or nitrocellulose membranes, preferably nylon. Anti-AAV capsid antibodies are then used as the primary anti-

bodies that bind to denatured capsid proteins, preferably an anti-AAV capsid monoclonal antibody, most preferably the B1 anti-AAV-2 monoclonal antibody (Wobus et al., *J. Virol.* (2000) 74:9281-9293). A secondary antibody is then used, one that binds to the primary antibody and contains a means for detecting binding with the primary antibody, more preferably an anti-IgG antibody containing a detection molecule covalently bound to it, most preferably a sheep anti-mouse IgG antibody covalently linked to horseradish peroxidase. A method for detecting binding is used to semi-quantitatively determine binding between the primary and secondary antibodies, preferably a detection method capable of detecting radioactive isotope emissions, electromagnetic radiation, or colorimetric changes, most preferably a chemiluminescence detection kit. For example, for SDS-PAGE, samples from column fractions can be taken and heated in SDS-PAGE loading buffer containing reducing agent (e.g., DTT), and capsid proteins were resolved on pre-cast gradient polyacrylamide gels (e.g., Novex). Silver staining may be performed using SilverXpress (Invitrogen, CA) according to the manufacturer's instructions or other suitable staining method, i.e. SYPRO ruby or coomassie stains. In one embodiment, the concentration of AAV vector genomes (vg) in column fractions can be measured by quantitative real time PCR (Q-PCR). Samples are diluted and digested with DNase I (or another suitable nuclease) to remove exogenous DNA. After inactivation of the nuclease, the samples are further diluted and amplified using primers and a TaqManTM fluorogenic probe specific for the DNA sequence between the primers. The number of cycles required to reach a defined level of fluorescence (threshold cycle, Ct) is measured for each sample on an Applied Biosystems Prism 7700 Sequence Detection System. Plasmid DNA containing identical sequences to that contained in the AAV vector is employed to generate a standard curve in the Q-PCR reaction. The cycle threshold (Ct) values obtained from the samples are used to determine vector genome titer by normalizing it to the Ct value of the plasmid standard curve. End-point assays based on the digital PCR can also be used.

In one aspect, an optimized q-PCR method is used which utilizes a broad spectrum serine protease, e.g., proteinase K (such as is commercially available from Qiagen). More particularly, the optimized qPCR genome titer assay is similar to a standard assay, except that after the DNase I digestion, samples are diluted with proteinase K buffer and treated with proteinase K followed by heat inactivation. Suitably samples are diluted with proteinase K buffer in an amount equal to the sample size. The proteinase K buffer may be concentrated to 2 fold or higher. Typically, proteinase K treatment is about 0.2 mg/mL, but may be varied from 0.1 mg/mL to about 1 mg/mL. The treatment step is generally conducted at about 55° C. for about 15 minutes, but may be performed at a lower temperature (e.g., about 37° C. to about 50° C.) over a longer time period (e.g., about 20 minutes to about 30 minutes), or a higher temperature (e.g., up to about 60° C.) for a shorter time period (e.g., about 5 to 10 minutes). Similarly, heat inactivation is generally at about 95° C. for about 15 minutes, but the temperature may be lowered (e.g., about 70 to about 90° C.) and the time extended (e.g., about 20 minutes to about 30 minutes). Samples are then diluted (e.g., 1000 fold) and subjected to TaqMan analysis as described in the standard assay.

Additionally, or alternatively, droplet digital PCR (ddPCR) may be used. For example, methods for determining single-stranded and self-complementary AAV vector genome titers by ddPCR have been described. See, e.g., M.

Lock et al, Hum Gene Ther Methods. 2014 April; 25 (2): 115-25. doi: 10.1089/hgtb.2013.131. Epub 2014 Feb. 14.

In brief, the method for separating rAAV particles having packaged genomic sequences from genome-deficient AAV intermediates involves subjecting a suspension comprising recombinant AAV viral particles and AAV capsid intermediates to fast performance liquid chromatography, wherein the AAV viral particles and AAV intermediates are bound to a strong anion exchange resin equilibrated at a high pH, and subjected to a salt gradient while monitoring eluate for ultraviolet absorbance at about 260 and about 280. The pH may be adjusted depending upon the AAV selected. See, e.g., WO2017/160360 (AAV9), WO2017/100704 (AAVrh10), WO 2017/100676 (e.g., AAV8), and WO 2017/100674 (AAV1)] which are incorporated by reference herein. In this method, the AAV full capsids are collected from a fraction which is eluted when the ratio of A260/A280 reaches an inflection point. In one example, for the Affinity Chromatography step, the diafiltered product may be applied to a Capture Select™ Poros-AAV2/9 affinity resin (Life Technologies) that efficiently captures the AAV2 serotype. Under these ionic conditions, a significant percentage of residual cellular DNA and proteins flow through the column, while AAV particles are efficiently captured.

III. Compositions and Uses

Provided herein are compositions containing at least one rAAV stock (e.g., an rAAV stock or a mutant rAAV stock) and an optional carrier, excipient and/or preservative. An rAAV stock refers to a plurality of rAAV vectors which are the same, e.g., such as in the amounts described below in the discussion of concentrations and dosage units.

As used herein, "carrier" includes any and all solvents, dispersion media, vehicles, coatings, diluents, antibacterial and antifungal agents, isotonic and absorption delaying agents, buffers, carrier solutions, suspensions, colloids, and the like. The use of such media and agents for pharmaceutical active substances is well known in the art. Supplementary active ingredients can also be incorporated into the compositions. The phrase "pharmaceutically-acceptable" refers to molecular entities and compositions that do not produce an allergic or similar untoward reaction when administered to a host. Delivery vehicles such as liposomes, nanocapsules, microparticles, microspheres, lipid particles, vesicles, and the like, may be used for the introduction of the compositions of the present invention into suitable host cells. In particular, the rAAV vector delivered transgenes may be formulated for delivery either encapsulated in a lipid particle, a liposome, a vesicle, a nanosphere, or a nanoparticle or the like.

In one embodiment, a composition includes a final formulation suitable for delivery to a subject, e.g., is an aqueous liquid suspension buffered to a physiologically compatible pH and salt concentration. Optionally, one or more surfactants are present in the formulation. In another embodiment, the composition may be transported as a concentrate which is diluted for administration to a subject. In other embodiments, the composition may be lyophilized and reconstituted at the time of administration.

A suitable surfactant, or combination of surfactants, may be selected from among non-ionic surfactants that are non-toxic. In one embodiment, a difunctional block copolymer surfactant terminating in primary hydroxyl groups is selected, e.g., such as Pluronic® F68 [BASF], also known as Poloxamer 188, which has a neutral pH, has an average molecular weight of 8400. Other surfactants and other Poloxamers may be selected, i.e., nonionic triblock copolymers composed of a central hydrophobic chain of polyoxy-

propylene (poly(propylene oxide)) flanked by two hydrophilic chains of polyoxyethylene (poly(ethylene oxide)), SOLUTOL HS 15 (Macrogol-15 Hydroxystearate), LABRASOL (Polyoxy caprylic glyceride), polyoxy 10 oleyl ether, TWEEN (polyoxyethylene sorbitan fatty acid esters), ethanol and polyethylene glycol. In one embodiment, the formulation contains a poloxamer. These copolymers are commonly named with the letter "P" (for poloxamer) followed by three digits: the first two digits×100 give the approximate molecular mass of the polyoxypropylene core, and the last digit×10 gives the percentage polyoxyethylene content. In one embodiment Poloxamer 188 is selected. The surfactant may be present in an amount up to about 0.0005% to about 0.001% of the suspension.

The vectors are administered in sufficient amounts to transfect the cells and to provide sufficient levels of gene transfer and expression to provide a therapeutic benefit without undue adverse effects, or with medically acceptable physiological effects, which can be determined by those skilled in the medical arts. Conventional and pharmaceutically acceptable routes of administration include, but are not limited to, direct delivery to a desired organ (e.g., the liver (optionally via the hepatic artery), lung, heart, eye, kidney), oral, inhalation, intranasal, intrathecal, intratracheal, intraarterial, direct delivery to the eye (optionally via ocular delivery, subretinal injection, intra-retinal injection, intravitreal, topical), intravenous, intramuscular, subcutaneous, intradermal, and other parental routes of administration. In one embodiment, the route of administration is subretinal or intravitreal injection. Routes of administration may be combined, if desired.

Dosages of the viral vector will depend primarily on factors such as the condition being treated, the age, weight and health of the patient, and may thus vary among patients. For example, a therapeutically effective human dosage of the viral vector is generally in the range of from about 25 to about 1000 microliters to about 100 mL of solution containing concentrations of from about 1×10^9 to 1×10^{16} genomes virus vector. The dosage will be adjusted to balance the therapeutic benefit against any side effects and such dosages may vary depending upon the therapeutic application for which the recombinant vector is employed. The levels of expression of the transgene can be monitored to determine the frequency of dosage resulting in viral vectors, preferably AAV vectors containing the minigene. Optionally, dosage regimens similar to those described for therapeutic purposes may be utilized for immunization using the compositions of the invention.

The replication-defective virus compositions can be formulated in dosage units to contain an amount of replication-defective virus that is in the range of about 1.0×10^9 GC to about 1.0×10^{16} GC (to treat an average subject of 70 kg in body weight) including all integers or fractional amounts within the range, and preferably 1.0×10^{12} GC to 1.0×10^{14} GC for a human patient. In one embodiment, the compositions are formulated to contain at least 1×10^9 , 2×10^9 , 3×10^9 , 4×10^9 , 5×10^9 , 6×10^9 , 7×10^9 , 8×10^9 , or 9×10^9 GC per dose including all integers or fractional amounts within the range. In another embodiment, the compositions are formulated to contain at least 1×10^{10} , 2×10^{10} , 3×10^{10} , 4×10^{10} , 5×10^{10} , 6×10^{10} , 7×10^{10} , 8×10^{10} , or 9×10^{10} GC per dose including all integers or fractional amounts within the range. In another embodiment, the compositions are formulated to contain at least 1×10^{11} , 2×10^{11} , 3×10^{11} , 4×10^{11} , 5×10^{11} , 6×10^{11} , 7×10^{11} , 8×10^{11} , or 9×10^{11} GC per dose including all integers or fractional amounts within the range. In another embodiment, the compositions are formulated to contain at least

1×10^{12} , 2×10^{12} , 3×10^{12} , 4×10^{12} , 5×10^{12} , 6×10^{12} , 7×10^{12} , 8×10^{12} , or 9×10^{12} GC per dose including all integers or fractional amounts within the range. In another embodiment, the compositions are formulated to contain at least 1×10^{13} , 2×10^{13} , 3×10^{13} , 4×10^{13} , 5×10^{13} , 6×10^{13} , 7×10^{13} , 8×10^{13} , or 9×10^{13} GC per dose including all integers or fractional amounts within the range. In another embodiment, the compositions are formulated to contain at least 1×10^{14} , 2×10^{14} , 3×10^{14} , 4×10^{14} , 5×10^{14} , 6×10^{14} , 7×10^{14} , 8×10^{14} , or 9×10^{14} GC per dose including all integers or fractional amounts within the range. In another embodiment, the compositions are formulated to contain at least 1×10^{15} , 2×10^{15} , 3×10^{15} , 4×10^{15} , 5×10^{15} , 6×10^{15} , 7×10^{15} , 8×10^{15} , or 9×10^{15} GC per dose including all integers or fractional amounts within the range. In one embodiment, for human application the dose can range from 1×10^{10} to about 1×10^{12} GC per dose including all integers or fractional amounts within the range.

These above doses may be administered in a variety of volumes of carrier, excipient or buffer formulation, ranging from about 25 to about 1000 microliters, or higher volumes, including all numbers within the range, depending on the size of the area to be treated, the viral titer used, the route of administration, and the desired effect of the method. In one embodiment, the volume of carrier, excipient or buffer is at least about 25 μL . In one embodiment, the volume is about 50 μL . In another embodiment, the volume is about 75 μL . In another embodiment, the volume is about 100 μL . In another embodiment, the volume is about 125 μL . In another embodiment, the volume is about 150 μL . In another embodiment, the volume is about 175 μL . In yet another embodiment, the volume is about 200 μL . In another embodiment, the volume is about 225 μL . In yet another embodiment, the volume is about 250 μL . In yet another embodiment, the volume is about 275 μL . In yet another embodiment, the volume is about 300 μL . In yet another embodiment, the volume is about 325 μL . In another embodiment, the volume is about 350 μL . In another embodiment, the volume is about 375 μL . In another embodiment, the volume is about 400 μL . In another embodiment, the volume is about 450 μL . In another embodiment, the volume is about 500 μL . In another embodiment, the volume is about 550 μL . In another embodiment, the volume is about 600 μL . In another embodiment, the volume is about 650 μL . In another embodiment, the volume is about 700 μL . In another embodiment, the volume is between about 700 and 1000 μL .

In certain embodiments, the dose may be in the range of about 1×10^9 GC/g brain mass to about 1×10^{12} GC/g brain mass. In certain embodiments, the dose may be in the range of about 3×10^{10} GC/g brain mass to about 3×10^{11} GC/g brain mass. In certain embodiments, the dose may be in the range of about 5×10^{10} GC/g brain mass to about 1.85×10^{11} GC/g brain mass.

In another aspect, an aqueous suspension suitable for administration to a subject is provided. In one embodiment, the suspension includes an aqueous suspending liquid and about 1×10^9 viral particles to about 1×10^{13} GC or viral particles per eye of a recombinant adeno-associated virus (rAAV) as described herein useful as a therapeutic for the treatment or prevention of ocular diseases. In one embodiment, the suspension is suitable for subretinal or intravitreal injection.

In one embodiment, the viral constructs may be delivered in doses of from at least about 1×10^9 GCs to about 1×10^{15} GC, or about 1×10^{11} GC to 5×10^{13} GC. Suitable volumes for delivery of these doses and concentrations may

be determined by one of skill in the art. For example, volumes of about 1 μL to 150 mL may be selected, with the higher volumes being selected for adults. Typically, for newborn infants a suitable volume is about 0.5 mL to about 10 mL, for older infants, about 0.5 mL to about 15 mL may be selected. For toddlers, a volume of about 0.5 mL to about 20 mL may be selected. For children, volumes of up to about 30 mL may be selected. For pre-teens and teens, volumes up to about 50 mL may be selected. In still other embodiments, a patient may receive an intrathecal administration in a volume of about 5 mL to about 15 mL are selected, or about 7.5 mL to about 10 mL. Other suitable volumes and dosages may be determined. The dosage will be adjusted to balance the therapeutic benefit against any side effects and such dosages may vary depending upon the therapeutic application for which the recombinant vector is employed.

The above-described recombinant vectors may be delivered to host cells according to published methods. The rAAV, preferably suspended in a physiologically compatible carrier, may be administered to a human or non-human mammalian patient. In certain embodiments, for administration to a human patient, the rAAV is suitably suspended in an aqueous solution containing saline, a surfactant, and a physiologically compatible salt or mixture of salts. Suitably, the formulation is adjusted to a physiologically acceptable pH, e.g., in the range of pH 6 to 9, or pH 6.5 to 7.5, pH 7.0 to 7.7, or pH 7.2 to 7.8. As the pH of the cerebrospinal fluid is about 7.28 to about 7.32, for intrathecal delivery, a pH within this range may be desired; whereas for intravenous, subretinal or intravitreal delivery, a pH of about 6.8 to about 7.2 may be desired. However, other pHs within the broadest ranges and these subranges may be selected for other route of delivery.

In another embodiment, the composition includes a carrier, diluent, excipient and/or adjuvant. Suitable carriers may be readily selected by one of skill in the art in view of the indication for which the transfer virus is directed. For example, one suitable carrier includes saline, which may be formulated with a variety of buffering solutions (e.g., phosphate buffered saline). Other exemplary carriers include sterile saline, lactose, sucrose, calcium phosphate, gelatin, dextran, agar, pectin, peanut oil, sesame oil, and water. The buffer/carrier should include a component that prevents the rAAV, from sticking to the infusion tubing but does not interfere with the rAAV binding activity *in vivo*. A suitable surfactant, or combination of surfactants, may be selected from among non-ionic surfactants that are nontoxic. In one embodiment, a difunctional block copolymer surfactant terminating in primary hydroxyl groups is selected, e.g., such as Pluronic® F68 [BASF], also known as Poloxamer 188, which has a neutral pH, has an average molecular weight of 8400. Other surfactants and other Poloxamers may be selected, i.e., nonionic triblock copolymers composed of a central hydrophobic chain of polyoxypropylene (poly(propylene oxide)) flanked by two hydrophilic chains of polyoxyethylene (poly(ethylene oxide)), SOLUTOL HS 15 (Macrogol-15 Hydroxystearate), LABRASOL (Polyoxy caprylic glyceride), polyoxy-oleyl ether, TWEEN (polyoxyethylene sorbitan fatty acid esters), ethanol and polyethylene glycol. In one embodiment, the formulation contains a poloxamer. These copolymers are commonly named with the letter "P" (for poloxamer) followed by three digits: the first two digits $\times 100$ give the approximate molecular mass of the polyoxypropylene core, and the last digit $\times 10$ gives the percentage polyoxyethylene content. In one embodiment Poloxamer 188 is selected. The surfactant may be present in an amount up to about 0.0005% to about 0.001% of the

suspension. In one example, the formulation may contain, e.g., buffered saline solution comprising one or more of sodium chloride, sodium bicarbonate, dextrose, magnesium sulfate (e.g., magnesium sulfate·7H₂O), potassium chloride, calcium chloride (e.g., calcium chloride·2H₂O), dibasic sodium phosphate, and mixtures thereof, in water. Suitably, for intrathecal delivery, the osmolarity is within a range compatible with cerebrospinal fluid (e.g., about 275 to about 290); see, e.g., emedicine_medscape_com/article/2093316-overview. Optionally, for intrathecal delivery, a commercially available diluent may be used as a suspending agent, or in combination with another suspending agent and other optional excipients. See, e.g., Elliotts B® solution [Lukare Medical]. In other embodiments, the formulation may contain one or more permeation enhancers. Examples of suitable permeation enhancers may include, e.g., mannitol, sodium glycocholate, sodium taurocholate, sodium deoxycholate, sodium salicylate, sodium caprylate, sodium caprate, sodium lauryl sulfate, polyoxyethylene-9-laurel ether, or EDTA.

Optionally, the compositions of the invention may contain, in addition to the rAAV and carrier(s), other conventional pharmaceutical ingredients, such as preservatives, or chemical stabilizers. Suitable exemplary preservatives include chlorobutanol, potassium sorbate, sorbic acid, sulfur dioxide, propyl gallate, the parabens, ethyl vanillin, glycerin, phenol, and parachlorophenol. Suitable chemical stabilizers include gelatin and albumin.

The compositions according to the present invention may comprise a pharmaceutically acceptable carrier, such as defined above. Suitably, the compositions described herein comprise an effective amount of one or more AAV suspended in a pharmaceutically suitable carrier and/or admixed with suitable excipients designed for delivery to the subject via injection, osmotic pump, intrathecal catheter, or for delivery by another device or route. In one example, the composition is formulated for intrathecal delivery.

As used herein, the terms “intrathecal delivery” or “intrathecal administration” refer to a route of administration for drugs via an injection into the spinal canal, more specifically into the subarachnoid space so that it reaches the cerebrospinal fluid (CSF). Intrathecal delivery may include lumbar puncture, intraventricular (including intracerebroventricular (ICV)), suboccipital/intracisternal, and/or C1-2 puncture. For example, material may be introduced for diffusion throughout the subarachnoid space by means of lumbar puncture. In another example, injection may be into the cisterna *magna*.

As used herein, the terms “intracisternal delivery” or “intracisternal administration” refer to a route of administration for drugs directly into the cerebrospinal fluid of the cisterna *magna* cerebellomedullaris, more specifically via a suboccipital puncture or by direct injection into the cisterna *magna* or via permanently positioned tube.

In one aspect, the vectors provided herein may be administered intrathecally via the method and/or the device. See, e.g., WO 2017/181113, which is incorporated by reference herein. Alternatively, other devices and methods may be selected. The method comprises the steps of advancing a spinal needle into the cisterna *magna* of a patient, connecting a length of flexible tubing to a proximal hub of the spinal needle and an output port of a valve to a proximal end of the flexible tubing, and after said advancing and connecting steps and after permitting the tubing to be self-primed with the patient's cerebrospinal fluid, connecting a first vessel containing an amount of isotonic solution to a flush inlet port of the valve and thereafter connecting a second vessel

containing an amount of a pharmaceutical composition to a vector inlet port of the valve. After connecting the first and second vessels to the valve, a path for fluid flow is opened between the vector inlet port and the outlet port of the valve and the pharmaceutical composition is injected into the patient through the spinal needle, and after injecting the pharmaceutical composition, a path for fluid flow is opened through the flush inlet port and the outlet port of the valve and the isotonic solution is injected into the spinal needle to flush the pharmaceutical composition into the patient.

This method and this device may each optionally be used for intrathecal delivery of the compositions provided herein. Alternatively, other methods and devices may be used for such intrathecal delivery.

It is to be noted that the term “a” or “an” refers to one or more. As such, the terms “a” (or “an”), “one or more,” and “at least one” are used interchangeably herein.

The words “comprise”, “comprises”, and “comprising” are to be interpreted inclusively rather than exclusively. The words “consist”, “consisting”, and its variants, are to be interpreted exclusively, rather than inclusively. While various embodiments in the specification are presented using “comprising” language, under other circumstances, a related embodiment is also intended to be interpreted and described using “consisting of” or “consisting essentially of” language.

As used herein, the term “about” means a variability of 10% (+10%) from the reference given, unless otherwise specified.

As used herein, “disease”, “disorder” and “condition” are used interchangeably, to indicate an abnormal state in a subject.

Unless defined otherwise in this specification, technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art and by reference to published texts, which provide one skilled in the art with a general guide to many of the terms used in the present application.

The term “expression” is used herein in its broadest meaning and comprises the production of RNA or of RNA and protein. With respect to RNA, the term “expression” or “translation” relates in particular to the production of peptides or proteins. Expression may be transient or may be stable.

As used herein, the term “NAb titer” a measurement of how much neutralizing antibody (e.g., anti-AAV Nab) is produced which neutralizes the physiologic effect of its targeted epitope (e.g., an AAV). Anti-AAV Nab titers may be measured as described in, e.g., Calcedo, R., et al., Worldwide Epidemiology of Neutralizing Antibodies to Adeno-Associated Viruses. *Journal of Infectious Diseases*, 2009. 199 (3): p. 381-390, which is incorporated by reference herein.

As used herein, an “expression cassette” refers to a nucleic acid molecule which comprises a coding sequence, promoter, and may include other regulatory sequences therefor, which cassette may be delivered via a genetic element (e.g., a plasmid) to a packaging host cell and packaged into the capsid of a viral vector (e.g., a viral particle). Typically, such an expression cassette for generating a viral vector contains the coding sequence for the gene product described herein flanked by packaging signals of the viral genome and other expression control sequences such as those described herein.

The abbreviation “sc” refers to self-complementary. “Self-complementary AAV” refers a construct in which a coding region carried by a recombinant AAV nucleic acid

sequence has been designed to form an intra-molecular double-stranded DNA template. Upon infection, rather than waiting for cell mediated synthesis of the second strand, the two complementary halves of scAAV will associate to form one double stranded DNA (dsDNA) unit that is ready for immediate replication and transcription. See, e.g., D M McCarty et al, "Self-complementary recombinant adeno-associated virus (scAAV) vectors promote efficient transduction independently of DNA synthesis", *Gene Therapy*, (August 2001), Vol 8, Number 16, Pages 1248-1254. Self-complementary AAVs are described in, e.g., U.S. Pat. Nos. 6,596,535; 7,125,717; and 7,456,683, each of which is incorporated herein by reference in its entirety.

As used herein, the term "operably linked" refers to both expression control sequences that are contiguous with the gene of interest and expression control sequences that act in trans or at a distance to control the gene of interest.

The term "heterologous" when used with reference to a protein or a nucleic acid indicates that the protein or the nucleic acid comprises two or more sequences or subsequences which are not found in the same relationship to each other in nature. For instance, the nucleic acid is typically recombinantly produced, having two or more sequences from unrelated genes arranged to make a new functional nucleic acid. For example, in one embodiment, the nucleic acid has a promoter from one gene arranged to direct the expression of a coding sequence from a different gene. Thus, with reference to the coding sequence, the promoter is heterologous.

A "replication-defective virus" or "viral vector" refers to a synthetic or artificial viral particle in which an expression cassette containing a gene of interest is packaged in a viral capsid or envelope, where any viral genomic sequences also packaged within the viral capsid or envelope are replication-deficient; i.e., they cannot generate progeny virions but retain the ability to infect target cells. In one embodiment, the genome of the viral vector does not include genes encoding the enzymes required to replicate (the genome can be engineered to be "gutless"-containing only the transgene of interest flanked by the signals required for amplification and packaging of the artificial genome), but these genes may be supplied during production. Therefore, it is deemed safe for use in gene therapy since replication and infection by progeny virions cannot occur except in the presence of the viral enzyme required for replication.

In many instances, rAAV particles are referred to as DNase resistant. However, in addition to this endonuclease (DNase), other endo- and exo-nucleases may also be used in the purification steps described herein, to remove contaminating nucleic acids. Such nucleases may be selected to degrade single stranded DNA and/or double-stranded DNA, and RNA. Such steps may contain a single nuclease, or mixtures of nucleases directed to different targets, and may be endonucleases or exonucleases.

The term "nuclease-resistant" indicates that the AAV capsid has fully assembled around the expression cassette which is designed to deliver a transgene to a host cell and protects these packaged genomic sequences from degradation (digestion) during nuclease incubation steps designed to remove contaminating nucleic acids which may be present from the production process.

The term "translation" in the context of the present invention relates to a process at the ribosome, wherein an mRNA strand controls the assembly of an amino acid sequence to generate a protein or a peptide.

As used throughout this specification and the claims, the terms "comprising" and "including" are inclusive of other

components, elements, integers, steps and the like. Conversely, the term "consisting" and its variants are exclusive of other components, elements, integers, steps and the like.

As described above, the term "about" when used to modify a numerical value means a variation of $\pm 10\%$, unless otherwise specified.

The following examples are illustrative only and are not intended to limit the present invention.

EXAMPLES

The following examples report the extensive deamidation of AAV8 and additional diverse AAV serotypes, with supporting evidence from structural, biochemical, and mass spectrometry approaches. The extent of deamidation at each site was dependent on the age of the vector and multiple primary-sequence and 3D-structural factors, but was largely independent of the conditions of vector recovery and purification. We demonstrate the potential for deamidation to impact vector transduction activity, and correlate an early timepoint loss in vector activity to rapidly progressing, spontaneous deamidation at several AAV8 asparagines. We explore mutational strategies that stabilize side-chain amides, improving vector transduction and reducing the lot-to-lot molecular variability that is a key concern in biologics manufacturing. This study illustrates a previously unknown aspect of AAV capsid heterogeneity and highlights its importance in the development of these vectors for gene therapy.

In Example 1 the characterization of post-translational modifications to the AAV8 vector capsid by one- and two-dimensional gel electrophoresis, mass spectrometry, and de novo structural modeling. Following the identification of a number of putative deamidation sites on the capsid surface, we evaluate their impact on capsid structure and function both in vitro and in vivo. Example 1 further extends this analysis to AAV9 to determine if this phenomenon applies to serotypes other than AAV8, confirming that deamidation of the AAV capsid is not serotype specific. Examples 2-5 show deamidation in distinct AAVs.

Example 1: Deamidation of Amino Acids on the Surface of Adeno-Associated Virus Capsids

A. Materials and Methods

1. 1D and 2D Gel Electrophoresis

For 1D SDS polyacrylamide gel electrophoresis (SDS-PAGE) analysis, we first denatured AAV vectors at 80°C . for 20 minutes in the presence of lithium dodecyl sulfate and reducing agent. Then, we ran them on a 4-12% Bis-Tris gel for 90 minutes at 200V and stained with Coomassie blue. For the data in FIG. 1A-FIG. 1D, Kendrick Laboratories, Inc. (Madison, WI) performed the 2D gel electrophoresis. For subsequent experiments, we performed 2D SDS-PAGE in-house. For this, we combined 3×10^{11} GCs of AAV vector with 500U turbonuclease marker (Accelagen, San Diego, CA) in 150 μL phosphate buffered saline (PBS) with 35 mM NaCl and 1 mM MgCl_2 and incubated at 37°C . for ten minutes. We next added nine volumes of absolute ethanol, vortexed the samples, and incubated them at -80°C . for at least two hours followed by incubation on ice for five minutes and centrifugation at maximum speed for 30 minutes at 15°C . We decanted the supernatant and air-dried the pellet, which we then resuspended in resuspension buffer #1 [0.15% SDS, 50 mM dithiothreitol (DTT), 10 mM Tris pH 7.5, and 1 μL pH6-9 ampholytes, ThermoFisher ZM0023, added day-of, in ddH_2O] and incubated undisturbed at room

temperature. After 30 minutes, we flicked the sample tubes to mix them, added lug chicken conalbumin marker (Sigma Aldrich, St. Louis, MO), and incubated the samples at 37° C. for 30 minutes, flicking to mix at 15 minutes. Samples were then transferred to 50° C. for 15-20 minutes, vortexed, incubated at 95° C. for 2.5 minutes, and allowed to cool before being centrifuged at maximum speed for one minute and briefly vortexed. We then mixed 10 µL of each sample with 140 µL resuspension buffer #2 (9.7M urea, 2% CHAPS, 0.002% bromophenol blue, and 0.05% ampholytes, described above, added day-of, in ddH₂O) and incubated at room temperature for ten minutes. We then applied the mixtures to pH 6-10 immobilized pH gradient (IPG) strips (ThermoFisher Waltham, MA) and ran them on the ZOOM IPGRunner system according to manufacturer's instructions. We used the following isoelectric focusing parameters: 100-1,000V for 120 minutes, 1,000-2,000V for 120 minutes, 2,000V for 120 minutes, limits of 0.1W and 0.05 mA per strip run. IPG strips were then reduced and loaded in a single-well 4-12% Bis-Tris gel and run in ID as described above. We determined the relative migration of AAV VPs by comparison to internal control proteins turbo-nuclease (Accelagen, 27 kDa) and chicken egg white conalbumin (Sigma Aldrich, 76 kDa, pI 6.0-6.6).

2. Vector Production

The University of Pennsylvania Vector Core produced recombinant AAV vectors for 1D and 2D gel electrophoresis and mass spectrometry experiments and purified them by cesium chloride or iodixanol gradients as previously described. (Lock M, et al. *Hum Gene Ther* 2010; 21 (10): 1259-71; Gao G P, et al. *Proc Natl Acad Sci USA*. 2002; 99 (18): 11854-9). We produced the affinity purified vectors as follows: We grew HEK293 cells in ten 36-layer hyperstack vessels (Corning), co-transfected them with a mixture of vector genome plasmid (pAAV-LSP-IVS2.hFIXco-WPRE-bGH), trans plasmid containing AAV2 rep and AAV8 cap genes, and adenovirus helper plasmid. We used PEIpro (PolyPlus) as the transfection reagent. Five days post transfection, the supernatant was harvested, clarified through Sartoguard PES Midicap filters (Sartorius Stedim), and treated with benzonase (Millipore), after which we added salt to bring it to 0.6M. The clarified bulk harvest material was concentrated ten-fold by tangential flow filtration (TFF) and then diafiltered against four volumes of affinity column loading buffer. We captured vectors on a POROS Capture-Select (ThermoFisher) affinity column and eluted the vector peak at low pH directly into neutralization buffer. We diluted the neutralized eluate into a high-pH binding buffer and loaded it onto an anion exchange polishing column (Cimul-tus QA-8; Bia Separations), where the preparation was enriched for genome-containing (full) particles. The full vector particles were eluted with a shallow salt elution gradient and neutralized immediately. Finally, we subjected the vector to a second round of TFF for final concentration and buffer exchange into formulation buffer (PBS+0.001% pluronic F-68).

We produced mutant vectors for in vitro assays by small-scale triple transfection of HEK293 cells in six-well plates. We mixed 5.6 µL of a 1 mg/ml polyethylenimine solution in 90 µL serum-free media with plasmid DNA (0.091 µg cis plasmid, 0.91 µg trans plasmid, 1.82 µg deltaF6 Ad-helper plasmid, in 90 µL serum-free media), incubated it at room temperature for 15 minutes, and added it to cells in and additional 0.8 mL of fresh serum-free media. The next day, we replaced 0.5 mL of the top media with full serum media. We harvested vector three days post-transfection by three freeze/thaw cycles followed by centrifugation to remove cell

debris and supernatant harvest. Cis plasmid contained a transgene cassette encoding the firefly luciferase transgene under the control of the chicken-beta actin (CB7) promoter with the Promega chimeric intron and rabbit beta-globin (RBG) polyadenylation signal. Trans plasmid encoded the wtAAV8 cap gene; to generate mutant AAV8 cap variants, we used the Quikchange Lightning Mutagenesis kit (Agilent Technologies, Wilmington, DE). Vector was titered as previously described. (Lock M, et al. *Hum Gene Ther* 2010; 21 (10): 1259-71).

For timecourse vector production experiments, we generated vector by medium-scale triple transfection of HEK293 cells in 15 cm tissue culture dishes. Per plate, we mixed 36 µL of a 1 mg/mL polyethylenimine solution in 2 mL serum-free media with plasmid DNA (0.6 µg cis plasmid, 5.8 µg trans plasmid, 11.6 µg deltaF6 Ad-helper plasmid), incubated it at room temperature for 15 minutes, and added it to cells at approximately 60% confluency on plates refreshed with 14 ml of serum-free media. The following day, we replaced 8 ml of the top media with fresh, full serum media. We harvested vector by collecting all top media, scraping cells from the dish and freezing this at -80° C. We recovered crude vector from the supernatant/cell mixture by applying 3 freeze/thaw cycles, and clarifying the lysate by centrifugation. We purified and concentrated the vector for mass spectrometry analysis by adding benzonase, 1M Tris pH7.5, and 5M NaCl to the clarified lysate to final concentrations of 20 mM Tris and 360 mM NaCl. We captured vectors on a 1 ml POROS CaptureSelect affinity column and eluted the vector peak at low pH directly into neutralization buffer. Fractions were analysed by absorbance at 280 nm, and the most concentrated fraction was subjected to mass spectrometry analysis.

For in vivo experiments, we produced vectors as previously described with a wtAAV8 capsid or with one of the 6 deamidation mutants; the transgene cassette included a CB7 promoter, PI intron, firefly luciferase transgene, and RBG polyadenylation signal (Lock M, et al. *Hum Gene Ther* 2010; 21 (10): 1259-71).

3. Mass Spec Run/Digest/Analysis

Materials: We purchased ammonium bicarbonate, DTT, iodoacetamide (IAM), and 18O-enriched water (97.1% purity) from Sigma (St. Louis, MO); and acetonitrile, formic acid, trifluoroacetic acid (TFA), 8M guanidine hydrochloride (GndHCl), and trypsin from Thermo Fischer Scientific (Rockford, IL).

Trypsin digestion: We prepared stock solutions of 1M DTT and 1.0M iodoacetamide. Capsid proteins were denatured and reduced at 90° C. for ten minutes in the presence of 10 mM DTT and 2M GndHCl. We allowed the samples to cool to room temperature and then alkylated them with 30 mM IAM at room temperature for 30 minutes in the dark. We quenched the alkylation reaction with the addition of 1 mL DTT. We added 20 mM ammonium bicarbonate (pH 7.5-8) to the denatured protein solution at a volume that diluted the final GndHCl concentration to 200 mM. We added trypsin solution for a 1:20 trypsin to protein ratio and incubated at 37° C. overnight. After digestion, we added TFA to a final concentration of 0.5% to quench the digestion reaction.

For 18O-water experiments, the capsid sample was first buffer exchanged into 100 mM ammonium bicarbonate prepared in 18O-water using Zeba spin desalting columns (Thermo Scientific, Rockford, IL). To ensure a complete removal of the water in the sample, we performed the buffer exchange twice. We prepared stock solutions of 1M DTT

and 1M IAM in 18O-water. We followed the same denaturation, alkylation, and digestion steps as above with 18O-water reagents and buffers.

Liquid chromatography tandem-mass spectrometry: We performed online chromatography with an Acclaim PepMap column (15 cm long, 300 μ m inner diameter) and a Thermo UltiMate 3000 RSLC system (Thermo Fisher Scientific) coupled to a Q Exactive HF with a NanoFlex source (Thermo Fisher Scientific). During online analysis, the column temperature was maintained at a temperature of 35° C. We separated peptides with a gradient of mobile phase A (MilliQ water with 0.1% formic acid) and mobile phase B (acetonitrile with 0.1% formic acid). We ran the gradient from 4% B to 6% B over 15 minutes, to 10% B for 25 minutes (40 minutes total), and then to 30% B for 46 minutes (86 minutes total). We loaded the samples directly to the column. The column size was 75 cm $\times$ 15 μ m I.D. and was packed with 2 micron C18 media (Acclaim PepMap). Due to the loading, lead-in, and washing steps, the total time for each liquid chromatography tandem-mass spectrometry run was about two hours.

We acquired mass spectrometry data using a data-dependent top-20 method on the Q Exactive HF mass spectrometer, dynamically choosing the most abundant not-yet-sequenced precursor ions from the survey scans (200-2000 m/z). We performed sequencing via higher energy collisional dissociation fragmentation with a target value of 1e5 ions determined with predictive automatic gain control; we performed isolation of precursors with a window of 4m/z. We acquired survey scans at a resolution of 120,000 at 200m/z. We set the resolution for HCD spectra to 30,000 at m/z200 with a maximum ion injection time of 50 ms and a normalized collision energy of 30. We set the S-lens RF level to 50, which gave optimal transmission of the m/z region occupied by the peptides from our digest. We excluded precursor ions with single, unassigned, or six and higher charge states from fragmentation selection.

Data processing: We used BioPharma Finder 1.0 software (Thermo Fischer Scientific) to analyze all data acquired. For peptide mapping, we performed searches using a single-entry protein FASTA database with carbamidomethylation set as a fixed modification, and oxidation, deamidation, and phosphorylation set as variable modifications. We used a 10 ppm mass accuracy, a high protease specificity, and a confidence level of 0.8 for tandem-mass spectrometry spectra. Mass spectrometric identification of deamidated peptides is relatively straightforward, as deamidation adds to the mass of intact molecule +0.984 Da (the mass difference between —OH and —NH₂ groups). We determined the percent deamidation of a particular peptide by dividing the mass area of the deamidated peptide by the sum of the area of the deamidated and native peptides. Considering the number of possible deamidation sites, isobaric species that are deamidated at different sites may co-migrate at a single peak. Consequently, fragment ions originating from peptides with multiple potential deamidation sites can be used to locate or differentiate multiple sites of deamidation. In these cases, the relative intensities within the observed isotope patterns can be used to specifically determine the relative abundance of the different deamidated peptide isomers. This method assumes that the fragmentation efficiency for all isomeric species is the same and independent of the site of deamidation. This approach allows the definition of the specific sites involved in deamidation and the potential combinations involved in deamidation.

Secondary data processing: Secondary analysis of raw mass spectrometry was performed at the University of

Maryland, Baltimore County using the following method. Peaks Studio v5.3 software (Bioinformatics Solutions Inc.) was used for all mass spectrometry analysis. Data refinement of the raw data files was performed with the following parameters: a precursor m/z tolerance of $\leq$ 10 ppm, and precursor charge state with a minimum of 2, maximum of 4. De novo sequencing of the input spectrum was performed using the Peaks algorithm with a precursor ion error tolerance of 10 ppm and product ion error tolerances of 0.1 Da. The digestion enzyme was set as trypsin, the variable modifications were oxidation, phosphorylation, and deamidation, and the fixed modification was carbamidomethylation of cysteine.

4. Structural Analysis of the AAV Capsid

We obtained the AAV8 atomic coordinates, structural factors, and associated capsid model from the RCSB Protein Data Bank (PDB ID: 3RA8). We performed structure refinement and generated an electron density independent of the primary amino acid sequence of AAV8 VP3 for use in three-dimensional (3D)) structural analysis of the capsid. We performed this analysis in order to observe the isoaspartic acid electron density in the AAV8 capsid that was not biased by the expected primary sequence of AAV8 VP3. Using the resulting structure, we modeled the four asparagines in the AAV8 VP3 primary sequence with N+1 glycines as isoaspartic acids and then refined the AAV8 capsid structure using Crystallography and NMR System (CNS) software by strictly imposing the icosahedral non-crystallographic matrices using the standard refinement protocol (Brunger A T, et al. *Acta Crystallogr D Biol Crystallogr* 1998; 54 (Pt 5): 905-21). We obtained a structural model of isoaspartic acid from the HIC-UP database, followed by generation of a molecular dictionary in PRODRG for structure refinement (Kleywegt G *J Acta Crystallogr D Biol Crystallogr* 2007; 63 (Pt 1): 94-100). We then calculated the average electron density map of the AAV8 capsid (also in CNS) and visualized it using COOT software, followed by minor adjustments of the resulting model to fit the modeled isoaspartic acid residues into the electron density map (Emsley P and Cowtan K *Acta Crystallogr D Biol Crystallogr* 2004; 60 (Pt 12 Pt 1): 2126-32). We repeated this protocol to additionally model N512 in the AAV9 VP3 primary sequence with N+1 glycines (PDB ID: 3UX1). We generated all figures using COOT, PyMol, and UCSF Chimera (Emsley P and Cowtan K *Acta Crystallogr D Biol Crystallogr* 2004; 60 (Pt 12 Pt 1): 2126-32; DeLano W L *PyMOL: An Open-Source Molecular Graphics Tool* Vol. 40, 2002:82-92; Pettersen E F, et al. *J Comput Chem* 2004; 25 (13): 1605-12). We obtained a number of structures of previously identified deamidated proteins (PDB IDs: 1DY5, 4E7G, 1RTU, 1W9V, 4E7D, and 1C9D) for comparison of their electron density map for deamidated isoaspartic acid residues with our modeled isoaspartic acid residues from AAV8 and AAV9 (Rao F V, et al. *Chem Biol* 2005; 12 (1): 65-76; Noguchi S, et al. *Biochemistry* 1995; 34 (47): 15583-91; Esposito L, et al. *J Mol Biol* 2000; 297 (3): 713-32).

We determined temperature factors for deamidated residues by averaging the temperature factors for each atom of each asparagine residue reported in the AAV8 or AAV9 crystal structure atomic coordinates (PDB ID: 3RA8, 3UX1).

5. Animal Studies

The Institutional Animal Care and Use Committee of the University of Pennsylvania approved all animal procedures. To evaluate vector performance, we injected eight-week-old C57BL/6 mice intravenously via tail vein injection with 3e10 GCs of wtAAV8 or capsid mutant vector in a volume

of 100 μ L. All mice were sacrificed at day 14. For in vivo evaluation of luciferase expression, mice (~20g) were anesthetized and injected intraperitoneally with 200 μ L or 15 mg/mL luciferin substrate (Perkin Elmer, Waltham, MA). Mice were imaged five minutes after luciferin administration and imaged via an IVIS Xenogen In Vivo Imaging System. We used Living Image 3.0 software to quantify signal in the described regions of interest. We took measurements at days 7 and 14.

6. Evaluation of Mutant Vector Titer and In Vitro Transduction Efficiency

We determined vector titers by qPCR of the DNaseI-resistant genomes. The qPCR primers anneal to the polyadenylation sequence of the packaged transgene. For in vitro evaluation of vector transduction efficiency by luciferase expression, we seeded 0.9e5 Huh7 cells/well in a black-walled 96-well plate in complete DMEM (10% fetal bovine serum, 1% penicillin/streptomycin). The next day, we removed the media and replaced it with 50 μ L crude or purified vector diluted in complete media. We tested 4 dilutions in a 3 fold dilution series for each crude vector

sample. After 48 hours, we prepared luciferin (Promega, Madison, WI) in complete media at 0.3 μ g/ μ L and added it to transduced cells in a volume of 50 μ L. Results were read on a Biotek Clarity luminometer. We find that luciferase activity/GC added to target cells is constant over a wide range of GCs, but can become saturated at high MOIs. Thus we inspect the dilution series data (luminescent units vs GC) for linearity, exclude the highest point if saturation is evident, and calculate an average Luciferase/GC for values in the linear range of each assay for each variant. This yields a transduction efficiency value. The data are normalized to simplify comparison by setting the wt control to a value of 1.

7. Biodistribution

We extracted DNA from liver samples using the QIAamp DNA Mini Kit (Qiagen, Hilden, Germany), and then analyzed the DNA for vector GC by real-time PCR as described previously with a primer/probe set designed against the RBG polyadenylation signal of the transgene cassette (Chen S J, et al. *Hum Gene Ther Clin Dev* 2013; 24 (4): 154-60). Primer Sequences for AAV8 Mutants

	Sequence	Description
SED ID NO: 56	CGACAACCGGGCAAACcagAATAGCAACTTTGCCTGG	QC mutagenic primers to change AAV8 N499 to Q
SED ID NO: 57	CCAGGCAAAGTTGCTATTCTGGTTTTCGCCGGTTGTCG	QC mutagenic primers to change AAV8 N499 to Q
SED ID NO: 58	GACAACCGGGCAAACgacAATAGCAACTTTGCCTG	QC mutagenic primers to change AAV8 N499 to D
SED ID NO: 59	CAGGCAAAGTTGCTATTGTCGTTTTCGCCGGTTGTC	QC mutagenic primers to change AAV8 N499 to D
SED ID NO: 60	GGAGGCACGGCAcagACGCAGACTCTGGG	qc mutagenic primers to change AAV8 N459 to Q
SED ID NO: 61	CCCAGAGTCTGCGTCTGTGCCGTGCCTCC	qc mutagenic primers to change AAV8 N459 to Q
SED ID NO: 62	CAGGAGGCACGGCAgacACGCAGACTCTGG	qc mutagenic primers to change AAV8 N459 to D
SED ID NO: 63	CCAGAGTCTGCGTATCTGCCGTGCCTCTG	qc mutagenic primers to change AAV8 N459 to D
SED ID NO: 64	ctectcccgatgtcgcgttgagatttgc	AAV8 NA263 F
SED ID NO: 65	gcaaatctccaacgcgacatcgggaggag	AAV8 NA263 R
SED ID NO: 66	cccacggcctgactagcgttggtgagtgtta	AAV8 NA385 F
SED ID NO: 67	taacactcaacaacgcctagtcaggccgtggg	AAV8 NA385 R
SED ID NO: 68	ggattagccaatgaatttcttgcatcagatggatttgggtcc	AAV8 NA514 F
SED ID NO: 69	ggaccaaataccatctgaatgaagaaattcattggctaattcc	AAV8 NA514 R
SED ID NO: 70	tttgccaaaaatcaggatcgcgcttactgggaaaaaacg	AAV8 NA540 F
SED ID NO: 71	cgtttttttccagtaacgcgcatcctgatttttggcaaa	AAV8 NA540 R
SED ID NO: 72	ggacccttcaacgcactcgacaagggg	AAV8 NA57 F
SED ID NO: 73	ccccttgctgagtgctgttgagggtcc	AAV8 NA57 R
SED ID NO: 74	tggctcctcccgatgtgctgttgagatttgccttg	AAV8 NS263 F
SED ID NO: 75	caagcaaatactccaacagcacatcgggaggagcca	AAV8 NS263 R
SED ID NO: 76	cccacggcctgactactgttggtgagtgtagg	AAV8 NS385 F
SED ID NO: 77	cctaacactcaacaacagtagtcaggccgtggg	AAV8 NS385 R

-continued

	Sequence	Description
SED ID NO: 78	ttagccaatgaatttctgctattcagatgggtatttgggtcccagcag	AAV8 NS514 F
SED ID NO: 79	ctgctgggaccaaataccatctgtaatagcagaaattcattggctaa	AAV8 NS514 R
SED ID NO: 80	ttgtttgccaaaaatcaggatgctgttactgggaaaaaacgctc	AAV8 NS540 F
SED ID NO: 81	gagcgtttttttcccagtaacagcatcctgatttttggcaaaciaa	AAV8 NS540 R
SED ID NO: 82	ctcccccttgctcaggctgttgaaggtccgag	AAV8 NS57 F
SED ID NO: 83	ctcggacccttcaacagcctcgacaagggggag	AAV8 NS57 R
SED ID NO: 84	cagcgactcatcaacGACaactggggattccg	QC primer for AAV8 N305D
SED ID NO: 85	ggaggcacggcagATacgcagactctgg	QC primer for AAV8 N459D
SED ID NO: 86	gacaaccgggcaaaacGACaatagcaactttgcctg	QC primer for AAV8 N499D
SED ID NO: 87	ccatctgaatggaagaGATtcattggctaatacctggcatc	QC primer for AAV8 N517D
SED ID NO: 88	cgaagcccaagccGACcagcaaaagcagg	QC primer for AAV8 N35D
SED ID NO: 89	gtacctgcggtatGACcagccgcagccc	QC primer for AAV8 N94D
SED ID NO: 90	gatgctgagaaccggcGACaacttccagtttacttac	QC primer for AAV8 N410D
SED ID NO: 91	cagactctgggcttcagcGATggtgggcctaatacaatg	QC primer for AAV8 Q467D
SED ID NO: 92	ccaatcaggcaaaagGACTggctgccaggac	QC primer for AAV8 N479D
SED ID NO: 93	cacggacggcGACTtccaccgtctc	QC primer for AAV8 N630D
SED ID NO: 94	gatcctgatcaagGACacgcctgtacctgcg	QC primer for AAV8 N653D
SED ID NO: 95	gtacctcggacccttcCAGggactcgacaagg	QC primer for AAV8 N57Q
SED ID NO: 96	ctacaagcaaatctccCAGgggacatcgggaggagc	QC primer for AAV8 N263Q
SED ID NO: 97	gctacctaacactcaacCAGggtagtcaggccgtgg	QC primer for AAV8 N385Q
SED ID NO: 98	gctgggaccaataaccatctgCAGggaagaaattcattggc	QC primer for AAV8 N514Q
SED ID NO: 99	ggagcgtttttttcccagtcAGgggacacctgatttttggc	QC primer for AAV8 N540Q
SED ID NO: 100	cggaatccccagttgtcgttgatgagtcgctg	QC primer for AAV8 N305D
SED ID NO: 101	ccagagtctgcgtatctgcctgcctcc	QC primer for AAV8 N459D
SED ID NO: 102	caggcaaagttgctattgtcgttttgcccggttgctc	QC primer for AAV8 N499D
SED ID NO: 103	gatgccaggattagccaatgaatctcttccattcagatgg	QC primer for AAV8 N517D
SED ID NO: 104	cctgcttttctggtcggcctttgggcttcg	QC primer for AAV8 N35D
SED ID NO: 105	ggcgtcggcgtggtcataccgcaggtag	QC primer for AAV8 N94D
SED ID NO: 106	gtaagtaaactggaagttgtcgcgggttctcagcatc	QC primer for AAV8 N410D
SED ID NO: 107	cattgtattagggccaccatcgctgaagcccagagtctg	QC primer for AAV8 Q467D
SED ID NO: 108	gtcctggcagccagtcctttgcctgattgg	QC primer for AAV8 N479D
SED ID NO: 109	gagacgggtggaagtcgccgtccgtg	QC primer for AAV8 N630D
SED ID NO: 110	cgcaggtagcaggcgtgtccttgatcaggatc	QC primer for AAV8 N653D
SED ID NO: 111	gcagcgactcatcaacGACaactggggattccggc	alternative longer primer to make AAV8 N305D by qc mutagenesis
SED ID NO: 112	GCCGGAATCCCAGTTGTCGTTGATGAGTCGCTGC	alternative longer primer to make AAV8 N305D by qc mutagenesis
SED ID NO: 113	cagcgactcatcaacGACaactggggattccggc	alternative longer primer to make AAV8 N305D by qc mutagenesis

-continued

	Sequence	Description
SED ID NO: 114	GCCGGAATCCCCAGTTGTCGTTGATGAGTCGCTG	alternative longer primer to make AAV8 N305D by qc mutagenesis
SED ID NO: 115	gcgactcatcaacGACaactggggattccg	alternative shorter primer to make AAV8 N305D by qc mutagenesis
SED ID NO: 116	CGGAATCCCCAGTTGTCGTTGATGAGTCGC	alternative shorter primer to make AAV8 N305D by qc mutagenesis
SED ID NO: 117	ctctggggttcagcGAAggtgggcctaatac	mutagenic QC primer to make AAV8 Q467E
SED ID NO: 118	GTATTAGGCCACCTTCGCTGAAGCCCAGAG	mutagenic QC primer to make AAV8 Q467E
SED ID NO: 119	cctcggacccttcGACggactcgacaagg	QC primer for AAV8 N57D
SED ID NO: 120	tacaagcaaatctccGACgggacatcgggaggag	QC primer for AAV8 N263D
SED ID NO: 121	ctacctaactcaacGACggtagtcaggccgtg	QC primer for AAV8 N385D
SED ID NO: 122	ctgggaccaaataccatctgGATggaagaattcattggctaatac	QC primer for AAV8 N514D
SED ID NO: 123	gagcgtttttttccagtgACgggatcctgatttttggc	QC primer for AAV8 N540D
SED ID NO: 124	ccttgctcagtcctcgaagggtccgagg	QC primer for AAV8 N57D
SED ID NO: 125	ctcctcccgatgtcccgtcggagatttgccttgta	QC primer for AAV8 N263D
SED ID NO: 126	cacggcctgactaccgtcgttgagtgttaggtag	QC primer for AAV8 N385D
SED ID NO: 127	gattagccaatgaattttcttccatccagatggtatttgggtcccag	QC primer for AAV8 N514D
SED ID NO: 128	gccaaaaatcaggatcccgctcactgggaaaaaacgcctc	QC primer for AAV8 N540D

B. Results

AAV8 Shows Substantial Charge Heterogeneity in its Capsid Proteins

To qualitatively assess the presence of post-translational modifications on the AAV8 vector capsid that could affect vector performance, we analyzed AAV8 total capsid protein purified by iodixanol gradient both by 1D and 2D gel electrophoresis. In a 1D reducing sodium dodecyl sulfate SDS gel, VP1, VP2, and VP3 resolved as single bands at the appropriate molecular weights (FIG. 1B) (Rose J A, et al. *J Virol* 1971; 8 (5): 766-70). When further evaluated by 2D gel electrophoresis, which separates proteins based on charge (FIG. 1C), each of the capsid proteins additionally resolved as a series of distinct spots with different isoelectric points (pIs) ranging from pH 6.3 to >7.0 dependent on the VP isoform (FIG. 1D). Individual spots on each VP were separated by discrete intervals of 0.1 pI units as measured as migration relative to the carbonic anhydrase isoform internal isoelectric point standards, suggesting a single residue charge change. The presence of these isoforms suggests that each VP has the potential to undergo many modifications, thereby causing them to migrate differently under isoelectric focusing.

Deamidation, in which a fraction of (typically asparagine) side-chain amide groups are converted to carboxylic acid (FIG. 1A), is a common source of charge heterogeneity in protein preparations. To determine if deamidation could be responsible for the distinct population of VP charge isoforms, we mutated two AAV8 asparagine residues individually to aspartate. These capsid mutations should shift the charge by an amount equivalent to the complete deamidation

of a single additional asparagine residue. 2D gel analysis of the mutants indicates the major spots for VP1, VP2, and VP3 shifted one spot location more acidic (0.1 pH units) than the equivalent spots in wild-type (wt) AAV8 (FIG. 1E-FIG. 1G). The magnitude of this shift is equivalent to the observed spacing between the wt VP charge isoforms. Thus, the 2D gel patterning of AAV capsid proteins is consistent with multi-site deamidation.

Spontaneous Deamidation Occurs on the AAV8 Vector Capsid

To identify modifications responsible for the discrete spotting pattern for each capsid protein, we analyzed a panel of AAV8 vectors by mass spectrometry. Coverage of the AAV8 capsid protein averaged >95% of the total VP1 sequence (data not shown). We detected extensive deamidation of a subset of asparagine and glutamine residues by mass spectroscopy, which showed an increase of ~1 Da in the observed mass of the individual peptides as compared to predicted values based on the sequence encoded by the DNA; we observed this pattern of deamidation in all preparations of AAV8 vectors (FIG. 2A-FIG. 2D).

To evaluate the global heterogeneity of deamidation between commonly used purification methods and to examine deamidation in the VP1 and VP2 unique regions, we selected nine lots of AAV8 produced by triple transfection in 293 cells and purified them by either cesium chloride gradient, iodixanol gradient, or affinity chromatography. Vectors also varied with respect to promoters and transgene cassettes. To determine if the presence of the vector genome had an impact on deamidation, we also evaluated an AAV8 prep produced by triple transfection in 293 cells in the

absence of cis plasmid (producing empty capsids only) and purified by iodixanol gradient.

A wide range of deamidation was present across asparagine and glutamine residues of the AAV8 capsid, ranging from undetectable to over 99% of individual amino acids being deamidated (FIG. 2E). The highest levels of deamidation (>75%) occurred at asparagine residues where the N+1 residue was glycine (i.e., NG pairs) (Table 1). We detected lower levels of deamidation (i.e., up to 17%) at additional asparagine residues where the N+1 was not glycine. The average deamidation for asparagines was largely consistent between preps. We also detected deamidation at glutamine residues but at a lower frequency than at asparagines; the highest percent we observed was <2% at Q467 (FIG. 7). This observation was inconsistent across preparations (data not shown). We observed the greatest preparation-to-preparation differences at residue N499 (N+1 residue is asparagine), with values ranging from <1% to over 50% deamidation. Regardless, the variations we observed in deamidation between preparations of vector did not appear to be related to purification method, transgene identity, or the presence of vector genome, suggesting that these factors do not impact deamidation rates.

TABLE 1

Characteristics of AAV8 deamidated residues of interest.					
	N + 1 residue	Structural topology	Structural motif	Average % deamidation	Temperature factor ($\text{\AA}^2$)
N35	Q	N/A	N/A	1	N/A
N57	G	N/A	N/A	80	N/A
N94	H	N/A	N/A	7	N/A
N254*	N	Surface exposed	Not assigned	9	35
N255*	H	Surface exposed	Not assigned	N/A	42
N263	G	Surface exposed	HVR I	99	51
N305	N	Buried	Alpha helix	8	33
N385	G	Surface exposed	HVR III	88	41
N410	N	Buried	Not assigned	3	33
N459	T	Surface exposed	HVR IV	7	65
N499	N	Surface exposed	HVR V	17	45
N514*	G	Surface exposed	HVR V	84	36
N517*	S	Surface exposed	HVR V	4	40
N540*	G	Buried	HVR VII	79	40
N630*	F	Buried	Not assigned	1	32
N653	T	Surface exposed	HI loop	1	35

Asterisks represent residues selected for further analysis.

Next, we ran a series of experiments to determine if sample handling contributed to the observed levels of deamidation in AAV8. Extreme temperature (70° C. for 7 days) or pH (pH 2 or pH 10 for 7 days) did not significantly induce additional deamidation in the AAV8 capsid (FIG. 4A and FIG. 4B). Given this resistance, we reason that it was unlikely that the deamidation observed occurred only in the purification phase, which was shorter and relatively mild in comparison. We attempted to perform mass spectrometry analysis on unpurified vector to determine the extent of deamidation before and after purification, but were unsuccessful.

Likewise, heavy water controls indicate that processing specific to our mass spectrometry workflow do not contribute additional deamidation events (FIG. 4C).

To validate our mass spectrometry workflow, we examined two recombinant proteins that have been evaluated previously for deamidation; our findings (FIG. 5A and FIG. 5B) agree with the published results [Henderson, LE, Henriksson, D, and Nyman, PO (1976). Primary structure of human carbonic anhydrase C. *The Journal of biological chemistry* 251:5457-5463 and Carvalho, RN, Solstad, T, Bjorgo, E, Barroso, J F, and Flatmark, T (2003). Deamidations in recombinant human phenylalanine hydroxylase. Identification of labile asparagine residues and functional characterization of Asn→Asp mutant forms. *The Journal of biological chemistry* 278:15142-1515]. Additionally, we engaged a secondary institution to evaluate our raw data from AAV8. This independent analysis identified the same sites as deamidated, with minimal variation in the extent of modification at each site attributable to software-to-software variations in peak detection and area calculation (FIG. 6). Structural Topology, Temperature Factor, and the Identity of the N+1 Amino Acid Contribute to Deamidation Frequency

As the structure of AAV8 has been solved and published (PDB identifier: 2QA0) (Nam H J, et al. *J Virol* 2011; 85 (22): 11791-99), we next examined the AAV8 capsid structure for evidence of favorable conditions for non-enzymatic deamidation and to correlate percent deamidation with established structural features (Nam H J, et al. *J Virol* 2007; 81 (22): 12260-71). We focused on asparagine residues exclusively, as the factors influencing asparagine deamidation are better characterized in the literature and asparagine deamidation events are far more common than glutamine deamidation events (Robinson, NE, and Robinson, AB (2001). Molecular clocks. *Proc Natl Acad Sci USA* 98:944-949). We also determined the temperature (or B) factor for each of these residues from the AAV8 crystal structure; temperature factor is a measure of the displacement of an atom from its mean position, with higher values indicating a larger displacement, higher thermal vibration, and therefore increased flexibility (Parthasarathy S and Murphy M R. *Protein Science: A Publication of the Protein Society* 1997; 6:2561-7). The majority of asparagines of interest were located in or near the surface-exposed HVRs (Table 1), which are structurally favorable for deamidation and provide a solvent-exposed environment (Govindasamy L, et al. *J Virol* 2013; 87 (20): 11187-99). We found that residues located in these flexible loop regions were, on average, more frequently deamidated than residues in less flexible regions such as beta strands and alpha helices. For example, the NG residue at position N263 is part of HVR I, has a high temperature factor, and was >98% deamidated on average (FIG. 7A and FIG. 6, Table 1). N514, which was deamidated ~85% of the time (FIG. 3 and FIG. 6, Table 1), is also in an HVR (HVR V) with an N+1 glycine; however, the local temperature factor is relatively low in comparison to that of N263 due to its interaction with residues on other VP monomers at the three-fold axis. Less-favorable+1 residues and lower local temperature factors correlated with lower deamidation, even for HVR residues. For example, N517 was on average only 4% deamidated (Table 1); this residue has an equivalent temperature factor to the highly deamidated N514, but its N+1 residue is a serine, decreasing the likelihood of deamidation events due to steric hindrance. This demonstrates that a number of factors cumulatively determine the extent of deamidation at a given capsid position, although the identity of the +1 residue is apparently the most influential factor.

To test the role of the +1 residue in asparagine deamidation, we generated mutant vectors in which AAV8 NG sites were individually mutated at the +1 position to either alanine or serine. Model peptide studies indicate that NG peptides deamidate with a half-life as short as 1 day, whereas NA or NS peptides typically deamidate 25- or 16-fold more slowly, respectively (Robinson N E and Robinson A B. *Proc Natl Acad Sci USA*. 2001; 98 (8): 4367-72). Mass spectrometry analysis of the vector mutants confirmed the central role of the +1 site in determining the extent of vector deamidation. NG sites in this set (>80% deamidation in wt) showed selective stabilization of the adjacent asparagine when the +1 site was changed to alanine (<5% deamidation) or serine (<14% deamidation) (Table 2).

TABLE 2

position\ variant	Extent of deamidation (%) at five AAV8 NG sites in wt and six +1 site mutants						
	WT (average)	G58S	G58A	G264A	G386S	G386A	G515A
N57	81.8	8.4	1.9	89.7	89.7	91.6	93.6
N263	99.3	98.2	98.9	4.8	100.0	94.5	97.2
N385	89.1	96.3	94.8	97.1	13.5	2.5	97.0
N514	85.2	100.0	98.0	98.8	100.0	100.0	2.2
N540	84.5	95.0	92.6	97.9	96.9	86.1	89.5

Residues that were at least partially buried and not readily exposed to solvent and/or were located in regions of low local flexibility in the intact, fully assembled AAV8 capsid had a lower frequency of deamidation compared to those located in a more favorable environment (Table 1). Despite this, a few of the residues in unfavorable conditions were deamidated. For example, N630 is at least partially buried but still had a detectable degree of deamidation. For this residue, the presence of phenylalanine as the N+1 residue suggests that this region could be a novel site of non-enzymatic autoproteolytic cleavage within the AAV8 VP3 protein.

Structural Modeling of AAV8 VP3 Confirms Deamidation Events

To provide direct evidence of deamidation in the context of an assembled capsid, we evaluated the crystal structure of AAV8 (Nam H-J, et al. *J Virol* 2011; 85 (22): 11791-9). The resolution of the available crystal structure (i.e., 2.7 Å) of this serotype is not high enough to identify the terminal atoms in the R groups and, therefore, is insufficient to directly distinguish between asparagine, aspartic and isoaspartic acid residues. Other aspects of the structure of the isomer of aspartic acid that forms under these conditions provided us an opportunity to determine deamidation from the 2.7 Å structure. This analysis was based on two assumptions: 1) The predominant product of spontaneous deamidation of an asparagine is isoaspartic rather than aspartic acid, which is generated at a 3:1 ratio (Geiger T and Clarke S. *J Biol Chem* 1987; 262 (2): 785-94), and 2) an asparagine or aspartic acid can be differentiated from an isoaspartic acid because the electron density map corresponding to the R group of isoaspartic acid is shorter in length. This shorter R group is created when the beta carbon from the R group of isoaspartic acid is lost when incorporated into the main chain of the AAV8 VP3 capsid protein backbone following resolution of the succinimidyl intermediate during the deamidation reaction.

We first refined the AAV8 structure itself, generating an AAV8 capsid electron density that was not biased by the

known AAV8 VP3 sequence. We then examined the refined AAV8 crystal structure for evidence of deamidation based on the presence of a shorter R group associated with isoaspartic acid (FIG. 3A-FIG. 3E). The electron density map confirmed a shorter R group for the highly deamidated N+1 glycine residues at positions 263 (FIG. 3C), 385 (not shown), 514 (FIG. 3D), and 540 (FIG. 3E) when compared to the asparagine at 410 that had no deamidation detected by mass spectrometry (FIG. 3B). The deamidation indicated by the electron density map is therefore consistent with the data generated by mass spectrometry at these sites with >75% deamidation. The resulting isoaspartic acid models were comparable to isoaspartic acid residues observed in the crystal structures of other known deamidated proteins, supporting the validity of our analysis of AAV8 (Rao F V, et al. *Chem Biol*. 2005; 12 (1): 65-76; Noguchi S, et al. *Biochemistry* 1995; 34 (47): 15583-91; Esposito L, et al. *J Mol Biol* 2000; 297 (3): 713-32). This structural analysis serves as an independent confirmation of the deamidation phenomena observed when analyzing the AAV8 capsid via mass spectrometry.

Deamidation of the AAV Capsid is not Serotype Specific

We investigated serotypes beyond AAV8 for evidence of capsid deamidation. We examined AAV9 vector preparations using 2D gel electrophoresis (FIG. 11A) and mass spectrometry (FIG. 11B), including controls for potential vector-processing effects (FIG. 11D-FIG. 11F). The pattern and extent of AAV9 deamidation was similar to that of AAV8. All four AAV9 NG sites were >85% deamidated; 13 non-NG sites were deamidated to lesser extent, with a few sites showing high lot-to-lot variability in % deamidation. Next, we applied our structural analysis workflow and refit existing AAV9 crystallographic data (FIG. 11C, Table 3). As with AAV8, isoaspartic acid fit better into the electron density of several NG sites in the AAV9 crystal structure. We extended our 2D gel analysis (data not shown) and mass spectrometry (summarized in Table 4) to five additional evolutionarily diverse serotypes (rh32.33, AAV7, AAV5, AAV4, AAV3B and AAV1). All of the capsids examined contain a similar pattern and extent of deamidation, indicating that this modification is widespread in clinically relevant AAV vectors, and is determined by similar underlying primary-sequence and structural factors.

TABLE 3

Characteristics of AAV9 deamidated residues of interest.					Temperature factor (Å ²)
	N + 1 residue	Structural topology	Structural motif	Average % deamidation	
N57	G	N/A	N/A	97	N/A
N94	H	N/A	N/A	5	N/A
N253	N	Surface exposed	Not assigned	9	41
N254	H	Surface exposed	Not assigned	2	50
N270	D	Surface exposed	HVR I	11	65
N304	N	Buried	Alpha helix	23	35
N329	G	Surface exposed	HVR II	94	89
N409	N	Buried	Not assigned	9	36
N452	G	Surface exposed	HVR IV	98	64
N477	Y	Buried	Not assigned	2	33

TABLE 3-continued

Characteristics of AAV9 deamidated residues of interest.					
	N + 1 residue	Structural topology	Structural motif	Average % deamidation	Temper- ature factor (Å ²)
N512	G	Surface exposed	HVR V	89	48
N515	S	Surface exposed	HVR V	3	47
N651	T	Buried	HI loop	1	38
N663	K	Surface exposed	HI loop	4	49
N668	S	Surface exposed	HI loop	13	52
N704	Y	Surface exposed	HVR IX	5	68
N709	N	Surface exposed	HVR IX	5	55

Conserved asparagine residues with homologous N + 1 residues (in comparison to AAV8) are denoted in *italics* (determined by alignment of the full-length amino acid sequences of AAV8 and AAV9 VP1).

TABLE 4

Extent of deamidation observed for diverse serotypes						
serotype	vector preps analyzed	Average % sequence Coverage by MS	# of NGs	average NG % deamidation	# of non NG sites observed deamidated	average non- NG % deamidation
AAV1	3	91.4	4	95.6	19	12.9
AAV3B	1	89.8	4	97.0	9	9.4
AAV4	3	84.7	4	96.2	15	15.3
AAV5	1	88.7	3	88.7	11	15.3
AAV7	1	90.9	4	92.1	9	13
AAV8	21	93.4	5	90.5	37	7.4
AAV9	7	90.2	4	95.5	26	5.3
rh32.33	1	100	3	97.4	14	16.2

Deamidation Events can Affect Capsid Assembly and Transduction Efficiency

One approach to testing the functional impact of deamidation is by substituting asparagine with aspartate by genetic mutation. We generated an aspartate mutant vector encoding a luciferase reporter for each deamidated AAV8 asparagine by small-scale triple transfection of 293 cells, and titrated the vectors by qPCR of DNaseI resistant genome copies (FIG. 8A). The mutations rarely affected capsid assembly relative to wtAAV8, and effects were limited to mostly buried, non-NG sites with low overall deamidation in the wt vector. Next, we assessed the mutation panel for in vitro transduction efficiency of human liver-derived Huh7 cells (FIG. 8B). Several mutants showed impaired transduction efficiency, with positions N57, N94, N263, N305, Q467, N479, and N653 exhibiting >10-fold transduction loss. We observed a similar number of sensitive sites for AAV9 (FIG. 11G and FIG. 11H). As typically only a fraction of residues at a given position are deamidated endogenously, this approach has the potential to overestimate functional loss for proteins such as capsids where the functional unit is a homomeric assembly; endogenous modification at one capsid site may be compensated for by a neighboring subunit with an intact residue. Nonetheless, we reasoned that the method could help prioritize deamidated residues for future monitoring during manufacturing or mutational stabilization. Functional data from populations of endogenously deamidating vectors will be required to place this loss-of-function mutagenesis data in the proper context.

Vector Activity Loss Through Time is Correlated with Progressive Deamidation

Given the apparently short half-life of NG deamidation, we reasoned that vector samples differing in age by as little as 1 day could show distinct deamidation profiles, thus providing an opportunity to correlate endogenous deamidation to function. Our large-scale vector preparation protocol calls for triple transfection of 293 cells followed by 5 days of incubation for vector production and 1-2 days for vector purification. To approximate this process, we prepared medium scale triple transfections (10×15 cm cell culture dishes each) of 293 cells with wt AAV8. We collected vector (2×15 cm cell culture dishes/day) at 1 day intervals for 5 days, preserving the timepoints until the end of the 5 day period by freezing vector at -80 C. Next, we assessed crude vector titer and in vitro transduction efficiency as described above. As expected, the number of assembled, DNaseI-resistant genome copies increased over time (FIG. 9A). We then quickly processed crude vector for early (day 1 and 2) and late (day 5) timepoints by affinity purification and

measured in vitro transduction efficiency of huh7 cells. Relative transduction efficiency of the vector dropped progressively over time (FIG. 9B). In terms of transgene expression per GC added to target cells, day 5 vector was only 40% as efficient as day 1 material. This activity drop was observed for crude material as well, indicating a change in molecular composition before purification (FIG.). We observed a similar trend in activity loss for AAV9 over 5 days, with approximately 40% reduction in vector potency (FIG. 11I-FIG. 11K).

Next we measured deamidation of the time course samples by mass spectrometry. NG site deamidation progressed substantially over every interval, with an average of 25% deamidation at day 1, and >60% of sites converted by day 5 (FIG. 9C). Non-NG site deamidation generally progressed over 5 days, although at much lower levels and with less consistency between days 2 and 5 (FIG. 9D). The data correlates endogenous vector deamidation to an early time-point decay in specific activity, and highlights a potential opportunity to capture more active vector by shortening the production cycle or finding capsid mutations that stabilize asparagines.

We note that the material used for mass spectrometry analysis in FIG. 2A-FIG. 2E was at least 7 days post-transfection, due to an additional 2 days for purification. The higher NG site deamidation in these samples (>80%) indicates that deamidation likely continues after the period of expression and during the recovery and purification processes at approximately the same rates until NG sites are

completely deamidated or the vector sample is frozen. Thus deamidation is largely determined by the age of the vector and is not a process that is exclusive to or caused by the recovery and purification process. The much lower deamidation values in the day 1 material vs the day 5 material (both affinity purified) underscore this point.

Stabilizing NG Asparagines can Improve Vector Performance

Given the correlation between vector NG deamidation and transduction efficiency loss, we reasoned that stabilizing NG amides by +1 site mutagenesis may improve vector function. We produced vector in small scale for AAV8 NG site mutants in which each +1 residue was individually converted to alanine or serine. Single+1 mutants were well tolerated in terms of vector assembly (FIG. 10A) and transduction efficiency (FIG. 10B). G386 substitutions, located near a previously defined “dead zone” on the capsid surface (Aydemir F, et al. *J Virol* July 2016; 90 (16): 7196-204), were defective for in vitro transduction. The loss of function for G386 mutants could indicate a preference for a deamidated asparagine at N385. Alternatively, the additional sidechain bulk at the +1 position may have a negative impact on function that is independent of amide-group stabilization. No single-site mutants significantly improved in vitro transduction, in spite of dramatic stabilization of their neighboring asparagines (Table 2). Because in vitro and in vivo transduction activities can be discordant, we tested a subset of the single-site+1 mutants for liver transduction in C57BL/6 mice. We performed intravenous tail vein injection (n=3 to 5) and examined luciferase expression by imaging weekly for 2 weeks (FIG. 10C). In vivo and in vitro transduction data were in agreement to within the associated errors of each assay (i.e., within the error range). G386 substitutions were defective for transduction, while+1 site mutations at other positions were largely tolerated, transducing liver at levels equivalent to but not exceeding wtAAV8.

Because stabilizing the amide at any one NG site may be necessary but not sufficient for functional restoration, we next evaluated vector variants with combinations of +1 site alanine substitutions. We recombined all 3 AAV8 NG sites for which the +1 alanine was highly functional (N263, N514, and N540). Some combinations, including the triple mutant G264A/G515A/G541A, assembled poorly and were dysfunctional for transduction. However, both pairwise combinations involving N263 (G246A/G515A and G264A/G541A) improved in vitro transduction efficiency (2.0- and 2.6-fold over wtAAV8, respectively) with no loss of titer (FIG. 10D). Because these mutations introduce at least two changes (N-amide stabilization and a +1 residue side chain substitution) these data do not conclusively link NG deamidation to functional loss. However, the data are consistent with the model established in the timecourse study in which NG site deamidation can impact in vitro transduction efficiency.

Functional Asparagine Substitutions Improve Lot-to-Lot Reproducibility in Vector Manufacturing

Another potentially problematic aspect of the vector deamidation profiles we report is the high lot-to-lot variability in deamidation at some positions. For wtAAV8, this variability is most pronounced for N459 (observed deamidation ranging from 0% to 31%) and N499 (observed deamidation ranging from 0% to 53%). Variability in post-translational modifications is typically de facto avoided during biologics development, either by avoiding clones altogether that exhibit this variability, carefully monitoring

and controlling production strains and conditions, or by protein engineering of the affected candidate.

As we were unable to determine the production or processing factors contributing to N459 and N499 deamidation variability (FIG. 2E), we sought functional amino acid substitutions at these positions. We first evaluated small scale vector preparations for conservative substitutions to glutamine at each position individually. Both N459Q and N499Q were assembled efficiently into vector, and were equivalent to the wtAAV8 reference for in vitro transduction efficiency (FIG. 7A). Next, we produced the mutants in large scale and performed mass spectrometry. Consistent with our observations of extremely rare glutamine deamidation, we observed selective and complete stabilization of the glutamine amides at positions 459 or 499 in these mutants (data not shown). We evaluated these mutant lots in vivo as above for liver transduction after tail vein injection in C57BL/6 mice (FIG. 7B and FIG. 7C). The wtAAV8 vector lot used as a control in this experiment was deamidated 16.8% at N499, but no deamidation was detected at N459 (data not shown). Liver transduction at day 14 for both mutants was equivalent to wtAAV8. This data demonstrates the potential for a protein engineering approach to address the molecular variability associated with deamidation in manufactured AAV vectors.

C. Discussion

We identified and evaluated non-enzymatic deamidation of asparagine and glutamine residues on the AAV8 capsid independently by 2D gel electrophoresis, mass spectrometry, de novo protein modeling, and functional studies both in vitro and in vivo. Deamidation has been shown to occur in a wide variety of proteins and to significantly impact the activity of biologics, including antibody-based therapeutics (Nebija D et al. *Int J Mol Sci* 2014; 15 (4): 6399-411) and peptide-based vaccines (Verma A et al. *Clin Vaccine Immunol.* 2016; 23 (5): 396-402). Other viral proteins, such as the VP6 protein of rotavirus, have been shown by mass spectrometry to undergo deamidation events (Emslie K R et al. *Funct Integr Genomics* 2000; 1 (1): 12-24).

The context in which these deamidations occurred in AAV8 suggested that they are the result of spontaneous non-enzymatic events. Asparagine residues are known to be more extensively deamidated than glutamine residues; the amino acid downstream of the asparagine substantially influences the rate of deamidation with an N+1 of glycine (i.e., NG) being the most efficiently deamidated. We observed remarkable confirmation of the role of the N+1 amino acid in deamidation of AAV capsids in that every NG present in VPI was deamidated at levels >75% while deamidation was never consistently >20% in any of the other asparagines or glutamines in the capsid. Virtually all NG motifs in the AAV8 and AAV9 capsids (i.e., 7/9) were also present on the surface of the capsid contained in HVR regions that are associated with high rates of conformational flexibility and thermal vibration. This is consistent with previous reports of NG motifs of other proteins that are located in regions where flexibility may be required for proper protein function and not in more ordered structures, such as alpha helices or beta sheets (Yan B X and Sun Y Q *J Biol Chem* 1997; 272 (6): 3190-4). The preference of NG motifs in surface exposed HVRs further enhances the rate of deamidation by providing solvent accessibility and conformational flexibility, thereby facilitating the formation of the succinimidyl intermediate. As predicted, less favorable environments lead to much lower rates of deamidation.

An important question regarding the biology of AAV and its use as a vector is the functional consequences of these

deamidations. Mutagenesis of the capsid DNA to convert an asparagine to an aspartic acid allows for an evaluation of capsids in which all amino acids at a particular site are represented as aspartic acids. However, no easy strategy exists to use mutagenesis to prevent deamidations other than potentially mutating the N+1 residue, which is confounded by direct consequences of the second site mutation. We studied a limited number of variants in which the asparagine residue was converted to an aspartic acid by mutagenesis. Functional analysis included capsid assembly and in vitro and in vivo transduction. The most substantial effects of mutagenesis on vector function were those involving asparagines that were incompletely deamidated at baseline and were not surface exposed. What was surprising, however, was that mutagenesis of the highly deamidated asparagine at 514 to an aspartic acid did have some effect on function. This result suggests that the presence of residual amounts of the corresponding amide may influence function. This could be due in part to the presence of hydrogen bond interactions between N514 and D531 of another three-fold related VP3 monomer (identified in the wtAAV8 crystal structure) that are lost upon conversion of this residue to aspartic acid following deamidation.

A better understanding of the factors that influence the extent of deamidation in AAV vectors is important when assessing the impact of these deamidations on the development of novel therapeutics. Incubation of vectors under extreme conditions, known to markedly accelerate deamidation kinetics, had little effect. Coupled with our isotope incorporation studies, this result suggests that deamidation occurs during capsid assembly and is not an artifact of vector processing or mass spectrometry analysis. Deamidations at NG sites are unlikely to have substantive impact on vector performance, as the reaction was virtually complete in every sample that we evaluated. However, our initial functional studies suggest that residual amounts of non-deamidated asparagines can contribute to function. We are more concerned about sites where deamidation was less complete, which in most cases was also associated with sample-to-sample variation. An example is the asparagine at position 499 that showed deamidation ranging from 0% to 53% with a mean of 17%. It is possible that subtle differences in the conditions of vector production could contribute to this heterogeneity. The striking similarity in deamidation in AAV8 and AAV9 suggests this is a property of this entire family of viruses.

In summary, we discovered substantial heterogeneity in the primary amino acid structure of AAV8 and AAV9 capsid proteins. These studies potentially impact the development of AAV as vectors in several ways. First, the actual amino acid sequences of the VP proteins are not what are predicted by the corresponding DNA sequences. Second, aspects of the production method could lead to variations in deamidation and corresponding changes in vector function. Until we have a handle on the factors that influence deamidation rates at non-NG sites and a better understanding of their functional consequences it may be necessary to include deamidation in the characterization of clinical-grade AAV vectors. 2D gel electrophoresis can provide an overall assessment of net deamidation, although mass spectrometry will be necessary to assess deamidation at specific residues.

Example 2: Deamidation AAV5.5.9

The novel sequences of AAV5.5.9 are provided in SEQ ID NO: 9 and 10, respectively. AAV5.5.9 vectors were assessed

for deamidation as described in Example 1 for AAV9. Highly deamidated residues are seen at N57, N319, N442, N502.

Modification	
SEQ ID NO: 10	
WL2019CS	
Enzyme	Trypsin
% Coverage	97.4
N35 + Deamidation	7.8
~N57 + Deamidation	99.7
N113 + Deamidation	3.6
~N204 + Deamidation	13.9
N217 + Deamidation	2.2
~N243 + Deamidation	19.0
Q249 + Deamidation	11.4
N293/294 + Deamidation	37.3
N304 + Deamidation	6.2
N309 + Deamidation	0.7
Q311 + Deamidation	0.3
~N319 + Deamidation	83.9
N399/400 + Deamidation	30.8
~N442 + Deamidation	97.7
N467 + Deamidation	2.6
N502 + Deamidation	100.0
N505 + Deamidation	18.6
~Q589 + Deamidation	21.1
N618 + Deamidation	6.6
~N641 + Deamidation	8.1
N653 + Deamidation	8.3
~N658 + Deamidation	21.7
N694 + Deamidation	0.6
~N699 + Deamidation	8.6

Example 3: Deamidation AAVrh79 (Clade E)

AAVrh79 was isolated from DNA extracted from small bowel tissue of rhesus macaque. It has been characterized phylogenetically as being within Clade E (FIG. 14A-14D). Its sequences are provided herein, with the nucleotide sequences being in SEQ ID NO: 1 and the amino acid sequence being in SEQ ID NO:2. An alignment of the amino acid sequences of AAVrh79, AAVrh.10 and AAVhu.37 are provided in FIG. 14A. An alignment of the nucleic acid sequences of AAVrh79, AAVrh.10 and AAVhu.37 are provided in FIG. 14B-14D.

AAVrh79 has three amino acid differences in its primary sequence. Whereas AAVhu37 has an Ala located at position 67 and a Lys at position 169 of its primary VPI sequence, AAVrh79 has a glutamic acid (E) at position 67 and an Arg at position 169. Differences in the DNA sequences of VPI among rh.79, hu.37, and hu.40 are shown in FIG. 11B. Vectors expressing eGFP based on the various clade E variants were prepared and evaluated for their relative infectivity of Huh7 cells (FIG. 11C). C57BL/6 mice were injected with two dosage levels (3×10^{10} and 3×10^{11} GC/mouse) of eGFP-expressing AAV8 or AAVrh.79 vectors and infectivity was assessed by fluorescence microscopy (Data not shown).

Vectors based on AAVrh79 were prepared using known production techniques using the AAVrh79 nucleotide sequence for production of the cap, such as previously described for AAV8 vectors. The results of the production yield and production purity assessments are provided in FIGS. 15A-15B and FIG. 16, respectively.

To assess expression levels using AAVrh79 containing a marker gene (firefly luciferase), male RAG KO mice at 6-8 weeks of age were injected intramuscularly with 3×10^{11} GC/mouse of vector performed using a Hamilton syringe. flLuc expression was visualized by whole-body bioluminescence.

cence imaging as previously described [Greig J A, Peng H, Ohlstein J, Medina-Jaszek C A, Ahonkhai O, Mentzinger A, et al. (2014) *Intramuscular Injection of AAV8 in Mice and Macaques Is Associated with Substantial Hepatic Targeting and Transgene Expression*. PLOS ONE 9 (11): el12268_doi_org/-10_1371/journal_pone_-0112268.] The results are provided in FIGS. 17A-17D.

Expression of AVV8triple, AAVhu68, AAV9, AAV8, and AAVrh79 vectors was compared following intramuscular administration of 10^{13} GC/kg AAVrh79 into male and female cynomolgus macaques (FIG. 17E).

Vectors expressing a secreted transgene (201lg IA) were administered intramuscularly into the gastrocnemius muscle of male RAG KO mice ($n=5/\text{group}$) (3×10^{10} or 3×10^{11} GC/mouse). The results indicated that AAV8triple expresses better following IM injection and at the lower dose tested the difference in expression from AAV8triple was substantial. At higher the higher dose, AVVrh79 expressed at levels at comparable to the other vectors tested (FIG. 17F).

A female cynomolgus macaque (RA2362) was pre-screened for NABs (FIG. 18A) and injected with

AAVG2.TBG.eGFP.WPRE.bGH (1×10^{13} ddGC/kg, intravenously). The animal was euthanatized 7 days following treatment, and a necropsy was performed to isolate the liver and other tissues for analysis. GFP expression in the liver and spleen was evaluated on day 7 (FIGS. 18B and 18C). Levels of vector detected in various tissues are shown in FIG. 18D. Levels of GFP expression were evaluated in the livers animals that received AAVG2 (RA2362) or AAV8 and AAVG3 vectors (FIG. 18E-FIG. 1J). FIG. 18K shows levels of vector detected in various tissues from these animals.

Various AAV8 and AAVrh79 vectors were generated and, in some instances, multiple lots were produced. The yields of these AAV9 and AAVrh79 vector lots were compared (FIG. 19).

AAVrh79 vectors were assessed for deamidation as described in Example 1 for AAV8 and AAV9. The results show that the vectors contain four amino acids which are highly deamidated (N57, N263, N385, N514), which correspond to asparagines in asparagine-glycine pairs, based on the numbering of AAVrh79 (SEQ ID NO: 1). Lower deamidation percentages are consistently observed in residues N94, N254, N410.

Modification AAVrh79 SEQ ID NO: 2	WL1781S	WL1784S	WL1785S	WL1781S	WL1784S	WL1785S
Enzyme	Trypsin	Trypsin	Trypsin	Chymotrypsin	Chymotrypsin	Chymotrypsin
% Coverage	89.6	93.9	92.4	91.3	88.7	89.9
N57 +	99.3	80.9	82.9	99.6	80.1	86.4
Deamidation						
N94 +	10.4	9.6	9.9	10.5	9.4	10.0
Deamidation						
~N254 +	16.0	15.8	16.4	15.3	16.3	16.7
Deamidation						
~N263 +	84.3	93.5	95.3	82.9	89.5	90.6
Deamidation						
~N305 +	3.2	2.5	2.4	3.2	2.6	2.3
Deamidation						
~N385 +	79.1	100.0	100.0	76.9	96.6	92.9
Deamidation						
~N410 +	2.0	17.8	23.9	2.0	17.6	23.0
Deamidation						
N479 +	2.0	2.0	1.9	2.0	2.0	2.0
Deamidation						
~N514 +	100.0	97.2	97.0	97.4	94.6	98.1
Deamidation						
~Q601 +	0.1					
Deamidation						
N653 +	1.3	1.1	1.4	1.3	1.1	1.5
Deamidation						
~R487 +	0.1	0.2	0.1	0.1	0.2	0.1
Methylation						
D97 +	1.3			1.2		
Isomerization						
S149 +	51.9	49.0	53.2	49.6	46.6	55.8
Phosphorylation						
~S153 +	59.7	54.3	51.0	59.8	51.7	48.5
Phosphorylation						
~S474 +		7.3	4.5		7.0	4.3
Phosphorylation						
~T570 +	46.3	36.4	21.6	45.9	35.9	21.4
Phosphorylation						
~S665 +	0.5	0.3	0.4	0.4	0.3	0.4
Phosphorylation						
W248 +			0.9			
Oxidation						
W307 +	1.7	1.2	1.5	1.8	1.3	1.5
Oxidation						
W307 +	0.3		0.4			
Oxidation to						
kynurenine						
M405 +	5.8			6.0		
Oxidation						
M437 +	15.0	5.2	95.4	15.1	5.3	12.0
Oxidation						

-continued

Modification AAVrh79 SEQ ID NO: 2	WL1781S	WL1784S	WL1785S	WL1781S	WL1784S	WL1785S
M473 + Oxidation	16.7	7.4	7.9	16.0	7.6	7.6
W480 + Oxidation	4.6	0.4	4.6	4.7	0.4	4.6
W480 + Oxidation to kynurenine			0.1			
W505 + Oxidation	2.4	2.2	1.3	2.5	2.1	1.3
M526 + Oxidation	19.0			18.6		
M544 + Oxidation	30.9	20.7		31.4	20.3	
M561 + Oxidation	15.6	7.7		16.0	7.5	
W621 + Oxidation to kynurenine			0.0			
M637 + Oxidation	6.9	12.5	4.5	6.9	12.6	4.7
W697 + Oxidation		0.5	0.6		0.5	0.6

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Example 4: Preparation of AAV8.2.08

As discussed in WO 2017/180854 (incorporated herein by reference), several AAV8 mutants were generated c41, c42, c46, g110, g113, g115 and g117 with mutations in the HVR.VIII region. As discussed in Gurda et al, the major ADK8 epitope lies in the HVR. VIII region (amino acids 586 to 591 using AAV8 vp1 numbering). Those mutants were tested in vitro for ADK8 resistance and some of them were tested in vivo for ADK8 resistance. See, e.g., Lochrie 2006 cited above.

AR2.1-9 were randomly picked. AR2.25-61 were selected based on frequency. The randomly picked variants, as well as those higher-frequency ones, are alive, in terms of 6-well plate yield and Huh7 transduction. FIG. 24—shows expression in various tissues of AAV8.AR2.08 (left most set of bars).

Example 5: Deamidation of AAV8.AR2.08

The novel sequences of AAV8.AR2.08 are provided in SEQ ID NO: 17 and 18, respectively, which were designed as in Example 4.

A. Modifications AAV8.AR2.08 vectors were produced and assessed for modifications as described in Example 1 for AAV8. The results show that the vectors contain five amino acids which are highly deamidated (N57, N263, N385, N514, and N540), which correspond to asparagines in asparagine-glycine pairs, based on the numbering of AAV8.AR2.08 (SEQ ID NO: 18). Lower deamidation percentages are consistently observed in residues N94, N254, N410. In contrast to AAV8, deamidation is not observed at position N459 (average 7% in AAV8) or N499 (average 17% in AAV8).

AAV8.AR2.08 Modification SEQ ID NO: 18	WL1846CS	WL1846CS
Enzyme	Trypsin	Chymotrypsin
% Coverage	97.4	92.3

-continued

AAV8.AR2.08 Modification SEQ ID NO: 18	WL1846CS	WL1846CS
N57 + Deamidation	90.7	89.5
N94 + Deamidation	9.0	9.3
~N254 + Deamidation	11.8	11.7
~N263 + Deamidation	88.6	86.3
~N305 + Deamidation	5.8	5.5
~N385 + Deamidation	86.1	83.3
~N514 + Deamidation	100.0	99.6
~N521 + Deamidation	2.0	2.1
~N540 + Deamidation	78.6	80.5
N590 + Deamidation	0.4	0.4
Q601 + Deamidation	0.5	0.6
N653 + Deamidation	0.8	0.8
N665 + Deamidation	1.2	1.2
D442 + Isomerization	12.0	
D584 + Isomerization	1.0	
~S149 + Phosphorylation	95.9	15.4
~T417 + Phosphorylation	0.0	
~T454 + Phosphorylation	0.1	
~T493 + Phosphorylation	0.1	
S600 + Phosphorylation	1.1	
~T663 + Phosphorylation	0.0	
~W22 + Oxidation	1.0	1.0
~M204 + Oxidation	0.1	0.1
~M212 + Oxidation	2.9	2.9
W248 + Oxidation	0.7	0.7
W307 + Oxidation	0.6	0.6
M405 + Oxidation	0.3	0.3
M437 + Oxidation	70.3	21.0
M473 + Oxidation	1.7	1.7
W480 + Oxidation	0.3	0.3
W505 + Oxidation	0.6	0.6
M526 + Oxidation	1.0	1.0
M561 + Oxidation	1.0	1.0
M607 + Oxidation	2.4	2.5
~W609 + Oxidation	0.1	0.1
W621 + Oxidation	0.8	0.8
M637 + Oxidation	2.9	3.0
W697 + Oxidation	0.2	0.2

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B. Single cell RNA-seq reveals tissue localization and transcriptional signatures of transduced hepatocytes isolated from non-human primates following treatment with AAV8

Single cell RNA sequencing has proven to be a powerful technique to characterize the cellular transcriptome with unprecedented, single cell resolution. In our current work, we utilize single cell RNA-seq to study the transcriptional landscape of primary hepatocytes isolated from rhesus macaques following treatment with an AAV8 vector expressing GFP. Transcriptome analysis of FACS-sorted GFP+ and GFP- cells reveals tissue localization of transduced cells within the hepatic lobule as well as genes and regulatory pathways involved in hepatocyte transduction and the regulation of transgene expression.

For our study design, rhesus macaques were treated with either 1×10¹³ ddGC/kg AAV8.TBG.EGFP.WPRE (n=1) or 1×10¹³ ddGC/kg AAV8.2.08.TBG.EGFP.WPRE (AAV8 variant, n=1). Animals were euthanatized 7 days following treatment, and necropsies were performed to isolate the liver from both animals. Following treatment with collagenase and gradient centrifugation, isolated hepatocytes were FACS sorted by GFP transgene expression onto BD Precise™ 96 well plates. 192 single cells were isolated from each animal (96 GFP+ and 96 GFP-) and were subsequently used to prepare single cell RNA-seq libraries following the standard BD Precise™ protocol. Data were analyzed using the Seurat, Scran, and Scater packages in R in order to determine differentially expressed transcripts between GFP- and GFP+ sorted cells and to perform spatial reconstruction of isolated cells within the hepatic lobule using established transcriptional expression signatures.

AAV8.AR2.08 was found to have an increased liver tropism and exhibited a 1.5-fold increase in transduction efficiency as compared to AAV8. Single cell transcriptome analysis of sorted hepatocytes reveals transgene-expressing cells are evenly distributed across the hepatic lobule, showing a slight preference for the periportal region, which was also observed by histopathology. Interestingly, a subpopulation of sorted GFP- cells are found to express the transgene transcript at levels comparable to sorted GFP+ cells, suggesting that these cells are in fact transduced and express transgene mRNA, despite the absence of the detectable levels of translated protein. Comparing the transcriptional profiles of GFP- and GFP+ cells reveals differentially expressed transcripts involved in viral mRNA translation, elucidating possible pathways involved in the regulation of transgene protein expression in transduced cells.

C. Isolation of an adeno-associated virus 8 variant with better liver transduction and higher liver specificity in nonhuman primates with directed evolution through a human liver xenograft model

To derive AAV variants with better transduction and higher specificity, we performed AAV-directed evolution, using saturation mutagenesis targeting surface exposed sites on the capsid of AAV8, the benchmark for liver gene

therapy, followed by two rounds of in vivo enrichment in human liver xenograft mouse model, and isolated an AAV8 variant called AAV8.2.08. After intravenous injection into nonhuman primates at a dose of 1e13 genome copies (GC)/kg body weight, vector genome copies delivered by AAV8.AR2.08 in various organs (including lung, heart, stomach, pancreas, kidney, and mesenteric lymph nodes) decreased while the delivery increased in liver, compared to AAV8, implying better liver transduction and higher tissue specificity. Next generation sequencing indicated significant enrichment of AAV8.AR2.08 during the in vivo selections, demonstrating the potential of the approach for isolating capsids with new and improved tropisms. Other comparisons are shown in FIGS. 21 to 27). FIG. 28 shows biodistribution of AAV8.AR2.08 and AAV8.

D. Barcoding

Black 6 mice were injected I.V. with 2×10¹² GC/mouse of a mixture of rAAVG3 derived from 12 preparations (FIG. 18B). Each preparation contains a separate barcode within the vector genome allowing identification of the specific preparation (FIG. 18A). After two weeks, animals were euthanized, and tissues harvested. As predicted, rAAVG3 expression was higher in liver than heart or muscle. FIG. 18C. Tissue distribution experiments show that actual frequencies match theoretical frequencies of barcodes in injected vector mix (FIG. 18D, total; FIG. 19A, 19B, muscle; FIG. 19C, 19D, heart; and FIG. 19E, 19F, liver), with slight anomalies in BC02 and BC06 (FIGS. 20A-20C). All documents cited in this specification are incorporated herein by reference. U.S. Provisional Patent Application Nos. 62/722,388 and 62/722,382, both filed Aug. 24, 2018, U.S. Provisional Patent Application Nos. 62/703,670 and 62/703,673, both filed Jul. 26, 2018, U.S. Provisional Patent Application Nos. 62/677,471 and 62/677,474, both filed May 29, 2018, U.S. Provisional Patent Application No. 62/667,585, filed May 29, 2018, and U.S. Provisional Patent Application No. 62/635,964, filed Feb. 27, 2018 are incorporated herein by reference. U.S. Provisional Patent Application No. 62/667,881, filed May 7, 2018, U.S. Provisional Patent Application No. 62/667,888, filed May 7, 2018, U.S. Provisional Patent Application No. 62/667,587, filed May 6, 2018, U.S. Provisional Patent Application No. 62/663,797, filed Apr. 27, 2018, U.S. Provisional Patent Application No. 62/663,788, filed Apr. 27, 2018, U.S. Provisional Patent application No. 62/635,968, filed Feb. 27, 2018 are incorporated by reference. The SEQ ID NO which are referenced herein and which appear in the appended Sequence Listing are incorporated by reference. While the invention has been described with reference to particular embodiments, it will be appreciated that modifications can be made without departing from the spirit of the invention. Such modifications are intended to fall within the scope of the appended claims.

SEQUENCE LISTING

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<210> SEQ ID NO 1

<211> LENGTH: 2214

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: AAVrh79

<220> FEATURE:

<221> NAME/KEY: CDS

<222> LOCATION: (1) .. (2214)

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<400> SEQUENCE: 1

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1 5 10 15	
 gag ggc att cgc gag tgg tgg gac ctg aaa cct gga gcc ccc aag ccc	96
Glu Gly Ile Arg Glu Trp Trp Asp Leu Lys Pro Gly Ala Pro Lys Pro	
20 25 30	
 aag gcc aac cag cag aag cag gac gac ggc cgg ggt ctg gtg ctt cct	144
Lys Ala Asn Gln Gln Lys Gln Asp Asp Gly Arg Gly Leu Val Leu Pro	
35 40 45	
 ggc tac aag tac ctc gga ccc ttc aac gga ctc gac aag ggg gag ccc	192
Gly Tyr Lys Tyr Leu Gly Pro Phe Asn Gly Leu Asp Lys Gly Glu Pro	
50 55 60	
 gtc aac gag gcg gac gcc gcg gcc ctc gag cac gac aag gcc tac gac	240
Val Asn Glu Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp	
65 70 75 80	
 cag cag ctc aaa gcg ggt gac aat ccg tac ctg cgg tat aac cac gcc	288
Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Arg Tyr Asn His Ala	
85 90 95	
 gac gcc gag ttt cag gag cgt ctg caa gaa gat acg tct ttt ggg ggc	336
Asp Ala Glu Phe Gln Glu Arg Leu Gln Glu Asp Thr Ser Phe Gly Gly	
100 105 110	
 aac ctc ggg cga gca gtc ttc cag gcc aag aag cgg gtt ctc gaa cct	384
Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Val Leu Glu Pro	
115 120 125	
 ctc ggt ctg gtt gag gaa gct gct aag acg gct cct gga aag aag aga	432
Leu Gly Leu Val Glu Glu Ala Lys Thr Ala Pro Gly Lys Lys Arg	
130 135 140	
 ccg gta gaa ccg tca cct cag cga tcc ccc gac tcc tcc acg ggc atc	480
Pro Val Glu Pro Ser Pro Gln Arg Ser Pro Asp Ser Ser Thr Gly Ile	
145 150 155 160	
 ggc aaa aaa ggc cag cag ccc gcg aga aag aga ctg aac ttt ggg cag	528
Gly Lys Lys Gly Gln Gln Pro Ala Arg Lys Arg Leu Asn Phe Gly Gln	
165 170 175	
 act ggc gac tca gag tca gtc ccc gac cct caa cca atc gga gaa cca	576
Thr Gly Asp Ser Glu Ser Val Pro Asp Pro Gln Pro Ile Gly Glu Pro	
180 185 190	
 cca gca ggc ccc tct ggt ctg gga tct ggt aca atg gct gca ggc ggt	624
Pro Ala Gly Pro Ser Gly Leu Gly Ser Gly Thr Met Ala Ala Gly Gly	
195 200 205	
 ggc gct cca atg gca gac aat aac gaa ggc gcc gac gga gtg ggt agt	672
Gly Ala Pro Met Ala Asp Asn Asn Glu Gly Ala Asp Gly Val Gly Ser	
210 215 220	
 tcc tca gga aat tgg cat tgc gat tcc aca tgg ctg ggc gac aga gtc	720
Ser Ser Gly Asn Trp His Cys Asp Ser Thr Trp Leu Gly Asp Arg Val	
225 230 235 240	
 atc acc acc agc acc cga acc tgg gcc ctg ccc acc tac aac aac cac	768
Ile Thr Thr Ser Thr Arg Thr Trp Ala Leu Pro Thr Tyr Asn Asn His	
245 250 255	
 ctc tac aag caa atc tcc aat ggg aca tcg gga gga agc acc aac gac	816
Leu Tyr Lys Gln Ile Ser Asn Gly Thr Ser Gly Gly Ser Thr Asn Asp	
260 265 270	
 aac acc tac ttc ggc tac agc acc ccc tgg ggg tat ttt gac ttc aac	864
Asn Thr Tyr Phe Gly Tyr Ser Thr Pro Trp Gly Tyr Phe Asp Phe Asn	
275 280 285	
 aga ttc cac tgt cac ttc tca cca cgt gac tgg cag aga ctc atc aac	912
Arg Phe His Cys His Phe Ser Pro Arg Asp Trp Gln Arg Leu Ile Asn	
290 295 300	

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aac aac tgg gga ttc cgg ccc aag aga ctc agc ttc aag ctc ttc aac	960
Asn Asn Trp Gly Phe Arg Pro Lys Arg Leu Ser Phe Lys Leu Phe Asn	
305 310 315 320	
atc cag gtt aag gag gtc acg cag aat gaa ggc acc aag acc atc gcc	1008
Ile Gln Val Lys Glu Val Thr Gln Asn Glu Gly Thr Lys Thr Ile Ala	
325 330 335	
aat aac ctt acc agc acg att cag gta ttt acg gac tcg gaa tac cag	1056
Asn Asn Leu Thr Ser Thr Ile Gln Val Phe Thr Asp Ser Glu Tyr Gln	
340 345 350	
ctg ccg tac gtc ctc ggc tcc gcg cac cag ggc tgc ctg cct ccg ttc	1104
Leu Pro Tyr Val Leu Gly Ser Ala His Gln Gly Cys Leu Pro Pro Phe	
355 360 365	
ccg gcg gat gtc ttc atg att ccc cag tac ggc tac ctg aca ctg aac	1152
Pro Ala Asp Val Phe Met Ile Pro Gln Tyr Gly Tyr Leu Thr Leu Asn	
370 375 380	
aac gga agt caa gcc gta ggc cgt tcc tca ttc tac tgc ctg gaa tat	1200
Asn Gly Ser Gln Ala Val Gly Arg Ser Ser Phe Tyr Cys Leu Glu Tyr	
385 390 395 400	
ttt cca tct caa atg ctg cgg act gga aac aac ttt gaa ttt agc tac	1248
Phe Pro Ser Gln Met Leu Arg Thr Gly Asn Asn Phe Glu Phe Ser Tyr	
405 410 415	
acc ttt gag gac gtg ccc ttc cac agc agc tac gca cac agc cag agc	1296
Thr Phe Glu Asp Val Pro Phe His Ser Ser Tyr Ala His Ser Gln Ser	
420 425 430	
ctg gac ccg ctg atg aac cct ctc atc gac cag tac ctg tat tac cta	1344
Leu Asp Arg Leu Met Asn Pro Leu Ile Asp Gln Tyr Leu Tyr Tyr Leu	
435 440 445	
tcc aga act cag tcc aca gga gga act caa ggt aca cag caa ttg tta	1392
Ser Arg Thr Gln Ser Thr Gly Gly Thr Gln Gly Thr Gln Gln Leu Leu	
450 455 460	
ttt tct caa gcc ggg cct gca aat atg tcg gct cag gcc aag aac tgg	1440
Phe Ser Gln Ala Gly Pro Ala Asn Met Ser Ala Gln Ala Lys Asn Trp	
465 470 475 480	
cta cct gga cct tgc tac cgg cag cag cga gtc tcc acg aca ctg tcg	1488
Leu Pro Gly Pro Cys Tyr Arg Gln Gln Arg Val Ser Thr Thr Leu Ser	
485 490 495	
caa aac aac aac agc aac ttt gct tgg act ggt gcc acg aaa tat cat	1536
Gln Asn Asn Asn Ser Asn Phe Ala Trp Thr Gly Ala Thr Lys Tyr His	
500 505 510	
ctg aac gga aga gac tct ttg gtg aat ccc ggt gtt gct atg gca acg	1584
Leu Asn Gly Arg Asp Ser Leu Val Asn Pro Gly Val Ala Met Ala Thr	
515 520 525	
cat aag gac gac gag gaa cgt ttc ttt cca tcg agc gga gtc ctg atg	1632
His Lys Asp Asp Glu Glu Arg Phe Phe Pro Ser Ser Gly Val Leu Met	
530 535 540	
ttt gga aaa cag ggt gct gga aga gac aat gtg gac tat agc agc gtt	1680
Phe Gly Lys Gln Gly Ala Gly Arg Asp Asn Val Asp Tyr Ser Ser Val	
545 550 555 560	
atg cta acc agc gag gaa gaa atc aag acc act aac cct gta gcc act	1728
Met Leu Thr Ser Glu Glu Glu Ile Lys Thr Thr Asn Pro Val Ala Thr	
565 570 575	
gaa caa tac ggc gtg gtg gct gat aac ttg cag caa acc aat aca gga	1776
Glu Gln Tyr Gly Val Val Ala Asp Asn Leu Gln Gln Thr Asn Thr Gly	
580 585 590	
cct atc gtg gga aat gtc aac agc caa gga gcc tta cct ggc atg gtc	1824
Pro Ile Val Gly Asn Val Asn Ser Gln Gly Ala Leu Pro Gly Met Val	
595 600 605	
tgg cag aac cga gac gtg tac ctg cag ggt ccc att tgg gcc aag att	1872
Trp Gln Asn Arg Asp Val Tyr Leu Gln Gly Pro Ile Trp Ala Lys Ile	
610 615 620	

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cct cac acg gac ggc aac ttt cac ccg tct cct ctg atg ggc ggc ttt Pro His Thr Asp Gly Asn Phe His Pro Ser Pro Leu Met Gly Gly Phe 625 630 635 640	1920
gga ctg aaa cac ccg cct cct caa atc ctg atc aag aac act ccc gtt Gly Leu Lys His Pro Pro Pro Gln Ile Leu Ile Lys Asn Thr Pro Val 645 650 655	1968
cct gcg gat cct cca acg acg ttc agc cag gcg aaa ttg gct tcc ttc Pro Ala Asp Pro Pro Thr Thr Phe Ser Gln Ala Lys Leu Ala Ser Phe 660 665 670	2016
atc acg cag tat agt acc ggc cag gtc agc gtg gag atc gag tgg gag Ile Thr Gln Tyr Ser Thr Gly Gln Val Ser Val Glu Ile Glu Trp Glu 675 680 685	2064
ctg cag aag gag aac agc aag cgc tgg aac cca gaa att cag tat act Leu Gln Lys Glu Asn Ser Lys Arg Trp Asn Pro Glu Ile Gln Tyr Thr 690 695 700	2112
tcc aac tac tac aaa tct aca aat gtg gac ttt gct gtc aat acc gag Ser Asn Tyr Tyr Lys Ser Thr Asn Val Asp Phe Ala Val Asn Thr Glu 705 710 715 720	2160
ggt aca tat tca gag cct cgc ccc att gga act cgt tac ctc acc cgt Gly Thr Tyr Ser Glu Pro Arg Pro Ile Gly Thr Arg Tyr Leu Thr Arg 725 730 735	2208
aat ctg Asn Leu	2214

<210> SEQ ID NO 2
 <211> LENGTH: 738
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 2

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Lys Ala Asn Gln Gln Lys Gln Asp Asp Gly Arg Gly Leu Val Leu Pro 35 40 45
Gly Tyr Lys Tyr Leu Gly Pro Phe Asn Gly Leu Asp Lys Gly Glu Pro 50 55 60
Val Asn Glu Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp 65 70 75 80
Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Arg Tyr Asn His Ala 85 90 95
Asp Ala Glu Phe Gln Glu Arg Leu Gln Glu Asp Thr Ser Phe Gly Gly 100 105 110
Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Val Leu Glu Pro 115 120 125
Leu Gly Leu Val Glu Glu Ala Ala Lys Thr Ala Pro Gly Lys Lys Arg 130 135 140
Pro Val Glu Pro Ser Pro Gln Arg Ser Pro Asp Ser Ser Thr Gly Ile 145 150 155 160
Gly Lys Lys Gly Gln Gln Pro Ala Arg Lys Arg Leu Asn Phe Gly Gln 165 170 175
Thr Gly Asp Ser Glu Ser Val Pro Asp Pro Gln Pro Ile Gly Glu Pro 180 185 190
Pro Ala Gly Pro Ser Gly Leu Gly Ser Gly Thr Met Ala Ala Gly Gly

195				200				205							
Gly	Ala	Pro	Met	Ala	Asp	Asn	Asn	Glu	Gly	Ala	Asp	Gly	Val	Gly	Ser
210					215					220					
Ser	Ser	Gly	Asn	Trp	His	Cys	Asp	Ser	Thr	Trp	Leu	Gly	Asp	Arg	Val
225				230						235					240
Ile	Thr	Thr	Ser	Thr	Arg	Thr	Trp	Ala	Leu	Pro	Thr	Tyr	Asn	Asn	His
				245					250					255	
Leu	Tyr	Lys	Gln	Ile	Ser	Asn	Gly	Thr	Ser	Gly	Gly	Ser	Thr	Asn	Asp
			260					265						270	
Asn	Thr	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro	Trp	Gly	Tyr	Phe	Asp	Phe	Asn
			275					280						285	
Arg	Phe	His	Cys	His	Phe	Ser	Pro	Arg	Asp	Trp	Gln	Arg	Leu	Ile	Asn
	290					295					300				
Asn	Asn	Trp	Gly	Phe	Arg	Pro	Lys	Arg	Leu	Ser	Phe	Lys	Leu	Phe	Asn
305					310					315					320
Ile	Gln	Val	Lys	Glu	Val	Thr	Gln	Asn	Glu	Gly	Thr	Lys	Thr	Ile	Ala
				325					330					335	
Asn	Asn	Leu	Thr	Ser	Thr	Ile	Gln	Val	Phe	Thr	Asp	Ser	Glu	Tyr	Gln
			340					345						350	
Leu	Pro	Tyr	Val	Leu	Gly	Ser	Ala	His	Gln	Gly	Cys	Leu	Pro	Pro	Phe
			355				360					365			
Pro	Ala	Asp	Val	Phe	Met	Ile	Pro	Gln	Tyr	Gly	Tyr	Leu	Thr	Leu	Asn
	370					375					380				
Asn	Gly	Ser	Gln	Ala	Val	Gly	Arg	Ser	Ser	Phe	Tyr	Cys	Leu	Glu	Tyr
385					390					395					400
Phe	Pro	Ser	Gln	Met	Leu	Arg	Thr	Gly	Asn	Asn	Phe	Glu	Phe	Ser	Tyr
			405						410					415	
Thr	Phe	Glu	Asp	Val	Pro	Phe	His	Ser	Ser	Tyr	Ala	His	Ser	Gln	Ser
			420					425						430	
Leu	Asp	Arg	Leu	Met	Asn	Pro	Leu	Ile	Asp	Gln	Tyr	Leu	Tyr	Tyr	Leu
			435				440					445			
Ser	Arg	Thr	Gln	Ser	Thr	Gly	Gly	Thr	Gln	Gly	Thr	Gln	Gln	Leu	Leu
	450					455					460				
Phe	Ser	Gln	Ala	Gly	Pro	Ala	Asn	Met	Ser	Ala	Gln	Ala	Lys	Asn	Trp
465					470					475					480
Leu	Pro	Gly	Pro	Cys	Tyr	Arg	Gln	Gln	Arg	Val	Ser	Thr	Thr	Leu	Ser
				485					490					495	
Gln	Asn	Asn	Asn	Ser	Asn	Phe	Ala	Trp	Thr	Gly	Ala	Thr	Lys	Tyr	His
			500					505						510	
Leu	Asn	Gly	Arg	Asp	Ser	Leu	Val	Asn	Pro	Gly	Val	Ala	Met	Ala	Thr
			515				520					525			
His	Lys	Asp	Asp	Glu	Glu	Arg	Phe								

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Pro His Thr Asp Gly Asn Phe His Pro Ser Pro Leu Met Gly Gly Phe
 625 630 635 640
 Gly Leu Lys His Pro Pro Pro Gln Ile Leu Ile Lys Asn Thr Pro Val
 645 650 655
 Pro Ala Asp Pro Pro Thr Thr Phe Ser Gln Ala Lys Leu Ala Ser Phe
 660 665 670
 Ile Thr Gln Tyr Ser Thr Gly Gln Val Ser Val Glu Ile Glu Trp Glu
 675 680 685
 Leu Gln Lys Glu Asn Ser Lys Arg Trp Asn Pro Glu Ile Gln Tyr Thr
 690 695 700
 Ser Asn Tyr Tyr Lys Ser Thr Asn Val Asp Phe Ala Val Asn Thr Glu
 705 710 715 720
 Gly Thr Tyr Ser Glu Pro Arg Pro Ile Gly Thr Arg Tyr Leu Thr Arg
 725 730 735
 Asn Leu

<210> SEQ ID NO 3
 <211> LENGTH: 2208
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: AAV9
 <220> FEATURE:
 <221> NAME/KEY: CDS
 <222> LOCATION: (1)..(2208)

<400> SEQUENCE: 3

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gaa gga att cgc gag tgg tgg gct ttg aaa cct gga gcc cct caa ccc	96
Glu Gly Ile Arg Glu Trp Trp Ala Leu Lys Pro Gly Ala Pro Gln Pro	
20 25 30	
aag gca aat caa caa cat caa gac aac gct cga ggt ctt gtg ctt ccg	144
Lys Ala Asn Gln Gln His Gln Asp Asn Ala Arg Gly Leu Val Leu Pro	
35 40 45	
ggt tac aaa tac ctt gga ccc ggc aac gga ctc gac aag ggg gag ccg	192
Gly Tyr Lys Tyr Leu Gly Pro Gly Asn Gly Leu Asp Lys Gly Glu Pro	
50 55 60	
gtc aac gca gca gac gcg gcg gcc ctc gag cac gac aag gcc tac gac	240
Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp	
65 70 75 80	
cag cag ctc aag gcc gga gac aac ccg tac ctc aag tac aac cac gcc	288
Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Lys Tyr Asn His Ala	
85 90 95	
gac gcc gag ttc cag gag cgg ctc aaa gaa gat acg tct ttt ggg ggc	336
Asp Ala Glu Phe Gln Glu Arg Leu Lys Glu Asp Thr Ser Phe Gly Gly	
100 105 110	
aac ctc ggg cga gca gtc ttc cag gcc aaa aag agg ctt ctt gaa cct	384
Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Leu Leu Glu Pro	
115 120 125	
ctt ggt ctg gtt gag gaa gcg gct aag acg gct cct gga aag aag agg	432
Leu Gly Leu Val Glu Ala Ala Lys Thr Ala Pro Gly Lys Lys Arg	
130 135 140	
cct gta gag cag tct cct cag gaa ccg gac tcc tcc gcg ggt att ggc	480
Pro Val Glu Gln Ser Pro Gln Glu Pro Asp Ser Ser Ala Gly Ile Gly	
145 150 155 160	
aaa tcg ggt gca cag ccc gct aaa aag aga ctc aat ttc ggt cag act	528
Lys Ser Gly Ala Gln Pro Ala Lys Lys Arg Leu Asn Phe Gly Gln Thr	

															165						170						175												
ggc	gac	aca	gag	tca	gtc	cca	gac	cct	caa	cca	atc	gga	gaa	cct	ccc	576																							
Gly	Asp	Thr	Glu	Ser	Val	Pro	Asp	Pro	Gln	Pro	Ile	Gly	Glu	Pro	Pro																								
															180						185						190												
gca	gcc	ccc	tca	ggg	gtg	gga	tct	ctt	aca	atg	gct	tca	ggg	ggg	ggc	624																							
Ala	Ala	Pro	Ser	Gly	Val	Gly	Ser	Leu	Thr	Met	Ala	Ser	Gly	Gly	Gly																								
															195						200						205												
gca	cca	gtg	gca	gac	aat	aac	gaa	ggg	gcc	gat	gga	gtg	ggg	agt	tcc	672																							
Ala	Pro	Val	Ala	Asp	Asn	Glu	Gly	Ala	Asp	Gly	Val	Gly	Gly	Ser	Ser																								
															210						215						220												
tcg	gga	aat	tgg	cat	tgc	gat	tcc	caa	tgg	ctg	ggg	gac	aga	gtc	atc	720																							
Ser	Gly	Asn	Trp	His	Cys	Asp	Ser	Gln	Trp	Leu	Gly	Asp	Arg	Val	Ile																								
															225						230						235						240						
acc	acc	agc	acc	cga	acc	tgg	gcc	ctg	ccc	acc	tac	aac	aat	cac	ctc	768																							
Thr	Thr	Ser	Thr	Arg	Thr	Trp	Ala	Leu	Pro	Thr	Tyr	Asn	Asn	His	Leu																								
															245						250						255												
tac	aag	caa	atc	tcc	aac	agc	aca	tct	gga	gga	tct	tca	aat	gac	aac	816																							
Tyr	Lys	Gln	Ile	Ser	Asn	Ser	Thr	Ser	Gly	Gly	Ser	Ser	Asn	Asp	Asn																								
															260						265						270												
gcc	tac	ttc	ggc	tac	agc	acc	ccc	tgg	ggg	tat	ttt	gac	ttc	aac	aga	864																							
Ala	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro	Trp	Gly	Tyr	Phe	Asp	Phe	Asn	Arg																								
															275						280						285												
ttc	cac	tgc	cac	ttc	tca	cca	cgt	gac	tgg	cag	cga	ctc	atc	aac	aac	912																							
Phe	His	Cys	His	Phe	Ser	Pro	Arg	Asp	Trp	Gln	Arg	Leu	Ile	Asn	Asn																								
															290						295						300												
aac	tgg	gga	ttc	cgg	cct	aag	cga	ctc	aac	ttc	aag	ctc	ttc	aac	att	960																							
Asn	Trp	Gly	Phe	Arg	Pro	Lys	Arg	Leu	Asn	Phe	Lys	Leu	Phe	Asn	Ile																								
															305						310						315						320						
cag	gtc	aaa	gag	gtt	acg	gac	aac	aat	gga	gtc	aag	acc	atc	gcc	aat	1008																							
Gln	Val	Lys	Glu	Val	Thr	Asp	Asn	Asn	Gly	Val	Lys	Thr	Ile	Ala	Asn																								
															325						330						335												
aac	ctt	acc	agc	acg	gtc	cag	gtc	ttc	acg	gac	tca	gac	tat	cag	ctc	1056																							
Asn	Leu	Thr	Ser	Thr	Val	Gln	Val	Phe	Thr	Asp	Ser	Asp	Tyr	Gln	Leu																								
															340						345						350												
ccg	tac	gtg	ctc	ggg	tcg	gct	cac	gag	ggc	tgc	ctc	ccg	ccg	ttc	cca	1104																							
Pro	Tyr	Val	Leu	Gly	Ser	Ala	His	Glu	Gly	Cys	Leu	Pro	Pro	Phe	Pro																								
															355						360						365												
gcg	gac	gtt	ttc	atg	att	cct	cag	tac	ggg	tat	ctg	acg	ctt	aat	gat	1152																							
Ala	Asp	Val	Phe	Met	Ile	Pro	Gln	Tyr	Gly	Tyr	Leu	Thr	Leu	Asn	Asp																								
															370						375						380												
gga	agc	cag	gcc	gtg	ggg	cgt	tcg	tcc	ttt	tac	tgc	ctg	gaa	tat	ttc	1200																							
Gly	Ser	Gln	Ala	Val	Gly	Arg	Ser	Ser	Phe	Tyr	Cys	Leu	Glu	Tyr	Phe																								
															385						390						395						400						

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Gly	Pro	Ser	Tyr	Arg	Gln	Gln	Arg	Val	Ser	Thr	Thr	Val	Thr	Gln	Asn	
				485					490					495		
aac	aac	agc	gaa	ttt	gct	tgg	cct	gga	gct	tct	tct	tgg	gct	ctc	aat	1536
Asn	Asn	Ser	Glu	Phe	Ala	Trp	Pro	Gly	Ala	Ser	Ser	Trp	Ala	Leu	Asn	
				500					505					510		
gga	cgt	aat	agc	ttg	atg	aat	cct	gga	cct	gct	atg	gcc	agc	cac	aaa	1584
Gly	Arg	Asn	Ser	Leu	Met	Asn	Pro	Gly	Pro	Ala	Met	Ala	Ser	His	Lys	
				515					520					525		
gaa	gga	gag	gac	cgt	tto	ttt	cct	ttg	tct	gga	tct	tta	att	ttt	ggc	1632
Glu	Gly	Glu	Asp	Arg	Phe	Phe	Pro	Leu	Ser	Gly	Ser	Leu	Ile	Phe	Gly	
				530					535					540		
aaa	caa	gga	act	gga	aga	gac	aac	gtg	gat	gcg	gac	aaa	gtc	atg	ata	1680
Lys	Gln	Gly	Thr	Gly	Arg	Asp	Asn	Val	Asp	Ala	Asp	Lys	Val	Met	Ile	
				545					550					555		
acc	aac	gaa	gaa	gaa	att	aaa	act	act	aac	ccg	gta	gca	acg	gag	tcc	1728
Thr	Asn	Glu	Glu	Glu	Ile	Lys	Thr	Thr	Asn	Pro	Val	Ala	Thr	Glu	Ser	
				565					570					575		
tat	gga	caa	gtg	gcc	aca	aac	cac	cag	agt	gcc	caa	gca	cag	gcg	cag	1776
Tyr	Gly	Gln	Val	Ala	Thr	Asn	His	Gln	Ser	Ala	Gln	Ala	Gln	Ala	Gln	
				580					585					590		
acc	ggc	tgg	gtt	caa	aac	caa	gga	ata	ctt	ccg	ggt	atg	gtt	tgg	cag	1824
Thr	Gly	Trp	Val	Gln	Asn	Gln	Gly	Ile	Leu	Pro	Gly	Met	Val	Trp	Gln	
				595					600					605		
gac	aga	gat	gtg	tac	ctg	caa	gga	ccc	att	tgg	gcc	aaa	att	cct	cac	1872
Asp	Arg	Asp	Val	Tyr	Leu	Gln	Gly	Pro	Ile	Trp	Ala	Lys	Ile	Pro	His	
				610					615					620		
acg	gac	ggc	aac	ttt	cac	cct	tct	ccg	ctg	atg	gga	ggg	ttt	gga	atg	1920
Thr	Asp	Gly	Asn	Phe	His	Pro	Ser	Pro	Leu	Met	Gly	Gly	Phe	Gly	Met	
				625					630					635		
aag	cac	ccg	cct	cct	cag	atc	ctc	atc	aaa	aac	aca	cct	gta	cct	gcg	1968
Lys	His	Pro	Pro	Pro	Gln	Ile	Leu	Ile	Lys	Asn	Thr	Pro	Val	Pro	Ala	
				645					650					655		
gat	cct	cca	acg	gcc	ttc	aac	aag	gac	aag	ctg	aac	tct	ttc	atc	acc	2016
Asp	Pro	Pro	Thr	Ala	Phe	Asn	Lys	Asp	Lys	Leu	Asn	Ser	Phe	Ile	Thr	
				660					665					670		
cag	tat	tct	act	ggc	caa	gtc	agc	gtg	gag	atc	gag	tgg	gag	ctg	cag	2064
Gln	Tyr	Ser	Thr	Gly	Gln	Val	Ser	Val	Glu	Ile	Glu	Trp	Glu	Leu	Gln	
				675					680					685		
aag	gaa	aac	agc	aag	cgc	tgg	aac	ccg	gag	atc	cag	tac	act	tcc	aac	2112
Lys	Glu	Asn	Ser	Lys	Arg	Trp	Asn	Pro	Glu	Ile	Gln	Tyr	Thr	Ser	Asn	
				690					695					700		
tat	tac	aag	tct	aat	aat	gtt	gaa	ttt	gct	gtt	aat	act	gaa	ggt	gta	2160
Tyr	Tyr	Lys	Ser	Asn	Asn	Val	Glu	Phe	Ala	Val	Asn	Thr	Glu	Gly	Val	
				705					710					715		
tat	agt	gaa	ccc	cgc	ccc	att	ggc	acc	aga	tac	ctg	act	cgt	aat	ctg	2208
Tyr	Ser	Glu	Pro	Arg	Pro	Ile	Gly	Thr	Arg	Tyr	Leu	Thr	Arg	Asn	Leu	
				725					730					735		

<210> SEQ ID NO 4
 <211> LENGTH: 736
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 4

Met	Ala	Ala	Asp	Gly	Tyr	Leu	Pro	Asp	Trp	Leu	Glu	Asp	Asn	Leu	Ser
1				5					10					15	
Glu	Gly	Ile	Arg	Glu	Trp	Trp	Ala	Leu	Lys	Pro	Gly	Ala	Pro	Gln	Pro
		20					25					30			

Lys	Ala	Asn	Gln	Gln	His	Gln	Asp	Asn	Ala	Arg	Gly	Leu	Val	Leu	Pro
	35						40					45			
Gly	Tyr	Lys	Tyr	Leu	Gly	Pro	Gly	Asn	Gly	Leu	Asp	Lys	Gly	Glu	Pro
	50					55					60				
Val	Asn	Ala	Ala	Asp	Ala	Ala	Ala	Leu	Glu	His	Asp	Lys	Ala	Tyr	Asp
	65				70					75					80
Gln	Gln	Leu	Lys	Ala	Gly	Asp	Asn	Pro	Tyr	Leu	Lys	Tyr	Asn	His	Ala
				85					90					95	
Asp	Ala	Glu	Phe	Gln	Glu	Arg	Leu	Lys	Glu	Asp	Thr	Ser	Phe	Gly	Gly
			100					105					110		
Asn	Leu	Gly	Arg	Ala	Val	Phe	Gln	Ala	Lys	Lys	Arg	Leu	Leu	Glu	Pro
							120					125			
Leu	Gly	Leu	Val	Glu	Glu	Ala	Ala	Lys	Thr	Ala	Pro	Gly	Lys	Lys	Arg
						135					140				
Pro	Val	Glu	Gln	Ser	Pro	Gln	Glu	Pro	Asp	Ser	Ser	Ala	Gly	Ile	Gly
					150					155					160
Lys	Ser	Gly	Ala	Gln	Pro	Ala	Lys	Lys	Arg	Leu	Asn	Phe	Gly	Gln	Thr
				165					170					175	
Gly	Asp	Thr	Glu	Ser	Val	Pro	Asp	Pro	Gln	Pro	Ile	Gly	Glu	Pro	Pro
			180					185					190		
Ala	Ala	Pro	Ser	Gly	Val	Gly	Ser	Leu	Thr	Met	Ala	Ser	Gly	Gly	Gly
		195					200					205			
Ala	Pro	Val	Ala	Asp	Asn	Asn	Glu	Gly	Ala	Asp	Gly	Val	Gly	Ser	Ser
		210				215					220				
Ser	Gly	Asn	Trp	His	Cys	Asp	Ser	Gln	Trp	Leu	Gly	Asp	Arg	Val	Ile
				230						235					240
Thr	Thr	Ser	Thr	Arg	Thr	Trp	Ala	Leu	Pro	Thr	Tyr	Asn	Asn	His	Leu
				245					250					255	
Tyr	Lys	Gln	Ile	Ser	Asn	Ser	Thr	Ser	Gly	Gly	Ser	Ser	Asn	Asp	Asn
			260					265					270		
Ala	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro	Trp	Gly	Tyr	Phe	Asp	Phe	Asn	Arg
		275					280					285			
Phe	His	Cys	His	Phe	Ser	Pro	Arg	Asp	Trp	Gln	Arg	Leu	Ile	Asn	Asn
						295					300				
Asn	Trp	Gly	Phe	Arg	Pro	Lys	Arg	Leu	Asn	Phe	Lys	Leu	Phe	Asn	Ile
					310					315					320
Gln	Val	Lys	Glu	Val	Thr	Asp	Asn	Asn	Gly	Val	Lys	Thr	Ile	Ala	Asn
				325					330					335	
Asn	Leu	Thr	Ser	Thr	Val	Gln	Val	Phe	Thr	Asp	Ser	Asp	Tyr	Gln	Leu
			340					345				350			
Pro	Tyr	Val	Leu	Gly	Ser	Ala	His	Glu	Gly	Cys	Leu	Pro	Pro	Phe	Pro
		355					360					365			
Ala	Asp	Val	Phe	Met	Ile	Pro	Gln	Tyr	Gly	Tyr	Leu	Thr	Leu	Asn	Asp
		370				375					380				
Gly	Ser	Gln	Ala	Val	Gly	Arg									

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450	455	460
Val Ala Gly Pro Ser Asn Met Ala Val Gln Gly Arg Asn Tyr Ile Pro		
465	470	475 480
Gly Pro Ser Tyr Arg Gln Gln Arg Val Ser Thr Thr Val Thr Gln Asn		
	485	490 495
Asn Asn Ser Glu Phe Ala Trp Pro Gly Ala Ser Ser Trp Ala Leu Asn		
	500	505 510
Gly Arg Asn Ser Leu Met Asn Pro Gly Pro Ala Met Ala Ser His Lys		
	515	520 525
Glu Gly Glu Asp Arg Phe Phe Pro Leu Ser Gly Ser Leu Ile Phe Gly		
	530	535 540
Lys Gln Gly Thr Gly Arg Asp Asn Val Asp Ala Asp Lys Val Met Ile		
545	550	555 560
Thr Asn Glu Glu Glu Ile Lys Thr Thr Asn Pro Val Ala Thr Glu Ser		
	565	570 575
Tyr Gly Gln Val Ala Thr Asn His Gln Ser Ala Gln Ala Gln Ala Gln		
	580	585 590
Thr Gly Trp Val Gln Asn Gln Gly Ile Leu Pro Gly Met Val Trp Gln		
	595	600 605
Asp Arg Asp Val Tyr Leu Gln Gly Pro Ile Trp Ala Lys Ile Pro His		
	610	615 620
Thr Asp Gly Asn Phe His Pro Ser Pro Leu Met Gly Gly Phe Gly Met		
625	630	635 640
Lys His Pro Pro Pro Gln Ile Leu Ile Lys Asn Thr Pro Val Pro Ala		
	645	650 655
Asp Pro Pro Thr Ala Phe Asn Lys Asp Lys Leu Asn Ser Phe Ile Thr		
	660	665 670
Gln Tyr Ser Thr Gly Gln Val Ser Val Glu Ile Glu Trp Glu Leu Gln		
	675	680 685
Lys Glu Asn Ser Lys Arg Trp Asn Pro Glu Ile Gln Tyr Thr Ser Asn		
	690	695 700
Tyr Tyr Lys Ser Asn Asn Val Glu Phe Ala Val Asn Thr Glu Gly Val		
705	710	715 720
Tyr Ser Glu Pro Arg Pro Ile Gly Thr Arg Tyr Leu Thr Arg Asn Leu		
	725	730 735

<210> SEQ ID NO 5

<211> LENGTH: 2211

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: AAVhu31

<220> FEATURE:

<221> NAME/KEY: CDS

<222> LOCATION: (1)..(2211)

<400> SEQUENCE: 5

atg gct gcc gat ggt tat ctt cca gat tgg ctc gag gac act ctc tct	48
Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Thr Leu Ser	
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gaa gga ata aga cag tgg tgg aag ctc aaa cct ggc ccc cca cca cca	96
Glu Gly Ile Arg Gln Trp Trp Lys Leu Lys Pro Gly Pro Pro Pro Pro	
20 25 30	
aag ccc gca gag cgg cat aag gac gac agc agg ggt ctt gtg ctt cct	144
Lys Pro Ala Glu Arg His Lys Asp Asp Ser Arg Gly Leu Val Leu Pro	
35 40 45	
ggg tac aag tac ctc gga ccc ggc aac gga ctc gac aag ggg gag ccg	192

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Gly	Tyr	Lys	Tyr	Leu	Gly	Pro	Gly	Asn	Gly	Leu	Asp	Lys	Gly	Glu	Pro	
50						55					60					
gtc	aac	gca	gca	gac	gcg	gcg	gcc	ctc	gag	cac	gac	aag	gcc	tac	gac	240
Val	Asn	Ala	Ala	Asp	Ala	Ala	Ala	Leu	Glu	His	Asp	Lys	Ala	Tyr	Asp	
65					70					75				80		
cag	cag	ctc	aag	gcc	gga	gac	aac	ccg	tac	ctc	aag	tac	aac	cac	gcc	288
Gln	Gln	Leu	Lys	Ala	Gly	Asp	Asn	Pro	Tyr	Leu	Lys	Tyr	Asn	His	Ala	
			85						90					95		
gac	gcc	gag	ttc	cag	gag	cgg	ctc	aaa	gaa	gat	acg	tct	ttt	ggg	ggc	336
Asp	Ala	Glu	Phe	Gln	Glu	Arg	Leu	Lys	Glu	Asp	Thr	Ser	Phe	Gly	Gly	
			100					105					110			
aac	ctc	ggg	cga	gca	gtc	ttc	cag	gcc	aaa	aag	agg	ctt	ctt	gaa	cct	384
Asn	Leu	Gly	Arg	Ala	Val	Phe	Gln	Ala	Lys	Lys	Arg	Leu	Leu	Glu	Pro	
		115					120					125				
ctt	ggt	ctg	gtt	gag	gaa	gcg	gct	aag	acg	gct	cct	gga	aag	aag	agg	432
Leu	Gly	Leu	Val	Glu	Glu	Ala	Ala	Lys	Thr	Ala	Pro	Gly	Lys	Lys	Arg	
	130					135					140					
cct	gta	gag	cag	tct	cct	cag	gaa	ccg	gac	tcc	tcc	gcg	ggt	att	ggc	480
Pro	Val	Glu	Gln	Ser	Pro	Gln	Glu	Pro	Asp	Ser	Ser	Ala	Gly	Ile	Gly	
145					150					155				160		
aaa	tcg	ggt	tca	cag	ccc	gct	aaa	aag	aaa	ctc	aat	ttc	ggt	cag	act	528
Lys	Ser	Gly	Ser	Gln	Pro	Ala	Lys	Lys	Lys	Leu	Asn	Phe	Gly	Gln	Thr	
				165					170					175		
ggc	gac	aca	gag	tca	gtc	ccc	gac	cct	caa	cca	atc	gga	gaa	cct	ccc	576
Gly	Asp	Thr	Glu	Ser	Val	Pro	Asp	Pro	Gln	Pro	Ile	Gly	Glu	Pro	Pro	
			180				185						190			
gca	gcc	ccc	tca	ggt	gtg	gga	tct	ctt	aca	atg	gct	tca	ggt	ggt	ggc	624
Ala	Ala	Pro	Ser	Gly	Val	Gly	Ser	Leu	Thr	Met	Ala	Ser	Gly	Gly	Gly	
		195				200						205				
gca	cca	gtg	gca	gac	aat	aac	gaa	ggt	gcc	gat	gga	gtg	ggt	agt	tcc	672
Ala	Pro	Val	Ala	Asp	Asn	Asn	Glu	Gly	Ala	Asp	Gly	Val	Gly	Ser	Ser	
		210			215						220					
tcg	gga	aat	tgg	cat	tgc	gat	tcc	caa	tgg	ctg	ggg	gac	aga	gtc	atc	720
Ser	Gly	Asn	Trp	His	Cys	Asp	Ser	Gln	Trp	Leu	Gly	Asp	Arg	Val	Ile	
225					230					235				240		
acc	acc	agc	acc	cga	acc	tgg	gcc	ctg	ccc	acc	tac	aac	aat	cac	ctc	768
Thr	Thr	Ser	Thr	Arg	Thr	Trp	Ala	Leu	Pro	Thr	Tyr	Asn	Asn	His	Leu	
				245					250					255		
tac	aag	caa	atc	tcc	aac	agc	aca	tct	gga	gga	tct	tca	aat	gac	aac	816
Tyr	Lys	Gln	Ile	Ser	Asn	Ser	Thr	Ser	Gly	Gly	Ser	Ser	Asn	Asp	Asn	
		260						265					270			
gcc	tac	ttc	ggc	tac	agc	acc	ccc	tgg	ggg	tat	ttt	gac	ttc	aac	aga	864
Ala	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro	Trp	Gly	Tyr	Phe	Asp	Phe	Asn	Arg	
		275					280					285				
ttc	cac	tgc	cac	ttc	tca	cca	cgt	gac	tgg	cag	cga	ctc	atc	aac	aac	912
Phe	His	Cys	His	Phe	Ser	Pro	Arg	Asp	Trp	Gln	Arg	Leu	Ile	Asn	Asn	
		290				295					300					
aac	tgg	gga	ttc	cgg	cct	aag	cga	ctc	aac	ttc	aag	ctc	ttc	aac	att	960
Asn	Trp	Gly	Phe	Arg	Pro	Lys	Arg	Leu	Asn	Phe	Lys	Leu	Phe	Asn	Ile	
		305			310					315				320		
cag	gtc	aaa	gag	gtt	acg	gac	aac	aat	gga	gtc	aag	acc	atc	gcc	aat	1008
Gln	Val	Lys	Glu	Val	Thr	Asp	Asn	Asn	Gly	Val	Lys	Thr	Ile	Ala	Asn	
			325						330					335		
aac	ctt	acc	agc	acg	gtc	cag	gtc	ttc	acg	gac	tca	gac	tat	cag	ctc	1056
Asn	Leu	Thr	Ser	Thr	Val	Gln	Val	Phe	Thr	Asp	Ser	Asp	Tyr	Gln	Leu	
			340					345					350			
ccg	tac	gtg	ctc	ggg	tcg	gct	cac	gag	ggc	tgc	ctc	ccg	ccg	ttc	cca	1104
Pro	Tyr	Val	Leu	Gly	Ser	Ala	His	Glu	Gly	Cys	Leu	Pro	Pro	Phe	Pro	
		355					360					365				

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gcg gac gtt ttc atg att cct cag tac ggg tat ctg acg ctt aat gat	1152
Ala Asp Val Phe Met Ile Pro Gln Tyr Gly Tyr Leu Thr Leu Asn Asp	
370 375 380	
gga ggc cag gcc gtg ggt cgt tcg tcc ttt tac tgc ctg gaa tat ttc	1200
Gly Gly Gln Ala Val Gly Arg Ser Ser Phe Tyr Cys Leu Glu Tyr Phe	
385 390 395 400	
ccg tcg caa atg cta aga acg ggt aac aac ttc cag ttc agc tac gag	1248
Pro Ser Gln Met Arg Thr Gly Asn Asn Phe Gln Phe Ser Tyr Glu	
405 410 415	
ttt gag aac gta cct ttc cat agc agc tac gct cac agc caa agc ctg	1296
Phe Glu Asn Val Pro Phe His Ser Ser Tyr Ala His Ser Gln Ser Leu	
420 425 430	
gac cga cta atg aat cca ctc atc gac caa tac ttg tac tat ctc tca	1344
Asp Arg Leu Met Asn Pro Leu Ile Asp Gln Tyr Leu Tyr Tyr Leu Ser	
435 440 445	
aag act att aac ggt tct gga cag aat caa caa acg cta aaa ttc agt	1392
Lys Thr Ile Asn Gly Ser Gly Gln Asn Gln Gln Thr Leu Lys Phe Ser	
450 455 460	
gtg gcc gga ccc agc aac atg gct gtc cag gga aga aac tac ata cct	1440
Val Ala Gly Pro Ser Asn Met Ala Val Gln Gly Arg Asn Tyr Ile Pro	
465 470 475 480	
gga ccc agc tac cga caa caa cgt gtc tca acc act gtg act caa aac	1488
Gly Pro Ser Tyr Arg Gln Gln Arg Val Ser Thr Thr Val Thr Gln Asn	
485 490 495	
aac aac agc gaa ttt gct tgg cct gga gct tct tct tgg gct ctc aat	1536
Asn Asn Ser Glu Phe Ala Trp Pro Gly Ala Ser Ser Trp Ala Leu Asn	
500 505 510	
gga cgt aat agc ttg atg aat cct gga cct gct atg gcc agc cac aaa	1584
Gly Arg Asn Ser Leu Met Asn Pro Gly Pro Ala Met Ala Ser His Lys	
515 520 525	
gaa gga gag gac cgt ttc ttt cct ttg tcc gga tct tta att ttt ggc	1632
Glu Gly Glu Asp Arg Phe Phe Pro Leu Ser Gly Ser Leu Ile Phe Gly	
530 535 540	
aaa caa gga act gga aga gac aac gtg gat gcg gac aaa gtc atg ata	1680
Lys Gln Gly Thr Gly Arg Asp Asn Val Asp Ala Asp Lys Val Met Ile	
545 550 555 560	
acc aac gaa gaa gaa att aaa act act aac ccg gta gca acg gag tcc	1728
Thr Asn Glu Glu Glu Ile Lys Thr Thr Asn Pro Val Ala Thr Glu Ser	
565 570 575	
tat gga caa gtg gcc aca aac cac cag agt gcc caa gca cag gcg cag	1776
Tyr Gly Gln Val Ala Thr Asn His Gln Ser Ala Gln Ala Gln Ala Gln	
580 585 590	
acc ggc tgg gtt caa aac caa gga ata ctt ccg ggt atg gtt tgg cag	1824
Thr Gly Trp Val Gln Asn Gln Gly Ile Leu Pro Gly Met Val Trp Gln	
595 600 605	
gac aga gat gtg tac ctg caa gga ccc att tgg gcc aaa att cct cac	1872
Asp Arg Asp Val Tyr Leu Gln Gly Pro Ile Trp Ala Lys Ile Pro His	
610 615 620	
acg gac ggc aac ttt cac cct tct ccg cta atg gga ggg ttt gga atg	1920
Thr Asp Gly Asn Phe His Pro Ser Pro Leu Met Gly Gly Phe Gly Met	
625 630 635 640	
aag cac ccg cct cct cag atc ctc atc aaa aac aca cct gta cct gcg	1968
Lys His Pro Pro Pro Gln Ile Leu Ile Lys Asn Thr Pro Val Pro Ala	
645 650 655	
gat cct cca acg gct ttc aat aag gac aag ctg aac tct ttc atc acc	2016
Asp Pro Pro Thr Ala Phe Asn Lys Asp Lys Leu Asn Ser Phe Ile Thr	
660 665 670	
cag tat tct act ggc caa gtc agc gtg gag att gag tgg gag ctg cag	2064
Gln Tyr Ser Thr Gly Gln Val Ser Val Glu Ile Glu Trp Glu Leu Gln	
675 680 685	

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aag gaa aac agc aag cgc tgg aac ccg gag atc cag tac act tcc aac      2112
Lys Glu Asn Ser Lys Arg Trp Asn Pro Glu Ile Gln Tyr Thr Ser Asn
   690                               695                               700

tat tac aag tct aat aat gtt gaa ttt gct gtt agt act gaa ggt gta      2160
Tyr Tyr Lys Ser Asn Asn Val Glu Phe Ala Val Ser Thr Glu Gly Val
   705                               710                               715                               720

tat agt gaa ccc cgc ccc att ggc acc aga tac ctg act cgt aat ctg      2208
Tyr Ser Glu Pro Arg Pro Ile Gly Thr Arg Tyr Leu Thr Arg Asn Leu
           725                               730                               735

taa                                                                    2211

<210> SEQ ID NO 6
<211> LENGTH: 736
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 6
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Glu Gly Ile Arg Gln Trp Trp Lys Leu Lys Pro Gly Pro Pro Pro Pro
                20                               25                               30
Lys Pro Ala Glu Arg His Lys Asp Asp Ser Arg Gly Leu Val Leu Pro
   35                               40                               45
Gly Tyr Lys Tyr Leu Gly Pro Gly Asn Gly Leu Asp Lys Gly Glu Pro
   50                               55                               60
Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp
 65                               70                               75                               80
Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Lys Tyr Asn His Ala
   85                               90                               95
Asp Ala Glu Phe Gln Glu Arg Leu Lys Glu Asp Thr Ser Phe Gly Gly
   100                               105                               110
Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Leu Leu Glu Pro
   115                               120                               125
Leu Gly Leu Val Glu Glu Ala Ala Lys Thr Ala Pro Gly Lys Lys Arg
   130                               135                               140
Pro Val Glu Gln Ser Pro Gln Glu Pro Asp Ser Ser Ala Gly Ile Gly
 145                               150                               155                               160
Lys Ser Gly Ser Gln Pro Ala Lys Lys Lys Leu Asn Phe Gly Gln Thr
   165                               170                               175
Gly Asp Thr Glu Ser Val Pro Asp Pro Gln Pro Ile Gly Glu Pro Pro
   180                               185                               190
Ala Ala Pro Ser Gly Val Gly Ser Leu Thr Met Ala Ser Gly Gly Gly
   195                               200                               205
Ala Pro Val Ala Asp Asn Asn Glu Gly Ala Asp Gly Val Gly Ser Ser
   210                               215                               220
Ser Gly Asn Trp His Cys Asp Ser Gln Trp Leu Gly Asp Arg Val Ile
 225                               230                               235                               240
Thr Thr Ser Thr Arg Thr Trp Ala Leu Pro Thr Tyr Asn Asn His Leu
   245                               250                               255
Tyr Lys Gln Ile Ser Asn Ser Thr Ser Gly Gly Ser Ser Asn Asp Asn
   260                               265                               270
Ala Tyr Phe Gly Tyr Ser Thr Pro Trp Gly Tyr Phe Asp Phe Asn Arg
   275                               280                               285

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Phe 290	Cys	His	Phe	Ser	Pro 295	Arg	Asp	Trp	Gln 300	Arg	Leu	Ile	Asn	Asn
Asn 305	Trp	Gly	Phe	Arg	Pro 310	Lys	Arg	Leu	Asn 315	Phe	Lys	Leu	Phe	Ile 320
Gln	Val	Lys	Glu	Val 325	Thr	Asp	Asn	Asn 330	Gly	Val	Lys	Thr	Ile	Ala 335
Asn	Leu	Thr	Ser	Thr 340	Val	Gln	Val 345	Phe	Thr	Asp	Ser	Asp	Tyr	Gln 350
Pro	Tyr	Val	Leu	Gly	Ser	Ala 360	His	Glu	Gly	Cys	Leu 365	Pro	Pro	Phe 370
Ala	Asp	Val	Phe	Met	Ile 375	Pro	Gln	Tyr	Gly	Tyr	Leu 380	Thr	Leu	Asn 385
Gly 385	Gly	Gln	Ala	Val 390	Gly	Arg	Ser	Ser	Phe	Tyr 395	Cys	Leu	Glu	Tyr 400
Pro	Ser	Gln	Met	Leu 405	Arg	Thr	Gly	Asn	Asn 410	Phe	Gln	Phe	Ser	Tyr 415
Phe	Glu	Asn	Val	Pro 420	Phe	His	Ser	Ser	Tyr	Ala	His	Ser	Gln	Ser 425
Asp	Arg	Leu	Met	Asn 435	Pro	Leu	Ile 440	Asp	Gln	Tyr	Leu 445	Tyr	Tyr	Leu 450
Lys 450	Thr	Ile	Asn	Gly	Ser	Gly 455	Gln	Asn	Gln	Gln	Thr 460	Leu	Lys	Phe 465
Val 465	Ala	Gly	Pro	Ser	Asn 470	Met	Ala	Val	Gln	Gly 475	Arg	Asn	Tyr	Ile 480
Gly	Pro	Ser	Tyr	Arg 485	Gln	Gln	Arg	Val	Ser 490	Thr	Thr	Val	Thr	Gln 495
Asn	Asn	Ser	Glu	Phe 500	Ala	Trp	Pro	Gly 505	Ala	Ser	Ser	Trp	Ala	Leu 510
Gly	Arg	Asn	Ser	Leu 515	Met	Asn	Pro 520	Gly	Pro	Ala	Met	Ala 525	Ser	His 530
Glu 530	Gly	Glu	Asp	Arg	Phe	Phe 535	Pro	Leu	Ser	Gly	Ser 540	Leu	Ile	Phe 545
Lys 545	Gln	Gly	Thr	Gly	Arg 550	Asp	Asn	Val	Asp	Ala 555	Asp	Lys	Val	Met 560
Thr	Asn	Glu	Glu	Glu 565	Ile	Lys	Thr	Thr	Asn 570	Pro	Val	Ala	Thr	Glu 575
Tyr	Gly	Gln	Val	Ala 580	Thr	Asn	His	Gln	Ser 585	Ala	Gln	Ala	Gln	Ala 590
Thr	Gly	Trp	Val	Gln 595	Asn	Gln	Gly 600	Ile	Leu	Pro	Gly 605	Met	Val	Trp 610
Asp 610	Arg	Asp	Val	Tyr	Leu	Gln 615	Gly	Pro	Ile	Trp	Ala 620	Lys	Ile	Pro 625
Thr 625	Asp	Gly	Asn	Phe 630	His	Pro	Ser	Pro	Leu 635	Met	Gly	Gly	Phe	Gly 640
Lys	His	Pro	Pro	Pro 645	Gln	Ile	Leu	Ile 650	Lys	Asn	Thr	Pro	Val	Pro 655
Asp	Pro	Pro	Thr	Ala 660	Phe	Asn	Lys	Asp 665	Lys	Leu	Asn	Ser	Phe	Ile 670
Gln	Tyr	Ser	Thr	Gly 675	Gln	Val	Ser	Val 680	Glu	Ile	Glu	Trp	Glu	Leu 685
Lys 690	Glu	Asn	Ser	Lys	Arg	Trp 695	Asn	Pro	Glu	Ile	Gln 700	Tyr	Thr	Ser 705
Tyr	Tyr	Lys	Ser	Asn	Asn	Val	Glu	Phe	Ala	Val	Ser	Thr	Glu	Gly 710

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705	710	715	720
Tyr Ser Glu Pro Arg Pro Ile Gly Thr Arg Tyr Leu Thr Arg Asn Leu	725	730	735
<210> SEQ ID NO 7 <211> LENGTH: 2211 <212> TYPE: DNA <213> ORGANISM: Artificial Sequence <220> FEATURE: <223> OTHER INFORMATION: AAVhu32 <220> FEATURE: <221> NAME/KEY: CDS <222> LOCATION: (1)..(2211)			
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atg gct gcc gat ggt tat ctt cca gat tgg ctc gag gac act ctc tct			48
Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Thr Leu Ser			
1 5 10 15			
gaa gga ata aga cag tgg tgg aag ctc aaa cct ggc cca cca cca cca			96
Glu Gly Ile Arg Gln Trp Trp Lys Leu Lys Pro Gly Pro Pro Pro Pro			
20 25 30			
aag ccc gca gag cgg cat aag gac gac agc agg ggt ctt gtg ctt cct			144
Lys Pro Ala Glu Arg His Lys Asp Asp Ser Arg Gly Leu Val Leu Pro			
35 40 45			
ggg tac aag tac ctc gga ccc ggc aac gga ctc gac aag ggg gag ccg			192
Gly Tyr Lys Tyr Leu Gly Pro Gly Asn Gly Leu Asp Lys Gly Glu Pro			
50 55 60			
gtc aac gca gca gac gcg gcg gcc ctc gag cac gac aag gcc tac gac			240
Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp			
65 70 75 80			
cag cag ctc aag gcc gga gac aac ccg tac ctc aag tac aac cac gcc			288
Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Lys Tyr Asn His Ala			
85 90 95			
gac gcc gag ttc cag gag cgg ctc aaa gaa gat acg tct ttt ggg ggc			336
Asp Ala Glu Phe Gln Glu Arg Leu Lys Glu Asp Thr Ser Phe Gly Gly			
100 105 110			
aac ctc ggg cga gca gtc ttc cag gcc aaa aag agg ctt ctt gaa cct			384
Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Leu Leu Glu Pro			
115 120 125			
ctt ggt ctg gtt gag gaa gcg gct aag acg gct cct gga aag aag agg			432
Leu Gly Leu Val Glu Glu Ala Ala Lys Thr Ala Pro Gly Lys Lys Arg			
130 135 140			
cct gta gag cag tct cct cag gaa ccg gac tcc tcc gcg ggt att ggc			480
Pro Val Glu Gln Ser Pro Gln Glu Pro Asp Ser Ser Ala Gly Ile Gly			
145 150 155 160			
aaa tcg ggt tca cag ccc gct aaa aag aaa ctc aat ttc ggt cag act			528
Lys Ser Gly Ser Gln Pro Ala Lys Lys Lys Leu Asn Phe Gly Gln Thr			
165 170 175			
ggc gac aca gag tca gtc ccc gac cct caa cca atc gga gaa cct ccc			576
Gly Asp Thr Glu Ser Val Pro Asp Pro Gln Pro Ile Gly Glu Pro Pro			
180 185 190			
gca gcc ccc tca ggt gtg gga tct ctt aca atg gct tca ggt ggt ggc			624
Ala Ala Pro Ser Gly Val Gly Ser Leu Thr Met Ala Ser Gly Gly Gly			
195 200 205			
gca cca gtg gca gac aat aac gaa ggt gcc gat gga gtg ggt agt tcc			672
Ala Pro Val Ala Asp Asn Asn Glu Gly Ala Asp Gly Val Gly Ser Ser			
210 215 220			
tcg gga aat tgg cat tgc gat tcc caa tgg ctg ggg gac aga gtc atc			720
Ser Gly Asn Trp His Cys Asp Ser Gln Trp Leu Gly Asp Arg Val Ile			
225 230 235 240			
acc acc agc acc cga acc tgg gcc ctg ccc acc tac aac aat cac ctc			768

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Thr	Thr	Ser	Thr	Arg	Thr	Trp	Ala	Leu	Pro	Thr	Tyr	Asn	Asn	His	Leu	
				245					250					255		
tac	aag	caa	atc	tcc	aac	agc	aca	tct	gga	gga	tct	tca	aat	gac	aac	816
Tyr	Lys	Gln	Ile	Ser	Asn	Ser	Thr	Ser	Gly	Gly	Ser	Ser	Asn	Asp	Asn	
			260					265					270			
gcc	tac	ttc	ggc	tac	agc	acc	ccc	tgg	ggg	tat	ttt	gac	ttc	aac	aga	864
Ala	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro	Trp	Gly	Tyr	Phe	Asp	Phe	Asn	Arg	
			275				280					285				
ttc	cac	tgc	cac	ttc	tca	cca	cgt	gac	tgg	cag	cga	ctc	atc	aac	aac	912
Phe	His	Cys	His	Phe	Ser	Pro	Arg	Asp	Trp	Gln	Arg	Leu	Ile	Asn	Asn	
	290					295				300						
aac	tgg	gga	ttc	cgg	cct	aag	cga	ctc	aac	ttc	aag	ctc	ttc	aac	att	960
Asn	Trp	Gly	Phe	Arg	Pro	Lys	Arg	Leu	Asn	Phe	Lys	Leu	Phe	Asn	Ile	
305					310					315				320		
cag	gtc	aaa	gag	gtt	acg	gac	aac	aat	gga	gtc	aag	acc	atc	gcc	aat	1008
Gln	Val	Lys	Glu	Val	Thr	Asp	Asn	Asn	Gly	Val	Lys	Thr	Ile	Ala	Asn	
			325						330					335		
aac	ctt	acc	agc	acg	gtc	cag	gtc	ttc	acg	gac	tca	gac	tat	cag	ctc	1056
Asn	Leu	Thr	Ser	Thr	Val	Gln	Val	Phe	Thr	Asp	Ser	Asp	Tyr	Gln	Leu	
			340				345						350			
ccg	tac	gtg	ctc	ggg	tcg	gct	cac	gag	ggc	tgc	ctc	ccg	ccg	ttc	cca	1104
Pro	Tyr	Val	Leu	Gly	Ser	Ala	His	Glu	Gly	Cys	Leu	Pro	Pro	Phe	Pro	
			355				360					365				
gcg	gac	gtt	ttc	atg	att	cct	cag	tac	ggg	tat	ctg	acg	ctt	aat	gat	1152
Ala	Asp	Val	Phe	Met	Ile	Pro	Gln	Tyr	Gly	Tyr	Leu	Thr	Leu	Asn	Asp	
	370					375					380					
ggg	agc	cag	gcc	gtg	ggc	cgt	tcg	tcc	ttt	tac	tgc	ctg	gaa	tat	ttc	1200
Gly	Ser	Gln	Ala	Val	Gly	Arg	Ser	Ser	Phe	Tyr	Cys	Leu	Glu	Tyr	Phe	
385					390					395				400		
ccg	tcg	caa	atg	cta	aga	acg	ggc	aac	aac	ttc	cag	ttc	agc	tac	gag	1248
Pro	Ser	Gln	Met	Leu	Arg	Thr	Gly	Asn	Asn	Phe	Gln	Phe	Ser	Tyr	Glu	
			405					410					415			
ttt	gag	aac	gta	cct	ttc	cat	agc	agc	tac	gct	cac	agc	caa	agc	ctg	1296
Phe	Glu	Asn	Val	Pro	Phe	His	Ser	Ser	Tyr	Ala	His	Ser	Gln	Ser	Leu	
		420					425						430			
gac	cga	cta	atg	aat	cca	ctc	atc	gac	caa	tac	ttg	tac	tat	ctc	tca	1344
Asp	Arg	Leu	Met	Asn	Pro	Leu	Ile	Asp	Gln	Tyr	Leu	Tyr	Tyr	Leu	Ser	
		435				440						445				
aag	act	att	aac	ggc	tct	gga	cag	aat	caa	caa	acg	cta	aaa	ttc	agc	1392
Lys	Thr	Ile	Asn	Gly	Ser	Gly	Gln	Asn	Gln	Gln	Thr	Leu	Lys	Phe	Ser	
	450					455					460					
gtg	gcc	gga	ccc	agc	aac	atg	gct	gtc	cag	gga	aga	aac	tac	ata	cct	1440
Val	Ala	Gly	Pro	Ser	Asn	Met	Ala	Val	Gln	Gly	Arg	Asn	Tyr	Ile	Pro	
465					470					475				480		
gga	ccc	agc	tac	cga	caa	caa	cgt	gtc	tca	acc	act	gtg	act	caa	aac	1488
Gly	Pro	Ser	Tyr	Arg	Gln	Gln	Arg	Val	Ser	Thr	Thr	Val	Thr	Gln	Asn	
			485					490					495			
aac	aac	agc	gaa	ttt	gct	tgg	cct	gga	gct	tct	tct	tgg	gct	ctc	aat	1536
Asn	Asn	Ser	Glu	Phe	Ala	Trp	Pro	Gly	Ala	Ser	Ser	Trp	Ala	Leu	Asn	
			500				505					510				
gga	cgt	aat	agc	ttg	atg	aat	cct	gga	cct	gct	atg	gcc	agc	cac	aaa	1584
Gly	Arg	Asn	Ser	Leu	Met	Asn	Pro	Gly	Pro	Ala	Met	Ala	Ser	His	Lys	
		515				520						525				
gaa	gga	gag	gac	cgt	ttc	ttt	cct	ttg	tct	gga	tct	tta	att	ttt	ggc	1632
Glu	Gly	Glu	Asp	Arg	Phe	Phe	Pro	Leu	Ser	Gly	Ser	Leu	Ile	Phe	Gly	
	530				535					540						
aaa	caa	gga	act	gga	aga	gac	aac	gtg	gat	gcg	gac	aaa	gtc	atg	ata	1680
Lys	Gln	Gly	Thr	Gly	Arg	Asp	Asn	Val	Asp	Ala	Asp	Lys	Val	Met	Ile	
545					550					555				560		

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acc aac gaa gaa gaa att aaa act act aac ccg gta gca acg gag tcc	1728
Thr Asn Glu Glu Glu Ile Lys Thr Thr Asn Pro Val Ala Thr Glu Ser	
565 570 575	
tat gga caa gtg gcc aca aac cac cag agt gcc caa gca cag gcg cag	1776
Tyr Gly Gln Val Ala Thr Asn His Gln Ser Ala Gln Ala Gln Ala Gln	
580 585 590	
acc ggc tgg gtt caa aac caa gga ata ctt ccg ggt atg gtt tgg cag	1824
Thr Gly Trp Val Gln Asn Gln Gly Ile Leu Pro Gly Met Val Trp Gln	
595 600 605	
gac aga gat gtg tac ctg caa gga ccc att tgg gcc aaa att cct cac	1872
Asp Arg Asp Val Tyr Leu Gln Gly Pro Ile Trp Ala Lys Ile Pro His	
610 615 620	
acg gac ggc aac ttt cac cct tct ccg cta atg gga ggg ttt gga atg	1920
Thr Asp Gly Asn Phe His Pro Ser Pro Leu Met Gly Gly Phe Gly Met	
625 630 635 640	
aag cac ccg cct cct cag atc ctc atc aaa aac aca cct gta cct gcg	1968
Lys His Pro Pro Pro Gln Ile Leu Ile Lys Asn Thr Pro Val Pro Ala	
645 650 655	
gat cct cca acg gct ttc aat aag gac aag ctg aac tct ttc atc acc	2016
Asp Pro Pro Thr Ala Phe Asn Lys Asp Lys Leu Asn Ser Phe Ile Thr	
660 665 670	
cag tat tct act ggc caa gtc agc gtg gag att gag tgg gag ctg cag	2064
Gln Tyr Ser Thr Gly Gln Val Ser Val Glu Ile Glu Trp Glu Leu Gln	
675 680 685	
aag gaa aac agc aag cgc tgg aac ccg gag atc cag tac act tcc aac	2112
Lys Glu Asn Ser Lys Arg Trp Asn Pro Glu Ile Gln Tyr Thr Ser Asn	
690 695 700	
tat tac aag tct aat aat gtt gaa ttt gct gtt aat act gaa ggt gta	2160
Tyr Tyr Lys Ser Asn Asn Val Glu Phe Ala Val Asn Thr Glu Gly Val	
705 710 715 720	
tat agt gaa ccc cgc ccc att ggc acc aga tac ctg act cgt aat ctg	2208
Tyr Ser Glu Pro Arg Pro Ile Gly Thr Arg Tyr Leu Thr Arg Asn Leu	
725 730 735	
taa	2211

<210> SEQ ID NO 8

<211> LENGTH: 736

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 8

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Lys Pro Ala Glu Arg His Lys Asp Asp Ser Arg Gly Leu Val Leu Pro
35 40 45

Gly Tyr Lys Tyr Leu Gly Pro Gly Asn Gly Leu Asp Lys Gly Glu Pro
50 55 60

Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp
65 70 75 80

Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Lys Tyr Asn His Ala
85 90 95

Asp Ala Glu Phe Gln Glu Arg Leu Lys Glu Asp Thr Ser Phe Gly Gly
100 105 110

Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Leu Leu Glu Pro
115 120 125

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Leu Gly Leu Val Glu Glu Ala Ala Lys Thr Ala Pro Gly Lys Lys Arg
 130 135 140
 Pro Val Glu Gln Ser Pro Gln Glu Pro Asp Ser Ser Ala Gly Ile Gly
 145 150 155 160
 Lys Ser Gly Ser Gln Pro Ala Lys Lys Lys Leu Asn Phe Gly Gln Thr
 165 170 175
 Gly Asp Thr Glu Ser Val Pro Asp Pro Gln Pro Ile Gly Glu Pro Pro
 180 185 190
 Ala Ala Pro Ser Gly Val Gly Ser Leu Thr Met Ala Ser Gly Gly Gly
 195 200 205
 Ala Pro Val Ala Asp Asn Asn Glu Gly Ala Asp Gly Val Gly Ser Ser
 210 215 220
 Ser Gly Asn Trp His Cys Asp Ser Gln Trp Leu Gly Asp Arg Val Ile
 225 230 235 240
 Thr Thr Ser Thr Arg Thr Trp Ala Leu Pro Thr Tyr Asn Asn His Leu
 245 250 255
 Tyr Lys Gln Ile Ser Asn Ser Thr Ser Gly Gly Ser Ser Asn Asp Asn
 260 265 270
 Ala Tyr Phe Gly Tyr Ser Thr Pro Trp Gly Tyr Phe Asp Phe Asn Arg
 275 280 285
 Phe His Cys His Phe Ser Pro Arg Asp Trp Gln Arg Leu Ile Asn Asn
 290 295 300
 Asn Trp Gly Phe Arg Pro Lys Arg Leu Asn Phe Lys Leu Phe Asn Ile
 305 310 315 320
 Gln Val Lys Glu Val Thr Asp Asn Asn Gly Val Lys Thr Ile Ala Asn
 325 330 335
 Asn Leu Thr Ser Thr Val Gln Val Phe Thr Asp Ser Asp Tyr Gln Leu
 340 345 350
 Pro Tyr Val Leu Gly Ser Ala His Glu Gly Cys Leu Pro Pro Phe Pro
 355 360 365
 Ala Asp Val Phe Met Ile Pro Gln Tyr Gly Tyr Leu Thr Leu Asn Asp
 370 375 380
 Gly Ser Gln Ala Val Gly Arg Ser Ser Phe Tyr Cys Leu Glu Tyr Phe
 385 390 395 400
 Pro Ser Gln Met Leu Arg Thr Gly Asn Asn Phe Gln Phe Ser Tyr Glu
 405 410 415
 Phe Glu Asn Val Pro Phe His Ser Ser Tyr Ala His Ser Gln Ser Leu
 420 425 430
 Asp Arg Leu Met Asn Pro Leu Ile Asp Gln Tyr Leu Tyr Tyr Leu Ser
 435 440 445
 Lys Thr Ile Asn Gly Ser Gly Gln Asn Gln Gln Thr Leu Lys Phe Ser
 450 455 460
 Val Ala Gly Pro Ser Asn Met Ala Val Gln Gly Arg Asn Tyr Ile Pro
 465 470 475 480
 Gly Pro Ser Tyr Arg Gln Gln Arg Val Ser Thr Thr Val Thr Gln Asn
 485 490 495
 Asn Asn Ser Glu Phe Ala Trp Pro Gly Ala Ser Ser Trp Ala Leu Asn
 500 505 510
 Gly Arg Asn Ser Leu Met Asn Pro Gly Pro Ala Met Ala Ser His Lys
 515 520 525
 Glu Gly Glu Asp Arg Phe Phe Pro Leu Ser Gly Ser Leu Ile Phe Gly
 530 535 540

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Lys Gln Gly Thr Gly Arg Asp Asn Val Asp Ala Asp Lys Val Met Ile
 545 550 555 560
 Thr Asn Glu Glu Glu Ile Lys Thr Thr Asn Pro Val Ala Thr Glu Ser
 565 570 575
 Tyr Gly Gln Val Ala Thr Asn His Gln Ser Ala Gln Ala Gln Ala Gln
 580 585 590
 Thr Gly Trp Val Gln Asn Gln Gly Ile Leu Pro Gly Met Val Trp Gln
 595 600 605
 Asp Arg Asp Val Tyr Leu Gln Gly Pro Ile Trp Ala Lys Ile Pro His
 610 615 620
 Thr Asp Gly Asn Phe His Pro Ser Pro Leu Met Gly Gly Phe Gly Met
 625 630 635 640
 Lys His Pro Pro Pro Gln Ile Leu Ile Lys Asn Thr Pro Val Pro Ala
 645 650 655
 Asp Pro Pro Thr Ala Phe Asn Lys Asp Lys Leu Asn Ser Phe Ile Thr
 660 665 670
 Gln Tyr Ser Thr Gly Gln Val Ser Val Glu Ile Glu Trp Glu Leu Gln
 675 680 685
 Lys Glu Asn Ser Lys Arg Trp Asn Pro Glu Ile Gln Tyr Thr Ser Asn
 690 695 700
 Tyr Tyr Lys Ser Asn Asn Val Glu Phe Ala Val Asn Thr Glu Gly Val
 705 710 715 720
 Tyr Ser Glu Pro Arg Pro Ile Gly Thr Arg Tyr Leu Thr Arg Asn Leu
 725 730 735

<210> SEQ ID NO 9
 <211> LENGTH: 2178
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: AAV5.5.9
 <220> FEATURE:
 <221> NAME/KEY: CDS
 <222> LOCATION: (1)..(2178)

<400> SEQUENCE: 9

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ggt ctt cgc gag ttt ttg ggc ctt gaa gcg ggc cca ccg aaa cca aaa	96
Gly Leu Arg Glu Phe Leu Gly Leu Glu Ala Gly Pro Pro Lys Pro Lys	
20 25 30	
ccc aat cag cag cat caa gat caa gcc cgt ggt ctt gtg ctg cct ggt	144
Pro Asn Gln Gln His Gln Asp Gln Ala Arg Gly Leu Val Leu Pro Gly	
35 40 45	
tat aac tat ctc gga ccc gga aac ggt ctc gat cga gga gag cct gtc	192
Tyr Asn Tyr Leu Gly Pro Gly Asn Gly Leu Asp Arg Gly Glu Pro Val	
50 55 60	
aac agg gca gac gag gtc gcg cga gag cac gac atc tcg tac aac gag	240
Asn Arg Ala Asp Glu Val Ala Arg Glu His Asp Ile Ser Tyr Asn Glu	
65 70 75 80	
cag ctt gag gcg gga gac aac ccc tac ctc aag tac aac cac gcg gac	288
Gln Leu Glu Ala Gly Asp Asn Pro Tyr Leu Lys Tyr Asn His Ala Asp	
85 90 95	
gcc gag ttt cag gag aag ctc gcc gac gac aca tcc ttc ggg gga aac	336
Ala Glu Phe Gln Glu Lys Leu Ala Asp Asp Thr Ser Phe Gly Gly Asn	
100 105 110	
ctc gga aag gca gtc ttt cag gcc aag aaa agg gtt ctc gaa cct ttt	384
Leu Gly Lys Ala Val Phe Gln Ala Lys Lys Arg Val Leu Glu Pro Phe	

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115	120	125	
ggc ctg gtt gaa gag ggt gct aag acg gcc cct acc gga aag cgg ata Gly Leu Val Glu Glu Gly Ala Lys Thr Ala Pro Thr Gly Lys Arg Ile 130 135 140			432
gac gac cac ttt cca aaa aga aag aag gcc cgg acc gaa gag gac tcc Asp Asp His Phe Pro Lys Arg Lys Lys Ala Arg Thr Glu Glu Asp Ser 145 150 155 160			480
aag cct tcc acc tcg tca gac gcc gaa gct gga ccc agc gga tcc cag Lys Pro Ser Thr Ser Ser Asp Ala Glu Ala Gly Pro Ser Gly Ser Gln 165 170 175			528
cag ctg caa atc cca gcc caa cca gcc tca agt ttg gga gct gat aca Gln Leu Gln Ile Pro Ala Gln Pro Ala Ser Ser Leu Gly Ala Asp Thr 180 185 190			576
atg gct tca ggt ggt gcc gca cca gtg gca gac aat aac gaa ggt gcc Met Ala Ser Gly Gly Gly Ala Pro Val Ala Asp Asn Asn Glu Gly Ala 195 200 205			624
gat gga gtg ggt agt tcc tcg gga aat tgg cat tgc gat tcc caa tgg Asp Gly Val Gly Ser Ser Ser Gly Asn Trp His Cys Asp Ser Gln Trp 210 215 220			672
ctg ggg gac aga gtc atc acc acc agc acc cga acc tgg gcc ctg ccc Leu Gly Asp Arg Val Ile Thr Thr Ser Thr Arg Thr Trp Ala Leu Pro 225 230 235 240			720
acc tac aac aat cac ctc tac aag caa atc tcc aac agc aca tct gga Thr Tyr Asn Asn His Leu Tyr Lys Gln Ile Ser Asn Ser Thr Ser Gly 245 250 255			768
gga tct tca aat gac aac gcc tac ttc ggc tac agc acc ccc tgg ggg Gly Ser Ser Asn Asp Asn Ala Tyr Phe Gly Tyr Ser Thr Pro Trp Gly 260 265 270			816
tat ttt gac ttc aac aga ttc cac tgc cac ttc tca cca cgt gac tgg Tyr Phe Asp Phe Asn Arg Phe His Cys His Phe Ser Pro Arg Asp Trp 275 280 285			864
cag cga ctc atc aac aac aac tgg gga ttc cgg cct aag cga ctc aac Gln Arg Leu Ile Asn Asn Asn Trp Gly Phe Arg Pro Lys Arg Leu Asn 290 295 300			912
ttc aag ctc ttc aac att cag gtc aaa gag gtt acg gac aac aat gga Phe Lys Leu Phe Asn Ile Gln Val Lys Glu Val Thr Asp Asn Asn Gly 305 310 315 320			960
gtc aag acc atc gcc aat aac ctt acc agc acg gtc cag gtc ttc acg Val Lys Thr Ile Ala Asn Asn Leu Thr Ser Thr Val Gln Val Phe Thr 325 330 335			1008
gac tca gac tat cag ctc ccg tac gtg ctc ggg tcg gct cac gag ggc Asp Ser Asp Tyr Gln Leu Pro Tyr Val Leu Gly Ser Ala His Glu Gly 340 345 350			1056
tgc ctc ccg ccg ttc cca gcg gac gtt ttc atg att cct cag tac ggg Cys Leu Pro Pro Phe Pro Ala Asp Val Phe Met Ile Pro Gln Tyr Gly 355 360 365			1104
tat ctg acg ctt aat gat gga agc cag gcc gtg ggt cgt tcg tcc ttt Tyr Leu Thr Leu Asn Asp Gly Ser Gln Ala Val Gly Arg Ser Ser Phe 370 375 380			1152
tac tgc ctg gaa tat ttc ccg tcg caa atg cta aga acg ggt aac aac Tyr Cys Leu Glu Tyr Phe Pro Ser Gln Met Leu Arg Thr Gly Asn Asn 385 390 395 400			1200
ttc cag ttc agc tac gag ttt gag aac gta cct ttc cat agc agc tac Phe Gln Phe Ser Tyr Glu Phe Glu Asn Val Pro Phe His Ser Ser Tyr 405 410 415			1248
gct cac agc caa agc ctg gac cga cta atg aat cca ctc atc gac caa Ala His Ser Gln Ser Leu Asp Arg Leu Met Asn Pro Leu Ile Asp Gln 420 425 430			1296
tac ttg tac tat ctc tca aag act att aac ggt tct gga cag aat caa			1344

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Tyr	Leu	Tyr	Tyr	Leu	Ser	Lys	Thr	Ile	Asn	Gly	Ser	Gly	Gln	Asn	Gln	
	435						440					445				
caa	acg	cta	aaa	ttc	agt	gtg	gcc	gga	ccc	agc	aac	atg	gct	gtc	cag	1392
Gln	Thr	Leu	Lys	Phe	Ser	Val	Ala	Gly	Pro	Ser	Asn	Met	Ala	Val	Gln	
	450					455				460						
gga	aga	aac	tac	ata	cct	gga	ccc	agc	tac	cga	caa	caa	cgt	gtc	tca	1440
Gly	Arg	Asn	Tyr	Ile	Pro	Gly	Pro	Ser	Tyr	Arg	Gln	Gln	Arg	Val	Ser	
465					470					475					480	
acc	act	gtg	act	caa	aac	aac	aac	agc	gaa	ttt	gct	tgg	cct	gga	gct	1488
Thr	Thr	Val	Thr	Gln	Asn	Asn	Asn	Ser	Glu	Phe	Ala	Trp	Pro	Gly	Ala	
			485					490						495		
tct	tct	tgg	gct	ctc	aat	gga	cgt	aat	agc	ttg	atg	aat	cct	gga	cct	1536
Ser	Ser	Trp	Ala	Leu	Asn	Gly	Arg	Asn	Ser	Leu	Met	Asn	Pro	Gly	Pro	
		500					505						510			
gct	atg	gcc	agc	cac	aaa	gaa	gga	gag	gac	cgt	ttc	ttt	cct	ttg	tct	1584
Ala	Met	Ala	Ser	His	Lys	Glu	Gly	Glu	Asp	Arg	Phe	Phe	Pro	Leu	Ser	
	515					520					525					
gga	tct	tta	att	ttt	ggc	aaa	caa	gga	act	gga	aga	gac	aac	gtg	gat	1632
Gly	Ser	Leu	Ile	Phe	Gly	Lys	Gln	Gly	Thr	Gly	Arg	Asp	Asn	Val	Asp	
	530				535					540						
gcg	gac	aaa	gtc	atg	ata	acc	aac	gaa	gaa	gaa	att	aaa	act	act	aac	1680
Ala	Asp	Lys	Val	Met	Ile	Thr	Asn	Glu	Glu	Glu	Ile	Lys	Thr	Thr	Asn	
545					550					555					560	
ccg	gta	gca	acg	gag	tcc	tat	gga	caa	gtg	gcc	aca	aac	cac	cag	agt	1728
Pro	Val	Ala	Thr	Glu	Ser	Tyr	Gly	Gln	Val	Ala	Thr	Asn	His	Gln	Ser	
		565					570							575		
gcc	caa	gca	cag	gcg	cag	acc	ggc	tgg	gtt	caa	aac	caa	gga	ata	ctt	1776
Ala	Gln	Ala	Gln	Ala	Gln	Thr	Gly	Trp	Val	Gln	Asn	Gln	Gly	Ile	Leu	
	580						585						590			
ccg	ggt	atg	gtt	tgg	cag	gac	aga	gat	gtg	tac	ctg	caa	gga	ccc	att	1824
Pro	Gly	Met	Val	Trp	Gln	Asp	Arg	Asp	Val	Tyr	Leu	Gln	Gly	Pro	Ile	
	595					600					605					
tgg	gcc	aaa	att	cct	cac	acg	gac	ggc	aac	ttt	cac	cct	tct	ccg	ctg	1872
Trp	Ala	Lys	Ile	Pro	His	Thr	Asp	Gly	Asn	Phe	His	Pro	Ser	Pro	Leu	
	610				615					620						
atg	gga	ggg	ttt	gga	atg	aag	cac	ccg	cct	cct	cag	atc	ctc	atc	aaa	1920
Met	Gly	Gly	Phe	Gly	Met	Lys	His	Pro	Pro	Pro	Gln	Ile	Leu	Ile	Lys	
625					630					635					640	
aac	aca	cct	gta	cct	gcg	gat	cct	cca	acg	gcc	ttc	aac	aag	gac	aag	1968
Asn	Thr	Pro	Val	Pro	Ala	Asp	Pro	Pro	Thr	Ala	Phe	Asn	Lys	Asp	Lys	
		645					650							655		
ctg	aac	tct	ttc	atc	acc	cag	tat	tct	act	ggc	caa	gtc	agc	gtg	gag	2016
Leu	Asn	Ser	Phe	Ile	Thr	Gln	Tyr	Ser	Thr	Gly	Gln	Val	Ser	Val	Glu	
		660					665							670		
atc	gag	tgg	gag	ctg	cag	aag	gaa	aac	agc	aag	cgc	tgg	aac	ccg	gag	2064
Ile	Glu	Trp	Glu	Leu	Gln	Lys	Glu	Asn	Ser	Lys	Arg	Trp	Asn	Pro	Glu	
	675					680						685				
atc	cag	tac	act	tcc	aac	tat	tac	aag	tct	aat	aat	gtt	gaa	ttt	gct	2112
Ile	Gln	Tyr	Thr	Ser	Asn	Tyr	Tyr	Lys	Ser	Asn	Asn	Val	Glu	Phe	Ala	
	690					695				700						
gtt	aat	act	gaa	ggt	gta	tat	agt	gaa	ccc	cgc	ccc	att	ggc	acc	aga	2160
Val	Asn	Thr	Glu	Gly	Val	Tyr	Ser	Glu	Pro	Arg	Pro	Ile	Gly	Thr	Arg	
705					710					715					720	
tac	ctg	act	cgt	aat	ctg											2178
Tyr	Leu	Thr	Arg	Asn	Leu											
			725													

<210> SEQ ID NO 10

<211> LENGTH: 726

<212> TYPE: PRT

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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 10

Met Ser Phe Val Asp His Pro Pro Asp Trp Leu Glu Glu Val Gly Glu
1          5          10          15

Gly Leu Arg Glu Phe Leu Gly Leu Glu Ala Gly Pro Pro Lys Pro Lys
20          25          30

Pro Asn Gln Gln His Gln Asp Gln Ala Arg Gly Leu Val Leu Pro Gly
35          40          45

Tyr Asn Tyr Leu Gly Pro Gly Asn Gly Leu Asp Arg Gly Glu Pro Val
50          55          60

Asn Arg Ala Asp Glu Val Ala Arg Glu His Asp Ile Ser Tyr Asn Glu
65          70          75          80

Gln Leu Glu Ala Gly Asp Asn Pro Tyr Leu Lys Tyr Asn His Ala Asp
85          90          95

Ala Glu Phe Gln Glu Lys Leu Ala Asp Asp Thr Ser Phe Gly Gly Asn
100         105         110

Leu Gly Lys Ala Val Phe Gln Ala Lys Lys Arg Val Leu Glu Pro Phe
115         120         125

Gly Leu Val Glu Glu Gly Ala Lys Thr Ala Pro Thr Gly Lys Arg Ile
130         135         140

Asp Asp His Phe Pro Lys Arg Lys Lys Ala Arg Thr Glu Glu Asp Ser
145         150         155         160

Lys Pro Ser Thr Ser Ser Asp Ala Glu Ala Gly Pro Ser Gly Ser Gln
165         170         175

Gln Leu Gln Ile Pro Ala Gln Pro Ala Ser Ser Leu Gly Ala Asp Thr
180         185         190

Met Ala Ser Gly Gly Gly Ala Pro Val Ala Asp Asn Asn Glu Gly Ala
195         200         205

Asp Gly Val Gly Ser Ser Ser Gly Asn Trp His Cys Asp Ser Gln Trp
210         215         220

Leu Gly Asp Arg Val Ile Thr Thr Ser Thr Arg Thr Trp Ala Leu Pro
225         230         235         240

Thr Tyr Asn Asn His Leu Tyr Lys Gln Ile Ser Asn Ser Thr Ser Gly
245         250         255

Gly Ser Ser Asn Asp Asn Ala Tyr Phe Gly Tyr Ser Thr Pro Trp Gly
260         265         270

Tyr Phe Asp Phe Asn Arg Phe His Cys His Phe Ser Pro Arg Asp Trp
275         280         285

Gln Arg Leu Ile Asn Asn Asn Trp Gly Phe Arg Pro Lys Arg Leu Asn
290         295         300

Phe Lys Leu Phe Asn Ile Gln Val Lys Glu Val Thr Asp Asn Asn Gly
305         310         315         320

Val Lys Thr Ile Ala Asn Asn Leu Thr Ser Thr Val Gln Val Phe Thr
325         330         335

Asp Ser Asp Tyr Gln Leu Pro Tyr Val Leu Gly Ser Ala His Glu Gly
340         345         350

Cys Leu Pro Pro Phe Pro Ala Asp Val Phe Met Ile Pro Gln Tyr Gly
355         360         365

Tyr Leu Thr Leu Asn Asp Gly Ser Gln Ala Val Gly Arg Ser Ser Phe
370         375         380

Tyr Cys Leu Glu Tyr Phe Pro Ser Gln Met Leu Arg Thr Gly Asn Asn

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385	390	395	400
Phe Gln Phe Ser Tyr Glu Phe Glu Asn Val Pro Phe His Ser Ser Tyr	405	410	415
Ala His Ser Gln Ser Leu Asp Arg Leu Met Asn Pro Leu Ile Asp Gln	420	425	430
Tyr Leu Tyr Tyr Leu Ser Lys Thr Ile Asn Gly Ser Gly Gln Asn Gln	435	440	445
Gln Thr Leu Lys Phe Ser Val Ala Gly Pro Ser Asn Met Ala Val Gln	450	455	460
Gly Arg Asn Tyr Ile Pro Gly Pro Ser Tyr Arg Gln Gln Arg Val Ser	465	470	475
Thr Thr Val Thr Gln Asn Asn Asn Ser Glu Phe Ala Trp Pro Gly Ala	485	490	495
Ser Ser Trp Ala Leu Asn Gly Arg Asn Ser Leu Met Asn Pro Gly Pro	500	505	510
Ala Met Ala Ser His Lys Glu Gly Glu Asp Arg Phe Phe Pro Leu Ser	515	520	525
Gly Ser Leu Ile Phe Gly Lys Gln Gly Thr Gly Arg Asp Asn Val Asp	530	535	540
Ala Asp Lys Val Met Ile Thr Asn Glu Glu Glu Ile Lys Thr Thr Asn	545	550	555
Pro Val Ala Thr Glu Ser Tyr Gly Gln Val Ala Thr Asn His Gln Ser	565	570	575
Ala Gln Ala Gln Ala Gln Thr Gly Trp Val Gln Asn Gln Gly Ile Leu	580	585	590
Pro Gly Met Val Trp Gln Asp Arg Asp Val Tyr Leu Gln Gly Pro Ile	595	600	605
Trp Ala Lys Ile Pro His Thr Asp Gly Asn Phe His Pro Ser Pro Leu	610	615	620
Met Gly Gly Phe Gly Met Lys His Pro Pro Pro Gln Ile Leu Ile Lys	625	630	635
Asn Thr Pro Val Pro Ala Asp Pro Pro Thr Ala Phe Asn Lys Asp Lys	645	650	655
Leu Asn Ser Phe Ile Thr Gln Tyr Ser Thr Gly Gln Val Ser Val Glu	660	665	670
Ile Glu Trp Glu Leu Gln Lys Glu Asn Ser Lys Arg Trp Asn Pro Glu	675	680	685
Ile Gln Tyr Thr Ser Asn Tyr Tyr Lys Ser Asn Asn Val Glu Phe Ala	690	695	700
Val Asn Thr Glu Gly Val Tyr Ser Glu Pro Arg Pro Ile Gly Thr Arg	705	710	715
Tyr Leu Thr Arg Asn Leu	725		

<210> SEQ ID NO 11
 <211> LENGTH: 2229
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: AAVPHP.B
 <220> FEATURE:
 <221> NAME/KEY: CDS
 <222> LOCATION: (1)..(2229)

<400> SEQUENCE: 11

atg gct gcc gat ggt tat ctt cca gat tgg ctc gag gac aac ctt agt

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Met	Ala	Ala	Asp	Gly	Tyr	Leu	Pro	Asp	Trp	Leu	Glu	Asp	Asn	Leu	Ser	
1				5					10					15		
gaa gga att cgc gag tgg tgg gct ttg aaa cct gga gcc cct caa ccc																96
Glu	Gly	Ile	Arg	Glu	Trp	Trp	Ala	Leu	Lys	Pro	Gly	Ala	Pro	Gln	Pro	
			20					25					30			
aag gca aat caa caa cat caa gac aac gct cga ggt ctt gtg ctt ccg																144
Lys	Ala	Asn	Gln	Gln	His	Gln	Asp	Asn	Ala	Arg	Gly	Leu	Val	Leu	Pro	
		35					40					45				
ggt tac aaa tac ctt gga ccc gcc aac gga ctc gac aag ggg gag ccg																192
Gly	Tyr	Lys	Tyr	Leu	Gly	Pro	Gly	Asn	Gly	Leu	Asp	Lys	Gly	Glu	Pro	
	50					55					60					
gtc aac gca gca gac gcg gcg gcc ctc gag cac gac aag gcc tac gac																240
Val	Asn	Ala	Ala	Asp	Ala	Ala	Ala	Leu	Glu	His	Asp	Lys	Ala	Tyr	Asp	
65					70					75					80	
cag cag ctc aag gcc gga gac aac ccg tac ctc aag tac aac cac gcc																288
Gln	Gln	Leu	Lys	Ala	Gly	Asp	Asn	Pro	Tyr	Leu	Lys	Tyr	Asn	His	Ala	
			85						90					95		
gac gcc gag ttc cag gag cgg ctc aaa gaa gat acg tct ttt ggg ggc																336
Asp	Ala	Glu	Phe	Gln	Glu	Arg	Leu	Lys	Glu	Asp	Thr	Ser	Phe	Gly	Gly	
			100					105					110			
aac ctc ggg cga gca gtc ttc cag gcc aaa aag agg ctt ctt gaa cct																384
Asn	Leu	Gly	Arg	Ala	Val	Phe	Gln	Ala	Lys	Lys	Arg	Leu	Leu	Glu	Pro	
		115					120					125				
ctt ggt ctg gtt gag gaa gcg gct aag acg gct cct gga aag aag agg																432
Leu	Gly	Leu	Val	Glu	Glu	Ala	Ala	Lys	Thr	Ala	Pro	Gly	Lys	Lys	Arg	
	130					135					140					
cct gta gag cag tct cct cag gaa ccg gac tcc tcc gcg ggt att ggc																480
Pro	Val	Glu	Gln	Ser	Pro	Gln	Glu	Pro	Asp	Ser	Ser	Ala	Gly	Ile	Gly	
145					150						155				160	
aaa tcg ggt gca cag ccc gct aaa aag aga ctc aat ttc ggt cag act																528
Lys	Ser	Gly	Ala	Gln	Pro	Ala	Lys	Lys	Arg	Leu	Asn	Phe	Gly	Gln	Thr	
			165						170				175			
ggc gac aca gag tca gtc cca gac cct caa cca atc gga gaa cct ccc																576
Gly	Asp	Thr	Glu	Ser	Val	Pro	Asp	Pro	Gln	Pro	Ile	Gly	Glu	Pro	Pro	
		180						185					190			
gca gcc ccc tca ggt gtg gga tct ctt aca atg gct tca ggt ggt ggc																624
Ala	Ala	Pro	Ser	Gly	Val	Gly	Ser	Leu	Thr	Met	Ala	Ser	Gly	Gly	Gly	
		195						200				205				
gca cca gtg gca gac aat aac gaa ggt gcc gat gga gtg ggt agt tcc																672
Ala	Pro	Val	Ala	Asp	Asn	Asn	Glu	Gly	Ala	Asp	Gly	Val	Gly	Ser	Ser	
		210				215					220					
tcg gga aat tgg cat tgc gat tcc caa tgg ctg ggg gac aga gtc atc																720
Ser	Gly	Asn	Trp	His	Cys	Asp	Ser	Gln	Trp	Leu	Gly	Asp	Arg	Val	Ile	
		225			230					235				240		
acc acc agc acc cga acc tgg gcc ctg ccc acc tac aac aat cac ctc																768
Thr	Thr	Ser	Thr	Arg	Thr	Trp	Ala	Leu	Pro	Thr	Tyr	Asn	Asn	His	Leu	
			245						250					255		
tac aag caa atc tcc aac agc aca tct gga gga tct tca aat gac aac																816
Tyr	Lys	Gln	Ile	Ser	Asn	Ser	Thr	Ser	Gly	Gly	Ser	Ser	Asn	Asp	Asn	
		260						265					270			
gcc tac ttc ggc tac agc acc ccc tgg ggg tat ttt gac ttc aac aga																864
Ala	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro	Trp	Gly	Tyr	Phe	Asp	Phe	Asn	Arg	
		275					280					285				
ttc cac tgc cac ttc tca cca cgt gac tgg cag cga ctc atc aac aac																912
Phe	His	Cys	His	Phe	Ser	Pro	Arg	Asp	Trp	Gln	Arg	Leu	Ile	Asn	Asn	
		290				295				300						
aac tgg gga ttc cgg cct aag cga ctc aac ttc aag ctc ttc aac att																960
Asn	Trp	Gly	Phe	Arg	Pro	Lys	Arg	Leu	Asn	Phe	Lys	Leu	Phe	Asn	Ile	
305					310					315				320		

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cag gtc aaa gag gtt acg gac aac aat gga gtc aag acc atc gcc aat	1008
Gln Val Lys Glu Val Thr Asp Asn Asn Gly Val Lys Thr Ile Ala Asn	
325 330 335	
aac ctt acc agc acg gtc cag gtc ttc acg gac tca gac tat cag ctc	1056
Asn Leu Thr Ser Thr Val Gln Val Phe Thr Asp Ser Asp Tyr Gln Leu	
340 345 350	
ccg tac gtg ctc ggg tcg gct cac gag ggc tgc ctc ccg ccg ttc cca	1104
Pro Tyr Val Leu Gly Ser Ala His Glu Gly Cys Leu Pro Pro Phe Pro	
355 360 365	
gcg gac gtt ttc atg att cct cag tac ggg tat ctg acg ctt aat gat	1152
Ala Asp Val Phe Met Ile Pro Gln Tyr Gly Tyr Leu Thr Leu Asn Asp	
370 375 380	
gga agc cag gcc gtg ggt cgt tcg tcc ttt tac tgc ctg gaa tat ttc	1200
Gly Ser Gln Ala Val Gly Arg Ser Ser Phe Tyr Cys Leu Glu Tyr Phe	
385 390 395 400	
ccg tcg caa atg cta aga acg ggt aac aac ttc cag ttc agc tac gag	1248
Pro Ser Gln Met Leu Arg Thr Gly Asn Asn Phe Gln Phe Ser Tyr Glu	
405 410 415	
ttt gag aac gta cct ttc cat agc agc tac gct cac agc caa agc ctg	1296
Phe Glu Asn Val Pro Phe His Ser Ser Tyr Ala His Ser Gln Ser Leu	
420 425 430	
gac cga cta atg aat cca ctc atc gac caa tac ttg tac tat ctc tct	1344
Asp Arg Leu Met Asn Pro Leu Ile Asp Gln Tyr Leu Tyr Tyr Leu Ser	
435 440 445	
aga act att aac ggt tct gga cag aat caa caa acg cta aaa ttc agt	1392
Arg Thr Ile Asn Gly Ser Gly Gln Asn Gln Gln Thr Leu Lys Phe Ser	
450 455 460	
gtg gcc gga ccc agc aac atg gct gtc cag gga aga aac tac ata cct	1440
Val Ala Gly Pro Ser Asn Met Ala Val Gln Gly Arg Asn Tyr Ile Pro	
465 470 475 480	
gga ccc agc tac cga caa caa cgt gtc tca acc act gtg act caa aac	1488
Gly Pro Ser Tyr Arg Gln Gln Arg Val Ser Thr Thr Val Thr Gln Asn	
485 490 495	
aac aac agc gaa ttt gct tgg cct gga gct tct tct tgg gct ctc aat	1536
Asn Asn Ser Glu Phe Ala Trp Pro Gly Ala Ser Ser Trp Ala Leu Asn	
500 505 510	
gga cgt aat agc ttg atg aat cct gga cct gct atg gcc agc cac aaa	1584
Gly Arg Asn Ser Leu Met Asn Pro Gly Pro Ala Met Ala Ser His Lys	
515 520 525	
gaa gga gag gac cgt ttc ttt cct ttg tct gga tct tta att ttt ggc	1632
Glu Gly Glu Asp Arg Phe Phe Pro Leu Ser Gly Ser Leu Ile Phe Gly	
530 535 540	
aaa caa ggt acc ggc aga gac aac gtg gat gcg gac aaa gtc atg ata	1680
Lys Gln Gly Thr Gly Arg Asp Asn Val Asp Ala Asp Lys Val Met Ile	
545 550 555 560	
acc aac gaa gaa gaa att aaa act act aac ccg gta gca acg gag tcc	1728
Thr Asn Glu Glu Glu Ile Lys Thr Thr Asn Pro Val Ala Thr Glu Ser	
565 570 575	
tat gga caa gtg gcc aca aac cac cag agt gcc caa act ttg gcg gtg	1776
Tyr Gly Gln Val Ala Thr Asn His Gln Ser Ala Gln Thr Leu Ala Val	
580 585 590	
cct ttt aag gca cag gcg cag acc ggt tgg gtt caa aac caa gga ata	1824
Pro Phe Lys Ala Gln Ala Gln Thr Gly Trp Val Gln Asn Gln Gly Ile	
595 600 605	
ctt ccg ggt atg gtt tgg cag gac aga gat gtg tac ctg caa gga ccc	1872
Leu Pro Gly Met Val Trp Gln Asp Arg Asp Val Tyr Leu Gln Gly Pro	
610 615 620	
att tgg gcc aaa att cct cac acg gac ggc aac ttt cac cct tct ccg	1920
Ile Trp Ala Lys Ile Pro His Thr Asp Gly Asn Phe His Pro Ser Pro	
625 630 635 640	

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ctg atg gga ggg ttt gga atg aag cac ccg cct cct cag atc ctc atc	1968
Leu Met Gly Gly Phe Gly Met Lys His Pro Pro Pro Gln Ile Leu Ile	
645 650 655	
aaa aac aca cct gta cct gcg gat cct cca acg gcc ttc aac aag gac	2016
Lys Asn Thr Pro Val Pro Ala Asp Pro Pro Thr Ala Phe Asn Lys Asp	
660 665 670	
aag ctg aac tct ttc atc acc cag tat tct act ggc caa gtc agc gtg	2064
Lys Leu Asn Ser Phe Ile Thr Gln Tyr Ser Thr Gly Gln Val Ser Val	
675 680 685	
gag atc gag tgg gag ctg cag aag gaa aac agc aag cgc tgg aac ccg	2112
Glu Ile Glu Trp Glu Leu Gln Lys Glu Asn Ser Lys Arg Trp Asn Pro	
690 695 700	
gag atc cag tac act tcc aac tat tac aag tct aat aat gtt gaa ttt	2160
Glu Ile Gln Tyr Thr Ser Asn Tyr Tyr Lys Ser Asn Asn Val Glu Phe	
705 710 715 720	
gct gtt aat act gaa ggt gta tat agt gaa ccc cgc ccc att ggc acc	2208
Ala Val Asn Thr Glu Gly Val Tyr Ser Glu Pro Arg Pro Ile Gly Thr	
725 730 735	
aga tac ctg act cgt aat ctg	2229
Arg Tyr Leu Thr Arg Asn Leu	
740	

<210> SEQ ID NO 12
 <211> LENGTH: 743
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 12

Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser	
1 5 10 15	
Glu Gly Ile Arg Glu Trp Trp Ala Leu Lys Pro Gly Ala Pro Gln Pro	
20 25 30	
Lys Ala Asn Gln Gln His Gln Asp Asn Ala Arg Gly Leu Val Leu Pro	
35 40 45	
Gly Tyr Lys Tyr Leu Gly Pro Gly Asn Gly Leu Asp Lys Gly Glu Pro	
50 55 60	
Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp	
65 70 75 80	
Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Lys Tyr Asn His Ala	
85 90 95	
Asp Ala Glu Phe Gln Glu Arg Leu Lys Glu Asp Thr Ser Phe Gly Gly	
100 105 110	
Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Leu Leu Glu Pro	
115 120 125	
Leu Gly Leu Val Glu Glu Ala Ala Lys Thr Ala Pro Gly Lys Lys Arg	
130 135 140	
Pro Val Glu Gln Ser Pro Gln Glu Pro Asp Ser Ser Ala Gly Ile Gly	
145 150 155 160	
Lys Ser Gly Ala Gln Pro Ala Lys Lys Arg Leu Asn Phe Gly Gln Thr	
165 170 175	
Gly Asp Thr Glu Ser Val Pro Asp Pro Gln Pro Ile Gly Glu Pro Pro	
180 185 190	
Ala Ala Pro Ser Gly Val Gly Ser Leu Thr Met Ala Ser Gly Gly Gly	
195 200 205	
Ala Pro Val Ala Asp Asn Asn Glu Gly Ala Asp Gly Val Gly Ser Ser	

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210					215					220					
Ser 225	Gly	Asn	Trp	His 245	Cys 230	Asp	Ser	Gln	Trp 250	Leu 235	Gly	Asp	Arg	Val	Ile 240
Thr	Thr	Ser	Thr	Arg	Thr	Trp	Ala	Leu	Pro	Thr	Tyr	Asn	Asn	His 255	Leu
Tyr	Lys	Gln	Ile 260	Ser	Asn	Ser	Thr	Ser 265	Gly	Gly	Ser	Ser	Asn	Asp	Asn
Ala	Tyr	Phe 275	Gly	Tyr	Ser	Thr	Pro 280	Trp	Gly	Tyr	Phe	Asp 285	Phe	Asn	Arg
Phe	His 290	Cys	His	Phe	Ser	Pro 295	Arg	Asp	Trp	Gln	Arg 300	Leu	Ile	Asn	Asn
Asn 305	Trp	Gly	Phe	Arg	Pro 310	Lys	Arg	Leu	Asn	Phe 315	Lys	Leu	Phe	Asn	Ile 320
Gln	Val	Lys	Glu 325	Val	Thr	Asp	Asn	Asn	Gly 330	Val	Lys	Thr	Ile	Ala 335	Asn
Asn	Leu	Thr	Ser 340	Thr	Val	Gln	Val	Phe 345	Thr	Asp	Ser	Asp	Tyr 350	Gln	Leu
Pro	Tyr	Val 355	Leu	Gly	Ser	Ala	His 360	Glu	Gly	Cys	Leu	Pro 365	Pro	Phe	Pro
Ala	Asp 370	Val	Phe	Met	Ile	Pro 375	Gln	Tyr	Gly	Tyr	Leu	Thr	Leu	Asn	Asp
Gly 385	Ser	Gln	Ala	Val	Gly 390	Arg	Ser	Ser	Phe	Tyr 395	Cys	Leu	Glu	Tyr	Phe 400
Pro	Ser	Gln	Met	Leu 405	Arg	Thr	Gly	Asn	Asn 410	Phe	Gln	Phe	Ser	Tyr 415	Glu
Phe	Glu	Asn	Val 420	Pro	Phe	His	Ser	Ser	Tyr 425	Ala	His	Ser	Gln	Ser	Leu
Asp	Arg	Leu 435	Met	Asn	Pro	Leu	Ile 440	Asp	Gln	Tyr	Leu	Tyr 445	Tyr	Leu	Ser
Arg	Thr 450	Ile	Asn	Gly	Ser	Gly 455	Gln	Asn	Gln	Gln	Thr 460	Leu	Lys	Phe	Ser
Val 465	Ala	Gly	Pro	Ser	Asn	Met 470	Ala	Val	Gln	Gly 475	Arg	Asn	Tyr	Ile	Pro 480
Gly	Pro	Ser	Tyr	Arg 485	Gln	Gln	Arg	Val	Ser 490	Thr	Thr	Val	Thr	Gln 495	Asn
Asn	Asn	Ser	Glu 500	Phe	Ala	Trp	Pro	Gly 505	Ala	Ser	Ser	Trp	Ala	Leu	Asn
Gly	Arg	Asn 515	Ser	Leu	Met	Asn	Pro 520	Gly	Pro	Ala	Met	Ala	Ser	His	Lys
Glu	Gly 530	Glu	Asp	Arg	Phe	Phe 535	Pro	Leu	Ser	Gly	Ser 540	Leu	Ile	Phe	Gly
Lys 545	Gln	Gly	Thr	Gly	Arg 550	Asp	Asn	Val	Asp	Ala 555	Asp	Lys	Val	Met	Ile 560
Thr	Asn	Glu	Glu	Glu 565	Ile	Lys	Thr	Thr	Asn 570	Pro	Val	Ala	Thr	Glu 575	Ser
Tyr	Gly	Gln	Val 580	Ala	Thr	Asn	His	Gln 585	Ser	Ala	Gln	Thr	Leu	Ala	Val
Pro	Phe	Lys 595	Ala	Gln	Ala	Gln	Thr 600	Gly	Trp	Val	Gln	Asn 605	Gln	Gly	Ile
Leu 610	Pro	Gly	Met	Val	Trp 615	Gln	Asp	Arg	Asp	Val	Tyr 620	Leu	Gln	Gly	Pro
Ile 625	Trp	Ala	Lys	Ile	Pro 630	His	Thr	Asp	Gly 635	Asn	Phe	His	Pro	Ser	Pro 640

Leu	Met	Gly	Gly	Phe	Gly	Met	Lys	His	Pro	Pro	Pro	Gln	Ile	Leu	Ile	
				645					650					655		
Lys	Asn	Thr	Pro	Val	Pro	Ala	Asp	Pro	Pro	Thr	Ala	Phe	Asn	Lys	Asp	
			660					665					670			
Lys	Leu	Asn	Ser	Phe	Ile	Thr	Gln	Tyr	Ser	Thr	Gly	Gln	Val	Ser	Val	
		675					680					685				
Glu	Ile	Glu	Trp	Glu	Leu	Gln	Lys	Glu	Asn	Ser	Lys	Arg	Trp	Asn	Pro	
		690				695					700					
Glu	Ile	Gln	Tyr	Thr	Ser	Asn	Tyr	Tyr	Lys	Ser	Asn	Asn	Val	Glu	Phe	
		705				710				715				720		
Ala	Val	Asn	Thr	Glu	Gly	Val	Tyr	Ser	Glu	Pro	Arg	Pro	Ile	Gly	Thr	
				725					730					735		
Arg	Tyr	Leu	Thr	Arg	Asn	Leu										
			740													
<210> SEQ ID NO 13																
<211> LENGTH: 2208																
<212> TYPE: DNA																
<213> ORGANISM: Artificial Sequence																
<220> FEATURE:																
<223> OTHER INFORMATION: AAVhu68																
<220> FEATURE:																
<221> NAME/KEY: CDS																
<222> LOCATION: (1)..(2208)																
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Met	Ala	Ala	Asp	Gly	Tyr	Leu	Pro	Asp	Trp	Leu	Glu	Asp	Asn	Leu	Ser	
1				5				10					15			
gaa	ggc	att	cgc	gag	tgg	tgg	gct	ttg	aaa	cct	gga	gcc	cct	caa	ccc	96
Glu	Gly	Ile	Arg	Glu	Trp	Trp	Ala	Leu	Lys	Pro	Gly	Ala	Pro	Gln	Pro	
			20					25				30				
aag	gca	aat	caa	caa	cat	caa	gac	aac	gct	cgg	ggt	ctt	gtg	ctt	cgg	144
Lys	Ala	Asn	Gln	Gln	His	Gln	Asp	Asn	Ala	Arg	Gly	Leu	Val	Leu	Pro	
		35					40					45				
ggg	tac	aaa	tac	ctt	gga	ccc	ggc	aac	gga	ctc	gac	aag	ggg	gag	cgg	192
Gly	Tyr	Lys	Tyr	Leu	Gly	Pro	Gly	Asn	Gly	Leu	Asp	Lys	Gly	Glu	Pro	
		50				55					60					
gtc	aac	gaa	gca	gac	gcg	gcg	gcc	ctc	gag	cac	gac	aag	gcc	tac	gac	240
Val	Asn	Glu	Ala	Asp	Ala	Ala	Ala	Leu	Glu	His	Asp	Lys	Ala	Tyr	Asp	
		65				70				75				80		
cag	cag	ctc	aag	gcc	gga	gac	aac	ccg	tac	ctc	aag	tac	aac	cac	gcc	288
Gln	Gln	Leu	Lys	Ala	Gly	Asp	Asn	Pro	Tyr	Leu	Lys	Tyr	Asn	His	Ala	
			85						90					95		
gac	gcc	gag	ttc	cag	gag	cgg	ctc	aaa	gaa	gat	acg	tct	ttt	ggg	ggc	336
Asp	Ala	Glu	Phe	Gln	Glu	Arg	Leu	Lys	Glu	Asp	Thr	Ser	Phe	Gly	Gly	
			100					105					110			
aac	ctc	ggg	cga	gca	gtc	ttc	cag	gcc	aaa	aag	agg	ctt	ctt	gaa	cct	384
Asn	Leu	Gly	Arg	Ala	Val	Phe	Gln	Ala	Lys	Lys	Arg	Leu	Leu	Glu	Pro	
		115						120				125				
ctt	ggg	ctg	gtt	gag	gaa	gcg	gct</									

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ggc gac aca gag tca gtc ccc gac cct caa cca atc gga gaa cct ccc	576
Gly Asp Thr Glu Ser Val Pro Asp Pro Gln Pro Ile Gly Glu Pro Pro	
180 185 190	
gca gcc ccc tca ggt gtg gga tct ctt aca atg gct tca ggt ggt ggc	624
Ala Ala Pro Ser Gly Val Gly Ser Leu Thr Met Ala Ser Gly Gly Gly	
195 200 205	
gca cca gtg gca gac aat aac gaa ggt gcc gat gga gtg ggt agt tcc	672
Ala Pro Val Ala Asp Asn Asn Glu Gly Ala Asp Gly Val Gly Ser Ser	
210 215 220	
tcg gga aat tgg cat tgc gat tcc caa tgg ctg ggg gac aga gtc atc	720
Ser Gly Asn Trp His Cys Asp Ser Gln Trp Leu Gly Asp Arg Val Ile	
225 230 235 240	
acc acc agc acc cga acc tgg gcc ctg ccc acc tac aac aat cac ctc	768
Thr Thr Ser Thr Arg Thr Trp Ala Leu Pro Thr Tyr Asn Asn His Leu	
245 250 255	
tac aag caa atc tcc aac agc aca tct gga gga tct tca aat gac aac	816
Tyr Lys Gln Ile Ser Asn Ser Thr Ser Gly Gly Ser Ser Asn Asp Asn	
260 265 270	
gcc tac ttc ggc tac agc acc ccc tgg ggg tat ttt gac ttc aac aga	864
Ala Tyr Phe Gly Tyr Ser Thr Pro Trp Gly Tyr Phe Asp Phe Asn Arg	
275 280 285	
ttc cac tgc cac ttc tca cca cgt gac tgg caa aga ctc atc aac aac	912
Phe His Cys His Phe Ser Pro Arg Asp Trp Gln Arg Leu Ile Asn Asn	
290 295 300	
aac tgg gga ttc cgg cct aag cga ctc aac ttc aag ctc ttc aac att	960
Asn Trp Gly Phe Arg Pro Lys Arg Leu Asn Phe Lys Leu Phe Asn Ile	
305 310 315 320	
cag gtc aaa gag gtt acg gac aac aat gga gtc aag acc atc gct aat	1008
Gln Val Lys Glu Val Thr Asp Asn Asn Gly Val Lys Thr Ile Ala Asn	
325 330 335	
aac ctt acc agc acg gtc cag gtc ttc acg gac tca gac tat cag ctc	1056
Asn Leu Thr Ser Thr Val Gln Val Phe Thr Asp Ser Asp Tyr Gln Leu	
340 345 350	
ccg tac gtg ctc ggg tgc gct cac gag ggc tgc ctc ccg ccg ttc cca	1104
Pro Tyr Val Leu Gly Ser Ala His Glu Gly Cys Leu Pro Pro Phe Pro	
355 360 365	
gcg gac gtt ttc atg att cct cag tac ggg tat cta acg ctt aat gat	1152
Ala Asp Val Phe Met Ile Pro Gln Tyr Gly Tyr Leu Thr Leu Asn Asp	
370 375 380	
gga agc caa gcc gtg ggt cgt tgc tcc ttt tac tgc ctg gaa tat ttc	1200
Gly Ser Gln Ala Val Gly Arg Ser Ser Phe Tyr Cys Leu Glu Tyr Phe	
385 390 395 400	
ccg tgc caa atg cta aga acg ggt aac aac ttc cag ttc agc tac gag	1248
Pro Ser Gln Met Leu Arg Thr Gly Asn Asn Phe Gln Phe Ser Tyr Glu	
405 410 415	
ttt gag aac gta cct ttc cat agc agc tat gct cac agc caa agc ctg	1296
Phe Glu Asn Val Pro Phe His Ser Ser Tyr Ala His Ser Gln Ser Leu	
420 425 430	
gac cga ctc atg aat cca ctc atc gac caa tac ttg tac tat ctc tca	1344
Asp Arg Leu Met Asn Pro Leu Ile Asp Gln Tyr Leu Tyr Tyr Leu Ser	
435 440 445	
aag act att aac ggt tct gga cag aat caa caa acg cta aaa ttc agt	1392
Lys Thr Ile Asn Gly Ser Gly Gln Asn Gln Gln Thr Leu Lys Phe Ser	
450 455 460	
gtg gcc gga ccc agc aac atg gct gtc cag gga aga aac tac ata cct	1440
Val Ala Gly Pro Ser Asn Met Ala Val Gln Gly Arg Asn Tyr Ile Pro	
465 470 475 480	
gga ccc agc tac cga caa caa cgt gtc tca acc act gtg act caa aac	1488
Gly Pro Ser Tyr Arg Gln Gln Arg Val Ser Thr Thr Val Thr Gln Asn	
485 490 495	

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aac aac agc gaa ttt gct tgg cct gga gct tct tct tgg gct ctc aat	1536
Asn Asn Ser Glu Phe Ala Trp Pro Gly Ala Ser Ser Trp Ala Leu Asn	
500 505 510	
gga cgt aat agc ttg atg aat cct gga cct gct atg gcc agc cac aaa	1584
Gly Arg Asn Ser Leu Met Asn Pro Gly Pro Ala Met Ala Ser His Lys	
515 520 525	
gaa gga gag gac cgt ttc ttt cct ttg tct gga tct tta att ttt ggc	1632
Glu Gly Glu Asp Arg Phe Phe Pro Leu Ser Gly Ser Leu Ile Phe Gly	
530 535 540	
aaa caa gga act gga aga gac aac gtg gat gcg gac aaa gtc atg ata	1680
Lys Gln Gly Thr Gly Arg Asp Asn Val Asp Ala Asp Lys Val Met Ile	
545 550 555 560	
acc aac gaa gaa gaa att aaa act acc aac cca gta gca acg gag tcc	1728
Thr Asn Glu Glu Ile Lys Thr Thr Asn Pro Val Ala Thr Glu Ser	
565 570 575	
tat gga caa gtg gcc aca aac cac cag agt gcc caa gca cag gcg cag	1776
Tyr Gly Gln Val Ala Thr Asn His Gln Ser Ala Gln Ala Gln Ala Gln	
580 585 590	
acc ggc tgg gtt caa aac caa gga ata ctt ccg ggt atg gtt tgg cag	1824
Thr Gly Trp Val Gln Asn Gln Gly Ile Leu Pro Gly Met Val Trp Gln	
595 600 605	
gac aga gat gtg tac ctg caa gga ccc att tgg gcc aaa att cct cac	1872
Asp Arg Asp Val Tyr Leu Gln Gly Pro Ile Trp Ala Lys Ile Pro His	
610 615 620	
acg gac ggc aac ttt cac cct tct ccg ctg atg gga ggg ttt gga atg	1920
Thr Asp Gly Asn Phe His Pro Ser Pro Leu Met Gly Gly Phe Gly Met	
625 630 635 640	
aag cac ccg cct cct cag atc ctc atc aaa aac aca cct gta cct gcg	1968
Lys His Pro Pro Pro Gln Ile Leu Ile Lys Asn Thr Pro Val Pro Ala	
645 650 655	
gat cct cca acg gct ttc aac aag gac aag ctg aac tct ttc atc acc	2016
Asp Pro Pro Thr Ala Phe Asn Lys Asp Lys Leu Asn Ser Phe Ile Thr	
660 665 670	
cag tat tct act ggc caa gtc agc gtg gag att gag tgg gag ctg cag	2064
Gln Tyr Ser Thr Gly Gln Val Ser Val Glu Ile Glu Trp Glu Leu Gln	
675 680 685	
aag gaa aac agc aag cgc tgg aac ccg gag atc cag tac act tcc aac	2112
Lys Glu Asn Ser Lys Arg Trp Asn Pro Glu Ile Gln Tyr Thr Ser Asn	
690 695 700	
tat tac aag tct aat aat gtt gaa ttt gct gtt aat act gaa ggt gtt	2160
Tyr Tyr Lys Ser Asn Asn Val Glu Phe Val Asn Thr Glu Gly Val	
705 710 715 720	
tat tct gaa ccc cgc ccc att ggc acc aga tac ctg act cgt aat ctg	2208
Tyr Ser Glu Pro Arg Pro Ile Gly Thr Arg Tyr Leu Thr Arg Asn Leu	
725 730 735	

<210> SEQ ID NO 14

<211> LENGTH: 736

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 14

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Glu Gly Ile Arg Glu Trp Trp Ala Leu Lys Pro Gly Ala Pro Gln Pro
20 25 30

Lys Ala Asn Gln Gln His Gln Asp Asn Ala Arg Gly Leu Val Leu Pro
35 40 45

Gly 50	Tyr	Lys	Tyr	Leu	Gly	Pro 55	Gly	Asn	Gly	Leu	Asp 60	Lys	Gly	Glu	Pro
Val 65	Asn	Glu	Ala	Asp	Ala 70	Ala	Ala	Leu	Glu	His 75	Asp	Lys	Ala	Tyr	Asp 80
Gln	Gln	Leu	Lys	Ala 85	Gly	Asp	Asn	Pro	Tyr 90	Leu	Lys	Tyr	Asn	His 95	Ala
Asp	Ala	Glu	Phe	Gln 100	Glu	Arg	Leu	Lys	Glu 105	Asp	Thr	Ser	Phe	Gly	Gly
Asn	Leu	Gly 115	Arg	Ala	Val	Phe	Gln 120	Ala	Lys	Lys	Arg	Leu 125	Leu	Glu	Pro
Leu 130	Gly	Leu	Val	Glu	Glu	Ala 135	Ala	Lys	Thr	Ala	Pro 140	Gly	Lys	Lys	Arg
Pro 145	Val	Glu	Gln	Ser	Pro 150	Gln	Glu	Pro	Asp	Ser 155	Ser	Val	Gly	Ile	Gly 160
Lys	Ser	Gly	Ala	Gln 165	Pro	Ala	Lys	Lys	Arg 170	Leu	Asn	Phe	Gly	Gln 175	Thr
Gly	Asp	Thr	Glu 180	Ser	Val	Pro	Asp	Pro 185	Gln	Pro	Ile	Gly	Glu	Pro 190	Pro
Ala	Ala	Pro 195	Ser	Gly	Val	Gly	Ser	Leu 200	Thr	Met	Ala	Ser 205	Gly	Gly	Gly
Ala 210	Pro	Val	Ala	Asp	Asn	Asn 215	Glu	Gly	Ala	Asp	Gly 220	Val	Gly	Ser	Ser
Ser 225	Gly	Asn	Trp	His	Cys 230	Asp	Ser	Gln	Trp	Leu 235	Gly	Asp	Arg	Val	Ile 240
Thr	Thr	Ser	Thr 245	Arg	Thr	Trp	Ala	Leu	Pro 250	Thr	Tyr	Asn	Asn	His 255	Leu
Tyr	Lys	Gln	Ile 260	Ser	Asn	Ser	Thr	Ser	Gly 265	Gly	Ser	Ser	Asn	Asp 270	Asn
Ala	Tyr	Phe 275	Gly	Tyr	Ser	Thr	Pro 280	Trp	Gly	Tyr	Phe	Asp 285	Phe	Asn	Arg
Phe 290	His	Cys	His	Phe	Ser	Pro 295	Arg	Asp	Trp	Gln	Arg 300	Leu	Ile	Asn	Asn
Asn 305	Trp	Gly	Phe	Arg	Pro 310	Lys	Arg	Leu	Asn	Phe 315	Lys	Leu	Phe	Asn	Ile 320
Gln	Val	Lys	Glu 325	Val	Thr	Asp	Asn	Asn	Gly 330	Val	Lys	Thr	Ile	Ala 335	Asn
Asn	Leu	Thr	Ser 340	Thr	Val	Gln	Val	Phe	Thr 345	Asp	Ser	Asp	Tyr 350	Gln	Leu
Pro	Tyr	Val 355	Leu	Gly	Ser	Ala	His 360	Glu	Gly	Cys	Leu 365	Pro	Pro	Phe	Pro
Ala 370	Asp	Val	Phe	Met	Ile	Pro 375	Gln	Tyr	Gly	Tyr	Leu 380	Thr	Leu	Asn	Asp
Gly 385	Ser	Gln	Ala	Val	Gly	Arg	Ser	Ser	Phe 395	Tyr	Cys	Leu	Glu	Tyr	Phe 400
Pro	Ser	Gln	Met 405	Leu	Arg	Thr	Gly	Asn	Asn 410	Phe	Gln	Phe	Ser	Tyr 415	Glu
Phe	Glu	Asn 420	Val	Pro	Phe	His	Ser	Ser	Tyr 425	Ala	His	Ser	Gln	Ser	Leu
Asp	Arg	Leu 435	Met	Asn	Pro	Leu	Ile 440	Asp	Gln	Tyr	Leu 445	Tyr	Tyr	Leu	Ser
Lys 450	Thr	Ile	Asn	Gly	Ser	Gly 455	Gln	Asn	Gln	Gln	Thr 460	Leu	Lys	Phe	Ser

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gtc aac gcg gcg gac gca gcg gcc ctc gag cac gac aag gcc tac gac Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp 65 70 75 80	240
cag cag ctg cag gcg ggt gac aat ccg tac ctg cgg tat aac cac gcc Gln Gln Leu Gln Ala Gly Asp Asn Pro Tyr Leu Arg Tyr Asn His Ala 85 90 95	288
gac gcc gag ttt cag gag cgt ctg caa gaa gat acg tct ttt ggg ggc Asp Ala Glu Phe Gln Glu Arg Leu Gln Glu Asp Thr Ser Phe Gly Gly 100 105 110	336
aac ctc ggg cga gca gtc ttc cag gcc aag aag cgg gtt ctc gaa cct Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Val Leu Glu Pro 115 120 125	384
ctc ggt ctg gtt gag gaa ggc gct aag acg gct cct gga aag aag aga Leu Gly Leu Val Glu Glu Gly Ala Lys Thr Ala Pro Gly Lys Lys Arg 130 135 140	432
ccg gta gag cca tca ccc cag cgt tct cca gac tcc tct acg ggc atc Pro Val Glu Pro Ser Pro Gln Arg Ser Pro Asp Ser Ser Thr Gly Ile 145 150 155 160	480
ggc aag aaa ggc caa cag ccc gcc aga aaa aga ctc aat ttt ggt cag Gly Lys Lys Gly Gln Gln Pro Ala Arg Lys Arg Leu Asn Phe Gly Gln 165 170 175	528
act ggc gac tca gag tca gtt cca gac cct caa cct ctc gga gaa cct Thr Gly Asp Ser Glu Ser Val Pro Asp Pro Gln Pro Leu Gly Glu Pro 180 185 190	576
cca gca gcg ccc tct ggt gtg gga cct aat aca atg gct gca ggc ggt Pro Ala Ala Pro Ser Gly Val Gly Pro Asn Thr Met Ala Ala Gly Gly 195 200 205	624
ggc gca cca atg gca gac aat aac gaa ggc gcc gac gga gtg ggt agt Gly Ala Pro Met Ala Asp Asn Asn Glu Gly Ala Asp Gly Val Gly Ser 210 215 220	672
tcc tcg gga aat tgg cat tgc gat tcc aca tgg ctg ggc gac aga gtc Ser Ser Gly Asn Trp His Cys Asp Ser Thr Trp Leu Gly Asp Arg Val 225 230 235 240	720
atc acc acc agc acc cga acc tgg gcc ctg ccc acc tac aac aac cac Ile Thr Thr Ser Thr Arg Thr Trp Ala Leu Pro Thr Tyr Asn Asn His 245 250 255	768
ctc tac aag caa atc tcc tct ggt act cat gga gcc acc aac gac aac Leu Tyr Lys Gln Ile Ser Ser Gly Thr His Gly Ala Thr Asn Asp Asn 260 265 270	816
acc tac ttc ggc tac agc acc ccc tgg ggg tat ttt gac ttt aac aga Thr Tyr Phe Gly Tyr Ser Thr Pro Trp Gly Tyr Phe Asp Phe Asn Arg 275 280 285	864
ttc cac tgc cac ttt tca cca cgt gac tgg cag cga ctc atc aac aac Phe His Cys His Phe Ser Pro Arg Asp Trp Gln Arg Leu Ile Asn Asn 290 295 300	912
aac tgg gga ttc cgg ccc aag aga ctc agc ttc aag ctc ttc aac atc Asn Trp Gly Phe Arg Pro Lys Arg Leu Ser Phe Lys Leu Phe Asn Ile 305 310 315 320	960
cag gtc aag gag gtc acg cag aat gaa ggc acc aag acc atc gcc aat Gln Val Lys Glu Val Thr Gln Asn Glu Gly Thr Lys Thr Ile Ala Asn 325 330 335	1008
aac ctc acc agc acc atc cag gtg ttt acg gac tcg gag tac cag ctg Asn Leu Thr Ser Thr Ile Gln Val Phe Thr Asp Ser Glu Tyr Gln Leu 340 345 350	1056
ccg tac gtt ctc ggc tct gcc cac cag ggc tgc ctg cct ccg ttc ccg Pro Tyr Val Leu Gly Ser Ala His Gln Gly Cys Leu Pro Pro Phe Pro 355 360 365	1104
gcg gac gtg ttc atg att ccc cag tac ggc tac cta aca ctc aac aac Ala Asp Val Phe Met Ile Pro Gln Tyr Gly Tyr Leu Thr Leu Asn Asn	1152

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370	375	380	
ggt agt cag gcc gtg gga cgc tcc tcc ttc tac tgc ctg gaa tac ttt Gly Ser Gln Ala Val Gly Arg Ser Ser Phe Tyr Cys Leu Glu Tyr Phe 385 390 395 400			1200
cct tcg cag atg ctg aga acc ggc aac aac ttc cag ttt act tac acc Pro Ser Gln Met Leu Arg Thr Gly Asn Asn Phe Gln Phe Thr Tyr Thr 405 410 415			1248
ttc gag gac gtg cct ttc cac agc agc tac gcc cac agc cag agc ttg Phe Glu Asp Val Pro Phe His Ser Ser Tyr Ala His Ser Gln Ser Leu 420 425 430			1296
gac cgg ctg atg aat cct ctg att gac cag tac ctg tac tac ttg tct Asp Arg Leu Met Asn Pro Leu Ile Asp Gln Tyr Leu Tyr Tyr Leu Ser 435 440 445			1344
cgg act caa aca aca ggt ggg agt agg cct acg cag act ctg ggc ttc Arg Thr Gln Thr Thr Gly Gly Ser Arg Pro Thr Gln Thr Leu Gly Phe 450 455 460			1392
agc caa ggt ggg cct aat aca atg gcc aat cag gca aag aac tgg ctg Ser Gln Gly Gly Pro Asn Thr Met Ala Asn Gln Ala Lys Asn Trp Leu 465 470 475 480			1440
cca gga ccc tgt tac cgc caa caa cgc gtc tca acg aca acc ggg caa Pro Gly Pro Cys Tyr Arg Gln Gln Arg Val Ser Thr Thr Thr Gly Gln 485 490 495			1488
aac aac aat agc aac ttt gcc tgg act gct ggg acc aaa tac cat ctg Asn Asn Asn Ser Asn Phe Ala Trp Thr Ala Gly Thr Lys Tyr His Leu 500 505 510			1536
aat gga aga aat tca ttg gct aat cct ggc atc gct atg gca aca cac Asn Gly Arg Asn Ser Leu Ala Asn Pro Gly Ile Ala Met Ala Thr His 515 520 525			1584
aaa gac gac gag gag cgt ttt ttt ccc agt aac ggg atc ctg att ttt Lys Asp Asp Glu Glu Arg Phe Phe Pro Ser Asn Gly Ile Leu Ile Phe 530 535 540			1632
ggc aaa caa aat gct gcc aga gac aat gcg gat tac agc gat gtc atg Gly Lys Gln Asn Ala Ala Arg Asp Asn Ala Asp Tyr Ser Asp Val Met 545 550 555 560			1680
ctc acc agc gag gaa gaa atc aaa acc act aac cct gtg gct aca gag Leu Thr Ser Glu Glu Glu Ile Lys Thr Thr Asn Pro Val Ala Thr Glu 565 570 575			1728
gaa tac ggt atc gtg ggt gat aac ttg cag ttg tat aac acg gct cct Glu Tyr Gly Ile Val Gly Asp Asn Leu Gln Leu Tyr Asn Thr Ala Pro 580 585 590			1776
ggt tcg gtg ttt gtc aac agc cag ggg gcc tta ccc ggt atg gtc tgg Gly Ser Val Phe Val Asn Ser Gln Gly Ala Leu Pro Gly Met Val Trp 595 600 605			1824
cag aac cgg gac gtg tac ctg cag ggt ccc atc tgg gcc aag att cct Gln Asn Arg Asp Val Tyr Leu Gln Gly Pro Ile Trp Ala Lys Ile Pro 610 615 620			1872
cac acg gac ggc aac ttc cac ccg tct ccg ctg atg ggc ggc ttt ggc His Thr Asp Gly Asn Phe His Pro Ser Pro Leu Met Gly Gly Phe Gly 625 630 635 640			1920
ctg aaa cat cct ccg cct cag atc ctg atc aag aac acg cct gta cct Leu Lys His Pro Pro Pro Gln Ile Leu Ile Lys Asn Thr Pro Val Pro 645 650 655			1968
gcg gat cct ccg acc acc ttc aac cag tca aag ctg aac tct ttc atc Ala Asp Pro Pro Thr Thr Phe Asn Gln Ser Lys Leu Asn Ser Phe Ile 660 665 670			2016
acg caa tac agc acc gga cag gtc agc gtg gaa att gaa tgg gag ctg Thr Gln Tyr Ser Thr Gly Gln Val Ser Val Glu Ile Glu Trp Glu Leu 675 680 685			2064
cag aag gaa aac agc aag cgc tgg aac ccc gag atc cag tac acc tcc			2112

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290	295	300
Asn Trp Gly Phe Arg Pro Lys Arg Leu Ser Phe Lys Leu Phe Asn Ile		
305	310	315 320
Gln Val Lys Glu Val Thr Gln Asn Glu Gly Thr Lys Thr Ile Ala Asn		
	325	330 335
Asn Leu Thr Ser Thr Ile Gln Val Phe Thr Asp Ser Glu Tyr Gln Leu		
	340	345 350
Pro Tyr Val Leu Gly Ser Ala His Gln Gly Cys Leu Pro Pro Phe Pro		
	355	360 365
Ala Asp Val Phe Met Ile Pro Gln Tyr Gly Tyr Leu Thr Leu Asn Asn		
	370	375 380
Gly Ser Gln Ala Val Gly Arg Ser Ser Phe Tyr Cys Leu Glu Tyr Phe		
385	390	395 400
Pro Ser Gln Met Leu Arg Thr Gly Asn Asn Phe Gln Phe Thr Tyr Thr		
	405	410 415
Phe Glu Asp Val Pro Phe His Ser Ser Tyr Ala His Ser Gln Ser Leu		
	420	425 430
Asp Arg Leu Met Asn Pro Leu Ile Asp Gln Tyr Leu Tyr Tyr Leu Ser		
	435	440 445
Arg Thr Gln Thr Thr Gly Gly Ser Arg Pro Thr Gln Thr Leu Gly Phe		
	450	455 460
Ser Gln Gly Gly Pro Asn Thr Met Ala Asn Gln Ala Lys Asn Trp Leu		
465	470	475 480
Pro Gly Pro Cys Tyr Arg Gln Gln Arg Val Ser Thr Thr Thr Gly Gln		
	485	490 495
Asn Asn Asn Ser Asn Phe Ala Trp Thr Ala Gly Thr Lys Tyr His Leu		
	500	505 510
Asn Gly Arg Asn Ser Leu Ala Asn Pro Gly Ile Ala Met Ala Thr His		
	515	520 525
Lys Asp Asp Glu Glu Arg Phe Phe Pro Ser Asn Gly Ile Leu Ile Phe		
	530	535 540
Gly Lys Gln Asn Ala Ala Arg Asp Asn Ala Asp Tyr Ser Asp Val Met		
545	550	555 560
Leu Thr Ser Glu Glu Glu Ile Lys Thr Thr Asn Pro Val Ala Thr Glu		
	565	570 575
Glu Tyr Gly Ile Val Gly Asp Asn Leu Gln Leu Tyr Asn Thr Ala Pro		
	580	585 590
Gly Ser Val Phe Val Asn Ser Gln Gly Ala Leu Pro Gly Met Val Trp		
	595	600 605
Gln Asn Arg Asp Val Tyr Leu Gln Gly Pro Ile Trp Ala Lys Ile Pro		
	610	615 620
His Thr Asp Gly Asn Phe His Pro Ser Pro Leu Met Gly Gly Phe Gly		
625	630	635 640
Leu Lys His Pro Pro Pro Gln Ile Leu Ile Lys Asn Thr Pro Val Pro		
	645	650 655
Ala Asp Pro Pro Thr Thr Phe Asn Gln Ser Lys Leu Asn Ser Phe Ile		
	660	665 670
Thr Gln Tyr Ser Thr Gly Gln Val Ser Val Glu Ile Glu Trp Glu Leu		
	675	680 685
Gln Lys Glu Asn Ser Lys Arg Trp Asn Pro Glu Ile Gln Tyr Thr Ser		
	690	695 700
Asn Tyr Tyr Lys Ser Thr Ser Val Asp Phe Ala Val Asn Thr Glu Gly		
705	710	715 720

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Val Tyr Ser Glu Pro Arg Pro Ile Gly Thr Arg Tyr Leu Thr Arg Asn
725 730 735

Leu

<210> SEQ ID NO 17
<211> LENGTH: 2214
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: AAV8.AR2.08
<220> FEATURE:
<221> NAME/KEY: CDS
<222> LOCATION: (1)..(2214)

<400> SEQUENCE: 17

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 gag ggc att cgc gag tgg tgg gcg ctg aaa cct gga gcc ccg aag ccc	 96
Glu Gly Ile Arg Glu Trp Trp Ala Leu Lys Pro Gly Ala Pro Lys Pro	
20 25 30	
 aaa gcc aac cag caa aag cag gac gac ggc cgg ggt ctg gtg ctt cct	 144
Lys Ala Asn Gln Gln Lys Gln Asp Asp Gly Arg Gly Leu Val Leu Pro	
35 40 45	
 ggc tac aag tac ctc gga ccc ttc aac gga ctc gac aag ggg gag ccc	 192
Gly Tyr Lys Tyr Leu Gly Pro Phe Asn Gly Leu Asp Lys Gly Glu Pro	
50 55 60	
 gtc aac gcg gcg gac gca gcg gcc ctc gag cac gac aag gcc tac gac	 240
Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp	
65 70 75 80	
 cag cag ctg cag gcg ggt gac aat ccg tac ctg cgg tat aac cac gcc	 288
Gln Gln Leu Gln Ala Gly Asp Asn Pro Tyr Leu Arg Tyr Asn His Ala	
85 90 95	
 gac gcc gag ttt cag gag cgt ctg caa gaa gat acg tct ttt ggg ggc	 336
Asp Ala Glu Phe Gln Glu Arg Leu Gln Glu Asp Thr Ser Phe Gly Gly	
100 105 110	
 aac ctc ggg cga gca gtc ttc cag gcc aag aag cgg gtt ctc gaa cct	 384
Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Val Leu Glu Pro	
115 120 125	
 ctc ggt ctg gtt gag gaa ggc gct aag acg gct cct gga aag aag aga	 432
Leu Gly Leu Val Glu Glu Gly Ala Lys Thr Ala Pro Gly Lys Lys Arg	
130 135 140	
 ccg gta gag cca tca ccc cag cgt tct cca gac tcc tct acg ggc atc	 480
Pro Val Glu Pro Ser Pro Gln Arg Ser Pro Asp Ser Ser Thr Gly Ile	
145 150 155 160	
 ggc aag aaa ggc caa cag ccc gcc aga aaa aga ctc aat ttt ggt cag	 528
Gly Lys Lys Gly Gln Gln Pro Ala Arg Lys Arg Leu Asn Phe Gly Gln	
165 170 175	
 act ggc gac tca gag tca gtt cca gac cct caa cct ctc gga gaa cct	 576
Thr Gly Asp Ser Glu Ser Val Pro Asp Pro Gln Pro Leu Gly Glu Pro	
180 185 190	
 cca gca gcg ccc tct ggt gtg gga cct aat aca atg gct gca gcc ggt	 624
Pro Ala Ala Pro Ser Gly Val Gly Pro Asn Thr Met Ala Ala Gly Gly	
195 200 205	
 ggc gca cca atg gca gac aat aac gaa ggc gcc gac gga gtg ggt agt	 672
Gly Ala Pro Met Ala Asp Asn Asn Glu Gly Ala Asp Gly Val Gly Ser	
210 215 220	
 tcc tcg gga aat tgg cat tgc gat tcc aca tgg ctg ggc gac aga gtc	 720
Ser Ser Gly Asn Trp His Cys Asp Ser Thr Trp Leu Gly Asp Arg Val	
225 230 235 240	

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atc acc acc agc acc cga acc tgg gcc ctg ccc acc tac aac aac cac	768
Ile Thr Thr Ser Thr Arg Thr Trp Ala Leu Pro Thr Tyr Asn Asn His	
245 250 255	
ctc tac aag caa atc tcc aac ggg aca tcg gga gga gcc acc aac gac	816
Leu Tyr Lys Gln Ile Ser Asn Gly Thr Ser Gly Gly Ala Thr Asn Asp	
260 265 270	
aac acc tac ttc ggc tac agc acc ccc tgg ggg tat ttt gac ttt aac	864
Asn Thr Tyr Phe Gly Tyr Ser Thr Pro Trp Gly Tyr Phe Asp Phe Asn	
275 280 285	
aga ttc cac tgc cac ttt tca cca cgt gac tgg cag cga ctc atc aac	912
Arg Phe His Cys His Phe Ser Pro Arg Asp Trp Gln Arg Leu Ile Asn	
290 295 300	
aac aac tgg gga ttc cgg ccc aag aga ctc agc ttc aag ctc ttc aac	960
Asn Asn Trp Gly Phe Arg Pro Lys Arg Leu Ser Phe Lys Leu Phe Asn	
305 310 315 320	
atc cag gtc aag gag gtc acg cag aat gaa ggc acc aag acc atc gcc	1008
Ile Gln Val Lys Glu Val Thr Gln Asn Glu Gly Thr Lys Thr Ile Ala	
325 330 335	
aat aac ctc acc agc acc atc cag gtg ttt acg gac tcg gag tac cag	1056
Asn Asn Leu Thr Ser Thr Ile Gln Val Phe Thr Asp Ser Glu Tyr Gln	
340 345 350	
ctg ccg tac gtt ctc ggc tct gcc cac cag ggc tgc ctg cct ccg ttc	1104
Leu Pro Tyr Val Leu Gly Ser Ala His Gln Gly Cys Leu Pro Pro Phe	
355 360 365	
ccg cgg gac gtg ttc atg att ccc cag tac ggc tac cta aca ctc aac	1152
Pro Ala Asp Val Phe Met Ile Pro Gln Tyr Gly Tyr Leu Thr Leu Asn	
370 375 380	
aac ggt agt cag gcc gtg gga cgc tcc tcc ttc tac tgc ctg gaa tac	1200
Asn Gly Ser Gln Ala Val Gly Arg Ser Ser Phe Tyr Cys Leu Glu Tyr	
385 390 395 400	
ttt cct tcg cag atg ctg aga acc ggc aac aac ttc cag ttt act tac	1248
Phe Pro Ser Gln Met Leu Arg Thr Gly Asn Asn Phe Gln Phe Thr Tyr	
405 410 415	
acc ttc gag gac gtg cct ttc cac agc agc tac gcc cac agc cag agc	1296
Thr Phe Glu Asp Val Pro Phe His Ser Ser Tyr Ala His Ser Gln Ser	
420 425 430	
ttg gac cgg ctg atg aat cct ctg att gac cag tac ctg tac tac ttg	1344
Leu Asp Arg Leu Met Asn Pro Leu Ile Asp Gln Tyr Leu Tyr Tyr Leu	
435 440 445	
tct cgg act caa aca aca gga ggc acg gca aat acg cag act ctg ggc	1392
Ser Arg Thr Gln Thr Thr Gly Gly Thr Ala Asn Thr Gln Thr Leu Gly	
450 455 460	
ttc agc caa ggt ggg cct aat aca atg gcc aat cag gca aag aac tgg	1440
Phe Ser Gln Gly Gly Pro Asn Thr Met Ala Asn Gln Ala Lys Asn Trp	
465 470 475 480	
ctg cca gga ccc tgt tac cgc caa caa cgc gtc tca acg aca acc ggg	1488
Leu Pro Gly Pro Cys Tyr Arg Gln Gln Arg Val Ser Thr Thr Thr Gly	
485 490 495	
caa aac aac aat agc aac ttt gcc tgg act gct ggg acc aaa tac cat	1536
Gln Asn Asn Asn Ser Asn Phe Ala Trp Thr Ala Gly Thr Lys Tyr His	
500 505 510	
ctg aat gga aga aat tca ttg gct aat cct ggc atc gct atg gca aca	1584
Leu Asn Gly Arg Asn Ser Leu Ala Asn Pro Gly Ile Ala Met Ala Thr	
515 520 525	
cac aaa gac gac gag gag cgt ttt ttt ccc agt aac ggg atc ctg att	1632
His Lys Asp Asp Glu Glu Arg Phe Phe Pro Ser Asn Gly Ile Leu Ile	
530 535 540	
ttt ggc aaa caa aat gct gcc aga gac aat gcg gat tac agc gat gtc	1680
Phe Gly Lys Gln Asn Ala Ala Arg Asp Asn Ala Asp Tyr Ser Asp Val	
545 550 555 560	

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atg ctc acc agc gag gaa gaa atc aaa acc act aac cct gtg gct aca	1728
Met Leu Thr Ser Glu Glu Ile Lys Thr Thr Asn Pro Val Ala Thr	
565 570 575	
gag gaa tac ggt atc gtg tgt gat aac ttg cag agt cgg aac acg gct	1776
Glu Glu Tyr Gly Ile Val Cys Asp Asn Leu Gln Ser Arg Asn Thr Ala	
580 585 590	
cct cgt gag gag att gtc aac agc cag ggg gcc tta ccc ggt atg gtc	1824
Pro Arg Glu Glu Ile Val Asn Ser Gln Gly Ala Leu Pro Gly Met Val	
595 600 605	
tgg cag aac cgg gac gtg tac ctg cag ggt ccc atc tgg gcc aag att	1872
Trp Gln Asn Arg Asp Val Tyr Leu Gln Gly Pro Ile Trp Ala Lys Ile	
610 615 620	
cct cac acg gac ggc aac ttc cac ccg tct ccg ctg atg ggc ggc ttt	1920
Pro His Thr Asp Gly Asn Phe His Pro Ser Pro Leu Met Gly Gly Phe	
625 630 635 640	
ggc ctg aaa cat cct ccg cct cag atc ctg atc aag aac acg cct gta	1968
Gly Leu Lys His Pro Pro Pro Gln Ile Leu Ile Lys Asn Thr Pro Val	
645 650 655	
cct gcg gat cct ccg acc acc ttc aac cag tca aag ctg aac tct ttc	2016
Pro Ala Asp Pro Pro Thr Thr Phe Asn Gln Ser Lys Leu Asn Ser Phe	
660 665 670	
atc acg caa tac agc acc gga cag gtc agc gtg gaa att gaa tgg gag	2064
Ile Thr Gln Tyr Ser Thr Gly Gln Val Ser Val Glu Ile Glu Trp Glu	
675 680 685	
ctg cag aag gaa aac agc aag cgc tgg aac ccc gag atc cag tac acc	2112
Leu Gln Lys Glu Asn Ser Lys Arg Trp Asn Pro Glu Ile Gln Tyr Thr	
690 695 700	
tcc aac tac tac aaa tct aca agt gtg gac ttt gct gtt aat aca gaa	2160
Ser Asn Tyr Tyr Lys Ser Thr Ser Val Asp Phe Ala Val Asn Thr Glu	
705 710 715 720	
ggc gtg tac tct gaa ccc cgc ccc att ggc acc cgt tac ctc acc cgt	2208
Gly Val Tyr Ser Glu Pro Arg Pro Ile Gly Thr Arg Tyr Leu Thr Arg	
725 730 735	
aat ctg	2214
Asn Leu	
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Lys Ala Asn Gln Gln Lys Gln Asp Asp Gly Arg Gly Leu Val Leu Pro	
35 40 45	
Gly Tyr Lys Tyr Leu Gly Pro Phe Asn Gly Leu Asp Lys Gly Glu Pro	
50 55 60	
Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp	
65 70 75 80	
Gln Gln Leu Gln Ala Gly Asp Asn Pro Tyr Leu Arg Tyr Asn His Ala	
85 90 95	
Asp Ala Glu Phe Gln Glu Arg Leu Gln Glu Asp Thr Ser Phe Gly Gly	
100 105 110	

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Asn	Leu	Gly	Arg	Ala	Val	Phe	Gln	Ala	Lys	Lys	Arg	Val	Leu	Glu	Pro
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Leu	Gly	Leu	Val	Glu	Glu	Gly	Ala	Lys	Thr	Ala	Pro	Gly	Lys	Lys	Arg
	130					135					140				
Pro	Val	Glu	Pro	Ser	Pro	Gln	Arg	Ser	Pro	Asp	Ser	Ser	Thr	Gly	Ile
145					150					155					160
Gly	Lys	Lys	Gly	Gln	Gln	Pro	Ala	Arg	Lys	Arg	Leu	Asn	Phe	Gly	Gln
			165						170					175	
Thr	Gly	Asp	Ser	Glu	Ser	Val	Pro	Asp	Pro	Gln	Pro	Leu	Gly	Glu	Pro
		180						185					190		
Pro	Ala	Ala	Pro	Ser	Gly	Val	Gly	Pro	Asn	Thr	Met	Ala	Ala	Gly	Gly
	195						200					205			
Gly	Ala	Pro	Met	Ala	Asp	Asn	Asn	Glu	Gly	Ala	Asp	Gly	Val	Gly	Ser
	210					215					220				
Ser	Ser	Gly	Asn	Trp	His	Cys	Asp	Ser	Thr	Trp	Leu	Gly	Asp	Arg	Val
225					230					235					240
Ile	Thr	Thr	Ser	Thr	Arg	Thr	Trp	Ala	Leu	Pro	Thr	Tyr	Asn	Asn	His
			245						250					255	
Leu	Tyr	Lys	Gln	Ile	Ser	Asn	Gly	Thr	Ser	Gly	Gly	Ala	Thr	Asn	Asp
			260					265					270		
Asn	Thr	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro	Trp	Gly	Tyr	Phe	Asp	Phe	Asn
	275						280					285			
Arg	Phe	His	Cys	His	Phe	Ser	Pro	Arg	Asp	Trp	Gln	Arg	Leu	Ile	Asn
	290					295					300				
Asn	Asn	Trp	Gly	Phe	Arg	Pro	Lys	Arg	Leu	Ser	Phe	Lys	Leu	Phe	Asn
305					310					315					320
Ile	Gln	Val	Lys	Glu	Val	Thr	Gln	Asn	Glu	Gly	Thr	Lys	Thr	Ile	Ala
			325						330					335	
Asn	Asn	Leu	Thr	Ser	Thr	Ile	Gln	Val	Phe	Thr	Asp	Ser	Glu	Tyr	Gln
		340						345					350		
Leu	Pro	Tyr	Val	Leu	Gly	Ser	Ala	His	Gln	Gly	Cys	Leu	Pro	Pro	Phe
	355						360					365			
Pro	Ala	Asp	Val	Phe	Met	Ile	Pro	Gln	Tyr	Gly	Tyr	Leu	Thr	Leu	Asn
	370					375					380				
Asn	Gly	Ser	Gln	Ala	Val	Gly	Arg	Ser	Ser	Phe	Tyr	Cys	Leu	Glu	Tyr
385					390					395					400
Phe	Pro	Ser	Gln	Met	Leu	Arg	Thr	Gly	Asn	Asn	Phe	Gln	Phe	Thr	Tyr
			405						410					415	
Thr	Phe	Glu	Asp	Val	Pro	Phe	His	Ser	Ser	Tyr	Ala	His	Ser	Gln	Ser
		420						425					430		
Leu	Asp	Arg	Leu	Met	Asn	Pro	Leu	Ile	Asp	Gln	Tyr	Leu	Tyr	Tyr	Leu
	435						440					445			
Ser	Arg	Thr	Gln	Thr	Thr	Gly	Gly	Thr	Ala	Asn	Thr	Gln	Thr	Leu	Gly
	450					455					460				
Phe	Ser	Gln	Gly	Gly	Pro	Asn	Thr	Met	Ala	Asn	Gln	Ala	Lys	Asn	Trp
465					470					475					480
Leu	Pro	Gly	Pro	Cys	Tyr	Arg	Gln	Gln	Arg	Val	Ser	Thr	Thr	Thr	Gly
			485						490					495	
Gln	Asn	Asn	Asn	Ser	Asn	Phe	Ala	Trp	Thr	Ala	Gly	Thr	Lys	Tyr	His
			500					505					510		
Leu	Asn	Gly	Arg	Asn	Ser	Leu	Ala	Asn	Pro	Gly	Ile	Ala	Met	Ala	Thr
	515						520					525			
His	Lys	Asp	Asp	Glu	Glu	Arg	Phe	Phe	Pro	Ser	Asn	Gly	Ile	Leu	Ile

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530	535	540	
Phe Gly Lys Gln Asn Ala Ala Arg Asp Asn Ala Asp Tyr Ser Asp Val 545 550 555 560			
Met Leu Thr Ser Glu Glu Glu Ile Lys Thr Thr Asn Pro Val Ala Thr 565 570 575			
Glu Glu Tyr Gly Ile Val Cys Asp Asn Leu Gln Ser Arg Asn Thr Ala 580 585 590			
Pro Arg Glu Glu Ile Val Asn Ser Gln Gly Ala Leu Pro Gly Met Val 595 600 605			
Trp Gln Asn Arg Asp Val Tyr Leu Gln Gly Pro Ile Trp Ala Lys Ile 610 615 620			
Pro His Thr Asp Gly Asn Phe His Pro Ser Pro Leu Met Gly Gly Phe 625 630 635 640			
Gly Leu Lys His Pro Pro Pro Gln Ile Leu Ile Lys Asn Thr Pro Val 645 650 655			
Pro Ala Asp Pro Pro Thr Thr Phe Asn Gln Ser Lys Leu Asn Ser Phe 660 665 670			
Ile Thr Gln Tyr Ser Thr Gly Gln Val Ser Val Glu Ile Glu Trp Glu 675 680 685			
Leu Gln Lys Glu Asn Ser Lys Arg Trp Asn Pro Glu Ile Gln Tyr Thr 690 695 700			
Ser Asn Tyr Tyr Lys Ser Thr Ser Val Asp Phe Ala Val Asn Thr Glu 705 710 715 720			
Gly Val Tyr Ser Glu Pro Arg Pro Ile Gly Thr Arg Tyr Leu Thr Arg 725 730 735			
Asn Leu			
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gag ggc att cgc gag tgg tgg gcg ctg aaa cct gga gcc ccg aag ccc Glu Gly Ile Arg Glu Trp Trp Ala Leu Lys Pro Gly Ala Pro Lys Pro 20 25 30			96
aaa gcc aac cag caa aag cag gac gac gcc cgg ggt ctg gtg ctt cct Lys Ala Asn Gln Gln Lys Gln Asp Asp Gly Arg Gly Leu Val Leu Pro 35 40 45			144
ggc tac aag tac ctc gga ccc ttc aac gga ctc gac aag ggg gag ccc Gly Tyr Lys Tyr Leu Gly Pro Phe Asn Gly Leu Asp Lys Gly Glu Pro 50 55 60			192
gtc aac gcg gcg gac gca gcg gcc ctc gag cac gac aag gcc tac gac Val Asn Ala Ala Asp Ala Ala Ala Leu His Asp Lys Ala Tyr Asp 65 70 75 80			240
cag cag ctg cag gcg ggt gac aat ccg tac ctg cgg tat aac cac gcc Gln Gln Leu Gln Ala Gly Asp Asn Pro Tyr Leu Arg Tyr Asn His Ala 85 90 95			288
gac gcc gag ttt cag gag cgt ctg caa gaa gat acg tct ttt ggg ggc Asp Ala Glu Phe Gln Glu Arg Leu Gln Glu Asp Thr Ser Phe Gly Gly			336

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100	105	110	
aac ctc ggg cga gca gtc ttc cag gcc aag aag cgg gtt ctc gaa cct			384
Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Val Leu Glu Pro			
115	120	125	
ctc ggt ctg gtt gag gaa ggc gct aag acg gct cct gga aag aag aga			432
Leu Gly Leu Val Glu Glu Gly Ala Lys Thr Ala Pro Gly Lys Lys Arg			
130	135	140	
ccg gta gag cca tca ccc cag cgt tct cca gac tcc tct acg ggc atc			480
Pro Val Glu Pro Ser Pro Gln Arg Ser Pro Asp Ser Ser Thr Gly Ile			
145	150	155	160
ggc aag aaa ggc caa cag ccc gcc aga aaa aga ctc aat ttt ggt cag			528
Gly Lys Lys Gly Gln Gln Pro Ala Arg Lys Arg Leu Asn Phe Gly Gln			
165	170	175	
act ggc gac tca gag tca gtt cca gac cct caa cct ctc gga gaa cct			576
Thr Gly Asp Ser Glu Ser Val Pro Asp Pro Gln Pro Leu Gly Glu Pro			
180	185	190	
cca gca gcg ccc tct ggt gtg gga cct aat aca atg gct gca ggc ggt			624
Pro Ala Ala Pro Ser Gly Val Gly Pro Asn Thr Met Ala Ala Gly Gly			
195	200	205	
ggc gca cca atg gca gac aat aac gaa ggc gcc gac gga gtg ggt agt			672
Gly Ala Pro Met Ala Asp Asn Asn Glu Gly Ala Asp Gly Val Gly Ser			
210	215	220	
tcc tcg gga aat tgg cat tgc gat tcc aca tgg ctg ggc gac aga gtc			720
Ser Ser Gly Asn Trp His Cys Asp Ser Thr Trp Leu Gly Asp Arg Val			
225	230	235	240
atc acc acc agc acc cga acc tgg gcc ctg ccc acc tac aac aac cac			768
Ile Thr Thr Ser Thr Arg Thr Trp Ala Leu Pro Thr Tyr Asn Asn His			
245	250	255	
ctc tac aag caa atc tcc aac ggg aca tcg gga gga gcc acc aac gac			816
Leu Tyr Lys Gln Ile Ser Asn Gly Thr Ser Gly Gly Ala Thr Asn Asp			
260	265	270	
aac acc tac ttc ggc tac agc acc ccc tgg ggg tat ttt gac ttt aac			864
Asn Thr Tyr Phe Gly Tyr Ser Thr Pro Trp Gly Tyr Phe Asp Phe Asn			
275	280	285	
aga ttc cac tgc cac ttt tca cca cgt gac tgg cag cga ctc atc aac			912
Arg Phe His Cys His Phe Ser Pro Arg Asp Trp Gln Arg Leu Ile Asn			
290	295	300	
aac aac tgg gga ttc cgg ccc aag aga ctc agc ttc aag ctc ttc aac			960
Asn Asn Trp Gly Phe Arg Pro Lys Arg Leu Ser Phe Lys Leu Phe Asn			
305	310	315	320
atc cag gtc aag gag gtc acg cag aat gaa ggc acc aag acc atc gcc			1008
Ile Gln Val Lys Glu Val Thr Gln Asn Glu Gly Thr Lys Thr Ile Ala			
325	330	335	
aat aac ctc acc agc acc atc cag gtg ttt acg gac tcg gag tac cag			1056
Asn Asn Leu Thr Ser Thr Ile Gln Val Phe Thr Asp Ser Glu Tyr Gln			
340	345	350	
ctg ccg tac gtt ctc ggc tct gcc cac cag ggc tgc ctg cct ccg ttc			1104
Leu Pro Tyr Val Leu Gly Ser Ala His Gln Gly Cys Leu Pro Pro Phe			
355	360	365	
ccg gcg gac gtg ttc atg att ccc cag tac ggc tac cta aca ctc aac			1152
Pro Ala Asp Val Phe Met Ile Pro Gln Tyr Gly Tyr Leu Thr Leu Asn			
370	375	380	
aac ggt agt cag gcc gtg gga cgc tcc tcc ttc tac tgc ctg gaa tac			1200
Asn Gly Ser Gln Ala Val Gly Arg Ser Ser Phe Tyr Cys Leu Glu Tyr			
385	390	395	400
ttt cct tcg cag atg ctg aga acc ggc aac aac ttc cag ttt act tac			1248
Phe Pro Ser Gln Met Leu Arg Thr Gly Asn Asn Phe Gln Phe Thr Tyr			
405	410	415	
acc ttc gag gac gtg cct ttc cac agc agc tac gcc cac agc cag agc			1296

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Thr	Phe	Glu	Asp	Val	Pro	Phe	His	Ser	Ser	Tyr	Ala	His	Ser	Gln	Ser	
			420					425					430			
ttg	gac	cgg	ctg	atg	aat	cct	ctg	att	gac	cag	tac	ctg	tac	tac	ttg	1344
Leu	Asp	Arg	Leu	Met	Asn	Pro	Leu	Ile	Asp	Gln	Tyr	Leu	Tyr	Tyr	Leu	
		435					440					445				
tct	cgg	act	caa	aca	aca	gga	ggc	acg	gca	aat	acg	cag	act	ctg	ggc	1392
Ser	Arg	Thr	Gln	Thr	Thr	Gly	Gly	Thr	Ala	Asn	Thr	Gln	Thr	Leu	Gly	
		450				455					460					
ttc	agc	caa	ggg	ggg	cct	aat	aca	atg	gcc	aat	cag	gca	aag	aac	tgg	1440
Phe	Ser	Gln	Gly	Gly	Pro	Asn	Thr	Met	Ala	Asn	Gln	Ala	Lys	Asn	Trp	
		465			470					475				480		
ctg	cca	gga	ccc	tgt	tac	cgc	caa	caa	cgc	gtc	tca	acg	aca	acc	ggg	1488
Leu	Pro	Gly	Pro	Cys	Tyr	Arg	Gln	Gln	Arg	Val	Ser	Thr	Thr	Thr	Gly	
			485						490					495		
caa	aac	aac	aat	agc	aac	ttt	gcc	tgg	act	gct	ggg	acc	aaa	tac	cat	1536
Gln	Asn	Asn	Asn	Ser	Asn	Phe	Ala	Trp	Thr	Ala	Gly	Thr	Lys	Tyr	His	
		500						505					510			
ctg	aat	gga	aga	aat	tca	ttg	gct	aat	cct	ggc	atc	gct	atg	gca	aca	1584
Leu	Asn	Gly	Arg	Asn	Ser	Leu	Ala	Asn	Pro	Gly	Ile	Ala	Met	Ala	Thr	
		515					520					525				
cac	aaa	gac	gac	gag	gag	cgt	ttt	ttt	ccc	agt	aac	ggg	atc	ctg	att	1632
His	Lys	Asp	Asp	Glu	Glu	Arg	Phe	Phe	Pro	Ser	Asn	Gly	Ile	Leu	Ile	
		530				535					540					
ttt	ggc	aaa	caa	aat	gct	gcc	aga	gac	aat	gcg	gat	tac	agc	gat	gtc	1680
Phe	Gly	Lys	Gln	Asn	Ala	Ala	Arg	Asp	Asn	Ala	Asp	Tyr	Ser	Asp	Val	
		545			550					555					560	
atg	ctc	acc	agc	gag	gaa	gaa	atc	aaa	acc	act	aac	cct	gtg	gct	aca	1728
Met	Leu	Thr	Ser	Glu	Glu	Glu	Ile	Lys	Thr	Thr	Asn	Pro	Val	Ala	Thr	
			565						570					575		
gag	gaa	tac	ggg	atc	gtg	gca	gat	aac	ttg	cag	cag	caa	aac	acg	gct	1776
Glu	Glu	Tyr	Gly	Ile	Val	Ala	Asp	Asn	Leu	Gln	Gln	Gln	Asn	Thr	Ala	
			580					585					590			
cct	caa	att	gga	act	gtc	aac	agc	cag	ggg	gcc	tta	ccc	ggg	atg	gtc	1824
Pro	Gln	Ile	Gly	Thr	Val	Asn	Ser	Gln	Gly	Ala	Leu	Pro	Gly	Met	Val	
		595				600						605				
tgg	cag	aac	cgg	gac	gtg	tac	ctg	cag	ggg	ccc	atc	tgg	gcc	aag	att	1872
Trp	Gln	Asn	Arg	Asp	Val	Tyr	Leu	Gln	Gly	Pro	Ile	Trp	Ala	Lys	Ile	
		610				615					620					
cct	cac	acg	gac	ggc	aac	ttc	cac	ccg	tct	ccg	ctg	atg	ggc	ggc	ttt	1920
Pro	His	Thr	Asp	Gly	Asn	Phe	His	Pro	Ser	Pro	Leu	Met	Gly	Gly	Phe	
		625			630					635				640		
ggc	ctg	aaa	cat	cct	ccg	cct	cag	atc	ctg	atc	aag	aac	acg	cct	gta	1968
Gly	Leu	Lys	His	Pro	Pro	Pro	Gln	Ile	Leu	Ile	Lys	Asn	Thr	Pro	Val	
			645						650					655		
cct	gcg	gat	cct	ccg	acc	acc	ttc	aac	cag	tca	aag	ctg	aac	tct	ttc	2016
Pro	Ala	Asp	Pro	Pro	Thr	Thr	Phe	Asn	Gln	Ser	Lys	Leu	Asn	Ser	Phe	
			660					665					670			
atc	acg	caa	tac	agc	acc	gga	cag	gtc	agc	gtg	gaa	att	gaa	tgg	gag	2064
Ile	Thr	Gln	Tyr	Ser	Thr	Gly	Gln	Val	Ser	Val	Glu	Ile	Glu	Trp	Glu	
		675					680					685				
ctg	cag	aag	gaa	aac	agc	aag	cgc	tgg	aac	ccc	gag	atc	cag	tac	acc	2112
Leu	Gln	Lys	Glu	Asn	Ser	Lys	Arg	Trp	Asn	Pro	Glu	Ile	Gln	Tyr	Thr	
		690				695				700						
tcc	aac	tac	tac	aaa	tct	aca	agt	gtg	gac	ttt	gct	gtt	aat	aca	gaa	2160
Ser	Asn	Tyr	Tyr	Lys	Ser	Thr	Ser	Val	Asp	Phe	Ala	Val	Asn	Thr	Glu	
		705			710					715					720	
ggc	gtg	tac	tct	gaa	ccc	cgc	ccc	att	ggc	acc	cgt	tac	ctc	acc	cgt	2208
Gly	Val	Tyr	Ser	Glu	Pro	Arg	Pro	Ile	Gly	Thr	Arg	Tyr	Leu	Thr	Arg	
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aat ctg 2214
Asn Leu

<210> SEQ ID NO 20
 <211> LENGTH: 738
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

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 20 25 30

Lys Ala Asn Gln Gln Lys Gln Asp Asp Gly Arg Gly Leu Val Leu Pro
 35 40 45

Gly Tyr Lys Tyr Leu Gly Pro Phe Asn Gly Leu Asp Lys Gly Glu Pro
 50 55 60

Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp
 65 70 75 80

Gln Gln Leu Gln Ala Gly Asp Asn Pro Tyr Leu Arg Tyr Asn His Ala
 85 90 95

Asp Ala Glu Phe Gln Glu Arg Leu Gln Glu Asp Thr Ser Phe Gly Gly
 100 105 110

Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Val Leu Glu Pro
 115 120 125

Leu Gly Leu Val Glu Glu Gly Ala Lys Thr Ala Pro Gly Lys Lys Arg
 130 135 140

Pro Val Glu Pro Ser Pro Gln Arg Ser Pro Asp Ser Ser Thr Gly Ile
 145 150 155 160

Gly Lys Lys Gly Gln Gln Pro Ala Arg Lys Arg Leu Asn Phe Gly Gln
 165 170 175

Thr Gly Asp Ser Glu Ser Val Pro Asp Pro Gln Pro Leu Gly Glu Pro
 180 185 190

Pro Ala Ala Pro Ser Gly Val Gly Pro Asn Thr Met Ala Ala Gly Gly
 195 200 205

Gly Ala Pro Met Ala Asp Asn Asn Glu Gly Ala Asp Gly Val Gly Ser
 210 215 220

Ser Ser Gly Asn Trp His Cys Asp Ser Thr Trp Leu Gly Asp Arg Val
 225 230 235 240

Ile Thr Thr Ser Thr Arg Thr Trp Ala Leu Pro Thr Tyr Asn Asn His
 245 250 255

Leu Tyr Lys Gln Ile Ser Asn Gly Thr Ser Gly Gly Ala Thr Asn Asp
 260 265 270

Asn Thr Tyr Phe Gly Tyr Ser Thr Pro Trp Gly Tyr Phe Asp Phe Asn
 275 280 285

Arg Phe His Cys His Phe Ser Pro Arg Asp Trp Gln Arg Leu Ile Asn
 290 295 300

Asn Asn Trp Gly Phe Arg Pro Lys Arg Leu Ser Phe Lys Leu Phe Asn
 305 310 315 320

Ile Gln Val Lys Glu Val Thr Gln Asn Glu Gly Thr Lys Thr Ile Ala
 325 330 335

Asn Asn Leu Thr Ser Thr Ile Gln Val Phe Thr Asp Ser Glu Tyr Gln
 340 345 350

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Leu Pro Tyr Val Leu Gly Ser Ala His Gln Gly Cys Leu Pro Pro Phe
   355                               360               365

Pro Ala Asp Val Phe Met Ile Pro Gln Tyr Gly Tyr Leu Thr Leu Asn
   370                               375               380

Asn Gly Ser Gln Ala Val Gly Arg Ser Ser Phe Tyr Cys Leu Glu Tyr
   385                               390               395               400

Phe Pro Ser Gln Met Leu Arg Thr Gly Asn Asn Phe Gln Phe Thr Tyr
   405                               410               415

Thr Phe Glu Asp Val Pro Phe His Ser Ser Tyr Ala His Ser Gln Ser
   420                               425               430

Leu Asp Arg Leu Met Asn Pro Leu Ile Asp Gln Tyr Leu Tyr Tyr Leu
   435                               440               445

Ser Arg Thr Gln Thr Thr Gly Gly Thr Ala Asn Thr Gln Thr Leu Gly
   450                               455               460

Phe Ser Gln Gly Gly Pro Asn Thr Met Ala Asn Gln Ala Lys Asn Trp
   465                               470               475               480

Leu Pro Gly Pro Cys Tyr Arg Gln Gln Arg Val Ser Thr Thr Thr Gly
   485                               490               495

Gln Asn Asn Asn Ser Asn Phe Ala Trp Thr Ala Gly Thr Lys Tyr His
   500                               505               510

Leu Asn Gly Arg Asn Ser Leu Ala Asn Pro Gly Ile Ala Met Ala Thr
   515                               520               525

His Lys Asp Asp Glu Glu Arg Phe Phe Pro Ser Asn Gly Ile Leu Ile
   530                               535               540

Phe Gly Lys Gln Asn Ala Ala Arg Asp Asn Ala Asp Tyr Ser Asp Val
   545                               550               555               560

Met Leu Thr Ser Glu Glu Glu Ile Lys Thr Thr Asn Pro Val Ala Thr
   565                               570               575

Glu Glu Tyr Gly Ile Val Ala Asp Asn Leu Gln Gln Gln Asn Thr Ala
   580                               585               590

Pro Gln Ile Gly Thr Val Asn Ser Gln Gly Ala Leu Pro Gly Met Val
   595                               600               605

Trp Gln Asn Arg Asp Val Tyr Leu Gln Gly Pro Ile Trp Ala Lys Ile
   610                               615               620

Pro His Thr Asp Gly Asn Phe His Pro Ser Pro Leu Met Gly Gly Phe
   625                               630               635               640

Gly Leu Lys His Pro Pro Pro Gln Ile Leu Ile Lys Asn Thr Pro Val
   645                               650               655

Pro Ala Asp Pro Pro Thr Thr Phe Asn Gln Ser Lys Leu Asn Ser Phe
   660                               665               670

Ile Thr Gln Tyr Ser Thr Gly Gln Val Ser Val Glu Ile Glu Trp Glu
   675                               680               685

Leu Gln Lys Glu Asn Ser Lys Arg Trp Asn Pro Glu Ile Gln Tyr Thr
   690                               695               700

Ser Asn Tyr Tyr Lys Ser Thr Ser Val Asp Phe Ala Val Asn Thr Glu
   705                               710               715               720

Gly Val Tyr Ser Glu Pro Arg Pro Ile Gly Thr Arg Tyr Leu Thr Arg
   725                               730               735

Asn Leu

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<210> SEQ ID NO 21

<211> LENGTH: 2214

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

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<220> FEATURE:
<223> OTHER INFORMATION: AAVhu37
<220> FEATURE:
<221> NAME/KEY: CDS
<222> LOCATION: (1) .. (2214)

<400> SEQUENCE: 21

atg gct gct gac ggt tat ctt cca gat tgg ctc gag gac aac ctc tct      48
Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser
1          5          10          15

gag ggc att cgc gag tgg tgg gac ctg aaa cct gga gcc ccc aag ccc      96
Glu Gly Ile Arg Glu Trp Trp Asp Leu Lys Pro Gly Ala Pro Lys Pro
          20          25          30

aag gcc aac cag cag aag cag gac gac ggc cgg ggt ctg gtg ctt cct     144
Lys Ala Asn Gln Gln Lys Gln Asp Asp Gly Arg Gly Leu Val Leu Pro
          35          40          45

ggc tac aag tac ctc gga ccc ttc aac gga ctc gac aag ggg gag ccc     192
Gly Tyr Lys Tyr Leu Gly Pro Phe Asn Gly Leu Asp Lys Gly Glu Pro
          50          55          60

gtc aac gcg gcg gac gca gcg gcc ctc gag cac gac aag gcc tac gac     240
Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp
          65          70          75          80

cag cag ctc aaa gcg ggt gac aat ccg tac ctg cgg tat aac cac gcc     288
Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Arg Tyr Asn His Ala
          85          90          95

gac gcc gag ttt cag gag cgt ctg caa gaa gat acg tct ttt ggg ggc     336
Asp Ala Glu Phe Gln Glu Arg Leu Gln Glu Asp Thr Ser Phe Gly Gly
          100          105          110

aac ctc ggg cga gca gtc ttc cag gcc aag aag cgg gtt ctc gaa cct     384
Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Val Leu Glu Pro
          115          120          125

ctc ggt ctg gtt gag gaa gct gct aag acg gct cct gga aag aag aga     432
Leu Gly Leu Val Glu Glu Ala Ala Lys Thr Ala Pro Gly Lys Lys Arg
          130          135          140

ccg gta gaa ccg tca cct cag cgt tcc ccc gac tcc tcc acg ggc atc     480
Pro Val Glu Pro Ser Pro Gln Arg Ser Pro Asp Ser Ser Thr Gly Ile
          145          150          155          160

ggc aag aaa ggc cag cag ccc gct aaa aag aga ctg aac ttt ggt cag     528
Gly Lys Lys Gly Gln Gln Pro Ala Lys Lys Arg Leu Asn Phe Gly Gln
          165          170          175

act ggc gac tca gag tca gtc ccc gac cct caa cca atc gga gaa cca     576
Thr Gly Asp Ser Glu Ser Val Pro Asp Pro Gln Pro Ile Gly Glu Pro
          180          185          190

cca gca ggc ccc tct ggt ctg gga tct ggt aca atg gct gca ggc ggt     624
Pro Ala Gly Pro Ser Gly Leu Gly Ser Gly Thr Met Ala Ala Gly Gly
          195          200          205

ggc gct cca atg gca gac aat aac gaa ggc gcc gac gga gtg ggt agt     672
Gly Ala Pro Met Ala Asp Asn Asn Glu Gly Ala Asp Gly Val Gly Ser
          210          215          220

tcc tca gga aat tgg cat tgc gat tcc aca tgg ctg ggc gac aga gtc     720
Ser Ser Gly Asn Trp His Cys Asp Ser Thr Trp Leu Gly Asp Arg Val
          225          230          235          240

atc acc acc agc acc cga acc tgg gcc ctg ccc acc tac aac aac cac     768
Ile Thr Thr Ser Thr Arg Thr Trp Ala Leu Pro Thr Tyr Asn Asn His
          245          250          255

ctc tac aag caa ata tcc aat ggg aca tcg gga gga agc acc aac gac     816
Leu Tyr Lys Gln Ile Ser Asn Gly Thr Ser Gly Gly Ser Thr Asn Asp
          260          265          270

aac acc tac ttc ggc tac agc acc ccc tgg ggg tat ttt gac ttc aac     864
Asn Thr Tyr Phe Gly Tyr Ser Thr Pro Trp Gly Tyr Phe Asp Phe Asn
          275          280          285

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aga ttc cac tgc cac ttc tca cca cgt gac tgg cag cga ctc atc aac Arg Phe His Cys His Phe Ser Pro Arg Asp Trp Gln Arg Leu Ile Asn 290 295 300	912
aac aac tgg gga ttc cgg cca aaa aga ctc agc ttc aag ctc ttc aac Asn Asn Trp Gly Phe Arg Pro Lys Arg Leu Ser Phe Lys Leu Phe Asn 305 310 315 320	960
atc cag gtc aag gag gtc acg cag aat gaa ggc acc aag acc atc gcc Ile Gln Val Lys Glu Val Thr Gln Asn Glu Gly Thr Lys Thr Ile Ala 325 330 335	1008
aat aac ctt acc agc acg att cag gta ttt acg gac tcg gaa tac cag Asn Asn Leu Thr Ser Thr Ile Gln Val Phe Thr Asp Ser Glu Tyr Gln 340 345 350	1056
ctg ccg tac gtc ctc ggc tcc gcg cac cag ggc tgc ctg cct ccg ttc Leu Pro Tyr Val Leu Gly Ser Ala His Gln Gly Cys Leu Pro Pro Phe 355 360 365	1104
ccg gcg gac gtc ttc atg att ccc cag tac ggc tac ctt aca ctg aac Pro Ala Asp Val Phe Met Ile Pro Gln Tyr Gly Tyr Leu Thr Leu Asn 370 375 380	1152
aat gga agt caa gcc gta ggc cgt tcc tcc ttc tac tgc ctg gaa tat Asn Gly Ser Gln Ala Val Gly Arg Ser Ser Phe Tyr Cys Leu Glu Tyr 385 390 395 400	1200
ttt cca tct caa atg ctg cga act gga aac aat ttt gaa ttc agc tac Phe Pro Ser Gln Met Leu Arg Thr Gly Asn Asn Phe Glu Phe Ser Tyr 405 410 415	1248
acc ttc gag gac gtg cct ttc cac agc agc tac gca cac agc cag agc Thr Phe Glu Asp Val Pro Phe His Ser Ser Tyr Ala His Ser Gln Ser 420 425 430	1296
ttg gac cga ctg atg aat cct ctc atc gac cag tac ctg tac tac tta Leu Asp Arg Leu Met Asn Pro Leu Ile Asp Gln Tyr Leu Tyr Tyr Leu 435 440 445	1344
tcc aga act cag tcc aca gga gga act caa ggt acc cag caa ttg tta Ser Arg Thr Gln Ser Thr Gly Gly Thr Gln Gly Thr Gln Gln Leu Leu 450 455 460	1392
ttt tct caa gct ggg cct gca aac atg tcg gct cag gct aag aac tgg Phe Ser Gln Ala Gly Pro Ala Asn Met Ser Ala Gln Ala Lys Asn Trp 465 470 475 480	1440
cta cct gga cct tgc tac cgg cag cag cga gtc tct acg aca ctg tcg Leu Pro Gly Pro Cys Tyr Arg Gln Gln Arg Val Ser Thr Thr Leu Ser 485 490 495	1488
caa aac aac aac agc aac ttt gct tgg act ggt gcc acc aaa tat cac Gln Asn Asn Asn Ser Asn Phe Ala Trp Thr Gly Ala Thr Lys Tyr His 500 505 510	1536
ctg aac gga aga gac tct ttg gta aat ccc ggt gtc gcc atg gca acc Leu Asn Gly Arg Asp Ser Leu Val Asn Pro Gly Val Ala Met Ala Thr 515 520 525	1584
cac aag gac gac gag gaa cgc ttc ttc ccg tcg agt gga gtc ctg atg His Lys Asp Asp Glu Glu Arg Phe Phe Pro Ser Ser Gly Val Leu Met 530 535 540	1632
ttc gga aaa cag ggt gct gga aga gac aat gtg gac tac agc agc gtt Phe Gly Lys Gln Gly Ala Gly Arg Asp Asn Val Asp Tyr Ser Ser Val 545 550 555 560	1680
atg cta acc agc gaa gaa gaa att aaa acc act aac ccc gta gcc aca Met Leu Thr Ser Glu Glu Glu Ile Lys Thr Thr Asn Pro Val Ala Thr 565 570 575	1728
gaa caa tac ggt gtg gtg gct gac aac ttg cag caa acc aat aca ggg Glu Gln Tyr Gly Val Val Ala Asp Asn Leu Gln Gln Thr Asn Thr Gly 580 585 590	1776
cct att gtg gga aat gtc aac agc caa gga gcc tta cct ggc atg gtc Pro Ile Val Gly Asn Val Asn Ser Gln Gly Ala Leu Pro Gly Met Val	1824

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595	600	605	
tgg cag aac cga gac gtg tac ctg cag ggt ccc atc tgg gcc aag att			1872
Trp Gln Asn Arg Asp Val Tyr Leu Gln Gly Pro Ile Trp Ala Lys Ile			
610	615	620	
cct cac acg gac ggc aac ttc cac cct tca ccg cta atg gga gga ttt			1920
Pro His Thr Asp Gly Asn Phe His Pro Ser Pro Leu Met Gly Gly Phe			
625	630	635	640
gga ctg aag cac cca cct cct cag atc ctg atc aag aac acg ccg gta			1968
Gly Leu Lys His Pro Pro Pro Gln Ile Leu Ile Lys Asn Thr Pro Val			
645	650	655	
cct gcg gat cct cca aca acg ttc agc cag gcg aaa ttg gct tcc ttc			2016
Pro Ala Asp Pro Pro Thr Thr Phe Ser Gln Ala Lys Leu Ala Ser Phe			
660	665	670	
att acg cag tac agc acc gga cag gtc agc gtg gaa atc gag tgg gag			2064
Ile Thr Gln Tyr Ser Thr Gly Gln Val Ser Val Glu Ile Glu Trp Glu			
675	680	685	
ctg cag aag gag aac agc aaa cgc tgg aac cca gag att cag tac act			2112
Leu Gln Lys Glu Asn Ser Lys Arg Trp Asn Pro Glu Ile Gln Tyr Thr			
690	695	700	
tca aac tac tac aaa tct aca aat gtg gac ttt gct gtc aat aca gag			2160
Ser Asn Tyr Tyr Lys Ser Thr Asn Val Asp Phe Ala Val Asn Thr Glu			
705	710	715	720
gga act tat tct gag cct cgc ccc att ggt act cgt tac ctc acc cgt			2208
Gly Thr Tyr Ser Glu Pro Arg Pro Ile Gly Thr Arg Tyr Leu Thr Arg			
725	730	735	
aat ctg			2214
Asn Leu			

<210> SEQ ID NO 22

<211> LENGTH: 738

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 22

Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser
1 5 10 15

Glu Gly Ile Arg Glu Trp Trp Asp Leu Lys Pro Gly Ala Pro Lys Pro
20 25 30

Lys Ala Asn Gln Gln Lys Gln Asp Asp Gly Arg Gly Leu Val Leu Pro
35 40 45

Gly Tyr Lys Tyr Leu Gly Pro Phe Asn Gly Leu Asp Lys Gly Glu Pro
50 55 60

Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp
65 70 75 80

Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Arg Tyr Asn His Ala
85 90 95

Asp Ala Glu Phe Gln Glu Arg Leu Gln Glu Asp Thr Ser Phe Gly Gly
100 105 110

Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Val Leu Glu Pro
115 120 125

Leu Gly Leu Val Glu Glu Ala Ala Lys Thr Ala Pro Gly Lys Lys Arg
130 135 140

Pro Val Glu Pro Ser Pro Gln Arg Ser Pro Asp Ser Ser Thr Gly Ile
145 150 155 160

Gly Lys Lys Gly Gln Gln Pro Ala Lys Lys Arg Leu Asn Phe Gly Gln
165 170 175

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Thr Gly Asp	Ser Glu Ser	Val Pro Asp	Pro Gln Pro	Ile Gly Glu	Pro
	180		185	190	
Pro Ala Gly	Pro Ser Gly	Leu Gly Ser	Gly Thr Met	Ala Ala Gly	Gly
	195		200	205	
Gly Ala Pro	Met Ala Asp	Asn Asn Glu	Gly Ala Asp	Gly Val Gly	Ser
	210		215	220	
Ser Ser Gly	Asn Trp His	Cys Asp Ser	Thr Trp Leu	Gly Asp Arg	Val
	225		230	235	240
Ile Thr Thr	Ser Thr Arg	Thr Trp Ala	Leu Pro Thr	Tyr Asn Asn	His
	245		250	255	
Leu Tyr Lys	Gln Ile Ser	Asn Gly Thr	Ser Gly Gly	Ser Thr Asn	Asp
	260		265	270	
Asn Thr Tyr	Phe Gly Tyr	Ser Thr Pro	Trp Gly Tyr	Phe Asp Phe	Asn
	275		280	285	
Arg Phe His	Cys His Phe	Ser Pro Arg	Asp Trp Gln	Arg Leu Ile	Asn
	290		295	300	
Asn Asn Trp	Gly Phe Arg	Pro Lys Arg	Leu Ser Phe	Lys Leu Phe	Asn
	305		310	315	320
Ile Gln Val	Lys Glu Val	Thr Gln Asn	Glu Gly Thr	Lys Thr Ile	Ala
	325		330	335	
Asn Asn Leu	Thr Ser Thr	Ile Gln Val	Phe Thr Asp	Ser Glu Tyr	Gln
	340		345	350	
Leu Pro Tyr	Val Leu Gly	Ser Ala His	Gln Gly Cys	Leu Pro Pro	Phe
	355		360	365	
Pro Ala Asp	Val Phe Met	Ile Pro Gln	Tyr Gly Tyr	Leu Thr Leu	Asn
	370		375	380	
Asn Gly Ser	Gln Ala Val	Gly Arg Ser	Ser Phe Tyr	Cys Leu Glu	Tyr
	385		390	395	400
Phe Pro Ser	Gln Met Leu	Arg Thr Gly	Asn Asn Phe	Glu Phe Ser	Tyr
	405		410	415	
Thr Phe Glu	Asp Val Pro	Phe His Ser	Ser Tyr Ala	His Ser Gln	Ser
	420		425	430	
Leu Asp Arg	Leu Met Asn	Pro Leu Ile	Asp Gln Tyr	Leu Tyr Tyr	Leu
	435		440	445	
Ser Arg Thr	Gln Ser Thr	Gly Gly Thr	Gln Gly Thr	Gln Gln Leu	Leu
	450		455	460	
Phe Ser Gln	Ala Gly Pro	Ala Asn Met	Ser Ala Gln	Ala Lys Asn	Trp
	465		470	475	480
Leu Pro Gly	Pro Cys Tyr	Arg Gln Gln	Arg Val Ser	Thr Thr Leu	Ser
	485		490	495	
Gln Asn Asn	Asn Ser Asn	Phe Ala Trp	Thr Gly Ala	Thr Lys Tyr	His
	500		505	510	
Leu Asn Gly	Arg Asp Ser	Leu Val Asn	Pro Gly Val	Ala Met Ala	Thr
	515		520	525	
His Lys Asp	Asp Glu Glu	Arg Phe Phe	Pro Ser Ser	Gly Val Leu	Met
	530		535	540	
Phe Gly Lys	Gln Gly Ala	Gly Arg Asp	Asn Val Asp	Tyr Ser Ser	Val
	545		550	555	560
Met Leu Thr	Ser Glu Glu	Glu Ile Lys	Thr Thr Asn	Pro Val Ala	Thr
	565		570	575	
Glu Gln Tyr	Gly Val Val	Ala Asp Asn	Leu Gln Gln	Thr Asn Thr	Gly
	580		585	590	

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Pro Ile Val Gly Asn Val Asn Ser Gln Gly Ala Leu Pro Gly Met Val
595 600 605

Trp Gln Asn Arg Asp Val Tyr Leu Gln Gly Pro Ile Trp Ala Lys Ile
610 615 620

Pro His Thr Asp Gly Asn Phe His Pro Ser Pro Leu Met Gly Gly Phe
625 630 635 640

Gly Leu Lys His Pro Pro Pro Gln Ile Leu Ile Lys Asn Thr Pro Val
645 650 655

Pro Ala Asp Pro Pro Thr Thr Phe Ser Gln Ala Lys Leu Ala Ser Phe
660 665 670

Ile Thr Gln Tyr Ser Thr Gly Gln Val Ser Val Glu Ile Glu Trp Glu
675 680 685

Leu Gln Lys Glu Asn Ser Lys Arg Trp Asn Pro Glu Ile Gln Tyr Thr
690 695 700

Ser Asn Tyr Tyr Lys Ser Thr Asn Val Asp Phe Ala Val Asn Thr Glu
705 710 715 720

Gly Thr Tyr Ser Glu Pro Arg Pro Ile Gly Thr Arg Tyr Leu Thr Arg
725 730 735

Asn Leu

<210> SEQ ID NO 23
 <211> LENGTH: 2214
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: AAVrh10
 <220> FEATURE:
 <221> NAME/KEY: CDS
 <222> LOCATION: (1)..(2214)

<400> SEQUENCE: 23

atg gct gcc gat ggt tat ctt cca gat tgg ctc gag gac aac ctc tct	48
Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser	
1 5 10 15	
gag ggc att cgc gag tgg tgg gac ttg aaa cct gga gcc ccg aaa ccc	96
Glu Gly Ile Arg Glu Trp Trp Asp Leu Lys Pro Gly Ala Pro Lys Pro	
20 25 30	
aaa gcc aac cag caa aag cag gac gac ggc cgg ggt ctg gtg ctt cct	144
Lys Ala Asn Gln Gln Lys Gln Asp Asp Gly Arg Gly Leu Val Leu Pro	
35 40 45	
ggc tac aag tac ctc gga ccc ttc aac gga ctc gac aag ggg gag ccc	192
Gly Tyr Lys Tyr Leu Gly Pro Phe Asn Gly Leu Asp Lys Gly Glu Pro	
50 55 60	
gtc aac gcg gcg gac gca gcg gcc ctc gag cac gac aag gcc tac gac	240
Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp	
65 70 75 80	
cag cag ctc aaa gcg ggt gac aat ccg tac ctg cgg tat aac cac gcc	288
Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Arg Tyr Asn His Ala	
85 90 95	
gac gcc gag ttt cag gag cgt ctg caa gaa gat acg tct ttt ggg ggc	336
Asp Ala Glu Phe Gln Glu Arg Leu Gln Glu Asp Thr Ser Phe Gly Gly	
100 105 110	
aac ctc ggg cga gca gtc ttc cag gcc aag aag cgg gtt ctc gaa cct	384
Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Val Leu Glu Pro	
115 120 125	
ctc ggt ctg gtt gag gaa ggc gct aag acg gct cct gga aag aag aga	432
Leu Gly Leu Val Glu Glu Gly Ala Lys Thr Ala Pro Gly Lys Lys Arg	
130 135 140	
ccg gta gag cca tca ccc cag cgt tct cca gac tcc tct acg ggc atc	480

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Pro Val Glu Pro Ser Pro Gln Arg Ser Pro Asp Ser Ser Thr Gly Ile 145 150 155 160	
ggc aag aaa ggc cag cag ccc gcg aaa aag aga ctc aac ttt ggg cag Gly Lys Lys Gly Gln Gln Pro Ala Lys Lys Arg Leu Asn Phe Gly Gln 165 170 175	528
act ggc gac tca gag tca gtg ccc gac cct caa cca atc gga gaa ccc Thr Gly Asp Ser Glu Ser Val Pro Asp Pro Gln Pro Ile Gly Glu Pro 180 185 190	576
ccc gca ggc ccc tct ggt ctg gga tct ggt aca atg gct gca ggc ggt Pro Ala Gly Pro Ser Gly Leu Gly Ser Gly Thr Met Ala Ala Gly Gly 195 200 205	624
ggc gct cca atg gca gac aat aac gaa ggc gcc gac gga gtg ggt agt Gly Ala Pro Met Ala Asp Asn Asn Glu Gly Ala Asp Gly Val Gly Ser 210 215 220	672
tcc tca gga aat tgg cat tgc gat tcc aca tgg ctg ggc gac aga gtc Ser Ser Gly Asn Trp His Cys Asp Ser Thr Trp Leu Gly Asp Arg Val 225 230 235 240	720
atc acc acc agc acc cga acc tgg gcc ctc ccc acc tac aac aac cac Ile Thr Thr Ser Thr Arg Thr Trp Ala Leu Pro Thr Tyr Asn Asn His 245 250 255	768
ctc tac aag caa atc tcc aac ggg act tcg gga gga agc acc aac gac Leu Tyr Lys Gln Ile Ser Asn Gly Thr Ser Gly Gly Ser Thr Asn Asp 260 265 270	816
aac acc tac ttc ggc tac agc acc ccc tgg ggg tat ttt gac ttt aac Asn Thr Tyr Phe Gly Tyr Ser Thr Pro Trp Gly Tyr Phe Asp Phe Asn 275 280 285	864
aga ttc cac tgc cac ttc tca cca cgt gac tgg cag cga ctc atc aac Arg Phe His Cys His Phe Ser Pro Arg Asp Trp Gln Arg Leu Ile Asn 290 295 300	912
aac aac tgg gga ttc cgg ccc aag aga ctc aac ttc aag ctc ttc aac Asn Asn Trp Gly Phe Arg Pro Lys Arg Leu Asn Phe Lys Leu Phe Asn 305 310 315 320	960
atc cag gtc aag gag gtc acg cag aat gaa ggc acc aag acc atc gcc Ile Gln Val Lys Glu Val Thr Gln Asn Glu Gly Thr Lys Thr Ile Ala 325 330 335	1008
aat aac ctt acc agc acg att cag gtc ttt acg gac tcg gaa tac cag Asn Asn Leu Thr Ser Thr Ile Gln Val Phe Thr Asp Ser Glu Tyr Gln 340 345 350	1056
ctc ccg tac gtc ctc ggc tct gcg cac cag ggc tgc ctg cct ccg ttc Leu Pro Tyr Val Leu Gly Ser Ala His Gln Gly Cys Leu Pro Pro Phe 355 360 365	1104
ccg gcg gac gtc ttc atg att cct cag tac ggg tac ctg act ctg aac Pro Ala Asp Val Phe Met Ile Pro Gln Tyr Gly Tyr Leu Thr Leu Asn 370 375 380	1152
aat ggc agt cag gcc gtg ggc cgt tcc tcc ttc tac tgc ctg gag tac Asn Gly Ser Gln Ala Val Gly Arg Ser Ser Phe Tyr Cys Leu Glu Tyr 385 390 395 400	1200
ttt cct tct caa atg ctg aga acg ggc aac aac ttt gag ttc agc tac Phe Pro Ser Gln Met Leu Arg Thr Gly Asn Asn Phe Glu Phe Ser Tyr 405 410 415	1248
cag ttt gag gac gtg cct ttt cac agc agc tac gcg cac agc caa agc Gln Phe Glu Asp Val Pro Phe His Ser Ser Tyr Ala His Ser Gln Ser 420 425 430	1296
ctg gac ccg ctg atg aac ccc ctc atc gac cag tac ctg tac tac ctg Leu Asp Arg Leu Met Asn Pro Leu Ile Asp Gln Tyr Leu Tyr Tyr Leu 435 440 445	1344
tct ccg act cag tcc acg gga ggt acc gca gga act cag cag ttg cta Ser Arg Thr Gln Ser Thr Gly Gly Thr Ala Gly Thr Gln Gln Leu Leu 450 455 460	1392

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t t t t c t c a g g c c g g g c c t a a t a a c a t g t c g g c t c a g g c c a a a a c t g g	1440
Phe Ser Gln Ala Gly Pro Asn Asn Met Ser Ala Gln Ala Lys Asn Trp	
465 470 475 480	
c t a c c c g g g c c c t g c t a c c g g c a a c g c g t c c a c g a c a c t g t g	1488
Leu Pro Gly Pro Cys Tyr Arg Gln Gln Arg Val Ser Thr Thr Leu Ser	
485 490 495	
c a a a a t a a c a a c a g c a a c t t t g c c t g g a c c g g t g c c a c c a a g t a t c a t	1536
Gln Asn Asn Asn Ser Asn Phe Ala Trp Thr Gly Ala Thr Lys Tyr His	
500 505 510	
c t g a a t g g c a g a g a c t c t c t g t a a a t c c c g g t g t c g c t a t g g c a a c c	1584
Leu Asn Gly Arg Asp Ser Leu Val Asn Pro Gly Val Ala Met Ala Thr	
515 520 525	
c a c a a g g a c g a c g a a g a g c g a t t t t c c g t c c a g c g g a g t c t t a a t g	1632
His Lys Asp Asp Glu Glu Arg Phe Phe Pro Ser Ser Gly Val Leu Met	
530 535 540	
t t t g g g a a a c a g g a g c t g g a a a g a c a a c g t g g a c t a t a g c a g c g t t	1680
Phe Gly Lys Gln Gly Ala Gly Lys Asp Asn Val Asp Tyr Ser Ser Val	
545 550 555 560	
a t g c t a a c a g t g a g g a a g a a t a a a a c c a c c a a c c c a g t g g c c a c a	1728
Met Leu Thr Ser Glu Glu Glu Ile Lys Thr Thr Asn Pro Val Ala Thr	
565 570 575	
g a a c a g t a c g g c g t g g t g c c g a t a a c c t g c a a c a g c a a a a c g c c g c t	1776
Glu Gln Tyr Gly Val Val Ala Asp Asn Leu Gln Gln Gln Asn Ala Ala	
580 585 590	
c c t a t t g t a g g g g c c g t c a a c a g t c a a g g a g c c t t a c c t g g c a t g g t c	1824
Pro Ile Val Gly Ala Val Asn Ser Gln Gly Ala Leu Pro Gly Met Val	
595 600 605	
t g g c a g a a c c g g g a c g t g t a c c t g c a g g g t c c t a t c t g g g c c a a g a t t	1872
Trp Gln Asn Arg Asp Val Tyr Leu Gln Gly Pro Ile Trp Ala Lys Ile	
610 615 620	
c c t c a c a c g g a c g g a a a c t t t c a t c c c t c g c c g c t g a t g g a g g c t t t	1920
Pro His Thr Asp Gly Asn Phe His Pro Ser Pro Leu Met Gly Gly Phe	
625 630 635 640	
g g a c t g a a a c a c c c g c c t c c t c a g a t c c t g a t t a a g a a t a c a c c t g t t	1968
Gly Leu Lys His Pro Pro Pro Gln Ile Leu Ile Lys Asn Thr Pro Val	
645 650 655	
c c c g c g g a t c c t c c a a c t a c c t t c a g t c a a g c t a a g c t g c g t c g t t c	2016
Pro Ala Asp Pro Pro Thr Thr Phe Ser Gln Ala Lys Leu Ala Ser Phe	
660 665 670	
a t c a c g c a g t a c a g c a c c g g a c a g g t c a g c g t g g a a t t g a a t g g g a g	2064
Ile Thr Gln Tyr Ser Thr Gly Gln Val Ser Val Glu Ile Glu Trp Glu	
675 680 685	
c t g c a g a a a g a a a c a g c a a a c g c t g g a a c c c a g a g a t t c a a t a c a c t	2112
Leu Gln Lys Glu Asn Ser Lys Arg Trp Asn Pro Glu Ile Gln Tyr Thr	
690 695 700	
t c c a a c t a c t a c a a a t c t a c a a t g t g g a c t t t g c t g t t a a c a c a g a t	2160
Ser Asn Tyr Tyr Lys Ser Thr Asn Val Asp Phe Ala Val Asn Thr Asp	
705 710 715 720	
g g c a c t t a t t c t g a g c c t c g c c c a t c g g c a c c c g t t a c c t c a c c c g t	2208
Gly Thr Tyr Ser Glu Pro Arg Pro Ile Gly Thr Arg Tyr Leu Thr Arg	
725 730 735	
a a t c t g	2214
Asn Leu	

<210> SEQ ID NO 24

<211> LENGTH: 738

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

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<400> SEQUENCE: 24

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Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser
1      5      10      15

Glu Gly Ile Arg Glu Trp Trp Asp Leu Lys Pro Gly Ala Pro Lys Pro
20      25      30

Lys Ala Asn Gln Gln Lys Gln Asp Asp Gly Arg Gly Leu Val Leu Pro
35      40      45

Gly Tyr Lys Tyr Leu Gly Pro Phe Asn Gly Leu Asp Lys Gly Glu Pro
50      55      60

Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp
65      70      75      80

Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Arg Tyr Asn His Ala
85      90      95

Asp Ala Glu Phe Gln Glu Arg Leu Gln Glu Asp Thr Ser Phe Gly Gly
100     105     110

Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Val Leu Glu Pro
115     120     125

Leu Gly Leu Val Glu Glu Gly Ala Lys Thr Ala Pro Gly Lys Lys Arg
130     135     140

Pro Val Glu Pro Ser Pro Gln Arg Ser Pro Asp Ser Ser Thr Gly Ile
145     150     155     160

Gly Lys Lys Gly Gln Gln Pro Ala Lys Lys Arg Leu Asn Phe Gly Gln
165     170     175

Thr Gly Asp Ser Glu Ser Val Pro Asp Pro Gln Pro Ile Gly Glu Pro
180     185     190

Pro Ala Gly Pro Ser Gly Leu Gly Ser Gly Thr Met Ala Ala Gly Gly
195     200     205

Gly Ala Pro Met Ala Asp Asn Asn Glu Gly Ala Asp Gly Val Gly Ser
210     215     220

Ser Ser Gly Asn Trp His Cys Asp Ser Thr Trp Leu Gly Asp Arg Val
225     230     235     240

Ile Thr Thr Ser Thr Arg Thr Trp Ala Leu Pro Thr Tyr Asn Asn His
245     250     255

Leu Tyr Lys Gln Ile Ser Asn Gly Thr Ser Gly Gly Ser Thr Asn Asp
260     265     270

Asn Thr Tyr Phe Gly Tyr Ser Thr Pro Trp Gly Tyr Phe Asp Phe Asn
275     280     285

Arg Phe His Cys His Phe Ser Pro Arg Asp Trp Gln Arg Leu Ile Asn
290     295     300

Asn Asn Trp Gly Phe Arg Pro Lys Arg Leu Asn Phe Lys Leu Phe Asn
305     310     315     320

Ile Gln Val Lys Glu Val Thr Gln Asn Glu Gly Thr Lys Thr Ile Ala
325     330     335

Asn Asn Leu Thr Ser Thr Ile Gln Val Phe Thr Asp Ser Glu Tyr Gln
340     345     350

Leu Pro Tyr Val Leu Gly Ser Ala His Gln Gly Cys Leu Pro Pro Phe
355     360     365

Pro Ala Asp Val Phe Met Ile Pro Gln Tyr Gly Tyr Leu Thr Leu Asn
370     375     380

Asn Gly Ser Gln Ala Val Gly Arg Ser Ser Phe Tyr Cys Leu Glu Tyr
385     390     395     400

Phe Pro Ser Gln Met Leu Arg Thr Gly Asn Asn Phe Glu Phe Ser Tyr
405     410     415

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Gln Phe Glu Asp Val Pro Phe His Ser Ser Tyr Ala His Ser Gln Ser
 420 425 430
 Leu Asp Arg Leu Met Asn Pro Leu Ile Asp Gln Tyr Leu Tyr Tyr Leu
 435 440 445
 Ser Arg Thr Gln Ser Thr Gly Gly Thr Ala Gly Thr Gln Gln Leu Leu
 450 455 460
 Phe Ser Gln Ala Gly Pro Asn Asn Met Ser Ala Gln Ala Lys Asn Trp
 465 470 475 480
 Leu Pro Gly Pro Cys Tyr Arg Gln Gln Arg Val Ser Thr Thr Leu Ser
 485 490 495
 Gln Asn Asn Asn Ser Asn Phe Ala Trp Thr Gly Ala Thr Lys Tyr His
 500 505 510
 Leu Asn Gly Arg Asp Ser Leu Val Asn Pro Gly Val Ala Met Ala Thr
 515 520 525
 His Lys Asp Asp Glu Glu Arg Phe Phe Pro Ser Ser Gly Val Leu Met
 530 535 540
 Phe Gly Lys Gln Gly Ala Gly Lys Asp Asn Val Asp Tyr Ser Ser Val
 545 550 555 560
 Met Leu Thr Ser Glu Glu Glu Ile Lys Thr Thr Asn Pro Val Ala Thr
 565 570 575
 Glu Gln Tyr Gly Val Val Ala Asp Asn Leu Gln Gln Gln Asn Ala Ala
 580 585 590
 Pro Ile Val Gly Ala Val Asn Ser Gln Gly Ala Leu Pro Gly Met Val
 595 600 605
 Trp Gln Asn Arg Asp Val Tyr Leu Gln Gly Pro Ile Trp Ala Lys Ile
 610 615 620
 Pro His Thr Asp Gly Asn Phe His Pro Ser Pro Leu Met Gly Gly Phe
 625 630 635 640
 Gly Leu Lys His Pro Pro Pro Gln Ile Leu Ile Lys Asn Thr Pro Val
 645 650 655
 Pro Ala Asp Pro Pro Thr Thr Phe Ser Gln Ala Lys Leu Ala Ser Phe
 660 665 670
 Ile Thr Gln Tyr Ser Thr Gly Gln Val Ser Val Glu Ile Glu Trp Glu
 675 680 685
 Leu Gln Lys Glu Asn Ser Lys Arg Trp Asn Pro Glu Ile Gln Tyr Thr
 690 695 700
 Ser Asn Tyr Tyr Lys Ser Thr Asn Val Asp Phe Ala Val Asn Thr Asp
 705 710 715 720
 Gly Thr Tyr Ser Glu Pro Arg Pro Ile Gly Thr Arg Tyr Leu Thr Arg
 725 730 735
 Asn Leu

<210> SEQ ID NO 25
 <211> LENGTH: 33
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: AAV9

<400> SEQUENCE: 25

Gln Arg Val Ser Thr Thr Val Thr Gln Asn Asn Asn Ser Glu Phe Ala
 1 5 10 15
 Trp Pro Gly Ala Ser Ser Trp Ala Leu Asn Gly Arg Asn Ser Leu Met
 20 25 30

-continued

Asn

<210> SEQ ID NO 26
<211> LENGTH: 33
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: AAV8

<400> SEQUENCE: 26

Gln Arg Val Ser Thr Thr Thr Gly Gln Asn Asn Asn Ser Asn Phe Ala
1 5 10 15

Trp Thr Ala Gly Thr Lys Tyr His Leu Asn Gly Arg Asn Ser Leu Ala
20 25 30

Asn

<210> SEQ ID NO 27
<211> LENGTH: 33
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: AAVrh10

<400> SEQUENCE: 27

Gln Arg Val Ser Thr Thr Leu Ser Gln Asn Asn Asn Ser Asn Phe Ala
1 5 10 15

Trp Thr Gly Ala Thr Lys Tyr His Leu Asn Gly Arg Asp Ser Leu Val
20 25 30

Asn

<210> SEQ ID NO 28
<211> LENGTH: 33
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: AAV3B

<400> SEQUENCE: 28

Gln Arg Leu Ser Lys Thr Ala Asn Asp Asn Asn Asn Ser Asn Phe Pro
1 5 10 15

Trp Thr Ala Ala Ser Lys Tyr His Leu Asn Gly Arg Asp Ser Leu Val
20 25 30

Asn

<210> SEQ ID NO 29
<211> LENGTH: 33
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: AAV2

<400> SEQUENCE: 29

Gln Arg Val Ser Lys Thr Ser Ala Asp Asn Asn Asn Ser Glu Tyr Ser
1 5 10 15

Trp Thr Gly Ala Thr Lys Tyr His Leu Asn Gly Arg Asp Ser Leu Val
20 25 30

Asn

<210> SEQ ID NO 30
<211> LENGTH: 33
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:
<223> OTHER INFORMATION: AAV9

<400> SEQUENCE: 30

Tyr	Gly	Gln	Val	Ala	Thr	Asn	His	Gln	Ser	Ala	Gln	Ala	Gln	Ala	Gln
1				5					10				15		
Thr	Gly	Trp	Val	Gln	Asn	Gln	Gly	Ile	Leu	Pro	Gly	Met	Val	Trp	Gln
			20					25					30		

Asp

<210> SEQ ID NO 31
<211> LENGTH: 33
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: AAV8

<400> SEQUENCE: 31

Tyr	Gly	Ile	Val	Ala	Asp	Asn	Leu	Gln	Gln	Gln	Asn	Thr	Ala	Pro	Gln
1				5					10					15	
Ile	Gly	Thr	Val	Asn	Ser	Gln	Gly	Ala	Leu	Pro	Gly	Met	Val	Trp	Gln
			20					25					30		

Asn

<210> SEQ ID NO 32
<211> LENGTH: 33
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: AAVrh10

<400> SEQUENCE: 32

Tyr	Gly	Val	Val	Ala	Asp	Asn	Leu	Gln	Gln	Gln	Asn	Ala	Ala	Pro	Ile
1				5					10					15	
Val	Gly	Ala	Val	Asn	Ser	Gln	Gly	Ala	Leu	Pro	Gly	Met	Val	Trp	Gln
			20					25					30		

Asn

<210> SEQ ID NO 33
<211> LENGTH: 33
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: AAV3B

<400> SEQUENCE: 33

Tyr	Gly	Thr	Val	Ala	Asn	Asn	Leu	Gln	Ser	Ser	Asn	Thr	Ala	Pro	Thr
1				5					10					15	
Thr	Arg	Thr	Val	Asn	Asp	Gln	Gly	Ala	Leu	Pro	Gly	Met	Val	Trp	Gln
			20					25					30		

Asp

<210> SEQ ID NO 34
<211> LENGTH: 33
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: AAV2

<400> SEQUENCE: 34

Tyr	Gly	Ser	Val	Ser	Thr	Asn	Leu	Gln	Arg	Gly	Asn	Arg	Gln	Ala	Ala
1				5					10					15	

-continued

Thr Ala Asp Val Asn Thr Gln Gly Val Leu Pro Gly Met Val Trp Gln
 20 25 30

Asp

<210> SEQ ID NO 35
 <211> LENGTH: 736
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: AAV1

<400> SEQUENCE: 35

Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser
 1 5 10 15
 Glu Gly Ile Arg Glu Trp Trp Asp Leu Lys Pro Gly Ala Pro Lys Pro
 20 25 30
 Lys Ala Asn Gln Gln Lys Gln Asp Asp Gly Arg Gly Leu Val Leu Pro
 35 40 45
 Gly Tyr Lys Tyr Leu Gly Pro Phe Asn Gly Leu Asp Lys Gly Glu Pro
 50 55 60
 Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp
 65 70 75 80
 Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Arg Tyr Asn His Ala
 85 90 95
 Asp Ala Glu Phe Gln Glu Arg Leu Gln Glu Asp Thr Ser Phe Gly Gly
 100 105 110
 Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Val Leu Glu Pro
 115 120 125
 Leu Gly Leu Val Glu Glu Gly Ala Lys Thr Ala Pro Gly Lys Lys Arg
 130 135 140
 Pro Val Glu Gln Ser Pro Gln Glu Pro Asp Ser Ser Ser Gly Ile Gly
 145 150 155 160
 Lys Thr Gly Gln Gln Pro Ala Lys Lys Arg Leu Asn Phe Gly Gln Thr
 165 170 175
 Gly Asp Ser Glu Ser Val Pro Asp Pro Gln Pro Leu Gly Glu Pro Pro
 180 185 190
 Ala Thr Pro Ala Ala Val Gly Pro Thr Thr Met Ala Ser Gly Gly Gly
 195 200 205
 Ala Pro Met Ala Asp Asn Asn Glu Gly Ala Asp Gly Val Gly Asn Ala
 210 215 220
 Ser Gly Asn Trp His Cys Asp Ser Thr Trp Leu Gly Asp Arg Val Ile
 225 230 235 240
 Thr Thr Ser Thr Arg Thr Trp Ala Leu Pro Thr Tyr Asn Asn His Leu
 245 250 255
 Tyr Lys Gln Ile Ser Ser Ala Ser Thr Gly Ala Ser Asn Asp Asn His
 260 265 270
 Tyr Phe Gly Tyr Ser Thr Pro Trp Gly Tyr Phe Asp Phe Asn Arg Phe
 275 280 285
 His Cys His Phe Ser Pro Arg Asp Trp Gln Arg Leu Ile Asn Asn Asn
 290 295 300
 Trp Gly Phe Arg Pro Lys Arg Leu Asn Phe Lys Leu Phe Asn Ile Gln
 305 310 315 320
 Val Lys Glu Val Thr Thr Asn Asp Gly Val Thr Thr Ile Ala Asn Asn
 325 330 335

Leu	Thr	Ser	Thr	Val	Gln	Val	Phe	Ser	Asp	Ser	Glu	Tyr	Gln	Leu	Pro
			340					345					350		
Tyr	Val	Leu	Gly	Ser	Ala	His	Gln	Gly	Cys	Leu	Pro	Pro	Phe	Pro	Ala
	355						360					365			
Asp	Val	Phe	Met	Ile	Pro	Gln	Tyr	Gly	Tyr	Leu	Thr	Leu	Asn	Asn	Gly
	370					375					380				
Ser	Gln	Ala	Val	Gly	Arg	Ser	Ser	Phe	Tyr	Cys	Leu	Glu	Tyr	Phe	Pro
385					390					395					400
Ser	Gln	Met	Leu	Arg	Thr	Gly	Asn	Asn	Phe	Thr	Phe	Ser	Tyr	Thr	Phe
			405						410					415	
Glu	Glu	Val	Pro	Phe	His	Ser	Ser	Tyr	Ala	His	Ser	Gln	Ser	Leu	Asp
		420						425					430		
Arg	Leu	Met	Asn	Pro	Leu	Ile	Asp	Gln	Tyr	Leu	Tyr	Tyr	Leu	Asn	Arg
	435						440					445			
Thr	Gln	Asn	Gln	Ser	Gly	Ser	Ala	Gln	Asn	Lys	Asp	Leu	Leu	Phe	Ser
	450				455						460				
Arg	Gly	Ser	Pro	Ala	Gly	Met	Ser	Val	Gln	Pro	Lys	Asn	Trp	Leu	Pro
465					470					475					480
Gly	Pro	Cys	Tyr	Arg	Gln	Gln	Arg	Val	Ser	Lys	Thr	Lys	Thr	Asp	Asn
			485						490					495	
Asn	Asn	Ser	Asn	Phe	Thr	Trp	Thr	Gly	Ala	Ser	Lys	Tyr	Asn	Leu	Asn
			500					505					510		
Gly	Arg	Glu	Ser	Ile	Ile	Asn	Pro	Gly	Thr	Ala	Met	Ala	Ser	His	Lys
	515						520					525			
Asp	Asp	Glu	Asp	Lys	Phe	Phe	Pro	Met	Ser	Gly	Val	Met	Ile	Phe	Gly
	530					535					540				
Lys	Glu	Ser	Ala	Gly	Ala	Ser	Asn	Thr	Ala	Leu	Asp	Asn	Val	Met	Ile
545					550				555						560
Thr	Asp	Glu	Glu	Glu	Ile	Lys	Ala	Thr	Asn	Pro	Val	Ala	Thr	Glu	Arg
			565						570					575	
Phe	Gly	Thr	Val	Ala	Val	Asn	Phe	Gln	Ser	Ser	Ser	Thr	Asp	Pro	Ala
			580					585					590		
Thr	Gly	Asp	Val	His	Ala	Met	Gly	Ala	Leu	Pro	Gly	Met	Val	Trp	Gln
	595						600					605			
Asp	Arg	Asp	Val	Tyr	Leu	Gln	Gly	Pro	Ile	Trp	Ala	Lys	Ile	Pro	His
	610					615					620				
Thr	Asp	Gly	His	Phe	His	Pro	Ser	Pro	Leu	Met	Gly	Gly	Phe	Gly	Leu
625					630				635						640
Lys	Asn	Pro	Pro	Pro	Gln	Ile	Leu	Ile	Lys	Asn	Thr	Pro	Val	Pro	Ala
			645						650					655	
Asn	Pro	Pro	Ala	Glu	Phe	Ser	Ala	Thr	Lys	Phe	Ala	Ser	Phe	Ile	Thr
			660					665					670		
Gln	Tyr	Ser	Thr	Gly	Gln	Val	Ser	Val	Glu	Ile	Glu	Trp	Glu	Leu	Gln
	675						680					685			
Lys	Glu	Asn	Ser												

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<210> SEQ ID NO 36
<211> LENGTH: 736
<212> TYPE: PRT
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<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: AAV3B

<400> SEQUENCE: 36

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Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser
 1             5             10             15

Glu Gly Ile Arg Glu Trp Trp Ala Leu Lys Pro Gly Val Pro Gln Pro
      20             25             30

Lys Ala Asn Gln Gln His Gln Asp Asn Arg Arg Gly Leu Val Leu Pro
      35             40             45

Gly Tyr Lys Tyr Leu Gly Pro Gly Asn Gly Leu Asp Lys Gly Glu Pro
      50             55             60

Val Asn Glu Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp
 65             70             75             80

Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Lys Tyr Asn His Ala
      85             90             95

Asp Ala Glu Phe Gln Glu Arg Leu Gln Glu Asp Thr Ser Phe Gly Gly
     100             105             110

Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Ile Leu Glu Pro
     115             120             125

Leu Gly Leu Val Glu Glu Ala Ala Lys Thr Ala Pro Gly Lys Lys Arg
     130             135             140

Pro Val Asp Gln Ser Pro Gln Glu Pro Asp Ser Ser Ser Gly Val Gly
    145             150             155             160

Lys Ser Gly Lys Gln Pro Ala Arg Lys Arg Leu Asn Phe Gly Gln Thr
     165             170             175

Gly Asp Ser Glu Ser Val Pro Asp Pro Gln Pro Leu Gly Glu Pro Pro
     180             185             190

Ala Ala Pro Thr Ser Leu Gly Ser Asn Thr Met Ala Ser Gly Gly Gly
     195             200             205

Ala Pro Met Ala Asp Asn Asn Glu Gly Ala Asp Gly Val Gly Asn Ser
     210             215             220

Ser Gly Asn Trp His Cys Asp Ser Gln Trp Leu Gly Asp Arg Val Ile
    225             230             235             240

Thr Thr Ser Thr Arg Thr Trp Ala Leu Pro Thr Tyr Asn Asn His Leu
     245             250             255

Tyr Lys Gln Ile Ser Ser Gln Ser Gly Ala Ser Asn Asp Asn His Tyr
     260             265             270

Phe Gly Tyr Ser Thr Pro Trp Gly Tyr Phe Asp Phe Asn Arg Phe His
     275             280             285

Cys His Phe Ser Pro Arg Asp Trp Gln Arg Leu Ile Asn Asn Asn Trp
     290             295             300

Gly Phe Arg Pro Lys Lys Leu Ser Phe Lys Leu Phe Asn Ile Gln Val
    305             310             315             320

Lys Glu Val Thr Gln Asn Asp Gly Thr Thr Thr Ile Ala Asn Asn Leu
     325             330             335

Thr Ser Thr Val Gln Val Phe Thr Asp Ser Glu Tyr Gln Leu Pro Tyr
     340             345             350

Val Leu Gly Ser Ala His Gln Gly Cys Leu Pro Pro Phe Pro Ala Asp
     355             360             365

Val Phe Met Val Pro Gln Tyr Gly Tyr Leu Thr Leu Asn Asn Gly Ser
     370             375             380

Gln Ala Val Gly Arg Ser Ser Phe Tyr Cys Leu Glu Tyr Phe Pro Ser

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385				390					395					400			
Gln	Met	Leu	Arg	Thr	Gly	Asn	Asn	Phe	Gln	Phe	Ser	Tyr	Thr	Phe	Glu		
				405					410					415			
Asp	Val	Pro	Phe	His	Ser	Ser	Tyr	Ala	His	Ser	Gln	Ser	Leu	Asp	Arg		
				420					425					430			
Leu	Met	Asn	Pro	Leu	Ile	Asp	Gln	Tyr	Leu	Tyr	Tyr	Leu	Asn	Arg	Thr		
				435					440					445			
Gln	Gly	Thr	Thr	Ser	Gly	Thr	Thr	Asn	Gln	Ser	Arg	Leu	Leu	Phe	Ser		
				450					455					460			
Gln	Ala	Gly	Pro	Gln	Ser	Met	Ser	Leu	Gln	Ala	Arg	Asn	Trp	Leu	Pro		
				465					470					475			
Gly	Pro	Cys	Tyr	Arg	Gln	Gln	Arg	Leu	Ser	Lys	Thr	Ala	Asn	Asp	Asn		
				485					490					495			
Asn	Asn	Ser	Asn	Phe	Pro	Trp	Thr	Ala	Ala	Ser	Lys	Tyr	His	Leu	Asn		
				500					505					510			
Gly	Arg	Asp	Ser	Leu	Val	Asn	Pro	Gly	Pro	Ala	Met	Ala	Ser	His	Lys		
				515					520					525			
Asp	Asp	Glu	Glu	Lys	Phe	Phe	Pro	Met	His	Gly	Asn	Leu	Ile	Phe	Gly		
				530					535					540			
Lys	Glu	Gly	Thr	Thr	Ala	Ser	Asn	Ala	Glu	Leu	Asp	Asn	Val	Met	Ile		
				545					550					555			
Thr	Asp	Glu	Glu	Glu	Ile	Arg	Thr	Thr	Asn	Pro	Val	Ala	Thr	Glu	Gln		
				565					570					575			
Tyr	Gly	Thr	Val	Ala	Asn	Asn	Leu	Gln	Ser	Ser	Asn	Thr	Ala	Pro	Thr		
				580					585					590			
Thr	Arg	Thr	Val	Asn	Asp	Gln	Gly	Ala	Leu	Pro	Gly	Met	Val	Trp	Gln		
				595					600					605			
Asp	Arg	Asp	Val	Tyr	Leu	Gln	Gly	Pro	Ile	Trp	Ala	Lys	Ile	Pro	His		
				610					615					620			
Thr	Asp	Gly	His	Phe	His	Pro	Ser	Pro	Leu	Met	Gly	Gly	Phe	Gly	Leu		
				625					630					635			
Lys	His	Pro	Pro	Pro	Gln	Ile	Met	Ile	Lys	Asn	Thr	Pro	Val	Pro	Ala		
				645					650					655			
Asn	Pro	Pro	Thr	Phe	Ser	Pro	Pro	Ala	Lys	Phe	Ala	Ser	Phe	Ile	Thr		
				660					665					670			
Gln	Tyr	Ser	Thr	Gly	Gln	Val	Ser	Val	Glu	Ile	Glu	Trp	Glu	Leu	Gln		
				675					680					685			
Lys	Glu	Asn	Ser	Lys	Arg	Trp	Asn	Pro	Glu	Ile	Gln	Tyr	Thr	Ser	Asn		
				690					695					700			
Tyr	Asn	Lys	Ser	Val	Asn	Val	Asp	Phe	Thr	Val	Asp	Thr	Asn	Gly	Val		
				705					710					715			
Tyr	Ser	Glu	Pro	Arg	Pro	Ile	Gly	Thr	Arg	Tyr	Leu	Thr	Arg	Asn	Leu		
				725					730					735			

Gly Leu Arg Glu Phe Leu Gly Leu Glu Ala Gly Pro Pro Lys Pro Lys

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20					25					30					
Pro	Asn	Gln	Gln	His	Gln	Asp	Gln	Ala	Arg	Gly	Leu	Val	Leu	Pro	Gly
		35					40				45				
Tyr	Asn	Tyr	Leu	Gly	Pro	Gly	Asn	Gly	Leu	Asp	Arg	Gly	Glu	Pro	Val
	50					55				60					
Asn	Arg	Ala	Asp	Glu	Val	Ala	Arg	Glu	His	Asp	Ile	Ser	Tyr	Asn	Glu
65					70					75					80
Gln	Leu	Glu	Ala	Gly	Asp	Asn	Pro	Tyr	Leu	Lys	Tyr	Asn	His	Ala	Asp
			85						90					95	
Ala	Glu	Phe	Gln	Glu	Lys	Leu	Ala	Asp	Asp	Thr	Ser	Phe	Gly	Gly	Asn
			100					105					110		
Leu	Gly	Lys	Ala	Val	Phe	Gln	Ala	Lys	Lys	Arg	Val	Leu	Glu	Pro	Phe
		115					120					125			
Gly	Leu	Val	Glu	Glu	Gly	Ala	Lys	Thr	Ala	Pro	Thr	Gly	Lys	Arg	Ile
	130					135					140				
Asp	Asp	His	Phe	Pro	Lys	Arg	Lys	Lys	Ala	Arg	Thr	Glu	Glu	Asp	Ser
145					150					155				160	
Lys	Pro	Ser	Thr	Ser	Ser	Asp	Ala	Glu	Ala	Gly	Pro	Ser	Gly	Ser	Gln
			165						170					175	
Gln	Leu	Gln	Ile	Pro	Ala	Gln	Pro	Ala	Ser	Ser	Leu	Gly	Ala	Asp	Thr
			180					185					190		
Met	Ser	Ala	Gly	Gly	Gly	Gly	Pro	Leu	Gly	Asp	Asn	Asn	Gln	Gly	Ala
	195						200				205				
Asp	Gly	Val	Gly	Asn	Ala	Ser	Gly	Asp	Trp	His	Cys	Asp	Ser	Thr	Trp
	210					215					220				
Met	Gly	Asp	Arg	Val	Val	Thr	Lys	Ser	Thr	Arg	Thr	Trp	Val	Leu	Pro
225					230					235				240	
Ser	Tyr	Asn	Asn	His	Gln	Tyr	Arg	Glu	Ile	Lys	Ser	Gly	Ser	Val	Asp
				245					250					255	
Gly	Ser	Asn	Ala	Asn	Ala	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro	Trp	Gly	Tyr
			260					265					270		
Phe	Asp	Phe	Asn	Arg	Phe	His	Ser	His	Trp	Ser	Pro	Arg	Asp	Trp	Gln
	275					280						285			
Arg	Leu	Ile	Asn	Asn	Tyr	Trp	Gly	Phe	Arg	Pro	Arg	Ser	Leu	Arg	Val
	290					295					300				
Lys	Ile	Phe	Asn	Ile	Gln	Val	Lys	Glu	Val	Thr	Val	Gln	Asp	Ser	Thr
305					310					315					320
Thr	Thr	Ile	Ala	Asn	Asn	Leu	Thr	Ser	Thr	Val	Gln	Val	Phe	Thr	Asp
			325						330					335	
Asp	Asp	Tyr	Gln	Leu	Pro	Tyr	Val	Val	Gly	Asn	Gly	Thr	Glu	Gly	Cys
		340						345					350		
Leu	Pro	Ala	Phe	Pro	Pro	Gln	Val	Phe	Thr	Leu	Pro	Gln	Tyr	Gly	Tyr
	355					360						365			
Ala	Thr	Leu	Asn	Arg	Asp	Asn	Thr	Glu	Asn	Pro	Thr	Glu	Arg	Ser	Ser
	370					375					380				
Phe	Phe	Cys	Leu	Glu	Tyr	Phe	Pro	Ser	Lys	Met	Leu	Arg	Thr	Gly	Asn
385					390					395					400
Asn	Phe	Glu	Phe	Thr	Tyr	Asn	Phe	Glu	Glu	Val	Pro	Phe	His	Ser	Ser
		405							410					415	
Phe	Ala	Pro	Ser	Gln	Asn	Leu	Phe	Lys	Leu	Ala	Asn	Pro	Leu	Val	Asp
		420						425					430		
Gln	Tyr	Leu	Tyr	Arg	Phe	Val	Ser	Thr	Asn	Asn	Thr	Gly	Gly	Val	Gln
	435						440					445			

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Phe Asn Lys Asn Leu Ala Gly Arg Tyr Ala Asn Thr Tyr Lys Asn Trp
 450 455 460
 Phe Pro Gly Pro Met Gly Arg Thr Gln Gly Trp Asn Leu Gly Ser Gly
 465 470 475 480
 Val Asn Arg Ala Ser Val Ser Ala Phe Ala Thr Thr Asn Arg Met Glu
 485 490 495
 Leu Glu Gly Ala Ser Tyr Gln Val Pro Pro Gln Pro Asn Gly Met Thr
 500 505 510
 Asn Asn Leu Gln Gly Ser Asn Thr Tyr Ala Leu Glu Asn Thr Met Ile
 515 520 525
 Phe Asn Ser Gln Pro Ala Asn Pro Gly Thr Thr Ala Thr Tyr Leu Glu
 530 535 540
 Gly Asn Met Leu Ile Thr Ser Glu Ser Glu Thr Gln Pro Val Asn Arg
 545 550 555 560
 Val Ala Tyr Asn Val Gly Gly Gln Met Ala Thr Asn Asn Gln Ser Ser
 565 570 575
 Thr Thr Ala Pro Ala Thr Gly Thr Tyr Asn Leu Gln Glu Ile Val Pro
 580 585 590
 Gly Ser Val Trp Met Glu Arg Asp Val Tyr Leu Gln Gly Pro Ile Trp
 595 600 605
 Ala Lys Ile Pro Glu Thr Gly Ala His Phe His Pro Ser Pro Ala Met
 610 615 620
 Gly Gly Phe Gly Leu Lys His Pro Pro Pro Met Met Leu Ile Lys Asn
 625 630 635 640
 Thr Pro Val Pro Gly Asn Ile Thr Ser Phe Ser Asp Val Pro Val Ser
 645 650 655
 Ser Phe Ile Thr Gln Tyr Ser Thr Gly Gln Val Thr Val Glu Met Glu
 660 665 670
 Trp Glu Leu Lys Lys Glu Asn Ser Lys Arg Trp Asn Pro Glu Ile Gln
 675 680 685
 Tyr Thr Asn Asn Tyr Asn Asp Pro Gln Phe Val Asp Phe Ala Pro Asp
 690 695 700
 Ser Thr Gly Glu Tyr Arg Thr Thr Arg Pro Ile Gly Thr Arg Tyr Leu
 705 710 715 720
 Thr Arg Pro Leu

<210> SEQ ID NO 38
 <211> LENGTH: 737
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: AAV7

<400> SEQUENCE: 38

Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser
 1 5 10 15
 Glu Gly Ile Arg Glu Trp Trp Asp Leu Lys Pro Gly Ala Pro Lys Pro
 20 25 30
 Lys Ala Asn Gln Gln Lys Gln Asp Asn Gly Arg Gly Leu Val Leu Pro
 35 40 45
 Gly Tyr Lys Tyr Leu Gly Pro Phe Asn Gly Leu Asp Lys Gly Glu Pro
 50 55 60
 Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp
 65 70 75 80

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Gln	Gln	Leu	Lys	Ala	Gly	Asp	Asn	Pro	Tyr	Leu	Arg	Tyr	Asn	His	Ala	
			85						90					95		
Asp	Ala	Glu	Phe	Gln	Glu	Arg	Leu	Gln	Glu	Asp	Thr	Ser	Phe	Gly	Gly	
			100					105					110			
Asn	Leu	Gly	Arg	Ala	Val	Phe	Gln	Ala	Lys	Lys	Arg	Val	Leu	Glu	Pro	
		115					120					125				
Leu	Gly	Leu	Val	Glu	Glu	Gly	Ala	Lys	Thr	Ala	Pro	Ala	Lys	Lys	Arg	
	130					135					140					
Pro	Val	Glu	Pro	Ser	Pro	Gln	Arg	Ser	Pro	Asp	Ser	Ser	Thr	Gly	Ile	
	145				150					155					160	
Gly	Lys	Lys	Gly	Gln	Gln	Pro	Ala	Arg	Lys	Arg	Leu	Asn	Phe	Gly	Gln	
			165						170					175		
Thr	Gly	Asp	Ser	Glu	Ser	Val	Pro	Asp	Pro	Gln	Pro	Leu	Gly	Glu	Pro	
		180						185					190			
Pro	Ala	Ala	Pro	Ser	Ser	Val	Gly	Ser	Gly	Thr	Val	Ala	Ala	Gly	Gly	
		195					200					205				
Gly	Ala	Pro	Met	Ala	Asp	Asn	Asn	Glu	Gly	Ala	Asp	Gly	Val	Gly	Asn	
	210					215					220					
Ala	Ser	Gly	Asn	Trp	His	Cys	Asp	Ser	Thr	Trp	Leu	Gly	Asp	Arg	Val	
	225				230					235					240	
Ile	Thr	Thr	Ser	Thr	Arg	Thr	Trp	Ala	Leu	Pro	Thr	Tyr	Asn	Asn	His	
			245						250					255		
Leu	Tyr	Lys	Gln	Ile	Ser	Ser	Glu	Thr	Ala	Gly	Ser	Thr	Asn	Asp	Asn	
		260						265					270			
Thr	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro	Trp	Gly	Tyr	Phe	Asp	Phe	Asn	Arg	
		275					280					285				
Phe	His	Cys	His	Phe	Ser	Pro	Arg	Asp	Trp	Gln	Arg	Leu	Ile	Asn	Asn	
	290					295					300					
Asn	Trp	Gly	Phe	Arg	Pro	Lys	Lys	Leu	Arg	Phe	Lys	Leu	Phe	Asn	Ile	
	305				310					315					320	
Gln	Val	Lys	Glu	Val	Thr	Thr	Asn	Asp	Gly	Val	Thr	Thr	Ile	Ala	Asn	
			325						330					335		
Asn	Leu	Thr	Ser	Thr	Ile	Gln	Val	Phe	Ser	Asp	Ser	Glu	Tyr	Gln	Leu	
		340						345					350			
Pro	Tyr	Val	Leu	Gly	Ser	Ala	His	Gln	Gly	Cys	Leu	Pro	Pro	Phe	Pro	
		355					360					365				
Ala	Asp	Val	Phe	Met	Ile	Pro	Gln	Tyr	Gly	Tyr	Leu	Thr	Leu	Asn	Asn	
	370					375					380					
Gly	Ser	Gln	Ser	Val	Gly	Arg	Ser	Ser	Phe	Tyr	Cys	Leu	Glu	Tyr	Phe	
	385				390					395					400	
Pro	Ser	Gln	Met	Leu	Arg	Thr	Gly	Asn	Asn	Phe	Glu	Phe	Ser	Tyr	Ser	
			405					410						415		
Phe	Glu	Asp	Val	Pro	Phe	His	Ser	Ser	Tyr	Ala	His	Ser	Gln	Ser	Leu	
		420						425					430			
Asp	Arg	Leu	Met	Asn	Pro	Leu	Ile	Asp	Gln	Tyr	Leu	Tyr	Tyr	Leu	Ala	
		435					440					445				
Arg	Thr	Gln	Ser	Asn	Pro	Gly	Gly	Thr	Ala	Gly	Asn	Arg	Glu	Leu	Gln	
	450					455					460					
Phe	Tyr	Gln	Gly	Gly	Pro	Ser	Thr	Met	Ala	Glu	Gln	Ala	Lys	Asn	Trp	
	465				470					475					480	
Leu	Pro	Gly	Pro	Cys	Phe	Arg	Gln	Gln	Arg	Val	Ser	Lys	Thr	Leu	Asp	
			485					490						495		
Gln	Asn	Asn	Asn	Ser	Asn	Phe	Ala	Trp	Thr	Gly	Ala	Thr	Lys	Tyr	His	

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500					505					510					
Leu	Asn	Gly	Arg	Asn	Ser	Leu	Val	Asn	Pro	Gly	Val	Ala	Met	Ala	Thr
	515						520					525			
His	Lys	Asp	Asp	Glu	Asp	Arg	Phe	Phe	Pro	Ser	Ser	Gly	Val	Leu	Ile
	530					535					540				
Phe	Gly	Lys	Thr	Gly	Ala	Thr	Asn	Lys	Thr	Thr	Leu	Glu	Asn	Val	Leu
545					550					555					560
Met	Thr	Asn	Glu	Glu	Glu	Ile	Arg	Pro	Thr	Asn	Pro	Val	Ala	Thr	Glu
			565						570						575
Glu	Tyr	Gly	Ile	Val	Ser	Ser	Asn	Leu	Gln	Ala	Ala	Asn	Thr	Ala	Ala
		580						585					590		
Gln	Thr	Gln	Val	Val	Asn	Asn	Gln	Gly	Ala	Leu	Pro	Gly	Met	Val	Trp
	595						600					605			
Gln	Asn	Arg	Asp	Val	Tyr	Leu	Gln	Gly	Pro	Ile	Trp	Ala	Lys	Ile	Pro
610						615					620				
His	Thr	Asp	Gly	Asn	Phe	His	Pro	Ser	Pro	Leu	Met	Gly	Gly	Phe	Gly
625					630					635					640
Leu	Lys	His	Pro	Pro	Pro	Gln	Ile	Leu	Ile	Lys	Asn	Thr	Pro	Val	Pro
			645					650							655
Ala	Asn	Pro	Pro	Glu	Val	Phe	Thr	Pro	Ala	Lys	Phe	Ala	Ser	Phe	Ile
		660						665					670		
Thr	Gln	Tyr	Ser	Thr	Gly	Gln	Val	Ser	Val	Glu	Ile	Glu	Trp	Glu	Leu
	675					680						685			
Gln	Lys	Glu	Asn	Ser	Lys	Arg	Trp	Asn	Pro	Glu	Ile	Gln	Tyr	Thr	Ser
690					695					700					
Asn	Phe	Glu	Lys	Gln	Thr	Gly	Val	Asp	Phe	Ala	Val	Asp	Ser	Gln	Gly
705				710					715						720
Val	Tyr	Ser	Glu	Pro	Arg	Pro	Ile	Gly	Thr	Arg	Tyr	Leu	Thr	Arg	Asn
			725					730						735	

Leu

<210> SEQ ID NO 39

<211> LENGTH: 733

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: AAVrh32.33

<400> SEQUENCE: 39

Met	Ala	Ala	Asp	Gly	Tyr	Leu	Pro	Asp	Trp	Leu	Glu	Asp	Asn	Leu	Ser
1			5						10					15	
Glu	Gly	Ile	Arg	Glu	Trp	Trp	Asp	Leu	Lys	Pro	Gly	Ala	Pro	Lys	Pro
	20						25					30			
Lys	Ala	Asn	Gln	Gln	Lys	Gln	Asp	Asp	Gly	Arg	Gly	Leu	Val	Leu	Pro
	35					40						45			
Gly	Tyr	Lys	Tyr	Leu	Gly	Pro	Phe	Asn	Gly	Leu	Asp	Lys	Gly	Glu	Pro
	50				55					60					
Val	Asn	Ala	Ala	Asp	Ala	Ala	Ala	Leu	Glu	His	Asp	Lys	Ala	Tyr	Asp
65				70					75					80	
Gln	Gln	Leu	Lys	Ala	Gly	Asp	Asn	Pro	Tyr	Leu	Arg	Tyr	Asn	His	Ala
		85						90						95	
Asp	Ala	Glu	Phe	Gln	Glu	Arg	Leu	Gln	Glu	Asp	Thr	Ser	Phe	Gly	Gly
		100					105						110		
Asn	Leu	Gly	Arg	Ala	Val	Phe	Gln	Ala	Lys	Lys	Arg	Val	Leu	Glu	Pro
	115						120					125			

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Leu Gly Leu Val Glu Glu Gly Ala Lys Thr Ala Pro Gly Lys Lys Arg
 130 135 140
 Pro Leu Glu Ser Pro Gln Glu Pro Asp Ser Ser Ser Gly Ile Gly Lys
 145 150 155 160
 Lys Gly Lys Gln Pro Ala Lys Lys Arg Leu Asn Phe Glu Glu Asp Thr
 165 170 175
 Gly Ala Gly Asp Gly Pro Pro Glu Gly Ser Asp Thr Ser Ala Met Ser
 180 185 190
 Ser Asp Ile Glu Met Arg Ala Ala Pro Gly Gly Asn Ala Val Asp Ala
 195 200 205
 Gly Gln Gly Ser Asp Gly Val Gly Asn Ala Ser Gly Asp Trp His Cys
 210 215 220
 Asp Ser Thr Trp Ser Glu Gly Lys Val Thr Thr Thr Ser Thr Arg Thr
 225 230 235 240
 Trp Val Leu Pro Thr Tyr Asn Asn His Leu Tyr Leu Arg Leu Gly Thr
 245 250 255
 Thr Ser Asn Ser Asn Thr Tyr Asn Gly Phe Ser Thr Pro Trp Gly Tyr
 260 265 270
 Phe Asp Phe Asn Arg Phe His Cys His Phe Ser Pro Arg Asp Trp Gln
 275 280 285
 Arg Leu Ile Asn Asn Asn Trp Gly Leu Arg Pro Lys Ala Met Arg Val
 290 295 300
 Lys Ile Phe Asn Ile Gln Val Lys Glu Val Thr Thr Ser Asn Gly Glu
 305 310 315 320
 Thr Thr Val Ala Asn Asn Leu Thr Ser Thr Val Gln Ile Phe Ala Asp
 325 330 335
 Ser Ser Tyr Glu Leu Pro Tyr Val Met Asp Ala Gly Gln Glu Gly Ser
 340 345 350
 Leu Pro Pro Phe Pro Asn Asp Val Phe Met Val Pro Gln Tyr Gly Tyr
 355 360 365
 Cys Gly Ile Val Thr Gly Glu Asn Gln Asn Gln Thr Asp Arg Asn Ala
 370 375 380
 Phe Tyr Cys Leu Glu Tyr Phe Pro Ser Gln Met Leu Arg Thr Gly Asn
 385 390 395 400
 Asn Phe Glu Met Ala Tyr Asn Phe Glu Lys Val Pro Phe His Ser Met
 405 410 415
 Tyr Ala His Ser Gln Ser Leu Asp Arg Leu Met Asn Pro Leu Leu Asp
 420 425 430
 Gln Tyr Leu Trp His Leu Gln Ser Thr Thr Ser Gly Glu Thr Leu Asn
 435 440 445
 Gln Gly Asn Ala Ala Thr Thr Phe Gly Lys Ile Arg Ser Gly Asp Phe
 450 455 460
 Ala Phe Tyr Arg Lys Asn Trp Leu Pro Gly Pro Cys Val Lys Gln Gln
 465 470 475 480
 Arg Phe Ser Lys Thr Ala Ser Gln Asn Tyr Lys Ile Pro Ala Ser Gly
 485 490 495
 Gly Asn Ala Leu Leu Lys Tyr Asp Thr His Tyr Thr Leu Asn Asn Arg
 500 505 510
 Trp Ser Asn Ile Ala Pro Gly Pro Pro Met Ala Thr Ala Gly Pro Ser
 515 520 525
 Asp Gly Asp Phe Ser Asn Ala Gln Leu Ile Phe Pro Gly Pro Ser Val
 530 535 540

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115	120	125	
ctc ggt ctg gtt gag gaa ggc gct aag acg gct cct gga aag aag aga Leu Gly Leu Val Glu Glu Gly Ala Lys Thr Ala Pro Gly Lys Lys Arg 130 135 140			432
ccg gta gag cca tca ccc cag cgt tct cca gac tcc tct acg ggc atc Pro Val Glu Pro Ser Pro Gln Arg Ser Pro Asp Ser Ser Thr Gly Ile 145 150 155 160			480
ggc aag aaa ggc caa cag ccc gcc aga aaa aga ctc aat ttt ggt cag Gly Lys Lys Gly Gln Gln Pro Ala Arg Lys Arg Leu Asn Phe Gly Gln 165 170 175			528
act ggc gac tca gag tca gtt cca gac cct caa cct ctc gga gaa cct Thr Gly Asp Ser Glu Ser Val Pro Asp Pro Gln Pro Leu Gly Glu Pro 180 185 190			576
cca gca gcg ccc tct ggt gtg gga cct aat aca atg gct gca ggc ggt Pro Ala Ala Pro Ser Gly Val Gly Pro Asn Thr Met Ala Ala Gly Gly 195 200 205			624
ggc gca cca atg gca gac aat aac gaa ggc gcc gac gga gtg ggt agt Gly Ala Pro Met Ala Asp Asn Asn Glu Gly Ala Asp Gly Val Gly Ser 210 215 220			672
tcc tcg gga aat tgg cat tgc gat tcc aca tgg ctg ggc gac aga gtc Ser Ser Gly Asn Trp His Cys Asp Ser Thr Trp Leu Gly Asp Arg Val 225 230 235 240			720
atc acc acc agc acc cga acc tgg gcc ctg ccc acc tac aac aac cac Ile Thr Thr Ser Thr Arg Thr Trp Ala Leu Pro Thr Tyr Asn Asn His 245 250 255			768
ctc tac aag caa atc tcc aac gcg aca tcg gga gga gcc acc aac gac Leu Tyr Lys Gln Ile Ser Asn Ala Thr Ser Gly Gly Ala Thr Asn Asp 260 265 270			816
aac acc tac ttc ggc tac agc acc ccc tgg ggg tat ttt gac ttt aac Asn Thr Tyr Phe Gly Tyr Ser Thr Pro Trp Gly Tyr Phe Asp Phe Asn 275 280 285			864
aga ttc cac tgc cac ttt tca cca cgt gac tgg cag cga ctc atc aac Arg Phe His Cys His Phe Ser Pro Arg Asp Trp Gln Arg Leu Ile Asn 290 295 300			912
aac aac tgg gga ttc cgg ccc aag aga ctc agc ttc aag ctc ttc aac Asn Asn Trp Gly Phe Arg Pro Lys Arg Leu Ser Phe Lys Leu Phe Asn 305 310 315 320			960
atc cag gtc aag gag gtc acg cag aat gaa ggc acc aag acc atc gcc Ile Gln Val Lys Glu Val Thr Gln Asn Glu Gly Thr Lys Thr Ile Ala 325 330 335			1008
aat aac ctc acc agc acc atc cag gtg ttt acg gac tcg gag tac cag Asn Asn Leu Thr Ser Thr Ile Gln Val Phe Thr Asp Ser Glu Tyr Gln 340 345 350			1056
ctg ccg tac gtt ctc ggc tct gcc cac cag ggc tgc ctg cct ccg ttc Leu Pro Tyr Val Leu Gly Ser Ala His Gln Gly Cys Leu Pro Pro Phe 355 360 365			1104
ccg gcg gac gtg ttc atg att ccc cag tac ggc tac cta aca ctc aac Pro Ala Asp Val Phe Met Ile Pro Gln Tyr Gly Tyr Leu Thr Leu Asn 370 375 380			1152
aac ggt agt cag gcc gtg gga cgc tcc tcc ttc tac tgc ctg gaa tac Asn Gly Ser Gln Ala Val Gly Arg Ser Ser Phe Tyr Cys Leu Glu Tyr 385 390 395 400			1200
ttt cct tcg cag atg ctg aga acc ggc aac aac ttc cag ttt act tac Phe Pro Ser Gln Met Leu Arg Thr Gly Asn Asn Phe Gln Phe Thr Tyr 405 410 415			1248
acc ttc gag gac gtg cct ttc cac agc agc tac gcc cac agc cag agc Thr Phe Glu Asp Val Pro Phe His Ser Ser Tyr Ala His Ser Gln Ser 420 425 430			1296
ttg gac cgg ctg atg aat cct ctg att gac cag tac ctg tac tac ttg			1344

Leu	Asp	Arg	Leu	Met	Asn	Pro	Leu	Ile	Asp	Gln	Tyr	Leu	Tyr	Tyr	Leu	
		435					440					445				
tct	cgg	act	caa	aca	aca	gga	ggc	acg	gca	aat	acg	cag	act	ctg	ggc	1392
Ser	Arg	Thr	Gln	Thr	Thr	Gly	Gly	Thr	Ala	Asn	Thr	Gln	Thr	Leu	Gly	
	450					455					460					
ttc	agc	caa	ggg	ggg	cct	aat	aca	atg	gcc	aat	cag	gca	aag	aac	tgg	1440
Phe	Ser	Gln	Gly	Gly	Pro	Asn	Thr	Met	Ala	Asn	Gln	Ala	Lys	Asn	Trp	
465					470					475					480	
ctg	cca	gga	ccc	tgt	tac	cgc	caa	caa	cgc	gtc	tca	acg	aca	acc	ggg	1488
Leu	Pro	Gly	Pro	Cys	Tyr	Arg	Gln	Gln	Arg	Val	Ser	Thr	Thr	Thr	Gly	
				485					490					495		
caa	aac	aac	aat	agc	aac	ttt	gcc	tgg	act	gct	ggg	acc	aaa	tac	cat	1536
Gln	Asn	Asn	Asn	Ser	Asn	Phe	Ala	Trp	Thr	Ala	Gly	Thr	Lys	Tyr	His	
			500					505					510			
ctg	aat	gca	aga	aat	tca	ttg	gct	aat	cct	ggc	atc	gct	atg	gca	aca	1584
Leu	Asn	Ala	Arg	Asn	Ser	Leu	Ala	Asn	Pro	Gly	Ile	Ala	Met	Ala	Thr	
		515					520				525					
cac	aaa	gac	gac	gag	gag	cgt	ttt	ttt	ccc	agt	aac	ggg	atc	ctg	att	1632
His	Lys	Asp	Asp	Glu	Glu	Arg	Phe	Phe	Pro	Ser	Asn	Gly	Ile	Leu	Ile	
	530					535					540					
ttt	ggc	aaa	caa	aat	gct	gcc	aga	gac	aat	cgc	gat	tac	agc	gat	gtc	1680
Phe	Gly	Lys	Gln	Asn	Ala	Ala	Arg	Asp	Asn	Ala	Asp	Tyr	Ser	Asp	Val	
545					550					555					560	
atg	ctc	acc	agc	gag	gaa	gaa	atc	aaa	acc	act	aac	cct	gtg	gct	aca	1728
Met	Leu	Thr	Ser	Glu	Glu	Glu	Ile	Lys	Thr	Thr	Asn	Pro	Val	Ala	Thr	
				565				570					575			
gag	gaa	tac	ggg	atc	gtg	gca	gat	aac	ttg	cag	cag	caa	aac	acg	gct	1776
Glu	Glu	Tyr	Gly	Ile	Val	Ala	Asp	Asn	Leu	Gln	Gln	Gln	Asn	Thr	Ala	
			580				585						590			
cct	caa	att	gga	act	gtc	aac	agc	cag	ggg	gcc	tta	ccc	ggg	atg	gtc	1824
Pro	Gln	Ile	Gly	Thr	Val	Asn	Ser	Gln	Gly	Ala	Leu	Pro	Gly	Met	Val	
		595					600					605				
tgg	cag	aac	cgg	gac	gtg	tac	ctg	cag	ggg	ccc	atc	tgg	gcc	aag	att	1872
Trp	Gln	Asn	Arg	Asp	Val	Tyr	Leu	Gln	Gly	Pro	Ile	Trp	Ala	Lys	Ile	
	610					615					620					
cct	cac	acg	gac	ggc	aac	ttc	cac	ccg	tct	ccg	ctg	atg	ggc	ggc	ttt	1920
Pro	His	Thr	Asp	Gly	Asn	Phe	His	Pro	Ser	Pro	Leu	Met	Gly	Gly	Phe	
625					630				635						640	
ggc	ctg	aaa	cat	cct	ccg	cct	cag	atc	ctg	atc	aag	aac	acg	cct	gta	1968
Gly	Leu	Lys	His	Pro	Pro	Pro	Gln	Ile								

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<210> SEQ ID NO 41
<211> LENGTH: 738
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 41

Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser
1          5          10          15

Glu Gly Ile Arg Glu Trp Trp Ala Leu Lys Pro Gly Ala Pro Lys Pro
20          25          30

Lys Ala Asn Gln Gln Lys Gln Asp Asp Gly Arg Gly Leu Val Leu Pro
35          40          45

Gly Tyr Lys Tyr Leu Gly Pro Phe Asn Gly Leu Asp Lys Gly Glu Pro
50          55          60

Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp
65          70          75          80

Gln Gln Leu Gln Ala Gly Asp Asn Pro Tyr Leu Arg Tyr Asn His Ala
85          90          95

Asp Ala Glu Phe Gln Glu Arg Leu Gln Glu Asp Thr Ser Phe Gly Gly
100         105         110

Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Val Leu Glu Pro
115         120         125

Leu Gly Leu Val Glu Glu Gly Ala Lys Thr Ala Pro Gly Lys Lys Arg
130         135         140

Pro Val Glu Pro Ser Pro Gln Arg Ser Pro Asp Ser Ser Thr Gly Ile
145         150         155         160

Gly Lys Lys Gly Gln Gln Pro Ala Arg Lys Arg Leu Asn Phe Gly Gln
165         170         175

Thr Gly Asp Ser Glu Ser Val Pro Asp Pro Gln Pro Leu Gly Glu Pro
180         185         190

Pro Ala Ala Pro Ser Gly Val Gly Pro Asn Thr Met Ala Ala Gly Gly
195         200         205

Gly Ala Pro Met Ala Asp Asn Asn Glu Gly Ala Asp Gly Val Gly Ser
210         215         220

Ser Ser Gly Asn Trp His Cys Asp Ser Thr Trp Leu Gly Asp Arg Val
225         230         235         240

Ile Thr Thr Ser Thr Arg Thr Trp Ala Leu Pro Thr Tyr Asn Asn His
245         250         255

Leu Tyr Lys Gln Ile Ser Asn Ala Thr Ser Gly Gly Ala Thr Asn Asp
260         265         270

Asn Thr Tyr Phe Gly Tyr Ser Thr Pro Trp Gly Tyr Phe Asp Phe Asn
275         280         285

Arg Phe His Cys His Phe Ser Pro Arg Asp Trp Gln Arg Leu Ile Asn
290         295         300

Asn Asn Trp Gly Phe Arg Pro Lys Arg Leu Ser Phe Lys Leu Phe Asn
305         310         315         320

Ile Gln Val Lys Glu Val Thr Gln Asn Glu Gly Thr Lys Thr Ile Ala
325         330         335

Asn Asn Leu Thr Ser Thr Ile Gln Val Phe Thr Asp Ser Glu Tyr Gln
340         345         350

Leu Pro Tyr Val Leu Gly Ser Ala His Gln Gly Cys Leu Pro Pro Phe
355         360         365

Pro Ala Asp Val Phe Met Ile Pro Gln Tyr Gly Tyr Leu Thr Leu Asn

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370					375					380				
Asn Gly Ser Gln Ala Val	Gly Arg Ser Ser Phe Tyr Cys Leu Glu Tyr													
385		390			395							400		
Phe Pro Ser Gln Met Leu Arg Thr Gly Asn Asn Phe Gln Phe Thr Tyr														
	405				410							415		
Thr Phe Glu Asp Val Pro Phe His Ser Ser Tyr Ala His Ser Gln Ser														
	420				425							430		
Leu Asp Arg Leu Met Asn Pro Leu Ile Asp Gln Tyr Leu Tyr Tyr Leu														
	435				440							445		
Ser Arg Thr Gln Thr Thr Gly Gly Thr Ala Asn Thr Gln Thr Leu Gly														
	450				455							460		
Phe Ser Gln Gly Gly Pro Asn Thr Met Ala Asn Gln Ala Lys Asn Trp														
	465				470							475		
Leu Pro Gly Pro Cys Tyr Arg Gln Gln Arg Val Ser Thr Thr Thr Gly														
		485										490		
Gln Asn Asn Asn Ser Asn Phe Ala Trp Thr Ala Gly Thr Lys Tyr His														
	500											505		
Leu Asn Ala Arg Asn Ser Leu Ala Asn Pro Gly Ile Ala Met Ala Thr														
	515											520		
His Lys Asp Asp Glu Glu Arg Phe Phe Pro Ser Asn Gly Ile Leu Ile														
	530											535		
Phe Gly Lys Gln Asn Ala Ala Arg Asp Asn Ala Asp Tyr Ser Asp Val														
	545											550		
Met Leu Thr Ser Glu Glu Glu Ile Lys Thr Thr Asn Pro Val Ala Thr														
		565										570		
Glu Glu Tyr Gly Ile Val Ala Asp Asn Leu Gln Gln Gln Asn Thr Ala														
	580											585		
Pro Gln Ile Gly Thr Val Asn Ser Gln Gly Ala Leu Pro Gly Met Val														
	595											600		
Trp Gln Asn Arg Asp Val Tyr Leu Gln Gly Pro Ile Trp Ala Lys Ile														
	610											615		
Pro His Thr Asp Gly Asn Phe His Pro Ser Pro Leu Met Gly Gly Phe														
	625											630		
Gly Leu Lys His Pro Pro Pro Gln Ile Leu Ile Lys Asn Thr Pro Val														
		645										650		
Pro Ala Asp Pro Pro Thr Thr Phe Asn Gln Ser Lys Leu Asn Ser Phe														
	660											665		
Ile Thr Gln Tyr Ser Thr Gly Gln Val Ser Val Glu Ile Glu Trp Glu														
	675											680		
Leu Gln Lys Glu Asn Ser Lys Arg Trp Asn Pro Glu Ile Gln Tyr Thr														
	690											695		
Ser Asn Tyr Tyr Lys Ser Thr Ser Val Asp Phe Ala Val Asn Thr Glu														
	705											710		
Gly Val Tyr Ser Glu Pro Arg Pro Ile Gly Thr Arg Tyr Leu Thr Arg														
		725										730		
													735	

Asn Leu

<210> SEQ ID NO 42
 <211> LENGTH: 2214
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: AVV mutant 8G264AG541A
 <220> FEATURE:
 <221> NAME/KEY: CDS

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<222> LOCATION: (1) .. (2214)

<400> SEQUENCE: 42

atg gct gcc gat ggt tat ctt cca gat tgg ctc gag gac aac ctc tct	48
Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser	
1 5 10 15	
 gag ggc att cgc gag tgg tgg gcg ctg aaa cct gga gcc ccg aag ccc	96
Glu Gly Ile Arg Glu Trp Trp Ala Leu Lys Pro Gly Ala Pro Lys Pro	
20 25 30	
 aaa gcc aac cag caa aag cag gac gac ggc cgg ggt ctg gtg ctt cct	144
Lys Ala Asn Gln Gln Lys Gln Asp Asp Gly Arg Gly Leu Val Leu Pro	
35 40 45	
 ggc tac aag tac ctc gga ccc ttc aac gga ctc gac aag ggg gag ccc	192
Gly Tyr Lys Tyr Leu Gly Pro Phe Asn Gly Leu Asp Lys Gly Glu Pro	
50 55 60	
 gtc aac gcg gcg gac gca gcg gcc ctc gag cac gac aag gcc tac gac	240
Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp	
65 70 75 80	
 cag cag ctg cag gcg ggt gac aat ccg tac ctg cgg tat aac cac gcc	288
Gln Gln Leu Gln Ala Gly Asp Asn Pro Tyr Leu Arg Tyr Asn His Ala	
85 90 95	
 gac gcc gag ttt cag gag cgt ctg caa gaa gat acg tct ttt ggg ggc	336
Asp Ala Glu Phe Gln Glu Arg Leu Gln Glu Asp Thr Ser Phe Gly Gly	
100 105 110	
 aac ctc ggg cga gca gtc ttc cag gcc aag aag cgg gtt ctc gaa cct	384
Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Val Leu Glu Pro	
115 120 125	
 ctc ggt ctg gtt gag gaa ggc gct aag acg gct cct gga aag aag aga	432
Leu Gly Leu Val Glu Glu Gly Ala Lys Thr Ala Pro Gly Lys Lys Arg	
130 135 140	
 ccg gta gag cca tca ccc cag cgt tct cca gac tcc tct acg ggc atc	480
Pro Val Glu Pro Ser Pro Gln Arg Ser Pro Asp Ser Ser Thr Gly Ile	
145 150 155 160	
 ggc aag aaa ggc caa cag ccc gcc aga aaa aga ctc aat ttt ggt cag	528
Gly Lys Lys Gly Gln Gln Pro Ala Arg Lys Arg Leu Asn Phe Gly Gln	
165 170 175	
 act ggc gac tca gag tca gtt cca gac cct caa cct ctc gga gaa cct	576
Thr Gly Asp Ser Glu Ser Val Pro Asp Pro Gln Pro Leu Gly Glu Pro	
180 185 190	
 cca gca gcg ccc tct ggt gtg gga cct aat aca atg gct gca ggc ggt	624
Pro Ala Ala Pro Ser Gly Val Gly Pro Asn Thr Met Ala Ala Gly Gly	
195 200 205	
 ggc gca cca atg gca gac aat aac gaa ggc gcc gac gga gtg ggt agt	672
Gly Ala Pro Met Ala Asp Asn Asn Glu Gly Ala Asp Gly Val Gly Ser	
210 215 220	
 tcc tcg gga aat tgg cat tgc gat tcc aca tgg ctg ggc gac aga gtc	720
Ser Ser Gly Asn Trp His Cys Asp Ser Thr Trp Leu Gly Asp Arg Val	
225 230 235 240	
 atc acc acc agc acc cga acc tgg gcc ctg ccc acc tac aac aac cac	768
Ile Thr Thr Ser Thr Arg Thr Trp Ala Leu Pro Thr Tyr Asn Asn His	
245 250 255	
 ctc tac aag caa atc tcc aac gcg aca tcg gga gga gcc acc aac gac	816
Leu Tyr Lys Gln Ile Ser Asn Ala Thr Ser Gly Gly Ala Thr Asn Asp	
260 265 270	
 aac acc tac ttc ggc tac agc acc ccc tgg ggg tat ttt gac ttt aac	864
Asn Thr Tyr Phe Gly Tyr Ser Thr Pro Trp Gly Tyr Phe Asp Phe Asn	
275 280 285	
 aga ttc cac tgc cac ttt tca cca cgt gac tgg cag cga ctc atc aac	912
Arg Phe His Cys His Phe Ser Pro Arg Asp Trp Gln Arg Leu Ile Asn	
290 295 300	

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aac aac tgg gga ttc cgg ccc aag aga ctc agc ttc aag ctc ttc aac Asn Asn Trp Gly Phe Arg Pro Lys Arg Leu Ser Phe Lys Leu Phe Asn 305 310 315 320	960
atc cag gtc aag gag gtc acg cag aat gaa ggc acc aag acc atc gcc Ile Gln Val Lys Glu Val Thr Gln Asn Glu Gly Thr Lys Thr Ile Ala 325 330 335	1008
aat aac ctc acc agc acc atc cag gtg ttt acg gac tcg gag tac cag Asn Asn Leu Thr Ser Thr Ile Gln Val Phe Thr Asp Ser Glu Tyr Gln 340 345 350	1056
ctg ccg tac gtt ctc ggc tct gcc cac cag ggc tgc ctg cct ccg ttc Leu Pro Tyr Val Leu Gly Ser Ala His Gln Gly Cys Leu Pro Pro Phe 355 360 365	1104
ccg gcg gac gtg ttc atg att ccc cag tac ggc tac cta aca ctc aac Pro Ala Asp Val Phe Met Ile Pro Gln Tyr Gly Tyr Leu Thr Leu Asn 370 375 380	1152
aac ggt agt cag gcc gtg gga cgc tcc tcc ttc tac tgc ctg gaa tac Asn Gly Ser Gln Ala Val Gly Arg Ser Ser Phe Tyr Cys Leu Glu Tyr 385 390 395 400	1200
ttt cct tcg cag atg ctg aga acc ggc aac aac ttc cag ttt act tac Phe Pro Ser Gln Met Leu Arg Thr Gly Asn Asn Phe Gln Phe Thr Tyr 405 410 415	1248
acc ttc gag gac gtg cct ttc cac agc agc tac gcc cac agc cag agc Thr Phe Glu Asp Val Pro Phe His Ser Ser Tyr Ala His Ser Gln Ser 420 425 430	1296
ttg gac ccg ctg atg aat cct ctg att gac cag tac ctg tac tac ttg Leu Asp Arg Leu Met Asn Pro Leu Ile Asp Gln Tyr Leu Tyr Tyr Leu 435 440 445	1344
tct ccg act caa aca aca gga ggc acg gca aat acg cag act ctg ggc Ser Arg Thr Gln Thr Thr Gly Gly Thr Ala Asn Thr Gln Thr Leu Gly 450 455 460	1392
ttc agc caa ggt ggg cct aat aca atg gcc aat cag gca aag aac tgg Phe Ser Gln Gly Gly Pro Asn Thr Met Ala Asn Gln Ala Lys Asn Trp 465 470 475 480	1440
ctg cca gga ccc tgt tac cgc caa caa cgc gtc tca acg aca acc ggg Leu Pro Gly Pro Cys Tyr Arg Gln Gln Arg Val Ser Thr Thr Thr Gly 485 490 495	1488
caa aac aac aat agc aac ttt gcc tgg act gct ggg acc aaa tac cat Gln Asn Asn Asn Ser Asn Phe Ala Trp Thr Ala Gly Thr Lys Tyr His 500 505 510	1536
ctg aat gga aga aat tca ttg gct aat cct ggc atc gct atg gca aca Leu Asn Gly Arg Asn Ser Leu Ala Asn Pro Gly Ile Ala Met Ala Thr 515 520 525	1584
cac aaa gac gac gag gag cgt ttt ttt ccc agt aac gcg atc ctg att His Lys Asp Asp Glu Glu Arg Phe Phe Pro Ser Asn Ala Ile Leu Ile 530 535 540	1632
ttt ggc aaa caa aat gct gcc aga gac aat gcg gat tac agc gat gtc Phe Gly Lys Gln Asn Ala Ala Arg Asp Asn Ala Asp Tyr Ser Asp Val 545 550 555 560	1680
atg ctc acc agc gag gaa gaa atc aaa acc act aac cct gtg gct aca Met Leu Thr Ser Glu Glu Glu Ile Lys Thr Thr Asn Pro Val Ala Thr 565 570 575	1728
gag gaa tac ggt atc gtg gca gat aac ttg cag cag caa aac acg gct Glu Glu Tyr Gly Ile Val Ala Asp Asn Leu Gln Gln Gln Asn Thr Ala 580 585 590	1776
cct caa att gga act gtc aac agc cag ggg gcc tta ccc ggt atg gtc Pro Gln Ile Gly Thr Val Asn Ser Gln Gly Ala Leu Pro Gly Met Val 595 600 605	1824
tgg cag aac ccg gac gtg tac ctg cag ggt ccc atc tgg gcc aag att Trp Gln Asn Arg Asp Val Tyr Leu Gln Gly Pro Ile Trp Ala Lys Ile	1872

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610	615	620	
cct cac acg gac ggc aac ttc cac ccg tct ccg ctg atg ggc ggc ttt Pro His Thr Asp Gly Asn Phe His Pro Ser Pro Leu Met Gly Gly Phe 625 630 635 640			1920
ggc ctg aaa cat cct ccg cct cag atc ctg atc aag aac acg cct gta Gly Leu Lys His Pro Pro Pro Gln Ile Leu Ile Lys Asn Thr Pro Val 645 650 655			1968
cct gcg gat cct ccg acc acc ttc aac cag tca aag ctg aac tct ttc Pro Ala Asp Pro Pro Thr Thr Phe Asn Gln Ser Lys Leu Asn Ser Phe 660 665 670			2016
atc acg caa tac agc acc gga cag gtc agc gtg gaa att gaa tgg gag Ile Thr Gln Tyr Ser Thr Gly Gln Val Ser Val Glu Ile Glu Trp Glu 675 680 685			2064
ctg cag aag gaa aac agc aag cgc tgg aac ccc gag atc cag tac acc Leu Gln Lys Glu Asn Ser Lys Arg Trp Asn Pro Glu Ile Gln Tyr Thr 690 695 700			2112
tcc aac tac tac aaa tct aca agt gtg gac ttt gct gtt aat aca gaa Ser Asn Tyr Tyr Lys Ser Thr Ser Val Asp Phe Ala Val Asn Thr Glu 705 710 715 720			2160
ggc gtg tac tct gaa ccc cgc ccc att ggc acc cgt tac ctc acc cgt Gly Val Tyr Ser Glu Pro Arg Pro Ile Gly Thr Arg Tyr Leu Thr Arg 725 730 735			2208
aat ctg Asn Leu			2214

<210> SEQ ID NO 43

<211> LENGTH: 738

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 43

Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser
1 5 10 15

Glu Gly Ile Arg Glu Trp Trp Ala Leu Lys Pro Gly Ala Pro Lys Pro
20 25 30

Lys Ala Asn Gln Gln Lys Gln Asp Asp Gly Arg Gly Leu Val Leu Pro
35 40 45

Gly Tyr Lys Tyr Leu Gly Pro Phe Asn Gly Leu Asp Lys Gly Glu Pro
50 55 60

Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp
65 70 75 80

Gln Gln Leu Gln Ala Gly Asp Asn Pro Tyr Leu Arg Tyr Asn His Ala
85 90 95

Asp Ala Glu Phe Gln Glu Arg Leu Gln Glu Asp Thr Ser Phe Gly Gly
100 105 110

Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Val Leu Glu Pro
115 120 125

Leu Gly Leu Val Glu Glu Gly Ala Lys Thr Ala Pro Gly Lys Lys Arg
130 135 140

Pro Val Glu Pro Ser Pro Gln Arg Ser Pro Asp Ser Ser Thr Gly Ile
145 150 155 160

Gly Lys Lys Gly Gln Gln Pro Ala Arg Lys Arg Leu Asn Phe Gly Gln
165 170 175

Thr Gly Asp Ser Glu Ser Val Pro Asp Pro Gln Pro Leu Gly Glu Pro
180 185 190

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Pro	Ala	Ala	Pro	Ser	Gly	Val	Gly	Pro	Asn	Thr	Met	Ala	Ala	Gly	Gly
	195						200					205			
Gly	Ala	Pro	Met	Ala	Asp	Asn	Asn	Glu	Gly	Ala	Asp	Gly	Val	Gly	Ser
	210					215					220				
Ser	Ser	Gly	Asn	Trp	His	Cys	Asp	Ser	Thr	Trp	Leu	Gly	Asp	Arg	Val
225					230					235					240
Ile	Thr	Thr	Ser	Thr	Arg	Thr	Trp	Ala	Leu	Pro	Thr	Tyr	Asn	Asn	His
				245					250					255	
Leu	Tyr	Lys	Gln	Ile	Ser	Asn	Ala	Thr	Ser	Gly	Gly	Ala	Thr	Asn	Asp
			260					265					270		
Asn	Thr	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro	Trp	Gly	Tyr	Phe	Asp	Phe	Asn
	275						280					285			
Arg	Phe	His	Cys	His	Phe	Ser	Pro	Arg	Asp	Trp	Gln	Arg	Leu	Ile	Asn
	290					295					300				
Asn	Asn	Trp	Gly	Phe	Arg	Pro	Lys	Arg	Leu	Ser	Phe	Lys	Leu	Phe	Asn
305					310					315					320
Ile	Gln	Val	Lys	Glu	Val	Thr	Gln	Asn	Glu	Gly	Thr	Lys	Thr	Ile	Ala
			325						330					335	
Asn	Asn	Leu	Thr	Ser	Thr	Ile	Gln	Val	Phe	Thr	Asp	Ser	Glu	Tyr	Gln
		340						345					350		
Leu	Pro	Tyr	Val	Leu	Gly	Ser	Ala	His	Gln	Gly	Cys	Leu	Pro	Pro	Phe
	355						360					365			
Pro	Ala	Asp	Val	Phe	Met	Ile	Pro	Gln	Tyr	Gly	Tyr	Leu	Thr	Leu	Asn
	370					375					380				
Asn	Gly	Ser	Gln	Ala	Val	Gly	Arg	Ser	Ser	Phe	Tyr	Cys	Leu	Glu	Tyr
385					390					395					400
Phe	Pro	Ser	Gln	Met	Leu	Arg	Thr	Gly	Asn	Asn	Phe	Gln	Phe	Thr	Tyr
			405						410					415	
Thr	Phe	Glu	Asp	Val	Pro	Phe	His	Ser	Ser	Tyr	Ala	His	Ser	Gln	Ser
		420						425					430		
Leu	Asp	Arg	Leu	Met	Asn	Pro	Leu	Ile	Asp	Gln	Tyr	Leu	Tyr	Tyr	Leu
	435						440					445			
Ser	Arg	Thr	Gln	Thr	Thr	Gly	Gly	Thr	Ala	Asn	Thr	Gln	Thr	Leu	Gly
	450					455					460				
Phe	Ser	Gln	Gly	Gly	Pro	Asn	Thr	Met	Ala	Asn	Gln	Ala	Lys	Asn	Trp
465					470					475					480
Leu	Pro	Gly	Pro	Cys	Tyr	Arg	Gln	Gln	Arg	Val	Ser	Thr	Thr	Thr	Gly
			485						490					495	
Gln	Asn	Asn	Asn	Ser	Asn	Phe	Ala	Trp	Thr	Ala	Gly	Thr	Lys	Tyr	His
			500					505					510		
Leu	Asn	Gly	Arg	Asn	Ser	Leu	Ala	Asn	Pro	Gly	Ile	Ala	Met	Ala	Thr
	515						520					525			
His	Lys	Asp	Asp	Glu	Glu	Arg	Phe	Phe	Pro	Ser	Asn	Ala	Ile	Leu	Ile
	530					535					540				
Phe	Gly	Lys	Gln	Asn	Ala	Ala	Arg	Asp	Asn	Ala	Asp	Tyr	Ser	Asp	Val
545					550					555					560
Met	Leu	Thr	Ser	Glu	Glu	Glu	Ile	Lys	Thr	Thr	Asn	Pro	Val	Ala	Thr
			565						570					575	
Glu	Glu	Tyr	Gly	Ile	Val	Ala	Asp	Asn	Leu	Gln	Gln	Gln	Asn	Thr	Ala
			580					585					590		
Pro	Gln	Ile	Gly	Thr	Val	Asn	Ser	Gln	Gly	Ala	Leu	Pro	Gly	Met	Val
	595					600						605			
Trp	Gln	Asn	Arg	Asp	Val	Tyr	Leu	Gln	Gly	Pro	Ile	Trp	Ala	Lys	Ile

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610	615	620	
Pro His Thr Asp Gly Asn Phe His Pro Ser Pro Leu Met Gly Gly Phe			
625	630	635	640
Gly Leu Lys His Pro Pro Pro Gln Ile Leu Ile Lys Asn Thr Pro Val			
645	650	655	
Pro Ala Asp Pro Pro Thr Thr Phe Asn Gln Ser Lys Leu Asn Ser Phe			
660	665	670	
Ile Thr Gln Tyr Ser Thr Gly Gln Val Ser Val Glu Ile Glu Trp Glu			
675	680	685	
Leu Gln Lys Glu Asn Ser Lys Arg Trp Asn Pro Glu Ile Gln Tyr Thr			
690	695	700	
Ser Asn Tyr Tyr Lys Ser Thr Ser Val Asp Phe Ala Val Asn Thr Glu			
705	710	715	720
Gly Val Tyr Ser Glu Pro Arg Pro Ile Gly Thr Arg Tyr Leu Thr Arg			
725	730	735	
Asn Leu			
<210> SEQ ID NO 44			
<211> LENGTH: 2214			
<212> TYPE: DNA			
<213> ORGANISM: Artificial Sequence			
<220> FEATURE:			
<223> OTHER INFORMATION: AVV mutant 8G515AG541A			
<220> FEATURE:			
<221> NAME/KEY: CDS			
<222> LOCATION: (1)..(2214)			
<400> SEQUENCE: 44			
atg gct gcc gat ggt tat ctt cca gat tgg ctc gag gac aac ctc tct			48
Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser			
1	5	10	15
gag ggc att cgc gag tgg tgg gcg ctg aaa cct gga gcc ccg aag ccc			96
Glu Gly Ile Arg Glu Trp Trp Ala Leu Lys Pro Gly Ala Pro Lys Pro			
20	25	30	
aaa gcc aac cag caa aag cag gac gac ggc cgg ggt ctg gtg ctt cct			144
Lys Ala Asn Gln Gln Lys Gln Asp Asp Gly Arg Gly Leu Val Leu Pro			
35	40	45	
ggc tac aag tac ctc gga ccc ttc aac gga ctc gac aag ggg gag ccc			192
Gly Tyr Lys Tyr Leu Gly Pro Phe Asn Gly Leu Asp Lys Gly Glu Pro			
50	55	60	
gtc aac gcg gcg gac gca gcg gcc ctc gag cac gac aag gcc tac gac			240
Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp			
65	70	75	80
cag cag ctg cag gcg ggt gac aat ccg tac ctg cgg tat aac cac gcc			288
Gln Gln Leu Gln Ala Gly Asp Asn Pro Tyr Leu Arg Tyr Asn His Ala			
85	90	95	
gac gcc gag ttt cag gag cgt ctg caa gaa gat acg tct ttt ggg ggc			336
Asp Ala Glu Phe Gln Glu Arg Leu Gln Glu Asp Thr Ser Phe Gly Gly			
100	105	110	
aac ctc ggg cga gca gtc ttc cag gcc aag aag cgg gtt ctc gaa cct			384
Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Val Leu Glu Pro			
115	120	125	
ctc ggt ctg gtt gag gaa ggc gct aag acg gct cct gga aag aag aga			432
Leu Gly Leu Val Glu Glu Gly Ala Lys Thr Ala Pro Gly Lys Lys Arg			
130	135	140	
ccg gta gag cca tca ccc cag cgt tct cca gac tcc tct acg ggc atc			480
Pro Val Glu Pro Ser Pro Gln Arg Ser Pro Asp Ser Ser Thr Gly Ile			
145	150	155	160
ggc aag aaa ggc caa cag ccc gcc aga aaa aga ctc aat ttt ggt cag			528

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Gly	Lys	Lys	Gly	Gln	Gln	Pro	Ala	Arg	Lys	Arg	Leu	Asn	Phe	Gly	Gln	
			165						170					175		
act	ggc	gac	tca	gag	tca	ggt	cca	gac	cct	caa	cct	ctc	gga	gaa	cct	576
Thr	Gly	Asp	Ser	Glu	Ser	Val	Pro	Asp	Pro	Gln	Pro	Leu	Gly	Glu	Pro	
			180					185					190			
cca	gca	gcg	ccc	tct	ggt	gtg	gga	cct	aat	aca	atg	gct	gca	ggc	ggt	624
Pro	Ala	Ala	Pro	Ser	Gly	Val	Gly	Pro	Asn	Thr	Met	Ala	Ala	Gly	Gly	
			195				200					205				
ggc	gca	cca	atg	gca	gac	aat	aac	gaa	ggc	gcc	gac	gga	gtg	ggt	agt	672
Gly	Ala	Pro	Met	Ala	Asp	Asn	Asn	Glu	Gly	Ala	Asp	Gly	Val	Gly	Ser	
			210			215					220					
tcc	tcg	gga	aat	tgg	cat	tgc	gat	tcc	aca	tgg	ctg	ggc	gac	aga	gtc	720
Ser	Ser	Gly	Asn	Trp	His	Cys	Asp	Ser	Thr	Trp	Leu	Gly	Asp	Arg	Val	
					230					235					240	
atc	acc	acc	agc	acc	cga	acc	tgg	gcc	ctg	ccc	acc	tac	aac	aac	cac	768
Ile	Thr	Thr	Ser	Thr	Arg	Thr	Trp	Ala	Leu	Pro	Thr	Tyr	Asn	Asn	His	
				245					250					255		
ctc	tac	aag	caa	atc	tcc	aac	ggg	aca	tcg	gga	gga	gcc	acc	aac	gac	816
Leu	Tyr	Lys	Gln	Ile	Ser	Asn	Gly	Thr	Ser	Gly	Gly	Ala	Thr	Asn	Asp	
			260				265					270				
aac	acc	tac	ttc	ggc	tac	agc	acc	ccc	tgg	ggg	tat	ttt	gac	ttt	aac	864
Asn	Thr	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro	Trp	Gly	Tyr	Phe	Asp	Phe	Asn	
			275				280					285				
aga	ttc	cac	tgc	cac	ttt	tca	cca	cgt	gac	tgg	cag	cga	ctc	atc	aac	912
Arg	Phe	His	Cys	His	Phe	Ser	Pro	Arg	Asp	Trp	Gln	Arg	Leu	Ile	Asn	
			290			295					300					
aac	aac	tgg	gga	ttc	cgg	ccc	aag	aga	ctc	agc	ttc	aag	ctc	ttc	aac	960
Asn	Asn	Trp	Gly	Phe	Arg	Pro	Lys	Arg	Leu	Ser	Phe	Lys	Leu	Phe	Asn	
			305		310					315				320		
atc	cag	gtc	aag	gag	gtc	acg	cag	aat	gaa	ggc	acc	aag	acc	atc	gcc	1008
Ile	Gln	Val	Lys	Glu	Val	Thr	Gln	Asn	Glu	Gly	Thr	Lys	Thr	Ile	Ala	
				325					330					335		
aat	aac	ctc	acc	agc	acc	atc	cag	gtg	ttt	acg	gac	tcg	gag	tac	cag	1056
Asn	Asn	Leu	Thr	Ser	Thr	Ile	Gln	Val	Phe	Thr	Asp	Ser	Glu	Tyr	Gln	
			340				345						350			
ctg	ccg	tac	gtt	ctc	ggc	tct	gcc	cac	cag	ggc	tgc	ctg	cct	ccg	ttc	1104
Leu	Pro	Tyr	Val	Leu	Gly	Ser	Ala	His	Gln	Gly	Cys	Leu	Pro	Pro	Phe	
			355				360					365				
ccg	gcg	gac	gtg	ttc	atg	att	ccc	cag	tac	ggc	tac	cta	aca	ctc	aac	1152
Pro	Ala	Asp	Val	Phe	Met	Ile	Pro	Gln	Tyr	Gly	Tyr	Leu	Thr	Leu	Asn	
			370			375					380					
aac	ggt	agt	cag	gcc	gtg	gga	cgc	tcc	tcc	ttc	tac	tgc	ctg	gaa	tac	1200
Asn	Gly	Ser	Gln	Ala	Val	Gly	Arg	Ser	Ser	Phe	Tyr	Cys	Leu	Glu	Tyr	
					390					395				400		
ttt	cct	tcg	cag	atg	ctg	aga	acc	ggc	aac	aac	ttc	cag	ttt	act	tac	1248
Phe	Pro	Ser	Gln	Met	Leu	Arg	Thr	Gly	Asn	Asn	Phe	Gln	Phe	Thr	Tyr	
				405					410					415		
acc	ttc	gag	gac	gtg	cct	ttc	cac	agc	agc	tac	gcc	cac	agc	cag	agc	1296
Thr	Phe	Glu	Asp	Val	Pro	Phe	His	Ser	Ser	Tyr	Ala	His	Ser	Gln	Ser	
				420				425					430			
ttg	gac	cgg	ctg	atg	aat	cct	ctg	att	gac	cag	tac	ctg	tac	tac	ttg	1344
Leu	Asp	Arg	Leu	Met	Asn	Pro	Leu	Ile	Asp	Gln	Tyr	Leu	Tyr	Tyr	Leu	
				435			440					445				
tct	cgg	act	caa	aca	aca	gga	ggc	acg	gca	aat	acg	cag	act	ctg	ggc	1392
Ser	Arg	Thr	Gln	Thr	Thr	Gly	Gly	Thr	Ala	Asn	Thr	Gln	Thr	Leu	Gly	
				450		455					460					
ttc	agc	caa	ggt	ggg	cct	aat	aca	atg	gcc	aat	cag	gca	aag	aac	tgg	1440
Phe	Ser	Gln	Gly	Gly	Pro	Asn	Thr	Met	Ala	Asn	Gln	Ala	Lys	Asn	Trp	
					470					475					480	

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ctg cca gga ccc tgt tac cgc caa caa cgc gtc tca acg aca acc ggg	1488
Leu Pro Gly Pro Cys Tyr Arg Gln Gln Arg Val Ser Thr Thr Thr Gly	
485 490 495	
caa aac aac aat agc aac ttt gcc tgg act gct ggg acc aaa tac cat	1536
Gln Asn Asn Asn Ser Asn Phe Ala Trp Thr Ala Gly Thr Lys Tyr His	
500 505 510	
ctg aat gca aga aat tca ttg gct aat cct ggc atc gct atg gca aca	1584
Leu Asn Ala Arg Asn Ser Leu Ala Asn Pro Gly Ile Ala Met Ala Thr	
515 520 525	
cac aaa gac gac gag gag cgt ttt ttt ccc agt aac gcg atc ctg att	1632
His Lys Asp Asp Glu Glu Arg Phe Phe Pro Ser Asn Ala Ile Leu Ile	
530 535 540	
ttt ggc aaa caa aat gct gcc aga gac aat gcg gat tac agc gat gtc	1680
Phe Gly Lys Gln Asn Ala Ala Arg Asp Asn Ala Asp Tyr Ser Asp Val	
545 550 555 560	
atg ctc acc agc gag gaa gaa atc aaa acc act aac cct gtg gct aca	1728
Met Leu Thr Ser Glu Glu Glu Ile Lys Thr Thr Asn Pro Val Ala Thr	
565 570 575	
gag gaa tac ggt atc gtg gca gat aac ttg cag cag caa aac acg gct	1776
Glu Glu Tyr Gly Ile Val Ala Asp Asn Leu Gln Gln Gln Asn Thr Ala	
580 585 590	
cct caa att gga act gtc aac agc cag ggg gcc tta ccc ggt atg gtc	1824
Pro Gln Ile Gly Thr Val Asn Ser Gln Gly Ala Leu Pro Gly Met Val	
595 600 605	
tgg cag aac cgg gac gtg tac ctg cag ggt ccc atc tgg gcc aag att	1872
Trp Gln Asn Arg Asp Val Tyr Leu Gln Gly Pro Ile Trp Ala Lys Ile	
610 615 620	
cct cac acg gac ggc aac ttc cac ccg tct ccg ctg atg ggc ggc ttt	1920
Pro His Thr Asp Gly Asn Phe His Pro Ser Pro Leu Met Gly Gly Phe	
625 630 635 640	
ggc ctg aaa cat cct ccg cct cag atc ctg atc aag aac acg cct gta	1968
Gly Leu Lys His Pro Pro Pro Gln Ile Leu Ile Lys Asn Thr Pro Val	
645 650 655	
cct gcg gat cct ccg acc acc ttc aac cag tca aag ctg aac tct ttc	2016
Pro Ala Asp Pro Pro Thr Thr Phe Asn Gln Ser Lys Leu Asn Ser Phe	
660 665 670	
atc acg caa tac agc acc gga cag gtc agc gtg gaa att gaa tgg gag	2064
Ile Thr Gln Tyr Ser Thr Gly Gln Val Ser Val Glu Ile Glu Trp Glu	
675 680 685	
ctg cag aag gaa aac agc aag cgc tgg aac ccc gag atc cag tac acc	2112
Leu Gln Lys Glu Asn Ser Lys Arg Trp Asn Pro Glu Ile Gln Tyr Thr	
690 695 700	
tcc aac tac tac aaa tct aca agt gtg gac ttt gct gtt aat aca gaa	2160
Ser Asn Tyr Tyr Lys Ser Thr Ser Val Asp Phe Ala Val Asn Thr Glu	
705 710 715 720	
ggc gtg tac tct gaa ccc cgc ccc att ggc acc cgt tac ctc acc cgt	2208
Gly Val Tyr Ser Glu Pro Arg Pro Ile Gly Thr Arg Tyr Leu Thr Arg	
725 730 735	
aat ctg	2214
Asn Leu	

<210> SEQ ID NO 45

<211> LENGTH: 738

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 45

Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser
1 5 10 15

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Glu Gly Ile Arg Glu Trp Trp Ala Leu Lys Pro Gly Ala Pro Lys Pro
 20 25 30
 Lys Ala Asn Gln Gln Lys Gln Asp Asp Gly Arg Gly Leu Val Leu Pro
 35 40 45
 Gly Tyr Lys Tyr Leu Gly Pro Phe Asn Gly Leu Asp Lys Gly Glu Pro
 50 55 60
 Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp
 65 70 75 80
 Gln Gln Leu Gln Ala Gly Asp Asn Pro Tyr Leu Arg Tyr Asn His Ala
 85 90 95
 Asp Ala Glu Phe Gln Glu Arg Leu Gln Glu Asp Thr Ser Phe Gly Gly
 100 105 110
 Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Val Leu Glu Pro
 115 120 125
 Leu Gly Leu Val Glu Glu Gly Ala Lys Thr Ala Pro Gly Lys Lys Arg
 130 135 140
 Pro Val Glu Pro Ser Pro Gln Arg Ser Pro Asp Ser Ser Thr Gly Ile
 145 150 155 160
 Gly Lys Lys Gly Gln Gln Pro Ala Arg Lys Arg Leu Asn Phe Gly Gln
 165 170 175
 Thr Gly Asp Ser Glu Ser Val Pro Asp Pro Gln Pro Leu Gly Glu Pro
 180 185 190
 Pro Ala Ala Pro Ser Gly Val Gly Pro Asn Thr Met Ala Ala Gly Gly
 195 200 205
 Gly Ala Pro Met Ala Asp Asn Asn Glu Gly Ala Asp Gly Val Gly Ser
 210 215 220
 Ser Ser Gly Asn Trp His Cys Asp Ser Thr Trp Leu Gly Asp Arg Val
 225 230 235 240
 Ile Thr Thr Ser Thr Arg Thr Trp Ala Leu Pro Thr Tyr Asn Asn His
 245 250 255
 Leu Tyr Lys Gln Ile Ser Asn Gly Thr Ser Gly Gly Ala Thr Asn Asp
 260 265 270
 Asn Thr Tyr Phe Gly Tyr Ser Thr Pro Trp Gly Tyr Phe Asp Phe Asn
 275 280 285
 Arg Phe His Cys His Phe Ser Pro Arg Asp Trp Gln Arg Leu Ile Asn
 290 295 300
 Asn Asn Trp Gly Phe Arg Pro Lys Arg Leu Ser Phe Lys Leu Phe Asn
 305 310 315 320
 Ile Gln Val Lys Glu Val Thr Gln Asn Glu Gly Thr Lys Thr Ile Ala
 325 330 335
 Asn Asn Leu Thr Ser Thr Ile Gln Val Phe Thr Asp Ser Glu Tyr Gln
 340 345 350
 Leu Pro Tyr Val Leu Gly Ser Ala His Gln Gly Cys Leu Pro Pro Phe
 355 360 365
 Pro Ala Asp Val Phe Met Ile Pro Gln Tyr Gly Tyr Leu Thr Leu Asn
 370 375 380
 Asn Gly Ser Gln Ala Val Gly Arg Ser Ser Phe Tyr Cys Leu Glu Tyr
 385 390 395 400
 Phe Pro Ser Gln Met Leu Arg Thr Gly Asn Asn Phe Gln Phe Thr Tyr
 405 410 415
 Thr Phe Glu Asp Val Pro Phe His Ser Ser Tyr Ala His Ser Gln Ser
 420 425 430

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Leu Asp Arg Leu Met Asn Pro Leu Ile Asp Gln Tyr Leu Tyr Tyr Leu
 435 440 445
 Ser Arg Thr Gln Thr Thr Gly Gly Thr Ala Asn Thr Gln Thr Leu Gly
 450 455 460
 Phe Ser Gln Gly Gly Pro Asn Thr Met Ala Asn Gln Ala Lys Asn Trp
 465 470 475 480
 Leu Pro Gly Pro Cys Tyr Arg Gln Gln Arg Val Ser Thr Thr Thr Gly
 485 490 495
 Gln Asn Asn Asn Ser Asn Phe Ala Trp Thr Ala Gly Thr Lys Tyr His
 500 505 510
 Leu Asn Ala Arg Asn Ser Leu Ala Asn Pro Gly Ile Ala Met Ala Thr
 515 520 525
 His Lys Asp Asp Glu Glu Arg Phe Phe Pro Ser Asn Ala Ile Leu Ile
 530 535 540
 Phe Gly Lys Gln Asn Ala Ala Arg Asp Asn Ala Asp Tyr Ser Asp Val
 545 550 555 560
 Met Leu Thr Ser Glu Glu Glu Ile Lys Thr Thr Asn Pro Val Ala Thr
 565 570 575
 Glu Glu Tyr Gly Ile Val Ala Asp Asn Leu Gln Gln Gln Asn Thr Ala
 580 585 590
 Pro Gln Ile Gly Thr Val Asn Ser Gln Gly Ala Leu Pro Gly Met Val
 595 600 605
 Trp Gln Asn Arg Asp Val Tyr Leu Gln Gly Pro Ile Trp Ala Lys Ile
 610 615 620
 Pro His Thr Asp Gly Asn Phe His Pro Ser Pro Leu Met Gly Gly Phe
 625 630 635 640
 Gly Leu Lys His Pro Pro Pro Gln Ile Leu Ile Lys Asn Thr Pro Val
 645 650 655
 Pro Ala Asp Pro Pro Thr Thr Phe Asn Gln Ser Lys Leu Asn Ser Phe
 660 665 670
 Ile Thr Gln Tyr Ser Thr Gly Gln Val Ser Val Glu Ile Glu Trp Glu
 675 680 685
 Leu Gln Lys Glu Asn Ser Lys Arg Trp Asn Pro Glu Ile Gln Tyr Thr
 690 695 700
 Ser Asn Tyr Tyr Lys Ser Thr Ser Val Asp Phe Ala Val Asn Thr Glu
 705 710 715 720
 Gly Val Tyr Ser Glu Pro Arg Pro Ile Gly Thr Arg Tyr Leu Thr Arg
 725 730 735
 Asn Leu

<210> SEQ ID NO 46
 <211> LENGTH: 2214
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: AVV mutant 8G264AG515AG541A
 <220> FEATURE:
 <221> NAME/KEY: CDS
 <222> LOCATION: (1) .. (2214)

<400> SEQUENCE: 46

atg gct gcc gat ggt tat ctt cca gat tgg ctc gag gac aac ctc tct	48
Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser	
1 5 10 15	
 gag ggc att cgc gag tgg tgg gcg ctg aaa cct gga gcc ccg aag ccc	96
Glu Gly Ile Arg Glu Trp Trp Ala Leu Lys Pro Gly Ala Pro Lys Pro	
20 25 30	

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aaa gcc aac cag caa aag cag gac gac ggc cgg ggt ctg gtg ctt cct Lys Ala Asn Gln Gln Lys Gln Asp Asp Gly Arg Gly Leu Val Leu Pro 35 40 45	144
ggc tac aag tac ctc gga ccc ttc aac gga ctc gac aag ggg gag ccc Gly Tyr Lys Tyr Leu Gly Pro Phe Asn Gly Leu Asp Lys Gly Glu Pro 50 55 60	192
gtc aac gcg gcg gac gca gcg gcc ctc gag cac gac aag gcc tac gac Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp 65 70 75 80	240
cag cag ctg cag gcg ggt gac aat ccg tac ctg cgg tat aac cac gcc Gln Gln Leu Gln Ala Gly Asp Asn Pro Tyr Leu Arg Tyr Asn His Ala 85 90 95	288
gac gcc gag ttt cag gag cgt ctg caa gaa gat acg tct ttt ggg ggc Asp Ala Glu Phe Gln Glu Arg Leu Gln Glu Asp Thr Ser Phe Gly Gly 100 105 110	336
aac ctc ggg cga gca gtc ttc cag gcc aag aag cgg gtt ctc gaa cct Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Val Leu Glu Pro 115 120 125	384
ctc ggt ctg gtt gag gaa ggc gct aag acg gct cct gga aag aag aga Leu Gly Leu Val Glu Glu Gly Ala Lys Thr Ala Pro Gly Lys Lys Arg 130 135 140	432
ccg gta gag cca tca ccc cag cgt tct cca gac tcc tct acg ggc atc Pro Val Glu Pro Ser Pro Gln Arg Ser Pro Asp Ser Ser Thr Gly Ile 145 150 155 160	480
ggc aag aaa ggc caa cag ccc gcc aga aaa aga ctc aat ttt ggt cag Gly Lys Lys Gly Gln Gln Pro Ala Arg Lys Arg Leu Asn Phe Gly Gln 165 170 175	528
act ggc gac tca gag tca gtt cca gac cct caa cct ctc gga gaa cct Thr Gly Asp Ser Glu Ser Val Pro Asp Pro Gln Pro Leu Gly Glu Pro 180 185 190	576
cca gca gcg ccc tct ggt gtg gga cct aat aca atg gct gca ggc ggt Pro Ala Ala Pro Ser Gly Val Gly Pro Asn Thr Met Ala Ala Gly Gly 195 200 205	624
ggc gca cca atg gca gac aat aac gaa ggc gcc gac gga gtg ggt agt Gly Ala Pro Met Ala Asp Asn Asn Glu Gly Ala Asp Gly Val Gly Ser 210 215 220	672
tcc tcg gga aat tgg cat tgc gat tcc aca tgg ctg ggc gac aga gtc Ser Ser Gly Asn Trp His Cys Asp Ser Thr Trp Leu Gly Asp Arg Val 225 230 235 240	720
atc acc acc agc acc cga acc tgg gcc ctg ccc acc tac aac aac cac Ile Thr Thr Ser Thr Arg Thr Trp Ala Leu Pro Thr Tyr Asn Asn His 245 250 255	768
ctc tac aag caa atc tcc aac gcg aca tcg gga gga gcc acc aac gac Leu Tyr Lys Gln Ile Ser Asn Ala Thr Ser Gly Gly Ala Thr Asn Asp 260 265 270	816
aac acc tac ttc ggc tac agc acc ccc tgg ggg tat ttt gac ttt aac Asn Thr Tyr Phe Gly Tyr Ser Thr Pro Trp Gly Tyr Phe Asp Phe Asn 275 280 285	864
aga ttc cac tgc cac ttt tca cca cgt gac tgg cag cga ctc atc aac Arg Phe His Cys His Phe Ser Pro Arg Asp Trp Gln Arg Leu Ile Asn 290 295 300	912
aac aac tgg gga ttc cgg ccc aag aga ctc agc ttc aag ctc ttc aac Asn Asn Trp Gly Phe Arg Pro Lys Arg Leu Ser Phe Lys Leu Phe Asn 305 310 315 320	960
atc cag gtc aag gag gtc acg cag aat gaa ggc acc aag acc atc gcc Ile Gln Val Lys Glu Val Thr Gln Asn Glu Gly Thr Lys Thr Ile Ala 325 330 335	1008
aat aac ctc acc agc acc atc cag gtg ttt acg gac tcg gag tac cag Asn Asn Leu Thr Ser Thr Ile Gln Val Phe Thr Asp Ser Glu Tyr Gln	1056

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340	345	350	
ctg ccg tac gtt ctc ggc tct gcc cac cag ggc tgc ctg cct ccg ttc Leu Pro Tyr Val Leu Gly Ser Ala His Gln Gly Cys Leu Pro Pro Phe 355 360 365			1104
ccg gcg gac gtg ttc atg att ccc cag tac ggc tac cta aca ctc aac Pro Ala Asp Val Phe Met Ile Pro Gln Tyr Gly Tyr Leu Thr Leu Asn 370 375 380			1152
aac ggt agt cag gcc gtg gga cgc tcc tcc ttc tac tgc ctg gaa tac Asn Gly Ser Gln Ala Val Gly Arg Ser Ser Phe Tyr Cys Leu Glu Tyr 385 390 395 400			1200
ttt cct tcg cag atg ctg aga acc ggc aac aac ttc cag ttt act tac Phe Pro Ser Gln Met Leu Arg Thr Gly Asn Asn Phe Gln Phe Thr Tyr 405 410 415			1248
acc ttc gag gac gtg cct ttc cac agc agc tac gcc cac agc cag agc Thr Phe Glu Asp Val Pro Phe His Ser Ser Tyr Ala His Ser Gln Ser 420 425 430			1296
ttg gac ccg ctg atg aat cct ctg att gac cag tac ctg tac tac ttg Leu Asp Arg Leu Met Asn Pro Leu Ile Asp Gln Tyr Leu Tyr Tyr Leu 435 440 445			1344
tct ccg act caa aca aca gga ggc acg gca aat acg cag act ctg ggc Ser Arg Thr Gln Thr Thr Gly Thr Ala Asn Thr Gln Thr Leu Gly 450 455 460			1392
ttc agc caa ggt ggg cct aat aca atg gcc aat cag gca aag aac tgg Phe Ser Gln Gly Gly Pro Asn Thr Met Ala Asn Gln Ala Lys Asn Trp 465 470 475 480			1440
ctg cca gga ccc tgt tac cgc caa caa cgc gtc tca acg aca acc ggg Leu Pro Gly Pro Cys Tyr Arg Gln Gln Arg Val Ser Thr Thr Thr Gly 485 490 495			1488
caa aac aac aat agc aac ttt gcc tgg act gct ggg acc aaa tac cat Gln Asn Asn Asn Ser Asn Phe Ala Trp Thr Thr Ala Gly Thr Lys Tyr His 500 505 510			1536
ctg aat gca aga aat tca ttg gct aat cct ggc atc gct atg gca aca Leu Asn Ala Arg Asn Ser Leu Ala Asn Pro Gly Ile Ala Met Ala Thr 515 520 525			1584
cac aaa gac gac gag gag cgt ttt ttt ccc agt aac gcg atc ctg att His Lys Asp Asp Glu Glu Arg Phe Phe Pro Ser Asn Ala Ile Leu Ile 530 535 540			1632
ttt ggc aaa caa aat gct gcc aga gac aat gcg gat tac agc gat gtc Phe Gly Lys Gln Asn Ala Ala Arg Asp Asn Ala Asp Tyr Ser Asp Val 545 550 555 560			1680
atg ctc acc agc gag gaa gaa atc aaa acc act aac cct gtg gct aca Met Leu Thr Ser Glu Glu Glu Ile Lys Thr Thr Asn Pro Val Ala Thr 565 570 575			1728
gag gaa tac ggt atc gtg gca gat aac ttg cag cag caa aac acg gct Glu Glu Tyr Gly Ile Val Ala Asp Asn Leu Gln Gln Gln Asn Thr Ala 580 585 590			1776
cct caa att gga act gtc aac agc cag ggg gcc tta ccc ggt atg gtc Pro Gln Ile Gly Thr Val Asn Ser Gln Gly Ala Leu Pro Gly Met Val 595 600 605			1824
tgg cag aac ccg gac gtg tac ctg cag ggt ccc atc tgg gcc aag att Trp Gln Asn Arg Asp Val Tyr Leu Gln Gly Pro Ile Trp Ala Lys Ile 610 615 620			1872
cct cac acg gac ggc aac ttc cac ccg tct ccg ctg atg ggc ggc ttt Pro His Thr Asp Gly Asn Phe His Pro Ser Pro Leu Met Gly Gly Phe 625 630 635 640			1920
ggc ctg aaa cat cct ccg cct cag atc ctg atc aag aac acg cct gta Gly Leu Lys His Pro Pro Pro Gln Ile Leu Ile Lys Asn Thr Pro Val 645 650 655			1968
cct gcg gat cct ccg acc acc ttc aac cag tca aag ctg aac tct ttc			2016

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Pro	Ala	Asp	Pro	Pro	Thr	Thr	Phe	Asn	Gln	Ser	Lys	Leu	Asn	Ser	Phe		
			660					665					670				
atc	acg	caa	tac	agc	acc	gga	cag	gtc	agc	gtg	gaa	att	gaa	tg	gag		2064
Ile	Thr	Gln	Tyr	Ser	Thr	Gly	Gln	Val	Ser	Val	Glu	Ile	Glu	Trp	Glu		
		675					680					685					
ctg	cag	aag	gaa	aac	agc	aag	cgc	tg	aac	ccc	gag	atc	cag	tac	acc		2112
Leu	Gln	Lys	Glu	Asn	Ser	Lys	Arg	Trp	Asn	Pro	Glu	Ile	Gln	Tyr	Thr		
		690				695				700							
tcc	aac	tac	tac	aaa	tct	aca	agt	gtg	gac	ttt	gct	gtt	aat	aca	gaa		2160
Ser	Asn	Tyr	Tyr	Lys	Ser	Thr	Ser	Val	Asp	Phe	Ala	Val	Asn	Thr	Glu		
		705			710					715					720		
ggc	gtg	tac	tct	gaa	ccc	cgc	ccc	att	ggc	acc	cgt	tac	ctc	acc	cgt		2208
Gly	Val	Tyr	Ser	Glu	Pro	Arg	Pro	Ile	Gly	Thr	Arg	Tyr	Leu	Thr	Arg		
			725					730					735				
aat	ctg																2214
Asn	Leu																

<210> SEQ ID NO 47
 <211> LENGTH: 738
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

 <400> SEQUENCE: 47

Met	Ala	Ala	Asp	Gly	Tyr	Leu	Pro	Asp	Trp	Leu	Glu	Asp	Asn	Leu	Ser		
1			5					10					15				
Glu	Gly	Ile	Arg	Glu	Trp	Trp	Ala	Leu	Lys	Pro	Gly	Ala	Pro	Lys	Pro		
		20					25					30					
Lys	Ala	Asn	Gln	Gln	Lys	Gln	Asp	Asp	Gly	Arg	Gly	Leu	Val	Leu	Pro		
		35				40						45					
Gly	Tyr	Lys	Tyr	Leu	Gly	Pro	Phe	Asn	Gly	Leu	Asp	Lys	Gly	Glu	Pro		
	50				55						60						
Val	Asn	Ala	Ala	Asp	Ala	Ala	Ala	Leu	Glu	His	Asp	Lys	Ala	Tyr	Asp		
65				70					75					80			
Gln	Gln	Leu	Gln	Ala	Gly	Asp	Asn	Pro	Tyr	Leu	Arg	Tyr	Asn	His	Ala		
			85					90						95			
Asp	Ala	Glu	Phe	Gln	Glu	Arg	Leu	Gln	Glu	Asp	Thr	Ser	Phe	Gly	Gly		
		100					105						110				
Asn	Leu	Gly	Arg	Ala	Val	Phe	Gln	Ala	Lys	Lys	Arg	Val	Leu	Glu	Pro		
		115					120					125					
Leu	Gly	Leu	Val	Glu	Glu	Gly	Ala	Lys	Thr	Ala	Pro	Gly	Lys	Lys	Arg		
		130				135					140						
Pro	Val	Glu	Pro	Ser	Pro	Gln	Arg	Ser	Pro	Asp	Ser	Ser	Thr	Gly	Ile		
145				150					155					160			
Gly	Lys	Lys	Gly	Gln	Gln	Pro	Ala	Arg	Lys	Arg	Leu	Asn	Phe	Gly	Gln		
			165					170						175			
Thr	Gly	Asp	Ser	Glu	Ser	Val	Pro	Asp	Pro	Gln	Pro	Leu	Gly	Glu	Pro		
		180						185					190				
Pro	Ala	Ala	Pro	Ser	Gly	Val	Gly	Pro	Asn	Thr	Met	Ala	Ala	Gly	Gly		
		195					200					205					
Gly	Ala	Pro	Met	Ala	Asp	Asn	Asn	Glu	Gly	Ala	Asp	Gly	Val	Gly	Ser		
		210				215					220						
Ser	Ser	Gly	Asn	Trp	His	Cys	Asp	Ser	Thr	Trp	Leu	Gly	Asp	Arg	Val		
225				230						235				240			
Ile	Thr	Thr	Ser	Thr	Arg	Thr	Trp	Ala	Leu	Pro	Thr	Tyr	Asn	Asn	His		
			245					250						255			

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Leu Tyr Lys Gln Ile Ser Asn Ala Thr Ser Gly Gly Ala Thr Asn Asp
 260 265 270
 Asn Thr Tyr Phe Gly Tyr Ser Thr Pro Trp Gly Tyr Phe Asp Phe Asn
 275 280 285
 Arg Phe His Cys His Phe Ser Pro Arg Asp Trp Gln Arg Leu Ile Asn
 290 295 300
 Asn Asn Trp Gly Phe Arg Pro Lys Arg Leu Ser Phe Lys Leu Phe Asn
 305 310 315 320
 Ile Gln Val Lys Glu Val Thr Gln Asn Glu Gly Thr Lys Thr Ile Ala
 325 330 335
 Asn Asn Leu Thr Ser Thr Ile Gln Val Phe Thr Asp Ser Glu Tyr Gln
 340 345 350
 Leu Pro Tyr Val Leu Gly Ser Ala His Gln Gly Cys Leu Pro Pro Phe
 355 360 365
 Pro Ala Asp Val Phe Met Ile Pro Gln Tyr Gly Tyr Leu Thr Leu Asn
 370 375 380
 Asn Gly Ser Gln Ala Val Gly Arg Ser Ser Phe Tyr Cys Leu Glu Tyr
 385 390 395 400
 Phe Pro Ser Gln Met Leu Arg Thr Gly Asn Asn Phe Gln Phe Thr Tyr
 405 410 415
 Thr Phe Glu Asp Val Pro Phe His Ser Ser Tyr Ala His Ser Gln Ser
 420 425 430
 Leu Asp Arg Leu Met Asn Pro Leu Ile Asp Gln Tyr Leu Tyr Tyr Leu
 435 440 445
 Ser Arg Thr Gln Thr Thr Gly Gly Thr Ala Asn Thr Gln Thr Leu Gly
 450 455 460
 Phe Ser Gln Gly Gly Pro Asn Thr Met Ala Asn Gln Ala Lys Asn Trp
 465 470 475 480
 Leu Pro Gly Pro Cys Tyr Arg Gln Gln Arg Val Ser Thr Thr Thr Gly
 485 490 495
 Gln Asn Asn Asn Ser Asn Phe Ala Trp Thr Ala Gly Thr Lys Tyr His
 500 505 510
 Leu Asn Ala Arg Asn Ser Leu Ala Asn Pro Gly Ile Ala Met Ala Thr
 515 520 525
 His Lys Asp Asp Glu Glu Arg Phe Phe Pro Ser Asn Ala Ile Leu Ile
 530 535 540
 Phe Gly Lys Gln Asn Ala Ala Arg Asp Asn Ala Asp Tyr Ser Asp Val
 545 550 555 560
 Met Leu Thr Ser Glu Glu Glu Ile Lys Thr Thr Asn Pro Val Ala Thr
 565 570 575
 Glu Glu Tyr Gly Ile Val Ala Asp Asn Leu Gln Gln Gln Asn Thr Ala
 580 585 590
 Pro Gln Ile Gly Thr Val Asn Ser Gln Gly Ala Leu Pro Gly Met Val
 595 600 605
 Trp Gln Asn Arg Asp Val Tyr Leu Gln Gly Pro Ile Trp Ala Lys Ile
 610 615 620
 Pro His Thr Asp Gly Asn Phe His Pro Ser Pro Leu Met Gly Gly Phe
 625 630 635 640
 Gly Leu Lys His Pro Pro Pro Gln Ile Leu Ile Lys Asn Thr Pro Val
 645 650 655
 Pro Ala Asp Pro Pro Thr Thr Phe Asn Gln Ser Lys Leu Asn Ser Phe
 660 665 670

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Ile Thr Gln Tyr Ser Thr Gly Gln Val Ser Val Glu Ile Glu Trp Glu
    675                                680                                685

Leu Gln Lys Glu Asn Ser Lys Arg Trp Asn Pro Glu Ile Gln Tyr Thr
    690                                695                                700

Ser Asn Tyr Tyr Lys Ser Thr Ser Val Asp Phe Ala Val Asn Thr Glu
    705                                710                                715                                720

Gly Val Tyr Ser Glu Pro Arg Pro Ile Gly Thr Arg Tyr Leu Thr Arg
    725                                730                                735

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Asn Leu

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<210> SEQ ID NO 48
<211> LENGTH: 2211
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: AVV mutant 9G330AG453A
<220> FEATURE:
<221> NAME/KEY: CDS
<222> LOCATION: (1)..(2211)

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<400> SEQUENCE: 48

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atg gct gcc gat ggt tat ctt cca gat tgg ctc gag gac aac ctt agt      48
Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser
1                                5                                10                                15

gaa gga att cgc gag tgg tgg gct ttg aaa cct gga gcc cct caa ccc      96
Glu Gly Ile Arg Glu Trp Trp Ala Leu Lys Pro Gly Ala Pro Gln Pro
20                                25                                30

aag gca aat caa caa cat caa gac aac gct cga ggt ctt gtg ctt ccg     144
Lys Ala Asn Gln Gln His Gln Asp Asn Ala Arg Gly Leu Val Leu Pro
35                                40                                45

ggt tac aaa tac ctt gga ccc ggc aac gga ctc gac aag ggg gag ccg     192
Gly Tyr Lys Tyr Leu Gly Pro Gly Asn Gly Leu Asp Lys Gly Glu Pro
50                                55                                60

gtc aac gca gca gac gcg gcg gcc ctc gag cac gac aag gcc tac gac     240
Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp
65                                70                                75                                80

cag cag ctc aag gcc gga gac aac ccg tac ctc aag tac aac cac gcc     288
Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Lys Tyr Asn His Ala
85                                90                                95

gac gcc gag ttc cag gag cgg ctc aaa gaa gat acg tct ttt ggg ggc     336
Asp Ala Glu Phe Gln Glu Arg Leu Lys Glu Asp Thr Ser Phe Gly Gly
100                                105                                110

aac ctc ggg cga gca gtc ttc cag gcc aaa aag agg ctt ctt gaa cct     384
Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Leu Leu Glu Pro
115                                120                                125

ctt ggt ctg gtt gag gaa gcg gct aag acg gct cct gga aag aag agg     432
Leu Gly Leu Val Glu Glu Ala Ala Lys Thr Ala Pro Gly Lys Lys Arg
130                                135                                140

cct gta gag cag tct cct cag gaa ccg gac tcc tcc gcg ggt att ggc     480
Pro Val Glu Gln Ser Pro Gln Glu Pro Asp Ser Ser Ala Gly Ile Gly
145                                150                                155                                160

aaa tcg ggt gca cag ccc gct aaa aag aga ctc aat ttc ggt cag act     528
Lys Ser Gly Ala Gln Pro Ala Lys Lys Arg Leu Asn Phe Gly Gln Thr
165                                170                                175

ggc gac aca gag tca gtc cca gac cct caa cca atc gga gaa cct ccc     576
Gly Asp Thr Glu Ser Val Pro Asp Pro Gln Pro Ile Gly Glu Pro Pro
180                                185                                190

gca gcc ccc tca ggt gtg gga tct ctt aca atg gct tca ggt ggt ggc     624
Ala Ala Pro Ser Gly Val Gly Ser Leu Thr Met Ala Ser Gly Gly Gly
195                                200                                205

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gca cca gtg gca gac aat aac gaa ggt gcc gat gga gtg ggt agt tcc Ala Pro Val Ala Asp Asn Asn Glu Gly Ala Asp Gly Val Gly Ser Ser 210 215 220	672
tcg gga aat tgg cat tgc gat tcc caa tgg ctg ggg gac aga gtc atc Ser Gly Asn Trp His Cys Asp Ser Gln Trp Leu Gly Asp Arg Val Ile 225 230 235 240	720
acc acc agc acc cga acc tgg gcc ctg ccc acc tac aac aat cac ctc Thr Thr Ser Thr Arg Thr Trp Ala Leu Pro Thr Tyr Asn Asn His Leu 245 250 255	768
tac aag caa atc tcc aac agc aca tct gga gga tct tca aat gac aac Tyr Lys Gln Ile Ser Asn Ser Thr Ser Gly Gly Ser Ser Asn Asp Asn 260 265 270	816
gcc tac ttc ggc tac agc acc ccc tgg ggg tat ttt gac ttc aac aga Ala Tyr Phe Gly Tyr Ser Thr Pro Trp Gly Tyr Phe Asp Phe Asn Arg 275 280 285	864
ttc cac tgc cac ttc tca cca cgt gac tgg cag cga ctc atc aac aac Phe His Cys His Phe Ser Pro Arg Asp Trp Gln Arg Leu Ile Asn Asn 290 295 300	912
aac tgg gga ttc cgg cct aag cga ctc aac ttc aag ctc ttc aac att Asn Trp Gly Phe Arg Pro Lys Arg Leu Asn Phe Lys Leu Phe Asn Ile 305 310 315 320	960
cag gtc aaa gag gtt acg gac aac aat gca gtc aag acc atc gcc aat Gln Val Lys Glu Val Thr Asp Asn Asn Ala Val Lys Thr Ile Ala Asn 325 330 335	1008
aac ctt acc agc acg gtc cag gtc ttc acg gac tca gac tat cag ctc Asn Leu Thr Ser Thr Val Gln Val Phe Thr Asp Ser Asp Tyr Gln Leu 340 345 350	1056
ccg tac gtg ctc ggg tgc gct cac gag ggc tgc ctc ccg ccg ttc cca Pro Tyr Val Leu Gly Ser Ala His Glu Gly Cys Leu Pro Pro Phe Pro 355 360 365	1104
gcg gac gtt ttc atg att cct cag tac ggg tat ctg acg ctt aat gat Ala Asp Val Phe Met Ile Pro Gln Tyr Gly Tyr Leu Thr Leu Asn Asp 370 375 380	1152
gga agc cag gcc gtg ggt cgt tgc tcc ttt tac tgc ctg gaa tat ttc Gly Ser Gln Ala Val Gly Arg Ser Ser Phe Tyr Cys Leu Glu Tyr Phe 385 390 395 400	1200
ccg tgc caa atg cta aga acg ggt aac aac ttc cag ttc agc tac gag Pro Ser Gln Met Leu Arg Thr Gly Asn Asn Phe Gln Phe Ser Tyr Glu 405 410 415	1248
ttt gag aac gta cct ttc cat agc agc tac gct cac agc caa agc ctg Phe Glu Asn Val Pro Phe His Ser Ser Tyr Ala His Ser Gln Ser Leu 420 425 430	1296
gac cga cta atg aat cca ctc atc gac caa tac ttg tac tat ctc tca Asp Arg Leu Met Asn Pro Leu Ile Asp Gln Tyr Leu Tyr Tyr Leu Ser 435 440 445	1344
aag act att aac gct tct gga cag aat caa caa acg cta aaa ttc agt Lys Thr Ile Asn Ala Ser Gly Gln Asn Gln Gln Thr Leu Lys Phe Ser 450 455 460	1392
gtg gcc gga ccc agc aac atg gct gtc cag gga aga aac tac ata cct Val Ala Gly Pro Ser Asn Met Ala Val Gln Gly Arg Asn Tyr Ile Pro 465 470 475 480	1440
gga ccc agc tac cga caa caa cgt gtc tca acc act gtg act caa aac Gly Pro Ser Tyr Arg Gln Gln Arg Val Ser Thr Thr Val Thr Gln Asn 485 490 495	1488
aac aac agc gaa ttt gct tgg cct gga gct tct tct tgg gct ctc aat Asn Asn Ser Glu Phe Ala Trp Pro Gly Ala Ser Ser Trp Ala Leu Asn 500 505 510	1536
gga cgt aat agc ttg atg aat cct gga cct gct atg gcc agc cac aaa Gly Arg Asn Ser Leu Met Asn Pro Gly Pro Ala Met Ala Ser His Lys 515 520 525	1584

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gaa gga gag gac cgt ttc ttt cct ttg tct gga tct tta att ttt ggc	1632
Glu Gly Glu Asp Arg Phe Phe Pro Leu Ser Gly Ser Leu Ile Phe Gly	
530 535 540	
aaa caa gga act gga aga gac aac gtg gat gcg gac aaa gtc atg ata	1680
Lys Gln Gly Thr Gly Arg Asp Asn Val Asp Ala Asp Lys Val Met Ile	
545 550 555 560	
acc aac gaa gaa gaa att aaa act act aac ccg gta gca acg gag tcc	1728
Thr Asn Glu Glu Glu Ile Lys Thr Thr Asn Pro Val Ala Thr Glu Ser	
565 570 575	
tat gga caa gtg gcc aca aac cac cag agt gcc caa gca cag gcg cag	1776
Tyr Gly Gln Val Ala Thr Asn His Gln Ser Ala Gln Ala Gln Ala Gln	
580 585 590	
acc ggc tgg gtt caa aac caa gga ata ctt ccg ggt atg gtt tgg cag	1824
Thr Gly Trp Val Gln Asn Gln Gly Ile Leu Pro Gly Met Val Trp Gln	
595 600 605	
gac aga gat gtg tac ctg caa gga ccc att tgg gcc aaa att cct cac	1872
Asp Arg Asp Val Tyr Leu Gln Gly Pro Ile Trp Ala Lys Ile Pro His	
610 615 620	
acg gac ggc aac ttt cac cct tct ccg ctg atg gga ggg ttt gga atg	1920
Thr Asp Gly Asn Phe His Pro Ser Pro Leu Met Gly Gly Phe Gly Met	
625 630 635 640	
aag cac ccg cct cct cag atc ctc atc aaa aac aca cct gta cct gcg	1968
Lys His Pro Pro Pro Gln Ile Leu Ile Lys Asn Thr Pro Val Pro Ala	
645 650 655	
gat cct cca acg gcc ttc aac aag gac aag ctg aac tct ttc atc acc	2016
Asp Pro Pro Thr Ala Phe Asn Lys Asp Lys Leu Asn Ser Phe Ile Thr	
660 665 670	
cag tat tct act ggc caa gtc agc gtg gag atc gag tgg gag ctg cag	2064
Gln Tyr Ser Thr Gly Gln Val Ser Val Glu Ile Glu Trp Glu Leu Gln	
675 680 685	
aag gaa aac agc aag cgc tgg aac ccg gag atc cag tac act tcc aac	2112
Lys Glu Asn Ser Lys Arg Trp Asn Pro Glu Ile Gln Tyr Thr Ser Asn	
690 695 700	
tat tac aag tct aat aat gtt gaa ttt gct gtt aat act gaa ggt gta	2160
Tyr Tyr Lys Ser Asn Asn Val Glu Phe Ala Val Asn Thr Glu Gly Val	
705 710 715 720	
tat agt gaa ccc cgc ccc att ggc acc aga tac ctg act cgt aat ctg	2208
Tyr Ser Glu Pro Arg Pro Ile Gly Thr Arg Tyr Leu Thr Arg Asn Leu	
725 730 735	
taa	2211

<210> SEQ ID NO 49

<211> LENGTH: 736

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 49

Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser
1 5 10 15

Glu Gly Ile Arg Glu Trp Trp Ala Leu Lys Pro Gly Ala Pro Gln Pro
20 25 30

Lys Ala Asn Gln Gln His Gln Asp Asn Ala Arg Gly Leu Val Leu Pro
35 40 45

Gly Tyr Lys Tyr Leu Gly Pro Gly Asn Gly Leu Asp Lys Gly Glu Pro
50 55 60

Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp
65 70 75 80

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Gln	Gln	Leu	Lys	Ala	Gly	Asp	Asn	Pro	Tyr	Leu	Lys	Tyr	Asn	His	Ala	
				85					90					95		
Asp	Ala	Glu	Phe	Gln	Glu	Arg	Leu	Lys	Glu	Asp	Thr	Ser	Phe	Gly	Gly	
			100					105					110			
Asn	Leu	Gly	Arg	Ala	Val	Phe	Gln	Ala	Lys	Lys	Arg	Leu	Leu	Glu	Pro	
		115						120				125				
Leu	Gly	Leu	Val	Glu	Glu	Ala	Ala	Lys	Thr	Ala	Pro	Gly	Lys	Lys	Arg	
	130					135					140					
Pro	Val	Glu	Gln	Ser	Pro	Gln	Glu	Pro	Asp	Ser	Ser	Ala	Gly	Ile	Gly	
145					150				155						160	
Lys	Ser	Gly	Ala	Gln	Pro	Ala	Lys	Lys	Arg	Leu	Asn	Phe	Gly	Gln	Thr	
			165						170					175		
Gly	Asp	Thr	Glu	Ser	Val	Pro	Asp	Pro	Gln	Pro	Ile	Gly	Glu	Pro	Pro	
			180					185					190			
Ala	Ala	Pro	Ser	Gly	Val	Gly	Ser	Leu	Thr	Met	Ala	Ser	Gly	Gly	Gly	
		195						200				205				
Ala	Pro	Val	Ala	Asp	Asn	Asn	Glu	Gly	Ala	Asp	Gly	Val	Gly	Ser	Ser	
	210					215					220					
Ser	Gly	Asn	Trp	His	Cys	Asp	Ser	Gln	Trp	Leu	Gly	Asp	Arg	Val	Ile	
225					230					235					240	
Thr	Thr	Ser	Thr	Arg	Thr	Trp	Ala	Leu	Pro	Thr	Tyr	Asn	Asn	His	Leu	
			245						250					255		
Tyr	Lys	Gln	Ile	Ser	Asn	Ser	Thr	Ser	Gly	Gly	Ser	Ser	Asn	Asp	Asn	
		260						265					270			
Ala	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro	Trp	Gly	Tyr	Phe	Asp	Phe	Asn	Arg	
		275					280					285				
Phe	His	Cys	His	Phe	Ser	Pro	Arg	Asp	Trp	Gln	Arg	Leu	Ile	Asn	Asn	
	290					295					300					
Asn	Trp	Gly	Phe	Arg	Pro	Lys	Arg	Leu	Asn	Phe	Lys	Leu	Phe	Asn	Ile	
305					310					315					320	
Gln	Val	Lys	Glu	Val	Thr	Asp	Asn	Asn	Ala	Val	Lys	Thr	Ile	Ala	Asn	
			325						330					335		
Asn	Leu	Thr	Ser	Thr	Val	Gln	Val	Phe	Thr	Asp	Ser	Asp	Tyr	Gln	Leu	
		340						345					350			
Pro	Tyr	Val	Leu	Gly	Ser	Ala	His	Glu	Gly	Cys	Leu	Pro	Pro	Phe	Pro	
		355					360					365				
Ala	Asp	Val	Phe	Met	Ile	Pro	Gln	Tyr	Gly	Tyr	Leu	Thr	Leu	Asn	Asp	
	370					375					380					
Gly	Ser	Gln	Ala	Val	Gly	Arg	Ser	Ser	Phe	Tyr	Cys	Leu	Glu	Tyr	Phe	
385					390					395					400	
Pro	Ser	Gln	Met	Leu	Arg	Thr	Gly	Asn	Asn	Phe	Gln	Phe	Ser	Tyr	Glu	
			405					410						415		
Phe	Glu	Asn	Val	Pro	Phe	His	Ser	Ser	Tyr	Ala	His	Ser	Gln	Ser	Leu	
			420					425					430			
Asp	Arg	Leu	Met	Asn	Pro	Leu	Ile	Asp	Gln	Tyr	Leu	Tyr	Tyr	Leu	Ser	
		435					440					445				
Lys	Thr	Ile	Asn	Ala	Ser	Gly	Gln	Asn	Gln	Gln	Thr	Leu	Lys	Phe	Ser	
	450					455					460					
Val	Ala	Gly	Pro	Ser	Asn	Met	Ala	Val	Gln	Gly	Arg	Asn	Tyr	Ile	Pro	
465					470					475					480	
Gly	Pro	Ser	Tyr	Arg	Gln	Gln	Arg	Val	Ser	Thr	Thr	Val	Thr	Gln	Asn	
				485					490					495		

Asn	Asn	Ser	Glu	Phe	Ala	Trp	Pro	Gly	Ala	Ser	Ser	Trp	Ala	Leu	Asn
			500					505					510		
Gly	Arg	Asn	Ser	Leu	Met	Asn	Pro	Gly	Pro	Ala	Met	Ala	Ser	His	Lys
	515						520					525			
Glu	Gly	Glu	Asp	Arg	Phe	Phe	Pro	Leu	Ser	Gly	Ser	Leu	Ile	Phe	Gly
	530					535					540				
Lys	Gln	Gly	Thr	Gly	Arg	Asp	Asn	Val	Asp	Ala	Asp	Lys	Val	Met	Ile
545					550					555					560
Thr	Asn	Glu	Glu	Glu	Ile	Lys	Thr	Thr	Asn	Pro	Val	Ala	Thr	Glu	Ser
				565					570					575	
Tyr	Gly	Gln	Val	Ala	Thr	Asn	His	Gln	Ser	Ala	Gln	Ala	Gln	Ala	Gln
			580					585					590		
Thr	Gly	Trp	Val	Gln	Asn	Gln	Gly	Ile	Leu	Pro	Gly	Met	Val	Trp	Gln
		595					600					605			
Asp	Arg	Asp	Val	Tyr	Leu	Gln	Gly	Pro	Ile	Trp	Ala	Lys	Ile	Pro	His
	610					615					620				
Thr	Asp	Gly	Asn	Phe	His	Pro	Ser	Pro	Leu	Met	Gly	Gly	Phe	Gly	Met
625					630					635					640
Lys	His	Pro	Pro	Pro	Gln	Ile	Leu	Ile	Lys	Asn	Thr	Pro	Val	Pro	Ala
				645					650					655	
Asp	Pro	Pro	Thr	Ala	Phe	Asn	Lys	Asp	Lys	Leu	Asn	Ser	Phe	Ile	Thr
			660					665					670		
Gln	Tyr	Ser	Thr	Gly	Gln	Val	Ser	Val	Glu	Ile	Glu	Trp	Glu	Leu	Gln
		675					680					685			
Lys	Glu	Asn	Ser	Lys	Arg	Trp	Asn	Pro	Glu	Ile	Gln	Tyr	Thr	Ser	Asn
	690					695					700				
Tyr	Tyr	Lys	Ser	Asn	Asn	Val	Glu	Phe	Ala	Val	Asn	Thr	Glu	Gly	Val
705				710						715					720
Tyr	Ser	Glu	Pro	Arg	Pro	Ile	Gly	Thr	Arg	Tyr	Leu	Thr	Arg	Asn	Leu
			725						730					735	

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<210> SEQ ID NO 50
<211> LENGTH: 2211
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: AVV mutant 9G330AG513A
<220> FEATURE:
<221> NAME/KEY: CDS
<222> LOCATION: (1)..(2211)
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<400> SEQUENCE: 50

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Met	Ala	Ala	Asp	Gly	Tyr	Leu	Pro	Asp	Trp	Leu	Glu	Asp	Asn	Leu	Ser	
1				5					10					15		
gaa	gga	att	cgc	gag	tgg	tgg	gct	ttg	aaa	cct	gga	gcc	cct	caa	ccc	96
Glu	Gly	Ile	Arg	Glu	Trp	Trp	Ala	Leu	Lys	Pro	Gly	Ala	Pro	Gln	Pro	
			20					25					30			
aag	gca	aat	caa	caa	cat	caa	gac	aac	gct	cga	ggt	ctt	gtg	ctt	ccg	144
Lys	Ala	Asn	Gln	Gln	His	Gln	Asp	Asn	Ala	Arg	Gly	Leu	Val	Leu	Pro	
		35					40					45				
ggt	tac	aaa	tac	ctt	gga	ccc	ggc	aac	gga	ctc	gac	aag	ggg	gag	ccg	192
Gly	Tyr	Lys	Tyr	Leu	Gly	Pro	Gly	Asn	Gly	Leu	Asp	Lys	Gly	Glu	Pro	
	50				55						60					
gtc	aac	gca	gca	gac	gcg	gcg	gcc	ctc	gag	cac	gac	aag	gcc	tac	gac	240
Val	Asn	Ala	Ala	Asp	Ala	Ala	Ala	Leu	Glu	His	Asp	Lys	Ala	Tyr	Asp	
65					70					75				80		
caq	caq	ctc	aaq	qcc	qqa	qac	aac	ccq	tac	ctc	aaq	tac	aac	cac	qcc	288

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Gln	Gln	Leu	Lys	Ala	Gly	Asp	Asn	Pro	Tyr	Leu	Lys	Tyr	Asn	His	Ala	
				85					90					95		
gac	gcc	gag	ttc	cag	gag	cgg	ctc	aaa	gaa	gat	acg	tct	ttt	ggg	ggc	336
Asp	Ala	Glu	Phe	Gln	Glu	Arg	Leu	Lys	Glu	Asp	Thr	Ser	Phe	Gly	Gly	
			100					105					110			
aac	ctc	ggg	cga	gca	gtc	ttc	cag	gcc	aaa	aag	agg	ctt	ctt	gaa	cct	384
Asn	Leu	Gly	Arg	Ala	Val	Phe	Gln	Ala	Lys	Lys	Arg	Leu	Leu	Glu	Pro	
		115						120					125			
ctt	ggt	ctg	gtt	gag	gaa	gcg	gct	aag	acg	gct	cct	gga	aag	aag	agg	432
Leu	Gly	Leu	Val	Glu	Glu	Ala	Ala	Lys	Thr	Ala	Pro	Gly	Lys	Lys	Arg	
	130					135					140					
cct	gta	gag	cag	tct	cct	cag	gaa	ccg	gac	tcc	tcc	gcg	ggt	att	ggc	480
Pro	Val	Glu	Gln	Ser	Pro	Gln	Glu	Pro	Asp	Ser	Ser	Ala	Gly	Ile	Gly	
	145					150				155					160	
aaa	tcg	ggt	gca	cag	ccc	gct	aaa	aag	aga	ctc	aat	ttc	ggt	cag	act	528
Lys	Ser	Gly	Ala	Gln	Pro	Ala	Lys	Lys	Arg	Leu	Asn	Phe	Gly	Gln	Thr	
			165						170					175		
ggc	gac	aca	gag	tca	gtc	cca	gac	cct	caa	cca	atc	gga	gaa	cct	ccc	576
Gly	Asp	Thr	Glu	Ser	Val	Pro	Asp	Pro	Gln	Pro	Ile	Gly	Glu	Pro	Pro	
		180						185					190			
gca	gcc	ccc	tca	ggt	gtg	gga	tct	ctt	aca	atg	gct	tca	ggt	ggt	ggc	624
Ala	Ala	Pro	Ser	Gly	Val	Gly	Ser	Leu	Thr	Met	Ala	Ser	Gly	Gly	Gly	
		195						200				205				
gca	cca	gtg	gca	gac	aat	aac	gaa	ggt	gcc	gat	gga	gtg	ggt	agt	tcc	672
Ala	Pro	Val	Ala	Asp	Asn	Asn	Glu	Gly	Ala	Asp	Gly	Val	Gly	Ser	Ser	
		210				215					220					
tcg	gga	aat	tgg	cat	tgc	gat	tcc	caa	tgg	ctg	ggg	gac	aga	gtc	atc	720
Ser	Gly	Asn	Trp	His	Cys	Asp	Ser	Gln	Trp	Leu	Gly	Asp	Arg	Val	Ile	
	225				230					235				240		
acc	acc	agc	acc	cga	acc	tgg	gcc	ctg	ccc	acc	tac	aac	aat	cac	ctc	768
Thr	Thr	Ser	Thr	Arg	Thr	Trp	Ala	Leu	Pro	Thr	Tyr	Asn	Asn	His	Leu	
			245						250					255		
tac	aag	caa	atc	tcc	aac	agc	aca	tct	gga	gga	tct	tca	aat	gac	aac	816
Tyr	Lys	Gln	Ile	Ser	Asn	Ser	Thr	Ser	Gly	Gly	Ser	Ser	Asn	Asp	Asn	
		260					265						270			
gcc	tac	ttc	ggc	tac	agc	acc	ccc	tgg	ggg	tat	ttt	gac	ttc	aac	aga	864
Ala	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro	Trp	Gly	Tyr	Phe	Asp	Phe	Asn	Arg	
		275					280					285				
ttc	cac	tgc	cac	ttc	tca	cca	cgt	gac	tgg	cag	cga	ctc	atc	aac	aac	912
Phe	His	Cys	His	Phe	Ser	Pro	Arg	Asp	Trp	Gln	Arg	Leu	Ile	Asn	Asn	
		290				295					300					
aac	tgg	gga	ttc	cgg	cct	aag	cga	ctc	aac	ttc	aag	ctc	ttc	aac	att	960
Asn	Trp	Gly	Phe	Arg	Pro	Lys	Arg	Leu	Asn	Phe	Lys	Leu	Phe	Asn	Ile	
		305			310					315				320		
cag	gtc	aaa	gag	gtt	acg	gac	aac	aat	gca	gtc	aag	acc	atc	gcc	aat	1008
Gln	Val	Lys	Glu	Val	Thr	Asp	Asn	Asn	Ala	Val	Lys	Thr	Ile	Ala	Asn	
			325						330					335		
aac	ctt	acc	agc	acg	gtc	cag	gtc	ttc	acg	gac	tca	gac	tat	cag	ctc	1056
Asn	Leu	Thr	Ser	Thr	Val	Gln	Val	Phe	Thr	Asp	Ser	Asp	Tyr	Gln	Leu	
		340					345						350			
ccg	tac	gtg	ctc	ggg	tcg	gct	cac	gag	ggc	tgc	ctc	ccg	ccg	ttc	cca	1104
Pro	Tyr	Val	Leu	Gly	Ser	Ala	His	Glu	Gly	Cys	Leu	Pro	Pro	Phe	Pro	
		355					360					365				
gcg	gac	gtt	ttc	atg	att	cct	cag	tac	ggg	tat	ctg	acg	ctt	aat	gat	1152
Ala	Asp	Val	Phe	Met	Ile	Pro	Gln	Tyr	Gly	Tyr	Leu	Thr	Leu	Asn	Asp	
	370					375					380					
gga	agc	cag	gcc	gtg	ggt	cgt	tcg	tcc	ttt	tac	tgc	ctg	gaa	tat	ttc	1200
Gly	Ser	Gln	Ala	Val	Gly	Arg	Ser	Ser	Phe	Tyr	Cys	Leu	Glu	Tyr	Phe	
	385				390					395					400	

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ccg tcg caa atg cta aga acg ggt aac aac ttc cag ttc agc tac gag	1248
Pro Ser Gln Met Leu Arg Thr Gly Asn Asn Phe Gln Phe Ser Tyr Glu	
405 410 415	
ttt gag aac gta cct ttc cat agc agc tac gct cac agc caa agc ctg	1296
Phe Glu Asn Val Pro Phe His Ser Ser Tyr Ala His Ser Gln Ser Leu	
420 425 430	
gac cga cta atg aat cca ctc atc gac caa tac ttg tac tat ctc tca	1344
Asp Arg Leu Met Asn Pro Leu Ile Asp Gln Tyr Leu Tyr Tyr Leu Ser	
435 440 445	
aag act att aac ggt tct gga cag aat caa caa acg cta aaa ttc agt	1392
Lys Thr Ile Asn Gly Ser Gly Gln Asn Gln Gln Thr Leu Lys Phe Ser	
450 455 460	
gtg gcc gga ccc agc aac atg gct gtc cag gga aga aac tac ata cct	1440
Val Ala Gly Pro Ser Asn Met Ala Val Gln Gly Arg Asn Tyr Ile Pro	
465 470 475 480	
gga ccc agc tac cga caa caa cgt gtc tca acc act gtg act caa aac	1488
Gly Pro Ser Tyr Arg Gln Gln Arg Val Ser Thr Thr Val Thr Gln Asn	
485 490 495	
aac aac agc gaa ttt gct tgg cct gga gct tct tct tgg gct ctc aat	1536
Asn Asn Ser Glu Phe Ala Trp Pro Gly Ala Ser Ser Trp Ala Leu Asn	
500 505 510	
gca cgt aat agc ttg atg aat cct gga cct gct atg gcc agc cac aaa	1584
Ala Arg Asn Ser Leu Met Asn Pro Gly Pro Ala Met Ala Ser His Lys	
515 520 525	
gaa gga gag gac cgt ttc ttt cct ttg tct gga tct tta att ttt ggc	1632
Glu Gly Glu Asp Arg Phe Phe Pro Leu Ser Gly Ser Leu Ile Phe Gly	
530 535 540	
aaa caa gga act gga aga gac aac gtg gat gcg gac aaa gtc atg ata	1680
Lys Gln Gly Thr Gly Arg Asp Asn Val Asp Ala Asp Lys Val Met Ile	
545 550 555 560	
acc aac gaa gaa gaa att aaa act act aac ccg gta gca acg gag tcc	1728
Thr Asn Glu Glu Glu Ile Lys Thr Thr Asn Pro Val Ala Thr Glu Ser	
565 570 575	
tat gga caa gtg gcc aca aac cac cag agt gcc caa gca cag gcg cag	1776
Tyr Gly Gln Val Ala Thr Asn His Gln Ser Ala Gln Ala Gln Ala Gln	
580 585 590	
acc ggc tgg gtt caa aac caa gga ata ctt ccg ggt atg gtt tgg cag	1824
Thr Gly Trp Val Gln Asn Gln Gly Ile Leu Pro Gly Met Val Trp Gln	
595 600 605	
gac aga gat gtg tac ctg caa gga ccc att tgg gcc aaa att cct cac	1872
Asp Arg Asp Val Tyr Leu Gln Gly Pro Ile Trp Ala Lys Ile Pro His	
610 615 620	
acg gac ggc aac ttt cac cct tct ccg ctg atg gga ggg ttt gga atg	1920
Thr Asp Gly Asn Phe His Pro Ser Pro Leu Met Gly Gly Phe Gly Met	
625 630 635 640	
aag cac ccg cct cct cag atc ctc atc aaa aac aca cct gta cct gcg	1968
Lys His Pro Pro Pro Gln Ile Leu Ile Lys Asn Thr Pro Val Pro Ala	
645 650 655	
gat cct cca acg gcc ttc aac aag gac aag ctg aac tct ttc atc acc	2016
Asp Pro Pro Thr Ala Phe Asn Lys Asp Lys Leu Asn Ser Phe Ile Thr	
660 665 670	
cag tat tct act ggc caa gtc agc gtg gag atc gag tgg gag ctg cag	2064
Gln Tyr Ser Thr Gly Gln Val Ser Val Glu Ile Glu Trp Glu Leu Gln	
675 680 685	
aag gaa aac agc aag cgc tgg aac ccg gag atc cag tac act tcc aac	2112
Lys Glu Asn Ser Lys Arg Trp Asn Pro Glu Ile Gln Tyr Thr Ser Asn	
690 695 700	
tat tac aag tct aat aat gtt gaa ttt gct gtt aat act gaa ggt gta	2160
Tyr Tyr Lys Ser Asn Asn Val Glu Phe Ala Val Asn Thr Glu Gly Val	
705 710 715 720	

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tat agt gaa ccc cgc ccc att ggc acc aga tac ctg act cgt aat ctg	2208
Tyr Ser Glu Pro Arg Pro Ile Gly Thr Arg Tyr Leu Thr Arg Asn Leu	
725 730 735	

taa	2211
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<210> SEQ ID NO 51
 <211> LENGTH: 736
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 51

Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser	
1 5 10 15	
Glu Gly Ile Arg Glu Trp Trp Ala Leu Lys Pro Gly Ala Pro Gln Pro	
20 25 30	
Lys Ala Asn Gln Gln His Gln Asp Asn Ala Arg Gly Leu Val Leu Pro	
35 40 45	
Gly Tyr Lys Tyr Leu Gly Pro Gly Asn Gly Leu Asp Lys Gly Glu Pro	
50 55 60	
Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp	
65 70 75 80	
Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Lys Tyr Asn His Ala	
85 90 95	
Asp Ala Glu Phe Gln Glu Arg Leu Lys Glu Asp Thr Ser Phe Gly Gly	
100 105 110	
Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Leu Leu Glu Pro	
115 120 125	
Leu Gly Leu Val Glu Glu Ala Ala Lys Thr Ala Pro Gly Lys Lys Arg	
130 135 140	
Pro Val Glu Gln Ser Pro Gln Glu Pro Asp Ser Ser Ala Gly Ile Gly	
145 150 155 160	
Lys Ser Gly Ala Gln Pro Ala Lys Lys Arg Leu Asn Phe Gly Gln Thr	
165 170 175	
Gly Asp Thr Glu Ser Val Pro Asp Pro Gln Pro Ile Gly Glu Pro Pro	
180 185 190	
Ala Ala Pro Ser Gly Val Gly Ser Leu Thr Met Ala Ser Gly Gly Gly	
195 200 205	
Ala Pro Val Ala Asp Asn Asn Glu Gly Ala Asp Gly Val Gly Ser Ser	
210 215 220	
Ser Gly Asn Trp His Cys Asp Ser Gln Trp Leu Gly Asp Arg Val Ile	
225 230 235 240	
Thr Thr Ser Thr Arg Thr Trp Ala Leu Pro Thr Tyr Asn Asn His Leu	
245 250 255	
Tyr Lys Gln Ile Ser Asn Ser Thr Ser Gly Gly Ser Ser Asn Asp Asn	
260 265 270	
Ala Tyr Phe Gly Tyr Ser Thr Pro Trp Gly Tyr Phe Asp Phe Asn Arg	
275 280 285	
Phe His Cys His Phe Ser Pro Arg Asp Trp Gln Arg Leu Ile Asn Asn	
290 295 300	
Asn Trp Gly Phe Arg Pro Lys Arg Leu Asn Phe Lys Leu Phe Asn Ile	
305 310 315 320	
Gln Val Lys Glu Val Thr Asp Asn Asn Ala Val Lys Thr Ile Ala Asn	
325 330 335	

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Asn Leu Thr Ser Thr Val Gln Val Phe Thr Asp Ser Asp Tyr Gln Leu
    340                      345                      350

Pro Tyr Val Leu Gly Ser Ala His Glu Gly Cys Leu Pro Pro Phe Pro
    355                      360                      365

Ala Asp Val Phe Met Ile Pro Gln Tyr Gly Tyr Leu Thr Leu Asn Asp
    370                      375                      380

Gly Ser Gln Ala Val Gly Arg Ser Ser Phe Tyr Cys Leu Glu Tyr Phe
    385                      390                      395                      400

Pro Ser Gln Met Leu Arg Thr Gly Asn Asn Phe Gln Phe Ser Tyr Glu
    405                      410                      415

Phe Glu Asn Val Pro Phe His Ser Ser Tyr Ala His Ser Gln Ser Leu
    420                      425                      430

Asp Arg Leu Met Asn Pro Leu Ile Asp Gln Tyr Leu Tyr Tyr Leu Ser
    435                      440                      445

Lys Thr Ile Asn Gly Ser Gly Gln Asn Gln Gln Thr Leu Lys Phe Ser
    450                      455                      460

Val Ala Gly Pro Ser Asn Met Ala Val Gln Gly Arg Asn Tyr Ile Pro
    465                      470                      475                      480

Gly Pro Ser Tyr Arg Gln Gln Arg Val Ser Thr Thr Val Thr Gln Asn
    485                      490                      495

Asn Asn Ser Glu Phe Ala Trp Pro Gly Ala Ser Ser Trp Ala Leu Asn
    500                      505                      510

Ala Arg Asn Ser Leu Met Asn Pro Gly Pro Ala Met Ala Ser His Lys
    515                      520                      525

Glu Gly Glu Asp Arg Phe Phe Pro Leu Ser Gly Ser Leu Ile Phe Gly
    530                      535                      540

Lys Gln Gly Thr Gly Arg Asp Asn Val Asp Ala Asp Lys Val Met Ile
    545                      550                      555                      560

Thr Asn Glu Glu Glu Ile Lys Thr Thr Asn Pro Val Ala Thr Glu Ser
    565                      570                      575

Tyr Gly Gln Val Ala Thr Asn His Gln Ser Ala Gln Ala Gln Ala Gln
    580                      585                      590

Thr Gly Trp Val Gln Asn Gln Gly Ile Leu Pro Gly Met Val Trp Gln
    595                      600                      605

Asp Arg Asp Val Tyr Leu Gln Gly Pro Ile Trp Ala Lys Ile Pro His
    610                      615                      620

Thr Asp Gly Asn Phe His Pro Ser Pro Leu Met Gly Gly Phe Gly Met
    625                      630                      635                      640

Lys His Pro Pro Pro Gln Ile Leu Ile Lys Asn Thr Pro Val Pro Ala
    645                      650                      655

Asp Pro Pro Thr Ala Phe Asn Lys Asp Lys Leu Asn Ser Phe Ile Thr
    660                      665                      670

Gln Tyr Ser Thr Gly Gln Val Ser Val Glu Ile Glu Trp Glu Leu Gln
    675                      680                      685

Lys Glu Asn Ser Lys Arg Trp Asn Pro Glu Ile Gln Tyr Thr Ser Asn
    690                      695                      700

Tyr Tyr Lys Ser Asn Asn Val Glu Phe Ala Val Asn Thr Glu Gly Val
    705                      710                      715                      720

Tyr Ser Glu Pro Arg Pro Ile Gly Thr Arg Tyr Leu Thr Arg Asn Leu
    725                      730                      735

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<210> SEQ ID NO 52

<211> LENGTH: 2211

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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: AVV mutant 9G453AG513A
<220> FEATURE:
<221> NAME/KEY: CDS
<222> LOCATION: (1)..(2211)

<400> SEQUENCE: 52

atg gct gcc gat ggt tat ctt cca gat tgg ctc gag gac aac ctt agt      48
Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser
1          5          10          15

gaa gga att cgc gag tgg tgg gct ttg aaa cct gga gcc cct caa ccc      96
Glu Gly Ile Arg Glu Trp Trp Ala Leu Lys Pro Gly Ala Pro Gln Pro
          20          25          30

aag gca aat caa caa cat caa gac aac gct cga ggt ctt gtg ctt ccg     144
Lys Ala Asn Gln Gln His Gln Asp Asn Ala Arg Gly Leu Val Leu Pro
          35          40          45

ggt tac aaa tac ctt gga ccc ggc aac gga ctc gac aag ggg gag ccg     192
Gly Tyr Lys Tyr Leu Gly Pro Gly Asn Gly Leu Asp Lys Gly Glu Pro
          50          55          60

gtc aac gca gca gac gcg gcg gcc ctc gag cac gac aag gcc tac gac     240
Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp
65          70          75          80

cag cag ctc aag gcc gga gac aac ccg tac ctc aag tac aac cac gcc     288
Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Lys Tyr Asn His Ala
          85          90          95

gac gcc gag ttc cag gag cgg ctc aaa gaa gat acg tct ttt ggg ggc     336
Asp Ala Glu Phe Gln Glu Arg Leu Lys Glu Asp Thr Ser Phe Gly Gly
          100          105          110

aac ctc ggg cga gca gtc ttc cag gcc aaa aag agg ctt ctt gaa cct     384
Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Leu Leu Glu Pro
          115          120          125

ctt ggt ctg gtt gag gaa gcg gct aag acg gct cct gga aag aag agg     432
Leu Gly Leu Val Glu Glu Ala Ala Lys Thr Ala Pro Gly Lys Lys Arg
          130          135          140

cct gta gag cag tct cct cag gaa ccg gac tcc tcc gcg ggt att ggc     480
Pro Val Glu Gln Ser Pro Gln Glu Pro Asp Ser Ser Ala Gly Ile Gly
145          150          155          160

aaa tcg ggt gca cag ccc gct aaa aag aga ctc aat ttc ggt cag act     528
Lys Ser Gly Ala Gln Pro Ala Lys Lys Arg Leu Asn Phe Gly Gln Thr
          165          170          175

ggc gac aca gag tca gtc cca gac cct caa cca atc gga gaa cct ccc     576
Gly Asp Thr Glu Ser Val Pro Asp Pro Gln Pro Ile Gly Glu Pro Pro
          180          185          190

gca gcc ccc tca ggt gtg gga tct ctt aca atg gct tca ggt ggt ggc     624
Ala Ala Pro Ser Gly Val Gly Ser Leu Thr Met Ala Ser Gly Gly Gly
          195          200          205

gca cca gtg gca gac aat aac gaa ggt gcc gat gga gtg ggt agt tcc     672
Ala Pro Val Ala Asp Asn Asn Glu Gly Ala Asp Gly Val Gly Ser Ser
          210          215          220

tcg gga aat tgg cat tgc gat tcc caa tgg ctg ggg gac aga gtc atc     720
Ser Gly Asn Trp His Cys Asp Ser Gln Trp Leu Gly Asp Arg Val Ile
225          230          235          240

acc acc agc acc cga acc tgg gcc ctg ccc acc tac aac aat cac ctc     768
Thr Thr Ser Thr Arg Thr Trp Ala Leu Pro Thr Tyr Asn Asn His Leu
          245          250          255

tac aag caa atc tcc aac agc aca tct gga gga tct tca aat gac aac     816
Tyr Lys Gln Ile Ser Asn Ser Thr Ser Gly Gly Ser Ser Asn Asp Asn
          260          265          270

gcc tac ttc ggc tac agc acc ccc tgg ggg tat ttt gac ttc aac aga     864

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Ala	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro	Trp	Gly	Tyr	Phe	Asp	Phe	Asn	Arg	
		275					280					285				
ttc	cac	tgc	cac	ttc	tca	cca	cgt	gac	tgg	cag	cga	ctc	atc	aac	aac	912
Phe	His	Cys	His	Phe	Ser	Pro	Arg	Asp	Trp	Gln	Arg	Leu	Ile	Asn	Asn	
		290				295					300					
aac	tgg	gga	ttc	cgg	cct	aag	cga	ctc	aac	ttc	aag	ctc	ttc	aac	att	960
Asn	Trp	Gly	Phe	Arg	Pro	Lys	Arg	Leu	Asn	Phe	Lys	Leu	Phe	Asn	Ile	
		305			310					315					320	
cag	gtc	aaa	gag	gtt	acg	gac	aac	aat	gga	gtc	aag	acc	atc	gcc	aat	1008
Gln	Val	Lys	Glu	Val	Thr	Asp	Asn	Asn	Gly	Val	Lys	Thr	Ile	Ala	Asn	
				325					330					335		
aac	ctt	acc	agc	acg	gtc	cag	gtc	ttc	acg	gac	tca	gac	tat	cag	ctc	1056
Asn	Leu	Thr	Ser	Thr	Val	Gln	Val	Phe	Thr	Asp	Ser	Asp	Tyr	Gln	Leu	
			340					345					350			
ccg	tac	gtg	ctc	ggg	tcg	gct	cac	gag	ggc	tgc	ctc	ccg	ccg	ttc	cca	1104
Pro	Tyr	Val	Leu	Gly	Ser	Ala	His	Glu	Gly	Cys	Leu	Pro	Pro	Phe	Pro	
		355				360					365					
gcg	gac	gtt	ttc	atg	att	cct	cag	tac	ggg	tat	ctg	acg	ctt	aat	gat	1152
Ala	Asp	Val	Phe	Met	Ile	Pro	Gln	Tyr	Gly	Tyr	Leu	Thr	Leu	Asn	Asp	
		370				375					380					
gga	agc	cag	gcc	gtg	ggt	cgt	tcg	tcc	ttt	tac	tgc	ctg	gaa	tat	ttc	1200
Gly	Ser	Gln	Ala	Val	Gly	Arg	Ser	Ser	Phe	Tyr	Cys	Leu	Glu	Tyr	Phe	
		385			390					395					400	
ccg	tcg	caa	atg	cta	aga	acg	ggt	aac	aac	ttc	cag	ttc	agc	tac	gag	1248
Pro	Ser	Gln	Met	Leu	Arg	Thr	Gly	Asn	Asn	Phe	Gln	Phe	Ser	Tyr	Glu	
			405					410						415		
ttt	gag	aac	gta	cct	ttc	cat	agc	agc	tac	gct	cac	agc	caa	agc	ctg	1296
Phe	Glu	Asn	Val	Pro	Phe	His	Ser	Ser	Tyr	Ala	His	Ser	Gln	Ser	Leu	
			420				425						430			
gac	cga	cta	atg	aat	cca	ctc	atc	gac	caa	tac	ttg	tac	tat	ctc	tca	1344
Asp	Arg	Leu	Met	Asn	Pro	Leu	Ile	Asp	Gln	Tyr	Leu	Tyr	Tyr	Leu	Ser	
		435				440					445					
aag	act	att	aac	gct	tct	gga	cag	aat	caa	caa	acg	cta	aaa	ttc	agt	1392
Lys	Thr	Ile	Asn	Ala	Ser	Gly	Gln	Asn	Gln	Gln	Thr	Leu	Lys	Phe	Ser	
		450				455					460					
gtg	gcc	gga	ccc	agc	aac	atg	gct	gtc	cag	gga	aga	aac	tac	ata	cct	1440
Val	Ala	Gly	Pro	Ser	Asn	Met	Ala	Val	Gln	Gly	Arg	Asn	Tyr	Ile	Pro	
		465			470					475				480		
gga	ccc	agc	tac	cga	caa	caa	cgt	gtc	tca	acc	act	gtg	act	caa	aac	1488
Gly	Pro	Ser	Tyr	Arg	Gln	Gln	Arg	Val	Ser	Thr	Thr	Val	Thr	Gln	Asn	
			485					490						495		
aac	aac	agc	gaa	ttt	gct	tgg	cct	gga	gct	tct	tct	tgg	gct	ctc	aat	1536
Asn	Asn	Ser	Glu	Phe	Ala	Trp	Pro	Gly	Ala	Ser	Ser	Trp	Ala	Leu	Asn	
			500				505						510			
gca	cgt	aat	agc	ttg	atg	aat	cct	gga	cct	gct	atg	gcc	agc	cac	aaa	1584
Ala	Arg	Asn	Ser	Leu	Met	Asn	Pro	Gly	Pro	Ala	Met	Ala	Ser	His	Lys	
		515					520					525				
gaa	gga	gag	gac	cgt	ttc	ttt	cct	ttg	tct	gga	tct	tta	att	ttt	ggc	1632
Glu	Gly	Glu	Asp	Arg	Phe	Phe	Pro	Leu	Ser	Gly	Ser	Leu	Ile	Phe	Gly	
		530				535					540					
aaa	caa	gga	act	gga	aga	gac	aac	gtg	gat	gcg	gac	aaa	gtc	atg	ata	1680
Lys	Gln	Gly	Thr	Gly	Arg	Asp	Asn	Val	Asp	Ala	Asp	Lys	Val	Met	Ile	
		545			550					555				560		
acc	aac	gaa	gaa	gaa	att	aaa	act	act	aac	ccg	gta	gca	acg	gag	tcc	1728
Thr	Asn	Glu	Glu	Glu	Ile	Lys	Thr	Thr	Asn	Pro	Val	Ala	Thr	Glu	Ser	
			565					570						575		
tat	gga	caa	gtg	gcc	aca	aac	cac	cag	agt	gcc	caa	gca	cag	gcg	cag	1776
Tyr	Gly	Gln	Val	Ala	Thr	Asn	His	Gln	Ser	Ala	Gln	Ala	Gln	Ala	Gln	
			580					585						590		

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acc ggc tgg gtt caa aac caa gga ata ctt ccg ggt atg gtt tgg cag	1824
Thr Gly Trp Val Gln Asn Gln Gly Ile Leu Pro Gly Met Val Trp Gln	
595 600 605	
gac aga gat gtg tac ctg caa gga ccc att tgg gcc aaa att cct cac	1872
Asp Arg Asp Val Tyr Leu Gln Gly Pro Ile Trp Ala Lys Ile Pro His	
610 615 620	
acg gac ggc aac ttt cac cct tct ccg ctg atg gga ggg ttt gga atg	1920
Thr Asp Gly Asn Phe His Pro Ser Pro Leu Met Gly Gly Phe Gly Met	
625 630 635 640	
aag cac ccg cct cct cag atc ctc atc aaa aac aca cct gta cct gcg	1968
Lys His Pro Pro Pro Gln Ile Leu Ile Lys Asn Thr Pro Val Pro Ala	
645 650 655	
gat cct cca acg gcc ttc aac aag gac aag ctg aac tct ttc atc acc	2016
Asp Pro Pro Thr Ala Phe Asn Lys Asp Lys Leu Asn Ser Phe Ile Thr	
660 665 670	
cag tat tct act ggc caa gtc agc gtg gag atc gag tgg gag ctg cag	2064
Gln Tyr Ser Thr Gly Gln Val Ser Val Glu Ile Glu Trp Glu Leu Gln	
675 680 685	
aag gaa aac agc aag cgc tgg aac ccg gag atc cag tac act tcc aac	2112
Lys Glu Asn Ser Lys Arg Trp Asn Pro Glu Ile Gln Tyr Thr Ser Asn	
690 695 700	
tat tac aag tct aat aat gtt gaa ttt gct gtt aat act gaa ggt gta	2160
Tyr Tyr Lys Ser Asn Asn Val Glu Phe Ala Val Asn Thr Glu Gly Val	
705 710 715 720	
tat agt gaa ccc cgc ccc att ggc acc aga tac ctg act cgt aat ctg	2208
Tyr Ser Glu Pro Arg Pro Ile Gly Thr Arg Tyr Leu Thr Arg Asn Leu	
725 730 735	
taa	2211

<210> SEQ ID NO 53

<211> LENGTH: 736

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 53

Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser	
1 5 10 15	
Glu Gly Ile Arg Glu Trp Trp Ala Leu Lys Pro Gly Ala Pro Gln Pro	
20 25 30	
Lys Ala Asn Gln Gln His Gln Asp Asn Ala Arg Gly Leu Val Leu Pro	
35 40 45	
Gly Tyr Lys Tyr Leu Gly Pro Gly Asn Gly Leu Asp Lys Gly Glu Pro	
50 55 60	
Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp	
65 70 75 80	
Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Lys Tyr Asn His Ala	
85 90 95	
Asp Ala Glu Phe Gln Glu Arg Leu Lys Glu Asp Thr Ser Phe Gly Gly	
100 105 110	
Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Leu Leu Glu Pro	
115 120 125	
Leu Gly Leu Val Glu Glu Ala Ala Lys Thr Ala Pro Gly Lys Lys Arg	
130 135 140	
Pro Val Glu Gln Ser Pro Gln Glu Pro Asp Ser Ser Ala Gly Ile Gly	
145 150 155 160	
Lys Ser Gly Ala Gln Pro Ala Lys Lys Arg Leu Asn Phe Gly Gln Thr	

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165								170				175			
Gly	Asp	Thr	Glu	Ser	Val	Pro	Asp	Pro	Gln	Pro	Ile	Gly	Glu	Pro	Pro
			180				185						190		
Ala	Ala	Pro	Ser	Gly	Val	Gly	Ser	Leu	Thr	Met	Ala	Ser	Gly	Gly	Gly
			195				200						205		
Ala	Pro	Val	Ala	Asp	Asn	Asn	Glu	Gly	Ala	Asp	Gly	Val	Gly	Ser	Ser
			210				215						220		
Ser	Gly	Asn	Trp	His	Cys	Asp	Ser	Gln	Trp	Leu	Gly	Asp	Arg	Val	Ile
			225				230						235		
Thr	Thr	Ser	Thr	Arg	Thr	Trp	Ala	Leu	Pro	Thr	Tyr	Asn	Asn	His	Leu
			245				250						255		
Tyr	Lys	Gln	Ile	Ser	Asn	Ser	Thr	Ser	Gly	Gly	Ser	Ser	Asn	Asp	Asn
			260				265						270		
Ala	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro	Trp	Gly	Tyr	Phe	Asp	Phe	Asn	Arg
			275				280						285		
Phe	His	Cys	His	Phe	Ser	Pro	Arg	Asp	Trp	Gln	Arg	Leu	Ile	Asn	Asn
			290				295						300		
Asn	Trp	Gly	Phe	Arg	Pro	Lys	Arg	Leu	Asn	Phe	Lys	Leu	Phe	Asn	Ile
			305				310						315		
Gln	Val	Lys	Glu	Val	Thr	Asp	Asn	Asn	Gly	Val	Lys	Thr	Ile	Ala	Asn
			325				330						335		
Asn	Leu	Thr	Ser	Thr	Val	Gln	Val	Phe	Thr	Asp	Ser	Asp	Tyr	Gln	Leu
			340				345						350		
Pro	Tyr	Val	Leu	Gly	Ser	Ala	His	Glu	Gly	Cys	Leu	Pro	Pro	Phe	Pro
			355				360						365		
Ala	Asp	Val	Phe	Met	Ile	Pro	Gln	Tyr	Gly	Tyr	Leu	Thr	Leu	Asn	Asp
			370				375						380		
Gly	Ser	Gln	Ala	Val	Gly	Arg	Ser	Ser	Phe	Tyr	Cys	Leu	Glu	Tyr	Phe
			385				390						395		
Pro	Ser	Gln	Met	Leu	Arg	Thr	Gly	Asn	Asn	Phe	Gln	Phe	Ser	Tyr	Glu
			405				410						415		
Phe	Glu	Asn	Val	Pro	Phe	His	Ser	Ser	Tyr	Ala	His	Ser	Gln	Ser	Leu
			420				425						430		
Asp	Arg	Leu	Met	Asn	Pro	Leu	Ile	Asp	Gln	Tyr	Leu	Tyr	Tyr	Leu	Ser
			435				440						445		
Lys	Thr	Ile	Asn	Ala	Ser	Gly	Gln	Asn	Gln	Gln	Thr	Leu	Lys	Phe	Ser
			450				455						460		
Val	Ala	Gly	Pro	Ser	Asn	Met	Ala	Val	Gln	Gly	Arg	Asn	Tyr	Ile	Pro
			465				470						475		
Gly	Pro	Ser	Tyr	Arg	Gln	Gln	Arg	Val	Ser	Thr	Thr	Val	Thr	Gln	Asn
			485				490						495		
Asn	Asn	Ser	Glu	Phe	Ala	Trp	Pro	Gly	Ala	Ser	Ser	Trp	Ala	Leu	Asn
			500				505						510		
Ala	Arg	Asn	Ser	Leu	Met	Asn	Pro	Gly	Pro	Ala	Met	Ala	Ser	His	Lys
			515				520						525		
Glu	Gly	Glu	Asp	Arg	Phe	Phe	Pro	Leu	Ser	Gly	Ser	Leu	Ile	Phe	Gly
			530				535						540		
Lys	Gln	Gly	Thr	Gly	Arg	Asp	Asn	Val	Asp	Ala	Asp	Lys	Val	Met	Ile
			545				550						555		
Thr	Asn	Glu	Glu	Glu	Ile	Lys	Thr	Thr	Asn	Pro	Val	Ala	Thr	Glu	Ser
			565				570						575		
Tyr	Gly	Gln	Val	Ala	Thr	Asn	His	Gln	Ser	Ala	Gln	Ala	Gln	Ala	Gln
			580				585						590		

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Thr Gly Trp Val Gln Asn Gln Gly Ile Leu Pro Gly Met Val Trp Gln
595 600 605

Asp Arg Asp Val Tyr Leu Gln Gly Pro Ile Trp Ala Lys Ile Pro His
610 615 620

Thr Asp Gly Asn Phe His Pro Ser Pro Leu Met Gly Gly Phe Gly Met
625 630 635 640

Lys His Pro Pro Pro Gln Ile Leu Ile Lys Asn Thr Pro Val Pro Ala
645 650 655

Asp Pro Pro Thr Ala Phe Asn Lys Asp Lys Leu Asn Ser Phe Ile Thr
660 665 670

Gln Tyr Ser Thr Gly Gln Val Ser Val Glu Ile Glu Trp Glu Leu Gln
675 680 685

Lys Glu Asn Ser Lys Arg Trp Asn Pro Glu Ile Gln Tyr Thr Ser Asn
690 695 700

Tyr Tyr Lys Ser Asn Asn Val Glu Phe Ala Val Asn Thr Glu Gly Val
705 710 715 720

Tyr Ser Glu Pro Arg Pro Ile Gly Thr Arg Tyr Leu Thr Arg Asn Leu
725 730 735

<210> SEQ ID NO 54
<211> LENGTH: 2211
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: AVV mutant 9G330AG453AG513A
<220> FEATURE:
<221> NAME/KEY: CDS
<222> LOCATION: (1)..(2211)

<400> SEQUENCE: 54

atg gct gcc gat ggt tat ctt cca gat tgg ctc gag gac aac ctt agt	48
Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser	
1 5 10 15	
gaa gga att cgc gag tgg tgg gct ttg aaa cct gga gcc cct caa ccc	96
Glu Gly Ile Arg Glu Trp Trp Ala Leu Lys Pro Gly Ala Pro Gln Pro	
20 25 30	
aag gca aat caa caa cat caa gac aac gct cga ggt ctt gtg ctt ccg	144
Lys Ala Asn Gln Gln His Gln Asp Asn Ala Arg Gly Leu Val Leu Pro	
35 40 45	
ggt tac aaa tac ctt gga ccc gcc aac gga ctc gac aag ggg gag ccg	192
Gly Tyr Lys Tyr Leu Gly Pro Gly Asn Gly Leu Asp Lys Gly Glu Pro	
50 55 60	
gtc aac gca gca gac gcg gcg gcc ctc gag cac gac aag gcc tac gac	240
Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp	
65 70 75 80	
cag cag ctc aag gcc gga gac aac ccg tac ctc aag tac aac cac gcc	288
Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Lys Tyr Asn His Ala	
85 90 95	
gac gcc gag ttc cag gag cgg ctc aaa gaa gat acg tct ttt ggg ggc	336
Asp Ala Glu Phe Gln Glu Arg Leu Lys Glu Asp Thr Ser Phe Gly Gly	
100 105 110	
aac ctc ggg cga gca gtc ttc cag gcc aaa aag agg ctt ctt gaa cct	384
Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Leu Leu Glu Pro	
115 120 125	
ctt ggt ctg gtt gag gaa gcg gct aag acg gct cct gga aag aag agg	432
Leu Gly Leu Val Glu Glu Ala Ala Lys Thr Ala Pro Gly Lys Lys Arg	
130 135 140	
cct gta gag cag tct cct cag gaa ccg gac tcc tcc gcg ggt att ggc	480
Pro Val Glu Gln Ser Pro Gln Glu Pro Asp Ser Ser Ala Gly Ile Gly	

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145	150	155	160	
aaa tcg ggt gca cag ccc gct aaa aag aga ctc aat ttc ggt cag act				528
Lys Ser Gly Ala Gln Pro Ala Lys Lys Arg Leu Asn Phe Gly Gln Thr	165	170	175	
ggc gac aca gag tca gtc cca gac cct caa cca atc gga gaa cct ccc				576
Gly Asp Thr Glu Ser Val Pro Asp Pro Gln Pro Ile Gly Glu Pro Pro	180	185	190	
gca gcc ccc tca ggt gtg gga tct ctt aca atg gct tca ggt ggt ggc				624
Ala Ala Pro Ser Gly Val Gly Ser Leu Thr Met Ala Ser Gly Gly Gly	195	200	205	
gca cca gtg gca gac aat aac gaa ggt gcc gat gga gtg ggt agt tcc				672
Ala Pro Val Ala Asp Asn Asn Glu Gly Ala Asp Gly Val Gly Ser Ser	210	215	220	
tcg gga aat tgg cat tgc gat tcc caa tgg ctg ggg gac aga gtc atc				720
Ser Gly Asn Trp His Cys Asp Ser Gln Trp Leu Gly Asp Arg Val Ile	225	230	235	240
acc acc agc acc cga acc tgg gcc ctg ccc acc tac aac aat cac ctc				768
Thr Thr Ser Thr Arg Thr Trp Ala Leu Pro Thr Tyr Asn Asn His Leu	245	250	255	
tac aag caa atc tcc aac agc aca tct gga gga tct tca aat gac aac				816
Tyr Lys Gln Ile Ser Asn Ser Thr Ser Gly Gly Ser Ser Asn Asp Asn	260	265	270	
gcc tac ttc ggc tac agc acc ccc tgg ggg tat ttt gac ttc aac aga				864
Ala Tyr Phe Gly Tyr Ser Thr Pro Trp Gly Tyr Phe Asp Phe Asn Arg	275	280	285	
ttc cac tgc cac ttc tca cca cgt gac tgg cag cga ctc atc aac aac				912
Phe His Cys His Phe Ser Pro Arg Asp Trp Gln Arg Leu Ile Asn Asn	290	295	300	
aac tgg gga ttc cgg cct aag cga ctc aac ttc aag ctc ttc aac att				960
Asn Trp Gly Phe Arg Pro Lys Arg Leu Asn Phe Lys Leu Phe Asn Ile	305	310	315	320
cag gtc aaa gag gtt acg gac aac aat gca gtc aag acc atc gcc aat				1008
Gln Val Lys Glu Val Thr Asp Asn Asn Ala Val Lys Thr Ile Ala Asn	325	330	335	
aac ctt acc agc acg gtc cag gtc ttc acg gac tca gac tat cag ctc				1056
Asn Leu Thr Ser Thr Val Gln Val Phe Thr Asp Ser Asp Tyr Gln Leu	340	345	350	
ccg tac gtg ctc ggg tgc gct cac gag ggc tgc ctc ccg ccg ttc cca				1104
Pro Tyr Val Leu Gly Ser Ala His Glu Gly Cys Leu Pro Pro Phe Pro	355	360	365	
gcg gac gtt ttc atg att cct cag tac ggg tat ctg acg ctt aat gat				1152
Ala Asp Val Phe Met Ile Pro Gln Tyr Gly Tyr Leu Thr Leu Asn Asp	370	375	380	
gga agc cag gcc gtg ggt cgt tgc tcc ttt tac tgc ctg gaa tat ttc				1200
Gly Ser Gln Ala Val Gly Arg Ser Ser Phe Tyr Cys Leu Glu Tyr Phe	385	390	395	400
ccg tcg caa atg cta aga acg ggt aac aac ttc cag ttc agc tac gag				1248
Pro Ser Gln Met Leu Arg Thr Gly Asn Asn Phe Gln Phe Ser Tyr Glu	405	410	415	
ttt gag aac gta cct ttc cat agc agc tac gct cac agc caa agc ctg				1296
Phe Glu Asn Val Pro Phe His Ser Ser Tyr Ala His Ser Gln Ser Leu	420	425	430	
gac cga cta atg aat cca ctc atc gac caa tac ttg tac tat ctc tca				1344
Asp Arg Leu Met Asn Pro Leu Ile Asp Gln Tyr Leu Tyr Tyr Leu Ser	435	440	445	
aag act att aac gct tct gga cag aat caa caa acg cta aaa ttc agt				1392
Lys Thr Ile Asn Ala Ser Gly Gln Asn Gln Gln Thr Leu Lys Phe Ser	450	455	460	
gtg gcc gga ccc agc aac atg gct gtc cag gga aga aac tac ata cct				1440

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Val	Ala	Gly	Pro	Ser	Asn	Met	Ala	Val	Gln	Gly	Arg	Asn	Tyr	Ile	Pro		
465					470					475					480		
gga ccc agc tac cga caa caa cgt gtc tca acc act gtg act caa aac 1488																	
Gly	Pro	Ser	Tyr	Arg	Gln	Gln	Arg	Val	Ser	Thr	Thr	Val	Thr	Gln	Asn		
				485				490						495			
aac aac agc gaa ttt gct tgg cct gga gct tct tct tgg gct ctc aat 1536																	
Asn	Asn	Ser	Glu	Phe	Ala	Trp	Pro	Gly	Ala	Ser	Ser	Trp	Ala	Leu	Asn		
			500					505					510				
gca cgt aat agc ttg atg aat cct gga cct gct atg gcc agc cac aaa 1584																	
Ala	Arg	Asn	Ser	Leu	Met	Asn	Pro	Gly	Pro	Ala	Met	Ala	Ser	His	Lys		
			515				520					525					
gaa gga gag gac cgt ttc ttt cct ttg tct gga tct tta att ttt ggc 1632																	
Glu	Gly	Glu	Asp	Arg	Phe	Phe	Pro	Leu	Ser	Gly	Ser	Leu	Ile	Phe	Gly		
			530			535					540						
aaa caa gga act gga aga gac aac gtg gat gcg gac aaa gtc atg ata 1680																	
Lys	Gln	Gly	Thr	Gly	Arg	Asp	Asn	Val	Asp	Ala	Asp	Lys	Val	Met	Ile		
545					550				555					560			
acc aac gaa gaa gaa att aaa act act aac ccg gta gca acg gag tcc 1728																	
Thr	Asn	Glu	Glu	Ile	Lys	Thr	Thr	Asn	Pro	Val	Ala	Thr	Glu	Ser			
			565					570					575				
tat gga caa gtg gcc aca aac cac cag agt gcc caa gca cag gcg cag 1776																	
Tyr	Gly	Gln	Val	Ala	Thr	Asn	His	Gln	Ser	Ala	Gln	Ala	Gln	Ala	Gln		
			580				585					590					
acc gcc tgg gtt caa aac caa gga ata ctt ccg ggt atg gtt tgg cag 1824																	
Thr	Gly	Trp	Val	Gln	Asn	Gln	Gly	Ile	Leu	Pro	Gly	Met	Val	Trp	Gln		
			595			600						605					
gac aga gat gtg tac ctg caa gga ccc att tgg gcc aaa att cct cac 1872																	
Asp	Arg	Asp	Val	Tyr	Leu	Gln	Gly	Pro	Ile	Trp	Ala	Lys	Ile	Pro	His		
			610			615					620						
acg gac gcc aac ttt cac cct tct ccg ctg atg gga ggg ttt gga atg 1920																	
Thr	Asp	Gly	Asn	Phe	His	Pro	Ser	Pro	Leu	Met	Gly	Gly	Phe	Gly	Met		
			625		630					635				640			
aag cac ccg cct cct cag atc ctc atc aaa aac aca cct gta cct gcg 1968																	
Lys	His	Pro	Pro	Pro	Gln	Ile	Leu	Ile	Lys	Asn	Thr	Pro	Val	Pro	Ala		
					645				650					655			
gat cct cca acg gcc ttc aac aag gac aag ctg aac tct ttc atc acc 2016																	
Asp	Pro	Pro	Thr	Ala	Phe	Asn	Lys	Asp	Lys	Leu	Asn	Ser	Phe	Ile	Thr		
			660				665						670				
cag tat tct act gcc caa gtc agc gtg gag atc gag tgg gag ctg cag 2064																	
Gln	Tyr	Ser	Thr	Gly	Gln	Val	Ser	Val	Glu	Ile	Glu	Trp	Glu	Leu	Gln		
			675			680					685						
aag gaa aac agc aag cgc tgg aac ccg gag atc cag tac act tcc aac 2112																	
Lys	Glu	Asn	Ser	Lys	Arg	Trp	Asn	Pro	Glu	Ile	Gln	Tyr	Thr	Ser	Asn		
			690			695					700						
tat tac aag tct aat aat gtt gaa ttt gct gtt aat act gaa ggt gta 2160																	
Tyr	Tyr	Lys	Ser	Asn	Asn	Val	Glu	Phe	Ala	Val	Asn	Thr	Glu	Gly	Val		
			705		710					715				720			
tat agt gaa ccc cgc ccc att gcc acc aga tac ctg act cgt aat ctg 2208																	
Tyr	Ser	Glu	Pro	Arg	Pro	Ile	Gly	Thr	Arg	Tyr	Leu	Thr	Arg	Asn	Leu		
				725				730						735			
taa 2211																	

<210> SEQ ID NO 55

<211> LENGTH: 736

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 55

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Met	Ala	Ala	Asp	Gly	Tyr	Leu	Pro	Asp	Trp	Leu	Glu	Asp	Asn	Leu	Ser	1	5	10	15
Glu	Gly	Ile	Arg	Glu	Trp	Trp	Ala	Leu	Lys	Pro	Gly	Ala	Pro	Gln	Pro	20	25	30	
Lys	Ala	Asn	Gln	Gln	His	Gln	Asp	Asn	Ala	Arg	Gly	Leu	Val	Leu	Pro	35	40	45	
Gly	Tyr	Lys	Tyr	Leu	Gly	Pro	Gly	Asn	Gly	Leu	Asp	Lys	Gly	Glu	Pro	50	55	60	
Val	Asn	Ala	Ala	Asp	Ala	Ala	Ala	Leu	Glu	His	Asp	Lys	Ala	Tyr	Asp	65	70	75	80
Gln	Gln	Leu	Lys	Ala	Gly	Asp	Asn	Pro	Tyr	Leu	Lys	Tyr	Asn	His	Ala	85	90	95	
Asp	Ala	Glu	Phe	Gln	Glu	Arg	Leu	Lys	Glu	Asp	Thr	Ser	Phe	Gly	Gly	100	105	110	
Asn	Leu	Gly	Arg	Ala	Val	Phe	Gln	Ala	Lys	Lys	Arg	Leu	Leu	Glu	Pro	115	120	125	
Leu	Gly	Leu	Val	Glu	Glu	Ala	Ala	Lys	Thr	Ala	Pro	Gly	Lys	Lys	Arg	130	135	140	
Pro	Val	Glu	Gln	Ser	Pro	Gln	Glu	Pro	Asp	Ser	Ser	Ala	Gly	Ile	Gly	145	150	155	160
Lys	Ser	Gly	Ala	Gln	Pro	Ala	Lys	Lys	Arg	Leu	Asn	Phe	Gly	Gln	Thr	165	170	175	
Gly	Asp	Thr	Glu	Ser	Val	Pro	Asp	Pro	Gln	Pro	Ile	Gly	Glu	Pro	Pro	180	185	190	
Ala	Ala	Pro	Ser	Gly	Val	Gly	Ser	Leu	Thr	Met	Ala	Ser	Gly	Gly	Gly	195	200	205	
Ala	Pro	Val	Ala	Asp	Asn	Asn	Glu	Gly	Ala	Asp	Gly	Val	Gly	Ser	Ser	210	215	220	
Ser	Gly	Asn	Trp	His	Cys	Asp	Ser	Gln	Trp	Leu	Gly	Asp	Arg	Val	Ile	225	230	235	240
Thr	Thr	Ser	Thr	Arg	Thr	Trp	Ala	Leu	Pro	Thr	Tyr	Asn	Asn	His	Leu	245	250	255	
Tyr	Lys	Gln	Ile	Ser	Asn	Ser	Thr	Ser	Gly	Gly	Ser	Ser	Asn	Asp	Asn	260	265	270	
Ala	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro	Trp	Gly	Tyr	Phe	Asp	Phe	Asn	Arg	275	280	285	
Phe	His	Cys	His	Phe	Ser	Pro	Arg	Asp	Trp	Gln	Arg	Leu	Ile	Asn	Asn	290	295	300	
Asn	Trp	Gly	Phe	Arg	Pro	Lys	Arg	Leu	Asn	Phe	Lys	Leu	Phe	Asn	Ile	305	310	315	320
Gln	Val	Lys	Glu	Val	Thr	Asp	Asn	Asn	Ala	Val	Lys	Thr	Ile	Ala	Asn	325	330	335	
Asn	Leu	Thr	Ser	Thr	Val	Gln	Val	Phe	Thr	Asp	Ser	Asp	Tyr	Gln	Leu	340	345	350	
Pro	Tyr	Val	Leu	Gly	Ser	Ala	His	Glu	Gly	Cys	Leu	Pro	Pro	Phe	Pro	355	360	365	
Ala	Asp	Val	Phe	Met	Ile	Pro	Gln	Tyr	Gly	Tyr	Leu	Thr	Leu	Asn	Asp	370	375	380	
Gly	Ser	Gln	Ala	Val	Gly	Arg	Ser	Ser	Phe	Tyr	Cys	Leu	Glu	Tyr	Phe	385	390	395	400
Pro	Ser	Gln	Met	Leu	Arg	Thr	Gly	Asn	Asn	Phe	Gln	Phe	Ser	Tyr	Glu	405	410	415	
Phe	Glu	Asn	Val	Pro	Phe	His	Ser	Ser	Tyr	Ala	His	Ser	Gln	Ser	Leu				

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420					425					430					
Asp	Arg	Leu	Met	Asn	Pro	Leu	Ile	Asp	Gln	Tyr	Leu	Tyr	Tyr	Leu	Ser
	435						440					445			
Lys	Thr	Ile	Asn	Ala	Ser	Gly	Gln	Asn	Gln	Gln	Thr	Leu	Lys	Phe	Ser
	450					455						460			
Val	Ala	Gly	Pro	Ser	Asn	Met	Ala	Val	Gln	Gly	Arg	Asn	Tyr	Ile	Pro
465					470					475					480
Gly	Pro	Ser	Tyr	Arg	Gln	Gln	Arg	Val	Ser	Thr	Thr	Val	Thr	Gln	Asn
			485						490					495	
Asn	Asn	Ser	Glu	Phe	Ala	Trp	Pro	Gly	Ala	Ser	Ser	Trp	Ala	Leu	Asn
			500						505					510	
Ala	Arg	Asn	Ser	Leu	Met	Asn	Pro	Gly	Pro	Ala	Met	Ala	Ser	His	Lys
		515					520					525			
Glu	Gly	Glu	Asp	Arg	Phe	Phe	Pro	Leu	Ser	Gly	Ser	Leu	Ile	Phe	Gly
	530					535					540				
Lys	Gln	Gly	Thr	Gly	Arg	Asp	Asn	Val	Asp	Ala	Asp	Lys	Val	Met	Ile
545					550					555					560
Thr	Asn	Glu	Glu	Glu	Ile	Lys	Thr	Thr	Asn	Pro	Val	Ala	Thr	Glu	Ser
			565						570					575	
Tyr	Gly	Gln	Val	Ala	Thr	Asn	His	Gln	Ser	Ala	Gln	Ala	Gln	Ala	Gln
		580						585					590		
Thr	Gly	Trp	Val	Gln	Asn	Gln	Gly	Ile	Leu	Pro	Gly	Met	Val	Trp	Gln
	595					600						605			
Asp	Arg	Asp	Val	Tyr	Leu	Gln	Gly	Pro	Ile	Trp	Ala	Lys	Ile	Pro	His
	610					615					620				
Thr	Asp	Gly	Asn	Phe	His	Pro	Ser	Pro	Leu	Met	Gly	Gly	Phe	Gly	Met
625				630						635					640
Lys	His	Pro	Pro	Pro	Gln	Ile	Leu	Ile	Lys	Asn	Thr	Pro	Val	Pro	Ala
			645						650					655	
Asp	Pro	Pro	Thr	Ala	Phe	Asn	Lys	Asp	Lys	Leu	Asn	Ser	Phe	Ile	Thr
		660						665					670		
Gln	Tyr	Ser	Thr	Gly	Gln	Val	Ser	Val	Glu	Ile	Glu	Trp	Glu	Leu	Gln
	675					680						685			
Lys	Glu	Asn	Ser	Lys	Arg	Trp	Asn	Pro	Glu	Ile	Gln	Tyr	Thr	Ser	Asn
	690					695					700				
Tyr	Tyr	Lys	Ser	Asn	Asn	Val	Glu	Phe	Ala	Val	Asn	Thr	Glu	Gly	Val
705				710						715					720
Tyr	Ser	Glu	Pro	Arg	Pro	Ile	Gly	Thr	Arg	Tyr	Leu	Thr	Arg	Asn	Leu
			725						730					735	

<210> SEQ ID NO 56
 <211> LENGTH: 38
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: primer sequence

<400> SEQUENCE: 56

cgacaaccgg gcaaaaccag aatagcaact ttgccttg

38

<210> SEQ ID NO 57
 <211> LENGTH: 38
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: primer sequence

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<400> SEQUENCE: 57

ccaggcaaaag ttgctattct ggttttgccc ggttgtcg 38

<210> SEQ ID NO 58

<211> LENGTH: 36

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: primer sequence

<400> SEQUENCE: 58

gacaaccggg caaaacgaca atagcaactt tgcctg 36

<210> SEQ ID NO 59

<211> LENGTH: 36

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: primer sequence

<400> SEQUENCE: 59

caggcaaaagt tgctattgtc gttttgcccc gttgtc 36

<210> SEQ ID NO 60

<211> LENGTH: 29

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: primer sequence

<400> SEQUENCE: 60

ggaggcacgg cacagacgca gactctggg 29

<210> SEQ ID NO 61

<211> LENGTH: 29

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: primer sequence

<400> SEQUENCE: 61

cccagagtct gcgtctgtgc cgtgcctcc 29

<210> SEQ ID NO 62

<211> LENGTH: 30

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: primer sequence

<400> SEQUENCE: 62

caggaggcac ggagatagc cagactctgg 30

<210> SEQ ID NO 63

<211> LENGTH: 30

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: primer sequence

<400> SEQUENCE: 63

ccagagtctg cgtatctgcc gtgcctctg 30

<210> SEQ ID NO 64

<211> LENGTH: 29

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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: primer sequence

<400> SEQUENCE: 64

ctctctccga tgctcggttg gagatttgc 29

<210> SEQ ID NO 65
<211> LENGTH: 29
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: primer sequence

<400> SEQUENCE: 65

gcaaatctcc aacgcgacat cgggaggag 29

<210> SEQ ID NO 66
<211> LENGTH: 31
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: primer sequence

<400> SEQUENCE: 66

cccacggcct gactagcggt gttgagtgtt a 31

<210> SEQ ID NO 67
<211> LENGTH: 31
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: primer sequence

<400> SEQUENCE: 67

taacactcaa caacgctagt caggccgtgg g 31

<210> SEQ ID NO 68
<211> LENGTH: 43
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: primer sequence

<400> SEQUENCE: 68

ggattagcca atgaatttct tgcattcaga tggatttgg tcc 43

<210> SEQ ID NO 69
<211> LENGTH: 43
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: primer sequence

<400> SEQUENCE: 69

ggaccaata ccatctgaat gcaagaaatt cattggctaa tcc 43

<210> SEQ ID NO 70
<211> LENGTH: 39
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: primer sequence

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<220> FEATURE:
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<400> SEQUENCE: 83

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<220> FEATURE:
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<220> FEATURE:

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<220> FEATURE:

<223> OTHER INFORMATION: primer sequence

<400> SEQUENCE: 94

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<220> FEATURE:
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<400> SEQUENCE: 100

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<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 101

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<210> SEQ ID NO 102
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<400> SEQUENCE: 106

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<220> FEATURE:
<223> OTHER INFORMATION: primer sequence

<400> SEQUENCE: 107

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<212> TYPE: DNA
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<400> SEQUENCE: 108

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<220> FEATURE:
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<220> FEATURE:
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<400> SEQUENCE: 118

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<220> FEATURE:
<223> OTHER INFORMATION: primer sequence

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<223> OTHER INFORMATION: primer sequence

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<210> SEQ ID NO 127
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<210> SEQ ID NO 128
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<212> TYPE: DNA
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The invention claimed is:

1. A recombinant adeno-associated virus (rAAV) which comprises an AAVrh79 capsid comprising AAVrh79 vp1 proteins, AAVrh79 vp2 proteins, and AAVrh79 vp3 proteins comprising glutamic acid (E) at position 67 in the AAVrh79 vp1 proteins and arginine (R) at position 169 in the AAVrh79 vp1 and AAVrh79 vp2 proteins, wherein the AAVrh79 vp1 proteins contain amino acid modifications comprising at least 50% to 100% deamidated asparagines (N) in asparagine-glycine pairs in each of positions N57, N263, N385 and N514 relative to SEQ ID NO: 2, as determined by mass

spectrometry, the AAVrh79 vp2 proteins contain amino acid modifications comprising at least 50% to 100% deamidated asparagines (N) in asparagine-glycine pairs in each of positions N263, N385 and N514 relative to SEQ ID NO: 2, as determined by mass spectrometry, and the AAVrh79 vp3 proteins contain amino acid modifications comprising at least 50% to 100% deamidated asparagines (N) in asparagine-glycine pairs in each of positions N57, N263, N385, and N514 relative to SEQ ID NO: 2, as determined by mass spectrometry, wherein the deamidated asparagines are

deamidated to an aspartic acid, an isoaspartic acid, an interconverting aspartic acid/isoaspartic acid pair, or combinations thereof,

wherein the rAAV further comprises a vector genome in the AAVrh79 capsid, the vector genome comprising a nucleic acid molecule comprising AAV inverted terminal repeat (ITR) sequences and a non-AAV nucleic acid sequence encoding a product operably linked to sequences which direct expression of the product in a host cell.

2. The rAAV according to claim 1, wherein the capsid further comprises deamidated glutamine(s) which are deamidated to (α)-glutamic acid, γ-glutamic acid, an interconverting (α)-glutamic acid/γ-glutamic acid pair, or combinations thereof.

3. The rAAV according to claim 1, wherein the AAVrh79 capsid comprises subpopulations having one or more of:

- (a) at least 75% of asparagines (N) in asparagine-glycine pairs located at position 57 of the vp1 proteins are deamidated, based on the numbering of SEQ ID NO:2;
- (b) at least 75% of N in asparagine-glycine pairs in position 263 of the vp1, vp2 and vp3 proteins are deamidated, based on the residue numbering of the amino acid sequence of SEQ ID NO: 2;
- (c) at least 70% of N in asparagine-glycine pairs in position 385 of the vp1, vp2 and vp3 proteins are deamidated, based on the residue numbering of the amino acid sequence of SEQ ID NO: 2; and/or
- (d) at least 85% of N in asparagine-glycine pairs in position 514 of the vp1, vp2 and vp3 proteins are deamidated, based on the residue numbering of the amino acid sequence of SEQ ID NO: 2.

4. The rAAV according to claim 1, wherein the rAAVrh79 capsid comprises a subpopulation of vp1 in which 75% to 100% of the N at position 57 of the vp1 proteins are deamidated, as determined using mass spectrometry.

5. The rAAV according to claim 1, wherein the rAAVrh79 capsid comprises subpopulations of vp1 proteins, vp2 proteins, and/or vp3 proteins in which 75% to 100% of the N at position 263, based on the numbering of SEQ ID NO:2, are deamidated as determined using mass spectrometry.

6. The rAAV according to claim 1, wherein the rAAVrh79 capsid comprises subpopulations of vp1 proteins, vp2 proteins, and/or vp3 proteins in which 75% to 100% of the N at position 385, based on the numbering of SEQ ID NO:2, are deamidated as determined using mass spectrometry.

7. The rAAV according to claim 1, wherein the rAAVrh79 capsid comprises subpopulations of vp1 proteins, vp2 proteins, and/or vp3 proteins in which 75% to 100% of the N at position 514, based on the numbering of SEQ ID NO:2, are deamidated.

8. The rAAV according to claim 1, wherein the nucleic acid sequence encoding the vp1 proteins, vp2 proteins and/or vp3 proteins is SEQ ID NO: 1, or a nucleic acid sequence at least 80% to at least 99% identical to SEQ ID NO: 1 which encodes the amino acid sequence of SEQ ID NO:2.

9. The rAAV according to claim 8, wherein the nucleic acid sequence is at least 80% to 97% identical to SEQ ID NO: 1.

10. The rAAV according to claim 1, wherein the rAAVrh79 capsid comprises subpopulations of vp1 proteins, vp2 proteins and/or vp3 proteins which further comprise 1% to about 40% deamidation in at least one or more of positions N94, N254, N305, N410, N479, N653, or combinations thereof.

11. The rAAV according to claim 1, wherein the rAAVrh79 capsid comprises subpopulations of vp1 proteins, vp2 proteins and/or vp3 proteins which further comprise one or more modifications consisting of: acetylated lysine, phosphorylated serine and/or threonine, isomerized aspartic acid, oxidized tryptophan and/or methionine, or an amidated amino acid.

12. The rAAV according to claim 1, wherein the AAV ITR sequences are a 5' ITR and a 3' ITR from AAV2.

13. A composition comprising an AAV according to claim 1 and a pharmaceutical carrier, excipient and/or diluent.

14. A production cell in vitro comprising the rAAV according to claim 1.

15. An rAAV production system useful for producing an rAAV according to claim 1 in a production cell, wherein the production system comprises:

- (a) an AAV capsid nucleic acid sequence encoding the amino acid sequence of SEQ ID NO:2;
- (b) a nucleic acid molecule suitable for packaging into the AAV capsid, said nucleic acid molecule comprising at least one AAV inverted terminal repeat (ITR), a functional rep gene and a non-AAV nucleic acid sequence encoding a gene product operably linked to sequences which direct expression of the product in a host cell; and
- (c) sufficient AAV rep functions and helper functions to permit packaging of the nucleic acid molecule into the AAV capsid.

16. The system according to claim 15, wherein the nucleic acid sequence of (a) comprises at least SEQ ID NO: 1, or a nucleic acid sequence at least 70% to at least 99% identical to SEQ ID NO: 1 which encodes the amino acid sequence of SEQ ID NO:2.

17. The system according to claim 15, wherein the production cell culture comprises human embryonic kidney 293 cells.

18. The system according to claim 15, wherein the AAV rep is from AAV2.

19. The rAAV according to claim 1, wherein the AAVrh79 capsid comprises less than 1% deamidation at positions other than N57, N263, N365 and/or N514.

20. The rAAV according to claim 1, wherein the AAVrh79 vp1 proteins contain amino acid modification comprising at least 70% to 100% deamidated asparagines (N) in asparagine-glycine pairs in each of positions N57, N263, N385 and N514 relative to SEQ ID NO: 2, as determined by mass spectrometry, the AAVrh79 vp2 proteins contain amino acid modification comprising at least 70% to 100% deamidated asparagines (N) in asparagine-glycine pairs in each of positions N263, N385 and N514 relative to SEQ ID NO: 2, as determined by mass spectrometry, and the AAVrh79 vp3 proteins contain amino acid modifications comprising at least 70% to 100% deamidated asparagines (N) in asparagine-glycine pairs in each of positions N263, N385, and N514 relative to SEQ ID NO: 2, as determined by mass spectrometry.

21. A recombinant adeno-associated virus (rAAV) which comprises an AAVrh79 capsid comprising AAVrh79 vp1 proteins, AAVrh79 vp2 proteins, and AAVrh79 vp3 proteins produced by expression of a nucleic acid sequence which encodes the amino acid sequence of 1 to 738 of SEQ ID NO: 2,

wherein the rAAV further comprises a vector genome in the AAVrh79 capsid, the vector genome comprising a nucleic acid molecule comprising AAV inverted terminal repeat (ITR) sequences and a non-AAV nucleic acid

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sequence encoding a product operably linked to sequences which direct expression of the product in a host cell.

22. The rAAV according to claim 21, wherein the AAVrh79 capsid comprising AAVrh79 vp1 proteins, AAVrh79 vp2 proteins, and AAVrh79 vp3 proteins is produced by expression from SEQ ID NO: 1.

23. A composition comprising an AAV according to claim 21 and a pharmaceutical carrier, excipient and/or diluent.

24. A method of generating a recombinant AAV comprising an AAV capsid comprising the steps of culturing a host cell containing:

(a) a nucleic acid sequence encoding an AAVrh79 vp1 capsid protein having amino acids 1 to 738 of SEQ ID NO: 2, wherein the AAVrh79 vp1 coding sequence is operably linked to regulatory sequences which permit expression of AAVrh79 vp1 proteins, AAVrh79 vp2 proteins and AAVrh79 vp3 proteins in the cultured host cell;

(b) a functional rep gene;

(c) a nucleic acid molecule which is a vector genome having a nucleic acid sequence comprising an AAV 5' inverted terminal repeats (ITRs), an expression cassette

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comprising a non-AAV nucleic acid sequence encoding a product, and an AAV 3' ITR sequence and a transgene; and

(d) sufficient helper functions to permit packaging of the vector genome into the AAV capsid.

25. A cultured host cell comprising a recombinant nucleic acid molecule comprising:

(a) a nucleic acid sequence encoding an AAVrh79 vp1 capsid protein having amino acids 1 to 738, of SEQ ID NO: 2, wherein the AAVrh79 vp1 coding sequence is operably linked to regulatory sequences which permit expression of AAVrh79 vp1 proteins, AAVrh79 vp2 proteins and AAVrh79 vp3 proteins in the cultured host cell;

(b) a functional rep gene;

(c) a nucleic acid molecule which is a vector genome having a nucleic acid sequence comprising an AAV 5' inverted terminal repeat (ITR), an expression cassette comprising a non-AAV nucleic acid sequence encoding a product, and an AAV 3' ITR sequence and a transgene; and

(d) sufficient helper functions to permit packaging of the vector genome into an AAV capsid.

* * * * *